

gilmore-lie-groups: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_F00001	10	■ 1.000	$U(2) \rightarrow H_4$	$U(2) \rightarrow H_4$	13.4 The contraction $U(2) \rightarrow H$
gilmore-lie-groups_F00002	10	■ 1.000	$S_0(3)$	$SO(3)$	14.5 Geometric symmetry $SO(3)$
gilmore-lie-groups_F00003	10	■ 1.000	$S_0(4)$	$SO(4)$	14.6 Dynamical symmetry $SO(4)$
gilmore-lie-groups_F00004	10	■ 0.997	$S_0(4,1)$	$SO(4,1)$	14.8 DeSitter symmetry $SO(4,1)$
gilmore-lie-groups_F00005	10	■ 1.000	$S_0(4,2)$	$SO(4,2)$	14.9 Conformal symmetry $SO(4,2)$
gilmore-lie-groups_F00006	15	■ 0.764	$(+, -, \times, \div)$	$(+, -, \times, \div)$	four standard operations of arithmetic $(+, -, \times, \div)$ as well as extraction of the n th
gilmore-lie-groups_F00007	15	■ 0.764	n	n	four standard operations of arithmetic $(+, -, \times, \div)$ as well as extraction of the n th
gilmore-lie-groups_F00008	17	■ 1.000	G	G	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00009	17	■ 1.000	$G = G_{\{0\}} \supset G_{\{1\}} \supset$	$G = G_0 \supset G_1 \supset$	if and only if its Galois group G contains a chain of subgroups $G = G_0 \supset G_1 \supset$
gilmore-lie-groups_F00010	17	■ 1.000	$\cdots \supset G_{\{\omega\}} = I$	$\cdots \supset G_{\omega} = I$	$\cdots \supset G_{\omega} = I$ with the properties:
gilmore-lie-groups_F00011	17	■ 1.000	$G_{\{i+1\}}$	G_{i+1}	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00012	17	■ 1.000	$G_{\{i\}}$	G_i	(i) G_{i+1} is an invariant subgroup of G_i ;
gilmore-lie-groups_F00013	17	■ 1.000	$G_{\{i\}} / G_{\{i+1\}}$	G_i / G_{i+1}	(ii) each factor group G_i / G_{i+1} is commutative.
gilmore-lie-groups_F00014	17	■ 0.998	$G = \{g_1, g_2, \dots\}$	$G = \{g_1, g_2, \dots\}$	A group G is defined as follows. It consists of a set of operations $G = \{g_1, g_2, \dots\}$,
gilmore-lie-groups_F00015	18	■ 1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G$	$g_i \in G, g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00016	18	■ 1.000	$g_{\{i\}} \cdot g_{\{j\}} \in G$	$g_i \cdot g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \cdot g_j \in G$.
gilmore-lie-groups_F00017	18	■ 1.000	$g_{\{i\}} \in G, g_{\{j\}} \in G, g_{\{k\}} \in G$	$g_i \in G, g_j \in G, g_k \in G$	(ii) Associativity: for all $g_i \in G, g_j \in G, g_k \in G$,
gilmore-lie-groups_F00018	18	■ 0.835	I	I	(iii) Identity: there is a group operation, I (identity operator), with the property that
gilmore-lie-groups_F00019	18	■ 1.000	$g_{\{i\}}$	g_i	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00020	18	■ 1.000	$g_{\{i\}}^{-1}$	g_i^{-1}	(iv) Inverse: every group operation g_i has an inverse (called g_i^{-1}):
gilmore-lie-groups_F00021	18	■ 1.000	$z_{\{1\}}, z_{\{2\}}, \dots, z_{\{n\}}$	$z_1, z_2, \dots, z_n$	is the group that permutes the roots $z_1, z_2, \dots, z_n$ among themselves and leaves the
gilmore-lie-groups_F00022	18	■ 0.658	$P_{\{n\}}$	P_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00023	18	■ 0.658	$S_{\{n\}}$	S_n	This group, called the permutation group P_n or the symmetric group S_n , has $n!$

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gilmore-lie-groups_F00024	18	0.658	$n!$	$n!$	This group, called the permutation group P_n or the symmetric group S_n , has $n!$
gilmore-lie-groups_F00025	18	0.996	$n \times n$	$n \times n$	of $n \times n$ matrices. These matrices have one nonzero element, $+1$, in each row
gilmore-lie-groups_F00026	18	1.000	$6=3!3 \times 3$	$6 = 3!3 \times 3$	and each column. For example, the $6 = 3!3 \times 3$ matrices for the permutation
gilmore-lie-groups_F00027	18	1.000	$S_{\{3\}}$	S_3	representation of S_3 are
gilmore-lie-groups_F00028	19	0.998	$z_{\{1\}}$	z_1	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00029	19	0.998	$z_{\{2\}}, z_{\{2\}}$	z_2, z_2	The symbol (123) means that the first root, z_1 , is replaced by z_2 , z_2 is replaced by
gilmore-lie-groups_F00030	19	0.999	$z_{\{3\}}$	z_3	z_3 , and z_3 is replaced by z_1
gilmore-lie-groups_F00031	19	0.768	$\times$	$\times$	stitutions in S_n) and $\times$ represents the multiplication operation among the matrices
gilmore-lie-groups_F00032	19	1.000	$G \rightarrow \Gamma(G)$	$G \rightarrow \Gamma(G)$	tion of a group, $G \rightarrow \Gamma(G)$, is that the product of matrices representing two group
gilmore-lie-groups_F00033	19	0.988	$\left(\Gamma(g_{\{i\}}) \times \Gamma(g_{\{j\}})\right)$	$(\Gamma(g_i) \times \Gamma(g_j))$	operations $(\Gamma(g_i) \times \Gamma(g_j))$ is equal to the matrix representing the product of these
gilmore-lie-groups_F00034	19	0.999	$\left(\Gamma(g_{\{i\}} \cdot g_{\{j\}})\right)$	$(\Gamma(g_i \cdot g_j))$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00035	19	0.999	$g_{\{i\}}, g_{\{j\}} \in G$	$g_i, g_j \in G$	operations in the group $(\Gamma(g_i \cdot g_j))$ for all group operations $g_i, g_j \in G$.
gilmore-lie-groups_F00036	19	0.874	3×3	3×3	since knowledge of the 3×3 matrix uniquely identifies the original group operation
gilmore-lie-groups_F00037	19	1.000	H	H	A subgroup H of the group G is a subset of group operations in G that is closed
gilmore-lie-groups_F00038	19	1.000	$A_{\{3\}}$	A_3	particular subgroup is denoted A_3 (alternating group). It consists of those oper-
gilmore-lie-groups_F00039	19	1.000	$A_{\{3\}} \subset S_{\{3\}}$	$A_3 \subset S_3$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00040	19	1.000	$S_{\{2\}}(ij)$	$S_2(ij)$	$A_3 \subset S_3$ and the three two-element subgroups $S_2(ij)$ of S_3 are illustrated in Fig. 1.1.
gilmore-lie-groups_F00041	20	0.706	$(123) \cdot (123) = (321)$	$(123) \cdot (123) = (321)$	of operations in this subset do not lie in this subset: $(123) \cdot (123) = (321)$, $(123) \cdot$
gilmore-lie-groups_F00042	20	1.000	$G, H_{\{1\}} \subset G$	$G, H_1 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00043	20	1.000	$H_{\{2\}} \subset G$	$H_2 \subset G$	Two subgroups of G , $H_1 \subset G$ and $H_2 \subset G$ are conjugate if there is a group
gilmore-lie-groups_F00044	20	1.000	$g \in G$	$g \in G$	element $g \in G$ with the property
gilmore-lie-groups_F00045	20	1.000	$S_{\{2\}}(12)$	$S_2(12)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00046	20	1.000	$S_{\{2\}}(13)$	$S_2(13)$	Example The subgroups $S_2(12)$ and $S_2(13)$ are conjugate in S_3 since
gilmore-lie-groups_F00047	20	1.000	$G=S_{\{3\}}$	$G = S_3$	operation in $G = S_3$ serves merely to permute the group operations in A_3 among
gilmore-lie-groups_F00048	20	1.000	$H \subset G$	$H \subset G$	A subgroup $H \subset G$ which is self-conjugate under all operations in G is called
gilmore-lie-groups_F00049	21	1.000	f	f	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-
gilmore-lie-groups_F00050	21	1.000	$g_{\{1\}}, g_{\{2\}}, \ldots$	$g_1, g_2, \dots$	A mapping f from a group G with group operations $g_1, g_2, \dots$ and group multi-

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gilmore-lie-groups_F00051	21	■ 0.999	$h_{\{1\}}, h_{\{2\}}, \dots$	$h_1, h_2, \dots$	plication $\cdot$ to a group H with group operations $h_1, h_2, \dots$ and group multiplication
gilmore-lie-groups_F00052	21	■ 1.000	$h_{\{i\}} \in H$	$h_i \in H$	ments in G may map to a single group element in H . Every element $h_i \in H$ has
gilmore-lie-groups_F00053	21	■ 1.000	$g_{\{j\}} \in G$	$g_j \in G$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups_F00054	21	■ 1.000	$h \in H$	$h \in H$	the same number of inverse images $g_j \in G$. If each group element $h \in H$ has a
gilmore-lie-groups_F00055	21	■ 0.933	$g \in G \left(h_{\{1\}} = f \left(g_{\{1\}} \right) \right)$	$g \in G (h_1 = f(g_1))$	unique inverse image $g \in G (h_1 = f(g_1))$ and $h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups_F00056	21	■ 0.933	$\left(h_{\{2\}} = f \left(g_{\{2\}} \right) \right), h_{\{1\}} = h_{\{2\}} \Rightarrow g_{\{1\}} = g_{\{2\}}$	$h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$	unique inverse image $g \in G (h_1 = f(g_1))$ and $h_2 = f(g_2), h_1 = h_2 \Rightarrow g_1 = g_2$ the
gilmore-lie-groups_F00057	21	■ 1.000	$S_{\{2\}}$	S_2	Example The 3:1 mapping f of S_3 onto S_2 given by
gilmore-lie-groups_F00058	22	■ 0.987	h	h	product of an element h in the subgroup H with a group element in a “quotient,”
gilmore-lie-groups_F00059	22	■ 1.000	G / H	G/H	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups_F00060	22	■ 1.000	$ G \left(S_{\{3\}} \right)$	$ G (S_3)$	or <i>coset</i> (denoted G/H). A coset is a subset of G . If the <i>order</i> of G is $ G $ (S_3
gilmore-lie-groups_F00061	22	■ 0.637	$3! = 6$	$3! = 6$	has $3! = 6$ group elements, so the order of S_3 is 6), then the order of G/H is
gilmore-lie-groups_F00062	22	■ 0.866	$ G / H = G / H $	$ G/H = G / H $	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00063	22	■ 0.866	$H = A_{\{3\}} = \{I$	$H = A_3 = \{I$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00064	22	■ 0.866	$\}$	$\}$	$ G/H = G / H $. For example, for subgroups $H = A_3 = \{I, (123), (321)\}$ and
gilmore-lie-groups_F00065	22	■ 0.992	$H = S_{\{2\}}(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$	$H = S_2(23) = \{I, (23)\}$ we have
gilmore-lie-groups_F00066	22	■ 1.000	$ G / H $	$ G / H $	The choice of the $ G / H $ group elements in the quotient space is not unique. For
gilmore-lie-groups_F00067	22	■ 0.692	$G / H = S_{\{3\}} / A_{\{3\}} = \{I$	$G/H = S_3/A_3 = \{I$	the subgroup A_3 we could equally well have chosen $G/H = S_3/A_3 = \{I, (13)\}$
gilmore-lie-groups_F00068	22	■ 0.997	$\{I, (23)\}$	$\{I, (23)\}$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00069	22	■ 0.997	$S_{\{2\}}(23)$	$S_2(23)$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00070	22	■ 0.997	$G / H = S_{\{3\}} / S_{\{2\}}(23) =$	$G/H = S_3/S_2(23) =$	or $\{I, (23)\}$; for $S_2(23)$ we could equally well have chosen $G/H = S_3/S_2(23) =$
gilmore-lie-groups_F00071	22	■ 0.556	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$	$\{I, (123), (321)\}$.
gilmore-lie-groups_F00072	22	■ 1.000	$S_{\{3\}} / A_{\{3\}}$	S_3/A_3	G/H . Since A_3 is an invariant subgroup of S_3 , the coset S_3/A_3 is a group, and
gilmore-lie-groups_F00073	22	■ 0.630	$\operatorname{group} G / H: G = G / H \times H$	group $G/H: G = G/H \times H$	$\operatorname{group} G/H: G = G/H \times H$.
gilmore-lie-groups_F00074	22	■ 1.000	k	k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00075	22	■ 1.000	$I_{\{k\}}$	I_k	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00076	22	■ 1.000	$z_{\{i\}}$	z_i	in terms of the k th order symmetric functions I_k of the roots z_i as follows:
gilmore-lie-groups_F00077	23	■ 0.992	$I_{\{k\}} (k=1, 2, \dots, n)$	$I_k (k = 1, 2, \dots, n)$	The n functions $I_k (k = 1, 2, \dots, n)$ of the n roots $(z_1, z_2, \dots, z_n)$ are symmetric:

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gilmore-lie-groups_F00078	23	■ 0.992	$\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$(z_1, z_2, \dots, z_n)$	The n functions I_k ($k = 1, 2, \dots, n$) of the n roots $\boxed{z_1, z_2, \dots, z_n}$ are symmetric:
gilmore-lie-groups_F00079	23	■ 1.000	$f\left(z_{\{1\}}, z_{\{2\}}, \ldots, z_{\{n\}}\right)$	$f(z_1, z_2, \dots, z_n)$	any function $f(z_1, z_2, \dots, z_n)$ that is invariant under S_n can be written as a function
gilmore-lie-groups_F00080	23	■ 1.000	$I_{\{1\}}, I_{\{2\}}, \ldots, I_{\{n\}}$	$I_1, I_2, \dots, I_n$	of the invariants $\boxed{I_1, I_2, \dots, I_n}$. The invariants are easily expressed in terms of
gilmore-lie-groups_F00081	23	■ 1.000	$G_{\{\omega-1\}} \supset G_{\{\omega\}}=I$	$G_{\omega-1} \supset G_{\omega} = I$	group-subgroup pair in this chain is isolated: $\boxed{G_{\omega-1} \supset G_{\omega} = I}$. The character
gilmore-lie-groups_F00082	23	■ 1.000	$G_{\{\omega-1\}} / G_{\{\omega\}}=G_{\{\omega-1\}}$	$G_{\omega-1}/G_{\omega} = G_{\omega-1}$	table for the commutative group $G_{\omega-1}/G_{\omega} = G_{\omega-1}$ is constructed. This lists
gilmore-lie-groups_F00083	23	■ 1.000	$\left G_{\{\omega-1\}}\right / \left G_{\{\omega\}}\right $	$ G_{\omega-1} / G_{\omega} $	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_F00084	23	■ 1.000	$G_{\{\omega-1\}}$	$G_{\omega-1}$	the $ G_{\omega-1} / G_{\omega} $ inequivalent one-dimensional representations of $G_{\omega-1}$. Linear
gilmore-lie-groups_F00085	23	■ 1.000	$G_{\{\omega\}}=I$	$G_{\omega} = I$	(i) symmetric under $\boxed{G_{\omega} = I}$,
gilmore-lie-groups_F00086	23	■ 1.000	$G_{\{\omega-2\}} \supset G_{\{\omega-1\}}$	$G_{\omega-2} \supset G_{\omega-1}$	Next, the next pair of groups $G_{\omega-2} \supset G_{\omega-1}$ is isolated. Starting from the set of
gilmore-lie-groups_F00087	23	■ 1.000	$G_{\{\omega-2\}}$	$G_{\omega-2}$	(ii) not all symmetric under $G_{\omega-2}$.
gilmore-lie-groups_F00088	23	■ 1.000	$G=G_{\{0\}} \supset$	$G = G_0 \supset$	This bootstrap procedure continues until the last group-subgroup pair $G = G_0 \supset$
gilmore-lie-groups_F00089	23	■ 1.000	$G_{\{1\}}$	G_1	G_1 is treated. At this stage the last set of functions can be solved by radicals.
gilmore-lie-groups_F00090	24	■ 1.000	$\Gamma^{\{1\}}, \Gamma^{\{2\}}$	Γ^1, Γ^2	ducible representations Γ^1, Γ^2 are
gilmore-lie-groups_F00091	24	■ 1.000	$r_{\{1\}}-r_{\{2\}}$	$r_1 - r_2$	That is, the function $\boxed{r_1 - r_2}$ is mapped into itself by the identity, and into its negative
gilmore-lie-groups_F00092	24	■ 0.899	$\left(r_{\{1\}}-r_{\{2\}}\right)$	$(r_1 - r_2)$	As a result, $\boxed{r_1 - r_2}$ is not symmetric under the action of the group S_2 . It transforms
gilmore-lie-groups_F00093	24	■ 1.000	$\Gamma^{\{2\}}$	Γ^2	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_F00094	24	■ 1.000	$\Gamma^{\{1\}}$	Γ^1	under the irreducible representation Γ^2 , not the identity representation Γ^1 .
gilmore-lie-groups_F00095	24	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)^{\{2\}}$	$(r_1 - r_2)^2$	Since the square $(r_1 - r_2)^2$ is symmetric (transforms under the identity repre-
gilmore-lie-groups_F00096	24	■ 0.999	$I_{\{1\}}, I_{\{2\}}$	I_1, I_2	sentation of S_2), it can be expressed in terms of the two invariants I_1, I_2 as follows
gilmore-lie-groups_F00097	25	■ 1.000	D	D	where D is the discriminant of the quadratic equation. Since $(r_1 - r_2) = \pm\sqrt{D}$,
gilmore-lie-groups_F00098	25	■ 1.000	$\left(r_{\{1\}}-r_{\{2\}}\right)=\pm\sqrt{D}$	$(r_1 - r_2) = \pm\sqrt{D}$	where D is the discriminant of the quadratic equation. Since $\boxed{r_1 - r_2} = \pm\sqrt{D}$,
gilmore-lie-groups_F00099	25	■ 1.000	x	x	to move the origin to the mean value of the roots by defining a new variable, x , in
gilmore-lie-groups_F00100	25	■ 1.000	z	z	terms of z (see Eq. (1.15)) by a Tschirnhaus transformation
gilmore-lie-groups_F00101	26	■ 1.000	$S_{\{3\}} / A_{\{3\}}=S_{\{2\}}$	$S_3/A_3 = S_2$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_F00102	26	■ 1.000	$A_{\{3\}} / I=A_{\{3\}}$	$A_3/I = A_3$	subgroups) is satisfied. Since $S_3/A_3 = S_2$ is commutative and $A_3/I = A_3$ is com-
gilmore-lie-groups_F00103	26	■ 1.000	$A_{\{3\}} \supset I$	$A_3 \supset I$	We begin the solution with the last group-subgroup pair in this chain: $A_3 \supset I$.

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gilmore-lie-groups_F00104	26	0.877	$v_{\{1\}}=s_{\{1\}}+s_{\{2\}}+s_{\{3\}}$	$v_1 = s_1 + s_2 + s_3$	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00105	26	0.877	ω	ω	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00106	26	0.877	ω^2	ω^2	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00107	26	0.877	$v_{\{2\}}=s_{\{1\}}+\omega s_{\{2\}}+\omega^2 s_{\{3\}}$	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00108	26	0.877	Γ^3	Γ^3	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00109	26	0.877	$v_{\{3\}}=s_{\{1\}}+\omega^2 s_{\{2\}}+\omega s_{\{3\}}$	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$	<table><tr><th></th><th>I</th><th>(123)</th><th>(321)</th><th>Basis functions</th></tr><tr><td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>$v_1 = s_1 + s_2 + s_3$</td></tr><tr><td>Γ^2</td><td>1</td><td>ω</td><td>ω^2</td><td>$v_2 = s_1 + \omega s_2 + \omega^2 s_3$</td></tr><tr><td>$\Gamma^3$</td><td>1</td><td>$\omega^2$</td><td>$\omega$</td><td>$v_3 = s_1 + \omega^2 s_2 + \omega s_3$</td></tr></table>		I	(123)	(321)	Basis functions	Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$	Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$	Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$
	I	(123)	(321)	Basis functions																					
Γ^1	1	1	1	$v_1 = s_1 + s_2 + s_3$																					
Γ^2	1	ω	ω^2	$v_2 = s_1 + \omega s_2 + \omega^2 s_3$																					
Γ^3	1	ω^2	ω	$v_3 = s_1 + \omega^2 s_2 + \omega s_3$																					
gilmore-lie-groups_F00110	26	1.000	$(123)^{-1}$	$(123)^{-1}$	For example, the action of $(123)^{-1}$ on v_2 is																				
gilmore-lie-groups_F00111	26	1.000	$v_{\{2\}}$	v_2	For example, the action of $(123)^{-1}$ on v_2 is																				
gilmore-lie-groups_F00112	27	1.000	$v_{\{1\}}$	v_1	Since v_1 is symmetric under both A_3 and S_3 , it can be expressed in terms of the																				
gilmore-lie-groups_F00113	27	1.000	$v_{\{3\}}$	v_3	The remaining functions, v_2 and v_3 , are symmetric under I but not under A_3 .																				
gilmore-lie-groups_F00114	27	1.000	$S_{\{3\}} \supset A_{\{3\}}$	$S_3 \supset A_3$	We now proceed to the next group-subgroup pair: $S_3 \supset A_3$. To construct func-																				
gilmore-lie-groups_F00115	27	1.000	$S_{\{2\}}=S_{\{3\}} / A_{\{3\}}$	$S_2 = S_3/A_3$	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the																				
gilmore-lie-groups_F00116	27	1.000	$v_{\{2\}}^3$	v_2^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the																				
gilmore-lie-groups_F00117	27	1.000	$v_{\{3\}}^3$	v_3^3	Since $S_2 = S_3/A_3$ permutes the functions v_2^3 and v_3^3 , it is the Galois group of the																				
gilmore-lie-groups_F00118	27	1.000	$J_{\{1\}}, J_{\{2\}}$	J_1, J_2	Since J_1, J_2 are symmetric under S_3 , they can be expressed in terms of the invariants																				
gilmore-lie-groups_F00119	27	0.995	$I_{\{1\}}, I_{\{2\}}, I_{\{3\}}$	I_1, I_2, I_3	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write																				

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gilmore-lie-groups_F00120	27	■ 0.995	$J_{\{1\}}$	J_1	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write																														
gilmore-lie-groups_F00121	27	■ 0.995	$J_{\{2\}}$	J_2	I_1, I_2, I_3 of the original cubic. Since J_1 has order 3 and J_2 has order 6, we can write																														
gilmore-lie-groups_F00122	27	■ 1.000	$I_{\{1\}}=s_{\{1\}}+s_{\{2\}}+$	$I_1 = s_1 + s_2 +$	These relations can be computed, but they simplify considerably if $\boxed{I_1 = s_1 + s_2 +}$																														
gilmore-lie-groups_F00123	27	■ 1.000	$s_{\{3\}}=0$	$s_3 = 0$	$s_3 = 0$. This can be accomplished by shifting the origin using a Tschirnhaus trans-																														
gilmore-lie-groups_F00124	28	■ 1.000	$J_{\{1\}}=v_{\{2\}}^{\{3\}}+v_{\{3\}}^{\{3\}}$	$J_1 = v_2^3 + v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'																														
gilmore-lie-groups_F00125	28	■ 1.000	$J_{\{2\}}=v_{\{2\}}^{\{3\}} \ v_{\{3\}}^{\{3\}}$	$J_2 = v_2^3 v_3^3$	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'																														
gilmore-lie-groups_F00126	28	■ 1.000	$I_{\{2\}}^{\{\backslash\prime\}}, \ I_{\{3\}}^{\{\backslash\prime\}}$	I_2', I_3'	The invariants $J_1 = v_2^3 + v_3^3$ and $J_2 = v_2^3 v_3^3$ can be expressed in terms of I_2', I_3'																														
gilmore-lie-groups_F00127	28	■ 1.000	$v_{\{2\}}^{\{3\}}, \ v_{\{3\}}^{\{3\}}$	v_2^3, v_3^3	The resolvent quadratic equation whose solution provides v_2^3, v_3^3 is																														
gilmore-lie-groups_F00128	28	■ 1.000	$s_{\{1\}}, \ s_{\{2\}}, \ s_{\{3\}}$	s_1, s_2, s_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by																														
gilmore-lie-groups_F00129	28	■ 1.000	$v_{\{1\}}, \ v_{\{2\}}, \ v_{\{3\}}$	v_1, v_2, v_3	Finally, the roots s_1, s_2, s_3 are linearly related to v_1, v_2, v_3 by																														
gilmore-lie-groups_F00130	29	■ 1.000	$z=z^{\{\backslash\prime\}}+\frac{1}{4} \ I_{\{1\}}$	$z = z' + \frac{1}{4} I_1$	of coordinates through the Tschirnhaus transformation $z = z' + \frac{1}{4} I_1$																														
gilmore-lie-groups_F00131	29	■ 1.000	$S_{\{4\}}$	S_4	The Galois group is S_4 . This has the subgroup chain shown in Fig. 1.5. The																														
gilmore-lie-groups_F00132	29	■ 1.000	$A_{\{4\}}$	A_4	alternating group A_4 consists of the twelve group operations that have determinant																														
gilmore-lie-groups_F00133	29	■ 0.957	$V_{\{4\}}$	V_4	group, Klein four-group) V_4 is $\{I, (12)(34), (13)(24), (14)(23)\}$. The chain																														
gilmore-lie-groups_F00134	29	■ 0.957	$\backslash\{I, (12) (34) , (13) (24) , (14) (23) \backslash\}$	$\{I, (12)(34), (13)(24), (14)(23)\}$	group, Klein four-group) V_4 is $\boxed{I, (12)(34), (13)(24), (14)(23)}$. The chain																														
gilmore-lie-groups_F00135	30	■ 1.000	$S_{\{4\}} \ / \ A_{\{4\}}=S_{\{2\}}$	$S_4/A_4 = S_2$	(i) A_4 is invariant in S_4 and $S_4/A_4 = S_2$,																														
gilmore-lie-groups_F00136	30	■ 0.987	$A_{\{4\}} \ / \ V_{\{4\}}=C_{\{3\}}=\backslash\{I$	$A_4/V_4 = C_3 = \{I$	(ii) V_4 is invariant in A_4 and $\boxed{A_4/V_4 = C_3 = \{I$ (234), (432)},																														
gilmore-lie-groups_F00137	30	■ 0.987	$V_{\{4\}} \ / \ I=V_{\{4\}}=\backslash\{I$	$V_4/I = V_4 = \{I$	(iii) I is invariant in V_4 and $V_4/I = V_4 = \{I, (12)(34), (13)(24), (14)(23)\}$.																														
gilmore-lie-groups_F00138	30	■ 0.669	$w_{\{1\}}=t_{\{1\}}+t_{\{2\}}+t_{\{3\}}+t_{\{4\}}$	$w_1 = t_1 + t_2 + t_3 + t_4$	<table> <tr> <th></th><th>I</th><th>$(12)(34)$</th><th>$(13)(24)$</th><th>$(14)(23)$</th><th>Basis functions</th></tr> <tr> <td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr> <tr> <td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr> <tr> <td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr> <tr> <td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr> </table>		I	$(12)(34)$	$(13)(24)$	$(14)(23)$	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	$(12)(34)$	$(13)(24)$	$(14)(23)$	Basis functions																														
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																														
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																														
Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																														
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																														
gilmore-lie-groups_F00139	30	■ 0.669	$w_{\{2\}}=t_{\{1\}}+t_{\{2\}}-t_{\{3\}}-t_{\{4\}}$	$w_2 = t_1 + t_2 - t_3 - t_4$	<table> <tr> <th></th><th>I</th><th>$(12)(34)$</th><th>$(13)(24)$</th><th>$(14)(23)$</th><th>Basis functions</th></tr> <tr> <td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr> <tr> <td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr> <tr> <td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr> <tr> <td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr> </table>		I	$(12)(34)$	$(13)(24)$	$(14)(23)$	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	$(12)(34)$	$(13)(24)$	$(14)(23)$	Basis functions																														
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																														
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																														
Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																														
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																														

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gilmore-lie-groups_ F00140	30	■ 0.669	$w_{\{3\}}=t_{\{1\}}-t_{\{2\}}+t_{\{3\}}-t_{\{4\}}$	$w_3 = t_1 - t_2 + t_3 - t_4$	<table> <tr> <th></th><th>I</th><th>(12)(34)</th><th>(13)(24)</th><th>(14)(23)</th><th>Basis functions</th></tr> <tr> <td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr> <tr> <td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr> <tr> <td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr> <tr> <td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr> </table>		I	(12)(34)	(13)(24)	(14)(23)	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	(12)(34)	(13)(24)	(14)(23)	Basis functions																														
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																														
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																														
Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																														
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																														
gilmore-lie-groups_ F00141	30	■ 0.669	Γ^4	Γ^4	<table> <tr> <th></th><th>I</th><th>(12)(34)</th><th>(13)(24)</th><th>(14)(23)</th><th>Basis functions</th></tr> <tr> <td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr> <tr> <td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr> <tr> <td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr> <tr> <td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr> </table>		I	(12)(34)	(13)(24)	(14)(23)	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	(12)(34)	(13)(24)	(14)(23)	Basis functions																														
Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$																														
Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$																														
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Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																														
gilmore-lie-groups_ F00142	30	■ 0.669	$w_{\{4\}}=t_{\{1\}}-t_{\{2\}}-t_{\{3\}}+t_{\{4\}}$	$w_4 = t_1 - t_2 - t_3 + t_4$	<table> <tr> <th></th><th>I</th><th>(12)(34)</th><th>(13)(24)</th><th>(14)(23)</th><th>Basis functions</th></tr> <tr> <td>Γ^1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>$w_1 = t_1 + t_2 + t_3 + t_4$</td></tr> <tr> <td>$\Gamma^2$</td><td>1</td><td>1</td><td>-1</td><td>-1</td><td>$w_2 = t_1 + t_2 - t_3 - t_4$</td></tr> <tr> <td>$\Gamma^3$</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>$w_3 = t_1 - t_2 + t_3 - t_4$</td></tr> <tr> <td>$\Gamma^4$</td><td>1</td><td>-1</td><td>-1</td><td>1</td><td>$w_4 = t_1 - t_2 - t_3 + t_4$</td></tr> </table>		I	(12)(34)	(13)(24)	(14)(23)	Basis functions	Γ^1	1	1	1	1	$w_1 = t_1 + t_2 + t_3 + t_4$	Γ^2	1	1	-1	-1	$w_2 = t_1 + t_2 - t_3 - t_4$	Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$	Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$
	I	(12)(34)	(13)(24)	(14)(23)	Basis functions																														
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Γ^3	1	-1	1	-1	$w_3 = t_1 - t_2 + t_3 - t_4$																														
Γ^4	1	-1	-1	1	$w_4 = t_1 - t_2 - t_3 + t_4$																														
gilmore-lie-groups_ F00143	30	■ 0.971	$w_{\{2\}}, w_{\{3\}}, w_{\{4\}}$	w_2, w_3, w_4	These basis vectors are symmetric under I but the basis vectors w_2, w_3, w_4 are not																														
gilmore-lie-groups_ F00144	30	■ 1.000	$A_{\{4\}} \supset V_{\{4\}}$	$A_4 \supset V_4$	We now advance to the next group-subgroup pair: $A_4 \supset V_4$. It is a simple matter																														
gilmore-lie-groups_ F00145	30	■ 1.000	$A_{\{4\}} / V_{\{4\}}$	A_4/V_4	(ii) permuted among themselves by A_4 and the group A_4/V_4 .																														
gilmore-lie-groups_ F00146	30	■ 0.999	$w_{\{1\}}=I_{\{1\}}$	$w_1 = I_1$	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the																														
gilmore-lie-groups_ F00147	30	■ 0.999	$w_{\{2\}}^{\wedge 2}, w_{\{3\}}^{\wedge 2}, w_{\{4\}}^{\wedge 2}$	w_2^2, w_3^2, w_4^2	These functions are $w_1 = I_1$ and w_2^2, w_3^2, w_4^2 . In the coordinate system in which the																														
gilmore-lie-groups_ F00148	30	■ 1.000	$w_{\{j\}}^{\wedge 2} (j=2, 3, 4)$	$w_j^2 (j = 2, 3, 4)$	It is clear that the three $w_j^2 (j = 2, 3, 4)$ are permuted among themselves by the																														
gilmore-lie-groups_ F00149	30	■ 1.000	$C_{\{3\}}=A_{\{4\}} / V_{\{4\}}$	$C_3 = A_4/V_4$	factor group $C_3 = A_4/V_4$, which is a subgroup of the Galois group of a resolvent																														
gilmore-lie-groups_ F00150	31	■ 1.000	$J_{\{k\}}$	J_k	Since the three J_k are invariant under C_3 , they can be expressed in terms of the																														
gilmore-lie-groups_ F00151	31	■ 1.000	$C_{\{3\}}$	C_3	Since the three J_k are invariant under C_3 , they can be expressed in terms of the																														
gilmore-lie-groups_ F00152	31	■ 1.000	$S_{\{4\}} \supset A_{\{4\}}$	$S_4 \supset A_4$	chain: $S_4 \supset A_4$, with $S_4/A_4 = S_2$. The cubic is solved by introducing the resolvent																														
gilmore-lie-groups_ F00153	31	■ 1.000	$y_{\{2\}}, y_{\{3\}}, y_{\{4\}}$	y_2, y_3, y_4	If the three solutions of the resolvent cubic equation are called y_2, y_3, y_4 , then																														
gilmore-lie-groups_ F00154	31	■ 1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}}=8 I_{\{3\}}^{\prime}$	$w_2 w_3 w_4 = 8I_3'$	A simple computation shows that $w_2 w_3 w_4 = 8I_3'$. The signs $\pm\sqrt{y_j}$ are chosen so																														
gilmore-lie-groups_ F00155	31	■ 1.000	$\pm\sqrt{y_{\{j\}}}$	$\pm\sqrt{y_j}$	A simple computation shows that $w_2 w_3 w_4 = 8I_3'$. The signs $\pm\sqrt{y_j}$ are chosen so																														
gilmore-lie-groups_ F00156	31	■ 1.000	$8 I_{\{3\}}^{\prime}$	$8I_3'$	that their product is $8I_3'$. The simple linear relation between the roots t_i and the																														
gilmore-lie-groups_ F00157	31	■ 1.000	$t_{\{i\}}$	t_i	that their product is $8I_3'$. The simple linear relation between the roots t_i and the																														
gilmore-lie-groups_ F00158	31	■ 1.000	$I_{\{1\}}$	I_1	invariants I_1 and functions $w_j(I')$ is easily inverted:																														
gilmore-lie-groups_ F00159	31	■ 1.000	$w_{\{j\}}\backslash\left(I^{\prime}\right)$	$w_j(I')$	invariants I_1 and functions $w_j(I')$ is easily inverted:																														

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gilmore-lie-groups_F00160	31	■ 1.000	$w_{\{j\}}$	w_j	where the w_j are square roots of the solutions of the resolvent cubic equation whose
gilmore-lie-groups_F00161	31	■ 1.000	$S_{\{5\}}$	S_5	is sufficient to study the structure of its Galois group S_5 . The alternating subgroup
gilmore-lie-groups_F00162	32	■ 1.000	$A_{\{5\}}$	A_5	A_5 of order 60 is an invariant subgroup. S_5 has no invariant subgroups except A_5
gilmore-lie-groups_F00163	32	■ 1.000	$S_{\{5\}} / A_{\{5\}} = S_{\{2\}}$	$S_5/A_5 = S_2$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_F00164	32	■ 1.000	$A_{\{5\}} / I = A_{\{5\}}$	$A_5/I = A_5$	Although $S_5/A_5 = S_2$ is commutative, $A_5/I = A_5$ is not. Therefore the quintic
gilmore-lie-groups_F00165	33	■ 1.000	$t_{\{1\}}^{\wedge\{\backslash\prime\}}=-3, t_{\{2\}}^{\wedge\{\backslash\prime\}}=-2, t_{\{3\}}^{\wedge\{\backslash\prime\}}=$	$t'_1 = -3, t'_2 = -2, t'_3 =$	Next, we introduce linear combinations of the four roots $t'_1 = -3, t'_2 = -2, t'_3 =$
gilmore-lie-groups_F00166	33	■ 0.961	$1, t_{\{4\}}^{\wedge\{\backslash\prime\}}=4$	$1, t'_4 = 4$	$1, t'_4 = 4$
gilmore-lie-groups_F00167	33	■ 1.000	$y=y^{\wedge\{\backslash\prime\}}+\frac{1}{3}J_1=y^{\wedge\{\backslash\prime\}}+\frac{1}{3}(4+16+100)=y^{\wedge\{\backslash\prime\}}+40$	$y = y' + \frac{1}{3}J_1 = y' + \frac{1}{3}(4 + 16 + 100) = y' + 40$	$y = y' + \frac{1}{3}J_1 = y' + \frac{1}{3}(4 + 16 + 100) = y' + 40$. The roots are now
gilmore-lie-groups_F00168	34	■ 0.990	$v_{\{2\}}^{\wedge\{3\}}+v_{\{3\}}^{\wedge\{3\}}, v_{\{2\}}^{\wedge\{3\}} v_{\{3\}}^{\wedge\{3\}}$	$v_2^3 + v_3^3, v_2^3 v_3^3$	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J'_2, J'_3 :
gilmore-lie-groups_F00169	34	■ 0.990	$J_{\{2\}}^{\wedge\{\backslash\prime\}}, J_{\{3\}}^{\wedge\{\backslash\prime\}}$	J'_2, J'_3	We can express $v_2^3 + v_3^3, v_2^3 v_3^3$ in terms of J'_2, J'_3 :
gilmore-lie-groups_F00170	34	■ 1.000	$x=x^{\wedge\{\backslash\prime\}}+\frac{1}{2}K_1$	$x = x' + \frac{1}{2}K_1$	A Tschirnhaus transformation $x = x' + \frac{1}{2}K_1$ produces the auxiliary quadratic
gilmore-lie-groups_F00171	35	■ 1.000	$y_{\{1\}}, y_{\{2\}}, y_{\{3\}}$	y_1, y_2, y_3	The values y_1, y_2, y_3 of the resolvent cubic are complex linear combinations of
gilmore-lie-groups_F00172	35	■ 0.982	$v_{\{2\}}, v_{\{3\}}$	v_2, v_3	v_2, v_3
gilmore-lie-groups_F00173	35	■ 1.000	$w_{\{2\}} w_{\{3\}} w_{\{4\}}=8 I_{\{3\}}^{\wedge\{\backslash\prime\}}=80$	$w_2 w_3 w_4 = 8I'_3 = 80$	Since $w_2 w_3 w_4 = 8I'_3 = 80$, an even number of these signs must be negative. The
gilmore-lie-groups_F00174	36	■ 1.000	$S_{\{4\}} / A_{\{4\}}, A_{\{4\}} / V_{\{4\}}, V_{\{4\}}$	$S_4/A_4, A_4/V_4, V_4$	1. Compute $S_4/A_4, A_4/V_4, V_4$ and show that they are commutative.
gilmore-lie-groups_F00175	36	■ 0.986	$V_{\{8\}}$	V_8	2. Construct the group V_8 with the property $S_4 \supset V_8 \supset V_4$ (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00176	36	■ 0.986	$S_{\{4\}} \supset V_{\{8\}} \supset V_{\{4\}}$	$S_4 \supset V_8 \supset V_4$	2. Construct the group V_8 with the property $S_4 \supset V_8 \supset V_4$ (see Fig. 1.5). (Hint: include
gilmore-lie-groups_F00177	36	■ 1.000	$z^{\wedge\{3\}}-7 z+6=0((z-1)(z-2)(z+3)=0)$	$z^3 - 7z + 6 = 0((z - 1)(z - 2)(z + 3) = 0)$	3. For the cubic equation $z^3 - 7z + 6 = 0((z - 1)(z - 2)(z + 3) = 0)$ show
gilmore-lie-groups_F00178	36	■ 1.000	$\left(x-v_{\{2\}}^{\wedge\{3\}}\right)\left(x-v_{\{3\}}^{\wedge\{3\}}\right)=x^{\wedge\{2\}}-162 x+$	$(x - v_2^3)(x - v_3^3) = x^2 - 162x +$	Show that the resolvent equation for v_2^3, v_3^3 is $(x - v_2^3)(x - v_3^3) = x^2 - 162x +$
gilmore-lie-groups_F00179	36	■ 0.996	$9261=0$	$9261 = 0$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00180	36	■ 0.996	$v_{\{2\}}^{\wedge\{3\}}, v_{\{3\}}^{\wedge\{3\}}=81 \pm i 30 \sqrt{3}$	$v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00181	36	■ 0.996	$v_{\{2\}}, v_{\{3\}}=\frac{1}{2}(3 \pm$	$v_2, v_3 = \frac{1}{2}(3 \pm$	$9261 = 0$. Solve this quadratic to find $v_2^3, v_3^3 = 81 \pm i30\sqrt{3}$, so that $v_2, v_3 = \frac{1}{2}(3 \pm$
gilmore-lie-groups_F00182	36	■ 1.000	$i 5 \sqrt{3}$	$i5\sqrt{3}$	$i5\sqrt{3}$). Invert Eq. (1.43) to determine the three roots of the original equation:
gilmore-lie-groups_F00183	36	■ 0.543	$x+y, x-y, x y$	$x + y, x - y, xy$	to construct intervals of lengths $x + y, x - y, xy$ and x/y using ruler and compass.
gilmore-lie-groups_F00184	36	■ 0.543	x / y	x/y	to construct intervals of lengths $x + y, x - y, xy$ and x/y using ruler and compass.
gilmore-lie-groups_F00185	36	■ 1.000	$\sqrt{x}$	$\sqrt{x}$	It is also possible to construct intervals of length $\sqrt{x}$ by these means. The set of all

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gilmore-lie-groups_F00186	36	■ 1.000	x+i y=(x, y)	$x + iy = (x, y)$	forms a subset of the numbers $x + iy = (x, y)$ in the complex plane. If a number is
gilmore-lie-groups_F00187	36	■ 1.000	K	K	number satisfies an algebraic equation of degree K with integer coefficients, where
gilmore-lie-groups_F00188	36	■ 1.000	K=2^{n}	$K = 2^n$	$K = 2^n$ must be some power of two.
gilmore-lie-groups_F00189	36	■ 1.000	\pi	π	a. Square a circle? For the circle of radius 1 the area is π . Squaring a circle means
gilmore-lie-groups_F00190	36	■ 1.000	x^{2}-\pi=0	$x^2 - \pi = 0$	finding an interval of length x , where $x^2 - \pi = 0$. This is of degree 2 but π is not
gilmore-lie-groups_F00191	36	■ 1.000	1^{3}=1	$1^3 = 1$	b. Double the cube? A cube with edge length 1 has volume $1^3 = 1$. A cube with
gilmore-lie-groups_F00192	36	■ 1.000	x^{3}-2=0	$x^3 - 2 = 0$	twice the volume has edge length x , where x satisfies $x^3 - 2 = 0$. Although the
gilmore-lie-groups_F00193	36	■ 1.000	3 \neq 2^{n}	$3 \neq 2^n$	coefficients are integers this equation is of degree $3 \neq 2^n$ for any integer n . Argue
gilmore-lie-groups_F00194	36	■ 1.000	3 \theta	3θ	c. Trisect an angle? If 3θ is some angle, the trigonometric functions of 3θ and
gilmore-lie-groups_F00195	36	■ 1.000	\frac{1}{3}(3 \theta)=\theta	$\frac{1}{3}(3\theta) = \theta$	$\frac{1}{3}(3\theta) = \theta$ are related by
gilmore-lie-groups_F00196	37	■ 1.000	\cos (3 \theta)	$\cos(3\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00197	37	■ 1.000	\cos (\theta)	$\cos(\theta)$	Whether $\cos(3\theta)$ is rational or irrational, the equation for $\cos(\theta)$:
gilmore-lie-groups_F00198	37	■ 1.000	\left(x^{2}+a x+b\right)(x+c)=0	$(x^2 + ax + b)(x + c) = 0$	cubic factors into the form $(x^2 + ax + b)(x + c) = 0$, where a, b, c are rational.
gilmore-lie-groups_F00199	37	■ 1.000	a, b, c	a, b, c	cubic factors into the form $(x^2 + ax + b)(x + c) = 0$, where a, b, c are rational.
gilmore-lie-groups_F00200	37	■ 1.000	\cos (3 \theta)=0, c=0	$\cos(3\theta) = 0, c = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00201	37	■ 1.000	a=0	$a = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00202	37	■ 1.000	b=-3 / 4	$b = -3/4$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00203	37	■ 1.000	\cos (\theta)=0	$\cos(\theta) = 0$	For example, if $\cos(3\theta) = 0, c = 0$ so that $a = 0$ and $b = -3/4$. Then $\cos(\theta) = 0$
gilmore-lie-groups_F00204	37	■ 0.986	\pm \sqrt{3} / 2	$\pm\sqrt{3}/2$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2 (+)$, $3\pi/2 (0)$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00205	37	■ 0.986	3 \theta=\pi / 2(+), 3 \pi / 2(0)	$3\theta = \pi/2(+), 3\pi/2(0)$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2 (+), 3\pi/2 (0)$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00206	37	■ 0.986	5 \pi / 2(-)	$5\pi/2(-)$	or $\pm\sqrt{3}/2$ for $3\theta = \pi/2 (+), 3\pi/2 (0)$, or $5\pi/2 (-)$.
gilmore-lie-groups_F00207	38	■ 1.000	g_{i}, g_{j}, g_{k}, \ldots	$g_i, g_j, g_k, \dots$	Definition A set $g_i, g_j, g_k, \dots$ (called group elements or group operations)
gilmore-lie-groups_F00208	38	■ 1.000	g_{i} \circ g_{j} \in G	$g_i \circ g_j \in G$	(i) Closure: if $g_i \in G, g_j \in G$, then $g_i \circ g_j \in G$.
gilmore-lie-groups_F00209	38	■ 1.000	e	e	(iii) Identity: there is an operator e (the identity operation) with the property that for
gilmore-lie-groups_F00210	38	■ 1.000	g_{i} \in G	$g_i \in G$	every group operation $g_i \in G$
gilmore-lie-groups_F00211	39	■ 1.000	2 \times 2	2×2	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:
gilmore-lie-groups_F00212	39	■ 1.000	S L(2 ; \mathbb{R})	$SL(2; \mathbb{R})$	Example We consider the set of real 2×2 matrices $SL(2; \mathbb{R})$:

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gilmore-lie-groups_F00213	39	■ 1.000	$\backslash\alpha, \backslash\beta, \backslash\gamma, \backslash\delta$	$\alpha, \beta, \gamma, \delta$	where $\alpha, \beta, \gamma, \delta$ are real numbers. This set forms a group under matrix multiplica-
gilmore-lie-groups_F00214	39	■ 0.797	A	A	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00215	39	■ 0.797	B	B	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00216	39	■ 0.797	$A \circ B = C$	$A \circ B = C$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00217	39	■ 0.797	$\circ$	$\circ$	(i) Closure if A and B are real 2×2 matrices, and $A \circ B = C$ (where $\circ$ now represents
gilmore-lie-groups_F00218	39	■ 0.922	C	C	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00219	39	■ 0.922	$\backslash\operatorname{det}(A)=+1$	$\det(A) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00220	39	■ 0.922	$\backslash\operatorname{det}(B)=+1$	$\det(B) = +1$	matrix multiplication), then C is a real 2×2 matrix. If $\det(A) = +1$ and $\det(B) = +1$,
gilmore-lie-groups_F00221	39	■ 1.000	$\backslash\operatorname{det}(C)=\backslash\operatorname{det}(A)\backslash\operatorname{det}(B)=+1$	$\det(C) = \det(A) \det(B) = +1$	then $\det(C) = \det(A) \det(B) = +1$.
gilmore-lie-groups_F00222	39	■ 0.998	$(A \circ B) \circ C$	$(A \circ B) \circ C$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups_F00223	39	■ 0.998	$A \circ (B \circ C)$	$A \circ (B \circ C)$	(ii) Associativity: $(A \circ B) \circ C$ and $A \circ (B \circ C)$ are given explicitly by
gilmore-lie-groups_F00224	39	■ 1.000	$g_i \rightarrow g(x)$	$g_i \rightarrow g(x)$	in the group with a point in some topological space: $g_i \rightarrow g(x)$. In other words, the
gilmore-lie-groups_F00225	39	■ 1.000	i	i	index i depends on one or more continuous real variables.
gilmore-lie-groups_F00226	39	■ 1.000	$S^2 \subset R^3: x^2+y^2+z^2=1$	$S^2 \subset R^3: x^2 + y^2 + z^2 = 1$	example, every point on the surface of a unit sphere $S^2 \subset R^3: x^2 + y^2 + z^2 = 1$,
gilmore-lie-groups_F00227	39	■ 1.000	R^2	R^2	has a neighborhood that looks, over small distances, like a piece of the plane R^2
gilmore-lie-groups_F00228	39	■ 1.000	S^2	S^2	(see Fig. 2.1). Locally, the two spaces S^2 and R^2 are topologically equivalent but
gilmore-lie-groups_F00229	39	■ 1.000	M^n	M^n	Definition An n -dimensional differentiable manifold M^n consists of the follow-
gilmore-lie-groups_F00230	39	■ 1.000	T	T	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups_F00231	39	■ 1.000	U_α	U_α	(i) A topological space T . This includes a collection of open sets U_α (a topology) that
gilmore-lie-groups_F00232	39	■ 0.896	$T: \cup_\alpha U_\alpha = T$	$T: \cup_\alpha U_\alpha = T$	cover $T: \cup_\alpha U_\alpha = T$.
gilmore-lie-groups_F00233	40	■ 1.000	p	p	Figure 2.1. Every point p on a sphere S^2 is surrounded by an open neighborhood
gilmore-lie-groups_F00234	40	■ 1.000	ϕ_α	ϕ_α	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups_F00235	40	■ 1.000	$\phi_\alpha \backslash\left(U_\alpha \backslash\right)=V_\alpha \subset R^n$	$\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$	(ii) A collection of charts ϕ_α , with $\phi_\alpha(U_\alpha) = V_\alpha \subset R^n$. Each ϕ_α is a homeomorphism of
gilmore-lie-groups_F00236	40	■ 1.000	V_α	V_α	U_α to V_α .
gilmore-lie-groups_F00237	40	■ 1.000	$\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta \left(U_\alpha \cap U_\beta\right) \rightarrow \phi_\alpha \left(U_\alpha \cap U_\beta\right)$	$\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$	(iii) Smoothness conditions. The homeomorphisms $\phi_\alpha \circ \phi_\beta^{-1}: \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta)$
gilmore-lie-groups_F00238	40	■ 0.988	U_β	U_β	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.
gilmore-lie-groups_F00239	40	■ 0.988	R^n	R^n	U_β) of open sets in R^n to open sets in R^n are 1:1, invertible, and differentiable.

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gilmore-lie-groups_ F00240	40	■ 1.000	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	$\phi_{\alpha} \circ \phi_{\beta}^{-1}$	functions” $\phi_{\alpha} \circ \phi_{\beta}^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups_ F00241	40	■ 1.000	$R^n \rightarrow R^n$	$R^n \rightarrow R^n$	functions” $\phi_{\alpha} \circ \phi_{\beta}^{-1}$ map $R^n \rightarrow R^n$, all the definitions of elementary multivariable
gilmore-lie-groups_ F00242	40	■ 0.697	C^k	C^k	replaced by other adjectives (C^k , smooth, analytic, ...) in the definition above.
gilmore-lie-groups_ F00243	40	■ 1.000	$x_1 \rightarrow 0$	$x_1 \rightarrow 0$	lem, as is clear when one asks: “what happens as $x_1 \rightarrow 0$?” We will consider this
gilmore-lie-groups_ F00244	40	■ 1.000	$\times S^1$	$\times S^1$	fold R^2 (plane) $\times S^1$ (circle) (see Fig. 2.2). This is not at all obvious, but will become
gilmore-lie-groups_ F00245	41	■ 0.999	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$	$z^2 - x^2 - y^2 = 1$. The rotation matrix is parameterized by a point on a circle. The
gilmore-lie-groups_ F00246	41	■ 1.000	$H^2 \times S^1$	$H^2 \times S^1$	parameterization manifold, $H^2 \times S^1$, is three dimensional.
gilmore-lie-groups_ F00247	41	■ 0.698	U_{α}	U_{α}	U_{α}) has a finite subcover: $\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$.
gilmore-lie-groups_ F00248	41	■ 0.698	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	$\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$	U_{α}) has a finite subcover: $\cup_{\alpha}^{\text{finite}} T \subset U_{\alpha}$.
gilmore-lie-groups_ F00249	41	■ 1.000	$ x - x' ^2 = x_1 - x_1' ^2 + \dots +$	$(x - x' ^2 = x_1 - x_1' ^2 + \dots +$	In spaces R^n with a Euclidean notion of distance $ x - x' ^2 = x_1 - x_1' ^2 + \dots +$
gilmore-lie-groups_ F00250	41	■ 1.000	$ x_n - x_n' ^2$	$ x_n - x_n' ^2$	$ x_n - x_n' ^2$), this definition is equivalent to an older definition of compact spaces:
gilmore-lie-groups_ F00251	42	■ 1.000	$(g(x), x \in M^n)$	$(g(x), x \in M^n)$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_ F00252	42	■ 1.000	$g(x) \circ g(y) =$	$g(x) \circ g(y) =$	operations $(g(x), x \in M^n)$ and a combinatorial operation defined by $g(x) \circ g(y) =$
gilmore-lie-groups_ F00253	42	■ 1.000	$g(z)$	$g(z)$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00254	42	■ 1.000	$z \in M^n$	$z \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00255	42	■ 1.000	$x \in M^n$	$x \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00256	42	■ 1.000	$y \in M^n$	$y \in M^n$	$g(z)$, where the coordinate $z \in M^n$ depends on the coordinates $x \in M^n$ and $y \in M^n$
gilmore-lie-groups_ F00257	42	■ 1.000	$z = \phi(x, y)$	$z = \phi(x, y)$	through a function $z = \phi(x, y)$.
gilmore-lie-groups_ F00258	42	■ 1.000	$g(x) \circ g(y) = g(z)$	$g(x) \circ g(y) = g(z)$	defined by $g(x) \circ g(y) = g(z)$, is differentiable.
gilmore-lie-groups_ F00259	42	■ 1.000	$y = \psi(x)$	$y = \psi(x)$	(ii) Smoothness of the group inversion map The group inversion map $y = \psi(x)$, defined
gilmore-lie-groups_ F00260	42	■ 1.000	$g(x)^{-1} = g(y)$	$g(x)^{-1} = g(y)$	by $g(x)^{-1} = g(y)$, is differentiable.
gilmore-lie-groups_ F00261	42	■ 1.000	$g(\phi(x, y))$	$g(\phi(x, y))$	$g(\phi(x, y))$
gilmore-lie-groups_ F00262	42	■ 1.000	ϕ	ϕ	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00263	42	■ 1.000	x_1	x_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00264	42	■ 1.000	y_1	y_1	The function ϕ is analytic in its two pairs of arguments provided x_1 and y_1 are
gilmore-lie-groups_ F00265	42	■ 1.000	$x_2 - x_3$	$x_2 - x_3$	bounded away from the $x_2 - x_3$ plane $x_1 = 0$ and the $y_2 - y_3$ plane $y_1 = 0$. In the
gilmore-lie-groups_ F00266	42	■ 1.000	$x_1 = 0$	$x_1 = 0$	bounded away from the $x_2 - x_3$ plane $x_1 = 0$ and the $y_2 - y_3$ plane $y_1 = 0$. In the

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gilmore-lie-groups_F00267	42	■ 1.000	$y_{\{2\}}-y_{\{3\}}$	$y_2 - y_3$	bounded away from the x_2 - x_3 plane $x_1 = 0$ and the y_2 - y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00268	42	■ 1.000	$y_{\{1\}}=0$	$y_1 = 0$	bounded away from the x_2 - x_3 plane $x_1 = 0$ and the y_2 - y_3 plane $y_1 = 0$. In the
gilmore-lie-groups_F00269	43	■ 0.983	$\left(y_{\{1\}}, y_{\{2\}}, y_{\{3\}}\right)$	(y_1, y_2, y_3)	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00270	43	■ 0.983	$\left[g\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)\right]^{-1}$	$[g(x_1, x_2, x_3)]^{-1}$	inverse. We can determine the coordinates (y_1, y_2, y_3) of $[g(x_1, x_2, x_3)]^{-1}$ by setting
gilmore-lie-groups_F00271	43	■ 0.998	$\left(z_{\{1\}}, z_{\{2\}}, z_{\{3\}}\right)=(1,0,0)$	$(z_1, z_2, z_3) = (1, 0, 0)$	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00272	43	■ 0.998	$\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)$	(x_1, x_2, x_3)	$(z_1, z_2, z_3) = (1, 0, 0)$ and solving for (y_1, y_2, y_3) in terms of (x_1, x_2, x_3) . Or more
gilmore-lie-groups_F00273	43	■ 1.000	$[g(x)]^{-1}=g(y)=g(\psi(x))$	$[g(x)]^{-1} = g(y) = g(\psi(x))$	The inverse mapping $[g(x)]^{-1} = g(y) = g(\psi(x))$ is
gilmore-lie-groups_F00274	43	■ 0.964	(x, y, z , θ)	(x, y, z , θ)	The rotation matrix is parameterized by a point on a circle. Two points $[x, y, z , \theta]$
gilmore-lie-groups_F00275	43	■ 0.882	$(-x, -y, - z , \theta+\pi)$	$(-x, -y, - z , \theta + \pi)$	and $[-x, -y, - z , \theta + \pi]$ map to the same matrix in $SL(2; \mathbb{R})$. The manifold that
gilmore-lie-groups_F00276	43	■ 1.000	$H^{2+} \times S^1$	$H^{2+} \times S^1$	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_F00277	43	■ 1.000	H^{2+}	H^{2+}	parameterizes $SL(2; \mathbb{R})$ is three dimensional. It is $H^{2+} \times S^1$, where H^{2+} is the
gilmore-lie-groups_F00278	43	■ 1.000	$\phi(x, y)$	$\phi(x, y)$	operation that obeys the group axioms. The combinatorial operation $\phi(x, y)$ de-
gilmore-lie-groups_F00279	43	■ 1.000	$g(x) \circ g(y)=g(z)=g(\phi(x, y))$	$g(x) \circ g(y) = g(z) = g(\phi(x, y))$	finied by $g(x) \circ g(y) = g(z) = g(\phi(x, y))$ is differentiable in both sets of variables.
gilmore-lie-groups_F00280	44	■ 1.000	$[g(x)]^{-1}=$	$[g(x)]^{-1} =$	In addition, the mapping $y = \psi(x)$ of a group operation to its inverse $[g(x)]^{-1} =$
gilmore-lie-groups_F00281	44	■ 1.000	$g(y)=g(\psi(x))$	$g(y) = g(\psi(x))$	$g(y) = g(\psi(x))$ is also differentiable.
gilmore-lie-groups_F00282	44	■ 0.863	$\operatorname{SL}(2 ; \mathbb{R})$	$SL(2; \mathbb{R})$	1. Construct the analytic mapping $\phi(x, y)$ for the parameterization of $SL(2; \mathbb{R})$ illus-
gilmore-lie-groups_F00283	44	■ 1.000	M	M	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_F00284	44	■ 1.000	$S L(n ; \mathbb{R})$	$SL(n; \mathbb{R})$	3. Convince yourself that every matrix M in the group $SL(n; \mathbb{R})$ can be written as the
gilmore-lie-groups_F00285	44	■ 1.000	S	S	product of an $n \times n$ real symmetric unimodular matrix S and an orthogonal matrix
gilmore-lie-groups_F00286	44	■ 0.755	O	O	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_F00287	44	■ 0.755	$S O(n): M=S O$	$SO(n) : M = SO$	O in $SO(n)$: $M = SO$. Devise an algorithm for constructing these matrices. Show
gilmore-lie-groups_F00288	44	■ 1.000	$S=\left(M M^t\right)^{1 / 2}$	$S = (MM^t)^{1/2}$	$S = (MM^t)^{1/2}$ and $O = S^{-1}M$. How do you compute the square root of a matrix?
gilmore-lie-groups_F00289	44	■ 1.000	$O=S^{-1} M$	$O = S^{-1}M$	$S = (MM^t)^{1/2}$ and $O = S^{-1}M$. How do you compute the square root of a matrix?
gilmore-lie-groups_F00290	44	■ 1.000	$(x, y, z) \rightarrow\left(x^{\prime}, y^{\prime}, z^{\prime}\right)$	$(x, y, z) \rightarrow (x', y', z')$	4. Construct the most general linear transformation $[x, y, z] \rightarrow [x', y', z']$ that leaves
gilmore-lie-groups_F00291	44	■ 1.000	$M_{\{1\}}$	M_1	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_F00292	44	■ 1.000	$M_{\{2\}}$	M_2	where the real symmetric matrices M_1 and M_2 satisfy
gilmore-lie-groups_F00293	44	■ 0.995	$[S O(1,1)]$	$[SO(1,1)]$	5. Construct the group of linear transformations $[SO(1, 1)]$ that leaves invariant the

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gilmore-lie-groups_F00294	44	■ 0.972	$(c\ t)^2 - x^2$	$(ct)^2 - x^2$	quantity $(ct)^2 - x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_F00295	44	■ 0.972	$S\ O(2)$	$SO(2)$	quantity $(ct)^2 - x^2$. Compare this with the group of linear transformations $[SO(2)]$
gilmore-lie-groups_F00296	44	■ 0.999	$x^2 + y^2$	$x^2 + y^2$	that leaves invariant the radius of the circle $x^2 + y^2$. (This comparison involves map-
gilmore-lie-groups_F00297	45	■ 1.000	$(c\ t)^2 - x^2 - y^2 - z^2$	$(ct)^2 - x^2 - y^2 - z^2$	$(ct)^2 - x^2 - y^2 - z^2$. This is the Lorentz group $O(3, 1)$. Four disconnected mani-
gilmore-lie-groups_F00298	45	■ 1.000	$O(3, 1)$	$O(3, 1)$	$(ct)^2 - x^2 - y^2 - z^2$. This is the Lorentz group $O(3, 1)$. Four disconnected mani-
gilmore-lie-groups_F00299	45	■ 0.535	$S\ L(2\ ;\ \mathbb{C})$	$SL(2; \mathbb{C})$	7. The group of 2×2 complex matrices with determinant $+1$ is named $SL(2; \mathbb{C})$. Ma-
gilmore-lie-groups_F00300	45	■ 0.788	$\left[\begin{array}{ll} \alpha & \beta \\ \gamma & \delta \end{array}\right]$	$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$	trices in this group have the structure $\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$, where $\alpha, \beta, \gamma, \delta$ are complex numbers
gilmore-lie-groups_F00301	45	■ 0.987	$\alpha\delta - \beta\gamma = 1$	$\alpha\delta - \beta\gamma = 1$	and $\alpha\delta - \beta\gamma = 1$. Define the matrix X by
gilmore-lie-groups_F00302	45	■ 0.987	X	X	and $\alpha\delta - \beta\gamma = 1$. Define the matrix X by
gilmore-lie-groups_F00303	45	■ 1.000	$\mathbf{x}$	$\mathbf{x}$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_F00304	45	■ 1.000	$\mathbf{x} = (x, y, z)$	$\mathbf{x} = (x, y, z)$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_F00305	45	■ 1.000	$\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$	$\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$	where $\mathbf{x}$ is the three vector $\mathbf{x} = (x, y, z)$ and $\sigma = (\sigma_1, \sigma_2, \sigma_3) = (\sigma_x, \sigma_y, \sigma_z)$ are the
gilmore-lie-groups_F00306	45	■ 1.000	$X^\dagger \equiv (X^\dagger)^* = X$	$X^\dagger \equiv (X^\dagger)^* = X$	a. Show that X is hermitian: $X^\dagger \equiv (X^\dagger)^* = X$.
gilmore-lie-groups_F00307	45	■ 1.000	$g \in SL(2; \mathbb{C})$	$g \in SL(2; \mathbb{C})$	c. If $g \in SL(2; \mathbb{C})$, show that $g^\dagger X g = X' = H(x', y', z', ct')$.
gilmore-lie-groups_F00308	45	■ 1.000	$g^\dagger X g = X' = H(x', y', z', ct')$	$g^\dagger X g = X' = H(x', y', z', ct')$	c. If $g \in SL(2; \mathbb{C})$, show that $g^\dagger X g = X' = H(x', y', z', ct')$.
gilmore-lie-groups_F00309	45	■ 0.964	(x', y', z', ct')	(x', y', z', ct')	d. How are the new space-time coordinates (x', y', z', ct') related to the original
gilmore-lie-groups_F00310	45	■ 0.918	(x, y, z, ct)	(x, y, z, ct)	coordinates (x, y, z, ct) ? (They are linearly related by coefficients that are bilinear
gilmore-lie-groups_F00311	45	■ 1.000	g	g	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_F00312	45	■ 1.000	$\alpha^*, \beta^*, \gamma^*, \delta^*$	$\alpha^*, \beta^*, \gamma^*, \delta^*$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_F00313	45	■ 1.000	$g^\dagger$	$g^\dagger$	in the matrix elements $\alpha, \beta, \gamma, \delta$ of g and $\alpha^*, \beta^*, \gamma^*, \delta^*$ of its adjoint matrix $g^\dagger$.)
gilmore-lie-groups_F00314	45	■ 0.990	$t' = t$	$t' = t$	e. Find the subgroup of $SL(2; \mathbb{C})$ that leaves $t' = t$. (It is $SU(2) \subset SL(2; \mathbb{C})$).
gilmore-lie-groups_F00315	45	■ 0.990	$SU(2) \subset SL(2; \mathbb{C})$	$SU(2) \subset SL(2; \mathbb{C})$	e. Find the subgroup of $SL(2; \mathbb{C})$ that leaves $t' = t$. (It is $SU(2) \subset SL(2; \mathbb{C})$).
gilmore-lie-groups_F00316	45	■ 0.952	$g = k h$	$g = k h$	f. For any $g \in SL(2; \mathbb{C})$ write $g = k h$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_F00317	45	■ 0.952	$h \in SU(2), h^\dagger = h^{-1}, h$	$h \in SU(2), h^\dagger = h^{-1}, h$	f. For any $g \in SL(2; \mathbb{C})$ write $g = k h$, where $h \in SU(2)$, $h^\dagger = h^{-1}$, h has the
gilmore-lie-groups_F00318	45	■ 0.461	$h = \exp\left(\frac{i}{2} \sigma \cdot \theta\right)$	$h = \exp\left(\frac{i}{2} \sigma \cdot \theta\right)$	form $h = \exp(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^\dagger = k^{+1}$, k has the form
gilmore-lie-groups_F00319	45	■ 0.461	$k \in SL(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$	$k \in SL(2; \mathbb{C})/SU(2), k^\dagger = k^{+1}, k$	form $h = \exp(\frac{i}{2} \sigma \cdot \theta)$ and $k \in SL(2; \mathbb{C})/SU(2)$, $k^\dagger = k^{+1}$, k has the form
gilmore-lie-groups_F00320	45	■ 1.000	$k = \exp\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \exp\left(\frac{1}{2} \sigma \cdot \mathbf{b}\right)$	$k = \exp(\frac{1}{2} \sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ

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gilmore-lie-groups_ F08321	45	■ 1.000	$\mathbf{b}$	$\mathbf{b}$	$k = \text{EXP}(\frac{1}{2}\sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08322	45	■ 1.000	θ	θ	$k = \text{EXP}(\frac{1}{2}\sigma \cdot \mathbf{b})$. The three vector $\mathbf{b}$ is called a boost vector. The three vectors θ
gilmore-lie-groups_ F08323	45	■ 0.899	$k^{\dagger}H(x,y,z,ct)k=H(x^{\prime},y^{\prime},z^{\prime},ct^{\prime})$	$k^{\dagger}H(x,y,z,ct)k=H(x^{\prime},y^{\prime},z^{\prime},ct^{\prime})$	and $\mathbf{b}$ are real. Construct $k^{\dagger}H(x,y,z,ct)k=H(x^{\prime},y^{\prime},z^{\prime},ct^{\prime})$. If this is too diffi-
gilmore-lie-groups_ F08324	45	■ 0.893	$\mathbf{b}=(0,0,b)$	$\mathbf{b}=(0,0,b)$	cult, choose $\mathbf{b}$ along the z -axis, $\mathbf{b}=(0,0,b)$.
gilmore-lie-groups_ F08325	45	■ 0.458	$k(b^{\prime})$	$k(b^{\prime})$	h. Applying $k(b^{\prime})$ after applying $k(b)$ results in (a) $k(b^{\prime}+b)$, (b) two successive
gilmore-lie-groups_ F08326	45	■ 0.458	$k(b)$	$k(b)$	h. Applying $k(b^{\prime})$ after applying $k(b)$ results in (a) $k(b^{\prime}+b)$, (b) two successive
gilmore-lie-groups_ F08327	45	■ 0.458	$(a)k(b^{\prime}+b)$	$(a)k(b^{\prime}+b)$	h. Applying $k(b^{\prime})$ after applying $k(b)$ results in (a) $k(b^{\prime}+b)$, (b) two successive
gilmore-lie-groups_ F08328	45	■ 0.981	$\mathbf{b}^{\prime}$	$\mathbf{b}^{\prime}$	i. If $\mathbf{b}$ and $\mathbf{b}^{\prime}$ are not colinear, $k(\mathbf{b}^{\prime})k(\mathbf{b})=k(\mathbf{b}^{\prime\prime})h(\theta)$. Compute $\mathbf{b}^{\prime\prime}$, θ . The angle θ is
gilmore-lie-groups_ F08329	45	■ 0.981	$k(b^{\prime})k(b)=k(b^{\prime\prime})h(\theta)$	$k(b^{\prime})k(b)=k(b^{\prime\prime})h(\theta)$	i. If $\mathbf{b}$ and $\mathbf{b}^{\prime}$ are not colinear, $k(\mathbf{b}^{\prime})k(\mathbf{b})=k(\mathbf{b}^{\prime\prime})h(\theta)$. Compute $\mathbf{b}^{\prime\prime}$, θ . The angle θ is
gilmore-lie-groups_ F08330	45	■ 0.981	$\mathbf{b}^{\prime\prime},\theta$	$\mathbf{b}^{\prime\prime},\theta$	i. If $\mathbf{b}$ and $\mathbf{b}^{\prime}$ are not colinear, $k(\mathbf{b}^{\prime})k(\mathbf{b})=k(\mathbf{b}^{\prime\prime})h(\theta)$. Compute $\mathbf{b}^{\prime\prime}$, θ . The angle θ is
gilmore-lie-groups_ F08331	45	■ 1.000	$\theta\rightarrow\theta^{\prime}=\theta+k+f(\theta)$	$\theta\rightarrow\theta^{\prime}=\theta+k+f(\theta)$	$\theta\rightarrow\theta^{\prime}=\theta+k+f(\theta)$, where k is a real number, $0\leq k<2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08332	45	■ 1.000	$0\leq k<2\pi$	$0\leq k<2\pi$	$\theta\rightarrow\theta^{\prime}=\theta+k+f(\theta)$, where k is a real number, $0\leq k<2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08333	45	■ 1.000	$f(\theta)$	$f(\theta)$	$\theta\rightarrow\theta^{\prime}=\theta+k+f(\theta)$, where k is a real number, $0\leq k<2\pi$, and $f(\theta)$ is periodic,
gilmore-lie-groups_ F08334	45	■ 0.991	$f(\theta+2\pi)=f(\theta)$	$f(\theta+2\pi)=f(\theta)$	$f(\theta+2\pi)=f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08335	45	■ 0.991	1: 1	1 : 1	$f(\theta+2\pi)=f(\theta)$. The mapping must be 1:1, so an additional condition is imposed
gilmore-lie-groups_ F08336	45	■ 1.000	$f(\theta):df(\theta)/d\theta>-1$	$f(\theta):df(\theta)/d\theta>-1$	on $f(\theta):df(\theta)/d\theta>-1$ everywhere. Does this set of transformations form a group?
gilmore-lie-groups_ F08337	46	■ 0.769	(a,b,c,d)	(a,b,c,d)	9. Rational fractional transformations (a,b,c,d) map points on the real line (real pro-
gilmore-lie-groups_ F08338	46	■ 1.000	RP^1	RP^1	jective line RP^1) to the real line as follows:
gilmore-lie-groups_ F08339	46	■ 1.000	$(\lambda a,\lambda b,\lambda c,\lambda d)=\lambda(a,b,c,d)(\lambda\neq 0)$	$(\lambda a,\lambda b,\lambda c,\lambda d)=\lambda(a,b,c,d)(\lambda\neq 0)$	The transformations (a,b,c,d) and $(\lambda a,\lambda b,\lambda c,\lambda d)=\lambda(a,b,c,d)(\lambda\neq 0)$ generate
gilmore-lie-groups_ F08340	46	■ 1.000	A,B,C,D	A,B,C,D	values of A,B,C,D .
gilmore-lie-groups_ F08341	46	■ 0.668	$(\lambda,0,0,\lambda)$	$(\lambda,0,0,\lambda)$	b. Show that the transformations $(\lambda,0,0,\lambda)$ map x to itself.
gilmore-lie-groups_ F08342	46	■ 0.999	$x^{\prime}\rightarrow x$	$x^{\prime}\rightarrow x$	c. Construct the inverse transformation $x^{\prime}\rightarrow x$, and show that it is $\lambda(d,-b,-c,a)$
gilmore-lie-groups_ F08343	46	■ 0.999	$\lambda(d,-b,-c,a)$	$\lambda(d,-b,-c,a)$	c. Construct the inverse transformation $x^{\prime}\rightarrow x$, and show that it is $\lambda(d,-b,-c,a)$
gilmore-lie-groups_ F08344	46	■ 1.000	$\lambda\neq 0$	$\lambda\neq 0$	provided $\lambda\neq 0$. Such transformations exist if $D=ad-bc\neq 0$.
gilmore-lie-groups_ F08345	46	■ 1.000	$D=ad-bc\neq 0$	$D=ad-bc\neq 0$	provided $\lambda\neq 0$. Such transformations exist if $D=ad-bc\neq 0$.
gilmore-lie-groups_ F08346	46	■ 1.000	$x^{\prime}=(a,b,c,d)x=\lambda(a,b,c,d)x$	$x^{\prime}=(a,b,c,d)x=\lambda(a,b,c,d)x$	d. Show that the transformation degeneracy $x^{\prime}=(a,b,c,d)x=\lambda(a,b,c,d)x$ can
gilmore-lie-groups_ F08347	46	■ 1.000	a,b,c,d	a,b,c,d	be lifted by requiring that the four parameters a,b,c,d describing these transfor-
gilmore-lie-groups_ F08348	46	■ 1.000	$D=ad-bc=1$	$D=ad-bc=1$	mations satisfy the constraint $D=ad-bc=1$.

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gilmore-lie-groups_F00349	46	■ 0.980	(y, z)	(y, z)	e. It is useful to introduce homogeneous coordinates (y, z) and define the real projec-
gilmore-lie-groups_F00350	46	■ 1.000	$x=y / z$	$x = y/z$	tive coordinate x as the ratio of these homogeneous coordinates: $x = y/z$. If the
gilmore-lie-groups_F00351	46	■ 0.477	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00352	46	■ 0.477	∞	∞	distinct points x_1, x_2, x_3 on the real line to the three standard positions $(0, 1, \infty)$,
gilmore-lie-groups_F00353	46	■ 0.996	$D=1$	$D = 1$	What matrix in $SL(2; \mathbb{R})$ describes this mapping? (Careful of the condition $D=1$.)
gilmore-lie-groups_F00354	46	■ 1.000	$\left(x_{\{1\}}^{\prime}, x_{\{2\}}^{\prime}, x_{\{3\}}^{\prime}\right)$	(x'_1, x'_2, x'_3)	points (x_1, x_2, x_3) to any other triple of distinct points (x'_1, x'_2, x'_3) .
gilmore-lie-groups_F00355	46	■ 1.000	$R P^{\{n\}}$	RP^n	10. The real projective space RP^n is the space of all straight lines through the origin
gilmore-lie-groups_F00356	46	■ 0.987	$R^{\{n+1\}}$	R^{n+1}	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00357	46	■ 0.987	$S L(n+1 ; \mathbb{R}) \operatorname{maps} x=\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n+1\}}\right) \in R^{\{n+1\}}$	$SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00358	46	■ 0.987	$x^{\{\prime\}}$ $\in R^{\{n+1\}}$	$x' \in R^{n+1}$	in R^{n+1} . The group $SL(n+1; \mathbb{R})$ maps $x = (x_1, x_2, \dots, x_{n+1}) \in R^{n+1}$ to $x' \in R^{n+1}$,
gilmore-lie-groups_F00359	46	■ 1.000	$x^{\{\prime\}} \neq 0 \rightarrow x \neq 0$	$x' \neq 0 \leftrightarrow x \neq 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_F00360	46	■ 1.000	$x^{\{\prime\}}=0 \rightarrow x=0$	$x' = 0 \leftrightarrow x = 0$	with $x' \neq 0 \leftrightarrow x \neq 0$ and $x' = 0 \leftrightarrow x = 0$. A straight line through the origin contains
gilmore-lie-groups_F00361	46	■ 1.000	$x \neq 0$	$x \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00362	46	■ 1.000	$y \neq 0$	$y \neq 0$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00363	46	■ 1.000	$y=\lambda x$	$y = \lambda x$	$x \neq 0$ and $y \neq 0$ if (and only if) $y = \lambda x$ for some real scale factor $\lambda \neq 0$. The scale
gilmore-lie-groups_F00364	46	■ 1.000	y	y	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_F00365	46	■ 1.000	$R^{\{n+1\}}: y \in S^{\{n\}} \subset R^{\{n+1\}}$	$R^{n+1}: y \in S^n \subset R^{n+1}$	factor can always be chosen so that y is in the unit sphere in R^{n+1} : $y \in S^n \subset R^{n+1}$.
gilmore-lie-groups_F00366	46	■ 1.000	λ	λ	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_F00367	46	■ 1.000	$\lambda=\pm 1 /\left(\sum_{i=1}^{n+1} x_{\{i\}}^{\{2\}}\right)^{1 / 2}$	$\lambda = \pm 1 / \left(\sum_{i=1}^{n+1} x_i^2\right)^{1/2}$	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_F00368	46	■ 1.000	$R^{\{3\}}$	R^3	In fact, two values of λ can be chosen: $\lambda = \pm 1/(\sum_{i=1}^{n+1} x_i^2)^{1/2}$. In R^3 the straight
gilmore-lie-groups_F00369	46	■ 0.983	(x, y, z)	(x, y, z)	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups_F00370	46	■ 0.983	$(X, Y)=$	$(X, Y) =$	line containing (x, y, z) can be represented by homogeneous coordinates $(X, Y) =$
gilmore-lie-groups_F00371	46	■ 1.000	$(x / z, y / z)$	$(x/z, y/z)$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups_F00372	46	■ 1.000	$z \neq 0$	$z \neq 0$	$(x/z, y/z)$ if $z \neq 0$. Straight lines through the origin of R^3 are mapped to straight lines
gilmore-lie-groups_F00373	46	■ 1.000	$x \rightarrow x^{\{\prime\}}=M x, M \in S L(3 ; \mathbb{R})$	$x \rightarrow x' = Mx, M \in SL(3; \mathbb{R})$	in R^3 by $x \rightarrow x' = Mx, M \in SL(3; \mathbb{R})$. Show that the homogeneous coordinates
gilmore-lie-groups_F00374	47	■ 1.000	$x^{\{\prime\}}$	x'	representing the two lines containing x and x' are related by the linear fractional
gilmore-lie-groups_F00375	47	■ 1.000	$R P^{\{n\}} \rightarrow R P^{\{n\}}$	$RP^n \rightarrow RP^n$	Generalize for linear fractional transformations $RP^n \rightarrow RP^n$.

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gilmore-lie-groups_ F00376	47	■ 0.989	$S L(2 ; \mathbb{R}) / S O(2) \simeq \left[\begin{array}{cc} z+x & y \\ y & z-x \end{array} \right]$	$SL(2; \mathbb{R})/SO(2) \simeq \begin{bmatrix} z+x & y \\ y & z-x \end{bmatrix}$	11. The hyperbolic two-space $SL(2; \mathbb{R})/SO(2) \simeq \begin{bmatrix} z+x & y \\ y & z-x \end{bmatrix}$ consists of the algebraic sub-
gilmore-lie-groups_ F00377	47	■ 1.000	$x=r \cos \phi, y=r \sin \phi$	$x = r \cos \phi, y = r \sin \phi$	b. Introduce polar coordinates $x = r \cos \phi$, $y = r \sin \phi$, and show
gilmore-lie-groups_ F00378	47	■ 0.633	$S O(3) / S O(2)$	$SO(3)/SO(2)$	d. Repeat this calculation for $SO(3)/SO(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups_ F00379	47	■ 0.633	$S^2 \subset R^3$	$S^2 \subset R^3$	d. Repeat this calculation for $SO(3)/SO(2)$. This space is a sphere $S^2 \subset R^3$: the
gilmore-lie-groups_ F00380	47	■ 1.000	$z^2+\left(x^2+y^2\right)=1$	$z^2 + (x^2 + y^2) = 1$	algebraic manifold in R^3 that satisfies $z^2 + (x^2 + y^2) = 1$ and inherits the metric
gilmore-lie-groups_ F00381	47	■ 1.000	$d s^2=d z^2+\left(d x^2+d y^2\right)$	$ds^2 = dz^2 + (dx^2 + dy^2)$	$ds^2 = dz^2 + (dx^2 + dy^2)$ from this Euclidean space. Show that the metric and
gilmore-lie-groups_ F00382	47	■ 1.000	H^2	H^2	measure on S^2 are obtained from the results above for H^2 by the substitutions
gilmore-lie-groups_ F00383	47	■ 1.000	$1+r^2 \rightarrow 1-r^2$	$1 + r^2 \rightarrow 1 - r^2$	$1 + r^2 \rightarrow 1 - r^2$. Show that the disk $0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$ maps onto the up-
gilmore-lie-groups_ F00384	47	■ 1.000	$0 \leq r \leq 1, 0 \leq \phi \leq 2 \pi$	$0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$	$1 + r^2 \rightarrow 1 - r^2$. Show that the disk $0 \leq r \leq 1, 0 \leq \phi \leq 2\pi$ maps onto the up-
gilmore-lie-groups_ F00385	47	■ 1.000	$r=0$	$r = 0$	per hemisphere of the sphere, with $r = 0$ mapping to the north pole and $r = 1$
gilmore-lie-groups_ F00386	47	■ 1.000	$r=1$	$r = 1$	per hemisphere of the sphere, with $r = 0$ mapping to the north pole and $r = 1$
gilmore-lie-groups_ F00387	47	■ 1.000	$\phi=0$	$\phi = 0$	equator along the longitude $\phi = 0$ is $s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$ and the volume of
gilmore-lie-groups_ F00388	47	■ 1.000	$s=\int_0^1 d r / \sqrt{1-r^2}=\pi / 2$	$s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$	equator along the longitude $\phi = 0$ is $s = \int_0^1 dr/\sqrt{1-r^2} = \pi/2$ and the volume of
gilmore-lie-groups_ F00389	47	■ 1.000	$V=\int_{r=0}^1 \int_{\phi=0}^{2 \pi} d V(r, \phi)=\int_0^1 r d r / \sqrt{1-r^2} \int_0^{2 \pi} d \phi=$	$V = \int_{r=0}^1 \int_{\phi=0}^{2\pi} dV(r, \phi) = \int_0^1 r dr/\sqrt{1-r^2} \int_0^{2\pi} d\phi =$	the hemisphere surface is $V = \int_{r=0}^1 \int_{\phi=0}^{2\pi} dV(r, \phi) = \int_0^1 r dr/\sqrt{1-r^2} \int_0^{2\pi} d\phi =$
gilmore-lie-groups_ F00390	47	■ 1.000	2π	2π	—
gilmore-lie-groups_ F00391	48	■ 0.969	$G L(n ; \mathbb{F})$	$GL(n; \mathbb{F})$	typically encountered. These include the general linear groups $GL(n; \mathbb{F})$
gilmore-lie-groups_ F00392	48	■ 1.000	$\mathbb{F}$	$\mathbb{F}$	of nonsingular $n \times n$ matrices over the fields $\mathbb{F}$ of real numbers, complex
gilmore-lie-groups_ F00393	48	■ 1.000	$H_1 \subset G$	$H_1 \subset G$	Theorem If $H_1 \subset G$ and $H_2 \subset G$ are subgroups of G then their intersection
gilmore-lie-groups_ F00394	48	■ 1.000	$H_{12}=H_1 \cap H_2$	$H_{12} = H_1 \cap H_2$	$H_{12} \equiv H_1 \cap H_2$ is a subgroup of G .
gilmore-lie-groups_ F00395	48	■ 1.000	$H_1 \cap H_2$	$H_1 \cap H_2$	Proof Verify that the four group axioms are satisfied for all operations in $H_1 \cap H_2$.
gilmore-lie-groups_ F00396	48	■ 0.950	H_1	H_1	Example If H_1 is the two-dimensional subgroup of $SL(2; \mathbb{R})$ described in (3.1)
gilmore-lie-groups_ F00397	48	■ 0.817	H_2	H_2	above and H_2 is the one-dimensional subgroup of 2×2 orthogonal matrices
gilmore-lie-groups_ F00398	49	■ 1.000	$\pm I_2$	$\pm I_2$	discrete group operations $\pm I_2$.
gilmore-lie-groups_ F00399	49	■ 0.950	$(\mathbb{F}=\mathbb{R})$	$(\mathbb{F} = \mathbb{R})$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups_ F00400	49	■ 0.950	$\mathbb{F}=\mathbb{C}$	$\mathbb{F} = \mathbb{C}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-
gilmore-lie-groups_ F00401	49	■ 0.950	$\mathbb{F}=\mathbb{Q}$	$\mathbb{F} = \mathbb{Q}$	$(\mathbb{F} = \mathbb{R})$, complex numbers $(\mathbb{F} = \mathbb{C})$, and quaternions $(\mathbb{F} = \mathbb{Q})$. The complex num-

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gilmore-lie-groups_F00402	49	■ 1.000	$1, \mathcal{I}, J, K$	$1, \mathcal{I}, J, K$	The four basis vectors $1, \mathcal{I}, J, K$ for this map are related to the four Pauli spin
gilmore-lie-groups_F00403	49	■ 0.900	$[\operatorname{det}(\mathrm{M})=+1]$	$[\det(M) = +1]$	$[\det(M) = +1]$ are imposed (12).
gilmore-lie-groups_F00404	49	■ 0.999	$G L(1 ; \mathbb{Q})$	$GL(1; \mathbb{Q})$	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups_F00405	49	■ 0.999	1×1	1×1	the real, complex, or quaternion fields. The group $GL(1; \mathbb{Q})$ consists of 1×1
gilmore-lie-groups_F00406	50	■ 1.000	$A_{\mathrm{i}}^{\mathrm{j}}$	A_i^j	The determinant of an $n \times n$ matrix A with matrix elements A_i^j is defined by
gilmore-lie-groups_F00407	50	■ 1.000	$\epsilon^{\mathrm{i}_1 \mathrm{i}_2 \cdots \mathrm{i}_n}$	$\epsilon^{i_1 i_2 \cdots i_n}$	Here $\epsilon^{i_1 i_2 \cdots i_n}$ and its covariant version are the Levi–Civita symbols: +1 for an even
gilmore-lie-groups_F00408	50	■ 0.719	$1, 2, \ldots, n ; -1$	$1, 2, \ldots, n; -1$	permutation of the integers $1, 2, \ldots, n; -1$ for an odd permutation; and 0 if two
gilmore-lie-groups_F00409	50	■ 1.000	$i_{\ast}$	i_*	or more values of the indices i_* are equal. With this definition there is no difficulty
gilmore-lie-groups_F00410	50	■ 1.000	$U T(p, q)$	$UT(p, q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups_F00411	50	■ 1.000	$n \times n (n=p+q)$	$n \times n (n = p + q)$	2. $UT(p, q)$. Upper triangular groups. The $n \times n$ ($n = p + q$) matrix is parti-
gilmore-lie-groups_F00412	50	■ 1.000	$U T(1, 1)$	$UT(1, 1)$	Example The action of transformations in $UT(1, 1)$ on the plane R^2 is as
gilmore-lie-groups_F00413	50	■ 0.991	$y=0$	$y = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($y = 0 \rightarrow y' = 0$),
gilmore-lie-groups_F00414	50	■ 0.991	$y=0 \rightarrow y^{\prime}=0$	$y = 0 \rightarrow y' = 0$	The x -axis $y = 0$ remains invariant. It is an invariant subspace ($y = 0 \rightarrow y' = 0$),
gilmore-lie-groups_F00415	50	■ 1.000	$x=0$	$x = 0$	mapped into itself by all group operations in $UT(1, 1)$. The y -axis $x = 0$ is not
gilmore-lie-groups_F00416	50	■ 1.000	$V_{\mathrm{p}} \oplus V_{\mathrm{q}}$	$V_p \oplus V_q$	invariant. More generally, if $UT(p, q)$ acts on the direct sum vector space $V_p \oplus V_q$,
gilmore-lie-groups_F00417	50	■ 1.000	V_{q}	V_q	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups_F00418	50	■ 1.000	V_{p}	V_p	the subspace V_q is invariant while V_p is not. For lower triangular matrices reverse
gilmore-lie-groups_F00419	50	■ 1.000	q	q	p and q .
gilmore-lie-groups_F00420	51	■ 1.000	$H T(p, q)$	$HT(p, q)$	3. $HT(p, q)$. This is a subgroup of $UT(p, q)$ obtained by imposing the additional
gilmore-lie-groups_F00421	51	■ 1.000	$H T(1, 1) \left(m_{22}=1 \right)$	$HT(1, 1) (m_{22} = 1)$	Example Affine transformations in $HT(1, 1) (m_{22} = 1)$ act on the x -axis by $x \rightarrow$
gilmore-lie-groups_F00422	51	■ 1.000	$x \rightarrow$	$x \rightarrow$	Example Affine transformations in $HT(1, 1) (m_{22} = 1)$ act on the x -axis by $x \rightarrow$
gilmore-lie-groups_F00423	51	■ 0.997	$x^{\prime}=a x+b$	$x' = ax + b$	$x' = ax + b$:
gilmore-lie-groups_F00424	51	■ 0.999	$U T(p, q, r)$	$UT(p, q, r)$	4. $UT(p, q, r)$. This matrix group consists of upper triangular matrices that are the
gilmore-lie-groups_F00425	51	■ 1.000	$U T(p, q+r) \cap U T(p+q, r)$	$UT(p, q + r) \cap UT(p + q, r)$	intersection of the matrix groups $UT(p, q + r) \cap UT(p + q, r)$.
gilmore-lie-groups_F00426	52	■ 0.460	$S U(1, 1)$	$SU(1, 1)$	where the 2×2 matrix $SU(1, 1)$ is defined below in (3.30). Matrix groups with the
gilmore-lie-groups_F00427	52	■ 1.000	$\operatorname{Sol}(n)=U T(1, 1, 1, \ldots, 1)$	$\operatorname{Sol}(n) = UT(1, 1, 1, \ldots, 1)$	5. $\operatorname{Sol}(n) = UT(1, 1, 1, \ldots, 1)$. Solvable groups are strictly upper triangular.
gilmore-lie-groups_F00428	52	■ 0.971	$U T(1, 1, 1)$	$UT(1, 1, 1)$	Example We consider the subgroup of 3×3 matrices in $UT(1, 1, 1)$ of the

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gilmore-lie-groups_ F00429	52	■ 1.000	$\hat{n}=a^{\dagger}a$	$\hat{n} = a^{\dagger}a$	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_ F00430	52	■ 1.000	$a^{\dagger}$	$a^{\dagger}$	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_ F00431	52	■ 1.000	a	a	of the photon number operator ($\hat{n} = a^{\dagger}a$), the creation ($a^{\dagger}$) and annihilation (a)
gilmore-lie-groups_ F00432	52	■ 0.994	$I=a^{\dagger}a-a a^{\dagger}$	$I = aa^{\dagger} - a^{\dagger}a = [a, a^{\dagger}]$	operators, and their commutator ($I = aa^{\dagger} - a^{\dagger}a = [a, a^{\dagger}]$). We will use this iden-
gilmore-lie-groups_ F00433	52	■ 1.000	$\operatorname{Nil}(n)$	$\operatorname{Nil}(n)$	6. $\operatorname{Nil}(n)$. Nilpotent groups are subgroups of $\operatorname{Sol}(n)$ whose diagonal matrix
gilmore-lie-groups_ F00434	52	■ 1.000	$\operatorname{Sol}(n)$	$\operatorname{Sol}(n)$	6. $\operatorname{Nil}(n)$. Nilpotent groups are subgroups of $\operatorname{Sol}(n)$ whose diagonal matrix
gilmore-lie-groups_ F00435	52	■ 0.998	$\operatorname{Nil}(3)$	$\operatorname{Nil}(3)$	Example Matrices in $\operatorname{Nil}(3)$ of the form
gilmore-lie-groups_ F00436	52	■ 0.976	$a^{\dagger}, a, I$	$a^{\dagger}, a, I$	are closely related to the photon creation and annihilation operators ($a^{\dagger}, a, I$) and
gilmore-lie-groups_ F00437	52	■ 0.716	$(p$	$(p$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00438	52	■ 0.716	$q)$	$q)$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00439	52	■ 0.716	$[p, q]=\hbar / i$	$[p, q] = \hbar/i$	$(p$ and $q)$ and their commutator $[p, q] = \hbar/i$. This 3×3 matrix group is called the
gilmore-lie-groups_ F00440	52	■ 1.000	$\langle p \mid q\rangle=\frac{1}{\sqrt{2}} e^{2 \pi i p q / \hbar}$	$\langle p \mid q\rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / \hbar}$	The set of change of basis transformations $\langle p q\rangle = \frac{1}{\sqrt{2}} e^{2\pi i p q / \hbar}$ encountered in
gilmore-lie-groups_ F00441	52	■ 0.995	$A(p, q)$	$A(p, q)$	7. $A(p, q)$. This group consists of matrices that are the sum of an identity matrix
gilmore-lie-groups_ F00442	52	■ 0.998	(p, q)	(p, q)	and the upper right-hand off-diagonal block of a (p, q) blocked matrix. Its matrix
gilmore-lie-groups_ F00443	53	■ 1.000	$AB=BA$	$AB = BA$	This group is abelian or commutative: $AB = BA$ for all elements (matrices) in this
gilmore-lie-groups_ F00444	53	■ 1.000	$A(1,1)$	$A(1,1)$	Example We consider the translation subgroup $A(1,1)$ of the affine group of
gilmore-lie-groups_ F00445	53	■ 0.798	$x \rightarrow x^{\prime}=x+a$	$x \rightarrow x' = x + a$	transformations of the x -axis (3.9): $x \rightarrow x' = x + a$. Successive transformations
gilmore-lie-groups_ F00446	53	■ 1.000	$M^{\dagger} G M=G$	$M^{\dagger} G M = G$	all satisfy the bilinear or quadratic constraint condition $M^{\dagger} G M = G$. If G is sym-
gilmore-lie-groups_ F00447	53	■ 0.679	$G=I_n$	$G = I_n$	metric positive-definite we can set $G = I_n$ (8). If G is nonsingular and symmetric
gilmore-lie-groups_ F00448	53	■ 0.573	$G=I_{p, q}$	$G = I_{p,q}$	but indefinite we can set $G = I_{p,q}$ (9). If G is nonsingular and antisymmetric, we
gilmore-lie-groups_ F00449	53	■ 1.000	I_n	I_n	which we can take as I_n
gilmore-lie-groups_ F00450	54	■ 0.936	$GL(n ; \mathbb{F}), \mathbb{F}=\mathbb{R}, \mathbb{C}, \mathbb{Q}$	$GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	terize the matrix groups $GL(n; \mathbb{F}), \mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.
gilmore-lie-groups_ F00451	54	■ 1.000	C^2	C^2	two-dimensional linear vector space C^2 (of spin states, for example), and quaternion
gilmore-lie-groups_ F00452	54	■ 1.000	$\mathbb{Q}$	$\mathbb{Q}$	over $\mathbb{Q}$
gilmore-lie-groups_ F00453	54	■ 0.808	$SU(1 ; \mathbb{Q})$	$SU(1; \mathbb{Q})$	The group $SU(1; \mathbb{Q})$ is the subgroup of $GL(1; \mathbb{Q})$ (3.5) subject to the condition
gilmore-lie-groups_ F00454	54	■ 1.000	R^4	R^4	in R^4
gilmore-lie-groups_ F00455	54	■ 0.992	$G=I_{p, q}, p+q=n$	$G = I_{p,q}, p + q = n$	$G = I_{p,q}, p + q = n$. This is a diagonal matrix with p elements $+1$ and q ele-

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gilmore-lie-groups_ F00456	54	■ 1.000	$p \neq 0, q \neq 0$	$p \neq 0, q \neq 0$	The manifolds that parameterize these groups are noncompact when $p \neq 0, q \neq 0$.
gilmore-lie-groups_ F00457	54	■ 1.000	$x^2+y^2+z^2-(ct)^2$	$x^2 + y^2 + z^2 - (ct)^2$	Example The Lorentz group preserves the invariant $x^2 + y^2 + z^2 - (ct)^2$ and is
gilmore-lie-groups_ F00458	55	■ 0.999	$N \times N$	$N \times N$	10. Antisymmetric metric-preserving groups The metric G is an $N \times N$ non-
gilmore-lie-groups_ F00459	55	■ 1.000	$\operatorname{det}(G)=\operatorname{det}\left(G^t\right)=\operatorname{det}(-G)=(-1)^N \operatorname{det}(G), N$	$\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$	Since $\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$ must be even: $N = 2n$. The
gilmore-lie-groups_ F00460	55	■ 1.000	$N=2 n$	$N = 2n$	Since $\det(G) = \det(G^t) = \det(-G) = (-1)^N \det(G), N$ must be even: $N = 2n$. The
gilmore-lie-groups_ F00461	55	■ 1.000	$i \sigma_y=\left[\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right]$	$i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	This consists of n copies of the matrix $i\sigma_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ along the diagonal. Symplectic
gilmore-lie-groups_ F00462	55	■ 0.990	$\operatorname{Sp}(2 n ; \mathbb{R})$	$\operatorname{Sp}(2n; \mathbb{R})$	transformations in $\operatorname{Sp}(2n; \mathbb{R})$ leave invariant the form of the classical hamiltonian
gilmore-lie-groups_ F00463	55	■ 0.935	$\operatorname{Sp}(2 ; \mathbb{R}) \subset \operatorname{GL}(2 ; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	Example The symplectic group $\operatorname{Sp}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$ satisfies the constraint
gilmore-lie-groups_ F00464	55	■ 0.510	$a d-b c=+1$	$ad - bc = +1$	The constraint is $ad - bc = +1$. Thus, $\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$.
gilmore-lie-groups_ F00465	55	■ 0.510	$\operatorname{Sp}(2 ; \mathbb{R})=\operatorname{SL}(2 ; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$	The constraint is $ad - bc = +1$. Thus, $\operatorname{Sp}(2; \mathbb{R}) = \operatorname{SL}(2; \mathbb{R})$.
gilmore-lie-groups_ F00466	55	■ 1.000	$M G M^t=G$	$MGM^t = G$	Partitioning M into blocks and imposing the condition $MGM^t = G$, we find
gilmore-lie-groups_ F00467	56	■ 0.999	$E(3)$	$E(3)$	Euclidean group $E(3)$ whose action on the coordinates (x, y, z) of a point in
gilmore-lie-groups_ F00468	56	■ 0.607	$\mathbf{t}$	$\mathbf{t}$	That is, the coordinates are rotated by the matrix A and translated by the vector $\mathbf{t}$. By
gilmore-lie-groups_ F00469	56	■ 0.792	$A \in \operatorname{SO}(3,1), A I_{3,1} A^t=I_{3,1}$	$A \in \operatorname{SO}(3,1), AI_{3,1}A^t = I_{3,1}$	($A \in \operatorname{SO}(3,1), AI_{3,1}A^t = I_{3,1}$ (3.17)) and space-time displacements is isomorphic
gilmore-lie-groups_ F00470	56	■ 0.608	$\mathbf{v}$	$\mathbf{v}$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00471	56	■ 0.608	$(\mathbf{t})$	$(\mathbf{t})$	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00472	56	■ 0.608	$\left(t_4\right)$	(t_4)	moving with velocity $\mathbf{v}$, and displacements of space ($\mathbf{t}$) and time (t_4) coordinates.
gilmore-lie-groups_ F00473	57	■ 0.991	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] \in \operatorname{SL}(2 ; \mathbb{R}) \subset \operatorname{GL}(2 ; \mathbb{R})$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \operatorname{SL}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$	traces $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \operatorname{SL}(2; \mathbb{R}) \subset \operatorname{GL}(2; \mathbb{R})$ satisfies the constraint $ad - bc = +1$, which
gilmore-lie-groups_ F00474	57	■ 0.998	$U(1,1)$	$U(1,1)$	taking the intersection of $U(1, 1)$ with $\operatorname{SL}(2; \mathbb{C})$:
gilmore-lie-groups_ F00475	57	■ 1.000	$a^* a-b^* b=+1$	$a^*a - b^*b = +1$	where $a^*a - b^*b = +1$.
gilmore-lie-groups_ F00476	57	■ 1.000	$U(n)$	$U(n)$	The unitary group $U(n)$ consists of $n \times n$ complex matrices that obey the constraint
gilmore-lie-groups_ F00477	57	■ 1.000	$U^{\dagger} U=I_n$	$U^\dagger U = I_n$	$U^\dagger U = I_n$. For some purposes it is useful to represent this group as a group of real
gilmore-lie-groups_ F00478	57	■ 1.000	$2 n \times 2 n$	$2n \times 2n$	is a real $2n \times 2n$ matrix M . This matrix inherits the constraint that comes with the
gilmore-lie-groups_ F00479	57	■ 1.000	$M^t M=I_{2 n}$	$M^t M = I_{2n}$	unitary group, $U^\dagger U = I_n$. This constraint now appears in the form $M^t M = I_{2n}$.
gilmore-lie-groups_ F00480	57	■ 1.000	$\{ \}^{\dagger}$	†	We have been able to replace † by t since the matrices are real, and must replace
gilmore-lie-groups_ F00481	57	■ 1.000	$\{ \}^t$	t	We have been able to replace † by t since the matrices are real, and must replace

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gilmore-lie-groups_F00482	57	■ 1.000	I_{2n}	I_{2n}	I_n by I_{2n} since the matrices are $2n \times 2n$. In other words, the matrices M obey the
gilmore-lie-groups_F00483	57	■ 0.712	$SO(2n)$	$SO(2n)$	an <i>orthogonal</i> representation of the <i>unitary</i> group. It is a subgroup of $SO(2n)$. This
gilmore-lie-groups_F00484	57	■ 1.000	$OU(2n)$	$OU(2n)$	matrix group is called $OU(2n)$. Symbolically,
gilmore-lie-groups_F00485	58	■ 0.999	$U(n; \mathbb{Q}) = Sp(n)$	$U(n; \mathbb{Q}) = Sp(n)$	For the metric-preserving quaternion group $U(n; \mathbb{Q}) = Sp(n)$ whose matrices obey
gilmore-lie-groups_F00486	58	■ 1.000	$M^{\dagger} M = I_{2n}$	$M^{\dagger} M = I_{2n}$	constraint in the form $M^{\dagger} M = I_{2n}$. In other words, the matrices M obey the condi-
gilmore-lie-groups_F00487	58	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	tion that determines unitary groups (over $\mathbb{C}$). This group of $2n \times 2n$ matrices forms
gilmore-lie-groups_F00488	58	■ 0.945	$SU(2n)$	$SU(2n)$	a <i>unitary</i> representation of the <i>symplectic</i> group. It is a subgroup of $SU(2n)$. This
gilmore-lie-groups_F00489	58	■ 1.000	$USp(2n)$	$USp(2n)$	matrix group is called $USp(2n)$. Symbolically,
gilmore-lie-groups_F00490	58	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	because they lack the geometric structure contributed by the continuous fields $\mathbb{R}$,
gilmore-lie-groups_F00491	58	■ 1.000	$GL(n; \mathbb{Z})$	$GL(n; \mathbb{Z})$	(i) $GL(n; \mathbb{Z})$: if $m \in GL(n; \mathbb{Z})$, $\det(m) = \pm 1$.
gilmore-lie-groups_F00492	58	■ 1.000	$m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$	$m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$	(i) $GL(n; \mathbb{Z})$: if $m \in GL(n; \mathbb{Z}), \det(m) = \pm 1$.
gilmore-lie-groups_F00493	58	■ 1.000	$SL(n; \mathbb{Z})$	$SL(n; \mathbb{Z})$	(ii) $SL(n; \mathbb{Z})$: if $m \in SL(n; \mathbb{Z})$, $\det(m) = +1$.
gilmore-lie-groups_F00494	58	■ 1.000	$m \in SL(n; \mathbb{Z}), \det(m) = +1$	$m \in SL(n; \mathbb{Z}), \det(m) = +1$	(ii) $SL(n; \mathbb{Z})$: if $m \in SL(n; \mathbb{Z}), \det(m) = +1$.
gilmore-lie-groups_F00495	58	■ 1.000	$PSL(n; \mathbb{Z}), n$	$PSL(n; \mathbb{Z}), n$	(iii) $PSL(n; \mathbb{Z}), n$ even: $PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$.
gilmore-lie-groups_F00496	58	■ 1.000	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	$PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$	(iii) $PSL(n; \mathbb{Z}), n$ even: $PSL(n; \mathbb{Z}) = SL(n; \mathbb{Z}) / \{I_n, -I_n\}$
gilmore-lie-groups_F00497	58	■ 0.999	$n=2$	$n = 2$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_F00498	58	■ 0.999	$\left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	For $n = 2$ these groups of matrices have the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with a, b, c, d all integers.
gilmore-lie-groups_F00499	58	■ 1.000	$\det(m) = n$	$\det(m) = n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_F00500	58	■ 1.000	$\det(m^{-1}) = 1/n$	$\det(m^{-1}) = 1/n$	If $\det(m) = n$, with n an integer, then $\det(m^{-1}) = 1/n$. Since the determinant of
gilmore-lie-groups_F00501	58	■ 1.000	$\det(m) =$	$\det(m) =$	any matrix composed of integers must be an integer, the condition is that $\det(m) =$
gilmore-lie-groups_F00502	58	■ 0.714	± 1	± 1	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_F00503	58	■ 0.714	$GL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_F00504	58	■ 0.714	$SL(2; \mathbb{Z}) \subset$	$SL(2; \mathbb{Z}) \subset$	± 1 . The subset of $GL(2; \mathbb{Z})$ with determinant $+1$ forms the subgroup $SL(2; \mathbb{Z}) \subset$
gilmore-lie-groups_F00505	58	■ 1.000	$PSL(2; \mathbb{Z})$	$PSL(2; \mathbb{Z})$	$GL(2; \mathbb{Z})$. The modular group $PSL(2; \mathbb{Z})$ is obtained by identifying each pair of
gilmore-lie-groups_F00506	58	■ 0.906	$SL(2; \mathbb{Z})$	$SL(2; \mathbb{Z})$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
gilmore-lie-groups_F00507	58	■ 0.906	$\left[\begin{array}{ll} -a & -b \\ -c & -d \end{array} \right] \simeq \left[\begin{array}{ll} a & b \\ c & d \end{array} \right]$	$\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	matrices in $SL(2; \mathbb{Z})$ of the form $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \simeq \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

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gilmore-lie-groups_F00508	59	■ 1.000	$F(n)$	$F(n)$	where $F(n)$ is the n th Fibonacci number, defined recursively by
gilmore-lie-groups_F00509	59	■ 1.000	$n=1(F(0)=0, F(1)=1)$	$n = 1(F(0) = 0, F(1) = 1)$	and by comparison of initial conditions for $n = 1$ ($F(0) = 0, F(1) = 1$). Many
gilmore-lie-groups_F00510	59	■ 0.998	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	$GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$	in the groups $GL(2; \mathbb{Z}), GL(3; \mathbb{Z})$, etc.
gilmore-lie-groups_F00511	59	■ 0.998	$GL(n; \mathbb{R})$	$GL(n; \mathbb{R})$	with $GL(n; \mathbb{R})$.
gilmore-lie-groups_F00512	59	■ 1.000	$M^t I_n M = I_n$	$M^t I_n M = I_n$	Imposing quadratic constraints, for example $M^t I_n M = I_n$, generates a sub-
gilmore-lie-groups_F00513	59	■ 1.000	$O(n; \mathbb{Z})$	$O(n; \mathbb{Z})$	group, $O(n; \mathbb{Z})$, consists of $n \times n$ matrices in which all but one matrix element
gilmore-lie-groups_F00514	59	■ 1.000	$\operatorname{det}(m)=+1$	$\det(m) = +1$	Finally, the multilinear condition $\det(m) = +1$ defines the unimodular subgroup
gilmore-lie-groups_F00515	60	■ 1.000	A_n	A_n	ple, the alternating group A_n consists of unimodular matrices in P_n :
gilmore-lie-groups_F00516	60	■ 0.932	$O(2; \mathbb{Z})$	$O(2; \mathbb{Z})$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups_F00517	60	■ 0.932	$8=2^2 \times 2!$	$8 = 2^2 \times 2!$	Example The group $O(2; \mathbb{Z})$ consists of the $8 = 2^2 \times 2!$ matrices
gilmore-lie-groups_F00518	60	■ 0.974	$O(3; \mathbb{Z})$	$O(3; \mathbb{Z})$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_F00519	60	■ 0.974	$2^3 \times 3! = 48$	$2^3 \times 3! = 48$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_F00520	60	■ 0.974	$6=3!$	$6 = 3!$	The group $O(3; \mathbb{Z})$ has order $2^3 \times 3! = 48$. Its subgroup S_3 of order $6 = 3!$ consists
gilmore-lie-groups_F00521	60	■ 1.000	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	$A_3 \subset S_3 \subset O(3; \mathbb{Z})$	Its alternating subgroup $A_3 \subset S_3 \subset O(3; \mathbb{Z})$ consists of the three matrices with
gilmore-lie-groups_F00522	60	■ 0.997	G_2, F_4, E_6, E_7, E_8	G_2, F_4, E_6, E_7, E_8	G_2, F_4, E_6, E_7, E_8 and their real forms (encountered in Chapters 10 and 11), and
gilmore-lie-groups_F00523	60	■ 1.000	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	$GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups_F00524	60	■ 1.000	$PSL(n; \mathbb{Z})$	$PSL(n; \mathbb{Z})$	integers, $GL(n; \mathbb{Z}), SL(n; \mathbb{Z})$, and $PSL(n; \mathbb{Z})$. Matrix groups over finite fields are
gilmore-lie-groups_F00525	61	■ 1.000	$\mathcal{I}J = -\mathcal{K}$	$\mathcal{I}J = -\mathcal{K}$	where $\mathcal{I}J = -\mathcal{K}$:
gilmore-lie-groups_F00526	61	■ 0.997	$\sigma_x, \sigma_y, \sigma_z$	$\sigma_x, \sigma_y, \sigma_z$	Here $\sigma_x, \sigma_y, \sigma_z$ are the Pauli spin matrices, and i is the usual square root of -1 . Show
gilmore-lie-groups_F00527	61	■ 0.924	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{I}\mathcal{J} + \mathcal{J}\mathcal{I} = 0$	$\{\mathcal{I}, \mathcal{J}\} = \mathcal{I}\mathcal{J} + \mathcal{J}\mathcal{I} = 0$	that any pair of the unit quaternions anticommute: i.e., $\{\mathcal{I}, \mathcal{J}\} = \mathcal{I}\mathcal{J} + \mathcal{J}\mathcal{I} = 0$.
gilmore-lie-groups_F00528	61	■ 1.000	$\mathcal{I}, J, K$	$\mathcal{I}, J, K$	2. Show that the unit quaternions $\mathcal{I}, J, K$ generate a group of order 8 under multi-
gilmore-lie-groups_F00529	61	■ 0.464	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	$SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$	3. Show that $SU(1; \mathbb{Q}) \sim SU(2; \mathbb{C})$.
gilmore-lie-groups_F00530	61	■ 1.000	$p+q=n$	$p + q = n$	in Example (9), for suitable positive integers p and q , with $p + q = n$.
gilmore-lie-groups_F00531	61	■ 0.996	$SL_i(n; \mathbb{C})$	$SL_i(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups_F00532	61	■ 0.996	$GL(n; \mathbb{C})$	$GL(n; \mathbb{C})$	6. Show that it is possible to define subgroups $SL_i(n; \mathbb{C})$ of $GL(n; \mathbb{C})$ by the con-
gilmore-lie-groups_F00533	61	■ 1.000	ϕ, λ, r	ϕ, λ, r	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups
gilmore-lie-groups_F00534	61	■ 1.000	$r \neq 0$	$r \neq 0$	where ϕ, λ, r are real and $r \neq 0$. Show that the dimensions of these three subgroups

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gilmore-lie-groups_F08535	61	1.000	$2n^2-1$	$2n^2-1$	are $2n^2-1$ and that $SL_3(n;\mathbb{C})$ is disconnected. It consists of two topologically iden-														
gilmore-lie-groups_F08536	61	1.000	$SL_3(n;\mathbb{C})$	$SL_3(n;\mathbb{C})$	are $2n^2-1$ and that $SL_3(n;\mathbb{C})$ is disconnected. It consists of two topologically iden-														
gilmore-lie-groups_F08537	61	1.000	$SL(n;\mathbb{C})=SL_1(n;\mathbb{C})\cap SL_2(n;\mathbb{C})$	$SL(n;\mathbb{C})=SL_1(n;\mathbb{C})\cap SL_2(n;\mathbb{C})$	Show that $SL(n;\mathbb{C})=SL_1(n;\mathbb{C})\cap SL_2(n;\mathbb{C})$. Do these results extend under the field														
gilmore-lie-groups_F08538	61	1.000	$\mathbb{C}\rightarrow\mathbb{R}$	$\mathbb{C}\rightarrow\mathbb{R}$	restriction $\mathbb{C}\rightarrow\mathbb{R}$? and the field extension $\mathbb{C}\rightarrow\mathbb{Q}$?														
gilmore-lie-groups_F08539	61	1.000	$\mathbb{C}\rightarrow\mathbb{Q}$	$\mathbb{C}\rightarrow\mathbb{Q}$	restriction $\mathbb{C}\rightarrow\mathbb{R}$? and the field extension $\mathbb{C}\rightarrow\mathbb{Q}$?														
gilmore-lie-groups_F08540	61	0.936	$\left[\begin{array}{cc}-1&a\\0&1\end{array}\right],a\in R$	$\left[\begin{array}{cc}-1&a\\0&1\end{array}\right],a\in R$	7. A subgroup of $UT(1,1)$ includes matrices of the form $\begin{bmatrix}-1&a\\0&1\end{bmatrix},a\in R$. Show that the														
gilmore-lie-groups_F08541	61	1.000	R^1	R^1	underlying group manifold consists of two copies of the real line R^1 . If matrices of														
gilmore-lie-groups_F08542	61	1.000	$\left[\begin{array}{cc}1&a\\0&-1\end{array}\right]$	$\left[\begin{array}{cc}1&a\\0&-1\end{array}\right]$	the form $\begin{bmatrix}1&a\\0&-1\end{bmatrix}$ are also included, then the parameterizing manifold consists of how														
gilmore-lie-groups_F08543	62	0.837	$p^2+q^2+p\,q$	$p^2+q^2+p\,q$	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>$p^2+q^2+p\,q$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+p\,q+p\,r+q\,r$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$p\,q$</td></tr></table>	Group	Dimension	$UT(p,q)$	$p^2+q^2+p\,q$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	$p\,q$
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$A(p,q)$	$p\,q$																		
gilmore-lie-groups_F08544	62	0.837	$p(p+q)$	$p(p+q)$	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>$p^2+q^2+p\,q$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+p\,q+p\,r+q\,r$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$p\,q$</td></tr></table>	Group	Dimension	$UT(p,q)$	$p^2+q^2+p\,q$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	$p\,q$
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gilmore-lie-groups_F08545	62	0.837	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>$p^2+q^2+p\,q$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+p\,q+p\,r+q\,r$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$p\,q$</td></tr></table>	Group	Dimension	$UT(p,q)$	$p^2+q^2+p\,q$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	$p\,q$
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$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	$p\,q$																		
gilmore-lie-groups_F08546	62	0.837	$n(n+1)/2$	$n(n+1)/2$	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>$p^2+q^2+p\,q$</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+p\,q+p\,r+q\,r$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>$p\,q$</td></tr></table>	Group	Dimension	$UT(p,q)$	$p^2+q^2+p\,q$	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	$p\,q$
Group	Dimension																		
$UT(p,q)$	$p^2+q^2+p\,q$																		
$HT(p,q)$	$p(p+q)$																		
$UT(p,q,r)$	$p^2+q^2+r^2+p\,q+p\,r+q\,r$																		
$Sol(n)$	$n(n+1)/2$																		
$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	$p\,q$																		

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gilmore-lie-groups_F00547	62	0.837	$n(n-1) / 2$	$n(n-1)/2$	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>p^2+q^2+pq</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+pq+pr+qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>pq</td></tr></table>	Group	Dimension	$UT(p,q)$	p^2+q^2+pq	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+pq+pr+qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
Group	Dimension																		
$UT(p,q)$	p^2+q^2+pq																		
$HT(p,q)$	$p(p+q)$																		
$UT(p,q,r)$	$p^2+q^2+r^2+pq+pr+qr$																		
$Sol(n)$	$n(n+1)/2$																		
$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	pq																		
gilmore-lie-groups_F00548	62	0.837	$p \ q$	pq	<table><tr><th>Group</th><th>Dimension</th></tr><tr><td>$UT(p,q)$</td><td>p^2+q^2+pq</td></tr><tr><td>$HT(p,q)$</td><td>$p(p+q)$</td></tr><tr><td>$UT(p,q,r)$</td><td>$p^2+q^2+r^2+pq+pr+qr$</td></tr><tr><td>$Sol(n)$</td><td>$n(n+1)/2$</td></tr><tr><td>$Nil(n)$</td><td>$n(n-1)/2$</td></tr><tr><td>$A(p,q)$</td><td>pq</td></tr></table>	Group	Dimension	$UT(p,q)$	p^2+q^2+pq	$HT(p,q)$	$p(p+q)$	$UT(p,q,r)$	$p^2+q^2+r^2+pq+pr+qr$	$Sol(n)$	$n(n+1)/2$	$Nil(n)$	$n(n-1)/2$	$A(p,q)$	pq
Group	Dimension																		
$UT(p,q)$	p^2+q^2+pq																		
$HT(p,q)$	$p(p+q)$																		
$UT(p,q,r)$	$p^2+q^2+r^2+pq+pr+qr$																		
$Sol(n)$	$n(n+1)/2$																		
$Nil(n)$	$n(n-1)/2$																		
$A(p,q)$	pq																		
gilmore-lie-groups_F00549	62	1.000	$\mathbf{F} = d \mathbf{p} / d t$	$\mathbf{F} = d\mathbf{p}/dt$	9. Newton's equations of motion are $\mathbf{F} = d\mathbf{p}/dt$. In the Lorentz gauge Maxwell's equa-														
gilmore-lie-groups_F00550	62	1.000	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_{\mu} = -\frac{4\pi}{c} j_{\mu}$	$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_{\mu} = -\frac{4\pi}{c} j_{\mu}$	<table><tr><th>Transformation</th><th>$\mathbf{F} = d\mathbf{p}/dt$</th><th>$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2})A_{\mu} = -\frac{4\pi}{c} j_{\mu}$</th></tr><tr><td>Galilean</td><td>invariant</td><td>not invariant</td></tr><tr><td>Poincaré</td><td>not invariant</td><td>invariant</td></tr></table>	Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2})A_{\mu} = -\frac{4\pi}{c} j_{\mu}$	Galilean	invariant	not invariant	Poincaré	not invariant	invariant					
Transformation	$\mathbf{F} = d\mathbf{p}/dt$	$(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2})A_{\mu} = -\frac{4\pi}{c} j_{\mu}$																	
Galilean	invariant	not invariant																	
Poincaré	not invariant	invariant																	
gilmore-lie-groups_F00551	62	0.814	$S \ U(2)$	$SU(2)$	10. Show that the group of 2×2 matrices $SU(2)$ is parameterized by two complex num-														
gilmore-lie-groups_F00552	62	1.000	$c_{\{1\}} = a_{\{1\}} + i \ b_{\{1\}}$	$c_1 = a_1 + ib_1$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that														
gilmore-lie-groups_F00553	62	1.000	$c_{\{2\}} = a_{\{2\}} + i \ b_{\{2\}}$	$c_2 = a_2 + ib_2$	bers $c_1 = a_1 + ib_1$ and $c_2 = a_2 + ib_2$, so that														
gilmore-lie-groups_F00554	62	1.000	$a_{\{1\}}^{\wedge 2} + b_{\{1\}}^{\wedge 2} + a_{\{2\}}^{\wedge 2} + b_{\{2\}}^{\wedge 2} = 1$	$a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 + a_2^2 + b_2^2 = 1$. Convince yourself (a) that topolog-														
gilmore-lie-groups_F00555	62	0.991	$S^{\wedge 3} \ \subset \ R^{\wedge 4}$	$S^3 \subset R^4$	$S^3 \subset R^4$; and (b) algebraically it is equivalent to $SU(1; \mathbb{Q})$ (cf. (3.16)).														
gilmore-lie-groups_F00556	62	1.000	$a_{\{1\}}^{\wedge 2} + b_{\{1\}}^{\wedge 2} - a_{\{2\}}^{\wedge 2} - b_{\{2\}}^{\wedge 2} = 1$	$a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$	subject to the condition $a_1^2 + b_1^2 - a_2^2 - b_2^2 = 1$. Identify the parameterizing manifold.														
gilmore-lie-groups_F00557	62	1.000	$\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$	$\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$	in the form $\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row														
gilmore-lie-groups_F00558	62	1.000	$m_{\{11\}}^{\wedge 2} + x^{\wedge 2} = 1$	$m_{11}^2 + x^2 = 1$	in the form $\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row														
gilmore-lie-groups_F00559	62	1.000	$m_{\{11\}} = \pm \sqrt{1-x^{\wedge 2}}$	$m_{11} = \pm\sqrt{1-x^2}$	in the form $\left[\begin{array}{cc} m_{11} & x \\ m_{21} & m_{22} \end{array}\right]$, where $m_{11}^2 + x^2 = 1$, so that $m_{11} = \pm\sqrt{1-x^2}$. The second row														
gilmore-lie-groups_F00560	63	1.000	$m_{\{21\}} \ m_{\{11\}} + m_{\{22\}} \ x = 0$	$m_{21}m_{11} + m_{22}x = 0$	is orthogonal to the first, so that $m_{21}m_{11} + m_{22}x = 0$. As a result, we find														
gilmore-lie-groups_F00561	63	0.994	$m_{\{i \ j\}} \ i \ \geq \ j$	$m_{ij}i \geq j$	$m_{ij}i \geq j$ in terms of the three parameters (x, y, z) .														
gilmore-lie-groups_F00562	63	0.495	$Z_{\{2\}}$	Z_2	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .														
gilmore-lie-groups_F00563	63	0.495	$Z_{\{1\}}$	Z_1	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .														
gilmore-lie-groups_F00564	63	0.495	$(x, \ y)$	(x, y)	Express the 2×2 and 1×1 submatrices Z_2 and Z_1 in terms of the coordinates (x, y) .														

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gilmore-lie-groups_F00565	63	■ 0.854	$M \in GL(n; \mathbb{Z})$	$M \in GL(n; \mathbb{Z})$	15. If $M \in GL(n; \mathbb{Z})$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F00566	63	■ 0.854	$\operatorname{det}(M)$	$\det(M)$	15. If $M \in GL(n; \mathbb{Z})$, show that $\det(M)$ must be ± 1 .
gilmore-lie-groups_F00567	63	■ 1.000	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	$O(n; \mathbb{Z}) \supset S_n \supset A_n$	16. Show that the orders of $O(n; \mathbb{Z}) \supset S_n \supset A_n$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F00568	63	■ 1.000	$2^n \times n!, n!, \frac{1}{2}n!$	$2^n \times n!, n!, \frac{1}{2}n!$	16. Show that the orders of $O(n; \mathbb{Z}) \supset S_n \supset A_n$ are $2^n \times n!$, $n!$, $\frac{1}{2}n!$.
gilmore-lie-groups_F00569	63	■ 1.000	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	$\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$	17. Estimate the Fibonacci number $F(n)$ from the eigenvalues $\lambda_{\pm} = \frac{1}{2}(1 \pm \sqrt{5})$ of the
gilmore-lie-groups_F00570	63	■ 0.999	$F(0)=0, F(1)=1$	$F(0) = 0, F(1) = 1$	(other than $F(0) = 0, F(1) = 1$) are introduced?
gilmore-lie-groups_F00571	63	■ 1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$	$GL(2; \mathbb{Z})$ (for example, $\left[\begin{array}{cc} 2 & -1 \\ -1 & 1 \end{array} \right]$) and other initial conditions.
gilmore-lie-groups_F00572	63	■ 1.000	$ n \ l \ m\rangle$	$ nlm\rangle$	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold
gilmore-lie-groups_F00573	63	■ 1.000	n^2	n^2	19. The energy levels $ nlm\rangle$ of the nonrelativistic hydrogen atom exhibit an n^2 -fold
gilmore-lie-groups_F00574	63	■ 1.000	$E(n \ l \ m) = -E_0 / n^2 \left(E_0 = 13.6 \text{ eV} \right)$	$E(nlm) = -E_0/n^2$ ($E_0 = 13.6\text{eV}$)	quantum number n have the same energy $E(nlm) = -E_0/n^2$ ($E_0 = 13.6\text{ eV}$). If the
gilmore-lie-groups_F00575	63	■ 1.000	$SO(4) \downarrow SO(3)$	$SO(4) \downarrow SO(3)$	symmetry reduction $SO(4) \downarrow SO(3)$. The representations of $SO(4)$ that enter into
gilmore-lie-groups_F00576	63	■ 0.993	$n(n=1,2,3, \ldots)$	$n(n = 1, 2, 3, \dots)$	quantum number n ($n = 1, 2, 3, \dots$). The $SO(4)$ representation with quantum num-
gilmore-lie-groups_F00577	63	■ 0.980	$l, l=0,1,2 \ldots, n-1$	$l, l = 0, 1, 2 \dots, n - 1$	number l ($l = 0, 1, 2, \dots, n - 1$), with $\sum_{l=0}^{l=n-1} (2l+1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00578	63	■ 0.980	$\sum_{l=0}^{l=n-1} (2l+1) = n^2$	$\sum_{l=0}^{l=n-1} (2l+1) = n^2$	number $l, l = 0, 1, 2 \dots, n - 1$, with $\sum_{l=0}^{l=n-1} (2l+1) = n^2$. The $SO(3)$ multiplet
gilmore-lie-groups_F00579	63	■ 1.000	l	l	with quantum number l is $2l+1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00580	63	■ 1.000	$2l+1$	$2l+1$	with quantum number l is $2l+1$ -fold degenerate. An empirical hamiltonian with
gilmore-lie-groups_F00581	63	■ 1.000	$+Ze$	$+Ze$	introduced into a Coulomb potential established by a central charge $+Ze$ can be
gilmore-lie-groups_F00582	64	■ 1.000	$\delta=0.28$	$\delta = 0.28$	trum with $\delta = 0.28$ provides the filling ordering that accounts for Mendelev's
gilmore-lie-groups_F00583	64	■ 0.608	$(n, l) \rightarrow 1s$	$(n, l) \rightarrow 1s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00584	64	■ 0.608	$2s$	$2s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00585	64	■ 0.608	$2p$	$2p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00586	64	■ 0.608	$3s$	$3s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00587	64	■ 0.608	$3p$	$3p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00588	64	■ 0.608	$4s$	$4s$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00589	64	■ 0.608	$3d$	$3d$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00590	64	■ 0.608	$4p$	$4p$	periodic table of the chemical elements: $(n, l) \rightarrow 1s; 2s, 2p; 3s, 3p; 4s, 3d, 4p;$
gilmore-lie-groups_F00591	64	■ 0.957	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$	$5s, 4d, 5p; 6s, 4f, 5d, 6p; 7s, 5f, 6d, 7p; 8s, 6f, 7d, 8p; \dots$.

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gilmore-lie-groups_F00592	64	■ 1.000	$G_{\{2\}}$	G_2	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00593	64	■ 1.000	$2n$	$2n$	21. G_1 and G_2 are two metrics on a real $2n$ -dimensional linear vector space that are
gilmore-lie-groups_F00594	64	■ 1.000	$M^{\{t\}} G_{\{i\}} M = G_{\{i\}}$	$M^t G_i M = G_i$	Show that the $2n \times 2n$ matrices M that satisfy the bilinear constraints $M^t G_i M = G_i$
gilmore-lie-groups_F00595	64	■ 1.000	$y^{\{j\}} = x^{\{i\}} M_{\{i\}}^{\{j\}}$	$y^j = x^i M_i^j$	by a linear transformation: $y^j = x^i M_i^j$. Show that the derivatives are related by the
gilmore-lie-groups_F00596	64	■ 0.990	$f(q, p)$	$f(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00597	64	■ 0.990	$g(q, p)$	$g(q, p)$	23. The Poisson brackets between two functions $f(q, p)$ and $g(q, p)$ on a classical phase
gilmore-lie-groups_F00598	65	■ 0.666	(Q, P)	(Q, P)	b. Introduce a new coordinate system (Q, P) , related to the original by a linear
gilmore-lie-groups_F00599	65	■ 1.000	$A^{\{t\}} C$	$A^t C$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00600	65	■ 1.000	$B^{\{t\}} D$	$B^t D$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00601	65	■ 1.000	$A^{\{t\}} D - B^{\{t\}} C = I_{\{n\}}$	$A^t D - B^t C = I_n$	Poisson brackets. Show $A^t C$ and $B^t D$ must be symmetric and $A^t D - B^t C = I_n$.
gilmore-lie-groups_F00602	65	■ 0.999	$\left[q_{\{j\}}, q_{\{k\}} \right] = \left[p_{\{j\}}, p_{\{k\}} \right] = 0$	$[q_j, q_k] = [p_j, p_k] = 0$	mechanical commutator bracket: $[q_i, q_k] = [p_i, p_k] = 0$ and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00603	65	■ 0.999	$\left[q_{\{j\}}, p_{\{k\}} \right] = i\hbar\delta_{\{j\} \{k\}}$	$[q_j, p_k] = i\hbar\delta_{jk}$	mechanical commutator bracket: $[q_j, q_k] = [p_j, p_k] = 0$ and $[q_j, p_k] = i\hbar\delta_{jk}$.
gilmore-lie-groups_F00604	65	■ 1.000	e^{+ikx}	e^{+ikx}	with a similar expression for the wavefunction on the right. The exponential e^{+ikx}
gilmore-lie-groups_F00605	65	■ 0.644	m	m	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00606	65	■ 0.644	$(+)$	$(+)$	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00607	65	■ 0.644	$\hbar k$	$\hbar k$	describes a particle of mass m moving to the right (+) with momentum $\hbar k$ and
gilmore-lie-groups_F00608	65	■ 1.000	$E = (\hbar k)^2 / 2m$	$E = (\hbar k)^2 / 2m$	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00609	65	■ 1.000	$A_{\{L\}}$	A_L	energy $E = (\hbar k)^2 / 2m$. The complex number A_L is the probability amplitude for
gilmore-lie-groups_F00610	65	■ 1.000	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	$\langle \hat{p} \rangle = \left(A_L ^2 - B_L ^2 \right) \hbar k$	momentum in the left-hand region is $\langle \hat{p} \rangle = (A_L ^2 - B_L ^2) \hbar k$, where the operator
gilmore-lie-groups_F00611	65	■ 1.000	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$	$\hat{p} = \frac{\hbar}{i} \frac{d}{dx}$.
gilmore-lie-groups_F00612	65	■ 1.000	$V_{\{L\}} = V_{\{R\}}$	$V_L = V_R$	constant, with $V_L = V_R$. The wavefunctions to the left and right of this
gilmore-lie-groups_F00613	65	■ 1.000	$\psi_{\{\sigma\}}(x) = A_{\{\sigma\}} e^{+ikx} + B_{\{\sigma\}} e^{-ikx}$	$\psi_{\sigma}(x) = A_{\sigma} e^{+ikx} + B_{\sigma} e^{-ikx}$	region are represented in the form $\psi_{\sigma}(x) = A_{\sigma} e^{+ikx} + B_{\sigma} e^{-ikx}$, where
gilmore-lie-groups_F00614	65	■ 0.969	$\sigma = L, R$	$\sigma = L, R$	$\sigma = L, R$.
gilmore-lie-groups_F00615	66	■ 1.000	$A_{\{L\}}, A_{\{R\}}$	A_L, A_R	b. Since the Schrödinger equation is second order the four amplitudes $A_L, A_R,$
gilmore-lie-groups_F00616	66	■ 1.000	$B_{\{L\}}, B_{\{R\}}$	B_L, B_R	B_L, B_R are not independent. Only two are independent. Two linear relations exist
gilmore-lie-groups_F00617	66	■ 1.000	E	E	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_F00618	66	■ 1.000	$T(E) \in U(1, 1)$	$T(E) \in U(1, 1)$	of energy E . Show that $T(E) \in U(1, 1)$.
gilmore-lie-groups_F00619	66	■ 1.000	$T \in SU(1, 1)$	$T \in SU(1, 1)$	c. Show that $T \in SU(1, 1)$ by appropriate choice of phase.

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gilmore-lie-groups_F00620	66	■ 1.000	$S \in U(2)$	$S \in U(2)$	a. Invoke conservation of momentum arguments to conclude $S \in U(2)$.
gilmore-lie-groups_F00621	66	■ 1.000	$S(E)$	$S(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00622	66	■ 1.000	$T(E)$	$T(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00623	66	■ 1.000	$t_{11}(E)$	$t_{11}(E)$	c. Show that the poles of $S(E)$ are the zeroes of $T(E)$, specifically of $t_{11}(E)$. Poles
gilmore-lie-groups_F00624	66	■ 1.000	$r_j / \left[\left(E - E_j \right) + i \left(\Gamma_j / 2 \right) \right]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_F00625	66	■ 1.000	E_j	E_j	$r_j / [(E - E_j) + i(\Gamma_j/2)]$ describe resonances at energy E_j with characteristic de-
gilmore-lie-groups_F00626	66	■ 1.000	$\Gamma_j / \hbar$	$\Gamma_j / \hbar$	cay time $\Gamma_j / \hbar$.
gilmore-lie-groups_F00627	66	■ 1.000	V_1	V_1	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_F00628	66	■ 1.000	V_2	V_2	26. Two interaction regions V_1 and V_2 on the line are characterized by transfer matrices
gilmore-lie-groups_F00629	66	■ 1.000	T_1	T_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00630	66	■ 1.000	T_2	T_2	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00631	66	■ 1.000	S_1	S_1	T_1 and T_2 , and also by S -matrices S_1 and S_2 (see Fig. 3.3). The outputs of one region
gilmore-lie-groups_F00632	67	■ 1.000	S_{Tot}	S_{Tot}	c. Show that S_{Tot} is unitary.
gilmore-lie-groups_F00633	67	■ 1.000	$1 / (1 - s_{12}s_{43})$	$1 / (1 - s_{12}s_{43})$	by expanding the fraction $1 / (1 - s_{12}s_{43})$ as a geometric sum and interpreting each
gilmore-lie-groups_F00634	67	■ 1.000	$V_1^{\prime}$	$V_1^{\prime}$	27. If the potentials V_1 and V_2 are modified to $V_1^{\prime}$ and $V_2^{\prime}$ their transfer matrices and
gilmore-lie-groups_F00635	67	■ 1.000	$V_2^{\prime}$	$V_2^{\prime}$	27. If the potentials V_1 and V_2 are modified to $V_1^{\prime}$ and $V_2^{\prime}$ their transfer matrices and
gilmore-lie-groups_F00636	67	■ 1.000	$T_i(E) \rightarrow T_i^{\prime}(E)$	$T_i(E) \rightarrow T_i^{\prime}(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i^{\prime}(E)$ and $S_i(E) \rightarrow S_i^{\prime}(E)$,
gilmore-lie-groups_F00637	67	■ 1.000	$S_i(E) \rightarrow S_i^{\prime}(E)$	$S_i(E) \rightarrow S_i^{\prime}(E)$	their scattering matrices will also be modified $T_i(E) \rightarrow T_i^{\prime}(E)$ and $S_i(E) \rightarrow S_i^{\prime}(E)$,
gilmore-lie-groups_F00638	67	■ 1.000	$i=1,2$	$i = 1, 2$	$i = 1, 2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00639	67	■ 1.000	$E, S_{\text{Tot}}^{\prime}(E) = S_{\text{Tot}}(E)$	$E, S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$	$i = 1, 2$. It is possible that for some energy E , $S'_{\text{Tot}}(E) = S_{\text{Tot}}(E)$. Find the set of
gilmore-lie-groups_F00640	67	■ 1.000	$S_1^{\prime}(E)$	$S_1^{\prime}(E)$	all modified scattering matrices $S_1^{\prime}(E)$ and $S_2^{\prime}(E)$ with the property that the modi-
gilmore-lie-groups_F00641	67	■ 1.000	$S_2^{\prime}(E)$	$S_2^{\prime}(E)$	all modified scattering matrices $S_1^{\prime}(E)$ and $S_2^{\prime}(E)$ with the property that the modi-
gilmore-lie-groups_F00642	67	■ 0.998	$S_{\text{Tot}}(E)$	$S_{\text{Tot}}(E)$	fied pair maps into the original total S -matrix $S_{\text{Tot}}(E)$. In fancy terms, find the fiber
gilmore-lie-groups_F00643	67	■ 0.999	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	$U(4) \supset U(2) \otimes U(2) \downarrow U(2)$	in $U(4) \supset U(2) \otimes U(2) \downarrow U(2)$. (Hint: if this seems daunting, note that to satisfy
gilmore-lie-groups_F00644	67	■ 0.997	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1^{\prime}(E)T_2^{\prime}(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1^{\prime}(E)T_2^{\prime}(E)$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1^{\prime}(E)T_2^{\prime}(E)$ we can take $T_1^{\prime}(E) = T_1(E)R$ and $T_2^{\prime}(E) =$
gilmore-lie-groups_F00645	67	■ 0.997	$T_1^{\prime}(E) = T_1(E)R$	$T_1^{\prime}(E) = T_1(E)R$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1^{\prime}(E)T_2^{\prime}(E)$ we can take $T_1^{\prime}(E) = T_1(E)R$ and $T_2^{\prime}(E) =$
gilmore-lie-groups_F00646	67	■ 0.997	$T_2^{\prime}(E) =$	$T_2^{\prime}(E) =$	$T_1(E)T_2(E) = T_{\text{Tot}}(E) = T_1^{\prime}(E)T_2^{\prime}(E)$ we can take $T_1^{\prime}(E) = T_1(E)R$ and $T_2^{\prime}(E) =$

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gilmore-lie-groups_ F00647	67	■ 1.000	$R^{-1} T_2(E)$	$R^{-1} T_2(E)$	$R^{-1} T_2(E)$ for any $R \in U(1, 1)$. The fiber in $U(2, 2) \supset U(1, 1) \otimes U(1, 1) \downarrow U(1, 1)$
gilmore-lie-groups_ F00648	67	■ 1.000	$R \in U(1, 1)$	$R \in U(1, 1)$	$R^{-1} T_2(E)$ for any $R \in U(1, 1)$. The fiber in $U(2, 2) \supset U(1, 1) \otimes U(1, 1) \downarrow U(1, 1)$
gilmore-lie-groups_ F00649	67	■ 1.000	$U(2, 2) \supset U(1, 1) \otimes U(1, 1) \downarrow U(1, 1)$	$U(2, 2) \supset U(1, 1) \otimes U(1, 1) \downarrow U(1, 1)$	$R^{-1} T_2(E)$ for any $R \in U(1, 1)$. The fiber in $U(2, 2) \supset U(1, 1) \otimes U(1, 1) \downarrow U(1, 1)$
gilmore-lie-groups_ F00650	67	■ 0.932	$T_{\text{Tot}}(E)$	$T_{\text{Tot}}(E)$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_ F00651	67	■ 0.932	$\left(T_1(E)R, R^{-1}T_2(E)\right)$	$(T_1(E)R, R^{-1}T_2(E))$	over $T_{\text{Tot}}(E)$ consists of the matrices $(T_1(E)R, R^{-1}T_2(E))$. Now map this into the
gilmore-lie-groups_ F00652	67	■ 0.850	$\left(S_1'(E), S_2'(E)\right)$	$(S_1'(E), S_2'(E))$	fiber $(S_1'(E), S_2'(E))$ over $S_{\text{Tot}}(E)$.
gilmore-lie-groups_ F00653	68	■ 0.987	$n_1, n_2, \dots, n_k$	$n_1, n_2, \dots, n_k$	with $n_1, n_2, \dots, n_k$ leads can be connected together by soldering some of the leads
gilmore-lie-groups_ F00654	68	■ 0.718	$k S$	kS	direct sum of k S matrices of sizes $n_j \times n_j$ ($j = 1, 2, \dots, k$). Through appropriate
gilmore-lie-groups_ F00655	68	■ 0.718	$n_j \times n_j (j=1, 2, \dots, k)$	$n_j \times n_j (j = 1, 2, \dots, k)$	direct sum of k S matrices of sizes $n_j \times n_j (j = 1, 2, \dots, k)$. Through appropriate
gilmore-lie-groups_ F00656	68	■ 1.000	Γ	Γ	The matrix Γ that relates internal outputs to internal inputs describes the topology, or
gilmore-lie-groups_ F00657	68	■ 0.999	$[o_e] = S_{\text{Network}}[i_e]$	$[o_e] = S_{\text{Network}}[i_e]$	a. Show that the S matrix that describes the network, defined by $[o_e] = S_{\text{Network}}[i_e]$,
gilmore-lie-groups_ F00658	68	■ 1.000	S_{Newtork}	S_{Newtork}	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_ F00659	68	■ 1.000	$S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$	$S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$	b. Show that S_{Newtork} is unitary: $S_{\text{Newtork}}^\dagger = S_{\text{Newtork}}, S_{\text{Newtork}} \subset U(d)$.
gilmore-lie-groups_ F00660	68	■ 1.000	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k \otimes U(n_j)$	$U\left(\sum_{j=1}^k n_j\right) \supset \Pi_{j=1}^k \otimes U(n_j)$	$U(\sum_{j=1}^k n_j) \supset \Pi_{j=1}^k \otimes U(n_j)$ and a projection to the total network scattering matrix
gilmore-lie-groups_ F00661	68	■ 1.000	$U(d)$	$U(d)$	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_ F00662	68	■ 1.000	d	d	in $U(d)$, where d is the number of the network's external leads. The connectivity is
gilmore-lie-groups_ F00663	68	■ 0.917	$\Pi_{j=1}^k \otimes U(n_j)$	$\Pi_{j=1}^k \otimes U(n_j)$	determined by the permutation matrix Γ . Determine the fiber in $\Pi_{j=1}^k \otimes U(n_j)$ over
gilmore-lie-groups_ F00664	68	■ 1.000	U	U	30. All the matrices in this problem are square $n \times n$, with: H hermitian; U unitary; A
gilmore-lie-groups_ F00665	70	■ 1.000	b	b	into the neighborhood of group element b by multiplying a , and every element in
gilmore-lie-groups_ F00666	70	■ 1.000	$b a^{-1}$	ba^{-1}	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_ F00667	70	■ 1.000	$a^{-1} b$	$a^{-1}b$	its neighborhood, on the left by group element ba^{-1} (or on the right by $a^{-1}b$). This
gilmore-lie-groups_ F00668	71	■ 0.953	(a, b, c)	(a, b, c)	done by allowing the parameters (a, b, c) to become infinitesimals and expanding
gilmore-lie-groups_ F00669	71	■ 1.000	$\phi(x, y) \rightarrow \phi(x, \theta + \delta y)$	$\phi(x, y) \rightarrow \phi(x, \theta + \delta y)$	arguments, say $\phi(x, y) \rightarrow \phi(x, \theta + \delta y)$. This generates a left-invariant vector field.
gilmore-lie-groups_ F00670	72	■ 1.000	ϵ	ϵ	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_ F00671	72	■ 1.000	$I + \epsilon X$	$I + \epsilon X$	given in (4.4). Then if ϵ is a small real number, $I + \epsilon X$ represents an element in
gilmore-lie-groups_ F00672	72	■ 0.744	X_a	X_a	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_ F00673	72	■ 0.744	X_b	X_b	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$

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gilmore-lie-groups_F00674	72	0.744	$X_{\{c\}}$	X_c	space of traceless 2×2 matrices spanned by the generators X_a, X_b, X_c of $SL(2; \mathbb{R})$
gilmore-lie-groups_F00675	72	1.000	$X^{\{n\}}$	X^n	With brute force, each of the matrices X^n is computed explicitly, a pattern is
gilmore-lie-groups_F00676	72	1.000	$X^{\{0\}}=I_{\{2\}}, X^{\{1\}}=X$	$X^0 = I_2, X^1 = X$	recognized, and the sum is carried out. The first few powers are $X^0 = I_2, X^1 = X$
gilmore-lie-groups_F00677	72	1.000	$X^{\{2\}}=\theta^2 I_{\{2\}}$	$X^2 = \theta^2 I_2$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00678	72	1.000	$X^{\{2\}}$	X^2	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00679	72	1.000	$X^{\{3\}}=X^{\{2\}} X^{\{1\}}$	$X^3 = X^2 X^1$	(given in (4.6)), and $X^2 = \theta^2 I_2$. Since X^2 is a multiple of the identity, $X^3 = X^2 X^1$
gilmore-lie-groups_F00680	72	0.991	$X \left(= \theta^2 X \right), X^{\{4\}}$	$X \left(= \theta^2 X \right), X^4$	must be proportional to $X (= \theta^2 X)$, X^4 is proportional to the identity, and so on.
gilmore-lie-groups_F00681	72	0.995	$X^{\{0\}}=I_{\{2\}}$	$X^0 = I_2$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00682	72	0.995	$X^{\{1\}}=X$	$X^1 = X$	equation of degree 2. Thus we can replace X^2 by a function of $X^0 = I_2$ and $X^1 = X$.
gilmore-lie-groups_F00683	72	1.000	$X^{\{3\}}$	X^3	Similarly, X^3 can be replaced by a linear combination of X^2 and X , and then X^2
gilmore-lie-groups_F00684	72	0.759	$I_{\{2\}}$	I_2	replaced by I_2 and X . By induction, any function of the 2×2 matrix X can be
gilmore-lie-groups_F00685	73	1.000	$f_{\{0\}}, f_{\{1\}}$	f_0, f_1	Furthermore, the functions f_0, f_1 are not arbitrary functions of the three parameters
gilmore-lie-groups_F00686	73	1.000	$\theta^2 = a^2 + bc$	$\theta^2 = a^2 + bc$	X is $\theta^2 = a^2 + bc$. As a result, we know from general and simple considerations
gilmore-lie-groups_F00687	73	1.000	$f_{\{0\}} \left(\theta^2 \right) = 1 + \theta^2 / 2! + \theta^4 / 4! + \theta^6 / 6! + \dots = \cosh \theta$ $/ 4! + \theta^6 / 6! + \dots = \cosh \theta$	$f_0 \left(\theta^2 \right) = 1 + \theta^2 / 2! + \theta^4 / 4! + \theta^6 / 6! + \dots = \cosh \theta$	The two functions are $f_0(\theta^2) = 1 + \theta^2 / 2! + \theta^4 / 4! + \theta^6 / 6! + \dots = \cosh \theta$ and
gilmore-lie-groups_F00688	73	1.000	$f_{\{1\}} \left(\theta^2 \right) = 1 + \theta^2 / 3! + \theta^4 / 5! + \theta^6 / 7! + \dots = \sinh(\theta) / \theta$ $/ 5! + \theta^6 / 7! + \dots = \sinh(\theta) / \theta$	$f_1 \left(\theta^2 \right) = 1 + \theta^2 / 3! + \theta^4 / 5! + \theta^6 / 7! + \dots = \sinh(\theta) / \theta$	$f_1(\theta^2) = 1 + \theta^2 / 3! + \theta^4 / 5! + \theta^6 / 7! + \dots = \sinh(\theta) / \theta$. These arguments are ap-
gilmore-lie-groups_F00689	73	1.000	Y	Y	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_F00690	73	1.000	$g_{\{1\}} = I + \epsilon X$	$g_1 = I + \epsilon X$	If X and Y are elements in the Lie algebra, then $g_1 = I + \epsilon X$ is an element in
gilmore-lie-groups_F00691	73	1.000	$I + \epsilon \alpha X$	$I + \epsilon \alpha X$	the Lie group near the identity for ϵ sufficiently small. In fact, so also is $I + \epsilon \alpha X$
gilmore-lie-groups_F00692	74	1.000	α	α	for any real number α . We can form the product
gilmore-lie-groups_F00693	74	1.000	ϵ, δ	ϵ, δ	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_F00694	74	1.000	$g_{\{1\}}(\epsilon) =$	$g_1(\epsilon) =$	If X and Y are in the Lie algebra, then for any ϵ, δ sufficiently small, $g_1(\epsilon) =$
gilmore-lie-groups_F00695	74	1.000	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_F00696	74	1.000	$g_{\{1\}}(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$	$\operatorname{EXP}(\epsilon X)$ and $g_1(\epsilon)^{-1} = \operatorname{EXP}(-\epsilon X)$ are group elements near the identity, as are
gilmore-lie-groups_F00697	74	1.000	$g_{\{2\}}(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$	$g_2(\delta)^{\pm 1} = \operatorname{EXP}(\pm \delta Y)$. Expanding the commutator to lowest order nonvanishing
gilmore-lie-groups_F00698	74	0.996	$g_{\{1\}}(\epsilon) = \operatorname{EXP}(\epsilon X)$	$g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$
gilmore-lie-groups_F00699	74	0.996	$g_{\{2\}}(\delta) =$	$g_2(\delta) =$	Therefore, the commutator of two group elements, $g_1(\epsilon) = \operatorname{EXP}(\epsilon X)$ and $g_2(\delta) =$

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gilmore-lie-groups_ F00700	74	■ 1.000	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$	$\operatorname{EXP}(\delta Y)$, which is in the group G , requires the commutator of the operators X and
gilmore-lie-groups_ F00701	74	■ 1.000	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F00702	74	■ 1.000	$\mathfrak{g}$	$\mathfrak{g}$	$Y, [X, Y] = (XY - YX)$, to be in its Lie algebra $\mathfrak{g}$
gilmore-lie-groups_ F00703	74	■ 1.000	$g_1 H g_1^{-1} \subset H$	$g_1 H g_1^{-1} \subset H$	If H is an invariant subgroup of G , then $g_1 H g_1^{-1} \subset H$. This means that if X is in
gilmore-lie-groups_ F00704	74	■ 1.000	X, Y, Z	X, Y, Z	If X, Y, Z are in the Lie algebra, then the Jacobi identity is satisfied
gilmore-lie-groups_ F00705	74	■ 0.996	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$	$([X, Y] = XY - YX)$ and showing that the 12 terms so obtained cancel pairwise.
gilmore-lie-groups_ F00706	74	■ 1.000	XY	XY	for which composition (e.g., XY is well defined, as are all other pairwise products)
gilmore-lie-groups_ F00707	75	■ 1.000	$X \in \mathfrak{g}$	$X \in \mathfrak{g}$	(i) It is a linear vector space under vector addition and scalar multiplication. If $X \in \mathfrak{g}$ and
gilmore-lie-groups_ F00708	75	■ 1.000	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$	$Y \in \mathfrak{g}$ then every linear combination of X and Y is in $\mathfrak{g}$:
gilmore-lie-groups_ F00709	75	■ 1.000	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	$X \in \mathfrak{g}, Y \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}, Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then
gilmore-lie-groups_ F00710	75	■ 1.000	$Z \in \mathfrak{g}$	$Z \in \mathfrak{g}$	(iii) The Jacobi identity is satisfied. If $X \in \mathfrak{g}, Y \in \mathfrak{g}$, and $Z \in \mathfrak{g}$, then
gilmore-lie-groups_ F00711	75	■ 0.996	$X_1, X_2, \dots, X_n$	$X_1, X_2, \dots, X_n$	$X_1, X_2, \dots, X_n$, then we can ask several additional questions such as: Is there a
gilmore-lie-groups_ F00712	75	■ 1.000	(X_i, X_j)	(X_i, X_j)	(X_i, X_j) ? We return to these questions shortly.
gilmore-lie-groups_ F00713	76	■ 1.000	C_{ij}^k	C_{ij}^k	The coefficients C_{ij}^k in this expansion are called structure constants . The structure
gilmore-lie-groups_ F00714	76	■ 0.865	X_a, X_b, X_c	X_a, X_b, X_c	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups_ F00715	76	■ 0.865	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	X_a, X_b, X_c (4.4) in the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ for the Lie group $SL(2; \mathbb{R})$ are, from
gilmore-lie-groups_ F00716	76	■ 1.000	Z	Z	change of basis affects the commutator of an arbitrary element Z in the algebra
gilmore-lie-groups_ F00717	76	■ 1.000	$R(Z)$	$R(Z)$	The association of a matrix $R(Z)$ with each element of a Lie algebra is called
gilmore-lie-groups_ F00718	77	■ 1.000	$\mathfrak{sl}(2; \mathbb{R})$	$\mathfrak{sl}(2; \mathbb{R})$	Example The regular representation of the Lie algebra $\mathfrak{sl}(2; \mathbb{R})$ is easily con-
gilmore-lie-groups_ F00719	77	■ 0.844	a, b	a, b	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups_ F00720	77	■ 0.844	c	c	The rows and columns of this 3×3 matrix are labeled by the indices a, b and c ,
gilmore-lie-groups_ F00721	78	■ 0.899	X_a, X_b	X_a, X_b	the three generators X_a, X_b , and X_c of $\mathfrak{sl}(2; \mathbb{R})$ to either the nonsemisimple or the
gilmore-lie-groups_ F00722	78	■ 0.997	$E(2)$	$E(2)$	$E(2)$ with structure
gilmore-lie-groups_ F00723	78	■ 1.000	P_x, P_y	P_x, P_y	where the rows and columns are labeled successively by the basis vectors P_x, P_y , and
gilmore-lie-groups_ F00724	78	■ 1.000	L_z	L_z	L_z . This regular representation has the block diagonal structure of a nonsemisimple
gilmore-lie-groups_ F00725	78	■ 0.985	A, B	A, B	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt
gilmore-lie-groups_ F00726	78	■ 0.985	$p \times q$	$p \times q$	very useful inner product when A, B are $p \times q$ matrices is the Hilbert–Schmidt

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gilmore-lie-groups_ F00727	79	■ 0.998	$X_{\{i\}}$	X_i	generators that commute with all operators in the Lie algebra (X_i , where $C_{i*}^* = 0$)
gilmore-lie-groups_ F00728	79	■ 0.998	$C_{\{i \star\}}^{\{ \}^*} = 0$	$C_{i*}^* = 0$	generators that commute with all operators in the Lie algebra (X_i , where $C_{i*}^* = 0$)
gilmore-lie-groups_ F00729	79	■ 1.000	$C_{\{* \star\}}^{\{i\}} = 0$	$C_{**}^i = 0$	generators (X_i , where $C_{**}^i = 0$).
gilmore-lie-groups_ F00730	79	■ 1.000	$V_{\{0\}}$	V_0	The subspace V_0 is a subalgebra of $\mathfrak{g}$. It is the largest nilpotent invariant subal-
gilmore-lie-groups_ F00731	79	■ 1.000	$V_{\{-\}}$	V_-	The subspace V_- is also a subalgebra of $\mathfrak{g}$. It consists of compact (a topological
gilmore-lie-groups_ F00732	79	■ 1.000	$V_{\{+\}}$	V_+	Finally, the subspace V_+ is not a subalgebra of $\mathfrak{g}$. It consists of noncompact
gilmore-lie-groups_ F00733	80	■ 1.000	$X_{\{ \pm\}} = X_{\{b\}} \pm X_{\{c\}}$	$X_{\pm} = X_b \pm X_c$	and $X_{+} = X_b \pm X_c$. In this basis the metric matrix is
gilmore-lie-groups_ F00734	80	■ 1.000	$X_{\{-\}}$	X_-	In this representation it is clear that the operator X_- spans a one-dimensional
gilmore-lie-groups_ F00735	80	■ 1.000	$X_{\{a\}}, X_{\{+\}}$	X_a, X_+	compact subalgebra in $\mathfrak{sl}(2; \mathbb{R})$ and the generators X_a, X_+ are noncompact.
gilmore-lie-groups_ F00736	81	■ 0.644	$\alpha^{\{1\}}, \alpha^{\{2\}}, \ldots, \alpha^{\{n\}}$	$\alpha^1, \alpha^2, \dots, \alpha^n$	ric and measure. We assume the coordinates of the identity are $(\alpha^1, \alpha^2, \dots, \alpha^n)$
gilmore-lie-groups_ F00737	81	■ 0.840	$\alpha^{\{1\}} + d \alpha^{\{1\}}, \alpha^{\{2\}} + d \alpha^{\{2\}}, \ldots, \alpha^{\{n\}} + d \alpha^{\{n\}}$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n$	and the coordinates of a point near the identity are $(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$
gilmore-lie-groups_ F00738	81	■ 0.866	$d \alpha^{\{n\}}$	$d\alpha^n$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00739	81	■ 0.866	$\left(x^{\{1\}}, x^{\{2\}}, \ldots, x^{\{n\}}\right)$	$(x^1, x^2, \dots, x^n)$	$d\alpha^n$). If $(x^1, x^2, \dots, x^n)$ represents some other group operation, then the point
gilmore-lie-groups_ F00740	81	■ 0.831	$\alpha^{\{1\}} + d \alpha^{\{1\}}, \alpha^{\{2\}} + d \alpha^{\{2\}}, \ldots, \alpha^{\{n\}} + d \alpha^{\{n\}}$	$\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00741	81	■ 0.831	$x^{\{1\}} + d x^{\{1\}}, x^{\{2\}} + d x^{\{2\}}$	$x^1 + dx^1, x^2 + d x^2$	$(\alpha^1 + d\alpha^1, \alpha^2 + d\alpha^2, \dots, \alpha^n + d\alpha^n)$ is mapped to the point $(x^1 + dx^1, x^2 +$
gilmore-lie-groups_ F00742	81	■ 1.000	$d x^{\{2\}}, \ldots, x^{\{n\}} + d x^{\{n\}}$	$dx^2, \dots, x^n + dx^n$	$(dx^2, \dots, x^n + dx^n)$ under left (right) multiplication by the group operation associ-
gilmore-lie-groups_ F00743	81	■ 0.997	$d x$	dx	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00744	81	■ 0.997	$d \alpha$	$d\alpha$	ated with $(x^1, x^2, \dots, x^n)$. The displacements dx and $d\alpha$ are related by a position-
gilmore-lie-groups_ F00745	81	■ 1.000	$d s$	ds	Suppose now that the distance ds between the identity and a point with coordi-
gilmore-lie-groups_ F00746	81	■ 1.000	$\alpha^{\{i\}} + d \alpha^{\{i\}}$	$\alpha^i + d\alpha^i$	nates $\alpha^i + d\alpha^i$ infinitesimally close to the identity is given by
gilmore-lie-groups_ F00747	81	■ 0.671	$x, g_{rs}(x)$	$x, g_{rs}(x)$	Killing inner product. Can we define a metric at $x, g_{rs}(x)$, with the property that
gilmore-lie-groups_ F00748	81	■ 1.000	$x, g(x)$	$x, g(x)$	In order to enforce the invariance condition, the metric at $x, g(x)$, must be related
gilmore-lie-groups_ F00749	82	■ 0.976	$\left(d \alpha^{\{1\}}, d \alpha^{\{2\}}, d \alpha^{\{3\}}\right)$	$(d\alpha^1, d\alpha^2, d\alpha^3)$	The linear relation between the infinitesimals $(d\alpha^1, d\alpha^2, d\alpha^3)$ in the neighborhood
gilmore-lie-groups_ F00750	82	■ 0.999	$(d x, d y, d z)$	(dx, dy, dz)	of the identity and the infinitesimals (dx, dy, dz) in the neighborhood of the group
gilmore-lie-groups_ F00751	83	■ 1.000	$[X, Y] = (X Y - Y X)$	$[X, Y] = (XY - YX)$	tional combinatorial operation, the commutator $[X, Y] = (XY - YX)$, and obey the
gilmore-lie-groups_ F00752	83	■ 1.000	$g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I - \epsilon X$	$g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I - \epsilon X$	1. Carry out the commutator calculation for $g_1 = (I + \epsilon X), g_1^{-1} = (I + \epsilon X)^{-1} = I - \epsilon X$

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gilmore-lie-groups_F00753	83	■ 1.000	$\epsilon X + \epsilon^2 X^2 - \dots$	$\epsilon X + \epsilon^2 X^2 - \dots$	$\epsilon X + \epsilon^2 X^2 - \dots$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_F00754	83	■ 1.000	$g_{\{2\}}$	g_2	$\epsilon X + \epsilon^2 X^2 - \dots$, with similar expressions for g_2 , to obtain the same result as in
gilmore-lie-groups_F00755	83	■ 1.000	$X^{\{\prime\}}$	X'	2. The inner product of two vectors X and X' in a linear vector space can be computed
gilmore-lie-groups_F00756	83	■ 1.000	$(X, X)=2\left(a^2+b c\right)$	$(X, X)=2\left(a^2+b c\right)$	c. Use the result from Eq. (4.43) that $(X, X)=2\left(a^2+b c\right)$ to show
gilmore-lie-groups_F00757	83	■ 1.000	$Y=e^{\{X\}}$	$Y=e^X$	a Lie algebra: $Y=e^X$. Show that “for Y sufficiently close to the identity” the matrix
gilmore-lie-groups_F00758	83	■ 0.570	$\lambda_{\{i\}}$	λ_i	ues λ_i of Y all satisfy $0<\lambda_i<+2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_F00759	83	■ 0.570	$0<\lambda_{\{i\}}<+2$	$0<\lambda_i<+2$	ues λ_i of Y all satisfy $0<\lambda_i<+2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_F00760	83	■ 0.570	$Y \in \operatorname{SL}(2 ; \mathbb{R})$	$Y \in SL(2; \mathbb{R})$	ues λ_i of Y all satisfy $0<\lambda_i<+2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_F00761	83	■ 0.570	$\operatorname{tr} Y<-2$	$\text{tr } Y < -2$	ues λ_i of Y all satisfy $0<\lambda_i<+2$. Show that if $Y \in SL(2; \mathbb{R})$ and $\text{tr } Y < -2$ this
gilmore-lie-groups_F00762	84	■ 1.000	$X \in \operatorname{sl}(2 ; \mathbb{R})$	$X \in \mathfrak{sl}(2; \mathbb{R})$	expansion does not converge. That is, there is no 2×2 matrix $X \in \mathfrak{sl}(2; \mathbb{R})$ with the
gilmore-lie-groups_F00763	84	■ 0.637	$\operatorname{tr} e^{\{X\}}<-2$	$\text{tr } e^X < -2$	property $\text{tr } e^X < -2$.
gilmore-lie-groups_F00764	84	■ 0.952	$\mathbf{L} =$	$\mathbf{L} =$	4. The Lie algebra of $SO(3)$ is spanned by three 3×3 antisymmetric matrices $\mathbf{L} =$
gilmore-lie-groups_F00765	84	■ 1.000	$\left(L_{\{1\}}, L_{\{2\}}, L_{\{3\}}\right)=\left(X_{\{23\}}, X_{\{31\}}, X_{\{12\}}\right)$	$(L_1, L_2, L_3)=\left(X_{23}, X_{31}, X_{12}\right)$	$(L_1, L_2, L_3)=\left(X_{23}, X_{31}, X_{12}\right)$, with
gilmore-lie-groups_F00766	84	■ 1.000	$\mathbf{X}=\theta \cdot \mathbf{L}$	$\mathbf{X}=\theta \cdot \mathbf{L}$	where θ is the single invariant that can be constructed from the matrix $\mathbf{X}=\theta \cdot \mathbf{L}$:
gilmore-lie-groups_F00767	84	■ 1.000	$\theta^2=\theta_1^2+\theta_2^2+\theta_3^2$	$\theta^2=\theta_1^2+\theta_2^2+\theta_3^2$	$\theta^2=\theta_1^2+\theta_2^2+\theta_3^2$. Show
gilmore-lie-groups_F00768	84	■ 1.000	$SO(n)$	$SO(n)$	5. The Lie algebra for the matrix group $SO(n)$ consists of antisymmetric $n \times n$ matrices.
gilmore-lie-groups_F00769	84	■ 1.000	$X_{\{i j\}}=-X_{\{j i\}}(1 \leq i \neq j \leq n)$	$X_{ij}=-X_{ji}(1 \leq i \neq j \leq n)$	$X_{ij}=-X_{ji}(1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha \beta}=\delta_{i \alpha} \delta_{j \beta}-\delta_{i \beta} \delta_{j \alpha}$.
gilmore-lie-groups_F00770	84	■ 1.000	$\left(X_{\{i j\}}\right)_{\alpha \beta}=\delta_{i \alpha} \delta_{j \beta}-\delta_{i \beta} \delta_{j \alpha}$	$(X_{ij})_{\alpha \beta}=\delta_{i \alpha} \delta_{j \beta}-\delta_{i \beta} \delta_{j \alpha}$	$X_{ij}=-X_{ji}(1 \leq i \neq j \leq n)$ with matrix elements $(X_{ij})_{\alpha \beta}=\delta_{i \alpha} \delta_{j \beta}-\delta_{i \beta} \delta_{j \alpha}$.
gilmore-lie-groups_F00771	84	■ 1.000	$\mathcal{X}_{\{i j\}}=x^{\{i\}} \partial_{\{j\}}-x^{\{j\}} \partial_{\{i\}}$	$\mathcal{X}_{ij}=x^i \partial_j-x^j \partial_i$	b. Show that the operators $\mathcal{X}_{ij}=x^i \partial_j-x^j \partial_i$ satisfy isomorphic commutation rela-
gilmore-lie-groups_F00772	84	■ 1.000	$\mathcal{B}_{\{i j\}} =$	$\mathcal{B}_{ij} =$	c. Show that bilinear products of boson creation and annihilation operators $\mathcal{B}_{ij} =$
gilmore-lie-groups_F00773	84	■ 1.000	$b_{\{i\}}^{\dagger} b_{\{j\}}-b_{\{j\}}^{\dagger} b_{\{i\}}(1 \leq i \neq j \leq n)$	$b_i^{\dagger} b_j-b_j^{\dagger} b_i(1 \leq i \neq j \leq n)$	$b_i^{\dagger} b_j-b_j^{\dagger} b_i(1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00774	84	■ 0.963	$\mathcal{F}_{\{i j\}} =$	$\mathcal{F}_{ij} =$	d. Show that bilinear products of fermion creation and annihilation operators $\mathcal{F}_{ij} =$
gilmore-lie-groups_F00775	84	■ 1.000	$f_{\{i\}}^{\dagger} f_{\{j\}}-f_{\{j\}}^{\dagger} f_{\{i\}}(1 \leq i \neq j \leq n)$	$f_i^{\dagger} f_j-f_j^{\dagger} f_i(1 \leq i \neq j \leq n)$	$f_i^{\dagger} f_j-f_j^{\dagger} f_i(1 \leq i \neq j \leq n)$ satisfy isomorphic commutation relations.
gilmore-lie-groups_F00776	84	■ 1.000	D, Y, Z	D, Y, Z	6. The Jacobi identity for operators D, Y, Z (replace X by D in Eq. (4.17)) can be
gilmore-lie-groups_F00777	84	■ 0.998	$\mathfrak{so}(4)$	$\mathfrak{so}(4)$	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00778	84	■ 0.998	4×4	4×4	7. For the matrix Lie algebra $\mathfrak{so}(4)$ the defining matrix representation consists of 4×4
gilmore-lie-groups_F00779	85	■ 1.000	$a_{\{i j\}} X_{\{i j\}}$	$a_{ij} X_{ij}$	element $a_{ij} X_{ij}$ in the Lie algebra:

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gilmore-lie-groups_F00780	85	1.000	$S_0(p, q)$	$SO(p, q)$	stant? How does this result extend to $SO(n)$? to $SO(p, q)$? Is there a simple relation																					
gilmore-lie-groups_F00781	85	1.000	$\operatorname{EXP}\left(V_{-}\right)$	$\operatorname{EXP}\left(V_{-}\right)$	$\operatorname{EXP}\left(V_{-}\right)$ is multiply periodic. Apply this construction to the noncompact groups																					
gilmore-lie-groups_F00782	85	0.546	$S_0(3, 1)$	$SO(3, 1)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}\left(V_{-}\right)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-																					
gilmore-lie-groups_F00783	85	0.546	$S_0(2, 2)$	$SO(2, 2)$	$SO(3, 1)$ and $SO(2, 2)$. Show $\operatorname{EXP}\left(V_{-}\right)$ is a two-sphere S^2 for $SO(3, 1)$ and a two-																					
gilmore-lie-groups_F00784	85	0.982	$T^{\wedge 2}$	T^2	torus T^2 for $SO(2, 2)$.																					
gilmore-lie-groups_F00785	85	0.752	$e^{\theta \cdot \mathbf{L}} \in S_0(3)$	$e^{\theta \cdot \mathbf{L}} \in SO(3)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric																					
gilmore-lie-groups_F00786	85	0.752	$\rho(\theta)$	$\rho(\theta)$	operator $e^{\theta \cdot \mathbf{L}} \in SO(3)$. From this construct the invariant density $\rho(\theta)$ and the metric																					
gilmore-lie-groups_F00787	85	1.000	$g_{\mu \nu}(\theta)$	$g_{\mu \nu}(\theta)$	tensor $g_{\mu \nu}(\theta)$. Give a reason for the strange behavior (singularities) that these invariant																					
gilmore-lie-groups_F00788	85	1.000	$d V(x)=\left g(x)\right ^{\wedge 1 / 2} d^{\wedge n} x$	$d V(x)=\ g(x)\ ^{1 / 2} d^n x$	standard result of Riemannian geometry that $d V(x)=\ g(x)\ ^{1 / 2} d^n x$. Discuss the																					
gilmore-lie-groups_F00789	85	0.623	$(\mathbf{x}, \mathbf{x})$	$(\mathbf{x}, \mathbf{x})$	12. An inner product is $(\mathbf{x}, \mathbf{x})$ is imposed on a real n -dimensional linear vector space.																					
gilmore-lie-groups_F00790	85	1.000	$g_{rs}=\left(\mathbf{e}_r, \mathbf{e}_s\right)$	$g_{rs}=\left(\mathbf{e}_r, \mathbf{e}_s\right)$	It is represented by a real symmetric nonsingular $n \times n$ matrix $g_{rs}=\left(\mathbf{e}_r, \mathbf{e}_s\right)$, where																					
gilmore-lie-groups_F00791	85	1.000	$\mathbf{x}=\mathbf{e}_i x^i$	$\mathbf{x}=\mathbf{e}_i x^i$	$\mathbf{x}=\mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.																					
gilmore-lie-groups_F00792	85	1.000	g^{rs}	g^{rs}	$\mathbf{x}=\mathbf{e}_i x^i$. The inverse matrix, g^{rs} , is well defined.																					
gilmore-lie-groups_F00793	85	1.000	$\mathbf{y}=G \mathbf{x},(\mathbf{y}, \mathbf{y})=(\mathbf{x}, \mathbf{x})$	$\mathbf{y}=G \mathbf{x},(\mathbf{y}, \mathbf{y})=(\mathbf{x}, \mathbf{x})$	a. Lie group G preserves inner products. If $\mathbf{y}=G \mathbf{x},(\mathbf{y}, \mathbf{y})=(\mathbf{x}, \mathbf{x})$. Show $G^t g G=g$.																					
gilmore-lie-groups_F00794	85	1.000	$G^t g G=g$	$G^t g G=g$	a. Lie group G preserves inner products. If $\mathbf{y}=G \mathbf{x},(\mathbf{y}, \mathbf{y})=(\mathbf{x}, \mathbf{x})$. Show $G^t g G=g$.																					
gilmore-lie-groups_F00795	85	1.000	$H^t g+g H=0$	$H^t g+g H=0$	b. Show the Lie algebra H of G satisfies $H^t g+g H=0$.																					
gilmore-lie-groups_F00796	85	1.000	$X_{rs}=g_{rt} x^t \partial_s-g_{st} x^t \partial_r$	$X_{rs}=g_{rt} x^t \partial_s-g_{st} x^t \partial_r$	c. Show that the infinitesimal generators of G are $X_{rs}=g_{rt} x^t \partial_s-g_{st} x^t \partial_r$.																					
gilmore-lie-groups_F00797	86	1.000	X_{rs}	X_{rs}	d. Show that the operators X_{rs} satisfy commutation relations																					
gilmore-lie-groups_F00798	86	0.887	$M=S_0$	$M=SO$	13. Every real unimodular 2×2 matrix M can be written in the form $M=SO$, where S																					
gilmore-lie-groups_F00799	86	1.000	$S^t=S^{\wedge+1}$	$S^t=S^{+1}$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S^t=S^{+1}$</td><td>$\det(S)=+1$</td><td>$S=e^{\Sigma}$</td></tr><tr><td>$O^t=O^{-1}$</td><td>$\det(O)=+1$</td><td>$O=e^A$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} \Sigma=0$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} A=0$</td></tr><tr><td></td><td></td><td>$\Sigma^t=+\Sigma$</td></tr><tr><td></td><td></td><td>$A^t=-A$</td></tr></table>	In group	Relation	In algebra	$S^t=S^{+1}$	$\det(S)=+1$	$S=e^{\Sigma}$	$O^t=O^{-1}$	$\det(O)=+1$	$O=e^A$			$\operatorname{Tr} \Sigma=0$			$\operatorname{Tr} A=0$			$\Sigma^t=+\Sigma$			$A^t=-A$
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gilmore-lie-groups_F00800	86	1.000	$\operatorname{det}(S)=+1$	$\det(S)=+1$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S^t=S^{+1}$</td><td>$\det(S)=+1$</td><td>$S=e^{\Sigma}$</td></tr><tr><td>$O^t=O^{-1}$</td><td>$\det(O)=+1$</td><td>$O=e^A$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} \Sigma=0$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} A=0$</td></tr><tr><td></td><td></td><td>$\Sigma^t=+\Sigma$</td></tr><tr><td></td><td></td><td>$A^t=-A$</td></tr></table>	In group	Relation	In algebra	$S^t=S^{+1}$	$\det(S)=+1$	$S=e^{\Sigma}$	$O^t=O^{-1}$	$\det(O)=+1$	$O=e^A$			$\operatorname{Tr} \Sigma=0$			$\operatorname{Tr} A=0$			$\Sigma^t=+\Sigma$			$A^t=-A$
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gilmore-lie-groups_F00801	86	1.000	$S=e^{\wedge \Sigma}$	$S=e^{\Sigma}$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S^t=S^{+1}$</td><td>$\det(S)=+1$</td><td>$S=e^{\Sigma}$</td></tr><tr><td>$O^t=O^{-1}$</td><td>$\det(O)=+1$</td><td>$O=e^A$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} \Sigma=0$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} A=0$</td></tr><tr><td></td><td></td><td>$\Sigma^t=+\Sigma$</td></tr><tr><td></td><td></td><td>$A^t=-A$</td></tr></table>	In group	Relation	In algebra	$S^t=S^{+1}$	$\det(S)=+1$	$S=e^{\Sigma}$	$O^t=O^{-1}$	$\det(O)=+1$	$O=e^A$			$\operatorname{Tr} \Sigma=0$			$\operatorname{Tr} A=0$			$\Sigma^t=+\Sigma$			$A^t=-A$
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gilmore-lie-groups_F00802	86	1.000	$\operatorname{Tr} \Sigma=0$	$\operatorname{Tr} \Sigma=0$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S^t=S^{+1}$</td><td>$\det(S)=+1$</td><td>$S=e^{\Sigma}$</td></tr><tr><td>$O^t=O^{-1}$</td><td>$\det(O)=+1$</td><td>$O=e^A$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} \Sigma=0$</td></tr><tr><td></td><td></td><td>$\operatorname{Tr} A=0$</td></tr><tr><td></td><td></td><td>$\Sigma^t=+\Sigma$</td></tr><tr><td></td><td></td><td>$A^t=-A$</td></tr></table>	In group	Relation	In algebra	$S^t=S^{+1}$	$\det(S)=+1$	$S=e^{\Sigma}$	$O^t=O^{-1}$	$\det(O)=+1$	$O=e^A$			$\operatorname{Tr} \Sigma=0$			$\operatorname{Tr} A=0$			$\Sigma^t=+\Sigma$			$A^t=-A$
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gilmore-lie-groups_ F08803	86	1.000	$\Sigma^t = +\Sigma$	$\Sigma^t = +\Sigma$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
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gilmore-lie-groups_ F08804	86	1.000	$O^t = O^{-1}$	$O^t = O^{-1}$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
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gilmore-lie-groups_ F08805	86	1.000	$\operatorname{det}(O) = +1$	$\det(O) = +1$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
In group	Relation	In algebra												
$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$												
$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$												
gilmore-lie-groups_ F08806	86	1.000	$O = e^A$	$O = e^A$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
In group	Relation	In algebra												
$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$												
$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$												
gilmore-lie-groups_ F08807	86	1.000	$\operatorname{Tr} A = 0$	$\operatorname{Tr} A = 0$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
In group	Relation	In algebra												
$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$												
$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$												
gilmore-lie-groups_ F08808	86	1.000	$A^t = -A$	$A^t = -A$	<table><tr><th>In group</th><th>Relation</th><th>In algebra</th></tr><tr><td>$S' = S^{+1}$</td><td>$\det(S) = +1$</td><td>$S = e^\Sigma$</td></tr><tr><td>$O' = O^{-1}$</td><td>$\det(O) = +1$</td><td>$O = e^A$</td></tr></table>	In group	Relation	In algebra	$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$	$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$
In group	Relation	In algebra												
$S' = S^{+1}$	$\det(S) = +1$	$S = e^\Sigma$												
$O' = O^{-1}$	$\det(O) = +1$	$O = e^A$												
gilmore-lie-groups_ F08809	86	0.999	$M M^t = S^2 = e^{2\Sigma}$	$MM^t = S^2 = e^{2\Sigma}$	a. Show that $MM^t = S^2 = e^{2\Sigma}$.									
gilmore-lie-groups_ F08810	86	1.000	$O = S^{-1}M = e^{-\Sigma}M$	$O = S^{-1}M = e^{-\Sigma}M$	b. Show $O = S^{-1}M = e^{-\Sigma}M$									
gilmore-lie-groups_ F08811	86	1.000	Σ	Σ	c. Write S as a power series expansion in Σ .									
gilmore-lie-groups_ F08812	86	1.000	$S - I_2$	$S - I_2$	d. Write Σ as a power series expansion in $S - I_2$.									
gilmore-lie-groups_ F08813	86	1.000	$M = HU$	$M = HU$	14. Extend the result of the previous problem to complex $n \times n$ matrices $M = HU$, with									
gilmore-lie-groups_ F08814	86	1.000	$H^\dagger = H^{+1}$	$H^\dagger = H^{+1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.									
gilmore-lie-groups_ F08815	86	1.000	$U^\dagger = U^{-1}$	$U^\dagger = U^{-1}$	M arbitrary but nonsingular, $H^\dagger = H^{+1}$ hermitian and $U^\dagger = U^{-1}$ unitary.									
gilmore-lie-groups_ F08816	86	1.000	$\hbar k_L$	$\hbar k_L$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)									
gilmore-lie-groups_ F08817	86	1.000	$-\hbar k_R$	$-\hbar k_R$	or right with momentum $\hbar k_L$ or $-\hbar k_R$, has the form (Gilmore, 2004)									
gilmore-lie-groups_ F08818	86	1.000	m_{ij}	m_{ij}	The real quantities m_{ij} are the four matrix elements of a group operation in $SL(2; \mathbb{R})$.									
gilmore-lie-groups_ F08819	86	1.000	$\hbar k_L = \hbar k_R$	$\hbar k_L = \hbar k_R$	They are energy dependent. By appropriate choice of $\hbar k_L = \hbar k_R$ and the matrix									
gilmore-lie-groups_ F08820	86	0.679	$\operatorname{SU}(1, 1)$	$SU(1, 1)$	Show that these three matrices span the Lie algebra of the group $SU(1, 1)$.									
gilmore-lie-groups_ F08821	87	1.000	κ_R	κ_R	The parameters κ_R and κ_L describe the decay length of the exponentially decaying									
gilmore-lie-groups_ F08822	87	1.000	κ_L	κ_L	The parameters κ_R and κ_L describe the decay length of the exponentially decaying									
gilmore-lie-groups_ F08823	87	1.000	$\kappa_L = \kappa_R$	$\kappa_L = \kappa_R$	energy dependent. By appropriate choice of $\kappa_L = \kappa_R$ and the matrix elements m_{ij} ,									

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gilmore-lie-groups_ F08824	87	■ 1.000	$k_*=\sqrt{2\,m\left(E-V_*\right)}/\hbar^{\{2\}}$	$k_*=\sqrt{2m\left(E-V_*\right)}/\hbar^2$	(Hint: $k_*=\sqrt{2m(E-V_*)}/\hbar^2$ for $E>V_*$ and $\kappa_*=\sqrt{2m(V_*-E)}/\hbar^2$ for $E<V_*$,																																																	
gilmore-lie-groups_ F08825	87	■ 1.000	$E>V_*$	$E>V_*$	(Hint: $k_*=\sqrt{2m(E-V_*)}/\hbar^2$ for $E>V_*$ and $\kappa_*=\sqrt{2m(V_*-E)}/\hbar^2$ for $E<V_*$,																																																	
gilmore-lie-groups_ F08826	87	■ 1.000	$\kappa_*=\sqrt{2\,m\left(V_*-E\right)}/\hbar^{\{2\}}$	$\kappa_*=\sqrt{2m\left(V_*-E\right)}/\hbar^2$	(Hint: $k_*=\sqrt{2m(E-V_*)}/\hbar^2$ for $E>V_*$ and $\kappa_*=\sqrt{2m(V_*-E)}/\hbar^2$ for $E<V_*$,																																																	
gilmore-lie-groups_ F08827	87	■ 1.000	$E<V_*$	$E<V_*$	(Hint: $k_*=\sqrt{2m(E-V_*)}/\hbar^2$ for $E>V_*$ and $\kappa_*=\sqrt{2m(V_*-E)}/\hbar^2$ for $E<V_*$,																																																	
gilmore-lie-groups_ F08828	87	■ 0.978	$*=L,\,R$	$*=L,R$	$*=L,R$.)																																																	
gilmore-lie-groups_ F08829	88	■ 0.998	$\operatorname{det}(M)\neq 0$	$\det(M)\neq 0$	groups $GL(n;\mathbb{F})$ have no defining condition (the only condition is $\det(M)\neq 0$),																																																	
gilmore-lie-groups_ F08830	88	■ 1.000	$I+\epsilon A$	$I+\epsilon A$	the constraint to matrices infinitesimally close to the identity: $I+\epsilon A$. The matrices																																																	
gilmore-lie-groups_ F08831	88	■ 1.000	$\mathfrak{gl}(n;\mathbb{F})$	$\mathfrak{gl}(n;\mathbb{F})$	1. $\mathfrak{gl}(n;\mathbb{F})$. This algebra consists of arbitrary $n\times n$ matrices over the field $\mathbb{F}$.																																																	
gilmore-lie-groups_ F08832	89	■ 1.000	$\mathfrak{ut}(p,q)$	$\mathfrak{ut}(p,q)$	2. $\mathfrak{ut}(p,q)$. Upper triangular algebras. The matrix algebra has the same structure																																																	
gilmore-lie-groups_ F08833	89	■ 0.998	$\mathfrak{ht}(p,q)$	$\mathfrak{ht}(p,q)$	3. $\mathfrak{ht}(p,q)$. The algebra for this class of groups is defined by the condition																																																	
gilmore-lie-groups_ F08834	90	■ 1.000	$(a,b)=(1,0)$	$(a,b)=(1,0)$	The identity is at $(a,b)=(1,0)$. Its algebra is spanned by the two operators																																																	
gilmore-lie-groups_ F08835	90	■ 0.999	$\left[X_{\{a\}},X_{\{b\}}\right]=X_{\{b\}}$	$[X_a,X_b]=X_b$	The commutation relations are given by $[X_a,X_b]=X_b$.																																																	
gilmore-lie-groups_ F08836	90	■ 1.000	$\mathfrak{ut}(p,q,r)$	$\mathfrak{ut}(p,q,r)$	4. $\mathfrak{ut}(p,q,r)$. This matrix algebra is identical in structure to the parent group.																																																	
gilmore-lie-groups_ F08837	90	■ 0.999	$\mathfrak{ut}(1,2,1)$	$\mathfrak{ut}(1,2,1)$	Example A very useful six-parameter subalgebra of $\mathfrak{ut}(1,2,1)$ is given by																																																	
gilmore-lie-groups_ F08838	90	■ 1.000	j	j	algebra are summarized in the table below. The operator in the i th row and j th																																																	
gilmore-lie-groups_ F08839	90	■ 1.000	$\left[X_{\{i\}},X_{\{j\}}\right]$	$[X_i,X_j]$	column is $[X_i,X_j]$.																																																	
gilmore-lie-groups_ F08840	90	■ 1.000	$X_{\{\eta\}}$	X_η	<table><tr><td></td><td>X_η</td><td>X_R</td><td>X_L</td><td>X_r</td><td>X_l</td><td>X_δ</td></tr><tr><td>X_η</td><td>0</td><td>$2X_R$</td><td>$-2X_L$</td><td>X_r</td><td>$-X_l$</td><td>0</td></tr><tr><td>X_R</td><td></td><td>0</td><td>$-4X_\eta$</td><td>0</td><td>$-2X_r$</td><td>0</td></tr><tr><td>X_L</td><td></td><td></td><td>0</td><td>$2X_l$</td><td>0</td><td>0</td></tr><tr><td>X_r</td><td></td><td></td><td></td><td>0</td><td>$-X_\delta$</td><td>0</td></tr><tr><td>X_l</td><td></td><td></td><td></td><td></td><td>0</td><td>0</td></tr><tr><td>X_δ</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr></table>		X_η	X_R	X_L	X_r	X_l	X_δ	X_η	0	$2X_R$	$-2X_L$	X_r	$-X_l$	0	X_R		0	$-4X_\eta$	0	$-2X_r$	0	X_L			0	$2X_l$	0	0	X_r				0	$-X_\delta$	0	X_l					0	0	X_δ						0
	X_η	X_R	X_L	X_r	X_l	X_δ																																																
X_η	0	$2X_R$	$-2X_L$	X_r	$-X_l$	0																																																
X_R		0	$-4X_\eta$	0	$-2X_r$	0																																																
X_L			0	$2X_l$	0	0																																																
X_r				0	$-X_\delta$	0																																																
X_l					0	0																																																
X_δ						0																																																
gilmore-lie-groups_ F08841	90	■ 1.000	$X_{\{R\}}$	X_R	<table><tr><td></td><td>X_η</td><td>X_R</td><td>X_L</td><td>X_r</td><td>X_l</td><td>X_δ</td></tr><tr><td>X_η</td><td>0</td><td>$2X_R$</td><td>$-2X_L$</td><td>X_r</td><td>$-X_l$</td><td>0</td></tr><tr><td>X_R</td><td></td><td>0</td><td>$-4X_\eta$</td><td>0</td><td>$-2X_r$</td><td>0</td></tr><tr><td>X_L</td><td></td><td></td><td>0</td><td>$2X_l$</td><td>0</td><td>0</td></tr><tr><td>X_r</td><td></td><td></td><td></td><td>0</td><td>$-X_\delta$</td><td>0</td></tr><tr><td>X_l</td><td></td><td></td><td></td><td></td><td>0</td><td>0</td></tr><tr><td>X_δ</td><td></td><td></td><td></td><td></td><td></td><td>0</td></tr></table>		X_η	X_R	X_L	X_r	X_l	X_δ	X_η	0	$2X_R$	$-2X_L$	X_r	$-X_l$	0	X_R		0	$-4X_\eta$	0	$-2X_r$	0	X_L			0	$2X_l$	0	0	X_r				0	$-X_\delta$	0	X_l					0	0	X_δ						0
	X_η	X_R	X_L	X_r	X_l	X_δ																																																
X_η	0	$2X_R$	$-2X_L$	X_r	$-X_l$	0																																																
X_R		0	$-4X_\eta$	0	$-2X_r$	0																																																
X_L			0	$2X_l$	0	0																																																
X_r				0	$-X_\delta$	0																																																
X_l					0	0																																																
X_δ						0																																																

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gilmore-lie-groups_ F08842	90	■ 1.000	$X_{\{L\}}$	X_L	
gilmore-lie-groups_ F08843	90	■ 1.000	$X_{\{r\}}$	X_r	
gilmore-lie-groups_ F08844	90	■ 1.000	$X_{\{l\}}$	X_l	
gilmore-lie-groups_ F08845	90	■ 1.000	$X_{\{\delta\}}$	X_δ	
gilmore-lie-groups_ F08846	90	■ 1.000	$2 X_{\{R\}}$	$2X_R$	
gilmore-lie-groups_ F08847	90	■ 1.000	$-2 X_{\{L\}}$	$-2X_L$	

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gilmore-lie-groups_ F08848	90	■ 1.000	$-X_{\{l\}}$	$-X_l$	$ \begin{array}{c cccccc} & X_{\eta} & X_R & X_L & X_r & X_l & X_{\delta} \\ \hline X_{\eta} & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_{\eta} & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_{\delta} & 0 \\ X_l & & & & & 0 & 0 \\ X_{\delta} & & & & & & 0 \end{array} $
gilmore-lie-groups_ F08849	90	■ 1.000	$-4 X_{\{\eta\}}$	$-4X_{\eta}$	$ \begin{array}{c cccccc} & X_{\eta} & X_R & X_L & X_r & X_l & X_{\delta} \\ \hline X_{\eta} & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_{\eta} & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_{\delta} & 0 \\ X_l & & & & & 0 & 0 \\ X_{\delta} & & & & & & 0 \end{array} $
gilmore-lie-groups_ F08850	90	■ 1.000	$-2 X_{\{r\}}$	$-2X_r$	$ \begin{array}{c cccccc} & X_{\eta} & X_R & X_L & X_r & X_l & X_{\delta} \\ \hline X_{\eta} & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_{\eta} & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_{\delta} & 0 \\ X_l & & & & & 0 & 0 \\ X_{\delta} & & & & & & 0 \end{array} $
gilmore-lie-groups_ F08851	90	■ 1.000	$2 X_{\{l\}}$	$2X_l$	$ \begin{array}{c cccccc} & X_{\eta} & X_R & X_L & X_r & X_l & X_{\delta} \\ \hline X_{\eta} & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_{\eta} & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_{\delta} & 0 \\ X_l & & & & & 0 & 0 \\ X_{\delta} & & & & & & 0 \end{array} $
gilmore-lie-groups_ F08852	90	■ 1.000	$-X_{\{\delta\}}$	$-X_{\delta}$	$ \begin{array}{c cccccc} & X_{\eta} & X_R & X_L & X_r & X_l & X_{\delta} \\ \hline X_{\eta} & 0 & 2X_R & -2X_L & X_r & -X_l & 0 \\ X_R & & 0 & -4X_{\eta} & 0 & -2X_r & 0 \\ X_L & & & 0 & 2X_l & 0 & 0 \\ X_r & & & & 0 & -X_{\delta} & 0 \\ X_l & & & & & 0 & 0 \\ X_{\delta} & & & & & & 0 \end{array} $
gilmore-lie-groups_ F08853	90	■ 0.600	$\hat{n}+\frac{1}{2} I$	$\hat{n} + \frac{1}{2}I$	$ \begin{array}{c cccccc} & \hat{n} + \frac{1}{2}I & a^{\dagger}a^{\dagger} & aa & a^{\dagger} & a & I \\ \hline \hat{n} + \frac{1}{2}I & 0 & 2a^{\dagger}a^{\dagger} & -2aa & a^{\dagger} & -a & 0 \\ a^{\dagger}a^{\dagger} & & 0 & -4(\hat{n} + \frac{1}{2}I) & 0 & -2a^{\dagger} & 0 \\ aa & & & 0 & 2a & 0 & 0 \\ a^{\dagger} & & & & 0 & -I & 0 \\ a & & & & & 0 & 0 \\ I & & & & & & 0 \end{array} $

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gilmore-lie-groups_ F00854	90	■ 0.600	$a^{\dagger} a^{\dagger}$	$a^{\dagger} a^{\dagger}$	
gilmore-lie-groups_ F00855	90	■ 0.600	$a a$	aa	
gilmore-lie-groups_ F00856	90	■ 0.600	$2 a^{\dagger} a^{\dagger}$	$2a^{\dagger} a^{\dagger}$	
gilmore-lie-groups_ F00857	90	■ 0.600	$-2 a a$	$-2aa$	
gilmore-lie-groups_ F00858	90	■ 0.600	$-a$	$-a$	
gilmore-lie-groups_ F00859	90	■ 0.600	$-4\left(\hat{n} + \frac{1}{2} I\right)$	$-4\left(\hat{n} + \frac{1}{2} I\right)$	

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gilmore-lie-groups_ F08860	90	0.600	$-2 a^{\dagger}$	$-2a^{\dagger}$	
gilmore-lie-groups_ F08861	90	0.600	$2 a$	$2a$	
gilmore-lie-groups_ F08862	90	0.600	$-I$	$-I$	
gilmore-lie-groups_ F08863	90	1.000	$\hat{n}=a^{\dagger}a+\frac{1}{2}$	$\hat{n} = a^{\dagger}a + \frac{1}{2}$	and the algebra of the photon energy operator $\hat{n} = a^{\dagger}a + \frac{1}{2}$, two-photon creation
gilmore-lie-groups_ F08864	91	0.996	$I=\left[a, a^{\dagger}\right]$	$I = [a, a^{\dagger}]$	operators $a^{\dagger}$ and a , and the identity operator $I = [a, a^{\dagger}]$. We observe that the 4×4
gilmore-lie-groups_ F08865	91	0.887	$\mathfrak{so}(n)=\mathfrak{ut}(1,1,1,\ldots,1)$	$\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$	5. $\mathfrak{so}(n) = \mathfrak{ut}(1, 1, 1, \dots, 1)$. This matrix algebra is also identical in structure to
gilmore-lie-groups_ F08866	91	1.000	$\mathfrak{ut}(1,1,1)$	$\mathfrak{ut}(1, 1, 1)$	its parent group. A very useful four-parameter subalgebra of $\mathfrak{ut}(1, 1, 1)$ is given by
gilmore-lie-groups_ F08867	91	0.936	$\mathfrak{nil}(n)$	$\mathfrak{nil}(n)$	6. $\mathfrak{nil}(n)$. Nilpotent matrices have an upper triangular structure, with $+1$ along the
gilmore-lie-groups_ F08868	91	1.000	$\eta=0$	$\eta = 0$	This is easily seen by setting $\eta = 0$ in (5.9). As a result, a number of difficult
gilmore-lie-groups_ F08869	91	1.000	$\mathfrak{a}(p,q)$	$\mathfrak{a}(p, q)$	7. $\mathfrak{a}(p, q)$. The matrix algebra for the commutative group of Example (7) in Chap-
gilmore-lie-groups_ F08870	92	0.998	$O(3)$	$O(3)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08871	92	0.998	$U(2)$	$U(2)$	antihermitian matrices. The Lie algebras for the groups $O(3)$ and $U(2)$ are
gilmore-lie-groups_ F08872	92	0.728	σ_{μ}	σ_{μ}	The four 2×2 matrices σ_{μ} are called Pauli spin matrices .
gilmore-lie-groups_ F08873	92	1.000	$\mathfrak{so}(2,1)$	$\mathfrak{so}(2, 1)$	example, the algebra $\mathfrak{so}(2, 1)$ for the Lorentz group in the plane is
gilmore-lie-groups_ F08874	93	1.000	$\mathfrak{so}(3,1)$	$\mathfrak{so}(3, 1)$	By an identical argument the matrix Lie algebra for the Lorentz group $\mathfrak{so}(3, 1)$ is
gilmore-lie-groups_ F08875	93	1.000	$M=-G^{-1}M^{\dagger}G$	$M = -G^{-1}M^{\dagger}G$	Since G is nonsingular, $M = -G^{-1}M^{\dagger}G$ and
gilmore-lie-groups_ F08876	93	1.000	$(p+q) \times (p+q)$	$(p + q) \times (p + q)$	In the case that the $(p + q) \times (p + q)$ matrix G has singular block diagonal struc-

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gilmore-lie-groups_ F00877	93	■ 0.983	$\left[\begin{smallmatrix} g & 0 \\ 0 & 0 \end{smallmatrix}\right]$	$\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F00878	93	■ 0.983	$\operatorname{det}(g) \neq 0$	$\det(g) \neq 0$	ture $\begin{bmatrix} g & 0 \\ 0 & 0 \end{bmatrix}$ with $\det(g) \neq 0$, this constraint reduces to
gilmore-lie-groups_ F00879	93	■ 1.000	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1,1,1,0)$	$\operatorname{diag}(1,1,1,0)$ the Lie algebra is
gilmore-lie-groups_ F00880	94	■ 1.000	θ_i	θ_i	Here the parameters θ_i describe rotations about the i th coordinate axis and the
gilmore-lie-groups_ F00881	94	■ 1.000	s_4	s_4	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4}t$. If s_4 is set to zero
gilmore-lie-groups_ F00882	94	■ 1.000	$t^{\prime} = e^{s_4}t$	$t' = e^{s_4}t$	parameter s_4 describes “scaling” of the time axis: $t' = e^{s_4}t$. If s_4 is set to zero
gilmore-lie-groups_ F00883	94	■ 1.000	$E(3) = ISO(3)$	$E(3) = ISO(3)$	Euclidean group $E(3) \equiv ISO(3)$ (inhomogeneous rotation group in R^3).
gilmore-lie-groups_ F00884	94	■ 0.981	$ISO(3,1)(3.26)$	$ISO(3,1)(3.26)$	The matrix Lie algebra for the Poincaré group $ISO(3,1)$ (3.26) is obtained by
gilmore-lie-groups_ F00885	94	■ 0.837	5×5	5×5	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1,1,1,-1,0)$. The Lie
gilmore-lie-groups_ F00886	94	■ 0.837	$G = \operatorname{diag}(1,1,1,-1,0)$	$G = \operatorname{diag}(1,1,1,-1,0)$	similar arguments using a singular 5×5 metric $G = \operatorname{diag}(1,1,1,-1,0)$. The Lie
gilmore-lie-groups_ F00887	94	■ 0.988	$ISO(3,1)$	$ISO(3,1)$	“contraction” (cf., Chapter 13) from the Lie algebra of $ISO(3,1)$:
gilmore-lie-groups_ F00888	94	■ 1.000	$\mathfrak{su}(n)$	$\mathfrak{su}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F00889	94	■ 1.000	$\mathfrak{u}(n)$	$\mathfrak{u}(n)$	Lie algebras. The important algebra $\mathfrak{su}(n)$ is obtained from the intersection of $\mathfrak{u}(n)$
gilmore-lie-groups_ F00890	94	■ 1.000	$\mathfrak{sl}(n; \mathbb{C})$	$\mathfrak{sl}(n; \mathbb{C})$	and $\mathfrak{sl}(n; \mathbb{C})$ (cf. (5.14)). For example
gilmore-lie-groups_ F00891	95	■ 1.000	$M_{ij} (i < j) (M \in \mathfrak{u}(n))$	$M_{ij} (i < j) (M \in \mathfrak{u}(n))$	each of the $n(n-1)/2$ complex matrix elements $M_{ij} (i < j) (M \in \mathfrak{u}(n))$ above the
gilmore-lie-groups_ F00892	95	■ 0.588	M_{ii}	M_{ii}	M_{ii} by a real 2×2 matrix representing an imaginary complex number ($a = 0$,
gilmore-lie-groups_ F00893	95	■ 0.761	M_{ij}	M_{ij}	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F00894	95	■ 0.761	$(i > j)$	$(i > j)$	b arbitrary in Eq. (3.3)). The matrix elements M_{ij} below the diagonal ($i > j$)
gilmore-lie-groups_ F00895	95	■ 0.999	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F00896	95	■ 0.999	$\mathfrak{o} \mathfrak{u}(2n)$	$\mathfrak{o} \mathfrak{u}(2n)$	$2n \times 2n$ matrix with the property $\mathfrak{u}(n) \rightarrow \mathfrak{ou}(2n) \subset \mathfrak{o}(2n)$. The dimension of $\mathfrak{ou}(2n)$
gilmore-lie-groups_ F00897	95	■ 0.640	$\mathcal{U}(n): 2 \times n(n-1) / 2 + 1 \times n = n^2$	$\mathcal{U}(n): 2 \times n(n-1)/2 + 1 \times n = n^2$	is the dimension of $\mathfrak{u}(n): 2 \times n(n-1)/2 + 1 \times n = n^2$.
gilmore-lie-groups_ F00898	95	■ 0.914	$Sp(n)$	$Sp(n)$	The Lie algebra of $Sp(n)$ consists of $n \times n$ antihermitian matrices M over $\mathbb{Q}$:
gilmore-lie-groups_ F00899	95	■ 0.791	$\operatorname{Sp}(n)$	$Sp(n)$	tained from the Lie algebra of $Sp(n)$ by replacing each of the $n(n-1)/2$ quaternion
gilmore-lie-groups_ F00900	95	■ 0.712	$M_{ij} (i < j) (M \in \mathfrak{sp}(n))$	$M_{ij} (i < j) (M \in \mathfrak{sp}(n))$	matrix elements $M_{ij} (i < j) (M \in \mathfrak{sp}(n))$ above the diagonal of M by a 2×2 com-
gilmore-lie-groups_ F00901	95	■ 0.998	$q_0 = 0, q_i$	$q_0 = 0, q_i$	matrix representing an imaginary quaternion ($q_0 = 0, q_i$ arbitrary in Eq. (3.4)). The
gilmore-lie-groups_ F00902	95	■ 0.615	$i > j$	$i > j$	matrix elements M_{ij} below the diagonal ($i > j$) are obtained from the antihermitic-
gilmore-lie-groups_ F00903	95	■ 1.000	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:

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gilmore-lie-groups_F00904	95	■ 1.000	$\mathfrak{usp}(2n)$	$\mathfrak{usp}(2n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups_F00905	95	■ 1.000	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n)$	$\mathfrak{sp}(n) \rightarrow \mathfrak{usp}(2n) \subset \mathfrak{su}(2n)$. The dimension of $\mathfrak{usp}(2n)$ is the dimension of $\mathfrak{sp}(n)$:
gilmore-lie-groups_F00906	95	■ 1.000	$4 \times n(n-1) / 2 + n \times 3 = 2n(2n+1) / 2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/2$	$4 \times n(n-1)/2 + n \times 3 = 2n(2n+1)/4$.
gilmore-lie-groups_F00907	96	■ 0.999	$(X, X)=0 \rightarrow X=$	$(X, X) = 0 \Rightarrow X =$	is usually very useful. This inner product is positive–definite: $(X, X) = 0 \Rightarrow X =$
gilmore-lie-groups_F00908	96	■ 0.949	$X, X^{\prime}$	X, X'	0. For example, for the algebra $\mathfrak{so}(2, 1)$ (Eq. (5.18)), if X, X' are two 3×3 matrices
gilmore-lie-groups_F00909	96	■ 0.811	(X, Y)	(X, Y)	Lie algebra. The Cartan–Killing inner product (X, Y) is specifically defined as the
gilmore-lie-groups_F00910	97	■ 1.000	$\exp(\theta X)$	$\exp(\theta X)$	In the compact case, the group element $\exp(\theta X)$ periodically returns to the identity
gilmore-lie-groups_F00911	97	■ 1.000	$-\pi \leq \theta \leq \pi$	$-\pi \leq \theta \leq +\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00912	97	■ 1.000	$-\pi$	$-\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00913	97	■ 1.000	$+\pi$	$+\pi$	parameter values: $-\pi \leq \theta \leq +\pi$, with $-\pi$ and $+\pi$ identified. On the other hand, in
gilmore-lie-groups_F00914	97	■ 1.000	$-\infty < \theta < +\infty$	$-\infty < \theta < +\infty$	the noncompact case the group is parameterized by the entire line $-\infty < \theta < +\infty$.
gilmore-lie-groups_F00915	97	■ 1.000	S^1	S^1	The underlying manifolds for the two groups are the circle S^1 and the line R^1 .
gilmore-lie-groups_F00916	97	■ 1.000	R^k	R^k	the entire subspace in the Lie algebra, R^k , where k is the number of noncompact
gilmore-lie-groups_F00917	97	■ 1.000	$G=I_n, I_{p, q}$	$G = I_n, I_{p,q}$	the metric-preserving groups ($G = I_n, I_{p,q}$, nonsingular antisymmetric, general
gilmore-lie-groups_F00918	97	■ 0.989	$(A, A)=$	$(A, A) =$	Show that in this matrix Lie algebra an inner product can be defined by $(A, A) =$
gilmore-lie-groups_F00919	97	■ 1.000	$\operatorname{tr}(A)^2=a^2+c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$	$\operatorname{tr}(A)^2 = a^2 + c^2$.
gilmore-lie-groups_F00920	97	■ 1.000	$\mathfrak{ut}(1,1)$	$\mathfrak{ut}(1,1)$	2. Show that the regular representation of the matrix Lie algebra $\mathfrak{ut}(1, 1)$ given in Prob-
gilmore-lie-groups_F00921	98	■ 1.000	$(A, A)=\operatorname{tr} R(A)^2=(a-c)^2$	$(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$	inner product in the regular representation is $(A, A) = \operatorname{tr} R(A)^2 = (a - c)^2$. The inner
gilmore-lie-groups_F00922	98	■ 0.998	X_a+X_c	$X_a + X_c$	product in the regular representation suggests that the linear combination $X_a + X_c$
gilmore-lie-groups_F00923	98	■ 1.000	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	$\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$	3. Write down the algebra inclusions $\mathfrak{gl}(1; \mathbb{R}) \subset \mathfrak{gl}(1; \mathbb{C}) \subset \mathfrak{gl}(1; \mathbb{Q})$ explicitly in terms
gilmore-lie-groups_F00924	98	■ 1.000	$X_\eta, X_r, X_l, X_\delta$	$X_\eta, X_r, X_l, X_\delta$	the matrices $X_\eta, X_r, X_l, X_\delta$ defined in (5.9). Show that the two Lie algebras are
gilmore-lie-groups_F00925	98	■ 1.000	$\hat{E}=\left(a^\dagger a+\frac{1}{2}\right) \hbar \omega \rightarrow a^\dagger a+\frac{1}{2}$	$\hat{E} = \left(a^\dagger a + \frac{1}{2}\right) \hbar \omega \rightarrow a^\dagger a + \frac{1}{2}$	is $\hat{E} = (a^\dagger a + \frac{1}{2})\hbar\omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar\omega = 1$.
gilmore-lie-groups_F00926	98	■ 1.000	$\hbar \omega=1$	$\hbar \omega = 1$	is $\hat{E} = (a^\dagger a + \frac{1}{2})\hbar\omega \rightarrow a^\dagger a + \frac{1}{2}$ for $\hbar\omega = 1$.
gilmore-lie-groups_F00927	98	■ 1.000	$X, X^{\prime} \in \mathfrak{h}$	$X, X' \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00928	98	■ 1.000	$Y, Y^{\prime} \in \mathfrak{p}$	$Y, Y' \in \mathfrak{p}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00929	98	■ 1.000	$\left[X, X^{\prime}\right] \in \mathfrak{h},\left[X, Y\right] \in \mathfrak{p},\left[Y, Y^{\prime}\right] \in \mathfrak{h}$	$[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$	This means that if $X, X' \in \mathfrak{h}$ and $Y, Y' \in \mathfrak{p}$, then $[X, X'] \in \mathfrak{h}, [X, Y] \in \mathfrak{p}, [Y, Y'] \in \mathfrak{h}$.
gilmore-lie-groups_F00930	98	■ 1.000	$\mathfrak{h}$	$\mathfrak{h}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what

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gilmore-lie-groups_F00931	98	■ 1.000	$\mathfrak{p}$	$\mathfrak{p}$	Conclude that the subspace $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$. Is $\mathfrak{p}$ a subalgebra (under what
gilmore-lie-groups_F00932	98	■ 0.995	$X=\left[\begin{array}{cc}A & \theta \\ 0 & D\end{array}\right]\in\mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00933	98	■ 0.995	$Y=\left[\begin{array}{cc}\theta & B \\ C & 0\end{array}\right]\in\mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$	If $X = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ C & 0 \end{bmatrix} \in \mathfrak{p}$, then
gilmore-lie-groups_F00934	98	■ 1.000	$(X, Y)=0$	$(X, Y) = 0$	Show that X and Y are orthogonal under this inner product: $(X, Y) = 0$.
gilmore-lie-groups_F00935	98	■ 0.970	$\mathfrak{so}(p, q)$	$\mathfrak{so}(p, q)$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00936	98	■ 0.970	$\left[\begin{array}{cc}A & B \\ B^t & C\end{array}\right]$	$\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00937	98	■ 0.970	$p \times p$	$p \times p$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00938	98	■ 0.970	$q \times q$	$q \times q$	7. The Lie algebra $\mathfrak{so}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00939	98	■ 0.986	$C^t=-C$	$C^t = -C$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00940	98	■ 0.986	$X=\left[\begin{array}{cc}A & \theta \\ 0 & C\end{array}\right]\in\mathfrak{h}$	$X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00941	98	■ 0.986	$Y=\left[\begin{array}{cc}\theta & B \\ B^t & 0\end{array}\right]\in\mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$	A and C satisfy $A^t = -A$ and $C^t = -C$. If $\tilde{X} = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and $Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, show
gilmore-lie-groups_F00942	98	■ 0.988	$(X, X) \leq 0, \quad \text{quad}(X, X)=0 \Rightarrow X=0$	$(X, X) \leq 0, \quad (X, X) = 0 \Rightarrow X = 0$	• $\boxed{(X, X) \leq 0. \quad (X, X) = 0 \Rightarrow X = 0}$
gilmore-lie-groups_F00943	98	■ 1.000	$(Y, Y) \geq 0, \quad \text{quad}(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) \geq 0, \quad (Y, Y) = 0 \Rightarrow Y = 0$	• $\boxed{(Y, Y) \geq 0. \quad (Y, Y) = 0 \Rightarrow Y = 0}$
gilmore-lie-groups_F00944	99	■ 0.818	$\mathfrak{su}(p, q)$	$\mathfrak{su}(p, q)$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00945	99	■ 0.818	$\left[\begin{array}{c}A \\ B^t \\ C\end{array}\right]$	$\begin{bmatrix} A \\ B^t \\ C \end{bmatrix}$	8. The Lie algebra $\mathfrak{su}(p, q)$ has the structure $\begin{bmatrix} A & B \\ B^t & C \end{bmatrix}$ where the $p \times p$ and $q \times q$ matrices
gilmore-lie-groups_F00946	99	■ 0.999	$A^\dagger=-A, \quad C^\dagger=-C$	$A^\dagger = -A, C^\dagger = -C$	A and C satisfy $\boxed{A^\dagger = -A, \quad C^\dagger = -C}$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00947	99	■ 0.999	$\text{operatorname{tr}}(A+C)=0$	$\text{tr}(A + C) = 0$	A and C satisfy $A^\dagger = -A, C^\dagger = -C$, and $\text{tr}(A + C) = 0$. If $X = \begin{bmatrix} A & 0 \\ 0 & C \end{bmatrix} \in \mathfrak{h}$ and
gilmore-lie-groups_F00948	99	■ 1.000	$Y=\left[\begin{array}{cc}\theta & B \\ B^t & 0\end{array}\right]\in\mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$	$Y = \begin{bmatrix} 0 & B \\ B^t & 0 \end{bmatrix} \in \mathfrak{p}$, then show once again that
gilmore-lie-groups_F00949	99	■ 1.000	$(X, X)=0 \Rightarrow X=0$	$(X, X) = 0 \Rightarrow X = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00950	99	■ 1.000	$(Y, Y)=0 \Rightarrow Y=0$	$(Y, Y) = 0 \Rightarrow Y = 0$	Show that $(X, X) = 0 \Rightarrow X = 0$ and $(Y, Y) = 0 \Rightarrow Y = 0$.
gilmore-lie-groups_F00951	99	■ 1.000	$\mathfrak{sl}(n; \mathbb{R})$	$\mathfrak{sl}(n; \mathbb{R})$	9. The Lie algebra for $\mathfrak{sl}(n; \mathbb{R})$ has a decomposition in terms of real antisymmetric and
gilmore-lie-groups_F00952	99	■ 1.000	$B^t=B$	$B^t = B$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\text{tr } B = 0$:
gilmore-lie-groups_F00953	99	■ 1.000	$\text{operatorname{tr}} B=0$	$\text{tr } B = 0$	traceless symmetric matrices $A^t = -A$ and $B^t = B$ with $\text{tr } B = 0$:
gilmore-lie-groups_F00954	99	■ 1.000	$(A, A)=0 \Rightarrow A=0$	$(A, A) = 0 \Rightarrow A = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00955	99	■ 1.000	$(B, B)=0 \Rightarrow B=0$	$(B, B) = 0 \Rightarrow B = 0$	Show that $(A, A) = 0 \Rightarrow A = 0$ and $(B, B) = 0 \Rightarrow B = 0$.
gilmore-lie-groups_F00956	99	■ 1.000	$A^\dagger=-A$	$A^\dagger = -A$	matrices $A^\dagger = -A$ and traceless hermitian matrices $H^\dagger = H$:

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gilmore-lie-groups_ F08957	99	1.000	$H^{\dagger}=H$	$H^{\dagger}=H$	matrices $A^{\dagger}=-A$ and traceless hermitian matrices $H^{\dagger}=H$:																								
gilmore-lie-groups_ F08958	99	1.000	$(H,H)=0 \Rightarrow H=0$	$(H,H)=0 \Rightarrow H=0$	Show that $(A,A)=0 \Rightarrow A=0$ and $(H,H)=0 \Rightarrow H=0$.																								
gilmore-lie-groups_ F08959	99	1.000	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	$\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$	11. Assume that $\mathfrak{g}$ is a Lie algebra with a Cartan decomposition $\mathfrak{g}=\mathfrak{h}+\mathfrak{p}$, with commu-																								
gilmore-lie-groups_ F08960	100	1.000	$\mathfrak{g}^{\prime}=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}^{\prime}$	$\mathfrak{g}^{\prime}=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}^{\prime}$	defined by $\mathfrak{g}=\mathfrak{h}+i\mathfrak{p}=\mathfrak{h}+\mathfrak{p}$ then																								
gilmore-lie-groups_ F08961	100	1.000	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	$\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$	nion algebra $\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$ with respect to the subalgebra $\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$.																								
gilmore-lie-groups_ F08962	100	1.000	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	$\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$	nion algebra $\mathfrak{g}=\mathfrak{sl}(n;\mathbb{Q})$ with respect to the subalgebra $\mathfrak{h}=\mathfrak{sl}(n;\mathbb{C})$.																								
gilmore-lie-groups_ F08963	100	0.817	μ	μ	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
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gilmore-lie-groups_ F08964	100	0.817	σ	σ	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
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gilmore-lie-groups_ F08965	100	0.817	$\mathfrak{so}(3,2)$	$\mathfrak{so}(3,2)$	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
μ	σ	Algebra	Singular subspace																										
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gilmore-lie-groups_ F08966	100	0.817	$\mathfrak{so}(4,1)$	$\mathfrak{so}(4,1)$	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
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gilmore-lie-groups_ F08967	100	0.817	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
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gilmore-lie-groups_ F08968	100	0.817	t_{μ}	t_{μ}	<table><tr><th>μ</th><th>σ</th><th>Algebra</th><th>Singular subspace</th></tr><tr><td>+1</td><td>+1</td><td>$\mathfrak{so}(3,2)$</td><td></td></tr><tr><td>-1</td><td>+1</td><td>$\mathfrak{so}(4,1)$</td><td></td></tr><tr><td>-1</td><td>-1</td><td>$\mathfrak{so}(5)$</td><td></td></tr><tr><td>+1</td><td>0</td><td>Poincare</td><td>translations t_{μ}</td></tr><tr><td>0</td><td>0</td><td>Galilei</td><td>translations t_{μ}, boosts $\mathbf{b}$</td></tr></table>	μ	σ	Algebra	Singular subspace	+1	+1	$\mathfrak{so}(3,2)$		-1	+1	$\mathfrak{so}(4,1)$		-1	-1	$\mathfrak{so}(5)$		+1	0	Poincare	translations t_{μ}	0	0	Galilei	translations t_{μ} , boosts $\mathbf{b}$
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gilmore-lie-groups-F00969	100	■ 1.000	$\mathfrak{g}=A$	$\mathfrak{g} = A$	14. Assume that $\mathfrak{g} = A$, where A is a Lie algebra of $n \times n$ matrices on which the in-
gilmore-lie-groups-F00970	100	■ 1.000	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	$\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$	ner product is negative-definite: $\operatorname{tr} A^2 < 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups-F00971	100	■ 1.000	$\operatorname{EXP}(tA)$	$\operatorname{EXP}(tA)$	ner product is negative-definite: $\operatorname{tr} A^2 \leq 0, = 0 \Rightarrow A = 0$. Then show that $\operatorname{EXP}(tA)$
gilmore-lie-groups-F00972	100	■ 1.000	t	t	returns to any neighborhood of the identity I_n if t becomes large enough. If the eigen-
gilmore-lie-groups-F00973	100	■ 1.000	$\lambda_i = \gamma n_i, n_i$	$\lambda_i = \gamma n_i, n_i$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups-F00974	100	■ 1.000	$\gamma \neq 0$	$\gamma \neq 0$	values of A are rationally related ($\lambda_i = \gamma n_i$, n_i are integers, $\gamma \neq 0$ is rational or
gilmore-lie-groups-F00975	100	■ 1.000	t_0	t_0	irrational), $\operatorname{EXP}(tA)$ returns periodically to I_n . What is t_0 , the minimum period in t ?
gilmore-lie-groups-F00976	100	■ 0.979	$\mathfrak{so}(3)$	$\mathfrak{so}(3)$	15. Use the parameterization of $\mathfrak{so}(3)$ given in Problem 3.14. Show that the differentials
gilmore-lie-groups-F00977	100	■ 1.000	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$	$(\delta x, \delta y, \delta z)$ in the neighborhood of the identity by
gilmore-lie-groups-F00978	100	■ 1.000	$g(x, y, z)$	$g(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups-F00979	100	■ 1.000	$d\mu(x, y, z)$	$d\mu(x, y, z)$	metric tensor $g(x, y, z)$ and the invariant measure $d\mu(x, y, z)$ on the group $SO(3)$
gilmore-lie-groups-F00980	101	■ 1.000	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	$(\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p})$	exist in the algebra $\mathfrak{g} \cap \mathfrak{h} = \mathfrak{h}, \mathfrak{g} \cap \mathfrak{p} = \mathfrak{p}$,
gilmore-lie-groups-F00981	102	■ 0.987	$b_i^\dagger$	$b_i^\dagger$	a set of n independent boson creation ($b_i^\dagger$) and annihilation (b_j) operators that obey
gilmore-lie-groups-F00982	102	■ 0.987	b_j	b_j	a set of n independent boson creation ($b_i^\dagger$) and annihilation (b_j) operators that obey
gilmore-lie-groups-F00983	102	■ 0.971	$[b_i, b_j], [b_i^\dagger, b_j^\dagger], [b_i, I], [b_j^\dagger, I]$	$[b_i, b_j], [b_i^\dagger, b_j^\dagger], [b_i, I], [b_j^\dagger, I]$	with all other commutators (e.g., $[b_i, b_j], [b_i^\dagger, b_j^\dagger], [b_i, I], [b_j^\dagger, I]$) equal to zero.
gilmore-lie-groups-F00984	102	■ 1.000	$\mathcal{A}$	$\mathcal{A}$	matrix A the operator $\mathcal{A}$ that is a linear combination of creation and annihilation
gilmore-lie-groups-F00985	103	■ 1.000	$[A, B] = C$	$[A, B] = C$	where $[A, B] = C$. This argument is invertible. An algebra of operators bilinear in
gilmore-lie-groups-F00986	103	■ 0.999	$2n+1$	$2n+1$	Remark The $2n+1$ operators $b_i, b_j^\dagger, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups-F00987	103	■ 0.999	$b_i, b_j^\dagger, I (1 \leq i, j \leq n)$	$b_i, b_j^\dagger, I (1 \leq i, j \leq n)$	Remark The $2n+1$ operators $b_i, b_j^\dagger, I$ ($1 \leq i, j \leq n$) span the Heisenberg
gilmore-lie-groups-F00988	103	■ 1.000	X_{ij}	X_{ij}	Any set of operators X_{ij} that satisfies isomorphic commutation relations
gilmore-lie-groups-F00989	103	■ 1.000	$b_i^\dagger b_j$	$b_i^\dagger b_j$	can be used in place of the bilinear combinations $b_i^\dagger b_j$:
gilmore-lie-groups-F00990	103	■ 0.970	$f_i^\dagger$	$f_i^\dagger$	creation ($f_i^\dagger$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups-F00991	103	■ 0.970	f_j	f_j	creation ($f_i^\dagger$) and annihilation (f_j) operators for n independent modes. These oper-
gilmore-lie-groups-F00992	103	■ 1.000	$\{f_i, f_j\}, \{f_i^\dagger, f_j^\dagger\}$	$\{f_i, f_j\}, \{f_i^\dagger, f_j^\dagger\}$	with all other bilinear anticommutators (e.g., $\{f_i, f_j\}, \{f_i^\dagger, f_j^\dagger\}$) equal to zero. Bi-
gilmore-lie-groups-F00993	104	■ 1.000	$SU(n)$	$SU(n)$	subgroup $SU(n)$. The mapping to a boson operator algebra greatly simplifies the

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gilmore-lie-groups_ F00994	104	■ 1.000	$\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$	$\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$	where $\mathcal{L}_1 = x_2 \partial_3 - x_3 \partial_2, \mathcal{L}_2 = x_3 \partial_1 - x_1 \partial_3, \mathcal{L}_3 = x_1 \partial_2 - x_2 \partial_1$. The two algebras
gilmore-lie-groups_ F00995	104	■ 0.705	L_i	L_i	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_ F00996	104	■ 0.705	$\mathcal{L}_i$	$\mathcal{L}_i$	where L_i are 3×3 matrices and $\mathcal{L}_i$ are first order differential operators.
gilmore-lie-groups_ F00997	105	■ 1.000	$E(2) = ISO(2)$	$E(2) = ISO(2)$	As another example, we treat the Lie algebra for the group $E(2) = ISO(2)$ of
gilmore-lie-groups_ F00998	105	■ 1.000	$x-y$	$x-y$	rigid motions (translations and rotations) in the $x-y$ plane, whose matrix algebra
gilmore-lie-groups_ F00999	105	■ 1.000	t_1	t_1	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_ F01000	105	■ 1.000	t_2	t_2	angle θ and displacements in the x and y directions by t_1 and t_2 . The associated
gilmore-lie-groups_ F01001	105	■ 1.000	$\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$	$\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$	where $\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$ and $\mathcal{T}_i = \partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_ F01002	105	■ 1.000	$\mathcal{T}_i = \partial_i$	$\mathcal{T}_i = \partial_i$	where $\mathcal{L}_z = x_1 \partial_2 - x_2 \partial_1$ and $\mathcal{T}_i = \partial_i$. The matrix algebra and operator algebra have
gilmore-lie-groups_ F01003	105	■ 0.867	(a, b)	(a, b)	the real line parameterized by points (a, b) in R^2 as follows
gilmore-lie-groups_ F01004	105	■ 1.000	$x(p)$	$x(p)$	is chosen a coordinate, $x(p)$, can be introduced and the function can be written
gilmore-lie-groups_ F01005	105	■ 1.000	$S^{\prime}$	$S^{\prime}$	If a new coordinate system $S^{\prime}$ is chosen, the value of the function at p remains
gilmore-lie-groups_ F01006	105	■ 1.000	$p, x^{\prime}(p)$	$p, x^{\prime}(p)$	unchanged but the new coordinate of $p, x^{\prime}(p)$, is different. Therefore the functions
gilmore-lie-groups_ F01007	105	■ 1.000	f_S	f_S	f_S and $f_{S^{\prime}}$ must be different. We ask: how is $f_{S^{\prime}}$ related to f_S ?
gilmore-lie-groups_ F01008	105	■ 1.000	$f_{S^{\prime}}$	$f_{S^{\prime}}$	f_S and $f_{S^{\prime}}$ must be different. We ask: how is $f_{S^{\prime}}$ related to f_S ?
gilmore-lie-groups_ F01009	105	■ 1.000	$x^{\prime}(p)$	$x^{\prime}(p)$	To answer this question, assume $x^{\prime}(p)$ and $x(p)$ are related by an infinitesimal
gilmore-lie-groups_ F01010	106	■ 1.000	$G(x, y)$	$G(x, y)$	As a second example we consider functions $G(x, y)$ defined on the plane R^2 that
gilmore-lie-groups_ F01011	106	■ 1.000	$(x^{\prime}, y^{\prime})$	$(x^{\prime}, y^{\prime})$	where $(x^{\prime}, y^{\prime})$ and (x, y) are related by
gilmore-lie-groups_ F01012	107	■ 1.000	$\mathcal{A} = \sum_i \sum_j A_{ij} X_{ij}$	$\mathcal{A} = \sum_i \sum_j A_{ij} X_{ij}$	ciating to each matrix A in a matrix Lie algebra an operator $\mathcal{A} = \sum_i \sum_j A_{ij} X_{ij}$.
gilmore-lie-groups_ F01013	107	■ 1.000	$[A, B] = C$	$[A, B] = C \Leftrightarrow [A, B] = C$	$[A, B] = C \Leftrightarrow [A, B] = C$. Under these conditions, complicated commutators in
gilmore-lie-groups_ F01014	107	■ 1.000	$e^A e^B = e^D$	$e^A e^B = e^D \Leftrightarrow e^A e^B = e^D$	with a little care: $e^A e^B = e^D \Leftrightarrow e^A e^B = e^D$.
gilmore-lie-groups_ F01015	107	■ 1.000	$a_i^{\dagger} a_j, 1 \leq i, j \leq 2$	$a_i^{\dagger} a_j, 1 \leq i, j \leq 2$	generate a four-dimensional Lie algebra with basis vectors $a_i^{\dagger} a_j, 1 \leq i, j \leq 2$.
gilmore-lie-groups_ F01016	107	■ 1.000	$\hat{n} = a_1^{\dagger} a_1 + a_2^{\dagger} a_2$	$\hat{n} = a_1^{\dagger} a_1 + a_2^{\dagger} a_2$	a. Show that $\hat{n} = a_1^{\dagger} a_1 + a_2^{\dagger} a_2$ commutes with all the operators in this set.
gilmore-lie-groups_ F01017	107	■ 1.000	$\hat{n}$	$\hat{n}$	b. If $\hat{n}$ is chosen as one basis vector in this four-dimensional space, the remaining
gilmore-lie-groups_ F01018	107	■ 1.000	$a_1^{\dagger} a_1 - a_2^{\dagger} a_2, a_1^{\dagger} a_2, a_2^{\dagger} a_1$	$a_1^{\dagger} a_1 - a_2^{\dagger} a_2, a_1^{\dagger} a_2, a_2^{\dagger} a_1$	three operators can be chosen as $a_1^{\dagger} a_1 - a_2^{\dagger} a_2, a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$. Construct their
gilmore-lie-groups_ F01019	107	■ 1.000	$a_2^{\dagger} a_1$	$a_2^{\dagger} a_1$	three operators can be chosen as $a_1^{\dagger} a_1 - a_2^{\dagger} a_2, a_1^{\dagger} a_2$, and $a_2^{\dagger} a_1$. Construct their

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gilmore-lie-groups_ F01020	108	■ 0.999	$\frac{1}{2}\left(a_1^\dagger a_1 - a_2^\dagger a_2\right), a_1^\dagger a_2$	$\frac{1}{2}\left(a_1^\dagger a_1 - a_2^\dagger a_2\right), a_1^\dagger a_2$	d. Show that the three operators $\frac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2)$, $a_1^\dagger a_2$, and $a_2^\dagger a_1$ satisfy commutation
gilmore-lie-groups_ F01021	108	■ 1.000	$J_z, J_\pm$	$J_z, J_\pm$	$J_z, J_\pm$. In particular, show
gilmore-lie-groups_ F01022	108	■ 1.000	$J^2 = J_x^2 + J_y^2 + J_z^2$	$J^2 = J_x^2 + J_y^2 + J_z^2$	e. Evaluate $J^2 = J_x^2 + J_y^2 + J_z^2$ in terms of the creation and annihilation operators,
gilmore-lie-groups_ F01023	108	■ 1.000	$i(i=1,2)$	$i(i=1,2)$	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01024	108	■ 1.000	$\left n_i\right\rangle$	$ n_i\rangle$	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01025	108	■ 1.000	n_i	n_i	modes. Assume that the quantum state of mode i ($i=1,2$) is $ n_i\rangle$, where n_i is
gilmore-lie-groups_ F01026	108	■ 1.000	$a_i^\dagger$	$a_i^\dagger$	operators $a_i^\dagger$ and a_i act on state $ n_i\rangle$ in the usual way:
gilmore-lie-groups_ F01027	108	■ 1.000	a_i	a_i	operators $a_i^\dagger$ and a_i act on state $ n_i\rangle$ in the usual way:
gilmore-lie-groups_ F01028	108	■ 1.000	$\left n_1\right\rangle \otimes\left n_2\right\rangle=\left n_1, n_2\right\rangle$	$ n_1\rangle \otimes n_2\rangle = n_1, n_2\rangle$	Choose as a set of basis vectors the direct product states $ n_1\rangle \otimes n_2\rangle \equiv n_1, n_2\rangle$. Define
gilmore-lie-groups_ F01029	108	■ 1.000	$\left n_1, n_2\right\rangle= j m\rangle$	$ n_1, n_2\rangle = jm\rangle$	a. Identify the lattice sites in Fig. 6.1 with the states $ n_1, n_2\rangle \equiv jm\rangle$, the diagonal
gilmore-lie-groups_ F01030	108	■ 0.994	$\frac{1}{2}\left(a_1^\dagger a_1 - a_2^\dagger a_2\right)$	$\frac{1}{2}\left(a_1^\dagger a_1 - a_2^\dagger a_2\right)$	operator $\frac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2)$ with the operator J_z , and the shift operators $a_1^\dagger a_2, a_2^\dagger a_1$
gilmore-lie-groups_ F01031	108	■ 0.994	J_z	J_z	operator $\frac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2)$ with the operator J_z , and the shift operators $a_1^\dagger a_2, a_2^\dagger a_1$
gilmore-lie-groups_ F01032	108	■ 0.994	$a_1^\dagger a_2, a_2^\dagger a_1$	$a_1^\dagger a_2, a_2^\dagger a_1$	operator $\frac{1}{2}(a_1^\dagger a_1 - a_2^\dagger a_2)$ with the operator J_z , and the shift operators $a_1^\dagger a_2, a_2^\dagger a_1$
gilmore-lie-groups_ F01033	108	■ 1.000	J_+	J_+	with J_+ and J_- .
gilmore-lie-groups_ F01034	108	■ 1.000	J_-	J_-	with J_+ and J_- .
gilmore-lie-groups_ F01035	108	■ 1.000	$a_i^\dagger a_j$	$a_i^\dagger a_j$	b. Show that the four operators $a_i^\dagger a_j$ leave invariant the sum $n_1 + n_2$.
gilmore-lie-groups_ F01036	108	■ 1.000	n_1+n_2	$n_1 + n_2$	b. Show that the four operators $a_i^\dagger a_j$ leave invariant the sum $n_1 + n_2$.
gilmore-lie-groups_ F01037	108	■ 1.000	$J^2 j m\rangle=j(j+1) j m\rangle$	$J^2 jm\rangle = j(j+1) jm\rangle$	c. $J^2 jm\rangle = j(j+1) jm\rangle$.
gilmore-lie-groups_ F01038	108	■ 1.000	$J_z j m\rangle=m j m\rangle$	$J_z jm\rangle = m jm\rangle$	d. $J_z jm\rangle = m jm\rangle$.
gilmore-lie-groups_ F01039	108	■ 1.000	$J_+ j m\rangle=a_1^\dagger a_2\left n_1, n_2\right\rangle=\sqrt{n_1+1} \sqrt{n_2}\left n_1+1, n_2-1\right\rangle=$	$J_+ jm\rangle = a_1^\dagger a_2 n_1, n_2\rangle = \sqrt{n_1+1} \sqrt{n_2} n_1+1, n_2-1\rangle =$	e. $J_+ jm\rangle = a_1^\dagger a_2 n_1, n_2\rangle = \sqrt{n_1+1} \sqrt{n_2} n_1+1, n_2-1\rangle =$
gilmore-lie-groups_ F01040	108	■ 0.999	$ j, m+1\rangle \sqrt{j+m+1} \sqrt{j-m}$	$ j, m+1\rangle \sqrt{j+m+1} \sqrt{j-m}$	$ j, m+1\rangle \sqrt{j+m+1} \sqrt{j-m}$.
gilmore-lie-groups_ F01041	108	■ 1.000	$J_- j m\rangle=a_2^\dagger a_1\left n_1, n_2\right\rangle=\sqrt{n_1} \sqrt{n_2+1}\left n_1-1, n_2+1\right\rangle=$	$J_- jm\rangle = a_2^\dagger a_1 n_1, n_2\rangle = \sqrt{n_1} \sqrt{n_2+1} n_1-1, n_2+1\rangle =$	f. $J_- jm\rangle = a_2^\dagger a_1 n_1, n_2\rangle = \sqrt{n_1} \sqrt{n_2+1} n_1-1, n_2+1\rangle =$
gilmore-lie-groups_ F01042	108	■ 0.999	$ j, m-1\rangle \sqrt{j+m} \sqrt{j-m+1}$	$ j, m-1\rangle \sqrt{j+m} \sqrt{j-m+1}$	$ j, m-1\rangle \sqrt{j+m} \sqrt{j-m+1}$.
gilmore-lie-groups_ F01043	108	■ 1.000	$J_\pm j m\rangle= j, m \pm 1\rangle \sqrt{(j \pm m+1)(j \mp m)}$	$J_\pm jm\rangle = j, m \pm 1\rangle \sqrt{(j \pm m+1)(j \mp m)}$	g. $J_\pm jm\rangle = j, m \pm 1\rangle \sqrt{(j \pm m+1)(j \mp m)}$. Note that $J_+ j, j\rangle = 0, J_- j, -j\rangle =$
gilmore-lie-groups_ F01044	108	■ 1.000	$J_+ j, j\rangle=0, J_- j,-j\rangle=$	$J_+ j, j\rangle = 0, J_- j, -j\rangle =$	g. $J_\pm jm\rangle = j, m \pm 1\rangle \sqrt{(j \pm m+1)(j \mp m)}$. Note that $J_+ j, j\rangle = 0, J_- j, -j\rangle =$

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gilmore-lie-groups_ F01045	108	■ 1.000	$\left\langle j^{\prime} m^{\prime} \left J_{\pm} \right j m \right\rangle = \sqrt{\left(j^{\prime} \pm m^{\prime} \right) \left(j \mp m \right)} \delta_{j^{\prime} j} \delta_{m^{\prime}, m \pm 1}$		h. $\langle j' m' J_{\pm} j m \rangle = \sqrt{(j' \pm m')(j \mp m)} \delta_{j' j} \delta_{m' m \pm 1}$.
gilmore-lie-groups_ F01046	108	■ 0.999	$\mathfrak{u}(3)$	$\mathfrak{u}(3)$	3. Basis vectors in the Lie algebra $\mathfrak{u}(3)$ for the group $U(3)$ have commutation relations
gilmore-lie-groups_ F01047	108	■ 0.999	$U(3)$	$U(3)$	3. Basis vectors in the Lie algebra $\mathfrak{u}(3)$ for the group $U(3)$ have commutation relations
gilmore-lie-groups_ F01048	108	■ 1.000	$1 \leq i, j \leq 3$	$1 \leq i, j \leq 3$	$1 \leq i, j \leq 3$. Choose a set of basis vectors for a matrix representation of this algebra of
gilmore-lie-groups_ F01049	108	■ 1.000	$\left n_1, n_2, n_3 \right\rangle = \left n_1 \right\rangle \otimes \left n_2 \right\rangle \otimes \left n_3 \right\rangle$	$ n_1, n_2, n_3\rangle = n_1\rangle \otimes n_2\rangle \otimes n_3\rangle$	the form $ n_1, n_2, n_3\rangle = n_1\rangle \otimes n_2\rangle \otimes n_3\rangle$, where for example $b_i n_i\rangle = n_i - 1\rangle \sqrt{n_i}$,
gilmore-lie-groups_ F01050	108	■ 1.000	$b_i \left n_i \right\rangle = \left n_i - 1 \right\rangle \sqrt{n_i}$	$b_i n_i\rangle = n_i - 1\rangle \sqrt{n_i}$	the form $ n_1, n_2, n_3\rangle = n_1\rangle \otimes n_2\rangle \otimes n_3\rangle$, where for example $b_i n_i\rangle = n_i - 1\rangle \sqrt{n_i}$,
gilmore-lie-groups_ F01051	109	■ 1.000	$N = \sum_{i=1}^3 n_i$	$N = \sum_{i=1}^3 n_i$	a. Show $N = \sum_{i=1}^3 n_i$ is not changed by the action of any of the nine operators in
gilmore-lie-groups_ F01052	109	■ 1.000	$D = (N+3-1)! / N! (3-$	$D = (N+3-1)! / N! (3-$	b. Show that the dimension, D , of this representation is $D = (N+3-1)! / N! (3-$
gilmore-lie-groups_ F01053	109	■ 0.953	N	N	sum is N (Bose–Einstein counting problem). In higher dimensions (n) replace 3 by
gilmore-lie-groups_ F01054	109	■ 0.953	(n)	(n)	sum is N (Bose–Einstein counting problem). In higher dimensions (n) replace 3 by
gilmore-lie-groups_ F01055	109	■ 0.852	$n \cdot D$	$n \cdot D$	$n \cdot D$ is also the number of monomials of degree N in the Taylor series expansion
gilmore-lie-groups_ F01056	109	■ 0.999	$f(x_1, x_2, \dots, x_n)$	$f(x_1, x_2, \dots, x_n)$	of a function $f(x_1, x_2, \dots, x_n)$ of n variables.
gilmore-lie-groups_ F01057	109	■ 1.000	I_D	I_D	d. Is there some operator in the Lie algebra that maps to the identity matrix, I_D , in
gilmore-lie-groups_ F01058	109	■ 1.000	$\mathcal{O}$	$\mathcal{O}$	What is $\mathcal{O}$?
gilmore-lie-groups_ F01059	109	■ 1.000	$f_i^{\dagger} f_j$	$f_i^{\dagger} f_j$	ators $f_i^{\dagger} f_j$. What is now the dimension of this representation?
gilmore-lie-groups_ F01060	109	■ 1.000	$d, d^{\dagger}$	$d, d^{\dagger}$	5. Construct operators $d, d^{\dagger}$ defined formally from the standard creation and annihilation
gilmore-lie-groups_ F01061	109	■ 0.996	$a, a^{\dagger}$	$a, a^{\dagger}$	operators $a, a^{\dagger}$ as follows:
gilmore-lie-groups_ F01062	110	■ 0.996	$[d, d^{\dagger}] = 1$	$[d, d^{\dagger}] = 1$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01063	110	■ 0.996	$[d, d] = [d^{\dagger}, d^{\dagger}] = 0$	$[d, d] = [d^{\dagger}, d^{\dagger}] = 0$	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01064	110	■ 0.996	AD	AD	$[d, d^{\dagger}] = 1$ and $[d, d] = [d^{\dagger}, d^{\dagger}] = 0$, the four matrix elements must satisfy $AD -$
gilmore-lie-groups_ F01065	110	■ 1.000	$BC = 1$	$BC = 1$	$BC = 1$.
gilmore-lie-groups_ F01066	110	■ 0.998	$\operatorname{Sp}(2; \mathbb{R}) =$	$\operatorname{Sp}(2; \mathbb{R}) =$	b. Argue that the commutation relations are invariant under the group $\operatorname{Sp}(2; \mathbb{R}) =$
gilmore-lie-groups_ F01067	110	■ 0.982	$\operatorname{Sp}(2; \mathbb{R})$	$\operatorname{Sp}(2; \mathbb{R})$	c. Show that under $\operatorname{Sp}(2; \mathbb{R})$, linear combinations of the coordinate and differential
gilmore-lie-groups_ F01068	110	■ 1.000	x, ∂	x, ∂	operators x, ∂ preserve the commutation relations. In particular, show that
gilmore-lie-groups_ F01069	110	■ 0.946	$a_1, a_2, \dots, a_n$	$a_1, a_2, \dots, a_n$	d. Replace a by $(a_1, a_2, \dots, a_n)$ and similarly for $a^{\dagger}$ and their images $d, d^{\dagger}$ under
gilmore-lie-groups_ F01070	110	■ 1.000	$ n_1, n_2, \dots, n_N\rangle$	$ n_1, n_2, \dots, n_N\rangle$	and eigenstates $ n_1, n_2, \dots, n_N\rangle$.

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gilmore-lie-groups_F01071	110	■ 1.000	$\hbar \omega(n +$	$\hbar \omega(n +$	a. Show that the degeneracy of the multiplet containing n quanta, with energy $\hbar \omega(n +$
gilmore-lie-groups_F01072	110	■ 0.531	$\frac{N}{2}$	$\frac{N}{2}$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_F01073	110	■ 0.531	$\operatorname{deg}(N, n) = (n + N - 1)!/n!(N - 1)!$	$\deg(N, n) = (n + N - 1)!/n!(N - 1)!$	$\frac{N}{2}$) is $\deg(N, n) = (n + N - 1)!/n!(N - 1)!$. This solution to the Bose–Einstein
gilmore-lie-groups_F01074	110	■ 0.999	$f(x_1, x_2, \ldots, x_N)$	$f(x_1, x_2, \ldots, x_N)$	Taylor series expansion of a function of N variables: $f(x_1, x_2, \ldots, x_N)$.
gilmore-lie-groups_F01075	110	■ 1.000	N^2	N^2	N^2 operators $a_i^\dagger a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^\dagger a_j] = 0$,
gilmore-lie-groups_F01076	110	■ 1.000	$\mathfrak{u}(N)$	$\mathfrak{u}(N)$	N^2 operators $a_i^\dagger a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^\dagger a_j] = 0$,
gilmore-lie-groups_F01077	110	■ 1.000	$[\mathcal{H}, a_i^\dagger a_j] = 0$	$[\mathcal{H}, a_i^\dagger a_j] = 0$	N^2 operators $a_i^\dagger a_j$. This is isomorphic to the Lie algebra $\mathfrak{u}(N)$. Since $[\mathcal{H}, a_i^\dagger a_j] = 0$,
gilmore-lie-groups_F01078	110	■ 1.000	$\mathfrak{su}(N)$	$\mathfrak{su}(N)$	this algebra is a direct sum of a simple Lie algebra, $\mathfrak{su}(N)$, plus the one-dimensional
gilmore-lie-groups_F01079	110	■ 1.000	$\mathcal{H}$	$\mathcal{H}$	algebra spanned by $\mathcal{H}$.
gilmore-lie-groups_F01080	110	■ 1.000	a_j	a_j	single creation and annihilation operators $a_i^\dagger$ and a_j , as well as their commutator I ,
gilmore-lie-groups_F01081	110	■ 1.000	$(N + 1)^2$	$(N + 1)^2$	the resulting set of operators closes to form an $(N + 1)^2$ dimensional Lie algebra
gilmore-lie-groups_F01082	110	■ 1.000	n'	n'	state in a multiplet with n' excitations.
gilmore-lie-groups_F01083	110	■ 1.000	$R, S, T, U, \ldots$	$R, S, T, U, \ldots$	7. The set of matrices $R, S, T, U, \ldots$ belong to a Lie algebra of $n \times n$ matrices, $a^\dagger =$
gilmore-lie-groups_F01084	110	■ 1.000	$a^\dagger =$	$a^\dagger =$	7. The set of matrices $R, S, T, U, \ldots$ belong to a Lie algebra of $n \times n$ matrices, $a^\dagger =$
gilmore-lie-groups_F01085	110	■ 0.999	$a_1^\dagger, a_2^\dagger, \ldots, a_n^\dagger$	$a_1^\dagger, a_2^\dagger, \ldots, a_n^\dagger$	$(a_1^\dagger, a_2^\dagger, \ldots, a_n^\dagger)$ is a row vector of creation operators for n boson modes, and a is its
gilmore-lie-groups_F01086	111	■ 1.000	$\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$	$\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$	adjoint, a column vector of annihilation operators. Define $\mathcal{R} = a^\dagger R a = a_i^\dagger R_{ij} a_j$, and
gilmore-lie-groups_F01087	111	■ 1.000	$\mathcal{S}, T, U, \ldots$	$\mathcal{S}, T, U, \ldots$	similarly for $\mathcal{S}, T, U, \ldots$.
gilmore-lie-groups_F01088	111	■ 1.000	$[R, S] = T \Leftrightarrow [\mathcal{R}, \mathcal{S}] = \mathcal{T}$	$[R, S] = T \Leftrightarrow [\mathcal{R}, \mathcal{S}] = \mathcal{T}$	a. $[R, S] = T \Leftrightarrow [\mathcal{R}, \mathcal{S}] = \mathcal{T}$
gilmore-lie-groups_F01089	111	■ 1.000	$e^R e^S = e^U \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$	$e^R e^S = e^U \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$	b. $e^R e^S = e^U \Leftrightarrow e^{\mathcal{R}} e^{\mathcal{S}} = e^{\mathcal{U}}$
gilmore-lie-groups_F01090	111	■ 0.994	$\left[\frac{d}{dx}, e^{-x^2/2}\right] = -x e^{-x^2/2}$	$\left[\frac{d}{dx}, e^{-x^2/2}\right] = -x e^{-x^2/2}$	a. Show $\left[\frac{d}{dx}, e^{-x^2/2}\right] = -x e^{-x^2/2}$.
gilmore-lie-groups_F01091	111	■ 1.000	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx}\right)$	$a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx}\right)$	d. Introduce the annihilation operator $a = \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx}\right)$, define the normalized ground
gilmore-lie-groups_F01092	111	■ 1.000	$\langle x 0 \rangle$	$\langle x 0 \rangle$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_F01093	111	■ 1.000	$a \langle x 0 \rangle = 0$	$a \langle x 0 \rangle = 0$	state $\langle x 0 \rangle$ by $a \langle x 0 \rangle = 0$. Solve this equation, normalize the solution, and show
gilmore-lie-groups_F01094	111	■ 0.990	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$	$\langle x 0 \rangle = e^{-x^2/2} / \sqrt{1\sqrt{\pi}}$.
gilmore-lie-groups_F01095	111	■ 1.000	$a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx}\right)$	$a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx}\right)$	e. Introduce the creation operator $a^\dagger = \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx}\right)$ and show
gilmore-lie-groups_F01096	111	■ 1.000	$\psi_n(x)$	$\psi_n(x)$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x n \rangle =$

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gilmore-lie-groups_F01097	111	■ 1.000	$\langle \text{angle } x \mid n \rangle =$	$\langle x \mid n \rangle =$	where $\psi_n(x)$ is the n th normalized harmonic oscillator eigenstate $\langle x \mid n \rangle =$
gilmore-lie-groups_F01098	111	■ 0.999	$\frac{\left(a^{\dagger} \right)^n}{\sqrt{n!}} \langle x \mid 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x \mid 0 \rangle$	$\frac{(a^\dagger)^n}{\sqrt{n!}} \langle x \mid 0 \rangle$.
gilmore-lie-groups_F01099	111	■ 0.749	$L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i$	$L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i$	$(L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i)$.
gilmore-lie-groups_F01100	111	■ 0.749	$Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i$	$Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i$	$(L_{ij} = a_i^{\dagger} a_j - a_j^{\dagger} a_i)$ and a quadrupole interaction $(Q_{ij} = a_i^{\dagger} a_j + a_j^{\dagger} a_i)$.
gilmore-lie-groups_F01101	111	■ 1.000	$i \leq j$	$i \leq j$	for $\leq$ the matrix elements are
gilmore-lie-groups_F01102	111	■ 0.999	$\Gamma_{ji}^* = \Gamma_{ij}$	$\Gamma_{ji}^* = \Gamma_{ij}$	with $\Gamma_{ji}^* = \Gamma_{ij}$.
gilmore-lie-groups_F01103	112	■ 0.992	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i \left(b_i^{\dagger} b_i + \frac{1}{2} \right) +$	$b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i \left(b_i^{\dagger} b_i + \frac{1}{2} \right) +$	annihilation operators $b_i = m_{ij} a_j : H = \sum_{i=1}^n \hbar \omega_i (b_i^{\dagger} b_i + \frac{1}{2}) + \text{constant}$. Express
gilmore-lie-groups_F01104	112	■ 1.000	$\Gamma(H)$	$\Gamma(H)$	the amplitudes m_{ij} in terms of the eigenvectors of $\Gamma(H)$.
gilmore-lie-groups_F01105	114	■ 1.000	$\text{operatorname{EXP}}(X)$	$\text{EXP}(X)$	Lie algebra onto the entire Lie group through a single mapping of the form $\text{EXP}(X)$.
gilmore-lie-groups_F01106	114	■ 1.000	$\text{operatorname{Tr}} X = 0$	$\text{Tr } X = 0$	Now choose S to diagonalize (7.2). Since $\text{Tr } X = 0$, the eigenvalues λ can only
gilmore-lie-groups_F01107	114	■ 1.000	$\pm \theta$	$\pm \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_F01108	114	■ 1.000	$\pm i \theta$	$\pm i \theta$	have the form $\pm \theta$ or $\pm i \theta$ (θ real)
gilmore-lie-groups_F01109	116	■ 0.997	$a = b = 0$	$a = b = 0$	To see this most easily set $a = b = 0$. Points on the c -axis map onto
gilmore-lie-groups_F01110	116	■ 1.000	$2 \pi n$	$2 \pi n$	and therefore points separated by $2 \pi n$ along the c -axis map onto the same group
gilmore-lie-groups_F01111	116	■ 0.651	$S O(2) \subset S L(2 ; \mathbb{R})$	$SO(2) \subset SL(2; \mathbb{R})$	operation in $SO(2) \subset SL(2; \mathbb{R})$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_F01112	116	■ 0.651	$a, b, 0$	$a, b, 0$	operation in $SO(2) \subset SL(2; \mathbb{R})$. The complementary subspace $(a, b, 0)$ maps onto
gilmore-lie-groups_F01113	116	■ 1.000	$z = \cosh r \geq 1$	$z = \cosh r \geq 1$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_F01114	116	■ 1.000	$(x, y) = (b, a) \sinh(r) / r, r^2 = a^2 +$	$(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$	For simplicity we have set $z = \cosh r \geq 1$ and $(x, y) = (b, a) \sinh(r)/r, r^2 = a^2 +$
gilmore-lie-groups_F01115	116	■ 1.000	b^2	b^2	b^2 . We observe that
gilmore-lie-groups_F01116	116	■ 1.000	H_{2+}^2	H_{2+}^2	which is just the upper sheet of the two-sheeted hyperboloid H_{2+}^2 , shown in
gilmore-lie-groups_F01117	116	■ 1.000	$-\pi \leq c \leq +\pi$	$-\pi \leq c \leq +\pi$	$-\pi \leq c \leq +\pi$ is required to map the subalgebra onto the subgroup $SO(2)$.
gilmore-lie-groups_F01118	117	■ 0.999	$S O(1,1)$	$SO(1, 1)$	$SO(2)$ and $SO(1, 1)$.
gilmore-lie-groups_F01119	117	■ 1.000	H_1^2	H_1^2	This equation describes the surface of the single-sheeted hyperboloid H_1^2 , shown
gilmore-lie-groups_F01120	118	■ 1.000	$c, 0 \leq c < 2 \pi$	$c, 0 \leq c < 2 \pi$	set parameterized by $c, 0 \leq c < 2 \pi$, is the coset. In the theory of Lie groups, cosets
gilmore-lie-groups_F01121	118	■ 0.999	$S L(2 ; \mathbb{R}) / S O(2)$	$SL(2; \mathbb{R})/SO(2)$	obtained from the coset decomposition $SL(2; \mathbb{R})/SO(2)$, the manifold parameteriz-
gilmore-lie-groups_F01122	118	■ 0.996	$R^2 \times S^1$	$R^2 \times S^1$	is the direct product $R^2 \times S^1$. A different parameterization (7.13) based on the coset

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gilmore-lie-groups_ F01123	118	■ 1.000	$[S L(2 ; \mathbb{R}) / S O(1,1)] \times S O(1,1) \left(S O(1,1) \simeq R^1 \right)$	$[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$	decomposition $[SL(2; \mathbb{R})/SO(1,1)] \times SO(1,1) (SO(1,1) \simeq R^1)$ shows that the
gilmore-lie-groups_ F01124	118	■ 1.000	$R^1 \times S^1$	$R^1 \times S^1$	boloid (equivalent to $R^1 \times S^1$) with R^1 . This product is once again $R^2 \times S^1$.
gilmore-lie-groups_ F01125	118	■ 1.000	$R^N, N=n^2$	$R^N, N = n^2$	the underlying manifold is a subset of $R^N, N = n^2$, that is defined by algebraic
gilmore-lie-groups_ F01126	119	■ 1.000	R^m	R^m	generators is R^m , for suitable m (m is the number of noncompact generators). The
gilmore-lie-groups_ F01127	119	■ 0.943	$S O(2,1)$	$SO(2,1)$	the groups $SO(2,1)$ and $SU(1,1)$ with Lie algebras
gilmore-lie-groups_ F01128	119	■ 0.993	$S O(2,1) / S O(2)$	$SO(2,1)/SO(2)$	The cosets $SO(2,1)/SO(2)$ and $SU(1,1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01129	119	■ 0.993	$S U(1,1) / U(1)$	$SU(1,1)/U(1)$	The cosets $SO(2,1)/SO(2)$ and $SU(1,1)/U(1)$ are both isomorphic to R^2 and have
gilmore-lie-groups_ F01130	119	■ 0.969	$U(1)$	$U(1)$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01131	119	■ 0.969	$2:1$	$2:1$	a 1:1 correspondence. The subgroups $SO(2)$ and $U(1)$ have a 2:1 correspondence.
gilmore-lie-groups_ F01132	119	■ 0.936	b_3	b_3	This can be seen by increasing b_3 by 2π and noticing that the 2×2 unitary matrix
gilmore-lie-groups_ F01133	119	■ 1.000	$U \left(b_3 + 2\pi \right) = -U \left(b_3 \right)$	$U(b_3 + 2\pi) = -U(b_3)$	in (7.19) goes to its negative: $U(b_3 + 2\pi) = -U(b_3)$. However, increasing a_3 by
gilmore-lie-groups_ F01134	119	■ 1.000	a_3	a_3	in (7.19) goes to its negative: $U(b_3 + 2\pi) = -U(b_3)$. However, increasing a_3 by
gilmore-lie-groups_ F01135	120	■ 0.984	$U(1) \subset S U(1,1)$	$U(1) \subset SU(1,1)$	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01136	120	■ 0.984	4π	4π	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01137	120	■ 0.984	$S O(2) \subset S O(2,1)$	$SO(2) \subset SO(2,1)$	$U(1) \subset SU(1,1)$ the result is 4π ; for the subgroup $SO(2) \subset SO(2,1)$ the result is
gilmore-lie-groups_ F01138	120	■ 0.811	$\operatorname{SO}(2,1)$	$SO(2,1)$	2π . Therefore, $SU(1,1)$ is “twice as large” as $SO(2,1)$. More formally, there is a
gilmore-lie-groups_ F01139	120	■ 0.992	$2 \rightarrow 1$	$2 \rightarrow 1$	$2 \rightarrow 1$ homomorphism of $SU(1,1)$ onto $SO(2,1)$.
gilmore-lie-groups_ F01140	120	■ 1.000	$\sqrt{a_1^2 + a_2^2 + a_3^2} \leq$	$\sqrt{a_1^2 + a_2^2 + a_3^2} \leq$	It can be checked that all points in the interior of a sphere of radius $\sqrt{a_1^2 + a_2^2 + a_3^2} \leq$
gilmore-lie-groups_ F01141	120	■ 0.994	$ \mathbf{a} = \pi$	$ \mathbf{a} = \pi$	π) map onto $SO(3)$ provided antipodal points at $ \mathbf{a} = \pi$ are identified
gilmore-lie-groups_ F01142	120	■ 0.999	$\operatorname{SU}(2)$	$SU(2)$	with θ the latitude, and ϕ the longitude on a sphere. For $SU(2)$ all points within a
gilmore-lie-groups_ F01143	120	■ 0.998	$2\pi \left(\sqrt{b_1^2 + b_2^2 + b_3^2} < 2\pi \right)$	$2\pi \left(\sqrt{b_1^2 + b_2^2 + b_3^2} < 2\pi \right)$	sphere of radius 2π ($\sqrt{b_1^2 + b_2^2 + b_3^2} < 2\pi$) are mapped onto distinct elements of
gilmore-lie-groups_ F01144	120	■ 0.986	$-I_2$	$-I_2$	$SU(2)$ and all points at a radius of 2π are mapped onto $-I_2$. There is an easier way
gilmore-lie-groups_ F01145	121	■ 0.652	$x^2 + y^2 + z^2 = 1$	$x^2 + y^2 + z^2 = 1$	(x, y, z) subject to $x^2 + y^2 + z^2 = 1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_ F01146	121	■ 0.652	$S U(2) / U(1)$	$SU(2)/U(1)$	(x, y, z) subject to $x^2 + y^2 + z^2 = 1$, and $SU(2)/U(1)$, parameterized by the real
gilmore-lie-groups_ F01147	121	■ 0.989	x', y', z'	x', y', z'	numbers (x', y', z') subject to $x'^2 + y'^2 + z'^2 = 1$, are in 1:1 correspondence with
gilmore-lie-groups_ F01148	121	■ 0.989	$x'^2 + y'^2 + z'^2 = 1$	$x'^2 + y'^2 + z'^2 = 1$	numbers (x', y', z') subject to $x'^2 + y'^2 + z'^2 = 1$, are in 1:1 correspondence with

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gilmore-lie-groups_ F01149	121	■ 0.604	$S U(2) \rightarrow S O(3)$	$SU(2) \rightarrow SO(3)$	of the mapping $SU(2) \rightarrow SO(3)$ can be seen from the 2:1 nature of the rotations
gilmore-lie-groups_ F01150	121	■ 1.000	$\bar{G}$	$\bar{G}$	group, $\bar{G}$, called the universal covering group . This group is simply connected:
gilmore-lie-groups_ F01151	121	■ 1.000	$\bar{G} / D$	$\bar{G}/D$	this simply connected Lie group, or else has the form of a quotient $\bar{G}/D$, where D
gilmore-lie-groups_ F01152	121	■ 1.000	$\bar{G}: g d_i = d_i g$	$\bar{G}: g d_i = d_i g$	is a discrete invariant subgroup of $\bar{G}$ whose elements commute with $\bar{G}$: $g d_i = d_i g$
gilmore-lie-groups_ F01153	121	■ 1.000	$d_i \in D$	$d_i \in D$	for $d_i \in D$ and $g \in G$. If $\bar{G}$ is compact it is useful to determine the largest such
gilmore-lie-groups_ F01154	121	■ 0.863	D_{MAX}	D_{MAX}	subgroup, D_{MAX} , of $\bar{G}$. Then all compact Lie groups with the same Lie algebra as
gilmore-lie-groups_ F01155	121	■ 0.979	D_{MAX}	D_{MAX}	$\bar{G}$ are obtained by “dividing” $\bar{G}$ by all possible subgroups of D_{MAX} , as shown in
gilmore-lie-groups_ F01156	121	■ 1.000	d_i	d_i	D is a simple matter. The only discrete group operations d_i that commute with all
gilmore-lie-groups_ F01157	121	■ 1.000	$G_1 = \bar{G} / D_1$	$G_1 = \bar{G}/D_1$	G_2 have the same Lie algebra, $G_1 = \bar{G}/D_1$ and G_1 is locally isomorphic with $\bar{G}$.
gilmore-lie-groups_ F01158	121	■ 0.848	$\bar{G} / D_{\mathrm{MAX}}$	$\bar{G}/D_{\mathrm{MAX}}$	G_1 . If $\bar{G}$ is compact, G_1 and G_2 are also locally isomorphic with $\bar{G}/D_{\mathrm{MAX}}$, which
gilmore-lie-groups_ F01159	122	■ 1.000	$SG = \bar{G}$	$SG = \bar{G}$	Lie algebras $\mathfrak{g}$ and simply connected Lie groups $SG = \bar{G}$. Every other Lie group
gilmore-lie-groups_ F01160	122	■ 1.000	D_i	D_i	discrete invariant subgroups D_i of $\bar{G}$.
gilmore-lie-groups_ F01161	122	■ 1.000	λI_2	λI_2	λI_2 that obey $\lambda^* \lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01162	122	■ 1.000	$\lambda^* \lambda = 1$	$\lambda^* \lambda = 1$	λI_2 that obey $\lambda^* \lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01163	122	■ 1.000	$\det(\lambda I_2) = +1$	$\det(\lambda I_2) = +1$	λI_2 that obey $\lambda^* \lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01164	122	■ 1.000	$\lambda = \pm 1 \cdot D$	$\lambda = \pm 1 \cdot D$	λI_2 that obey $\lambda^* \lambda = 1$ and $\det(\lambda I_2) = +1$, so that $\lambda = \pm 1$. D is the two-element
gilmore-lie-groups_ F01165	122	■ 1.000	$D = \{I_2, -I_2\}$	$D = \{I_2, -I_2\}$	subgroup $D = \{I_2, -I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D = \lambda I_3$
gilmore-lie-groups_ F01166	122	■ 1.000	$SO(3), D = \lambda I_3$	$SO(3), D = \lambda I_3$	subgroup $D = \{I_2, -I_2\}$. For the locally isomorphic Lie group $SO(3)$, $D = \lambda I_3$
gilmore-lie-groups_ F01167	122	■ 0.874	$\lambda = +1$	$\lambda = +1$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group
gilmore-lie-groups_ F01168	122	■ 0.874	$S U(2) / \{I_2, -I_2\} = S O(3) / I_3 = S O(3)$	$SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$	with $\lambda = +1$. As a result $SU(2)/\{I_2, -I_2\} = SO(3)/I_3 = SO(3)$. For each group
gilmore-lie-groups_ F01169	122	■ 1.000	$\overline{SO(2,1)} =$	$\overline{SO(2,1)} =$	is R^1 , the group of translations of the line. The covering group $\overline{SO(2,1)} =$
gilmore-lie-groups_ F01170	122	■ 1.000	$\overline{SU(1,1)}$	$\overline{SU(1,1)}$	$\overline{SU(1,1)}$ has no compact subgroup at all. Its underlying group manifold
gilmore-lie-groups_ F01171	122	■ 1.000	$\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$	$\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$	is $\overline{SO(2,1)/SO(2)} \times \overline{SO(2)} = \overline{SU(1,1)/U(1)} \times \overline{U(1)} = [SO(2,1)/SO(2)] \times$
gilmore-lie-groups_ F01172	122	■ 1.000	$\overline{SO(2)} = SU(1,1)/U(1) \times \overline{U(1)} = R^2 \times R^1$	$\overline{SO(2)} = SU(1,1)/U(1) \times \overline{U(1)} = R^2 \times R^1$	$\overline{SO(2)} = SU(1,1)/U(1) \times \overline{U(1)} = R^2 \times R^1$. It is the only group we will encounter
gilmore-lie-groups_ F01173	122	■ 1.000	$\overline{SO(2,1)} = \overline{SU(1,1)}$	$\overline{SO(2,1)} = \overline{SU(1,1)}$	in this book that is not a matrix group. The covering group $\overline{SO(2,1)} = \overline{SU(1,1)}$ has
gilmore-lie-groups_ F01174	123	■ 1.000	R_+^2	R_+^2	right half-plane R_+^2 into the group operator

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gilmore-lie-groups_F01175	123	■ 0.974	(w, z)	(w, z)	The second maps a point (w, z) in R^2 into the group under the EXPonential map
gilmore-lie-groups_F01176	123	■ 1.000	$R_{+}^2(x > 0, y)$	$R_{+}^2(x > 0, y)$	We ask: is there some mapping of the half-plane $R_{+}^2(x > 0, y)$ into $R^2(w, z)$
gilmore-lie-groups_F01177	123	■ 1.000	$R^2(w, z)$	$R^2(w, z)$	We ask: is there some mapping of the half-plane $R_{+}^2(x > 0, y)$ into $R^2(w, z)$
gilmore-lie-groups_F01178	123	■ 0.995	$x > 0$	$x > 0$	These transformations are analytic for $x > 0$.
gilmore-lie-groups_F01179	124	■ 0.981	$a, a^{\dagger}, I$	$a, a^{\dagger}, I$	The single-mode photon operators $a, a^{\dagger}, I$ obey isomorphic commutation relations
gilmore-lie-groups_F01180	124	■ 1.000	$l=l^{\prime \prime}, r=r^{\prime \prime}, \delta^{\prime}=\delta+\frac{1}{2} l r=\delta^{\prime \prime}+l^{\prime \prime} r^{\prime \prime}$	$l=l^{\prime \prime}, r=r^{\prime \prime}, \delta^{\prime}=\delta+\frac{1}{2} l r=\delta^{\prime \prime}+l^{\prime \prime} r^{\prime \prime}$	We see immediately that $\boxed{=l^{\prime}=l^{\prime \prime}, r=r^{\prime}=r^{\prime \prime}, \delta^{\prime}=\delta+\frac{1}{2} l r=\delta^{\prime \prime}+l^{\prime \prime} r^{\prime \prime}}$, and
gilmore-lie-groups_F01181	124	■ 1.000	$\delta=0$	$\delta=0$	obtain the Heisenberg identity (for $\delta=0$)
gilmore-lie-groups_F01182	125	■ 1.000	$\hat{n}=a^{\dagger} a, a, a^{\dagger}, I$	$\hat{n}=a^{\dagger} a, a, a^{\dagger}, I$	photon operators $\boxed{\hat{n}=a^{\dagger} a, a, a^{\dagger}, I}$ under the identification
gilmore-lie-groups_F01183	125	■ 1.000	$\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$	$\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$	If for some reason $\operatorname{EXP}\left(\eta a^{\dagger} a+r a^{\dagger}+l a\right)$ needed to be rewritten in the more con-
gilmore-lie-groups_F01184	125	■ 1.000	$\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$	$\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$	veniently ordered form $\operatorname{EXP}\left(r^{\prime} a^{\dagger}\right) \operatorname{EXP}\left(\eta^{\prime} a^{\dagger} a+\delta^{\prime} I\right) \operatorname{EXP}\left(l^{\prime} a\right)$, then the reparame-
gilmore-lie-groups_F01185	125	■ 0.948	$e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	$e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $	Since $e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $ and $e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$, the expectation value is
gilmore-lie-groups_F01186	125	■ 0.948	$e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$	$e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$	Since $e^{l^{\prime} a} 0\rangle= 0\rangle,\langle 0 e^{r^{\prime} a^{\dagger}}=\langle 0 $ and $e^{\eta^{\prime} a^{\dagger} a} 0\rangle= 0\rangle$, the expectation value is
gilmore-lie-groups_F01187	125	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	As a final example we treat the Lie algebra $\mathfrak{su}(2)$. First, we show how to compute
gilmore-lie-groups_F01188	125	■ 1.000	$(j,-j\rangle)$	$(j,-j\rangle)$	$(j,-j\rangle)$
gilmore-lie-groups_F01189	126	■ 1.000	$\theta_z^{\prime}$	$\theta_z^{\prime}$	The only problem that remains is to compute $\theta_z^{\prime}$ as a function of θ . To do this
gilmore-lie-groups_F01190	126	■ 1.000	$\mathbf{J} \rightarrow \frac{1}{2} \sigma$	$\mathbf{J} \rightarrow \frac{1}{2} \sigma$	representation $\mathbf{J} \rightarrow \frac{1}{2} \sigma$, where σ are the Pauli spin matrices (5.14):
gilmore-lie-groups_F01191	126	■ 1.000	$\theta_{\pm}=\theta_1 \pm i \theta_2$	$\theta_{\pm}=\theta_1 \pm i \theta_2$	where $\boxed{\theta_{\pm}=\theta_1 \pm i \theta_2}$. Comparison of the two matrices gives immediately
gilmore-lie-groups_F01192	127	■ 1.000	$e^{\alpha J_{+}} e^{\beta J_{-}}$	$e^{\alpha J_{+}} e^{\beta J_{-}}$	The operator product $e^{\alpha J_{+}} e^{\beta J_{-}}$ is written in normally ordered form $\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$
gilmore-lie-groups_F01193	127	■ 1.000	$\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$	$\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$	The operator product $e^{\alpha J_{+}} e^{\beta J_{-}}$ is written in normally ordered form $\operatorname{EXP}\left(\beta^{\prime} J_{-}\right)$
gilmore-lie-groups_F01194	127	■ 0.999	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_F01195	127	■ 0.999	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\alpha^{\prime}, \beta^{\prime}, n^{\prime}$	$\operatorname{EXP}\left(n^{\prime} J_z\right) \operatorname{EXP}\left(\alpha^{\prime} J_{+}\right)$ and the parameters $\alpha^{\prime}, \beta^{\prime}, n^{\prime}$ computed. We find
gilmore-lie-groups_F01196	127	■ 1.000	$(1+\alpha \beta)^{2 j}$	$(1+\alpha \beta)^{2 j}$	By expanding $(1+\alpha \beta)^{2 j}$ and invoking analyticity, we find
gilmore-lie-groups_F01197	127	■ 1.000	$\mathcal{A}, \mathcal{B}$	$\mathcal{A}, \mathcal{B}$	For example, if operators $\mathcal{A}, \mathcal{B}$ belong to two complementary subspaces in some
gilmore-lie-groups_F01198	127	■ 0.995	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	operator Lie algebra $\mathfrak{g}$ then the operator product $e^{\mathcal{A}} e^{\mathcal{B}}$ can be reparameterized as

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gilmore-lie-groups_ F01199	127	■ 0.999	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}} \left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime} \right)$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}} (\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}} (\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01200	127	■ 0.999	$\left(\mathcal{A}^{\prime}, \mathcal{B}^{\prime} \right)$	$\mathcal{A}, \mathcal{B}$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}} (\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$ different operators in the same subspaces as $\mathcal{A}, \mathcal{B}$ by
gilmore-lie-groups_ F01201	127	■ 1.000	$e^{\mathcal{A}} e^{\mathcal{B}}$	$e^{\mathcal{A}} e^{\mathcal{B}}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$,
gilmore-lie-groups_ F01202	127	■ 1.000	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$	$e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$	(iii) Carrying out the matrix calculations $e^{\mathcal{A}} e^{\mathcal{B}}$ and $e^{\mathcal{B}^{\prime}} e^{\mathcal{A}^{\prime}}$,
gilmore-lie-groups_ F01203	127	■ 1.000	$\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$	$\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$	(iv) determining the matrices $\mathcal{A}^{\prime}, \mathcal{B}^{\prime}$ by comparing matrix elements; and
gilmore-lie-groups_ F01204	127	■ 1.000	$\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$	$\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$	(v) using the isomorphism $\mathcal{A}^{\prime} \leftrightarrow \mathcal{A}^{\prime} \mathcal{B}^{\prime} \leftrightarrow \mathcal{B}^{\prime}$.
gilmore-lie-groups_ F01205	127	■ 1.000	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	$(\mathcal{A}, \mathcal{B}) \leftrightarrow$	This procedure will produce a local analytic reparameterization $(\mathcal{A}, \mathcal{B}) \leftrightarrow$
gilmore-lie-groups_ F01206	127	■ 0.998	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$	$(\mathcal{A}^{\prime}, \mathcal{B}^{\prime})$. If the matrix group used to construct this reparameterization is simply
gilmore-lie-groups_ F01207	127	■ 1.000	$\mathcal{A}, \mathcal{B}, \dots$	$\mathcal{A}, \mathcal{B}, \dots$	groups of the operator group and the matrix group. When the operators $\mathcal{A}, \mathcal{B}, \dots$ are
gilmore-lie-groups_ F01208	128	■ 1.000	$\mathcal{A}, \mathcal{B}, \dots$	$\mathcal{A}, \mathcal{B}, \dots$	related to matrices $\mathcal{A}, \mathcal{B}, \dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01209	128	■ 1.000	$\mathcal{A} \leftrightarrow A$	$\mathcal{A} \leftrightarrow A$	related to matrices $\mathcal{A}, \mathcal{B}, \dots$ by a matrix–operator mapping (see Chapter 6) $\mathcal{A} \leftrightarrow A$,
gilmore-lie-groups_ F01210	128	■ 1.000	$t + \delta t$	$t + \delta t$	The state of the system at time $t + \delta t$ is related to the state at time t by
gilmore-lie-groups_ F01211	128	■ 1.000	$ \psi(t_f)\rangle$	$ \psi(t_f)\rangle$	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01212	128	■ 1.000	t_f	t_f	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01213	128	■ 1.000	$ \psi(t_f)\rangle =$	$ \psi(t_f)\rangle =$	$ \psi(t_f)\rangle$ at some final time t_f is related to the state at initial time t_i by $ \psi(t_f)\rangle =$
gilmore-lie-groups_ F01214	128	■ 0.996	$U(t_f, t_i) \psi(t_i)\rangle$	$U(t_f, t_i) \psi(t_i)\rangle$	$U(t_f, t_i) \psi(t_i)\rangle$. The finite time unitary operator is built up from small dis-
gilmore-lie-groups_ F01215	128	■ 1.000	$H(t')$	$H(t')$	Care must be taken with the formal integration in this equation, as in general $H(t')$
gilmore-lie-groups_ F01216	128	■ 1.000	$H(t), t' \neq t$	$H(t), t' \neq t$	does not commute with $H(t)$, $t' \neq t$. It is for this reason that the symbol “ T ”
gilmore-lie-groups_ F01217	129	■ 1.000	$2j + 1$	$2j + 1$	Example 1. A Hamiltonian acts in a $2j + 1$ dimensional space through a set of
gilmore-lie-groups_ F01218	129	■ 1.000	$ j, m_j\rangle$	$ j, m_j\rangle$	wish to determine the evolution of some particular state $ j, m_j\rangle$. The Hamiltonian
gilmore-lie-groups_ F01219	129	■ 1.000	$g(t) \in SU(2)$	$g(t) \in SU(2)$	$g(t) \in SU(2)$ evolves, and then construct its unitary representation, than it is to de-
gilmore-lie-groups_ F01220	129	■ 1.000	$(2j + 1) \times (2j + 1)$	$(2j + 1) \times (2j + 1)$	termine the time evolution of the $(2j + 1) \times (2j + 1)$ unitary matrix. Specifically,
gilmore-lie-groups_ F01221	129	■ 1.000	$a(t)$	$a(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of
gilmore-lie-groups_ F01222	129	■ 1.000	$b(t)$	$b(t)$	the complex coefficients $a(t)$ and $b(t)$ or three equations for the real coefficients of
gilmore-lie-groups_ F01223	129	■ 1.000	$\sigma_1, \sigma_2, \sigma_3$	$\sigma_1, \sigma_2, \sigma_3$	the Pauli spin matrices $\sigma_1, \sigma_2, \sigma_3$. These are first order equations and can be solved
gilmore-lie-groups_ F01224	129	■ 1.000	$a(t_i) =$	$a(t_i) =$	by standard integration methods (e.g., RK4). The initial conditions are $a(t_i) =$
gilmore-lie-groups_ F01225	129	■ 0.999	$1, b(t_i) = 0$	$1, b(t_i) = 0$	$1, b(t_i) = 0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is

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gilmore-lie-groups_ F01226	129	■ 0.999	$a\left(t_f\right), b\left(t_f\right)$	$a\left(t_f\right), b\left(t_f\right)$	$1, b(t_i) = 0$. The final 2×2 unitary matrix is determined by $a(t_f), b(t_f)$. This is
gilmore-lie-groups_ F01227	129	■ 0.971	$(2 j+1) \times$	$(2 j+1) \times$	a group operation in $SU(2)$ that can subsequently be mapped into the $(2 j+1) \times$
gilmore-lie-groups_ F01228	129	■ 1.000	$(2 j+1)$	$(2 j+1)$	$(2 j+1)$ unitary irreducible representation of this group. At this point the problem
gilmore-lie-groups_ F01229	129	■ 1.000	$\left \psi\right\rangle\left(t_i\right)$	$\left \psi\left(t_i\right)\right\rangle$	is solved, independent of the initial state $\left \psi\left(t_i\right)\right\rangle$.
gilmore-lie-groups_ F01230	130	■ 1.000	$\left n\right\rangle, n=0,1,2, \ldots$	$\left n\right\rangle, n=0,1,2, \ldots$	space with basis vectors $\left n\right\rangle, n=0,1,2, \ldots$. The matrix on the right is a faith-
gilmore-lie-groups_ F01231	130	■ 1.000	$U=$	$U=$	general unitary operator that can be constructed from these operators is $U=$
gilmore-lie-groups_ F01232	130	■ 1.000	$\operatorname{EXP}\left(i\left[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I\right]\right)$	$\operatorname{EXP}\left(i\left[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I\right]\right)$	$\operatorname{EXP}(i[n(t) a^{\dagger} a+r(t) a^{\dagger}+r^{*}(t) a+d(t) I])$. This exponential is easy to compute
gilmore-lie-groups_ F01233	130	■ 0.982	$\left(n, r, r^{*}, d\right)$	(n, r, r^{*}, d)	tions in four real variables (n, r, r^{*}, d) that can be integrated by standard methods.
gilmore-lie-groups_ F01234	130	■ 1.000	$d \omega(t) / d t=0$	$d \omega(t) / d t=0$	In the case that $d \omega(t) / d t=0$ the equations simplify considerably, and can almost
gilmore-lie-groups_ F01235	130	■ 1.000	ρ	ρ	density operator ρ
gilmore-lie-groups_ F01236	130	■ 1.000	$\rho=e^{-\beta H} / Z$	$\rho=e^{-\beta H} / Z$	hamiltonian describing the system as $\rho=e^{-\beta H} / Z$, where the normalization con-
gilmore-lie-groups_ F01237	130	■ 1.000	$Z=\operatorname{tr} e^{-\beta H}$	$Z=\operatorname{tr} e^{-\beta H}$	stant, or partition function, is $Z=\operatorname{tr} e^{-\beta H}$ and $\beta=1 / k_B T$, k_B is the Boltzmann
gilmore-lie-groups_ F01238	130	■ 1.000	$\beta=1 / k_B T, k_B$	$\beta=1 / k_B T, k_B$	stant, or partition function, is $Z=\operatorname{tr} e^{-\beta H}$ and $\beta=1 / k_B T$, k_B is the Boltzmann
gilmore-lie-groups_ F01239	131	■ 1.000	$\left\langle e^{\Lambda}\right\rangle$	$\left\langle e^{\Lambda}\right\rangle$	generating functions for these expectation values. To do this we compute $\left\langle e^{\Lambda}\right\rangle$,
gilmore-lie-groups_ F01240	131	■ 1.000	$\Lambda=\lambda \cdot \mathbf{J}$	$\Lambda=\lambda \cdot \mathbf{J}$	where $\Lambda=\lambda \cdot \mathbf{J}$. All symmetric moments can be constructed by taking derivatives
gilmore-lie-groups_ F01241	131	■ 1.000	$H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$	$H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$	In these expressions $H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$, and similarly for $ H =\sqrt{(H, H)}$
gilmore-lie-groups_ F01242	131	■ 1.000	$ H =\sqrt{(H, H)}$	$ H =\sqrt{(H, H)}$	In these expressions $H \cdot \Lambda=(H, \Lambda)=\frac{1}{2} \operatorname{tr} H \Lambda$, and similarly for $ H =\sqrt{(H, H)}$
gilmore-lie-groups_ F01243	131	■ 1.000	$ \Lambda =\sqrt{(\Lambda, \Lambda)}$	$ \Lambda =\sqrt{(\Lambda, \Lambda)}$	and $ \Lambda =\sqrt{(\Lambda, \Lambda)}$.
gilmore-lie-groups_ F01244	131	■ 1.000	2^N	2^N	If N two-level atoms are acting incoherently, the trace over the 2^N states of all
gilmore-lie-groups_ F01245	131	■ 1.000	$2 J+1$	$2 J+1$	N atoms are acting coherently, there are $2 J+1$ states, where $N=2 J$. The trace
gilmore-lie-groups_ F01246	131	■ 1.000	$N=2 J$	$N=2 J$	N atoms are acting coherently, there are $2 J+1$ states, where $N=2 J$. The trace
gilmore-lie-groups_ F01247	131	■ 1.000	$\mu(H, \Lambda, T)$	$\mu(H, \Lambda, T)$	where $\mu(H, \Lambda, T)$ is determined from Eq. (7.65). The thermodynamic generating
gilmore-lie-groups_ F01248	131	■ 1.000	$\left\langle J_{-}\right\rangle$	$\left\langle J_{-}\right\rangle$	To construct explicit expectation values (e.g., $\left\langle J_{-}\right\rangle$) it is sufficient to differenti-
gilmore-lie-groups_ F01249	131	■ 1.000	$\frac{\partial}{\partial \lambda^{*}}\left\langle e^{\Lambda}\right\rangle /\left\langle e^0\right\rangle$	$\frac{\partial}{\partial \lambda^{*}}\left\langle e^{\Lambda}\right\rangle /\left\langle e^0\right\rangle$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^{*}}\left\langle e^{\Lambda}\right\rangle /\left\langle e^0\right\rangle$) and evaluate the result at $\Lambda=0$.
gilmore-lie-groups_ F01250	131	■ 1.000	$\Lambda=0$	$\Lambda=0$	ate the generating function (e.g., $\frac{\partial}{\partial \lambda^{*}}\left\langle e^{\Lambda}\right\rangle /\left\langle e^0\right\rangle$) and evaluate the result at $\Lambda=0$.
gilmore-lie-groups_ F01251	131	■ 1.000	$\left.\frac{\partial}{\partial \lambda^{*}} \log \left(\left\langle e^{\Lambda}\right\rangle\right)\right _{\Lambda=0}$	$\left.\frac{\partial}{\partial \lambda^{*}} \log \left(\left\langle e^{\Lambda}\right\rangle\right)\right _{\Lambda=0}$	$\left.\frac{\partial}{\partial \lambda^{*}} \log \left(\left\langle e^{\Lambda}\right\rangle\right)\right _{\Lambda=0}$.

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gilmore-lie-groups_ F01252	132	■ 0.999	$\rho = e^{\{-\beta \hbar \omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I\}} / Z$	$\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$	The density operator is $\rho = e^{-\beta(\hbar\omega a^\dagger a + \alpha a^\dagger + \alpha^* a + \delta I)} / Z$. The generating function for
gilmore-lie-groups_ F01253	132	■ 1.000	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	$\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$	operator expectation values s $\chi(H, \Lambda, T) = \text{tr} e^{-\beta H} e^{\lambda_n a^\dagger + \lambda a^\dagger + \lambda^* a + dI} / Z = \langle e^\Lambda \rangle$.
gilmore-lie-groups_ F01254	132	■ 0.967	$ 0\rangle, 1\rangle, 2\rangle, \dots$	$ 0\rangle, 1\rangle, 2\rangle, \dots$	(0), (1), (2), It would be insane to attempt to compute this expectation value
gilmore-lie-groups_ F01255	132	■ 0.989	H, Λ	H, Λ	(i) Write each of the operators H, Λ in the 3×3 matrix representation M (cf., Eq.
gilmore-lie-groups_ F01256	132	■ 0.999	$Z_{\{l\}}$	Z_l	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_ F01257	132	■ 0.999	$Z_{\{r\}}$	Z_r	(iv) Find a similarity transformation, S , that zeroes out Z_l and Z_r :
gilmore-lie-groups_ F01258	133	■ 1.000	$A < 0$	$A < 0$	(vi) Take the trace. Assuming $A < 0$ the sum converges to
gilmore-lie-groups_ F01259	133	■ 1.000	$\lambda_n =$	$\lambda_n =$	The generating function for only the creation and annihilation operators ($\lambda_n =$
gilmore-lie-groups_ F01260	133	■ 1.000	$d = 0$	$d = 0$	$d = 0$) is considerably simpler.
gilmore-lie-groups_ F01261	133	■ 1.000	$D = \text{Id}, G$	$D = \text{Id}, G$	D is a discrete invariant subgroup of $\overline{G}$. If $D = \text{Id}, G$ is isomorphic to $\overline{G}$, otherwise
gilmore-lie-groups_ F01262	133	■ 1.000	λI_n	λI_n	of the identity matrix, λI_n , and is simple to compute. If G_1 and G_2 have isomorphic
gilmore-lie-groups_ F01263	134	■ 0.940	$it / \hbar \leftrightarrow 1 / k_B T$	$it / \hbar \leftrightarrow 1 / k_B T$	(dynamics and statics) are related by a “Wick rotation”: $it / \hbar \leftrightarrow 1 / k_B T$. We have
gilmore-lie-groups_ F01264	134	■ 1.000	$\phi((x_1, y_1), (x_2, y_2))$	$\phi((x_1, y_1), (x_2, y_2))$	1. Construct the analytic group mapping $\phi((x_1, y_1), (x_2, y_2))$ for the parameterization
gilmore-lie-groups_ F01265	134	■ 1.000	$\phi((w_1, z_1), (w_2, z_2))$	$\phi((w_1, z_1), (w_2, z_2))$	(7.24) of the affine group. Construct the mapping $\phi((w_1, z_1), (w_2, z_2))$ for the param-
gilmore-lie-groups_ F01266	134	■ 1.000	$a^2 + b^2 - c^2 < 0$	$a^2 + b^2 - c^2 < 0$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_ F01267	134	■ 1.000	$SO(2) \subset$	$SO(2) \subset$	inside the light cone $a^2 + b^2 - c^2 < 0$ (Eq. (7.7)) maps onto the subgroup $SO(2) \subset$
gilmore-lie-groups_ F01268	134	■ 0.999	c_T	c_T	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0$ ($\sqrt{a^2 + b^2} = \beta \times c, \beta < 1$), the basic
gilmore-lie-groups_ F01269	134	■ 0.999	$a^2 + b^2 > 0 \left(\sqrt{a^2 + b^2} = \beta \times c, \beta < 1 \right)$	$a^2 + b^2 > 0 \left(\sqrt{a^2 + b^2} = \beta \times c, \beta < 1 \right)$	c_T , in the subgroup is 2π but if $a^2 + b^2 > 0 (\sqrt{a^2 + b^2} = \beta \times c, \beta < 1)$, the basic
gilmore-lie-groups_ F01270	134	■ 0.999	$2\pi\gamma$	$2\pi\gamma$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_ F01271	134	■ 0.999	$\gamma = 1 / \sqrt{1 - \beta^2}$	$\gamma = 1 / \sqrt{1 - \beta^2}$	repetition period in the c -direction is increased to $2\pi\gamma$, where $\gamma = 1/\sqrt{1 - \beta^2}$ and
gilmore-lie-groups_ F01272	134	■ 1.000	$\beta^2 = (a^2 + b^2) / c^2$	$\beta^2 = (a^2 + b^2) / c^2$	$\beta^2 = (a^2 + b^2) / c^2$. Compare this renormalization of periodicity with “time dilation.”
gilmore-lie-groups_ F01273	134	■ 0.947	$SU(3)$	$SU(3)$	3. Compute the maximal discrete invariant subgroup D_{MAX} of $SU(3)$ and show that it
gilmore-lie-groups_ F01274	134	■ 0.954	$\{I_3, \lambda I_3, \lambda^2 I_3\}$	$\{I_3, \lambda I_3, \lambda^2 I_3\}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01275	134	■ 0.954	$\lambda = e^{2\pi i/3}$	$\lambda = e^{2\pi i/3}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01276	134	■ 0.954	$SU(3) / D_{\text{MAX}}$	$SU(3) / D_{\text{MAX}}$	is $\{I_3, \lambda I_3, \lambda^2 I_3\}$, where $\lambda = e^{2\pi i/3}$. Next, show that $SU(3)/D_{\text{MAX}}$ is isomorphic to
gilmore-lie-groups_ F01277	134	■ 0.229	$[\text{Re } \mathfrak{g}(\mathfrak{su}(3))]$	$[\text{Reg}(\mathfrak{su}(3))]$	the group of real 8×8 matrices $\text{EXP}[\text{Reg}(\mathfrak{su}(3))]$ (“eight-fold way”).

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gilmore-lie-groups_F01278	134	■ 1.000	$\backslash\epsilon I_n, \backslash\epsilon=e^{2\pi i/n}$	$\epsilon I_n, \epsilon = e^{2\pi i/n}$	and show that it is the cyclic group of order n generated by $\epsilon I_n, \epsilon = e^{2\pi i/n}$. What real
gilmore-lie-groups_F01279	134	■ 0.988	$SU(n)/D_{\text{MAX}}$	$SU(n)/D_{\text{MAX}}$	matrix group is $SU(n)/D_{\text{MAX}}$ equivalent to?
gilmore-lie-groups_F01280	135	■ 1.000	$a^{\dagger}, a$	$a^{\dagger}, a$	6. It is convenient to introduce the creation and annihilation operators $a^{\dagger}, a$ to study the
gilmore-lie-groups_F01281	135	■ 1.000	$\backslash\left\langle n^{\prime}\right\rangle\left x^k\right\rangle\left n\right\rangle$	$\langle n' x^k n\rangle$	tor basis, $\langle n' x^k n\rangle$, can be simplified using disentangling theorems. This problem
gilmore-lie-groups_F01282	135	■ 1.000	$e^{\backslash\lambda x}$	$e^{\lambda x}$	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_F01283	135	■ 1.000	x^k	x^k	a. The function $e^{\lambda x}$ is a generating function for matrix elements of x^k . Show that
gilmore-lie-groups_F01284	135	■ 1.000	$\backslash\left[a, a^{\dagger}\right]=I$	$[a, a^{\dagger}] = I$	ators and their commutator $[a, a^{\dagger}] = I$ to show
gilmore-lie-groups_F01285	135	■ 0.955	$e^{ra^{\dagger}}e^{\delta I}e^{la}$	$e^{ra^{\dagger}}e^{\delta I}e^{la}$	$e^{ra^{\dagger}}e^{\delta I}e^{la}$ by constructing the matrix product of these three operators:
gilmore-lie-groups_F01286	135	■ 0.999	$\backslash\left\langle n^{\prime}\right\rangle\left x^4\right\rangle\left n\right\rangle$	$\langle n' x^4 n\rangle$	e. Use the disentangling theorem in d to compute $\langle n' x^4 n\rangle$. In particular, show
gilmore-lie-groups_F01287	135	■ 1.000	$\backslash\left\langle e^{ikx}\right\rangle$	$\langle e^{ikx}\rangle$	potential it is necessary to compute a structure factor $\langle e^{ikx}\rangle$. The expectation value is
gilmore-lie-groups_F01288	136	■ 1.000	$P_n \backslash\sim e^{-n \beta \hbar \omega}$	$P_n \simeq e^{-n\beta\hbar\omega}$	thermal: $P_n \simeq e^{-n\beta\hbar\omega}$. This expectation value can be written in algebraic form as
gilmore-lie-groups_F01289	136	■ 1.000	$k \rightarrow 0$	$k \rightarrow 0$	We concentrate on the numerator, as the denominator is obtained in the limit $k \rightarrow 0$.
gilmore-lie-groups_F01290	136	■ 1.000	$\backslash\left\langle e^{ikx}\right\rangle=e^{\delta}$	$\langle e^{ikx}\rangle = e^{\delta}$	As a result $\langle e^{ikx}\rangle = e^{\delta}$.
gilmore-lie-groups_F01291	136	■ 0.976	δ	δ	c. Compute δ using 3×3 nonunitary matrix multiplications to carry out multi-
gilmore-lie-groups_F01292	136	■ 1.000	$\infty \times \infty$	$\infty \times \infty$	plications <i>in the group</i> rather than in an $\infty \times \infty$ unitary representation of the
gilmore-lie-groups_F01293	136	■ 0.999	$\omega^{\backslash\prime}=\omega$	$\omega' = \omega$	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01294	136	■ 0.999	α, β	α, β	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01295	136	■ 0.999	γ	γ	Show that $\omega' = \omega$. Determine α, β , and compute γ . Show
gilmore-lie-groups_F01296	136	■ 1.000	$\backslash\left[X_i, X_j\right]=\sum_{k=1}^N C_{ij}^k X_k$	$[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$	8. A finite set of operators X_i closes under commutation: $[X_i, X_j] = \sum_{k=1}^N C_{ij}^k X_k$.
gilmore-lie-groups_F01297	136	■ 1.000	R	R	this set of operators has two representations R and S with the following properties:
gilmore-lie-groups_F01298	136	■ 1.000	$\backslash\left(R\left(a^i X_i\right)\right)^{\dagger}=\left(a^i R\left(X_i\right)\right)^{\dagger}=\left(a^i\right)^{*} R^{\dagger}\left(X_i\right)$	$(R(a^i X_i))^{\dagger} = (a^i R(X_i))^{\dagger} = (a^i)^{*} R^{\dagger}(X_i)$	• R is hermitian: $(R(a^i X_i))^{\dagger} = (a^i R(X_i))^{\dagger} = (a^i)^{*} R^{\dagger}(X_i)$
gilmore-lie-groups_F01299	136	■ 1.000	$S\left(a^i X_i\right)=0 \backslash\Rightarrow a^i=0$	$S(a^i X_i) = 0 \Rightarrow a^i = 0$	• S is faithful: $S(a^i X_i) = 0 \Rightarrow a^i = 0$.
gilmore-lie-groups_F01300	136	■ 1.000	$\mathcal{H}=R\left(a^i X_i\right)$	$\mathcal{H} = R(a^i X_i)$	a. It happens frequently that $\mathcal{H} = R(a^i X_i)$ describes the physics of some quantum
gilmore-lie-groups_F01301	136	■ 0.999	$H_1, H_2, \ldots, H_r \in \mathfrak{g}$	$H_1, H_2, \dots, H_r \in \mathfrak{g}$	mechanical system. Show that if $H_1, H_2, \dots, H_r \in \mathfrak{d}$ span a maximal commutative
gilmore-lie-groups_F01302	136	■ 1.000	$\backslash\left[H_i, H_j\right]=0, 1 \leq i, j \leq r$	$[H_i, H_j] = 0, 1 \leq i, j \leq r$	subspace, so that $[H_i, H_j] = 0, 1 \leq i, j \leq r$, then the hermitian operators $R(H_i)$
gilmore-lie-groups_F01303	136	■ 1.000	$R\left(H_i\right)$	$R(H_i)$	subspace, so that $[H_i, H_j] = 0, 1 \leq i, j \leq r$, then the hermitian operators $R(H_i)$

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gilmore-lie-groups_ F01304	136	■ 1.000	$\left[R\left(H_i\right)\right]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$	$[R\left(H_i\right)]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$	representation: $[R\left(H_i\right)]_{\alpha \beta}=r_{\alpha}(i) \delta_{\alpha \beta}$.
gilmore-lie-groups_ F01305	137	■ 1.000	$\left[S\left(H_i\right), S\left(H_j\right)\right]=0$	$[S\left(H_i\right), S\left(H_j\right)]=0$	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_ F01306	137	■ 1.000	r	r	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_ F01307	137	■ 1.000	$S\left(H_i\right)$	$S\left(H_i\right)$	b. Show that $[S(H_i), S(H_j)] = 0$, but show by example that the r matrices $S(H_i)$
gilmore-lie-groups_ F01308	137	■ 1.000	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H} t}\right) R\left(\mathcal{H}\right) t$	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H} t}\right)=e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$	$U(t)=R\left(e^{-\frac{i}{\hbar} \mathcal{H} t}\right)=e^{-\frac{i}{\hbar} R(\mathcal{H}) t}$.
gilmore-lie-groups_ F01309	137	■ 1.000	$\rho(T)=e^{-\beta \mathcal{H}} / Z=$	$\rho(T)=e^{-\beta \mathcal{H}} / Z=$	d. Show that the density operator for thermal expectation values is $\rho(T)=e^{-\beta \mathcal{H}} / Z=$
gilmore-lie-groups_ F01310	137	■ 0.975	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$	$R\left(e^{-\beta \mathcal{H}}\right) / Z=e^{-\beta R(\mathcal{H})} / Z$. What is Z ?
gilmore-lie-groups_ F01311	137	■ 1.000	$U(t)$	$U(t)$	e. Show that the unitary time evolution operator $U(t)$ and the hermitian density
gilmore-lie-groups_ F01312	137	■ 1.000	$\rho(T)$	$\rho(T)$	operator $\rho(T)$ are related by a Wick rotation $it / \hbar \leftrightarrow \beta=1 / k_B T$.
gilmore-lie-groups_ F01313	137	■ 1.000	$it / \hbar \leftrightarrow \beta=1 / k_B T$	$it / \hbar \leftrightarrow \beta=1 / k_B T$	operator $\rho(T)$ are related by a Wick rotation $it / \hbar \leftrightarrow \beta=1 / k_B T$.
gilmore-lie-groups_ F01314	137	■ 1.000	$G=e^{\mathfrak{g}}$	$G=e^{\mathfrak{g}}$	g. The operator product in the numerator is in the group $G=e^{\mathfrak{g}}$ or its complex exten-
gilmore-lie-groups_ F01315	137	■ 1.000	H_i	H_i	in terms of the operators H_i) the trace can easily be constructed. Show that for
gilmore-lie-groups_ F01316	137	■ 1.000	x^i	x^i	x^i sufficiently small it is always possible to construct a similarity transformation
gilmore-lie-groups_ F01317	137	■ 1.000	$S=e^{y^k X_k}$	$S=e^{y^k X_k}$	$S=e^{y^k X_k}$ with the property
gilmore-lie-groups_ F01318	137	■ 1.000	$e^{x^i X_i}$	$e^{x^i X_i}$	function $e^{x^i X_i}$, and the Wick rotation.
gilmore-lie-groups_ F01319	137	■ 1.000	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)}$	$U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)}$	The unitary transformation has the form $U(\alpha)=e^{\left(\alpha a^{\dagger}-\alpha^* a\right)}$, where $a^{\dagger}$ and a are the
gilmore-lie-groups_ F01320	138	■ 0.896	$a^{\dagger} n\rangle=$	$a^{\dagger} n\rangle=$	Hilbert space and out of the nonunitary 3×3 matrix representation), $a^{\dagger} n\rangle=$
gilmore-lie-groups_ F01321	138	■ 1.000	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$	$ n+1\rangle \sqrt{n+1}$ to conclude
gilmore-lie-groups_ F01322	138	■ 1.000	$\langle\beta \alpha\rangle$	$\langle\beta \alpha\rangle$	e. Compute the inner product $\langle\beta \alpha\rangle$ and show $\langle\alpha \alpha\rangle=1$.
gilmore-lie-groups_ F01323	138	■ 1.000	$\langle\alpha \alpha\rangle=1$	$\langle\alpha \alpha\rangle=1$	e. Compute the inner product $\langle\beta \alpha\rangle$ and show $\langle\alpha \alpha\rangle=1$.
gilmore-lie-groups_ F01324	138	■ 1.000	$a \alpha\rangle=\alpha \alpha\rangle$	$a \alpha\rangle=\alpha \alpha\rangle$	f. Show $a \alpha\rangle=\alpha \alpha\rangle$.
gilmore-lie-groups_ F01325	138	■ 1.000	$\langle\alpha x \alpha\rangle=\left(\alpha^*+\alpha\right) / \sqrt{2}$	$\langle\alpha x \alpha\rangle=\left(\alpha^*+\alpha\right) / \sqrt{2}$	g. Show $\langle\alpha x \alpha\rangle=\left(\alpha^*+\alpha\right) / \sqrt{2}$
gilmore-lie-groups_ F01326	138	■ 0.966	$e^{i \phi J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \phi}$	$e^{i \phi J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \phi}$	by a phase angle: $e^{i \phi J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \phi}$. This simply “renormalizes” the fiducial
gilmore-lie-groups_ F01327	138	■ 1.000	$e^{i\left(\theta_x J_x+\theta_y J_y\right)}$	$e^{i\left(\theta_x J_x+\theta_y J_y\right)}$	c. Rewrite $e^{i\left(\theta_x J_x+\theta_y J_y\right)}$ in the form $e^{i \alpha_+ J_+} e^{i \alpha_z J_z} e^{i \alpha_- J_-}$ and compute the analytic relation
gilmore-lie-groups_ F01328	138	■ 1.000	$e^{i \alpha_+ J_+} e^{i \alpha_z J_z} e^{i \alpha_- J_-}$	$e^{i \alpha_+ J_+} e^{i \alpha_z J_z} e^{i \alpha_- J_-}$	c. Rewrite $e^{i\left(\theta_x J_x+\theta_y J_y\right)}$ in the form $e^{i \alpha_+ J_+} e^{i \alpha_z J_z} e^{i \alpha_- J_-}$ and compute the analytic relation
gilmore-lie-groups_ F01329	138	■ 0.859	$e^{i \alpha_- J_-}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle$	$e^{i \alpha_- J_-}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle$	d. Show $e^{i \alpha_- J_-}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle$.
gilmore-lie-groups_ F01330	138	■ 0.952	$e^{i \alpha_z J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \alpha_z}$	$e^{i \alpha_z J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \alpha_z}$	e. Show $e^{i \alpha_z J_z}\left j_{-j}\right\rangle=\left j_{-j}\right\rangle e^{-i j \alpha_z}$.

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gilmore-lie-groups_ F01331	139	0.815	$\left.\left.J_{-}\right _{\theta_x\theta_y}\right\rangle^j$	$J_{-} _{\theta_x\theta_y}^j\rangle$	g. Show that $J_{-} _{\theta_x\theta_y}^j\rangle$ cannot be proportional to $ _{\theta_x\theta_y}^j\rangle$ because the state $ _{+j}^j\rangle$ is not
gilmore-lie-groups_ F01332	139	0.815	$\left \begin{array}{c} j \\ \theta_x\theta_y \end{array}\right\rangle$	$\left \begin{array}{c} j \\ \theta_x\theta_y \end{array}\right\rangle$	g. Show that $J_{-} _{\theta_x\theta_y}^j\rangle$ cannot be proportional to $ _{\theta_x\theta_y}^j\rangle$ because the state $ _{+j}^j\rangle$ is not
gilmore-lie-groups_ F01333	139	0.815	$\left \begin{array}{c} j \\ +j \end{array}\right\rangle$	$\left \begin{array}{c} j \\ +j \end{array}\right\rangle$	g. Show that $J_{-} _{\theta_x\theta_y}^j\rangle$ cannot be proportional to $ _{\theta_x\theta_y}^j\rangle$ because the state $ _{+j}^j\rangle$ is not
gilmore-lie-groups_ F01334	139	1.000	$H_{\{4\}}$	H_4	irreducible representations and the harmonic oscillator group H_4 is not compact
gilmore-lie-groups_ F01335	139	1.000	$e^{-i\phi}=\left(\theta_x-i\theta_y\right)/\theta$	$e^{-i\phi}=(\theta_x-i\theta_y)/\theta$	where $e^{-i\phi}=(\theta_x-i\theta_y)/\theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_ F01336	139	1.000	$\theta^{\prime}$	θ'	where $e^{-i\phi}=(\theta_x-i\theta_y)/\theta$, and similarly for θ' (cf., Eq. (7.46)).
gilmore-lie-groups_ F01337	139	1.000	$ u\rangle$	$ u\rangle$	sometimes does) represent a charge, and $ u\rangle$ is an eigenfunction. In this toy example,
gilmore-lie-groups_ F01338	139	1.000	$J_{\{3\}}$	J_3	the operators J_3 and J_1 are assumed to belong to the Lie algebra $\mathfrak{su}(2)$ and the equation
gilmore-lie-groups_ F01339	139	1.000	$\left(E^{\prime} J_3-Z\right) v\rangle=0$	$(E'J_3-Z) v\rangle=0$	$E'J_3-Z v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_ F01340	139	1.000	$E^{\prime}=\sqrt{E^2+p^2}$	$E'=\sqrt{E^2+p^2}$	$(E'J_3-Z) v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_ F01341	139	1.000	$ v\rangle=U u\rangle$	$ v\rangle=U u\rangle$	$(E'J_3-Z) v\rangle=0$, where $E'=\sqrt{E^2+p^2}$ and $ v\rangle=U u\rangle$.
gilmore-lie-groups_ F01342	139	1.000	$E=\pm\sqrt{\left(Z / m\right)^2-p^2}$	$E=\pm\sqrt{(Z/m)^2-p^2}$	b. Show that $E=\pm\sqrt{(Z/m)^2-p^2}$
gilmore-lie-groups_ F01343	139	0.871	$(p \rightarrow 0)$	$(p \rightarrow 0)$	c. Compare this spectrum with the unperturbed spectrum ($p \rightarrow 0$).
gilmore-lie-groups_ F01344	139	1.000	j, p, Z	j, p, Z	d. Under what conditions on j, p, Z are these solutions valid?
gilmore-lie-groups_ F01345	139	0.986	$ u\rangle=e^{i\theta J_2}\left _m^j\right\rangle$	$ u\rangle=e^{i\theta J_2} _m^j\rangle$	and show that $ u\rangle=e^{i\theta J_2} _m^j\rangle$. Compute θ for each E .
gilmore-lie-groups_ F01346	139	1.000	$z=\cos (\beta)$	$z=\cos (\beta)$	where $z=\cos (\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_ F01347	139	1.000	$D_{m n}^j$	D_{mn}^j	where $z=\cos (\beta)$. The Wigner matrix elements D_{mn}^j are related to the Jacobi poly-
gilmore-lie-groups_ F01348	139	1.000	$j=l$	$j=l$	nomials when $j=l$, where l is an integer.
gilmore-lie-groups_ F01349	140	1.000	$I_{\{2\}} \in S U(2)$	$I_2 \in S U(2)$	a. Geodesics through $I_2 \in S U(2)$ focus at $-I_2$ and geodesics through $I_3 \in S O(3)$
gilmore-lie-groups_ F01350	140	1.000	$I_{\{3\}} \in S O(3)$	$I_3 \in S O(3)$	a. Geodesics through $I_2 \in S U(2)$ focus at $-I_2$ and geodesics through $I_3 \in S O(3)$
gilmore-lie-groups_ F01351	140	0.835	$I_{\{3\}}$	I_3	focus at I_3 . Conclude that $S U(2)$ is a two-fold covering group of $S O(3)$.
gilmore-lie-groups_ F01352	140	0.999	$S U(2) / U(1)(z=1, x=y=0)$	$S U(2) / U(1)(z=1, x=y=0)$	b. Geodesics through the “north pole” of $S U(2) / U(1)(z=1, x=y=0)$ focus at
gilmore-lie-groups_ F01353	140	0.793	$z=-1, x=y=0$	$z=-1, x=y=0$	its “south pole” ($z=-1, x=y=0$) and geodesics through the north pole of
gilmore-lie-groups_ F01354	140	0.701	$S O(3) / S O(2)(z=1, x=y=0)$	$S O(3) / S O(2)(z=1, x=y=0)$	$S O(3) / S O(2)(z=1, x=y=0)$ focus at its south pole ($z=-1, x=y=0$).
gilmore-lie-groups_ F01355	140	0.701	$(z=-1, x=y=0)$	$(z=-1, x=y=0)$	$S O(3) / S O(2)(z=1, x=y=0)$ focus at its south pole ($z=-1, x=y=0$).
gilmore-lie-groups_ F01356	140	1.000	$S U(2) / U(1)=S^2=S O(3) / S O(2)$	$S U(2) / U(1)=S^2=S O(3) / S O(2)$	c. Conclude that $S U(2) / U(1)=S^2=S O(3) / S O(2)$ and the $2 \rightarrow 1$ nature of the

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gilmore-lie-groups_F01357	140	■ 0.986	$S U(2) \searrow S O(3)$	$SU(2) \downarrow SO(3)$	covering $SU(2) \downarrow SO(3)$ is contained in the subgroup of rotations about the z -
gilmore-lie-groups_F01358	140	■ 0.918	$U(1) \searrow S O(2)$	$U(1) \downarrow SO(2)$	axis $U(1) \downarrow SO(2)$:
gilmore-lie-groups_F01359	140	■ 1.000	$e^{2 \pi i k / r} I_n$	$e^{2 \pi i k / r} I_n$	order r , with group elements $e^{2 \pi i k / r} I_n$, with n / r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_F01360	140	■ 1.000	n / r	n / r	order r , with group elements $e^{2 \pi i k / r} I_n$, with n / r integer. Compute the foci in $SU(n)$.
gilmore-lie-groups_F01361	140	■ 1.000	$e^{2 \pi i k / n} I_n$	$e^{2 \pi i k / n} I_n$	How are the foci related to the group operations of the form $e^{2 \pi i k / n} I_n$?
gilmore-lie-groups_F01362	140	■ 0.999	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$	$\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$	14. Show that the matrix $\left[\begin{array}{cc} -\lambda & 0 \\ 0 & -1 / \lambda \end{array}\right]$ in $SL(2 ; \mathbb{R})$ cannot be reached by exponentiating
gilmore-lie-groups_F01363	140	■ 1.000	$\lambda > 1$	$\lambda > 1$	any element in the Lie algebra if $\lambda > 1$. Show that it can be reached by following a
gilmore-lie-groups_F01364	140	■ 1.000	$\hbar \omega$	$\hbar \omega$	photon has energy $\hbar \omega$. The hamiltonian is chosen as
gilmore-lie-groups_F01365	140	■ 1.000	$\sigma_z^{(i)}$	$\sigma_z^{(i)}$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_F01366	140	■ 1.000	$a^{\dagger} a$	$a^\dagger a$	The operator $\sigma_z^{(i)}$ describes the two states of atom i and the operator $a^\dagger a$ describes
gilmore-lie-groups_F01367	140	■ 0.999	$\sigma_+^{(j)} \left(\sigma_{\pm}^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	$\sigma_+^{(j)} \left(\sigma_{\pm}^{(j)} = \frac{1}{2} \left(\sigma_x^{(j)} \pm i \sigma_y^{(j)} \right) \right)$	the number of photons in the field mode. The operator $\sigma_+^{(j)}$ ($\sigma_-^{(j)} = \frac{1}{2}(\sigma_x^{(j)} \pm i \sigma_y^{(j)})$)
gilmore-lie-groups_F01368	140	■ 1.000	$\sigma_-^{(j)} a^\dagger$	$\sigma_-^{(j)} a^\dagger$	operator $\sigma_-^{(j)} a^\dagger$, describes deexcitation of an atom with emission (creation, $a^\dagger$) of a
gilmore-lie-groups_F01369	140	■ 1.000	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	$\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$	b. If all the atoms behave cooperatively it is possible to replace $\sum_{i=1}^N \frac{1}{2} \sigma_z^{(i)} \rightarrow J_z$,
gilmore-lie-groups_F01370	140	■ 0.999	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$. Show that the operators $J_z, J_{\pm}$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_F01371	140	■ 0.999	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	$\sum_{i=1}^N \sigma_{\pm}^{(i)} \rightarrow J_{\pm}$. Show that the operators $J_z, J_{\pm}$ satisfy the usual $\mathfrak{su}(2)$ commuta-
gilmore-lie-groups_F01372	141	■ 1.000	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_z \rightarrow \langle J_z(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$	$J_- \rightarrow \langle J_-(t) \rangle, J_+ \rightarrow \langle J_+(t) \rangle, J_- \rightarrow \langle J_-(t) \rangle = \langle J_+(t) \rangle^*$. Show that this semiclas-
gilmore-lie-groups_F01373	141	■ 1.000	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	$ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$	of the electromagnetic field: $ \alpha(t)\rangle = U(\alpha(t)) 0\rangle = e^{\alpha a^\dagger - \alpha^* a} 0\rangle$. Use the disentan-
gilmore-lie-groups_F01374	141	■ 1.000	$\alpha(t)$	$\alpha(t)$	gling theorems to compute the relation between the coherent state parameter $\alpha(t)$
gilmore-lie-groups_F01375	141	■ 1.000	$\langle J_z(t) \rangle$	$\langle J_z(t) \rangle$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.
gilmore-lie-groups_F01376	141	■ 1.000	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	$\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$	and the classical driving fields $\langle J_z(t) \rangle$ and $\langle J_+(t) \rangle = \langle J_-(t) \rangle^*$.
gilmore-lie-groups_F01377	141	■ 1.000	$ \beta\rangle$	$ \beta\rangle$	state $ \beta\rangle$, the state obtained by the action of the classical current is still a coherent
gilmore-lie-groups_F01378	141	■ 1.000	β	β	state. How are the parameters β , describing the initial condition, and α , describing
gilmore-lie-groups_F01379	141	■ 1.000	$a^\dagger \rightarrow \langle a(t) \rangle^*$	$a^\dagger \rightarrow \langle a(t) \rangle^*$	considered classically. Show that this amounts to the substitutions $a^\dagger \rightarrow \langle a(t) \rangle^*$,
gilmore-lie-groups_F01380	141	■ 1.000	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.
gilmore-lie-groups_F01381	141	■ 1.000	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$	$a \rightarrow \langle a(t) \rangle$, and $a^\dagger a \rightarrow \langle a(t) \rangle^* \langle a(t) \rangle$.

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gilmore-lie-groups_ F01382	141	■ 1.000	$m = -\frac{1}{2}$	$m = -\frac{1}{2}$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_ F01383	141	■ 1.000	$M = -J$, $J = N / 2$	$M = -J$, $J = N/2$	ground state ($m = -\frac{1}{2}$ for each atom, or $M = -J$, $J = N/2$ for the ensemble of
gilmore-lie-groups_ F01384	141	■ 0.997	$ \theta(t)\rangle = e^{i\theta(t)\cdot\mathbf{J}} J, -J\rangle$	$ \theta(t)\rangle = e^{i\theta(t)\cdot\mathbf{J}} J, -J\rangle$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\cdot\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_ F01385	141	■ 0.997	$J = N / 2$	$J = N/2$	evolve into a coherent state: $ \theta(t)\rangle = e^{i\theta(t)\cdot\mathbf{J}} J, -J\rangle$, where $J = N/2$. How are the
gilmore-lie-groups_ F01386	141	■ 1.000	$\theta(t)$	$\theta(t)$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_ F01387	141	■ 1.000	$\langle a(t) \rangle$	$\langle a(t) \rangle$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_ F01388	141	■ 1.000	$\langle a(t) \rangle^*$	$\langle a(t) \rangle^*$	angles $\theta(t)$ related to the classical field variables $\langle a(t) \rangle$ and $\langle a(t) \rangle^*$?
gilmore-lie-groups_ F01389	141	■ 1.000	$\langle a \rangle$	$\langle a \rangle$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_ F01390	141	■ 1.000	$\langle a \rangle^*$	$\langle a \rangle^*$	field expectation values $\langle a \rangle$ and $\langle a \rangle^*$. Conversely, the semiclassical hamiltonian
gilmore-lie-groups_ F01391	141	■ 1.000	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$	$\langle J_+ \rangle = \langle J_- \rangle^*$. Construct a self-consistent model by requiring that both sets of
gilmore-lie-groups_ F01392	141	■ 1.000	$T(\beta = 1/k_B T)$	$T(\beta = 1/k_B T)$	equilibrium at temperature T ($\beta = 1/k_B T$).
gilmore-lie-groups_ F01393	141	■ 1.000	$\langle \sigma_+^{(i)} \rangle_T$	$\langle \sigma_+^{(i)} \rangle_T$	a. Assume $\langle \sigma_+^{(i)} \rangle_T$ has some fixed unknown value, and similarly for the other atomic
gilmore-lie-groups_ F01394	141	■ 1.000	$a^\dagger, a, a^\dagger a$	$a^\dagger, a, a^\dagger a$	expectation values for the operators $a^\dagger, a, a^\dagger a$.
gilmore-lie-groups_ F01395	142	■ 1.000	$\sigma_z, \sigma_+, \sigma_-$	$\sigma_z, \sigma_+, \sigma_-$	operators $\sigma_z, \sigma_+, \sigma_-$.
gilmore-lie-groups_ F01396	142	■ 1.000	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_ F01397	142	■ 1.000	$\langle J_+ \rangle_T \neq 0$	$\langle J_+ \rangle_T \neq 0$	$\epsilon, \hbar\omega, \lambda, N$ is there a nontrivial solution (e.g., $\langle J_+ \rangle_T \neq 0$)?
gilmore-lie-groups_ F01398	142	■ 1.000	$\lambda^2 / \epsilon \hbar \omega$	$\lambda^2 / \epsilon \hbar \omega$	e. Show that a thermodynamic phase transition occurs as $\lambda^2 / \epsilon \hbar \omega$ increases through
gilmore-lie-groups_ F01399	142	■ 1.000	$a(t), b(t)$	$a(t), b(t)$	17. The two complex parameters $a(t), b(t)$ in the evolution equation (7.58) can be ex-
gilmore-lie-groups_ F01400	142	■ 1.000	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	$a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$	pressed in terms of their real and imaginary parts. These obey $a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$
gilmore-lie-groups_ F01401	142	■ 0.542	(a_r, a_i, b_r, b_i)	(a_r, a_i, b_r, b_i)	compute the length of the vector (a_r, a_i, b_r, b_i) and scale this length back to +1.
gilmore-lie-groups_ F01402	142	■ 1.000	$\lambda_3 \rightarrow 0$	$\lambda_3 \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_ F01403	142	■ 1.000	$\lambda_n \rightarrow 0$	$\lambda_n \rightarrow 0$	Simplify (7.67) by taking the limit $\lambda_3 \rightarrow 0$. Simplify (7.70) by taking the limit $\lambda_n \rightarrow 0$
gilmore-lie-groups_ F01404	142	■ 1.000	$d = \delta = 0$	$d = \delta = 0$	and setting $d = \delta = 0$.
gilmore-lie-groups_ F01405	142	■ 1.000	$\langle J_+ J_- + J_- J_+ \rangle$	$\langle J_+ J_- + J_- J_+ \rangle$	<i>metric</i> operator products such as $J_+ J_- + J_- J_+$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01406	142	■ 1.000	$\langle a a^\dagger + a^\dagger a \rangle$	$\langle a a^\dagger + a^\dagger a \rangle$	<i>metric</i> operator products such as $\langle J_+ J_- + J_- J_+ \rangle$ or $\langle a a^\dagger + a^\dagger a \rangle$ than it is to construct
gilmore-lie-groups_ F01407	142	■ 1.000	$\langle J_+ J_- \rangle$	$\langle J_+ J_- \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show
gilmore-lie-groups_ F01408	142	■ 1.000	$\langle a^\dagger a \rangle$	$\langle a^\dagger a \rangle$	generating functions for <i>ordered</i> products of operators such as $\langle J_+ J_- \rangle$ or $\langle a^\dagger a \rangle$. Show

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gilmore-lie-groups_ F01489	143	■ 0.749	$X_{\{j\}}=A_{\{j\}}^{\{s\}} Y_{\{s\}}$	$X_j = A_j^s Y_s$	Under a change of basis $X_j = A_j^s Y_s$ this $n \times n$ matrix undergoes a similarity trans-
gilmore-lie-groups_ F01410	144	■ 1.000	$N=p \ q$	$N = pq$	This is an $n \times n$ ($n = p + q$) matrix algebra which is $N = pq$ dimensional. The
gilmore-lie-groups_ F01411	144	■ 1.000	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	3. $\mathfrak{so}(n)$ Upper triangular In this case nonzero elements occur on and above the
gilmore-lie-groups_ F01412	145	■ 1.000	$\hat{n} =$	$\hat{n} =$	Example We consider the algebra consisting of the six photon operators $\hat{n} =$
gilmore-lie-groups_ F01413	145	■ 0.993	$\frac{1}{2}\{a, a^{\dagger}\} = a^{\dagger}a + \frac{1}{2}, a^{\dagger 2}, a^2, a^{\dagger}, a, I = [a, a^{\dagger}]$	$\frac{1}{2}\{a, a^{\dagger}\} = a^{\dagger}a + \frac{1}{2}, a^{\dagger 2}, a^2, a^{\dagger}, a, I = [a, a^{\dagger}]$	$\frac{1}{2}\{a, a^{\dagger}\} = a^{\dagger}a + \frac{1}{2}, a^{\dagger 2}, a^2, a^{\dagger}, a, I = [a, a^{\dagger}]$. This is isomorphic to the algebra
gilmore-lie-groups_ F01414	145	■ 1.000	$X_{\{i \ j\}}=-X_{\{j \ i\}}, 1 \leq i, j \leq 4$	$X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$	Example The Lie algebra $\mathfrak{so}(4)$ has six generators $X_{ij} = -X_{ji}, 1 \leq i, j \leq 4$ and
gilmore-lie-groups_ F01415	146	■ 1.000	$\mathfrak{so}(n), n > 4$	$\mathfrak{so}(n), n > 4$	other orthogonal Lie algebras, $\mathfrak{so}(n), n > 4$. We will have to wait until Chapter 10
gilmore-lie-groups_ F01416	146	■ 0.999	$\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n > 4)$	$\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n > 4)$	Example The Lie algebras $\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n > 4)$, and $\mathfrak{sp}(n)(n \geq 1)$ are
gilmore-lie-groups_ F01417	146	■ 0.999	$\mathfrak{sp}(n)(n \geq 1)$	$\mathfrak{sp}(n)(n \geq 1)$	Example The Lie algebras $\mathfrak{su}(n)(n \geq 2), \mathfrak{so}(n)(n > 4)$, and $\mathfrak{sp}(n)(n \geq 1)$ are
gilmore-lie-groups_ F01418	147	■ 0.999	$\left[Z, X_{\{i\}} \right]$	$[Z, X_i]$	This means that $[Z, X_i]$ can be expressed as a linear combination of basis vectors
gilmore-lie-groups_ F01419	147	■ 1.000	$X_{\{j\}}$	X_j	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01420	147	■ 1.000	$j \geq i$	$j \geq i$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01421	147	■ 1.000	$X_{\{i\}}, X_{\{i+1\}}, \ldots, X_{\{n\}}$	$X_i, X_{i+1}, \dots, X_n$	X_j with $j \geq i$. This in turn means that the basis vectors $X_i, X_{i+1}, \dots, X_n$ span a
gilmore-lie-groups_ F01422	147	■ 0.593	$i=1,2, \ldots, n$	$i = 1, 2, \dots, n$	subalgebra for each value of $i = 1, 2, \dots, n$. Since this subalgebra is mapped onto
gilmore-lie-groups_ F01423	147	■ 1.000	$V_{\{i\}}$	V_i	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01424	147	■ 1.000	$V_{\{j\}}, i > j$	$V_j, i > j$	Each V_i is an invariant subalgebra in $V_j, i > j$. The original algebra is V_1 .
gilmore-lie-groups_ F01425	148	■ 1.000	$\left[V_{\{2\}}, V_{\{2\}} \right] \subseteq V_{\{2\}}, V_{\{2\}}$	$[V_2, V_2] \subseteq V_2, V_2$	Since in particular $[V_2, V_2] \subseteq V_2, V_2$ is a subalgebra in the original algebra. Further,
gilmore-lie-groups_ F01426	148	■ 1.000	$V_{\{2\}}, V_{\{2\}}$	V_2, V_2	since the commutator of anything in the original algebra with V_2 is in V_2, V_2 is an
gilmore-lie-groups_ F01427	149	■ 1.000	$\mathfrak{g}' = \mathfrak{g} - V_{\{0\}}$	$\mathfrak{g}' = \mathfrak{g} - V_0$	(iv) Form the difference $\mathfrak{g}' = \mathfrak{g} - V_0$. This is a Lie algebra (under the “mod” operation: set
gilmore-lie-groups_ F01428	149	■ 1.000	$\mathfrak{g}'$	$\mathfrak{g}'$	Cartan–Killing inner product for $\mathfrak{g}'$.
gilmore-lie-groups_ F01429	150	■ 1.000	$V_{\{0\}}^{\prime} = 0$	$V'_0 = 0$	In this case $V'_0 = 0$ and the two photon algebra has the decomposition
gilmore-lie-groups_ F01430	150	■ 1.000	$a^{\dagger 2}$	$a^{\dagger 2}$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01431	150	■ 1.000	a^2	a^2	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01432	150	■ 1.000	$a^{\dagger 2} + a^2$	$a^{\dagger 2} + a^2$	binations of the operators $a^{\dagger 2}$ and a^2 . Then $a^{\dagger 2} + a^2$ is a compact generator, since
gilmore-lie-groups_ F01433	150	■ 1.000	$a^{\dagger}a + \frac{1}{2}$	$a^{\dagger}a + \frac{1}{2}$	the Cartan–Killing form is negative-definite on it. The other two operators, $a^{\dagger}a + \frac{1}{2}$
gilmore-lie-groups_ F01434	150	■ 1.000	$a^{\dagger 2} - a^2$	$a^{\dagger 2} - a^2$	and $a^{\dagger 2} - a^2$, are noncompact.

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gilmore-lie-groups_ F01435	151	■ 1.000	$\hat{n}, a^{\dagger}, a, I$	$\hat{n}, a^{\dagger}, a, I$	a. The photon algebra $\hat{n}, a^{\dagger}, a, I$ (Eq. (8.5)).
gilmore-lie-groups_ F01436	151	■ 0.786	$\left[b_i, b_j^{\dagger}\right]=I \delta_{ij}, 1 \leq i, j \leq n$	$\left[b_i, b_j^{\dagger}\right]=I \delta_{ij}, 1 \leq i, j \leq n$	$\left[b_i, b_j^{\dagger}\right]=I \delta_{ij}, 1 \leq i, j \leq n$. Commutators involving bilinear (trilinear, ...) products
gilmore-lie-groups_ F01437	151	■ 1.000	$b_{\{i}, b_{\{j}^{\dagger}\dagger}, I$	$b_i, b_j^{\dagger}, I$	a. $b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01438	151	■ 1.000	$b_{\{i}^{\dagger}\dagger} b_{\{j}, b_{\{i}, b_{\{j}^{\dagger}\dagger}, I$	$b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$	c. $b_i^{\dagger} b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01439	151	■ 1.000	$b_{\{i}^{\dagger}\dagger} b_{\{j}^{\dagger}\dagger}, b_{\{i}^{\dagger}\dagger} b_{\{j}+\frac{1}{2} \delta_{ij}, b_{\{i} b_{\{j}$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j+\frac{1}{2} \delta_{ij}, b_i b_j$	d. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j+\frac{1}{2} \delta_{ij}, b_i b_j$.
gilmore-lie-groups_ F01440	151	■ 0.998	$b_{\{i}^{\dagger}\dagger} b_{\{j}^{\dagger}\dagger}, b_{\{i}^{\dagger}\dagger} b_{\{j}+\frac{1}{2} \delta_{ij}, b_{\{i} b_{\{j}, b_{\{i}, b_{\{j}^{\dagger}\dagger}, I$	$b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j+\frac{1}{2} \delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$	e. $b_i^{\dagger} b_j^{\dagger}, b_i^{\dagger} b_j+\frac{1}{2} \delta_{ij}, b_i b_j, b_i, b_j^{\dagger}, I$.
gilmore-lie-groups_ F01441	151	■ 1.000	$b, b^{\dagger\dagger} b, b^{\dagger\dagger} b^{\dagger\dagger} b$	$b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$	f. $b, b^{\dagger} b, b^{\dagger} b^{\dagger} b$.
gilmore-lie-groups_ F01442	151	■ 0.788	$b^{\dagger\dagger}$	$b^{\dagger}$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01443	151	■ 0.788	$b^{\dagger\dagger} b, b^{\dagger\dagger} b b$	$b^{\dagger} b, b^{\dagger} b b$	g. $b^{\dagger}, b^{\dagger} b, b^{\dagger} b b$.
gilmore-lie-groups_ F01444	151	■ 1.000	$\left\{f_i, f_j^{\dagger}\right\}=\delta_{ij}$	$\left\{f_i, f_j^{\dagger}\right\}=\delta_{ij}$	$\left\{f_i, f_j^{\dagger}\right\}=\delta_{ij}$, but their bilinear combinations close under the same commutation re-
gilmore-lie-groups_ F01445	151	■ 0.985	$f_{\{i}^{\dagger}\dagger} f_{\{j}^{\dagger}\dagger}$	$f_i^{\dagger} f_j^{\dagger}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j+\frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01446	151	■ 0.985	$f_{\{i}^{\dagger}\dagger} f_{\{j}+\frac{1}{2} \delta_{ij}$	$f_i^{\dagger} f_j+\frac{1}{2} \delta_{ij}$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j+\frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01447	151	■ 0.985	$f_{\{i} f_{\{j}$	$f_i f_j$	b. $f_i^{\dagger} f_j^{\dagger}, f_i^{\dagger} f_j+\frac{1}{2} \delta_{ij}, f_i f_j$.
gilmore-lie-groups_ F01448	151	■ 1.000	$\left(x_i\right)$	$\left(x_i\right)$	of the position (x_i) and momentum (∂_j) operators for n independent directions:
gilmore-lie-groups_ F01449	151	■ 1.000	$\left(\partial_j\right)$	$\left(\partial_j\right)$	of the position (x_i) and momentum (∂_j) operators for n independent directions:
gilmore-lie-groups_ F01450	151	■ 1.000	$x_{\{i}, \partial_{\{j}, I$	x_i, ∂_j, I	a. x_i, ∂_j, I .
gilmore-lie-groups_ F01451	151	■ 1.000	$x_{\{i} \partial_{\{j}$	$x_i \partial_j$	b. $x_i \partial_j$.
gilmore-lie-groups_ F01452	151	■ 1.000	$x_{\{i} \partial_{\{j}, x_{\{i}, \partial_{\{j}, I$	$x_i \partial_j, x_i, \partial_j, I$	c. $x_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01453	151	■ 0.991	$x_{\{i} x_{\{j}, x_{\{i} \partial_{\{j}+\frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	$x_i x_j, x_i \partial_j+\frac{1}{2} I \delta_{ij}, \partial_i \partial_j$	d. $x_i x_j, x_i \partial_j+\frac{1}{2} I \delta_{ij}, \partial_i \partial_j$.
gilmore-lie-groups_ F01454	151	■ 1.000	$x_{\{i} x_{\{j}, x_{\{i} \partial_{\{j}, \partial_{\{i} \partial_{\{j}, \partial_{\{i} \partial_{\{j}, x_{\{i}, \partial_{\{j}, I$	$x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$	e. $x_i x_j, x_i \partial_j, \partial_i \partial_j, x_i, \partial_j, I$.
gilmore-lie-groups_ F01455	151	■ 1.000	$\frac{d}{d x}, x \frac{d}{d x}, x^2 \frac{d}{d x}$	$\frac{d}{d x}, x \frac{d}{d x}, x^2 \frac{d}{d x}$	f. $\frac{d}{d x}, x \frac{d}{d x}, x^2 \frac{d}{d x}$.
gilmore-lie-groups_ F01456	151	■ 1.000	$x, x \frac{d}{d x}, x \frac{d^2}{d x^2}$	$x, x \frac{d}{d x}, x \frac{d^2}{d x^2}$	g. $x, x \frac{d}{d x}, x \frac{d^2}{d x^2}$.
gilmore-lie-groups_ F01457	151	■ 0.799	$2+1$	$2+1$	6. The Lorentz, Poincaré, and Galilei groups in $2+1$ dimensions (x, y and t) have Lie
gilmore-lie-groups_ F01458	151	■ 0.799	$(x, y$	$(x, y$	6. The Lorentz, Poincaré, and Galilei groups in $2+1$ dimensions (x, y and t) have Lie
gilmore-lie-groups_ F01459	151	■ 0.799	$t)$	$t)$	6. The Lorentz, Poincaré, and Galilei groups in $2+1$ dimensions (x, y and t) have Lie

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gilmore-lie-groups_ F01460	151	■ 1.000	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Lorentz $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Poincare $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline 0 & 0 & 0 \end{array}\right]$ Galilei
gilmore-lie-groups_ F01461	151	■ 1.000	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline v_1 & v_2 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline v_1 & v_2 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Lorentz $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Poincare $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline 0 & 0 & 0 \end{array}\right]$ Galilei
gilmore-lie-groups_ F01462	151	■ 1.000	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline 0 & 0 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} 0 & \theta & v_1 & t_1 \\ -\theta & 0 & v_2 & t_2 \\ \hline 0 & 0 & 0 & t_3 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Lorentz $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline v_1 & v_2 & 0 \end{array}\right]$ Poincare $\left[\begin{array}{cc c} 0 & \theta & v_1 \\ -\theta & 0 & v_2 \\ \hline 0 & 0 & 0 \end{array}\right]$ Galilei
gilmore-lie-groups_ F01463	152	■ 1.000	$x_{\{i\}}, \partial_{\{j\}}$	x_i, ∂_j	d. For each Lie algebra, express the generators in terms of the operators x_i, ∂_j .
gilmore-lie-groups_ F01464	152	■ 0.781	$b_{\{i\}}^{\dagger}, b_{\{j\}}$	$b_i^{\dagger}, b_j$	e. For each Lie algebra, express the generators in terms of the boson operators $b_i^{\dagger}, b_j$,
gilmore-lie-groups_ F01465	152	■ 1.000	$g_{\{i\ j\}} = C_{\{i\ r\}}^{\{s\}} C_{\{j\ s\}}^{\{r\}}$	$g_{ij} = C_{ir}^s C_{js}^r$	7. In a semisimple Lie algebra the Cartan–Killing metric $g_{ij} = C_{ir}^s C_{js}^r$ is nonsingular and
gilmore-lie-groups_ F01466	152	■ 1.000	$g^{\{i\ j\}}$	g^{ij}	therefore the contravariant metric g^{ij} is well defined. Show that the bilinear operator
gilmore-lie-groups_ F01467	152	■ 0.940	$\mathcal{C}^2 = g^{\{i\ j\}} X_{\{i\}} X_{\{j\}}$	$\mathcal{C}^2 = g^{ij} X_i X_j$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups_ F01468	152	■ 0.940	$\left[\mathcal{C}^2, X_k\right] = 0$	$[\mathcal{C}^2, X_k] = 0$	$\mathcal{C}^2 = g^{ij} X_i X_j$ satisfies $[\mathcal{C}^2, X_k] = 0$. If there is one quadratic Casimir operator, it must
gilmore-lie-groups_ F01469	152	■ 1.000	$\mathcal{C}^2$	$\mathcal{C}^2$	therefore be proportional to $\mathcal{C}^2$.
gilmore-lie-groups_ F01470	152	■ 1.000	$C_{\{i\ j\ k\}} = C_{\{i\ j\}}^{\{r\}} g_{\{r\ k\}}$	$C_{ijk} = C_{ij}^r g_{rk}$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups_ F01471	152	■ 1.000	$C_{\{i\ j\ k\}} = C_{\{j\ k\ i\}} = C_{\{k\ i\ j\}} =$	$C_{ijk} = C_{jki} = C_{kij} =$	8. Show that $C_{ijk} = C_{ij}^r g_{rk}$ is a third order antisymmetric tensor: $C_{ijk} = C_{jki} = C_{kij} =$
gilmore-lie-groups_ F01472	152	■ 1.000	$-C_{\{k\ j\ i\}} = -C_{\{j\ i\ k\}} = -C_{\{i\ k\ j\}}$	$-C_{kji} = -C_{jik} = -C_{ikj}$	$-C_{kji} = -C_{jik} = -C_{ikj}$. (Hint: use the Jacobi identity.)
gilmore-lie-groups_ F01473	154	■ 0.989	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$ have the same complex extension and have their different commutation
gilmore-lie-groups_ F01474	154	■ 1.000	Z, X	Z, X	It would be very useful to find vectors Z, X in the Lie algebra that obeyed the
gilmore-lie-groups_ F01475	154	■ 1.000	$X = \sum_{i=1}^N a^{\{i\}} X_{\{i\}}$	$X = \sum_{i=1}^N a^i X_i$	we write $X = \sum_{i=1}^N a^i X_i$, where X_i form a basis set. Then
gilmore-lie-groups_ F01476	154	■ 1.000	$a^{\{i\}}$	a^i	This equation has a nonzero solution for the coefficients a^i provided the secular
gilmore-lie-groups_ F01477	154	■ 1.000	$\phi_{\{j\}}(Z)$	$\phi_j(Z)$	coefficients $\phi_j(Z)$ are homogeneous polynomials of degree j in the coefficients
gilmore-lie-groups_ F01478	154	■ 0.973	$z^{\{i\}} \left(Z = \sum z^{\{i\}} X_{\{i\}} \right)$	$z^i \left(Z = \sum z^i X_i \right)$	$z^i (Z = \sum z^i X_i)$ that describe Z :
gilmore-lie-groups_ F01479	155	■ 1.000	$a^{\dagger}, a, I = \left[a, a^{\dagger} \right]$	$a^{\dagger}, a, I = [a, a^{\dagger}]$	$a^{\dagger}, a, I = [a, a^{\dagger}]$ is

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gilmore-lie-groups_ F01480	155	■ 1.000	$(-\lambda)^N=0$	$(-\lambda)^N = 0$	The converse is true. If the secular equation of an $N \times N$ matrix is $(-\lambda)^N = 0$, then
gilmore-lie-groups_ F01481	155	■ 0.852	$X=\sum a_{\mathrm{i}} X_{\mathrm{i}} \in \mathfrak{su}(2)$	$X = \sum a_i X_i \in \mathfrak{su}(2)$	Example For $X = \sum a_i X_i \in \mathfrak{su}(2)$ the defining 2×2 matrix representation
gilmore-lie-groups_ F01482	155	■ 0.953	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$	$\mathfrak{def}(X)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(X)$ are
gilmore-lie-groups_ F01483	155	■ 0.953	$\mathfrak{Reg}(X)$	$\mathfrak{Reg}(X)$	$\mathfrak{def}(X)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(X)$ are
gilmore-lie-groups_ F01484	155	■ 1.000	$\phi_2(\mathbf{a}) \geq 0$	$\phi_2(\mathbf{a}) \geq 0$	Since $\phi_2(\mathbf{a}) \geq 0$, this secular equation cannot be solved over the field of real num-
gilmore-lie-groups_ F01485	155	■ 0.760	$\mathfrak{su}(2))$	$\mathfrak{su}(2))$	ization to find the three (three is the dimension of $\mathfrak{su}(2))$ roots: $\lambda = 0, \lambda = \pm ia$,
gilmore-lie-groups_ F01486	155	■ 0.760	$\lambda=0, \lambda=\pm ia$	$\lambda = 0, \lambda = \pm ia$	ization to find the three (three is the dimension of $\mathfrak{su}(2))$ roots: $\lambda=0, \lambda=\pm ia$,
gilmore-lie-groups_ F01487	155	■ 0.999	$a^2=+a_1^2+a_2^2+a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$	$a^2 = +a_1^2 + a_2^2 + a_3^2$.
gilmore-lie-groups_ F01488	156	■ 0.861	$Y=\sum b_{\mathrm{i}} Y_{\mathrm{i}} \in \mathfrak{su}(1,1)$	$Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$	Example For $Y = \sum b_i Y_i \in \mathfrak{su}(1,1)$ the defining 2×2 matrix representation
gilmore-lie-groups_ F01489	156	■ 0.976	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups_ F01490	156	■ 0.976	$\mathfrak{Reg}(Y)$	$\mathfrak{Reg}(Y)$	$\mathfrak{def}(Y)$ and the regular 3×3 matrix representation $\mathfrak{Reg}(Y)$ are
gilmore-lie-groups_ F01491	156	■ 0.999	$\left(a_{\mathrm{1}}, a_{\mathrm{2}}, a_{\mathrm{3}}\right) \rightarrow\left(i b_{\mathrm{1}}, i b_{\mathrm{2}}, b_{\mathrm{3}}\right)$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$, the coefficient $\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$ of the secular
gilmore-lie-groups_ F01492	156	■ 0.999	$\phi_2(\mathbf{a})=a_1^2+a_2^2+a_3^2$	$\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$	$(a_1, a_2, a_3) \rightarrow (ib_1, ib_2, b_3)$, the coefficient $\phi_2(\mathbf{a}) = a_1^2 + a_2^2 + a_3^2$ of the secular
gilmore-lie-groups_ F01493	156	■ 0.953	$\phi_2(\mathbf{b})=-b_1^2-b_2^2+b_3^2$	$\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2$	equation for $\mathfrak{su}(2)$ maps to $\phi_2(\mathbf{b}) = -b_1^2 - b_2^2 + b_3^2$ for $\mathfrak{su}(1,1)$. This same rotation
gilmore-lie-groups_ F01494	156	■ 1.000	ϕ_2	ϕ_2	For each algebra the functional forms of the nonzero coefficient ϕ_2 in the secular
gilmore-lie-groups_ F01495	157	■ 0.515	$\mathfrak{Reg}$	$\mathfrak{Reg}$	equation of its regular representation, $\mathfrak{Reg}$. Since the number of independent roots
gilmore-lie-groups_ F01496	157	■ 1.000	$\phi_j\left(z^i\right)$	$\phi_j\left(z^i\right)$	of the secular equation is equal to the number of independent coefficients $\phi_j(z^i)$,
gilmore-lie-groups_ F01497	157	■ 1.000	$X=Z$	$X = Z$	least one zero root (put $X = Z$ in (9.1)). For simple Lie algebras of dimension N ,
gilmore-lie-groups_ F01498	157	■ 1.000	$l^2 \sim N$	$l^2 \sim N$	$l^2 \sim N$, so describing commutation relations in terms of rank rather than dimension
gilmore-lie-groups_ F01499	157	■ 1.000	$\mathfrak{m}$	$\mathfrak{m}$	If $\mathfrak{m}$ is some matrix Lie algebra of $n \times n$ matrices, then any operator in $\mathfrak{m}$ can be
gilmore-lie-groups_ F01500	158	■ 1.000	$\phi_j(X)$	$\phi_j(X)$	Figure 9.2. Invariant operators $\phi_j(X)$ expressed in terms of the fully antisymmetric
gilmore-lie-groups_ F01501	158	■ 0.955	a^r_{s}	a^r_s	the coordinates a^r_s by the operators X^r_s in the coefficients ϕ_j of the secular equation.
gilmore-lie-groups_ F01502	158	■ 0.955	X^r_{s}	X^r_s	the coordinates a^r_s by the operators X^r_s in the coefficients ϕ_j of the secular equation.
gilmore-lie-groups_ F01503	158	■ 0.955	ϕ_j	ϕ_j	the coordinates a^r_s by the operators X^r_s in the coefficients ϕ_j of the secular equation.
gilmore-lie-groups_ F01504	158	■ 0.836	$X=a_{\mathrm{s}}^r X^r_{\mathrm{s}}$	$X = a_s^r X^r_s$	Here the general element in the Lie algebra is $X = a_s^r X^r_s$.
gilmore-lie-groups_ F01505	158	■ 0.751	$R^3,=\epsilon_{i j k}=+1$	$R^3, = \epsilon_{ijk} = +1$	(e.g., in $R^3, = \epsilon_{ijk} = +1$ for (ijk) a cyclic permutation of (123), -1 for a cyclic
gilmore-lie-groups_ F01506	158	■ 0.751	$(i j k)$	(ijk)	(e.g., in $R^3, = \epsilon_{ijk} = +1$ for (ijk) a cyclic permutation of (123), -1 for a cyclic

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gilmore-lie-groups_F01507	159	■ 1.000	$\phi_{\{j\}}\left(X^{\{r\}}_{\{s\}}\right)$	$\phi_j(X^r_s)$	these invariant operators $\phi_j(X^r_s)$ remain invariant when the matrices are replaced
gilmore-lie-groups_F01508	159	■ 1.000	$o(n)$	$O(n)$	Example The orthogonal groups $O(n)$ and their subgroups $SO(n)$ have Lie alge-
gilmore-lie-groups_F01509	159	■ 0.651	$d \times d$	$d \times d$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n - 1)/2$) regular matrix representation
gilmore-lie-groups_F01510	159	■ 0.651	$\mathfrak{so}(n)$	$\mathfrak{so}(n)$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n - 1)/2$) regular matrix representation
gilmore-lie-groups_F01511	159	■ 0.651	$d=n(n-1) / 2$	$d = n(n - 1)/2$	the $d \times d$ (the dimension of $\mathfrak{so}(n)$ is $d = n(n - 1)/2$) regular matrix representation
gilmore-lie-groups_F01512	159	■ 1.000	$\mathfrak{def}(X)^t = -\mathfrak{def}(X)$	$\mathfrak{def}(X)^t = -\mathfrak{def}(X)$	Lie algebra consists of antisymmetric matrices, $\mathfrak{def}(X)^t = -\mathfrak{def}(X)$, and we find
gilmore-lie-groups_F01513	159	■ 0.986	$[n / 2]$	$[n/2]$	even coefficients. Therefore the algebra $\mathfrak{so}(n)$ has rank $[n/2]$.
gilmore-lie-groups_F01514	159	■ 0.996	$j=2$	$j = 2$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01515	159	■ 0.996	$X_{\{i\}j} = -X_{\{j\}i}$	$X_{ij} = -X_{ji}$	setting $j = 2$ in Fig. 9.2 for the generators X_{ij} of $SO(n)$. Since $X_{ij} = -X_{ji}$, it is
gilmore-lie-groups_F01516	159	■ 1.000	$\mathfrak{so}(5)$	$\mathfrak{so}(5)$	higher order Casimir invariant operators. For example, for $\mathfrak{so}(5)$ the fourth order
gilmore-lie-groups_F01517	159	■ 1.000	$X^{\{r\}}_{\{s\}} = -X^{\{s\}}_{\{r\}}$	$X^r_s = -X^s_r$	Figure 9.3. If the operators X are antisymmetric, $X^r_s = -X^s_r$, contractions in the
gilmore-lie-groups_F01518	160	■ 0.940	$v^{\{m\}} = \epsilon^{\{ijklm\}} X_{\{i\}j} X_{\{k\}l}$	$v^m = \epsilon^{ijklm} X_{ij} X_{kl}$	where the components of the five-vector $\mathbf{v}$ are $v^m = \epsilon^{ijklm} X_{ij} X_{kl}$, for example
gilmore-lie-groups_F01519	160	■ 1.000	$n / 2$	$n/2$	square. Its square root, of order $n/2$, should be taken as an appropriate functionally
gilmore-lie-groups_F01520	160	■ 1.000	$Z=0$	$Z = 0$	the opposite extreme, the choice $Z = 0$ is not clever since all X obey the same
gilmore-lie-groups_F01521	160	■ 1.000	$[Z, X]=0 \ X$	$[Z, X] = 0X$	eigenvalue equation $[Z, X] = 0X$.)
gilmore-lie-groups_F01522	160	■ 1.000	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^{\dagger}\}$	$\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$	dimensional algebra of two photon operators spanned by $\hat{n} + \frac{1}{2} = \frac{1}{2} \{a, a^\dagger\}$,
gilmore-lie-groups_F01523	160	■ 1.000	$a^{\dagger 2}, a^\dagger, I = \left[a, a^{\dagger} \right], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$	$a^{\dagger 2}, a^\dagger, I = [a, a^\dagger], a, a^2$. A useful choice for Z is
gilmore-lie-groups_F01524	161	■ 1.000	$\pm 2z_1, \pm z_1, 0, 0$	$\pm 2z_1, \pm z_1, 0, 0$	representation are $\pm 2z_1, \pm z_1, 0, 0$ and the commutation relations can be summa-
gilmore-lie-groups_F01525	161	■ 0.999	$X_{\{2,0\}} = a^{\dagger 2}, X_{\{1,0\}} = a^\dagger, X_{\{0,0\}} = \hat{n} + \frac{1}{2} I, X_{\{-1,0\}} = a, X_{\{-2,0\}} = a^2$	$X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$	where $X_{(2,0)} = a^{\dagger 2}, X_{(1,0)} = a^\dagger, X_{(0,0)} = \hat{n} + \frac{1}{2} I, X_{(-1,0)} = a, X_{(-2,0)} = a^2$. We
gilmore-lie-groups_F01526	161	■ 1.000	$k, l \in \{-2, -1, 0, +1, +2\}$	$k, l \in \{-2, -1, 0, +1, +2\}$	also see that if $k, l \in \{-2, -1, 0, +1, +2\}$
gilmore-lie-groups_F01527	161	■ 1.000	$k+l$	$k + l$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01528	161	■ 1.000	$\{-2, -1, 0, +1, +2\}$	$\{-2, -1, 0, +1, +2\}$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01529	161	■ 1.000	$k+l=0$	$k + l = 0$	if $k + l$ is in the set $\{-2, -1, 0, +1, +2\}$ and zero otherwise. If $k + l = 0$ the
gilmore-lie-groups_F01530	161	■ 0.968	$(0,0): \hat{n} + \frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$	$(0,0): \hat{n} + \frac{1}{2}$ and I .
gilmore-lie-groups_F01531	162	■ 1.000	$\alpha_1, \alpha_2, \dots, \alpha_l$	$\alpha_1, \alpha_2, \dots, \alpha_l$	(ii) the number of independent roots $\alpha_1, \alpha_2, \dots, \alpha_l$ of the secular equation; these l inde-
gilmore-lie-groups_F01532	162	■ 0.570	$\left(\alpha_1, \alpha_2, \dots, \alpha_l \right)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$	$(\alpha_1, \alpha_2, \dots, \alpha_l)$ in a root space ;

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gilmore-lie-groups_F01533	162	■ 1.000	V_{β}	V_{β}	(i) If α and β are roots with subspaces V_{α} and V_{β} in the Lie algebra, then
gilmore-lie-groups_F01534	162	■ 1.000	$V_{\alpha+\beta}$	$V_{\alpha+\beta}$	That is, the commutator of any vector in V_{α} with any vector in V_{β} is a vector in $V_{\alpha+\beta}$.
gilmore-lie-groups_F01535	162	■ 1.000	$\alpha+\beta$	$\alpha+\beta$	If $\alpha+\beta$ is not a root, the commutator vanishes.
gilmore-lie-groups_F01536	162	■ 1.000	$H_1, H_2, \dots, H_l$	$H_1, H_2, \dots, H_l$	(ii) The l basis vectors $H_1, H_2, \dots, H_l$ in the l -dimensional subspace V_0 commute:
gilmore-lie-groups_F01537	162	■ 1.000	$V_{\alpha} (\alpha \neq 0)$	$V_{\alpha} (\alpha \neq 0)$	(iii) Each subspace V_{α} ($\alpha \neq 0$) is one dimensional. Therefore each subspace V_{α} is spanned
gilmore-lie-groups_F01538	162	■ 0.995	E_{α}	E_{α}	by an operator E_{α} that can be labeled by the root α . As a result (i.e., $[V_0, V_{\alpha}] \subset V_{\alpha}$),
gilmore-lie-groups_F01539	162	■ 0.995	$\left[V_0, V_{\alpha}\right] \subset V_{\alpha}$	$[V_0, V_{\alpha}] \subset V_{\alpha}$	by an operator E_{α} that can be labeled by the root α . As a result (i.e., $[V_0, V_{\alpha}] \subset V_{\alpha}$),
gilmore-lie-groups_F01540	162	■ 1.000	$-\alpha$	$-\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01541	162	■ 1.000	$c \alpha$	$c\alpha$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01542	162	■ 1.000	$ c =1$	$ c = 1$	(iv) If α is a root, $-\alpha$ is a root. If α is a root and $c\alpha$ is a root, then $ c = 1$. Thus, nonzero
gilmore-lie-groups_F01543	162	■ 1.000	$E_{-\alpha}$	$E_{-\alpha}$	(v) The commutator of E_{α} and $E_{-\alpha}$ is in V_0 , so can be expanded as a linear superposition
gilmore-lie-groups_F01544	162	■ 1.000	α^i	α^i	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij} \alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01545	162	■ 1.000	α_j	α_j	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij} \alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01546	162	■ 1.000	$\alpha^i = h^{ij} \alpha_j$	$\alpha^i = h^{ij} \alpha_j$	(vi) An inner product relating α^i and α_j by $\alpha^i = h^{ij} \alpha_j$ can be introduced in this root space
gilmore-lie-groups_F01547	163	■ 1.000	$h^{ij} = \delta^{ij}$	$h^{ij} = \delta^{ij}$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01548	163	■ 1.000	$\alpha_i: \alpha^i = \alpha_i$	$\alpha_i: \alpha^i = \alpha_i$	then $h^{ij} = \delta^{ij}$ and we can identify α^i with α_i : $\alpha^i = \alpha_i$.
gilmore-lie-groups_F01549	163	■ 1.000	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$	$N_{\alpha, \beta}$ when $\alpha + \beta$ is a nonzero root.
gilmore-lie-groups_F01550	163	■ 1.000	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$E_{\alpha}, E_{\beta}, E_{\gamma}$	$E_{\alpha}, E_{\beta}, E_{\gamma}$ of three nonzero roots that sum to zero
gilmore-lie-groups_F01551	163	■ 1.000	$\alpha, \beta+k \alpha$	$\alpha, \beta+k \alpha$	are not roots. By applying the Jacobi identity to roots $\alpha, \beta+k \alpha$, and $-\alpha$ we obtain
gilmore-lie-groups_F01552	164	■ 1.000	$\left[E_{\alpha}, E_{\beta}\right]=N_{\alpha, \beta} E_{\alpha+\beta}$	$[E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha+\beta}$	in commutators $[E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha+\beta}$.
gilmore-lie-groups_F01553	164	■ 1.000	$N_{\alpha, \beta+n \alpha}=0$	$N_{\alpha, \beta+n \alpha}=0$	The initial condition $N_{\alpha, \beta+n \alpha}=0$ leads to
gilmore-lie-groups_F01554	164	■ 1.000	$N_{-\alpha, \beta-m \alpha}^2=N_{\alpha, \beta-(m+1) \alpha}^2=0$	$N_{-\alpha, \beta-m \alpha}^2=N_{\alpha, \beta-(m+1) \alpha}^2=0$	The other boundary condition $N_{-\alpha, \beta-m \alpha}^2=N_{\alpha, \beta-(m+1) \alpha}^2=0$ leads to
gilmore-lie-groups_F01555	164	■ 1.000	$N_{\alpha, \beta+k \alpha}^2=(n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$	$N_{\alpha, \beta+k \alpha}^2=(n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$	(i) $N_{\alpha, \beta+k \alpha}^2=(n-k)(m+k+1)(\alpha \cdot \alpha) / 2 \geq 0$. We use this expression because it shows
gilmore-lie-groups_F01556	164	■ 1.000	$\alpha \cdot \beta>0$	$\alpha \cdot \beta>0$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta>0$ when $m -$
gilmore-lie-groups_F01557	164	■ 1.000	$m -$	$m -$	clearly how the boundary conditions are imposed. We note that $\alpha \cdot \beta>0$ when $m -$
gilmore-lie-groups_F01558	164	■ 1.000	$n>0$	$n>0$	$n>0$ and $\alpha \cdot \beta<0$ when $m-n<0$.
gilmore-lie-groups_F01559	164	■ 1.000	$\alpha \cdot \beta<0$	$\alpha \cdot \beta<0$	$n>0$ and $\alpha \cdot \beta<0$ when $m-n<0$.

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gilmore-lie-groups_ F01560	164	■ 1.000	$m-n<0$	$m-n<0$	$n>0$ and $\alpha\cdot\beta<0$ when $m-n<0$.
gilmore-lie-groups_ F01561	165	■ 0.998	$A_{\{2\}}, B_{\{2\}}, C_{\{2\}}$	A_2, B_2, C_2	Figure 9.6. Two-dimensional root space diagrams. Top: A_2, B_2, C_2 . Bottom: D_2, G_2 .
gilmore-lie-groups_ F01562	165	■ 0.998	$D_{\{2\}}, G_{\{2\}}$	D_2, G_2	Figure 9.6. Two-dimensional root space diagrams. Top: A_2, B_2, C_2 . Bottom: D_2, G_2 .
gilmore-lie-groups_ F01563	165	■ 0.456	$l(l$	$l(l$	constants of a semisimple Lie algebra. The basis vectors are the l (l is the rank)
gilmore-lie-groups_ F01564	165	■ 0.977	$\mathbf{H}=\left(H_{\{1\}}, H_{\{2\}}, \ldots, H_{\{l\}}\right)$	$\mathbf{H}=(H_1, H_2, \dots, H_l)$	operators $\mathbf{H}=(H_1, H_2, \dots, H_l)$ and the “shift” operators E_α , one corresponding
gilmore-lie-groups_ F01565	165	■ 0.994	$\alpha=\left(\alpha_{\{1\}}, \alpha_{\{2\}}, \ldots, \alpha_{\{l\}}\right)$	$\alpha=(\alpha_1, \alpha_2, \dots, \alpha_l)$	to each nonzero root. The root vector $\alpha=(\alpha_1, \alpha_2, \dots, \alpha_l)$ has l components. The
gilmore-lie-groups_ F01566	166	■ 1.000	$C_{\{2\}}$	C_2	Figure 9.7. Root space C_2 . The roots are expressed in terms of a Cartesian
gilmore-lie-groups_ F01567	166	■ 1.000	$\mathbf{e}_{\{1\}}, \mathbf{e}_{\{2\}}$	$\mathbf{e}_1, \mathbf{e}_2$	(orthogonal) set $\mathbf{e}_1, \mathbf{e}_2$ and a fundamental set α_1, α_2 .
gilmore-lie-groups_ F01568	166	■ 1.000	$\alpha_{\{1\}}, \alpha_{\{2\}}$	α_1, α_2	(orthogonal) set $\mathbf{e}_1, \mathbf{e}_2$ and a fundamental set α_1, α_2 .
gilmore-lie-groups_ F01569	166	■ 1.000	$\mathbf{e}_{\{1\}}$	$\mathbf{e}_1$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F01570	166	■ 1.000	$\mathbf{e}_{\{2\}}$	$\mathbf{e}_2$	If we choose orthogonal vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ in a two-dimensional Euclidean space,
gilmore-lie-groups_ F01571	166	■ 1.000	$\pm 2 \mathbf{e}_{\{1\}}, \pm 2 \mathbf{e}_{\{2\}}, \pm \mathbf{e}_{\{1\}} \pm \mathbf{e}_{\{2\}}$	$\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm\mathbf{e}_1 \pm \mathbf{e}_2$	the nonzero roots for C_2 are $\pm 2\mathbf{e}_1, \pm 2\mathbf{e}_2, \pm\mathbf{e}_1 \pm \mathbf{e}_2$. The 10 basis vectors in the Lie
gilmore-lie-groups_ F01572	166	■ 0.986	$H_{\{i\}}, i=1,2$	$H_i, i=1,2$	algebra are $H_i, i=1,2$, and E_α , with α the eight nonzero roots. We normalize
gilmore-lie-groups_ F01573	166	■ 1.000	$\sum \alpha \cdot \alpha=2$	$\sum \alpha \cdot \alpha=2$	these roots by $\sum \alpha \cdot \alpha=2$ so that
gilmore-lie-groups_ F01574	167	■ 0.608	$\left[H_{\{1\}}, H_{\{2\}}\right]$	$[H_1, H_2]$	$ \begin{array}{ll} [H_1, H_2] & = 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] & = (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] & = (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] & = (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] & = (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] & = (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] & = *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] & = (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] & = (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] & = (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] & = (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] & = (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} $

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gilmore-lie-groups_ F01575	167	■ 0.608	$\left[\mathbf{H}, E_{\pm 2\mathbf{e}_1}\right]$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	$ \begin{array}{lcl} [H_1, H_2] & = & 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] & = & (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] & = & (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] & = & (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] & = & (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] & = & (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] & = & *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} $
gilmore-lie-groups_ F01576	167	■ 0.608	$(\pm 2/\sqrt{12}, 0) E_{\pm 2\mathbf{e}_1}$	$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$	$ \begin{array}{lcl} [H_1, H_2] & = & 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] & = & (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] & = & (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] & = & (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] & = & (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] & = & (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] & = & *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01577	167	■ 0.608	$\left[\mathbf{H}, E_{\pm 2} \mathbf{e}_2\right]$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	$ \begin{array}{lcl} [H_1, H_2] & = & 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] & = & (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] & = & (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] & = & (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] & = & (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] & = & (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] & = & *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} $
gilmore-lie-groups_ F01578	167	■ 0.608	$(0, \pm 2/\sqrt{12}) E_{\pm 2} \mathbf{e}_2$	$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$	$ \begin{array}{lcl} [H_1, H_2] & = & 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] & = & (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] & = & (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] & = & (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] & = & (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] & = & (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] & = & *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] & = & *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] & = & (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] & = & (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] & = & (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] & = & (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01579	167	■ 0.608	$\left[\mathbf{H}, E_{\pm \sqrt{12}} \mathbf{e}_1 \pm \sqrt{12} \mathbf{e}_2\right]$	$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01580	167	■ 0.608	$(\pm \sqrt{12}, \pm \sqrt{12}) E_{\pm \sqrt{12} \mathbf{e}_1 \pm \sqrt{12} \mathbf{e}_2}$	$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01581	167	■ 0.608	$\left[E_{+2} \mathrm{e}_1, E_{-2} \mathrm{e}_1\right]$	$[E_{+2e_1}, E_{-2e_1}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $
gilmore-lie-groups_ F01582	167	■ 0.608	$(2 / \sqrt{12}) H_1$	$(2/\sqrt{12})H_1$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $

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gilmore-lie-groups_ F01583	167	■ 0.608	$\left[E_{+2} \mathrm{e}_2, E_{-2} \mathrm{e}_2\right]$	$[E_{+2e_2}, E_{-2e_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $
gilmore-lie-groups_ F01584	167	■ 0.608	$(2 / \sqrt{12}) H_2$	$(2/\sqrt{12})H_2$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01585	167	■ 0.608	$\left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}\right]$	$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01586	167	■ 0.608	$(1/\sqrt{12})\left(\pm H_1 \pm H_2\right)$	$(1/\sqrt{12})(\pm H_1 \pm H_2)$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01587	167	■ 0.608	$\left[E_{+2} \mathbf{e}_1, E_{-\left(\mathbf{e}_1 + \mathbf{e}_2\right)} \mathbf{e}_2\right]$	$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01588	167	■ 0.608	$*(1/\sqrt{6}) E_{\mathbf{e}_1 - \mathbf{e}_2}$	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01589	167	■ 0.608	$\left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2} \right]$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01590	167	■ 0.608	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01591	167	■ 0.608	$\left[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}\right]$	$[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01592	167	■ 0.608	$(1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2}$	$(1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01593	167	■ 0.608	$\left[E_{-2} \mathbf{e}_1, E_{\mathbf{e}_1} - \mathbf{e}_2\right]$	$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01594	167	■ 0.608	$(-1/\sqrt{6}) E_{-\mathbf{e}_1 - \mathbf{e}_2}$	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01595	167	■ 0.608	$\left[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01596	167	■ 0.608	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01597	167	■ 0.608	$\left[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01598	167	■ 0.608	$(-1/\sqrt{6}) E_{+\mathbf{e}_1-\mathbf{e}_2}$	$(-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01599	167	■ 0.608	$\left[E_{+2} \mathbf{e}_2, E_{-1} \mathbf{e}_1 - \mathbf{e}_2\right]$	$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01600	167	■ 0.608	$*(1/\sqrt{6}) E_{-\mathbf{e}_1 + \mathbf{e}_2}$	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01601	167	■ 0.608	$\left[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}\right]$	$[E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01602	167	■ 0.608	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01603	167	■ 0.608	$\left[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}\right]$	$[E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01604	167	■ 0.608	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ [E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01605	167	■ 0.608	$\left[E_{-2} \mathbf{e}_2\right], E_{+} \mathbf{e}_1 + \mathbf{e}_2\right]$	$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $
gilmore-lie-groups_ F01606	167	■ 0.608	$\left[E_{+} \mathbf{e}_1 + \mathbf{e}_2\right], E_{-} \mathbf{e}_1 - \mathbf{e}_2\right]$	$[E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ [\mathbf{H}, E_{\pm 2\mathbf{e}_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ [\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ [E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1+\mathbf{e}_2)}] &= *(1/\sqrt{6})E_{\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_1}] &= (1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1-\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1-\mathbf{e}_2}] &= *(1/\sqrt{6})E_{-\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-\mathbf{e}_1-\mathbf{e}_2}, E_{+\mathbf{e}_1-\mathbf{e}_2}] &= (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ [E_{+\mathbf{e}_1-\mathbf{e}_2}, E_{+2\mathbf{e}_2}] &= (1/\sqrt{6})E_{+\mathbf{e}_1+\mathbf{e}_2} \\ [E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+\mathbf{e}_1-\mathbf{e}_2} \\ [E_{+\mathbf{e}_1+\mathbf{e}_2}, E_{-\mathbf{e}_1+\mathbf{e}_2}] &= (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ [E_{-\mathbf{e}_1+\mathbf{e}_2}, E_{-2\mathbf{e}_2}] &= (-1/\sqrt{6})E_{-\mathbf{e}_1-\mathbf{e}_2} \end{aligned} $

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gilmore-lie-groups_ F01607	167	■ 0.608	$(-1 / \sqrt{6}) E_{+2} \mathrm{e}_{\{2\}}$	$(-1/\sqrt{6})E_{+2e_2}$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $
gilmore-lie-groups_ F01608	167	■ 0.608	$\left[E_{-\mathbf{e}_{\{1\}}+\mathbf{e}_{\{2\}}}, E_{-2\mathbf{e}_{\{2\}}}\right]$	$[E_{-e_1+e_2}, E_{-2e_2}]$	$ \begin{aligned} [H_1, H_2] &= 0 \\ [\mathbf{H}, E_{\pm 2e_1}] &= (\pm 2/\sqrt{12}, 0)E_{\pm 2e_1} \\ [\mathbf{H}, E_{\pm 2e_2}] &= (0, \pm 2/\sqrt{12})E_{\pm 2e_2} \\ [\mathbf{H}, E_{\pm e_1 \pm e_2}] &= (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm e_1 \pm e_2} \\ [E_{+2e_1}, E_{-2e_1}] &= (2/\sqrt{12})H_1 \\ [E_{+2e_2}, E_{-2e_2}] &= (2/\sqrt{12})H_2 \\ [E_{\pm e_1 \pm e_2}, E_{-(\pm e_1 \pm e_2)}] &= (1/\sqrt{12})(\pm H_1 \pm H_2) \\ [E_{+2e_1}, E_{-(e_1+e_2)}] &= *(1/\sqrt{6})E_{e_1-e_2} \\ [E_{-e_1+e_2}, E_{-e_1-e_2}] &= (1/\sqrt{6})E_{-2e_1} \\ [E_{-e_1-e_2}, E_{+2e_1}] &= (1/\sqrt{6})E_{+e_1-e_2} \\ [E_{-2e_1}, E_{e_1-e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \\ [E_{+e_1-e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_1} \\ [E_{+e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+2e_2}, E_{-e_1-e_2}] &= *(1/\sqrt{6})E_{-e_1+e_2} \\ [E_{-e_1-e_2}, E_{+e_1-e_2}] &= (1/\sqrt{6})E_{-2e_2} \\ [E_{+e_1-e_2}, E_{+2e_2}] &= (1/\sqrt{6})E_{+e_1+e_2} \\ [E_{-2e_2}, E_{+e_1+e_2}] &= (-1/\sqrt{6})E_{+e_1-e_2} \\ [E_{+e_1+e_2}, E_{-e_1+e_2}] &= (-1/\sqrt{6})E_{+2e_2} \\ [E_{-e_1+e_2}, E_{-2e_2}] &= (-1/\sqrt{6})E_{-e_1-e_2} \end{aligned} $
gilmore-lie-groups_ F01609	168	■ 1.000	$\mathfrak{sl}(2; \mathbb{C}))$	$\mathfrak{sl}(2; \mathbb{C}))$	extension $\mathfrak{sl}(2; \mathbb{C}))$. Root spaces classify commutation relations of these complex
gilmore-lie-groups_ F01610	168	■ 1.000	α_i	α_i	real subalgebras of these complex algebras – some roots α_i and structure constants
gilmore-lie-groups_ F01611	168	■ 1.000	$\frac{1}{2}\{a^\dagger, a\}$	$\frac{1}{2}\{a^\dagger, a\}$	1. Construct the regular representation for the two-photon operator algebra: $\frac{1}{2}\{a^\dagger, a\}$,
gilmore-lie-groups_ F01612	168	■ 1.000	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$	$a^{\dagger^2}, a^\dagger, I, a, a^2$. Determine the secular equation for this matrix. Determine the rank
gilmore-lie-groups_ F01613	168	■ 1.000	$X_{\{i\ j\}}=x^{\{i\}}\partial_{\{j\}}-x^{\{j\}}\partial_{\{i\}}(1\leq i<j\leq 4)$	$X_{ij}=x^i\partial_j-x^j\partial_i(1\leq i<j\leq 4)$	ical basis $X_{ij}=x^i\partial_j-x^j\partial_i(1\leq i<j\leq 4)$ these operators are $\mathcal{C}_2=\sum_{i<j}X_{ij}^2$ and
gilmore-lie-groups_ F01614	168	■ 1.000	$\mathcal{C}_2=\sum_{i<j}X_{\{i\ j\}}^2$	$\mathcal{C}_2=\sum_{i<j}X_{ij}^2$	ical basis $X_{ij}=x^i\partial_j-x^j\partial_i(1\leq i<j\leq 4)$ these operators are $\mathcal{C}_2=\sum_{i<j}X_{ij}^2$ and

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gilmore-lie-groups_F01615	168	■ 0.980	$\mathcal{C}_2^{\prime} = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$C_2' = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$	$C_2' = X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23}$.
gilmore-lie-groups_F01616	168	■ 0.994	$\mathfrak{su}(4)$	$\mathfrak{su}(4)$	3. The Lie algebra $\mathfrak{su}(4)$ has a 4×4 defining matrix representation and a 15×15 regular
gilmore-lie-groups_F01617	168	■ 1.000	$\mathfrak{so}(2n+1)$	$\mathfrak{so}(2n+1)$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree $2, 4, \dots, 2n$.
gilmore-lie-groups_F01618	168	■ 1.000	$2, 4, \dots, 2n$	$2, 4, \dots, 2n$	4. For $\mathfrak{so}(2n+1)$ the invariant operators (Casimir operators) are of degree $2, 4, \dots, 2n$.
gilmore-lie-groups_F01619	168	■ 1.000	$\mathfrak{so}(2n)$	$\mathfrak{so}(2n)$	This is true also for $\mathfrak{so}(2n)$, with one difference: the invariant operator of degree $2n$
gilmore-lie-groups_F01620	168	■ 0.600	$\mathcal{C}_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	$C_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$	$C_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out C_2' for $\mathfrak{so}(4)$ and C_3' for $\mathfrak{so}(6)$.
gilmore-lie-groups_F01621	168	■ 0.600	$\mathcal{C}_2'$	C_2'	$C_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out C_2' for $\mathfrak{so}(4)$ and C_3' for $\mathfrak{so}(6)$.
gilmore-lie-groups_F01622	168	■ 0.600	$\mathcal{C}_3'$	C_3'	$C_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out C_2' for $\mathfrak{so}(4)$ and C_3' for $\mathfrak{so}(6)$.
gilmore-lie-groups_F01623	168	■ 0.600	$\mathfrak{so}(6)$	$\mathfrak{so}(6)$	$C_n' = \epsilon^{i_1 i_2 \dots i_{2n}} X_{i_1 i_2} X_{i_3 i_4} \dots X_{i_{2n-1} i_{2n}}$. Explicitly write out C_2' for $\mathfrak{so}(4)$ and C_3' for $\mathfrak{so}(6)$.
gilmore-lie-groups_F01624	168	■ 1.000	$\mathfrak{su}(4) = \mathfrak{so}(6)$	$\mathfrak{su}(4) = \mathfrak{so}(6)$	5. In Chapter 11 we will show that $\mathfrak{su}(4) = \mathfrak{so}(6)$. Both Lie algebras have invariant
gilmore-lie-groups_F01625	168	■ 1.000	$a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$	$a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$	two modes. This algebra is ten dimensional. It contains the four operators $a_i^\dagger a_j + \frac{1}{2} \delta_{ij}$
gilmore-lie-groups_F01626	169	■ 0.999	$(1 \leq i, j \leq 2)$	$(1 \leq i, j \leq 2)$	$1 \leq i, j \leq 2$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_F01627	169	■ 0.999	$a_i^\dagger a_j^\dagger$	$a_i^\dagger a_j^\dagger$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_F01628	169	■ 0.999	$a_i a_j (a_i a_j = a_j a_i)$	$a_i a_j (a_i a_j = a_j a_i)$	$(1 \leq i, j \leq 2)$ and the two pairs of three operators $a_i^\dagger a_j^\dagger$ and $a_i a_j$ ($a_i a_j = a_j a_i$).
gilmore-lie-groups_F01629	169	■ 1.000	$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+e_i+e_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+e_i-e_j} (i \neq j)$	$a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+e_i+e_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+e_i-e_j} (i \neq j)$	identification is: $a_i^\dagger a_i + \frac{1}{2} \leftrightarrow H_i, a_i^\dagger a_j^\dagger \leftrightarrow E_{+e_i+e_j} (i \neq j), a_i^\dagger a_j \leftrightarrow E_{+e_i-e_j} (i \neq j)$,
gilmore-lie-groups_F01630	169	■ 1.000	$a_i a_j \leftrightarrow E_{-e_i-e_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2e_i}, a_i a_i \leftrightarrow E_{-2e_i}$	$a_i a_j \leftrightarrow E_{-e_i-e_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2e_i}, a_i a_i \leftrightarrow E_{-2e_i}$	$a_i a_j \leftrightarrow E_{-e_i-e_j} (i \neq j), a_i^\dagger a_i^\dagger \leftrightarrow E_{+2e_i}, a_i a_i \leftrightarrow E_{-2e_i}$.
gilmore-lie-groups_F01631	169	■ 1.000	D_2	D_2	is six dimensional. Show that the resulting root space diagram is D_2 (Fig. 9.6). Why
gilmore-lie-groups_F01632	169	■ 0.995	$f_i^\dagger f_i^\dagger = 0$	$f_i^\dagger f_i^\dagger = 0$	the difference? (Hint: $f_i^\dagger f_i^\dagger = 0$.)
gilmore-lie-groups_F01633	169	■ 1.000	$X, Y \in \mathfrak{su}(2)$	$X, Y \in \mathfrak{su}(2)$	ular representation for the former. Choose $X, Y \in \mathfrak{su}(2)$ and compute $(X, Y) =$
gilmore-lie-groups_F01634	169	■ 0.577	$\text{tr}[\mathfrak{def}(X)\mathfrak{def}(Y)]$	$\text{tr}[\mathfrak{def}(X)\mathfrak{def}(Y)]$	$\text{tr}[\mathfrak{def}(X)\mathfrak{def}(Y)]$ by taking the trace of the 2×2 matrices in $\mathfrak{su}(2)$ that represent
gilmore-lie-groups_F01635	169	■ 0.569	$\mathfrak{su}(2) : (X, Y) = \text{tr}[\mathfrak{Reg}(X)\mathfrak{Reg}(Y)]$	$\mathfrak{su}(2) : (X, Y) = \text{tr}[\mathfrak{Reg}(X)\mathfrak{Reg}(Y)]$	the regular matrix representation of $\mathfrak{su}(2)$: $(X, Y) = \text{tr}[\mathfrak{Reg}(X)\mathfrak{Reg}(Y)]$. Show that
gilmore-lie-groups_F01636	169	■ 0.995	$(n^2 - 1) \times (n^2 - 1)$	$(n^2 - 1) \times (n^2 - 1)$	the $n \times n$ defining matrix representation and in the $(n^2 - 1) \times (n^2 - 1)$ regular matrix
gilmore-lie-groups_F01637	169	■ 1.000	$Y = X$	$Y = X$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)

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gilmore-lie-groups_F01638	169	■ 1.000	$X=H_1$	$X = H_1$	constant? (Hint: set $Y = X$ and choose a special X , for example $X = H_1$.)
gilmore-lie-groups_F01639	169	■ 0.999	$1 \leq i, j \leq N$	$1 \leq i, j \leq N$	form (b boson operators; f fermion operators; $1 \leq i, j \leq N$):
gilmore-lie-groups_F01640	169	■ 1.000	$\mathbf{R}=\frac{1}{2} \sum_{\alpha>0} \alpha$	$\mathbf{R} = \frac{1}{2} \sum_{\alpha>0} \alpha$	11. Compute $\mathbf{R} = \frac{1}{2} \sum_{\alpha>0} \alpha$, half the sum over all positive roots, in each of the simple Lie
gilmore-lie-groups_F01641	169	■ 1.000	$B_{\{n\}}, R_{\{i\}}=\frac{1}{2}(2n+1)-i$	$B_n, R_i = \frac{1}{2}(2n+1) - i$	algebras. For example, for $B_n, R_i = \frac{1}{2}(2n+1) - i$ and the spectrum is
gilmore-lie-groups_F01642	169	■ 0.996	$\mathbf{M}$	$\mathbf{M}$	where $\mathbf{M}$ is the highest weight in the representation. For the $(2j+1)$ dimensional
gilmore-lie-groups_F01643	169	■ 1.000	$\mathfrak{so}(3), \mathbf{M}=j, \mathbf{R}=R_1=\frac{1}{2}$	$\mathfrak{so}(3), \mathbf{M} = j, \mathbf{R} = R_1 = \frac{1}{2}$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_F01644	169	■ 1.000	$\mathcal{C}^2(j)=\left(j+\frac{1}{2}\right)^2-\left(0+\frac{1}{2}\right)^2=$	$\mathcal{C}^2(j) = \left(j + \frac{1}{2}\right)^2 - \left(0 + \frac{1}{2}\right)^2 =$	representation of $\mathfrak{so}(3)$, $\mathbf{M} = j$, $\mathbf{R} = R_1 = \frac{1}{2}$ and $\mathcal{C}^2(j) = (j + \frac{1}{2})^2 - (0 + \frac{1}{2})^2 =$
gilmore-lie-groups_F01645	169	■ 1.000	$j(j+1)$	$j(j+1)$	$j(j+1)$.
gilmore-lie-groups_F01646	169	■ 1.000	$A_{\{n-1\}}$	A_{n-1}	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_F01647	169	■ 1.000	$D_{\{n\}}$	D_n	a. Show that the Weyl group for A_{n-1} is of order $n!$, the Weyl group for D_n is of
gilmore-lie-groups_F01648	169	■ 0.919	$2^{n-1} n!$	$2^{n-1} n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01649	169	■ 0.919	$B_{\{n\}}$	B_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01650	169	■ 0.919	$C_{\{n\}}$	C_n	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01651	169	■ 0.919	$2^n n!$	$2^n n!$	order $2^{n-1} n!$, and the Weyl groups for B_n and C_n are of order $2^n n!$.
gilmore-lie-groups_F01652	170	■ 1.000	$f_{\{r\}}(\Gamma)$	$f_r(\Gamma)$	the index $f_r(\Gamma)$.
gilmore-lie-groups_F01653	170	■ 1.000	$\mathcal{C}^r$	$\mathcal{C}^r$	In this expression $\mathcal{C}^r$ is the value of the r th Casimir invariant in the representation
gilmore-lie-groups_F01654	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$	15. The matrix Lie algebras $\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where
gilmore-lie-groups_F01655	170	■ 1.000	$\sum_{ij} a^{ij} M_{ij}$	$\sum_{ij} a^{ij} M_{ij}$	15. The matrix Lie algebras $\mathfrak{so}(2n), \mathfrak{so}(2n+1), \mathfrak{sp}(2n)$ have the form $\sum_{ij} a^{ij} M_{ij}$, where
gilmore-lie-groups_F01656	170	■ 1.000	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	$\mathfrak{so}(2n), \mathfrak{sp}(2n)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable
gilmore-lie-groups_F01657	170	■ 1.000	$(2n+1) \times (2n+1)$	$(2n+1) \times (2n+1)$	M is $2n \times 2n$ for $\mathfrak{so}(2n), \mathfrak{sp}(2n)$ and $(2n+1) \times (2n+1)$ for $\mathfrak{so}(2n+1)$, and suitable
gilmore-lie-groups_F01658	170	■ 1.000	a^{ij}	a^{ij}	reality restrictions are imposed on the coefficients a^{ij} .
gilmore-lie-groups_F01659	170	■ 1.000	$\phi_{\{r\}}(a^{ij})$	$\phi_r(a^{ij})$	b. Write down the coefficients $\phi_r(a^{ij})$ that occur in the secular equation for each of
gilmore-lie-groups_F01660	170	■ 0.999	$\epsilon_{i_1 i_2 \cdots i_l}$	$\epsilon_{i_1 i_2 \cdots i_l}$	d. Express the even coefficients in terms of the Levi–Civita skew tensors $\epsilon_{i_1 i_2 \cdots i_l}$
gilmore-lie-groups_F01661	170	■ 1.000	$(l=2n, 2n+1, 2n)$	$(l = 2n, 2n+1, 2n)$	$l = 2n, 2n+1, 2n$.
gilmore-lie-groups_F01662	170	■ 1.000	$\phi_{2n}(a^{ij})$	$\phi_{2n}(a^{ij})$	g. Show that $\phi_{2n}(a^{ij})$ is a perfect square for $\mathfrak{so}(2n)$; write down its square root; show
gilmore-lie-groups_F01663	171	■ 0.999	$\operatorname{tr} M^2 = -2 \theta \cdot \theta$	$\operatorname{tr} M^2 = -2\theta \cdot \theta$	a. Show $\operatorname{tr} M^2 = -2\theta \cdot \theta$.
gilmore-lie-groups_F01664	171	■ 0.987	$\operatorname{tr} \mathcal{M}^2 = -2 \mathbf{J} \cdot \mathbf{J}$	$\operatorname{tr} \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$	b. Show $\operatorname{tr} \mathcal{M}^2 = -2\mathbf{J} \cdot \mathbf{J}$.

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gilmore-lie-groups_F01665	171	■ 1.000	$\left[\mathbf{J}, \operatorname{tr} \mathcal{M}^2\right]=0$	$[\mathbf{J}, \operatorname{tr} \mathcal{M}^2] = 0$	c. Show $[\mathbf{J}, \operatorname{tr} \mathcal{M}^2] = 0$.																																								
gilmore-lie-groups_F01666	171	■ 0.999	$\operatorname{tr} \mathcal{M}^{2 n+1}=0$	$\operatorname{tr} \mathcal{M}^{2 n+1} = 0$	d. Show $\operatorname{tr} \mathcal{M}^{2 n+1} = 0$ and $\operatorname{tr} \mathcal{M}^{2 n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.																																								
gilmore-lie-groups_F01667	171	■ 0.999	$\operatorname{tr} \mathcal{M}^{2 n}=(-2)^n\left(\mathbf{J} \cdot \mathbf{J}\right)^n$	$\operatorname{tr} \mathcal{M}^{2 n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$	d. Show $\operatorname{tr} \mathcal{M}^{2 n+1} = 0$ and $\operatorname{tr} \mathcal{M}^{2 n} = (-2)^n (\mathbf{J} \cdot \mathbf{J})^n$.																																								
gilmore-lie-groups_F01668	171	■ 1.000	$\left[X_i, X_j\right]=C_{i j}^k X_k$	$[X_i, X_j] = C_{ij}^k X_k$	commutation relations $[X_i, X_j] = C_{ij}^k X_k$. There are two linear vector spaces, $V^{(1)}$																																								
gilmore-lie-groups_F01669	171	■ 1.000	$V^{(1)}$	$V^{(1)}$	commutation relations $[X_i, X_j] = C_{ij}^k X_k$. There are two linear vector spaces, $V^{(1)}$																																								
gilmore-lie-groups_F01670	171	■ 1.000	$V^{(2)}$	$V^{(2)}$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$																																								
gilmore-lie-groups_F01671	171	■ 1.000	$X_i \rightarrow \Gamma^{(1)}\left(X_i\right)=$	$X_i \rightarrow \Gamma^{(1)}(X_i) =$	and $V^{(2)}$, that carry irreducible representations of this Lie algebra: $X_i \rightarrow \Gamma^{(1)}(X_i) =$																																								
gilmore-lie-groups_F01672	171	■ 1.000	Y_i	Y_i	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with																																								
gilmore-lie-groups_F01673	171	■ 1.000	$X_i \rightarrow \Gamma^{(2)}\left(X_i\right)=Z_i$	$X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with																																								
gilmore-lie-groups_F01674	171	■ 1.000	$g^{i j} Y_i Z_j$	$g^{ij} Y_i Z_j$	Y_i and $X_i \rightarrow \Gamma^{(2)}(X_i) = Z_i$. Show that the Casimir covariant $g^{ij} Y_i Z_j$ commutes with																																								
gilmore-lie-groups_F01675	171	■ 0.857	$(Y+Z)_k$	$(Y + Z)_k$	$(Y + Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).																																								
gilmore-lie-groups_F01676	171	■ 0.857	$\Gamma^{(1)}\left(X_i\right) \otimes I_{\dim V^{(2)}}+I_{\dim V^{(1)}} \otimes \Gamma^{(2)}\left(X_i\right)$	$\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$	$(Y + Z)_k$ (more accurately, with $\Gamma^{(1)}(X_i) \otimes I_{\dim V^{(2)}} + I_{\dim V^{(1)}} \otimes \Gamma^{(2)}(X_i)$).																																								
gilmore-lie-groups_F01677	171	■ 1.000	$X^0=I_n$	$X^0 = I_n$	of X , starting at $X^0 = I_n$:																																								
gilmore-lie-groups_F01678	171	■ 1.000	$\lambda_i(X)$	$\lambda_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation																																								
gilmore-lie-groups_F01679	171	■ 1.000	$\phi_i(X)$	$\phi_i(X)$	eigenvalues $\lambda_i(X)$ or the independent coefficients $\phi_i(X)$ of the secular equation																																								
gilmore-lie-groups_F01680	171	■ 0.967	$e^{i \phi J_z}$	$e^{i \phi J_z}$	d. Compute $e^{i \phi J_z}$ for the $(2 j + 1)$ dimensional matrix representations of $SU(2)$ by																																								
gilmore-lie-groups_F01681	171	■ 0.999	$j=\frac{1}{2}, 1, \frac{3}{2}, 2$	$j = \frac{1}{2}, 1, \frac{3}{2}, 2$	computing the vanderMonde matrices. Show that for $j = \frac{1}{2}, 1, \frac{3}{2}, 2$ the resulting																																								
gilmore-lie-groups_F01682	172	■ 1.000	$f_j(\phi)$	$f_j(\phi)$	e. Invert each of these van der Monde matrices and determine the functions $f_j(\phi)$ in																																								
gilmore-lie-groups_F01683	172	■ 1.000	e^X	e^X	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show																																								
gilmore-lie-groups_F01684	172	■ 1.000	$X \in \mathfrak{su}(2)$	$X \in \mathfrak{su}(2)$	the expansions of e^X for $X \in \mathfrak{su}(2)$. In particular, show																																								
gilmore-lie-groups_F01685	172	■ 1.000	$(i \phi)^j f_j$	$(i \phi)^j f_j$	<table><tr><th colspan="5">Representation</th></tr><tr><th></th><th>$\frac{1}{2}$</th><th>1</th><th>$\frac{3}{2}$</th><th>2</th></tr><tr><th>$(i \phi)^j f_j$</th><th>$\frac{1}{2}$</th><th>3</th><th>4</th><th>5</th></tr><tr><td>f_0</td><td>$\cos (\phi / 2)$</td><td>1</td><td>$\frac{9}{8} \cos \left(\frac{\phi}{2}\right)-\frac{1}{8} \cos \left(\frac{3 \phi}{2}\right)$</td><td>1</td></tr><tr><td>$(i \phi)^1 f_1$</td><td>$2 i \sin (\phi / 2)$</td><td>$i \sin (\phi)$</td><td>$\frac{9 i}{4} \sin \left(\frac{\phi}{2}\right)-\frac{i}{12} \sin \left(\frac{3 \phi}{2}\right)$</td><td>$\frac{i}{3} \sin (\phi)-\frac{i}{6} \sin (2 \phi)$</td></tr><tr><td>$(i \phi)^2 f_2$</td><td></td><td>$\cos (\phi)-1$</td><td>$-\frac{1}{2} \cos \left(\frac{\phi}{2}\right)+\frac{1}{2} \cos \left(\frac{3 \phi}{2}\right)$</td><td>$-\frac{5}{4}+\frac{1}{3} \cos (\phi)-\frac{1}{12} \cos (2 \phi)$</td></tr><tr><td>$(i \phi)^3 f_3$</td><td></td><td></td><td>$-i \sin \left(\frac{\phi}{2}\right)+\frac{i}{3} \sin \left(\frac{3 \phi}{2}\right)$</td><td>$-\frac{i}{3} \sin (\phi)+\frac{i}{6} \sin (2 \phi)$</td></tr><tr><td>$(i \phi)^4 f_4$</td><td></td><td></td><td></td><td>$\frac{1}{4}-\frac{1}{3} \cos (\phi)+\frac{1}{12} \cos (2 \phi)$</td></tr></table>	Representation						$\frac{1}{2}$	1	$\frac{3}{2}$	2	$(i \phi)^j f_j$	$\frac{1}{2}$	3	4	5	f_0	$\cos (\phi / 2)$	1	$\frac{9}{8} \cos \left(\frac{\phi}{2}\right)-\frac{1}{8} \cos \left(\frac{3 \phi}{2}\right)$	1	$(i \phi)^1 f_1$	$2 i \sin (\phi / 2)$	$i \sin (\phi)$	$\frac{9 i}{4} \sin \left(\frac{\phi}{2}\right)-\frac{i}{12} \sin \left(\frac{3 \phi}{2}\right)$	$\frac{i}{3} \sin (\phi)-\frac{i}{6} \sin (2 \phi)$	$(i \phi)^2 f_2$		$\cos (\phi)-1$	$-\frac{1}{2} \cos \left(\frac{\phi}{2}\right)+\frac{1}{2} \cos \left(\frac{3 \phi}{2}\right)$	$-\frac{5}{4}+\frac{1}{3} \cos (\phi)-\frac{1}{12} \cos (2 \phi)$	$(i \phi)^3 f_3$			$-i \sin \left(\frac{\phi}{2}\right)+\frac{i}{3} \sin \left(\frac{3 \phi}{2}\right)$	$-\frac{i}{3} \sin (\phi)+\frac{i}{6} \sin (2 \phi)$	$(i \phi)^4 f_4$				$\frac{1}{4}-\frac{1}{3} \cos (\phi)+\frac{1}{12} \cos (2 \phi)$
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gilmore-lie-groups_ F01686	172	1.000	$\frac{1}{2}$	$\frac{1}{2}$	<table><tr><th colspan="5">Representation</th></tr><tr><th>$(i\phi)^j f_j$</th><th>$\frac{1}{2}$</th><th>1</th><th>$\frac{3}{2}$</th><th>2</th></tr><tr><td>f_0</td><td>$\cos(\phi/2)$</td><td>1</td><td>$\frac{9}{8}\cos(\frac{\phi}{2}) - \frac{1}{8}\cos(\frac{3\phi}{2})$</td><td>1</td></tr><tr><td>$(i\phi)^1 f_1$</td><td>$2i\sin(\phi/2)$</td><td>$i\sin(\phi)$</td><td>$\frac{9i}{4}\sin(\frac{\phi}{2}) - \frac{i}{12}\sin(\frac{3\phi}{2})$</td><td>$\frac{i}{3}\sin(\phi) - \frac{i}{6}\sin(2\phi)$</td></tr><tr><td>$(i\phi)^2 f_2$</td><td></td><td>$\cos(\phi) - 1$</td><td>$-\frac{1}{2}\cos(\frac{\phi}{2}) + \frac{1}{2}\cos(\frac{3\phi}{2})$</td><td>$-\frac{5}{4} + \frac{1}{3}\cos(\phi) - \frac{1}{12}\cos(2\phi)$</td></tr><tr><td>$(i\phi)^3 f_3$</td><td></td><td></td><td>$-i\sin(\frac{\phi}{2}) + \frac{i}{3}\sin(\frac{3\phi}{2})$</td><td>$-\frac{i}{3}\sin(\phi) + \frac{i}{6}\sin(2\phi)$</td></tr><tr><td>$(i\phi)^4 f_4$</td><td></td><td></td><td></td><td>$\frac{1}{4} - \frac{1}{3}\cos(\phi) + \frac{1}{12}\cos(2\phi)$</td></tr></table>	Representation					$(i\phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2	f_0	$\cos(\phi/2)$	1	$\frac{9}{8}\cos(\frac{\phi}{2}) - \frac{1}{8}\cos(\frac{3\phi}{2})$	1	$(i\phi)^1 f_1$	$2i\sin(\phi/2)$	$i\sin(\phi)$	$\frac{9i}{4}\sin(\frac{\phi}{2}) - \frac{i}{12}\sin(\frac{3\phi}{2})$	$\frac{i}{3}\sin(\phi) - \frac{i}{6}\sin(2\phi)$	$(i\phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2}\cos(\frac{\phi}{2}) + \frac{1}{2}\cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3}\cos(\phi) - \frac{1}{12}\cos(2\phi)$	$(i\phi)^3 f_3$			$-i\sin(\frac{\phi}{2}) + \frac{i}{3}\sin(\frac{3\phi}{2})$	$-\frac{i}{3}\sin(\phi) + \frac{i}{6}\sin(2\phi)$	$(i\phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3}\cos(\phi) + \frac{1}{12}\cos(2\phi)$
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gilmore-lie-groups_ F01687	172	1.000	$\frac{3}{2}$	$\frac{3}{2}$	<table><tr><th colspan="5">Representation</th></tr><tr><th>$(i\phi)^j f_j$</th><th>$\frac{1}{2}$</th><th>1</th><th>$\frac{3}{2}$</th><th>2</th></tr><tr><td>f_0</td><td>$\cos(\phi/2)$</td><td>1</td><td>$\frac{9}{8}\cos(\frac{\phi}{2}) - \frac{1}{8}\cos(\frac{3\phi}{2})$</td><td>1</td></tr><tr><td>$(i\phi)^1 f_1$</td><td>$2i\sin(\phi/2)$</td><td>$i\sin(\phi)$</td><td>$\frac{9i}{4}\sin(\frac{\phi}{2}) - \frac{i}{12}\sin(\frac{3\phi}{2})$</td><td>$\frac{i}{3}\sin(\phi) - \frac{i}{6}\sin(2\phi)$</td></tr><tr><td>$(i\phi)^2 f_2$</td><td></td><td>$\cos(\phi) - 1$</td><td>$-\frac{1}{2}\cos(\frac{\phi}{2}) + \frac{1}{2}\cos(\frac{3\phi}{2})$</td><td>$-\frac{5}{4} + \frac{1}{3}\cos(\phi) - \frac{1}{12}\cos(2\phi)$</td></tr><tr><td>$(i\phi)^3 f_3$</td><td></td><td></td><td>$-i\sin(\frac{\phi}{2}) + \frac{i}{3}\sin(\frac{3\phi}{2})$</td><td>$-\frac{i}{3}\sin(\phi) + \frac{i}{6}\sin(2\phi)$</td></tr><tr><td>$(i\phi)^4 f_4$</td><td></td><td></td><td></td><td>$\frac{1}{4} - \frac{1}{3}\cos(\phi) + \frac{1}{12}\cos(2\phi)$</td></tr></table>	Representation					$(i\phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2	f_0	$\cos(\phi/2)$	1	$\frac{9}{8}\cos(\frac{\phi}{2}) - \frac{1}{8}\cos(\frac{3\phi}{2})$	1	$(i\phi)^1 f_1$	$2i\sin(\phi/2)$	$i\sin(\phi)$	$\frac{9i}{4}\sin(\frac{\phi}{2}) - \frac{i}{12}\sin(\frac{3\phi}{2})$	$\frac{i}{3}\sin(\phi) - \frac{i}{6}\sin(2\phi)$	$(i\phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2}\cos(\frac{\phi}{2}) + \frac{1}{2}\cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3}\cos(\phi) - \frac{1}{12}\cos(2\phi)$	$(i\phi)^3 f_3$			$-i\sin(\frac{\phi}{2}) + \frac{i}{3}\sin(\frac{3\phi}{2})$	$-\frac{i}{3}\sin(\phi) + \frac{i}{6}\sin(2\phi)$	$(i\phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3}\cos(\phi) + \frac{1}{12}\cos(2\phi)$
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$(i\phi)^3 f_3$			$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$	$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$																																				
$(i\phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$																																				
gilmore-lie-groups_ F01704	172	■ 1.000	$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$	$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$	<table><tr><th colspan="5">Representation</th></tr><tr><th>$(i\phi)^j f_j$</th><th>$\frac{1}{2}$</th><th>1</th><th>$\frac{3}{2}$</th><th>2</th></tr><tr><td>f_0</td><td>$\cos(\phi/2)$</td><td>1</td><td>$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$</td><td>1</td></tr><tr><td>$(i\phi)^1 f_1$</td><td>$2i \sin(\phi/2)$</td><td>$i \sin(\phi)$</td><td>$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$</td><td>$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$</td></tr><tr><td>$(i\phi)^2 f_2$</td><td></td><td>$\cos(\phi) - 1$</td><td>$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$</td><td>$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$</td></tr><tr><td>$(i\phi)^3 f_3$</td><td></td><td></td><td>$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$</td><td>$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$</td></tr><tr><td>$(i\phi)^4 f_4$</td><td></td><td></td><td></td><td>$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$</td></tr></table>	Representation					$(i\phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2	f_0	$\cos(\phi/2)$	1	$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$	1	$(i\phi)^1 f_1$	$2i \sin(\phi/2)$	$i \sin(\phi)$	$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$	$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$	$(i\phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$	$(i\phi)^3 f_3$			$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$	$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$	$(i\phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$
Representation																																								
$(i\phi)^j f_j$	$\frac{1}{2}$	1	$\frac{3}{2}$	2																																				
f_0	$\cos(\phi/2)$	1	$\frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2})$	1																																				
$(i\phi)^1 f_1$	$2i \sin(\phi/2)$	$i \sin(\phi)$	$\frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2})$	$\frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi)$																																				
$(i\phi)^2 f_2$		$\cos(\phi) - 1$	$-\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2})$	$-\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi)$																																				
$(i\phi)^3 f_3$			$-i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2})$	$-\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi)$																																				
$(i\phi)^4 f_4$				$\frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi)$																																				
gilmore-lie-groups_ F01705	172	■ 1.000	$j = \frac{1}{2}$	$j = \frac{1}{2}$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:																																			
gilmore-lie-groups_ F01706	172	■ 1.000	$l = 1$	$l = 1$	f. Recover the two well-known expansions for $j = \frac{1}{2}$ and $l = 1$:																																			
gilmore-lie-groups_ F01707	173	■ 1.000	$l + 1$	$l + 1$	grams to construct rank $l + 1$ root space diagrams. However, it is not easy																																			
gilmore-lie-groups_ F01708	174	■ 1.000	m, n	m, n	The coefficients $N_{\alpha, \beta}$ are defined in terms of the nonnegative integers m, n by																																			
gilmore-lie-groups_ F01709	174	■ 1.000	$\beta + k \alpha$	$\beta + k \alpha$	where $\beta + k \alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by																																			
gilmore-lie-groups_ F01710	174	■ 1.000	$k = -m, \dots, +n$	$k = -m, \dots, +n$	where $\beta + k \alpha$ is a root only for $k = -m, \dots, +n$. The roots are normalized by																																			

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																								
gilmore-lie-groups_ F01711	174	■ 0.999	$\beta^{\prime}$	β'	The root β' is obtained by reflecting β in the hyperplane orthogonal to α . These two																																								
gilmore-lie-groups_ F01712	174	■ 1.000	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$	$2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are																																								
gilmore-lie-groups_ F01713	174	■ 1.000	$2(\alpha \cdot \beta) / (\beta \cdot \beta) = n^{\prime}$	$2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$	If we write $2(\alpha \cdot \beta) / (\alpha \cdot \alpha) = n$ and $2(\alpha \cdot \beta) / (\beta \cdot \beta) = n'$, where n and n' are																																								
gilmore-lie-groups_ F01714	174	■ 0.992	$\pm \mathbf{e}_1$	$\pm \mathbf{e}_1$	vectors $\pm \mathbf{e}_1$. To construct rank-two root spaces, add a noncollinear vector to this																																								
gilmore-lie-groups_ F01715	174	■ 1.000	$A_{\{2\}}, B_{\{2\}} = C_{\{2\}}, D_{\{2\}}$	$A_2, B_2 = C_2, D_2$	are $A_2, B_2 = C_2, D_2$ and G_2 , as shown in Fig. 9.6.																																								
gilmore-lie-groups_ F01716	175	■ 0.995	$\cos^2(\alpha, \beta)$	$\cos^2(\alpha, \beta)$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 3	3^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 2	2^{+1}	0	$\frac{\pi}{2}$	0	0	–					
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01717	175	■ 0.995	$\theta(\alpha, \beta)$	$\theta(\alpha, \beta)$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01718	175	■ 0.995	$n = \frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 1$</td><td>$\pm 3$</td><td>$3^{+1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 1$</td><td>$\pm 2$</td><td>$2^{+1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 1	± 3	3^{+1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 1	± 2	2^{+1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																																									
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gilmore-lie-groups_ F01719	175	■0.995	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	<table><tr><th>$\cos ^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$																															
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gilmore-lie-groups_ F01720	175	■0.995	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	<table><tr><th>$\cos ^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$																															
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gilmore-lie-groups_ F01721	175	■0.995	$\frac{\pi}{2} \pm \frac{\pi}{2}$	$\frac{\pi}{2} \pm \frac{\pi}{2}$	<table><tr><th>$\cos ^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$																															
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gilmore-lie-groups_ F01722	175	■0.995	± 2	± 2	<table><tr><th>$\cos ^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos ^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n=\frac{2 \alpha \cdot \beta}{\alpha \cdot \alpha}$	$n^{\prime}=\frac{2 \alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta}=\frac{n^{\prime}}{n}$																															
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gilmore-lie-groups_ F01723	175	■ 0.995	$\frac{\pi}{2} \pm \frac{\pi}{3}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01724	175	■ 0.995	± 3	± 3	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01725	175	■ 0.995	3^{-1}	3^{-1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01726	175	■ 0.995	3^{+1}	3^{+1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01727	175	0.995	$\frac{2}{4}$	$\frac{2}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
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gilmore-lie-groups_ F01728	175	0.995	$\frac{\pi}{2} \pm \frac{\pi}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
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gilmore-lie-groups_ F01729	175	0.995	2^{-1}	2^{-1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01730	175	0.995	2^{+1}	2^{+1}	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01731	175	■0.995	$\frac{1}{4}$	$\frac{1}{4}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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gilmore-lie-groups_ F01732	175	■0.995	$\frac{\pi}{2} \pm \frac{\pi}{6}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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0	$\frac{\pi}{2}$	0	0	–																															
gilmore-lie-groups_ F01733	175	■0.995	$\frac{\pi}{2}$	$\frac{\pi}{2}$	<table><tr><th>$\cos^2(\alpha, \beta)$</th><th>$\theta(\alpha, \beta)$</th><th>$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$</th><th>$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$</th><th>$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$</th></tr><tr><td>1</td><td>$\frac{\pi}{2} \pm \frac{\pi}{2}$</td><td>$\pm 2$</td><td>$\pm 2$</td><td>1</td></tr><tr><td>$\frac{3}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{3}$</td><td>$\pm 3$</td><td>$\pm 1$</td><td>$3^{-1}$</td></tr><tr><td>$\frac{2}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{4}$</td><td>$\pm 2$</td><td>$\pm 1$</td><td>$2^{-1}$</td></tr><tr><td>$\frac{1}{4}$</td><td>$\frac{\pi}{2} \pm \frac{\pi}{6}$</td><td>$\pm 1$</td><td>$\pm 1$</td><td>1</td></tr><tr><td>0</td><td>$\frac{\pi}{2}$</td><td>0</td><td>0</td><td>–</td></tr></table>	$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$	1	$\frac{\pi}{2} \pm \frac{\pi}{2}$	± 2	± 2	1	$\frac{3}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{3}$	± 3	± 1	3^{-1}	$\frac{2}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{4}$	± 2	± 1	2^{-1}	$\frac{1}{4}$	$\frac{\pi}{2} \pm \frac{\pi}{6}$	± 1	± 1	1	0	$\frac{\pi}{2}$	0	0	–
$\cos^2(\alpha, \beta)$	$\theta(\alpha, \beta)$	$n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha}$	$n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta}$	$\frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n}$																															
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0	$\frac{\pi}{2}$	0	0	–																															
gilmore-lie-groups_ F01734	175	■0.998	$(l-1)$	$(l-1)$	this way from rank- $(l-1)$ root space diagrams. We find by this building-up process																														
gilmore-lie-groups_ F01735	175	■0.998	$D_{\{l\}}$	D_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by																														
gilmore-lie-groups_ F01736	175	■0.998	$\pm \left(\mathbf{e}_{\{i\}} + \mathbf{e}_{\{j\}} \right)$	$\pm (\mathbf{e}_i + \mathbf{e}_j)$	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by																														
gilmore-lie-groups_ F01737	175	■0.998	$A_{\{l-1\}}$	A_{l-1}	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by																														
gilmore-lie-groups_ F01738	175	■0.998	$B_{\{l\}}, C_{\{l\}}$	B_l, C_l	D_l is constructed by adding roots $\pm(\mathbf{e}_i + \mathbf{e}_j)$ to A_{l-1} , and B_l, C_l are constructed by																														
gilmore-lie-groups_ F01739	175	■1.000	$\pm \mathbf{e}_{\{i\}}, \pm 2 \mathbf{e}_{\{i\}}$	$\pm \mathbf{e}_i, \pm 2\mathbf{e}_i$	adding roots $\pm \mathbf{e}_i, \pm 2\mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent																														

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gilmore-lie-groups_ F01746	175	■ 1.000	A_{l-1}, D_l, B_l, C_l	A_{l-1}, D_l, B_l, C_l	adding roots $\pm \mathbf{e}_i, \pm 2\mathbf{e}_i$ to D_l . The root spaces A_{l-1}, D_l, B_l, C_l are all inequivalent																														
gilmore-lie-groups_ F01741	176	■ 1.000	$A_{\{3\}}, D_{\{3\}}$	A_3, D_3	Figure 10.1. Rank-three root space diagrams. Top: A_3, D_3 . Bottom: B_3, C_3 .																														
gilmore-lie-groups_ F01742	176	■ 1.000	$B_{\{3\}}, C_{\{3\}}$	B_3, C_3	Figure 10.1. Rank-three root space diagrams. Top: A_3, D_3 . Bottom: B_3, C_3 .																														
gilmore-lie-groups_ F01743	178	■ 1.000	$A_{\{1\}}$	A_1	describe the rank-one algebra A_1 .																														
gilmore-lie-groups_ F01744	178	■ 1.000	$SL(l; \mathbb{C})$	$SL(l; \mathbb{C})$	space has the same complex extension, for example, $SL(l; \mathbb{C})$ for A_{l-1} .																														
gilmore-lie-groups_ F01745	178	■ 0.997	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	$\pm(\mathbf{e}_1 - \mathbf{e}_2)$	Remark 2. The root space D_2 consists of two orthogonal sets of roots $\pm(\mathbf{e}_1 - \mathbf{e}_2)$																														
gilmore-lie-groups_ F01746	178	■ 1.000	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	$\pm(\mathbf{e}_1 + \mathbf{e}_2)$	and $\pm(\mathbf{e}_1 + \mathbf{e}_2)$. The decomposition is shown in Fig. 10.3. Orthogonal root spaces																														
gilmore-lie-groups_ F01747	178	■ 0.964	$B_{\{2\}}$	B_2	Remark 3. The root spaces B_2 and C_2 are equivalent, as is easily seen by ro-																														
gilmore-lie-groups_ F01748	178	■ 0.513	$SO(5)$	$SO(5)$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,																														
gilmore-lie-groups_ F01749	178	■ 0.513	$Sp(2) = U(2; \mathbb{Q})$	$Sp(2) = U(2; \mathbb{Q})$	tation. The root space B_2 describes $SO(5)$ while C_2 describes $Sp(2) = U(2; \mathbb{Q})$,																														
gilmore-lie-groups_ F01750	178	■ 1.000	$U(2; \mathbb{Q})$	$U(2; \mathbb{Q})$	have a four-dimensional “spinor” representation based on $U(2; \mathbb{Q})$ in the same way																														
gilmore-lie-groups_ F01751	178	■ 0.975	$SO(3)(B_1)$	$SO(3)(B_1)$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or																														
gilmore-lie-groups_ F01752	178	■ 0.975	$U(1; \mathbb{Q})$	$U(1; \mathbb{Q})$	that $SO(3)(B_1)$ has a two-dimensional spinor representation based on $U(1; \mathbb{Q})$ or																														
gilmore-lie-groups_ F01753	178	■ 0.995	$SU(2)(A_1)$	$SU(2)(A_1)$	$SU(2)(A_1)$.																														
gilmore-lie-groups_ F01754	179	■ 1.000	$\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$	$\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$	Figure 10.4. Root space for G_2 . Fundamental roots are $\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$ and $\alpha_2 =$																														
gilmore-lie-groups_ F01755	179	■ 1.000	$\alpha_2 =$	$\alpha_2 =$	Figure 10.4. Root space for G_2 . Fundamental roots are $\alpha_1 = \mathbf{e}_1 - \mathbf{e}_2$ and $\alpha_2 =$																														
gilmore-lie-groups_ F01756	179	■ 1.000	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$. All roots are orthogonal to $\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$.																														
gilmore-lie-groups_ F01757	179	■ 1.000	$\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$	$\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$	$-\mathbf{e}_1 + 2\mathbf{e}_2 - \mathbf{e}_3$. All roots are orthogonal to $\mathbf{R} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$.																														
gilmore-lie-groups_ F01758	180	■ 1.000	$G_{\{2\}} \oplus B_{\{3\}}$	$G_2 \oplus B_3$	Here the disconnected diagram describes $G_2 \oplus B_3$.																														
gilmore-lie-groups_ F01759	180	■ 0.508	$\alpha_{\{1\}}$	α_1	<table> <tr> <td></td><td>α_1</td><td>α_2</td><td>α_3</td><td>α_{l-1}</td><td>α_l</td></tr> <tr> <td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>$\dots$</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td></tr> <tr> <td>$D_l$</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> <tr> <td>B_l</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> <tr> <td>C_l</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> </table>		α_1	α_2	α_3	α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	$\dots$	$\mathbf{e}_{l-1} - \mathbf{e}_l$	D_l	”	”	”	$\dots$	”	B_l	”	”	”	$\dots$	”	C_l	”	”	”	$\dots$	”
	α_1	α_2	α_3	α_{l-1}	α_l																														
A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	$\dots$	$\mathbf{e}_{l-1} - \mathbf{e}_l$																														
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B_l	”	”	”	$\dots$	”																														
C_l	”	”	”	$\dots$	”																														
gilmore-lie-groups_ F01760	180	■ 0.508	$\alpha_{\{2\}}$	α_2	<table> <tr> <td></td><td>α_1</td><td>α_2</td><td>α_3</td><td>α_{l-1}</td><td>α_l</td></tr> <tr> <td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>$\dots$</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td></tr> <tr> <td>$D_l$</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> <tr> <td>B_l</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> <tr> <td>C_l</td><td>”</td><td>”</td><td>”</td><td>$\dots$</td><td>”</td></tr> </table>		α_1	α_2	α_3	α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	$\dots$	$\mathbf{e}_{l-1} - \mathbf{e}_l$	D_l	”	”	”	$\dots$	”	B_l	”	”	”	$\dots$	”	C_l	”	”	”	$\dots$	”
	α_1	α_2	α_3	α_{l-1}	α_l																														
A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	$\dots$	$\mathbf{e}_{l-1} - \mathbf{e}_l$																														
D_l	”	”	”	$\dots$	”																														
B_l	”	”	”	$\dots$	”																														
C_l	”	”	”	$\dots$	”																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01769	180	0.508	$B_{\{l\}}$	B_l	$ \begin{array}{cccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_l \\ C_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & 2\mathbf{e}_l \end{array} $
gilmore-lie-groups_ F01770	180	0.508	$\mathbf{e}_{\{l\}}$	$\mathbf{e}_l$	$ \begin{array}{cccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_l \\ C_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & 2\mathbf{e}_l \end{array} $
gilmore-lie-groups_ F01771	180	0.508	$C_{\{l\}}$	C_l	$ \begin{array}{cccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_l \\ C_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & 2\mathbf{e}_l \end{array} $
gilmore-lie-groups_ F01772	180	0.508	$2 \mathbf{e}_{\{l\}}$	$2\mathbf{e}_l$	$ \begin{array}{cccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & \mathbf{e}_l \\ C_l & \text{"} & \text{"} & \text{"} & \cdots & \text{"} & 2\mathbf{e}_l \end{array} $
gilmore-lie-groups_ F01773	180	1.000	$n_{\{i \ j\}}$	n_{ij}	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01774	180	1.000	$0, 1, 2$	$0, 1, 2$	where n_{ij} is 0, 1, 2, or 3. Each fundamental root is represented by a dot. Dots i
gilmore-lie-groups_ F01775	180	1.000	$G_{\{2\}} + B_{\{3\}}$	$G_2 + B_3$	represented by orthogonal root spaces $G_2 + B_3$ is shown in Fig. 10.5.
gilmore-lie-groups_ F01776	180	0.665	$\mathbf{v}_{\{i\}}$	$\mathbf{v}_i$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01777	180	0.665	$\mathbf{u}$	$\mathbf{u}$	Observation 2 If $\mathbf{v}_i$ is an orthonormal system of vectors in the root space and $\mathbf{u}$ is a unit
gilmore-lie-groups_ F01778	180	1.000	$\mathbf{u} \cdot \mathbf{v}_{\{i\}}$	$\mathbf{u} \cdot \mathbf{v}_i$	vector, then the direction cosines $\mathbf{u} \cdot \mathbf{v}_i$ obey
gilmore-lie-groups_ F01779	181	1.000	$\mathbf{u}_{\{i\}} = \alpha_{\{i\}} / \alpha_{\{i\}} $	$\mathbf{u}_i = \alpha_i / \alpha_i $	vertices. With $\mathbf{u}_i = \alpha_i / \alpha_i $ the inner product
gilmore-lie-groups_ F01780	181	1.000	$2 \mathbf{u}_{\{i\}} \cdot \mathbf{u}_{\{j\}} \leq -1$	$2\mathbf{u}_i \cdot \mathbf{u}_j \leq -1$	cannot be positive since $2\mathbf{u}_i \cdot \mathbf{u}_j \leq -1$ if α_i and α_j are connected.
gilmore-lie-groups_ F01781	181	0.979	(B, C, F)	(B, C, F)	For the diagrams (B, C, F) with a single double link, the Schwarz inequality
gilmore-lie-groups_ F01782	182	1.000	$\mathbf{u}_{\{i\}}, \mathbf{v}_{\{j\}}$	$\mathbf{u}_i, \mathbf{v}_j$	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_i = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01783	182	1.000	$\mathbf{v}_{\{i\}} = \alpha_{\{j\}} / \alpha_{\{j\}} $	$\mathbf{v}_i = \alpha_j / \alpha_j $	where $\mathbf{u}_i, \mathbf{v}_j$ are unit vectors $\mathbf{u}_i = \alpha_i / \alpha_i $ and $\mathbf{v}_i = \alpha_j / \alpha_j $, can be transformed to
gilmore-lie-groups_ F01784	182	0.985	$p \geq q$	$p \geq q$	This has the following solutions with $p \geq q$
gilmore-lie-groups_ F01785	182	0.657	(D, E)	(D, E)	For the diagrams (D, E) Observation 2 applied to the vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ defined
gilmore-lie-groups_ F01786	182	0.997	$p \geq q \geq r \geq 2$	$p \geq q \geq r \geq 2$	This has the following solutions with $p \geq q \geq r \geq 2$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																				
gilmore-lie-groups_ F01787	182	0.999	D_{p+2}	D_{p+2}	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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3	3	2	E_6																						
4	3	2	E_7																						
5	3	2	E_8																						
gilmore-lie-groups_ F01788	182	0.999	E_{6}	E_6	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
p	q	r	Root space																						
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4	3	2	E_7																						
5	3	2	E_8																						
gilmore-lie-groups_ F01789	182	0.999	E_{7}	E_7	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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gilmore-lie-groups_ F01790	182	0.999	E_{8}	E_8	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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5	3	2	E_8																						
gilmore-lie-groups_ F01791	182	1.000	l-1	l-1	inductively from each root space diagram of rank l-1. The completeness of this																				
gilmore-lie-groups_ F01792	184	1.000	f_{1}^{\dag} f_{1}^{\dag}=	f_1^{\dagger} f_1^{\dagger} =	(b) Roots of D_2 are identified with products of fermion operators. Note that f_1^{\dagger} f_1^{\dagger} =																				
gilmore-lie-groups_ F01793	184	1.000	U(l)	U(l)	group U(l):																				
gilmore-lie-groups_ F01794	184	1.000	b_{i}^{\dag} b_{i}	b_i^{\dagger} b_i	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>b_i^{\dagger} b_i</td><td>x^i \partial_i</td><td>f_i^{\dagger} f_i</td></tr><tr><td>E_{+\mathbf{e}_i-\mathbf{e}_j}</td><td>b_i^{\dagger} b_j</td><td>x^i \partial_j</td><td>f_i^{\dagger} f_j</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	b_i^{\dagger} b_i	x^i \partial_i	f_i^{\dagger} f_i	E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j								
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E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j																						
gilmore-lie-groups_ F01795	184	1.000	x^{i} \ \partial_{i}	x^i \partial_i	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>b_i^{\dagger} b_i</td><td>x^i \partial_i</td><td>f_i^{\dagger} f_i</td></tr><tr><td>E_{+\mathbf{e}_i-\mathbf{e}_j}</td><td>b_i^{\dagger} b_j</td><td>x^i \partial_j</td><td>f_i^{\dagger} f_j</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	b_i^{\dagger} b_i	x^i \partial_i	f_i^{\dagger} f_i	E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j								
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E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j																						
gilmore-lie-groups_ F01796	184	1.000	f_{i}^{\dag} f_{i}	f_i^{\dagger} f_i	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>b_i^{\dagger} b_i</td><td>x^i \partial_i</td><td>f_i^{\dagger} f_i</td></tr><tr><td>E_{+\mathbf{e}_i-\mathbf{e}_j}</td><td>b_i^{\dagger} b_j</td><td>x^i \partial_j</td><td>f_i^{\dagger} f_j</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	b_i^{\dagger} b_i	x^i \partial_i	f_i^{\dagger} f_i	E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j								
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E_{+\mathbf{e}_i-\mathbf{e}_j}	b_i^{\dagger} b_j	x^i \partial_j	f_i^{\dagger} f_j																						

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gilmore-lie-groups_ F01797	184	■ 1.000	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																											
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$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																																									
gilmore-lie-groups_ F01798	184	■ 1.000	$x^{\{i} \ \partial_{j\}}$	$x^i \partial_j$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																											
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gilmore-lie-groups_ F01799	184	■ 0.960	$b_{\{i}^{\dagger} \ b_{i\}+\frac{1}{2}}$	$b_i^\dagger b_i + \frac{1}{2}$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i+\mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ F01800	184	■ 0.960	$f_{\{i}^{\dagger} \ f_{i\}+\frac{1}{2}}$	$f_i^\dagger f_i + \frac{1}{2}$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i+\mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ F01801	184	■ 0.960	$x^{\{i} \ \partial_{i\}+\frac{1}{2}}$	$x^i \partial_i + \frac{1}{2}$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i+\mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ F01807	184	0.960	$\backslash\mathrm{partial}_{\{i\}}\ \backslash\mathrm{partial}_{\{j\}}$	$\partial_i \partial_j$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i + \mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i + \mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i - \mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ F01811	184	■ 0.960	$E_{-2 \ \mathbf{e}_{\{i\}}}$	$E_{-2\mathbf{e}_i}$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i+\mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
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gilmore-lie-groups_ F01813	184	■ 0.960	$\partial_{\{i} \ \partial_{\{i}$	$\partial_i \partial_i$	<table><tr><th rowspan="2">Canonical form</th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td></tr><tr><td>$E_{+\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td></tr><tr><td>$E_{+\mathbf{e}_i+\mathbf{e}_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td></tr><tr><td>$E_{-\mathbf{e}_i-\mathbf{e}_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td></tr><tr><td>$E_{+2\mathbf{e}_i}$</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td></tr><tr><td>$E_{-2\mathbf{e}_i}$</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td></tr></table>	Canonical form	C_l		D_l		Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives	H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$	$E_{+\mathbf{e}_i-\mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$	$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$	$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$	$E_{+2\mathbf{e}_i}$	$b_i^\dagger b_i^\dagger$	$x^i x^i$			$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$		
Canonical form	C_l		D_l																																									
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H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$																																								
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$E_{+\mathbf{e}_i+\mathbf{e}_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$																																								
$E_{-\mathbf{e}_i-\mathbf{e}_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$																																								
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$E_{-2\mathbf{e}_i}$	$b_i b_i$	$\partial_i \partial_i$																																										
gilmore-lie-groups_ F01814	185	■ 0.852	D, E	D, E	Dynkin diagrams of type (D, E) to obtain the inequality of Eq. (10.19).																																							
gilmore-lie-groups_ F01815	185	■ 1.000	$1 \leq i, \ j \leq n$	$1 \leq i, j \leq n$	8. A Lie algebra is spanned by n^2 operators of the form $a_i^\dagger a_j$, with $1 \leq i, j \leq n$. Show																																							
gilmore-lie-groups_ F01816	185	■ 1.000	n=3	$n = 3$	For $n = 3$ the subspaces transform like an angular momentum vector and a quadrupole																																							
gilmore-lie-groups_ F01817	185	■ 1.000	$\hat{\mathbf{Q}}$	$\hat{\mathbf{Q}}$	traceless tensor $\hat{\mathbf{Q}}$ and a scalar N :																																							

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gilmore-lie-groups_ F01818	185	■ 1.000	$\hat{\mathbf{N}}$	$\hat{N}$	The operator $\hat{N}$ commutes with all operators $a_i^\dagger a_j$. Interpret these commutation re-
gilmore-lie-groups_ F01819	185	■ 0.998	$\mathbf{L} + \hat{\mathbf{Q}}$	$\mathbf{L} + \hat{\mathbf{Q}}$	simple Lie algebra $\mathbf{L} + \mathbf{Q}$).
gilmore-lie-groups_ F01820	185	■ 0.623	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	$\mathbf{L}(3 \rightarrow n(n-1)/2)$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_ F01821	185	■ 0.623	$\mathbf{Q}$	$\mathbf{Q}$	the discussion of Problem 8 are the dimensions of the spaces $\mathbf{L}(3 \rightarrow n(n-1)/2)$, $\mathbf{Q}$
gilmore-lie-groups_ F01822	185	■ 1.000	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_ F01823	185	■ 1.000	$\hat{\mathbf{N}}(3 \rightarrow n)$	$\hat{N}(3 \rightarrow n)$	$(6 \rightarrow n(n+1)/2)$, and the definition of $\hat{N}(3 \rightarrow n)$.
gilmore-lie-groups_ F01824	186	■ 1.000	$Z = r^i X_i$	$Z = r^i X_i$	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_ F01825	186	■ 1.000	r^i	r^i	Lie algebra with elements $Z = r^i X_i$ (r^i are real numbers, X_i the generators of the
gilmore-lie-groups_ F01826	186	■ 1.000	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	$\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$	of the form $\sum \sum r^i [R_i^j(Z) - \lambda \delta_i^j] X_j = 0$. This equation cannot be solved in
gilmore-lie-groups_ F01827	186	■ 1.000	h^i	h^i	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_ F01828	186	■ 1.000	e^α	e^α	where h^i and e^α are complex numbers and the “diagonal” and “shift” operators
gilmore-lie-groups_ F01829	187	■ 1.000	A_{n-1}, D_n, B_n, C_n	A_{n-1}, D_n, B_n, C_n	classical simple Lie algebras A_{n-1}, D_n, B_n, C_n . These decompositions are: block
gilmore-lie-groups_ F01830	187	■ 1.000	a_1, a_2, a_3	a_1, a_2, a_3	for certain choices of the real parameters a_1, a_2, a_3 while in the compact case this
gilmore-lie-groups_ F01831	187	■ 1.000	(a_1, a_2, a_3)	(a_1, a_2, a_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_ F01832	187	■ 1.000	(b_1, b_2, b_3)	(b_1, b_2, b_3)	is not possible. If the real parameters (a_1, a_2, a_3) and (b_1, b_2, b_3) are allowed to
gilmore-lie-groups_ F01833	188	■ 0.998	H_i, E_α	H_i, E_α	The Cartan–Killing inner product for the basis vectors H_i, E_α is
gilmore-lie-groups_ F01834	188	■ 1.000	$\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$	of basis vectors $\frac{1}{\sqrt{2}}(E_\alpha \pm E_{-\alpha})$:
gilmore-lie-groups_ F01835	189	■ 0.826	$H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	$H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01836	189	■ 0.826	$\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01837	189	■ 0.826	$\alpha \neq 0$	$\alpha \neq 0$	If we restrict the coefficients of $H_i, \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$, and $\frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha})$ (all $\alpha \neq 0$)
gilmore-lie-groups_ F01838	189	■ 1.000	$\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	$\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$	generators H_i and $\frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha})$ span a noncompact subspace.
gilmore-lie-groups_ F01839	189	■ 0.999	$A_1, \mathfrak{sl}(2; \mathbb{R})$	$A_1, \mathfrak{sl}(2; \mathbb{R})$	Two real forms of $A_1, \mathfrak{sl}(2; \mathbb{R})$ and $\mathfrak{su}(1, 1)$, are obtained as follows
gilmore-lie-groups_ F01840	189	■ 1.000	$h_r, h_i; a_r, a_i; b_r, b_i$	$h_r, h_i; a_r, a_i; b_r, b_i$	Here the six parameters $h_r, h_i; a_r, a_i; b_r, b_i$ are real.
gilmore-lie-groups_ F01841	190	■ 0.850	n_+, n_-	n_+, n_-	It is useful to specify how compact a real form is by specifying its index (n_+, n_-) ,
gilmore-lie-groups_ F01842	190	■ 1.000	n_+	n_+	where n_+ is the dimension of the subspace on which the nonsingular Cartan–
gilmore-lie-groups_ F01843	190	■ 1.000	n_-	n_-	Killing inner product is positive-definite and n_- is the dimension of the subspace

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gilmore-lie-groups_ F01844	190	■ 1.000	$\chi=n_{+}-n_{-}$	$\chi = n_{+} - n_{-}$	be abbreviated to a single integer, the character $\chi = n_{+} - n_{-}$, to describe a real
gilmore-lie-groups_ F01845	190	■ 1.000	$i\,H_i, i\,\frac{1}{\sqrt{2}}\left(E_{\alpha}+E_{-\alpha}\right)$	$iH_i, i\frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$	the compact real form spanned by real linear combinations of $iH_i, i\frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$,
gilmore-lie-groups_ F01846	190	■ 0.917	$g_{\mu,\,\,\nu}=\operatorname{diag}(1,1,1,-1)$	$g_{\mu,\nu} = \operatorname{diag}(1,1,1,-1)$	mation of space-time $(x,\,y,\,z,\,ct)$ with indefinite metric $g_{\mu,\nu} = \operatorname{diag}(1,\,1,\,1,\,-1)$ to
gilmore-lie-groups_ F01847	190	■ 0.870	$x,\,y,\,z,\,i\,c\,t$	x,y,z,ict	space-time with imaginary time $(x,\,y,\,z,\,ict)$ and positive-definite metric $g_{\mu,\nu} =$
gilmore-lie-groups_ F01848	190	■ 0.870	$g_{\mu,\,\,\nu}=$	$g_{\mu,\nu} =$	space-time with imaginary time $(x,\,y,\,z,\,ict)$ and positive-definite metric $g_{\mu,\nu} =$
gilmore-lie-groups_ F01849	190	■ 0.672	$\operatorname{diag}(1,1,1,1)$	$\operatorname{diag}(1,1,1,1)$	diag(1, 1, 1, 1).
gilmore-lie-groups_ F01850	190	■ 1.000	$\mathfrak{p}\rightarrow\mathfrak{p}^{\prime}=i\mathfrak{p}$	$\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$	“Minkowski’s trick”: $\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$. The mapping, commutation relations, and
gilmore-lie-groups_ F01851	191	■ 1.000	$\mathfrak{g}^{\prime},\,\mathfrak{h}$	$\mathfrak{g}', \mathfrak{h}$	In $\mathfrak{g}'$, $\mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact
gilmore-lie-groups_ F01852	191	■ 1.000	$\mathfrak{p}^{\prime}$	$\mathfrak{p}'$	In $\mathfrak{g}'$, $\mathfrak{h}$ is the maximal compact subalgebra and $\mathfrak{p}'$ consists of all the noncompact
gilmore-lie-groups_ F01853	191	■ 1.000	$T^2=I$	$T^2 = I$	compact real form of that Lie algebra to itself that obey $T^2 = I$.
gilmore-lie-groups_ F01854	192	■ 1.000	$\mathfrak{u}(n,\,\mathbb{F})$	$\mathfrak{u}(n, \mathbb{F})$	In a block matrix decomposition the compact Lie algebras $\mathfrak{u}(n, \mathbb{F})$ have the form
gilmore-lie-groups_ F01855	192	■ 0.949	$A_{\mathfrak{p}}=-A_{\mathfrak{p}}^{\dagger},\,A_{\mathfrak{q}}=-A_{\mathfrak{q}}^{\dagger}$	$A_p = -A_p^{\dagger}, A_q = -A_q^{\dagger}$	where $A_p = -A_p^{\dagger}$, $A_q = -A_q^{\dagger}$, and B is an arbitrary $p \times q$ matrix. Under the
gilmore-lie-groups_ F01856	192	■ 0.999	$I_{\mathfrak{p}+\mathfrak{q}}$	I_{p+q}	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01857	192	■ 0.999	$\mathfrak{u}(n\,;\,\mathbb{F})$	$\mathfrak{u}(n; \mathbb{F})$	i . This is equivalent to changing the metric I_{p+q} that is preserved by $\mathfrak{u}(n; \mathbb{F})$ to the
gilmore-lie-groups_ F01858	192	■ 1.000	$I_{\mathfrak{p},\,\mathfrak{q}}$	$I_{p,q}$	metric $I_{p,q}$ that is preserved by $\mathfrak{u}(p,\,q; \mathbb{F})$, where $p+q=n$. The factor i can be
gilmore-lie-groups_ F01859	192	■ 1.000	$\mathfrak{u}(p,\,q\,;\,\mathbb{F})$	$\mathfrak{u}(p,q; \mathbb{F})$	metric $I_{p,q}$ that is preserved by $\mathfrak{u}(p,q; \mathbb{F})$, where $p+q=n$. The factor i can be
gilmore-lie-groups_ F01860	192	■ 0.995	$\mathbb{F}=\mathbb{R},\,\mathbb{C},\,\mathbb{Q}$	$\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces $(D,\,B)$, $A,\,C$ the real forms
gilmore-lie-groups_ F01861	192	■ 0.995	$(D,\,B),\,A,\,C$	$(D,B), A, C$	For the fields $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{Q}$ related to the root spaces $(D,\,B)$, $A,\,C$ the real forms
gilmore-lie-groups_ F01862	192	■ 0.998	$\mathfrak{sp}(2n\,;\,\mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$.
gilmore-lie-groups_ F01863	193	■ 1.000	$\mathfrak{sp}(n)=\mathfrak{u}(n\,;\,\mathbb{Q})$	$\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$	The transformation from $\mathfrak{sp}(n) = \mathfrak{u}(n; \mathbb{Q})$ to $\mathfrak{sp}(2n; \mathbb{R})$ is somewhat less familiar.
gilmore-lie-groups_ F01864	193	■ 1.000	$\alpha,\,\beta,\,\gamma$	α, β, γ	where α, β, γ and δ are real. With these replacements the Lie algebra of $n \times n$
gilmore-lie-groups_ F01865	193	■ 0.999	$\mathfrak{o}(2n)$	$\mathfrak{ou}(2n)$	these matrices $\mathfrak{ou}(2n)$, since they form an <u>orthogonal</u> representation of the <u>unitary</u>
gilmore-lie-groups_ F01866	193	■ 1.000	$\underline{2n}\times 2n$	$\underline{2n} \times 2n$	algebra in terms of $\underline{2n} \times 2n$ matrices. Similarly, the Lie algebra of $n \times n$ quaternion
gilmore-lie-groups_ F01867	194	■ 1.000	R^{2n}	R^{2n}	metric metric in R^{2n} . It is possible to choose coordinates $p_1,\,q_1,\,p_2,\,q_2,\,\dots,\,p_n,\,q_n$
gilmore-lie-groups_ F01868	194	■ 1.000	$p_1,\,q_1,\,p_2,\,q_2,\,\ldots,\,p_n,\,q_n$	$p_1, q_1, p_2, q_2, \dots, p_n, q_n$	metric metric in R^{2n} . It is possible to choose coordinates $p_1,\,q_1,\,p_2,\,q_2,\,\dots,\,p_n,\,q_n$
gilmore-lie-groups_ F01869	194	■ 1.000	$v_i^{\prime}\,G_{ij}\,v_j$	$v_i' G_{ij} v_j$	in this space so that the inner product between two vectors $v_i' G_{ij} v_j$ is
gilmore-lie-groups_ F01870	194	■ 0.967	$M\in\operatorname{Sp}(2n\,;\,\mathbb{R})$	$M \in \operatorname{Sp}(2n; \mathbb{R})$	Then symplectic transformations $M \in \operatorname{Sp}(2n; \mathbb{R})$ leave this metric matrix G invari-

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gilmore-lie-groups_ F01871	194	■ 1.000	$M^{\mathrm{t}} \text{ G M} = \text{G}$	$M^t G M = G$	ant: $M^t G M = G$. Symplectic transformations leave invariant the canonical form																															
gilmore-lie-groups_ F01872	194	■ 1.000	$\mathfrak{p} \rightarrow i \mathfrak{p}$	$\mathfrak{p} \rightarrow i \mathfrak{p}$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.																															
gilmore-lie-groups_ F01873	194	■ 1.000	$\mathfrak{s} \mathfrak{o}^{\wedge *} (2 \ n)$	$\mathfrak{so}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.																															
gilmore-lie-groups_ F01874	194	■ 1.000	$\mathfrak{s} \mathfrak{u}^{\wedge *} (2 \ n)$	$\mathfrak{su}^*(2n)$	Application of the map $\mathfrak{p} \rightarrow i \mathfrak{p}$ produces the real forms $\mathfrak{so}^*(2n)$ and $\mathfrak{su}^*(2n)$.																															
gilmore-lie-groups_ F01875	194	■ 0.991	$A_{2 \ n-1}$	A_{2n-1}	Remark The real forms $\mathfrak{so}^*(2n)$ of D_n and $\mathfrak{su}^*(2n)$ of A_{2n-1} do not occur explicitly																															
gilmore-lie-groups_ F01876	195	■ 1.000	$\mathfrak{s} \mathfrak{u}(2)), B_1(\mathfrak{s} \mathfrak{o}(3))$	$\mathfrak{su}(2)), B_1(\mathfrak{so}(3))$	the compact real form is $\mathfrak{su}(2)), B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2																															
gilmore-lie-groups_ F01877	195	■ 1.000	$C_1(\mathfrak{s} \mathfrak{p}(1))$	$C_1(\mathfrak{sp}(1))$	the compact real form is $\mathfrak{su}(2)), B_1(\mathfrak{so}(3))$, and $C_1(\mathfrak{sp}(1))$ are equivalent, as are B_2																															
gilmore-lie-groups_ F01878	195	■ 1.000	$(\mathfrak{s} \mathfrak{o}(5))$	$(\mathfrak{so}(5))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are																															
gilmore-lie-groups_ F01879	195	■ 1.000	$C_2(\mathfrak{s} \mathfrak{p}(2))$	$C_2(\mathfrak{sp}(2))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are																															
gilmore-lie-groups_ F01880	195	■ 1.000	$A_3(\mathfrak{s} \mathfrak{u}(4))$	$A_3(\mathfrak{su}(4))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are																															
gilmore-lie-groups_ F01881	195	■ 1.000	$D_3(\mathfrak{s} \mathfrak{o}(6))$	$D_3(\mathfrak{so}(6))$	$(\mathfrak{so}(5))$ and $C_2(\mathfrak{sp}(2))$. So also are $A_3(\mathfrak{su}(4))$ and $D_3(\mathfrak{so}(6))$. As a result, there are																															
gilmore-lie-groups_ F01882	195	■ 0.899	$\mathfrak{s} \mathfrak{o}(p, q)$	$\mathfrak{so}(p, q)$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}	
Mapping	Real form	Root space	Condition																																	
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gilmore-lie-groups_ F01883	195	■ 0.899	$p+q=2 \ n$	$p + q = 2n$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}	
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gilmore-lie-groups_ F01884	195	■ 0.899	$p+q=2 \ n+1$	$p + q = 2n + 1$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}	
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gilmore-lie-groups_ F01885	195	0.899	$\mathfrak{s} \mathfrak{u}(p, q)$	$\mathfrak{su}(p, q)$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}																																																												
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gilmore-lie-groups_ F01886	195	0.899	$\mathfrak{s} \mathfrak{p}(p, q)$	$\mathfrak{sp}(p, q)$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}																																																												
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gilmore-lie-groups_ F01887	195	0.899	$\mathfrak{s} \mathfrak{p}(2n; \mathbb{R})$	$\mathfrak{sp}(2n; \mathbb{R})$	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td rowspan="4">Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td rowspan="2">Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td rowspan="2">Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$	$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$	$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$	$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}		$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n		$\mathfrak{su}^*(2n)$	A_{2n-1}																																																												
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gilmore-lie-groups_ F01888	195	0.500	B_1	B_1	<table><tr><td>A_1</td><td>$\sim$</td><td>B_1</td><td>$\sim$</td><td>C_1</td></tr><tr><td>$\mathfrak{su}(2)$</td><td>$\sim$</td><td>$\mathfrak{so}(3)$</td><td>$\sim$</td><td>$\mathfrak{sp}(1) = \mathfrak{usp}(2)$</td></tr><tr><td>$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$</td><td>$\sim$</td><td>$\mathfrak{so}(2, 1)$</td><td>$\sim$</td><td>$\mathfrak{sp}(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>$=$</td><td>$A_1$</td><td>$+$</td><td>$A_1$</td></tr><tr><td>$\mathfrak{so}(4)$</td><td>$=$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(3)$</td></tr><tr><td>$\mathfrak{so}^*(4)$</td><td>$\sim$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>$\mathfrak{so}(3, 1)$</td><td>$\sim$</td><td>$\mathfrak{sl}(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(2, 2)$</td><td>$\sim$</td><td>$\mathfrak{so}(2, 1)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5)$</td><td>$\sim$</td><td>$\mathfrak{sp}(2) = \mathfrak{usp}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 1)$</td><td>$\sim$</td><td>$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 2)$</td><td>$\sim$</td><td>$\mathfrak{sp}(4; \mathbb{R})$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5, 1)$</td><td>$\sim$</td><td>$\mathfrak{su}^*(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}^*(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(3, 1)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 2)$</td><td>$\sim$</td><td>$\mathfrak{su}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 3)$</td><td>$\sim$</td><td>$\mathfrak{sl}(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1	$\sim$	B_1	$\sim$	C_1	$\mathfrak{su}(2)$	$\sim$	$\mathfrak{so}(3)$	$\sim$	$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$	$\sim$	$\mathfrak{so}(2, 1)$	$\sim$	$\mathfrak{sp}(2; \mathbb{R})$	D_2	$=$	A_1	$+$	A_1	$\mathfrak{so}(4)$	$=$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(3)$	$\mathfrak{so}^*(4)$	$\sim$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(3, 1)$	$\sim$	$\mathfrak{sl}(2; \mathbb{C})$			$\mathfrak{so}(2, 2)$	$\sim$	$\mathfrak{so}(2, 1)$	$+$	$\mathfrak{so}(2, 1)$	B_2	$=$	C_2			$\mathfrak{so}(5)$	$\sim$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$			$\mathfrak{so}(4, 1)$	$\sim$	$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$			$\mathfrak{so}(3, 2)$	$\sim$	$\mathfrak{sp}(4; \mathbb{R})$			D_3	$=$	A_3			$\mathfrak{so}(6)$	$\sim$	$\mathfrak{su}(4)$			$\mathfrak{so}(5, 1)$	$\sim$	$\mathfrak{su}^*(4)$			$\mathfrak{so}^*(6)$	$\sim$	$\mathfrak{su}(3, 1)$			$\mathfrak{so}(4, 2)$	$\sim$	$\mathfrak{su}(2, 2)$			$\mathfrak{so}(3, 3)$	$\sim$	$\mathfrak{sl}(4; \mathbb{R})$		
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gilmore-lie-groups_ F01889	195	■ 0.500	C_1	C_1	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01890	195	■ 0.500	$\frac{\mathfrak{s}}{\mathfrak{o}}(3)$	$\mathfrak{so}(3)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01891	195	■ 0.500	$\frac{\mathfrak{s}}{\mathfrak{p}}(1) = \frac{\mathfrak{u}}{\mathfrak{s}} \frac{\mathfrak{s}}{\mathfrak{p}}(2)$	$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01892	195	■ 0.500	$\mathfrak{su}(1,1)=\mathfrak{sl}(2;\mathbb{R})$	$\mathfrak{su}(1,1)=\mathfrak{sl}(2;\mathbb{R})$	
gilmore-lie-groups_ F01893	195	■ 0.500	$\mathfrak{so}(2,1)$	$\mathfrak{so}(2,1)$	
gilmore-lie-groups_ F01894	195	■ 0.500	$\mathfrak{sp}(2;\mathbb{R})$	$\mathfrak{sp}(2;\mathbb{R})$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01895	195	■ 0.500	$\mathfrak{so}(4)$	$\mathfrak{so}(4)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01896	195	■ 0.500	$\mathfrak{so}^*(4)$	$\mathfrak{so}^*(4)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01897	195	■ 0.500	$\mathfrak{so}(3, 1)$	$\mathfrak{so}(3, 1)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F01898	195	■ 0.500	$\mathfrak{sl}(2; \mathbb{C})$	$\mathfrak{sl}(2; \mathbb{C})$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01899	195	■ 0.500	$\mathfrak{so}(2, 2)$	$\mathfrak{so}(2, 2)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $
gilmore-lie-groups_ F01900	195	■ 0.500	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; \mathbb{R}) & \\ D_3 & = & A_3 & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & \end{array} $

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gilmore-lie-groups_ F01901	195	■ 0.500	$\frac{s}{p}(1,1)=\frac{u}{s}\frac{s}{p}(2,2)$	$\mathfrak{sp}(1,1) = \mathfrak{usp}(2,2)$	
gilmore-lie-groups_ F01902	195	■ 0.500	$\frac{s}{p}(4 ; R)$	$\mathfrak{sp}(4; R)$	
gilmore-lie-groups_ F01903	195	■ 0.500	$D_{\{3\}}$	D_3	

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gilmore-lie-groups_ F01984	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(6)$	$\mathfrak{so}(6)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01985	195	■ 0.500	$\mathfrak{s} \mathfrak{u}(4)$	$\mathfrak{su}(4)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01986	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(5, 1)$	$\mathfrak{so}(5, 1)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2, 1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2, 1) \\ \mathfrak{so}(3, 1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2, 2) & \sim & \mathfrak{so}(2, 1) & + & \mathfrak{so}(2, 1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4, 1) & \sim & \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2) & & \\ \mathfrak{so}(3, 2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5, 1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3, 1) & & \\ \mathfrak{so}(4, 2) & \sim & \mathfrak{su}(2, 2) & & \\ \mathfrak{so}(3, 3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $

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gilmore-lie-groups_ F01907	195	■ 0.500	$\mathfrak{s} \mathfrak{u}^*(4)$	$\mathfrak{su}^*(4)$	$A_1 \sim B_1 \sim C_1$ $\mathfrak{su}(2) \sim \mathfrak{so}(3) \sim \mathfrak{sp}(1) = \mathfrak{usp}(2)$ $\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) \sim \mathfrak{so}(2, 1) \sim \mathfrak{sp}(2; \mathbb{R})$ $D_2 = A_1 + A_1$ $\mathfrak{so}(4) = \mathfrak{so}(3) + \mathfrak{so}(3)$ $\mathfrak{so}^*(4) \sim \mathfrak{so}(3) + \mathfrak{so}(2, 1)$ $\mathfrak{so}(3, 1) \sim \mathfrak{sl}(2; \mathbb{C})$ $\mathfrak{so}(2, 2) \sim \mathfrak{so}(2, 1) + \mathfrak{so}(2, 1)$ $B_2 = C_2$ $\mathfrak{so}(5) \sim \mathfrak{sp}(2) = \mathfrak{usp}(4)$ $\mathfrak{so}(4, 1) \sim \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$ $\mathfrak{so}(3, 2) \sim \mathfrak{sp}(4; R)$ $D_3 = A_3$ $\mathfrak{so}(6) \sim \mathfrak{su}(4)$ $\mathfrak{so}(5, 1) \sim \mathfrak{su}^*(4)$ $\mathfrak{so}^*(6) \sim \mathfrak{su}(3, 1)$ $\mathfrak{so}(4, 2) \sim \mathfrak{su}(2, 2)$ $\mathfrak{so}(3, 3) \sim \mathfrak{sl}(4; \mathbb{R})$
gilmore-lie-groups_ F01908	195	■ 0.500	$\mathfrak{s} \mathfrak{o}^*(6)$	$\mathfrak{so}^*(6)$	$A_1 \sim B_1 \sim C_1$ $\mathfrak{su}(2) \sim \mathfrak{so}(3) \sim \mathfrak{sp}(1) = \mathfrak{usp}(2)$ $\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) \sim \mathfrak{so}(2, 1) \sim \mathfrak{sp}(2; \mathbb{R})$ $D_2 = A_1 + A_1$ $\mathfrak{so}(4) = \mathfrak{so}(3) + \mathfrak{so}(3)$ $\mathfrak{so}^*(4) \sim \mathfrak{so}(3) + \mathfrak{so}(2, 1)$ $\mathfrak{so}(3, 1) \sim \mathfrak{sl}(2; \mathbb{C})$ $\mathfrak{so}(2, 2) \sim \mathfrak{so}(2, 1) + \mathfrak{so}(2, 1)$ $B_2 = C_2$ $\mathfrak{so}(5) \sim \mathfrak{sp}(2) = \mathfrak{usp}(4)$ $\mathfrak{so}(4, 1) \sim \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$ $\mathfrak{so}(3, 2) \sim \mathfrak{sp}(4; R)$ $D_3 = A_3$ $\mathfrak{so}(6) \sim \mathfrak{su}(4)$ $\mathfrak{so}(5, 1) \sim \mathfrak{su}^*(4)$ $\mathfrak{so}^*(6) \sim \mathfrak{su}(3, 1)$ $\mathfrak{so}(4, 2) \sim \mathfrak{su}(2, 2)$ $\mathfrak{so}(3, 3) \sim \mathfrak{sl}(4; \mathbb{R})$
gilmore-lie-groups_ F01909	195	■ 0.500	$\mathfrak{s} \mathfrak{u}(3, 1)$	$\mathfrak{su}(3, 1)$	$A_1 \sim B_1 \sim C_1$ $\mathfrak{su}(2) \sim \mathfrak{so}(3) \sim \mathfrak{sp}(1) = \mathfrak{usp}(2)$ $\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R}) \sim \mathfrak{so}(2, 1) \sim \mathfrak{sp}(2; \mathbb{R})$ $D_2 = A_1 + A_1$ $\mathfrak{so}(4) = \mathfrak{so}(3) + \mathfrak{so}(3)$ $\mathfrak{so}^*(4) \sim \mathfrak{so}(3) + \mathfrak{so}(2, 1)$ $\mathfrak{so}(3, 1) \sim \mathfrak{sl}(2; \mathbb{C})$ $\mathfrak{so}(2, 2) \sim \mathfrak{so}(2, 1) + \mathfrak{so}(2, 1)$ $B_2 = C_2$ $\mathfrak{so}(5) \sim \mathfrak{sp}(2) = \mathfrak{usp}(4)$ $\mathfrak{so}(4, 1) \sim \mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$ $\mathfrak{so}(3, 2) \sim \mathfrak{sp}(4; R)$ $D_3 = A_3$ $\mathfrak{so}(6) \sim \mathfrak{su}(4)$ $\mathfrak{so}(5, 1) \sim \mathfrak{su}^*(4)$ $\mathfrak{so}^*(6) \sim \mathfrak{su}(3, 1)$ $\mathfrak{so}(4, 2) \sim \mathfrak{su}(2, 2)$ $\mathfrak{so}(3, 3) \sim \mathfrak{sl}(4; \mathbb{R})$

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gilmore-lie-groups_ F01910	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(4,2)$	$\mathfrak{so}(4,2)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1,1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2,1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01911	195	■ 0.500	$\mathfrak{s} \mathfrak{u}(2,2)$	$\mathfrak{su}(2,2)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1,1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2,1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $
gilmore-lie-groups_ F01912	195	■ 0.500	$\mathfrak{s} \mathfrak{o}(3,3)$	$\mathfrak{so}(3,3)$	$ \begin{array}{llll} A_1 & \sim & B_1 & \sim & C_1 \\ \mathfrak{su}(2) & \sim & \mathfrak{so}(3) & \sim & \mathfrak{sp}(1) = \mathfrak{usp}(2) \\ \mathfrak{su}(1,1) = \mathfrak{sl}(2; \mathbb{R}) & \sim & \mathfrak{so}(2,1) & \sim & \mathfrak{sp}(2; \mathbb{R}) \\ D_2 & = & A_1 & + & A_1 \\ \mathfrak{so}(4) & = & \mathfrak{so}(3) & + & \mathfrak{so}(3) \\ \mathfrak{so}^*(4) & \sim & \mathfrak{so}(3) & + & \mathfrak{so}(2,1) \\ \mathfrak{so}(3,1) & \sim & \mathfrak{sl}(2; \mathbb{C}) & & \\ \mathfrak{so}(2,2) & \sim & \mathfrak{so}(2,1) & + & \mathfrak{so}(2,1) \\ B_2 & = & C_2 & & \\ \mathfrak{so}(5) & \sim & \mathfrak{sp}(2) = \mathfrak{usp}(4) & & \\ \mathfrak{so}(4,1) & \sim & \mathfrak{sp}(1,1) = \mathfrak{usp}(2,2) & & \\ \mathfrak{so}(3,2) & \sim & \mathfrak{sp}(4; R) & & \\ D_3 & = & A_3 & & \\ \mathfrak{so}(6) & \sim & \mathfrak{su}(4) & & \\ \mathfrak{so}(5,1) & \sim & \mathfrak{su}^*(4) & & \\ \mathfrak{so}^*(6) & \sim & \mathfrak{su}(3,1) & & \\ \mathfrak{so}(4,2) & \sim & \mathfrak{su}(2,2) & & \\ \mathfrak{so}(3,3) & \sim & \mathfrak{sl}(4; \mathbb{R}) & & \end{array} $

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gilmore-lie-groups_ F01913	195	■ 0.500	$\mathfrak{sl}(4; \mathbb{R})$	$\mathfrak{sl}(4; \mathbb{R})$	<table><tr><td>A_1</td><td>$\sim$</td><td>B_1</td><td>$\sim$</td><td>C_1</td></tr><tr><td>$\mathfrak{su}(2)$</td><td>$\sim$</td><td>$\mathfrak{so}(3)$</td><td>$\sim$</td><td>$\mathfrak{sp}(1) = \mathfrak{usp}(2)$</td></tr><tr><td>$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$</td><td>$\sim$</td><td>$\mathfrak{so}(2, 1)$</td><td>$\sim$</td><td>$\mathfrak{sp}(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>$=$</td><td>$A_1$</td><td>$+$</td><td>$A_1$</td></tr><tr><td>$\mathfrak{so}(4)$</td><td>$=$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(3)$</td></tr><tr><td>$\mathfrak{so}^*(4)$</td><td>$\sim$</td><td>$\mathfrak{so}(3)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>$\mathfrak{so}(3, 1)$</td><td>$\sim$</td><td>$\mathfrak{sl}(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(2, 2)$</td><td>$\sim$</td><td>$\mathfrak{so}(2, 1)$</td><td>$+$</td><td>$\mathfrak{so}(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5)$</td><td>$\sim$</td><td>$\mathfrak{sp}(2) = \mathfrak{usp}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 1)$</td><td>$\sim$</td><td>$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 2)$</td><td>$\sim$</td><td>$\mathfrak{sp}(4; \mathbb{R})$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(5, 1)$</td><td>$\sim$</td><td>$\mathfrak{su}^*(4)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}^*(6)$</td><td>$\sim$</td><td>$\mathfrak{su}(3, 1)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(4, 2)$</td><td>$\sim$</td><td>$\mathfrak{su}(2, 2)$</td><td></td><td></td></tr><tr><td>$\mathfrak{so}(3, 3)$</td><td>$\sim$</td><td>$\mathfrak{sl}(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1	$\sim$	B_1	$\sim$	C_1	$\mathfrak{su}(2)$	$\sim$	$\mathfrak{so}(3)$	$\sim$	$\mathfrak{sp}(1) = \mathfrak{usp}(2)$	$\mathfrak{su}(1, 1) = \mathfrak{sl}(2; \mathbb{R})$	$\sim$	$\mathfrak{so}(2, 1)$	$\sim$	$\mathfrak{sp}(2; \mathbb{R})$	D_2	$=$	A_1	$+$	A_1	$\mathfrak{so}(4)$	$=$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(3)$	$\mathfrak{so}^*(4)$	$\sim$	$\mathfrak{so}(3)$	$+$	$\mathfrak{so}(2, 1)$	$\mathfrak{so}(3, 1)$	$\sim$	$\mathfrak{sl}(2; \mathbb{C})$			$\mathfrak{so}(2, 2)$	$\sim$	$\mathfrak{so}(2, 1)$	$+$	$\mathfrak{so}(2, 1)$	B_2	$=$	C_2			$\mathfrak{so}(5)$	$\sim$	$\mathfrak{sp}(2) = \mathfrak{usp}(4)$			$\mathfrak{so}(4, 1)$	$\sim$	$\mathfrak{sp}(1, 1) = \mathfrak{usp}(2, 2)$			$\mathfrak{so}(3, 2)$	$\sim$	$\mathfrak{sp}(4; \mathbb{R})$			D_3	$=$	A_3			$\mathfrak{so}(6)$	$\sim$	$\mathfrak{su}(4)$			$\mathfrak{so}(5, 1)$	$\sim$	$\mathfrak{su}^*(4)$			$\mathfrak{so}^*(6)$	$\sim$	$\mathfrak{su}(3, 1)$			$\mathfrak{so}(4, 2)$	$\sim$	$\mathfrak{su}(2, 2)$			$\mathfrak{so}(3, 3)$	$\sim$	$\mathfrak{sl}(4; \mathbb{R})$		
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gilmore-lie-groups_ F01914	196	■ 1.000	$\{ \}_{{\text{rank}}(\text{character})}$	$\text{rank}(\text{character})$	<table><tr><th colspan="4">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Class_{rank(character)}</th><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Maximal compact subgroup				Root space	Class _{rank(character)}	Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120																										
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gilmore-lie-groups_ F01915	196	1.000	$G_{2(-14)}$	$G_{2(-14)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01918	196	1.000	$F_{\{4\}}$	F_4	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01919	196	1.000	$F_{4(-52)}$	$F_{4(-52)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01920	196	1.000	$F_{4(-20)}$	$F_{4(-20)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01924	196	■ 1.000	$E_{\{6(-78)\}}$	$E_{6(-78)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01926	196	1.000	$E_{\{6(-14)\}}$	$E_{6(-14)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01928	196	1.000	$E_{\{6(+2)\}}$	$E_{6(+2)}$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01929	196	1.000	$A_{\{5\}}+A_{\{1\}}$	$A_5 + A_1$	<table><thead><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01930	196	1.000	$E_{\{6(+6)\}}$	$E_{6(+6)}$	<table><thead><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01931	196	1.000	C_{4}	C ₄	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G₂</td><td>G₂₍₋₁₄₎</td><td>G₂</td><td>14</td></tr><tr><td>G₂₍₊₂₎</td><td>A₁ + A₁</td><td>6</td></tr><tr><td rowspan="3">F₄</td><td>F₄₍₋₅₂₎</td><td>F₄</td><td>52</td></tr><tr><td>F₄₍₋₂₀₎</td><td>B₄</td><td>36</td></tr><tr><td>F₄₍₊₄₎</td><td>C₃ + A₁</td><td>24</td></tr><tr><td rowspan="5">E₆</td><td>E₆₍₋₇₈₎</td><td>E₆</td><td>78</td></tr><tr><td>E₆₍₋₂₆₎</td><td>F₄</td><td>52</td></tr><tr><td>E₆₍₋₁₄₎</td><td>D₅ + D₁</td><td>46</td></tr><tr><td>E₆₍₊₂₎</td><td>A₅ + A₁</td><td>38</td></tr><tr><td>E₆₍₊₆₎</td><td>C₄</td><td>36</td></tr><tr><td rowspan="4">E₇</td><td>E₇₍₋₁₃₃₎</td><td>E₇</td><td>133</td></tr><tr><td>E₇₍₋₂₅₎</td><td>E₆ + D₁</td><td>79</td></tr><tr><td>E₇₍₋₅₎</td><td>D₆ + A₁</td><td>69</td></tr><tr><td>E₇₍₊₇₎</td><td>A₇</td><td>63</td></tr><tr><td rowspan="3">E₈</td><td>E₈₍₋₂₄₈₎</td><td>E₈</td><td>248</td></tr><tr><td>E₈₍₋₂₄₎</td><td>E₇ + A₁</td><td>136</td></tr><tr><td>E₈₍₊₈₎</td><td>D₈</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G ₂	G ₂₍₋₁₄₎	G ₂	14	G ₂₍₊₂₎	A ₁ + A ₁	6	F ₄	F ₄₍₋₅₂₎	F ₄	52	F ₄₍₋₂₀₎	B ₄	36	F ₄₍₊₄₎	C ₃ + A ₁	24	E ₆	E ₆₍₋₇₈₎	E ₆	78	E ₆₍₋₂₆₎	F ₄	52	E ₆₍₋₁₄₎	D ₅ + D ₁	46	E ₆₍₊₂₎	A ₅ + A ₁	38	E ₆₍₊₆₎	C ₄	36	E ₇	E ₇₍₋₁₃₃₎	E ₇	133	E ₇₍₋₂₅₎	E ₆ + D ₁	79	E ₇₍₋₅₎	D ₆ + A ₁	69	E ₇₍₊₇₎	A ₇	63	E ₈	E ₈₍₋₂₄₈₎	E ₈	248	E ₈₍₋₂₄₎	E ₇ + A ₁	136	E ₈₍₊₈₎	D ₈	120
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gilmore-lie-groups_ F01932	196	1.000	E_{7(-133)}	E ₇₍₋₁₃₃₎	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G₂</td><td>G₂₍₋₁₄₎</td><td>G₂</td><td>14</td></tr><tr><td>G₂₍₊₂₎</td><td>A₁ + A₁</td><td>6</td></tr><tr><td rowspan="3">F₄</td><td>F₄₍₋₅₂₎</td><td>F₄</td><td>52</td></tr><tr><td>F₄₍₋₂₀₎</td><td>B₄</td><td>36</td></tr><tr><td>F₄₍₊₄₎</td><td>C₃ + A₁</td><td>24</td></tr><tr><td rowspan="5">E₆</td><td>E₆₍₋₇₈₎</td><td>E₆</td><td>78</td></tr><tr><td>E₆₍₋₂₆₎</td><td>F₄</td><td>52</td></tr><tr><td>E₆₍₋₁₄₎</td><td>D₅ + D₁</td><td>46</td></tr><tr><td>E₆₍₊₂₎</td><td>A₅ + A₁</td><td>38</td></tr><tr><td>E₆₍₊₆₎</td><td>C₄</td><td>36</td></tr><tr><td rowspan="4">E₇</td><td>E₇₍₋₁₃₃₎</td><td>E₇</td><td>133</td></tr><tr><td>E₇₍₋₂₅₎</td><td>E₆ + D₁</td><td>79</td></tr><tr><td>E₇₍₋₅₎</td><td>D₆ + A₁</td><td>69</td></tr><tr><td>E₇₍₊₇₎</td><td>A₇</td><td>63</td></tr><tr><td rowspan="3">E₈</td><td>E₈₍₋₂₄₈₎</td><td>E₈</td><td>248</td></tr><tr><td>E₈₍₋₂₄₎</td><td>E₇ + A₁</td><td>136</td></tr><tr><td>E₈₍₊₈₎</td><td>D₈</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G ₂	G ₂₍₋₁₄₎	G ₂	14	G ₂₍₊₂₎	A ₁ + A ₁	6	F ₄	F ₄₍₋₅₂₎	F ₄	52	F ₄₍₋₂₀₎	B ₄	36	F ₄₍₊₄₎	C ₃ + A ₁	24	E ₆	E ₆₍₋₇₈₎	E ₆	78	E ₆₍₋₂₆₎	F ₄	52	E ₆₍₋₁₄₎	D ₅ + D ₁	46	E ₆₍₊₂₎	A ₅ + A ₁	38	E ₆₍₊₆₎	C ₄	36	E ₇	E ₇₍₋₁₃₃₎	E ₇	133	E ₇₍₋₂₅₎	E ₆ + D ₁	79	E ₇₍₋₅₎	D ₆ + A ₁	69	E ₇₍₊₇₎	A ₇	63	E ₈	E ₈₍₋₂₄₈₎	E ₈	248	E ₈₍₋₂₄₎	E ₇ + A ₁	136	E ₈₍₊₈₎	D ₈	120
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gilmore-lie-groups_ F01934	196	1.000	$E_{\{6\}+D_{\{1\}}}$	$E_6 + D_1$	<table><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01941	196	1.000	$E_{\{7\}}+A_{\{1\}}$	$E_7 + A_1$	<table><thead><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_ F01942	196	1.000	$E_{\{8(+8)\}}$	$E_{8(+8)}$	<table><thead><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_F01943	196	1.000	$D_{\{8\}}$	D_8	<table><thead><tr><th rowspan="2">Root space</th><th rowspan="2">Class_{rank(character)}</th><th colspan="2">Maximal compact subgroup</th></tr><tr><th>Root space</th><th>Dimension</th></tr></thead><tbody><tr><td rowspan="2">G_2</td><td>$G_{2(-14)}$</td><td>G_2</td><td>14</td></tr><tr><td>$G_{2(+2)}$</td><td>$A_1 + A_1$</td><td>6</td></tr><tr><td rowspan="3">F_4</td><td>$F_{4(-52)}$</td><td>F_4</td><td>52</td></tr><tr><td>$F_{4(-20)}$</td><td>B_4</td><td>36</td></tr><tr><td>$F_{4(+4)}$</td><td>$C_3 + A_1$</td><td>24</td></tr><tr><td rowspan="5">E_6</td><td>$E_{6(-78)}$</td><td>E_6</td><td>78</td></tr><tr><td>$E_{6(-26)}$</td><td>F_4</td><td>52</td></tr><tr><td>$E_{6(-14)}$</td><td>$D_5 + D_1$</td><td>46</td></tr><tr><td>$E_{6(+2)}$</td><td>$A_5 + A_1$</td><td>38</td></tr><tr><td>$E_{6(+6)}$</td><td>C_4</td><td>36</td></tr><tr><td rowspan="4">E_7</td><td>$E_{7(-133)}$</td><td>E_7</td><td>133</td></tr><tr><td>$E_{7(-25)}$</td><td>$E_6 + D_1$</td><td>79</td></tr><tr><td>$E_{7(-5)}$</td><td>$D_6 + A_1$</td><td>69</td></tr><tr><td>$E_{7(+7)}$</td><td>A_7</td><td>63</td></tr><tr><td rowspan="3">E_8</td><td>$E_{8(-248)}$</td><td>E_8</td><td>248</td></tr><tr><td>$E_{8(-24)}$</td><td>$E_7 + A_1$</td><td>136</td></tr><tr><td>$E_{8(+8)}$</td><td>D_8</td><td>120</td></tr></tbody></table>	Root space	Class _{rank(character)}	Maximal compact subgroup		Root space	Dimension	G_2	$G_{2(-14)}$	G_2	14	$G_{2(+2)}$	$A_1 + A_1$	6	F_4	$F_{4(-52)}$	F_4	52	$F_{4(-20)}$	B_4	36	$F_{4(+4)}$	$C_3 + A_1$	24	E_6	$E_{6(-78)}$	E_6	78	$E_{6(-26)}$	F_4	52	$E_{6(-14)}$	$D_5 + D_1$	46	$E_{6(+2)}$	$A_5 + A_1$	38	$E_{6(+6)}$	C_4	36	E_7	$E_{7(-133)}$	E_7	133	$E_{7(-25)}$	$E_6 + D_1$	79	$E_{7(-5)}$	$D_6 + A_1$	69	$E_{7(+7)}$	A_7	63	E_8	$E_{8(-248)}$	E_8	248	$E_{8(-24)}$	$E_7 + A_1$	136	$E_{8(+8)}$	D_8	120
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gilmore-lie-groups_F01944	196	1.000	$T(\frac{p}{})=-\frac{p}{}$	$T(\mathfrak{p}) = -\mathfrak{p}$	that obeys $T(\mathfrak{p}) = -\mathfrak{p}$ is analytically continued by $\mathfrak{p} \rightarrow \mathfrak{p}' = i\mathfrak{p}$; the subspace $\mathfrak{h}$																																																														
gilmore-lie-groups_F01945	196	1.000	$T(\frac{h}{})=+\frac{h}{}$	$T(\mathfrak{h}) = +\mathfrak{h}$	of $\mathfrak{g}$ that obeys $T(\mathfrak{h}) = +\mathfrak{h}$ is the maximal compact subalgebra of the noncompact																																																														
gilmore-lie-groups_F01946	196	1.000	$\frac{g}{}^{\prime}:\frac{g}{}=\frac{h}{}+\frac{p}{}\rightarrow\frac{g}{}^{\prime}=\frac{h}{}+i\frac{p}{}$	$\mathfrak{g}' : \mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}'$	real form $\mathfrak{g}'$: $\mathfrak{g} = \mathfrak{h} + \mathfrak{p} \rightarrow \mathfrak{g}' = \mathfrak{h} + i\mathfrak{p}$. For the simple classical (matrix) Lie alge-																																																														
gilmore-lie-groups_F01947	197	1.000	$1 \leq i$	$1 \leq i$	1. Four operators $a_i^\dagger a_j$ can be constructed from boson operators for two modes $\leq$,																																																														
gilmore-lie-groups_F01948	197	1.000	$j \leq 2$	$j \leq 2$	$j \leq 2$. These operators close under commutation.																																																														
gilmore-lie-groups_F01949	197	1.000	$a_{\{1\}}^{\dagger} a_{\{1\}}+a_{\{2\}}^{\dagger} a_{\{2\}}$	$a_1^\dagger a_1 + a_2^\dagger a_2$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$																																																														
gilmore-lie-groups_F01950	197	1.000	$a_{\{1\}}^{\dagger} a_{\{2\}}-$	$a_1^\dagger a_2 -$	Conclude that $a_1^\dagger a_1 + a_2^\dagger a_2$ spans the maximum commutative subalgebra, $a_1^\dagger a_2 -$																																																														
gilmore-lie-groups_F01951	197	1.000	$a_{\{1\}}^{\dagger} a_{\{1\}}-a_{\{2\}}^{\dagger} a_{\{2\}}$	$a_1^\dagger a_1 - a_2^\dagger a_2$	$a_2^\dagger a_1$ spans the maximal compact subalgebra, and the two generators $a_1^\dagger a_1 - a_2^\dagger a_2$																																																														
gilmore-lie-groups_F01952	197	1.000	$a_{\{1\}}^{\dagger} a_{\{2\}}+a_{\{2\}}^{\dagger} a_{\{1\}}$	$a_1^\dagger a_2 + a_2^\dagger a_1$	and $a_1^\dagger a_2 + a_2^\dagger a_1$ are noncompact.																																																														
gilmore-lie-groups_F01953	197	1.000	$i \sigma_{\{j\}}$	$i\sigma_j$	Show that the three matrices that result are exactly $i\sigma_j$, where σ_j are the Pauli spin																																																														
gilmore-lie-groups_F01954	197	1.000	$\sigma_{\{j\}}$	σ_j	Show that the three matrices that result are exactly $i\sigma_j$, where σ_j are the Pauli spin																																																														
gilmore-lie-groups_F01955	197	0.982	$k \rightarrow 1$	$k \rightarrow 1$	2. The classical matrix groups $SO(n)$ are not simply connected, so they are $k \rightarrow 1$																																																														
gilmore-lie-groups_F01956	197	0.844	$\overline{SO(n)}=Spin(n)$	$\overline{SO(n)} = Spin(n)$	images of their universal covering groups $\overline{SO(n)} = Spin(n)$, for some integer k .																																																														
gilmore-lie-groups_F01957	197	1.000	$\operatorname{Spin}(n)$	$Spin(n)$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,																																																														
gilmore-lie-groups_F01958	197	1.000	$n=3,4,5,6$	$n = 3, 4, 5, 6$	Show that the covering groups $Spin(n)$ are classical matrix groups for $n = 3, 4, 5, 6$,																																																														
gilmore-lie-groups_F01959	197	1.000	$n>6$	$n > 6$	Show that for $n > 6$ the groups $Spin(n)$ are not equal to any classical matrix Lie																																																														

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gilmore-lie-groups_ F01960	198	■ 0.968	$H_{\{i\}}:\left n_{\{1\}}, n_{\{2\}}, \ldots, n_{\{r\}}\right\rangle$	$H_i: n_1,n_2,\ldots,n_r\rangle$	eigenvalues under the operators $H_i: n_1,n_2,\ldots,n_r\rangle$,
gilmore-lie-groups_ F01961	198	■ 1.000	$H_{\{j\}} \rightarrow$	$H_j \rightarrow$	c. For the orthogonal groups $SO(n)$, impose suitable reality conditions (i.e., $H_j \rightarrow$
gilmore-lie-groups_ F01962	198	■ 1.000	$i H_{\{j\}}$	iH_j	iH_j , etc.), choose a Hilbert space containing the state $ l, 0, \ldots, 0\rangle$ and show that
gilmore-lie-groups_ F01963	198	■ 1.000	$ l, 0, \ldots, 0\rangle$	$ l, 0, \ldots, 0\rangle$	iH_j , etc.), choose a Hilbert space containing the state 7, 0, . . . , 0 and show that
gilmore-lie-groups_ F01964	198	■ 0.989	$-m^{\{2\}}$	$-m^2$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01965	198	■ 0.989	$e^{\{i\} m \phi}$	$e^{im\phi}$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01966	198	■ 0.989	$-l(l+1)$	$-l(l+1)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01967	198	■ 0.989	$Y_{\{m\}}^{\{l\}}(\theta, \phi)$	$Y_m^l(\theta, \phi)$	known spectrum $-m^2$ for $SO(2)$ (on $e^{im\phi}$) and $-l(l+1)$ for $SO(3)$ (on $Y_m^l(\theta, \phi)$).
gilmore-lie-groups_ F01968	198	■ 1.000	$J_{\{i\}}, i=1,2,3$	$J_i, i=1,2,3$	matrix elements of a <i>hermitian</i> representation of U, t = 1, 2, 3 . Exponentials of the
gilmore-lie-groups_ F01969	198	■ 0.926	$\operatorname{EXP}\left(i r^{\{k\}} J_{\{k\}}\right)$	$\operatorname{EXP}\left(i r^k J_k\right)$	form $\operatorname{EXP}\left(i r^k J_k\right)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_ F01970	198	■ 0.926	$r^{\{k\}}$	r^k	form $\operatorname{EXP}\left(i r^k J_k\right)$, with r^k real, provide <i>unitary</i> representations of the compact Lie
gilmore-lie-groups_ F01971	198	■ 1.000	$\operatorname{mathfrak{s}} \operatorname{mathfrak{u}}(1,1)$	$\mathfrak{su}(1,1)$	$\mathfrak{su}(1,1)$.
gilmore-lie-groups_ F01972	199	■ 0.826	$K_{\{3\}}$	K_3	$SU(2)$ with J_3 diagonal, or in any matrix representation of $SU(1,1)$ with K_3
gilmore-lie-groups_ F01973	199	■ 1.000	$e^{\{i\} m \phi} \delta_{m' m}$	$e^{im\phi} \delta_{m' m}$	diagonal, the matrix is diagonal with matrix elements $e^{im\phi} \delta_{m' m}$. Show that the
gilmore-lie-groups_ F01974	199	■ 1.000	$\phi \rightarrow \phi+4 \pi$	$\phi \rightarrow \phi+4 \pi$	single-valuedness condition under $\phi \rightarrow \phi+4 \pi$ requires that $m=\frac{1}{2}\left(n_1-n_2\right)$ is
gilmore-lie-groups_ F01975	199	■ 1.000	$m=\frac{1}{2}\left(n_{\{1\}}-n_{\{2\}}\right)$	$m=\frac{1}{2}\left(n_1-n_2\right)$	single-valuedness condition under $\phi \rightarrow \phi+4 \pi$ requires that $m=\frac{1}{2}\left(n_1-n_2\right)$ is
gilmore-lie-groups_ F01976	199	■ 1.000	$J_{\{ \pm\}}$	$J_{\pm}$	integer or half-integer. Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_ F01977	199	■ 1.000	$K_{\{ \pm\}}$	$K_{\pm}$	integer or half-integer. Show that the shift operators $J_{\pm}$ and $K_{\pm}$ require that all m
gilmore-lie-groups_ F01978	199	■ 1.000	$n_{\{1\}}, n_{\{2\}}$	n_1, n_2	c. Relax the assumption that all indices n_1, n_2 in the basis states $ n_1, n_2\rangle= n_1\rangle \otimes$
gilmore-lie-groups_ F01979	199	■ 1.000	$\left n_{\{1\}}, n_{\{2\}}\right\rangle=\left n_{\{1\}}\right\rangle \otimes$ $\otimes$	$ n_1, n_2\rangle= n_1\rangle \otimes$	c. Relax the assumption that all indices n_1, n_2 in the basis states $n_1, n_2\rangle= n_1\rangle \otimes$
gilmore-lie-groups_ F01980	199	■ 1.000	$\left n_{\{2\}}\right\rangle$	$ n_2\rangle$	$ n_2\rangle$ must be nonnegative integers. Construct the matrix elements of the operators
gilmore-lie-groups_ F01981	199	■ 0.999	$J_{\{3\}}, J_{\{ \pm\}}, K_{\{3\}}, K_{\{ \pm\}}$	$J_3, J_{\pm}, K_3, K_{\pm}$	$J_3, J_{\pm}, K_3, K_{\pm}$ under this relaxed assumption. Show that all commutation relations
gilmore-lie-groups_ F01982	199	■ 1.000	$J_{\{x\}}$	J_x	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_ F01983	199	■ 1.000	$J_{\{y\}}$	J_y	d. Show that the matrices for J_x and J_y are also hermitian provided that
gilmore-lie-groups_ F01984	199	■ 0.749	$(0,0)$	$(0,0)$	Show that the lattice sites in quadrants I and III of Fig. 11.2, with vertices at $(0,0)$
gilmore-lie-groups_ F01985	199	■ 0.975	$\left \left\{\right\}_{\{m\}}^{\{j\}}\right\rangle=\left n_{\{1\}}, n_{\{2\}}\right\rangle$ $\otimes$ $\frac{1}{2}\left(n_{\{1\}}+n_{\{2\}}\right), m=\frac{1}{2}\left(n_{\{1\}}-n_{\{2\}}\right)$	$ _m^j\rangle= n_1, n_2\rangle, j=\frac{1}{2}\left(n_1+n_2\right), m=\frac{1}{2}\left(n_1-n_2\right)$	e. With the identification $_m^j\rangle= n_1, n_2\rangle, j=\frac{1}{2}\left(n_1+n_2\right), m=\frac{1}{2}\left(n_1-n_2\right)$, show

Identifier	Page	Conf.	LaTeX source	Rendered	Image															
gilmore-lie-groups_F01986	199	■1.000	n_{1}	n_1	successive application of these operators to remain in QI or QIII, n_1 and n_2 must															
gilmore-lie-groups_F01987	199	■1.000	n_{2}	n_2	successive application of these operators to remain in QI or QIII, n_1 and n_2 must															
gilmore-lie-groups_F01988	199	■1.000	n_{2}=0	$n_2 = 0$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
Quadrant	Operator	Edge																		
I	J_+	$n_2 = 0$																		
I	J_-	$n_1 = 0$																		
III	J_+	$n_1 = -1$																		
III	J_-	$n_2 = -1$																		
gilmore-lie-groups_F01989	199	■1.000	n_{1}=0	$n_1 = 0$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
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I	J_+	$n_2 = 0$																		
I	J_-	$n_1 = 0$																		
III	J_+	$n_1 = -1$																		
III	J_-	$n_2 = -1$																		
gilmore-lie-groups_F01990	199	■1.000	n_{1}=-1	$n_1 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
Quadrant	Operator	Edge																		
I	J_+	$n_2 = 0$																		
I	J_-	$n_1 = 0$																		
III	J_+	$n_1 = -1$																		
III	J_-	$n_2 = -1$																		
gilmore-lie-groups_F01991	199	■1.000	n_{2}=-1	$n_2 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
Quadrant	Operator	Edge																		
I	J_+	$n_2 = 0$																		
I	J_-	$n_1 = 0$																		
III	J_+	$n_1 = -1$																		
III	J_-	$n_2 = -1$																		
gilmore-lie-groups_F01992	200	■0.982	n_{1}+n_{2}=	$n_1 + n_2 =$	mapped to other points along diagonals of the form $n_1 + n_2 = \text{constant}$. The															
gilmore-lie-groups_F01993	200	■0.872	-1<n_{1}<0	$-1 < n_1 < 0$	and $SU(1, 1)$ are separated by a “no man’s land” defined by $-1 < n_1 < 0$															
gilmore-lie-groups_F01994	200	■1.000	-1<n_{2}<0	$-1 < n_2 < 0$	and $-1 < n_2 < 0$ (wavy lines). All points labeled x belong to the principal															
gilmore-lie-groups_F01995	200	■0.997	n_{1}+n_{2}=-\frac{1}{2}+i\beta	$n_1 + n_2 = -\frac{1}{2} + i\beta$	series of representations of $SU(1, 1)$ with $n_1 + n_2 = -\frac{1}{2} + i\beta$.															
gilmore-lie-groups_F01996	200	■0.998	\left n_{1}+k, n_{2}-k\right\rangle	$ n_1 + k, n_2 - k\rangle$	i. In order to ensure that a set of states $ n_1 + k, n_2 - k\rangle$ (k integer) mapped into each															
gilmore-lie-groups_F01997	200	■1.000	\left n_{1}, n_{2}=-1\right\rangle	$ n_1, n_2 = -1\rangle$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $n_1 = 0, 1, 2, \dots$. The basis states for															
gilmore-lie-groups_F01998	200	■1.000	n_{1}=0,1,2, \ldots	$n_1 = 0, 1, 2, \dots$	(lowest m) state must be $ n_1, n_2 = -1\rangle$ for $n_1 = 0, 1, 2, \dots$. The basis states for															
gilmore-lie-groups_F01999	200	■0.995	\left \left\{ \right\}_m^j\right\rangle=\left n_{1}, n_{2}\right\rangle	$ _m^j\rangle = n_1, n_2\rangle$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$															
gilmore-lie-groups_F02000	200	■0.995	j+\frac{1}{2}=	$j + \frac{1}{2} =$	this bounded discrete series of representations are $ _m^j\rangle = n_1, n_2\rangle$ with $j + \frac{1}{2} =$															
gilmore-lie-groups_F02001	200	■0.993	0, \frac{1}{2}, 1, \frac{3}{2}, \ldots	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the															
gilmore-lie-groups_F02002	200	■0.993	2j+1=0,1,2,3, \ldots	$2j + 1 = 0, 1, 2, 3, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the															
gilmore-lie-groups_F02003	200	■0.993	m=j+1,j+2, \ldots	$m = j + 1, j + 2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = j + 1, j + 2, \dots$. This is the															

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gilmore-lie-groups_ F02004	200	■ 1.000	$\mathcal{D}_{+}^{j}$	$\mathcal{D}_{+}^j$	discrete series of representations that is bounded below: $\mathcal{D}_{+}^j$.
gilmore-lie-groups_ F02005	200	■ 0.831	$\mathbf{j}$	$\mathbf{j}$	$\mathbf{j}$. In order to ensure that a set of states $ n_1 + k, n_2 - k\rangle$ (k integer) mapped into
gilmore-lie-groups_ F02006	200	■ 1.000	$\left n_{\{1\}}=-1, n_{\{2\}}\right\rangle$	$ n_1 = -1, n_2\rangle$	(highest m) state must be $n_1 = -1, n_2$ for $n_2 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_ F02007	200	■ 1.000	$n_{\{2\}}=0,1,2, \ldots$	$n_2 = 0, 1, 2, \dots$	(highest m) state must be $ n_1 = -1, n_2\rangle$ for $n_2 = 0, 1, 2, \dots$. The basis states for
gilmore-lie-groups_ F02008	200	■ 1.000	$m=-j-1,-j-2, \ldots$	$m = -j - 1, -j - 2, \dots$	$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ or $2j + 1 = 0, 1, 2, 3, \dots$ and $m = -j - 1, -j - 2, \dots$. This is
gilmore-lie-groups_ F02009	200	■ 0.999	$\mathcal{D}_{-}^{j}$	$\mathcal{D}_{-}^j$	the discrete series of representations that is bounded above: $\mathcal{D}_{-}^j$.
gilmore-lie-groups_ F02010	201	■ 0.998	$\left(\mathcal{D}_{+}^{j}\right)$	$\left(\mathcal{D}_{+}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02011	201	■ 0.998	$\left(\mathcal{D}_{-}^{j}\right)$	$\left(\mathcal{D}_{-}^j\right)$	$(\mathcal{D}_{+}^j)$ or from QII $(\mathcal{D}_{-}^j)$.
gilmore-lie-groups_ F02012	201	■ 0.999	$\mid n_{\{1\}}+k, n_{\{2\}}-$	$ n_1 + k, n_2 -$	$\mathbf{I}$. Now relax the condition that n_1 and n_2 are integers. The set of states $ n_1 + k, n_2 -$
gilmore-lie-groups_ F02013	201	■ 1.000	$k\rangle(k=\ldots,-2,-1,0,+1,+2, \ldots)$	$k\rangle(k = \dots, -2, -1, 0, +1, +2, \dots)$	$k\rangle(k = \dots, -2, -1, 0, +1, +2, \dots)$ connected by $K_{\pm}$ carries a hermitian repre-
gilmore-lie-groups_ F02014	201	■ 1.000	$\mid p, q\rangle$	$ p, q\rangle$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $-1 \leq p, q \leq 0$ then the
gilmore-lie-groups_ F02015	201	■ 1.000	$-1 \leq p, q \leq 0$	$-1 \leq p, q \leq 0$	vertices of the four quadrants. If this state is $ p, q\rangle$, with $-1 \leq p, q \leq 0$ then the
gilmore-lie-groups_ F02016	201	■ 1.000	$\frac{1}{2}(p-q)=$	$\frac{1}{2}(p - q) =$	single-valuedness condition requires $\frac{1}{2}(p - q) =$ integer or half-integer. In the lat-
gilmore-lie-groups_ F02017	201	■ 0.891	$p-q=0$	$p - q = 0$	values of the $U(1)$ index m . Therefore $p - q = 0$ and $-1 \leq p = q \leq 0$. The states
gilmore-lie-groups_ F02018	201	■ 0.891	$-1 \leq p=q \leq 0$	$-1 \leq p = q \leq 0$	values of the $U(1)$ index m . Therefore $p - q = 0$ and $-1 \leq p = q \leq 0$. The states
gilmore-lie-groups_ F02019	201	■ 0.679	$\left _{m}^{j}\right\rangle$	$ _m^j\rangle$	$ _m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02020	201	■ 0.679	$-\frac{1}{2} \leq j+\frac{1}{2} \leq+\frac{1}{2}$	$-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$	$ _m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02021	201	■ 0.679	$\mathcal{D}^p$	$\mathcal{D}^p$	$ _m^j\rangle$ with m integer, j real and $-\frac{1}{2} \leq j + \frac{1}{2} \leq +\frac{1}{2}$ carry representations $\mathcal{D}^p$ of the
gilmore-lie-groups_ F02022	201	■ 1.000	$j+\frac{1}{2}=i \beta$	$j + \frac{1}{2} = i\beta$	$\mathbf{m}$. By setting $j + \frac{1}{2} = i\beta$ (β real) the matrix elements in Eq. (11.30) become
gilmore-lie-groups_ F02023	201	■ 1.000	$j^{\prime}<-\frac{1}{2}$	$j' < -\frac{1}{2}$	Show that states with $j' < -\frac{1}{2}$ obtained by reflection through the diagonal contain-
gilmore-lie-groups_ F02024	201	■ 0.693	$\left(n_{\{1\}}, n_{\{2\}}\right)=\left(-\frac{1}{2},-\frac{1}{2}\right)$	$(n_1, n_2) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	ing the central point in the shaded square with coordinates $(n_1, n_2) = (-\frac{1}{2}, -\frac{1}{2})$
gilmore-lie-groups_ F02025	201	■ 1.000	$j>-\frac{1}{2}$	$j > -\frac{1}{2}$	support representations equivalent to those with index $j > -\frac{1}{2}$. The relation among
gilmore-lie-groups_ F02026	201	■ 0.311	$j+\frac{1}{2}=-\left(j^{\prime}+\frac{1}{2}\right)$	$j + \frac{1}{2} = -\left(j' + \frac{1}{2}\right)$	indices is $j + \frac{1}{2} = -(j' + \frac{1}{2})$ and the relation among states is $ _{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$.
gilmore-lie-groups_ F02027	201	■ 0.311	$\left k_{m^{\prime}}^{\prime}\right\rangle \simeq\left _{m=m^{\prime}}^j\right\rangle$	$ k_{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$	indices is $j + \frac{1}{2} = -(j' + \frac{1}{2})$ and the relation among states is $ _{m'}^{j'}\rangle \simeq _{m=m'}^j\rangle$.
gilmore-lie-groups_ F02028	201	■ 0.996	$X=h^i H_i+e^{\alpha} E_{\alpha}$	$X = h^i H_i + e^{\alpha} E_{\alpha}$	$X = h^i H_i + e^{\alpha} E_{\alpha}$, with h^i, e^{α} real. Show that the real subalgebra spanned by the
gilmore-lie-groups_ F02029	201	■ 0.996	h^i, e^{α}	h^i, e^{α}	$X = h^i H_i + e^{\alpha} E_{\alpha}$, with h^i, e^{α} real. Show that the real subalgebra spanned by the

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gilmore-lie-groups_ F02030	201	■ 1.000	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha - E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_ F02031	201	■ 1.000	$\left(E_{\alpha}-E_{-\alpha}\right) / \sqrt{2}$	$(E_\alpha - E_{-\alpha})/\sqrt{2}$	$\frac{1}{2}(n-l)$ linear combinations of the form $(E_\alpha - E_{-\alpha})/\sqrt{2}$ is the maximal compact
gilmore-lie-groups_ F02032	202	■ 1.000	$\mathfrak{s p}(p, q)$	$\mathfrak{sp}(p, q)$	6. The noncompact real form $\mathfrak{sp}(p, q)$ of the symplectic algebra was constructed from
gilmore-lie-groups_ F02033	202	■ 1.000	$\mathfrak{s p}(p+q)$	$\mathfrak{sp}(p+q)$	the compact real form $\mathfrak{sp}(p+q)$ by “Minkowski’s trick,” or analytic continuation.
gilmore-lie-groups_ F02034	203	■ 0.999	$\mathfrak{p}, i \mathfrak{p}$	$\mathfrak{p}, i\mathfrak{p}$	countered subspaces $\mathfrak{p}, i\mathfrak{p}$ on which the Cartan–Killing inner product was
gilmore-lie-groups_ F02035	203	■ 1.000	$i \mathfrak{p}$	$i\mathfrak{p}$	negative-definite (on $\mathfrak{p}$) or positive-definite (on $i\mathfrak{p}$). In both cases these
gilmore-lie-groups_ F02036	203	■ 1.000	g_{ij}	g_{ij}	metric g_{ij} is definite, either negative or positive. Manifolds with a definite
gilmore-lie-groups_ F02037	203	■ 1.000	$\operatorname{EXP}(\mathfrak{p})$	$\operatorname{EXP}(\mathfrak{p})$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(i\mathfrak{p})$ is the image of the origin
gilmore-lie-groups_ F02038	203	■ 1.000	$\operatorname{EXP}(i \mathfrak{p})$	$\operatorname{EXP}(i\mathfrak{p})$	each point in the space $\operatorname{EXP}(\mathfrak{p})$ or $\operatorname{EXP}(i\mathfrak{p})$ is the image of the origin
gilmore-lie-groups_ F02039	203	■ 0.997	$S^2 \sim S U(2) / U(1)$	$S^2 \sim SU(2)/U(1)$	These were $S^2 \sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2 = SL(2; \mathbb{R})/$
gilmore-lie-groups_ F02040	203	■ 0.997	$H_{2+}^2=S L(2 ; \mathbb{R}) /$	$H_{2+}^2 = SL(2; \mathbb{R})/$	These were $S^2 \sim SU(2)/U(1)$, which is compact, and its dual $H_{2+}^2 = SL(2; \mathbb{R})/$
gilmore-lie-groups_ F02041	203	■ 1.000	$S O(2)=S U(1,1) / U(1)$	$SO(2) = SU(1,1)/U(1)$	$SO(2) = SU(1, 1)/U(1)$, which is the upper sheet of the two-sheeted hyperboloid.
gilmore-lie-groups_ F02042	203	■ 1.000	$H_1^2=S L(2 ; \mathbb{R}) / S O(1,1)$	$H_1^2 = SL(2; \mathbb{R})/SO(1, 1)$	“Between” these two spaces occurs $H_1^2 = SL(2; \mathbb{R})/SO(1, 1)$, which is the single-
gilmore-lie-groups_ F02043	203	■ 0.943	$\mathfrak{s u}(2)-\mathfrak{u}(1)$	$\mathfrak{su}(2) - \mathfrak{u}(1)$	The Cartan–Killing inner product in the linear vector subspace $\mathfrak{su}(2) - \mathfrak{u}(1)$ is
gilmore-lie-groups_ F02044	203	■ 0.999	$S U(2) / U(1) \sim S^2$	$SU(2)/U(1) \sim S^2$	Killing metric on the space $SU(2)/U(1) \sim S^2$, the sphere. On S^2 the Cartan–Killing
gilmore-lie-groups_ F02045	203	■ 0.991	$\mathfrak{s u}(1,1)-\mathfrak{u}(1) \simeq \mathfrak{s l}(2 ; \mathbb{R})-\mathfrak{s o}(2)$	$\mathfrak{su}(1, 1) - \mathfrak{u}(1) \simeq \mathfrak{sl}(2; \mathbb{R}) - \mathfrak{so}(2)$	The Cartan–Killing inner product on $\mathfrak{su}(1, 1) - \mathfrak{u}(1) \simeq \mathfrak{sl}(2; \mathbb{R}) - \mathfrak{so}(2)$ is
gilmore-lie-groups_ F02046	203	■ 1.000	$H_{2+}^2=S U(1,1) / S O(2)$	$H_{2+}^2 = SU(1,1)/SO(2)$	positive-definite. It maps to a positive-definite metric on $H_{2+}^2 = SU(1, 1)/SO(2)$.
gilmore-lie-groups_ F02047	204	■ 1.000	$S^2=S O(3) / S O(2)=S U(2) / U(1), \quad H_{2+}^2=S O(2,1) / S O(2)=$	$S^2 = SO(3)/SO(2) = SU(2)/U(1), \quad H_{2+}^2 = SO(2,1)/SO(2) =$	Figure 12.1. $S^2 = SO(3)/SO(2) = SU(2)/U(1), \quad H_{2+}^2 = SO(2,1)/SO(2) =$
gilmore-lie-groups_ F02048	204	■ 1.000	$S U(1,1) / U(1), H_1^2=S O(2,1) / S O(1,1)=S L(2 ; \mathbb{R}) / S O(1,1)$	$SU(1,1)/U(1), H_1^2 = SO(2,1)/SO(1,1) = SL(2; \mathbb{R})/SO(1, 1)$	$SU(1,1)/U(1), H_1^2 = SO(2,1)/SO(1,1) = SL(2; \mathbb{R})/SO(1, 1)$. The first two
gilmore-lie-groups_ F02049	204	■ 0.998	$\mathfrak{s u}(1,1)-\mathfrak{s o}(1,1)$	$\mathfrak{su}(1, 1) - \mathfrak{so}(1, 1)$	obtained by exponentiating $\mathfrak{su}(1, 1) - \mathfrak{so}(1, 1)$. The Cartan–Killing inner product
gilmore-lie-groups_ F02050	204	■ 0.966	$\operatorname{EXP}[\mathfrak{s u}(1,1)-\mathfrak{s o}(1,1)]=S U(1,1) / S O(1,1)$	$\operatorname{EXP}[\mathfrak{su}(1, 1) - \mathfrak{so}(1, 1)] = SU(1,1)/SO(1, 1)$	the topological space $\operatorname{EXP}[\mathfrak{su}(1, 1) - \mathfrak{so}(1, 1)] = SU(1, 1)/SO(1, 1)$ is indefinite.
gilmore-lie-groups_ F02051	204	■ 0.952	$\mathfrak{s o}(n), \mathfrak{s u}(n), \mathfrak{s p}(n)$	$\mathfrak{so}(n), \mathfrak{su}(n), \mathfrak{sp}(n)$	algebra $\mathfrak{g}$ (i.e., $\mathfrak{so}(n), \mathfrak{su}(n), \mathfrak{sp}(n)$) the Cartan decompositions have the form (11.10)
gilmore-lie-groups_ F02052	205	■ 1.000	$\mathfrak{p}(i \mathfrak{p})$	$\mathfrak{p}(i\mathfrak{p})$	On the linear vector space $\mathfrak{p}$ ($i\mathfrak{p}$) the Cartan–Killing inner product is negative
gilmore-lie-groups_ F02053	205	■ 1.000	$P=G / H=\operatorname{EXP}(\mathfrak{p})$	$P = G/H = \operatorname{EXP}(\mathfrak{p})$	The space $P = G/H = \operatorname{EXP}(\mathfrak{p})$ (e.g., S^2) is compact. The exponential of a
gilmore-lie-groups_ F02054	205	■ 1.000	P	P	identity. The space P is not topologically equivalent to any Euclidean space, in
gilmore-lie-groups_ F02055	205	■ 0.999	$P^{\prime}=G^{\prime} / H=\operatorname{EXP}(i \mathfrak{p})$	$P' = G'/H = \operatorname{EXP}(i\mathfrak{p})$	The space $P' = G'/H = \operatorname{EXP}(i\mathfrak{p})$ (i.e., H_{2+}^2) is noncompact. The exponential of
gilmore-lie-groups_ F02056	205	■ 1.000	$P^{\prime}=\operatorname{EXP}(i \mathfrak{p})$	$P' = \operatorname{EXP}(i\mathfrak{p})$	simply goes away from this point without ever returning. The space $P' = \operatorname{EXP}(i\mathfrak{p})$ is

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gilmore-lie-groups_F02657	205	1.000	$n=\operatorname{dim} i \, \mathfrak{p}$	$n = \dim i\mathfrak{p}$	topologically equivalent to a Euclidean space R^n , where $n = \dim i\mathfrak{p}$. Geometrically																																				
gilmore-lie-groups_F02658	205	1.000	$P=\operatorname{EXP}(\mathfrak{p})$	$P = \operatorname{EXP}(\mathfrak{p})$	The Riemannian spaces $P = \operatorname{EXP}(\mathfrak{p})$ and $P' = \operatorname{EXP}(i\mathfrak{p})$ are symmetric but not																																				
gilmore-lie-groups_F02659	205	1.000	$\mathfrak{g}=\mathfrak{k}+\mathfrak{p}$	$\mathfrak{g} = \mathfrak{k} + \mathfrak{p}$	If $\mathfrak{g}$ is simple with a Cartan decomposition of the form $\mathfrak{g} = \mathfrak{k} + \mathfrak{p}$, with standard																																				
gilmore-lie-groups_F02660	205	1.000	$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}, [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$	$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P =$																																				
gilmore-lie-groups_F02661	205	1.000	$[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$	$[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P =$																																				
gilmore-lie-groups_F02662	205	1.000	$P=$	$P =$	commutation relations $[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}$, and $[\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}$, the quotient coset $P =$																																				
gilmore-lie-groups_F02663	205	1.000	G / K	G/K	G/K is a globally symmetric space as every point “looks like” every other point.																																				
gilmore-lie-groups_F02664	205	1.000	$P=G / H$	$P = G/H$	of points (coset representatives $P = G/H$ or $P' = G'/H$) in the Lie group.																																				
gilmore-lie-groups_F02665	205	1.000	$P^{\prime}=G^{\prime} / H$	$P' = G'/H$	of points (coset representatives $P = G/H$ or $P' = G'/H$) in the Lie group.																																				
gilmore-lie-groups_F02666	205	0.993	$S O(p, q) / S O(p) \times S O(q)$	$SO(p, q)/SO(p) \times SO(q)$	resentation. The secular equations for the spaces $SO(p, q)/SO(p) \times SO(q)$,																																				
gilmore-lie-groups_F02667	206	1.000	$B^{\dagger} B$	$B^{\dagger} B$	It is easy to check that the function ϕ_j depends on the $q \times q$ matrix $B^{\dagger} B$ or the																																				
gilmore-lie-groups_F02668	206	1.000	$B B^{\dagger}$	$BB^{\dagger}$	$p \times p$ matrix $BB^{\dagger}$, whichever is smaller. The rank of these spaces is $\min(p, q)$.																																				
gilmore-lie-groups_F02669	206	1.000	$\min (p, q)$	$\min(p, q)$	$p \times p$ matrix $BB^{\dagger}$, whichever is smaller. The rank of these spaces is $\min(p, q)$.																																				
gilmore-lie-groups_F02670	206	1.000	$P\left(P^{\prime}\right)$	$P\left(P'\right)$	(iv) the number of invariant (Laplace–Beltrami) operators defined over the space $P\left(P'\right)$;																																				
gilmore-lie-groups_F02671	206	0.999	$G^{\prime} / H$	G'/H	form G'/H , where G' is simple and noncompact and H is the maximal compact																																				
gilmore-lie-groups_F02672	206	0.999	$G^{\prime}$	G'	form G'/H , where G' is simple and noncompact and H is the maximal compact																																				
gilmore-lie-groups_F02673	206	0.877	A_{p+q-1}	A_{p+q-1}	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02674	206	0.877	$S U(p, q) / S[U(p) \otimes U(q)]$	$SU(p, q)/S[U(p) \otimes U(q)]$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02675	206	0.877	$2\ p\ q$	$2pq$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S\ [U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\ Sp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S\ [U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\ Sp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02676	206	0.877	$S\ L(n\ ;\ \mathbb{R})\ /\ S\ 0(n)$	$SL(n; \mathbb{R})/SO(n)$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S\ [U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\ Sp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S\ [U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\ Sp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02677	206	0.877	$\frac{1}{2}(n+2)(n-1)$	$\frac{1}{2}(n+2)(n-1)$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S\ [U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\ Sp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S\ [U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\ Sp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02678	206	0.877	$n-1$	$n-1$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S\ [U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\ Sp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S\ [U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\ Sp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02679	206	0.877	$S\ U^{*}(2\ n)\ /\ U\ \operatorname{Sp}(2\ n)$	$SU^*(2n)/U\ Sp(2n)$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S\ [U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/U\ Sp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S\ [U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/U\ Sp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$U\ Sp(2p, 2q)/U\ Sp(2p) \otimes U\ Sp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																				
gilmore-lie-groups_F02686	206	0.877	$(2n+1)(n-1)$	$(2n+1)(n-1)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02681	206	0.877	$B_{\{p+q\}}$	B_{p+q}	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02682	206	0.877	$SO(p,q)/SO(p)\otimes SO(q)$	$SO(p,q)/SO(p)\otimes SO(q)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02683	206	0.877	$D_{\{p+q\}}$	D_{p+q}	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02684	206	0.877	$SO^{\{*\}}(2n)/U(n)$	$SO^*(2n)/U(n)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																				
gilmore-lie-groups_F02685	206	0.877	$n(n-1)$	$n(n-1)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02686	206	0.877	$C_{\{p+q\}}$	C_{p+q}	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02687	206	0.877	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_F02689	206	0.877	$Sp(2n;\mathbb{R})/U(n)$	$Sp(2n;\mathbb{R})/U(n)$	<table><thead><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr></thead><tbody><tr><td>A_{p+q-1}</td><td>$SU(p,q)/S[U(p)\otimes U(q)]$</td><td>$2pq$</td><td>$\min(p,q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n;\mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p,q)/SO(p)\otimes SO(q)$</td><td>$pq$</td><td>$\min(p,q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p,2q)/USp(2p)\otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p,q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n;\mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></tbody></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p,q)/S[U(p)\otimes U(q)]$	$2pq$	$\min(p,q)$	A_{n-1}	$SL(n;\mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_{p+q}	$SO(p,q)/SO(p)\otimes SO(q)$	pq	$\min(p,q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p,2q)/USp(2p)\otimes USp(2q)$	$4pq$	$\min(p,q)$	C_n	$Sp(2n;\mathbb{R})/U(n)$	$n(n+1)$	n
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gilmore-lie-groups_ F02092	207	0.979	$\operatorname{Dim} H$	$\operatorname{Dim} H$	<table><tr><th>Root space</th><th>G'/H</th><th>$\operatorname{Dim} G^{\prime}$</th><th>$\operatorname{Dim} H$</th><th>$\operatorname{Dim} P$</th><th>Rank P</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1\oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3\oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5\oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5\oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6\oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6\oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7\oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table>	Root space	G'/H	$\operatorname{Dim} G^{\prime}$	$\operatorname{Dim} H$	$\operatorname{Dim} P$	Rank P	G_2	$G_{2(+2)}/(A_1\oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3\oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5\oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5\oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6\oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6\oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7\oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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gilmore-lie-groups_ F02099	207	■ 0.979	$E_{6(+2)} / \left(A_5 \oplus A_1 \right)$	$E_{6(+2)} / (A_5 \oplus A_1)$	<table> <tr> <th>Root space</th><th>G'/H</th><th>Dim G'</th><th>Dim H</th><th>Dim P</th><th>Rank P</th></tr> <tr> <td>G_2</td><td>$G_{2(+2)} / (A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr> <tr> <td>F_4</td><td>$F_{4(-20)} / B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr> <tr> <td></td><td>$F_{4(+4)} / (C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr> <tr> <td>E_6</td><td>$E_{6(-26)} / F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr> <tr> <td></td><td>$E_{6(-14)} / (D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr> <tr> <td></td><td>$E_{6(+2)} / (A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr> <tr> <td></td><td>$E_{6(+6)} / C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr> <tr> <td>E_7</td><td>$E_{7(-25)} / (E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr> <tr> <td></td><td>$E_{7(-5)} / (D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr> <tr> <td></td><td>$E_{7(+7)} / A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr> <tr> <td>E_8</td><td>$E_{8(-24)} / (E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr> <tr> <td></td><td>$E_{8(+8)} / D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr> </table>	Root space	G'/H	Dim G'	Dim H	Dim P	Rank P	G_2	$G_{2(+2)} / (A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)} / B_4$	52	36	16	1		$F_{4(+4)} / (C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)} / F_4$	78	52	26	2		$E_{6(-14)} / (D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)} / (A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)} / C_4$	78	36	42	6	E_7	$E_{7(-25)} / (E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)} / (D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)} / A_7$	133	63	70	7	E_8	$E_{8(-24)} / (E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)} / D_8$	248	120	128	8
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gilmore-lie-groups-F02107	207	■ 1.000	$S O(p+q) / S O(p) \otimes S O(q)$	$SO(p+q)/SO(p) \otimes SO(q)$	example, $SO(p,q)/SO(p) \otimes SO(q)$ and $SO(p+q)/SO(p) \otimes SO(q)$ are dual.
gilmore-lie-groups-F02108	207	■ 1.000	$P, P^{\prime}$	P, P'	The metric tensor on the spaces P, P' is computed by defining a metric at the
gilmore-lie-groups-F02109	207	■ 0.987	$d x(\mathrm{Id})$	$dx(\mathrm{Id})$	If $dx(\mathrm{Id})$ are infinitesimal displacements at the identity that are translated to
gilmore-lie-groups-F02110	207	■ 1.000	$d x(p)$	$dx(p)$	infinitesimal displacements $dx(p)$ at point p , then these two sets of infinitesimals
gilmore-lie-groups-F02111	208	■ 1.000	$M^{\{i\}}_{\{\mu\}}(p)$	$M^i_{\mu}(p)$	The matrix $M^i_{\mu}(p)$ is not easy to compute in general. For the rank-one spaces
gilmore-lie-groups-F02112	208	■ 0.998	$S O(n,1) / S O(n), S U(n,1) / U(n), S p(n,1) / S p(n) \otimes S p(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$	$SO(n,1)/SO(n), SU(n,1)/U(n), Sp(n,1)/Sp(n) \times Sp(1)$ defined by
gilmore-lie-groups-F02113	208	■ 1.000	$M^{\{i\}}_{\{\mu\}}(X)$	$M^i_{\mu}(X)$	the matrix $M^i_{\mu}(X)$ is determined from
gilmore-lie-groups-F02114	208	■ 1.000	W^{-1}	W^{-1}	The matrix $M^i_{\mu}(X)$ is given by W^{-1} . Since the Cartan–Killing inner product is I_n
gilmore-lie-groups-F02115	208	■ 1.000	$H_2^2=S O(2,1) / S O(2)$	$H_2^2 = SO(2,1)/SO(2)$	the noncompact hyperboloid $H_2^2 = SO(2,1)/SO(2)$ and the compact sphere
gilmore-lie-groups-F02116	209	■ 0.999	$S^2=S O(3) / S O(2)$	$S^2 = SO(3)/SO(2)$	$S^2 = SO(3)/SO(2)$. The metric tensors in the two cases are the 2×2 matrices
gilmore-lie-groups-F02117	209	■ 1.000	$S O(2+1) / S O(2)$	$SO(2+1)/SO(2)$	ric space $SO(2+1)/SO(2)$ is parameterized by the interior of the unit circle
gilmore-lie-groups-F02118	209	■ 0.999	$Y^2=1-\left(x^2+y^2\right) \geq 0$	$Y^2 = 1 - \left(x^2 + y^2\right) \geq 0$	$Y^2 = 1 - \left(x^2 + y^2\right) > 0$.
gilmore-lie-groups-F02119	209	■ 1.000	x, y	x, y	for the compact case, the sphere. Instead of using the pair x, y as coordinates,
gilmore-lie-groups-F02120	209	■ 1.000	$x^{\{i\}}, i=1,2, \ldots, N$	$x^i, i = 1, 2, \dots, N$	we use indexed coordinates $x^i, i = 1, 2, \dots, N$, and set $N = 2$ at the end of this
gilmore-lie-groups-F02121	209	■ 1.000	$N=2$	$N = 2$	we use indexed coordinates $x^i, i = 1, 2, \dots, N$, and set $N = 2$ at the end of this
gilmore-lie-groups-F02122	210	■ 1.000	$g^{\{i j\}} \simeq \delta^{\{i j\}}-x^{\{i\}} x^{\{j\}}$	$g^{ij} \simeq \delta^{ij} - x^i x^j$	The inverse (contravariant metric) is $g^{ij} \simeq \delta^{ij} - x^i x^j$, but we will not need this
gilmore-lie-groups-F02123	210	■ 0.900	$g^{\{i j\}} \rightarrow \delta^{\{i j\}}$	$g^{ij} \rightarrow \delta^{ij}$	result. In the neighborhood of the identity ($g^{ij} \rightarrow \delta^{ij}$)
gilmore-lie-groups-F02124	210	■ 1.000	$R_{\{\mu \sigma, \alpha \beta\}}=$	$R_{\mu\sigma,\alpha\beta} =$	delta function at the origin, and the resulting fully covariant metric tensor $R_{\mu\sigma,\alpha\beta} =$
gilmore-lie-groups-F02125	210	■ 1.000	$\delta_{\alpha \mu} \delta_{\beta \sigma}-\delta_{\alpha \sigma} \delta_{\beta \mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$	$\delta_{\alpha\mu}\delta_{\beta\sigma} - \delta_{\alpha\sigma}\delta_{\beta\mu}$ exhibits the full spectrum of expected symmetries.
gilmore-lie-groups-F02126	210	■ 0.874	$R=2$	$R = 2$	For $N = 2$ (sphere S^2), $R = 2$.
gilmore-lie-groups-F02127	210	■ 1.000	H_2^2	H_2^2	The computation can be carried out just as easily for the noncompact space H_2^2 .
gilmore-lie-groups-F02128	210	■ 0.848	$\Gamma_{\mu \nu}^{\sigma} \rightarrow-\delta_{\mu \nu} x^{\sigma}$	$\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$	for example $\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups-F02129	210	■ 0.848	$R=-2$	$R = -2$	for example $\Gamma_{\mu\nu}^{\sigma} \rightarrow -\delta_{\mu\nu} x^{\sigma}$. The end result for H_2^2 is that $R = -2$.
gilmore-lie-groups-F02130	211	■ 0.998	$d s^2=g_{\{\mu \nu\}} d x^{\{\mu\}} d x^{\{\nu\}}>0$	$ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$	positive-definite ($ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups-F02131	211	■ 0.998	$\Rightarrow d x=0$	$\Rightarrow dx = 0$	positive-definite ($ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu} > 0$, equality $\Rightarrow dx = 0$) nor negative-
gilmore-lie-groups-F02132	211	■ 0.958	$\left(d s^2<0\right)$	$(ds^2 < 0)$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-

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gilmore-lie-groups_ F02133	211	■ 0.958	$\ g\ \neq 0$	$\ g\ \neq 0$	definite ($ds^2 < 0$), but which is nonsingular ($\ g\ \neq 0$) are called pseudo-																
gilmore-lie-groups_ F02134	211	■ 1.000	$\mathfrak{g}^{\{\prime \prime\}}$	$\mathfrak{g}''$	To make these statements more explicit, assume a Lie algebra $\mathfrak{g}''$ (noncompact)																
gilmore-lie-groups_ F02135	211	■ 1.000	$\mathfrak{h}^{\{\prime \prime\}}$	$\mathfrak{h}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner																
gilmore-lie-groups_ F02136	211	■ 1.000	$\mathfrak{p}^{\{\prime \prime\}}$	$\mathfrak{p}''$	Then $\mathfrak{h}''$ and $\mathfrak{p}''$ are orthogonal subspaces in $\mathfrak{g}''$ under the Cartan–Killing inner																
gilmore-lie-groups_ F02137	211	■ 1.000	$H^{\{\prime \prime\}}=$	$H'' =$	indefinite. The space is curved and typically multiply connected. The space $H'' =$																
gilmore-lie-groups_ F02138	211	■ 1.000	$\operatorname{EXP}\left(\mathfrak{h}^{\{\prime \prime\}}\right)$	$\operatorname{EXP}(\mathfrak{h}'')$	$\operatorname{EXP}(\mathfrak{h}'')$ is also an interesting pseudo-Riemannian symmetric space.																
gilmore-lie-groups_ F02139	211	■ 1.000	T_1, T_2	T_1, T_2	spaces. Begin with a compact simple Lie algebra $\mathfrak{g}$ and suppose T_1, T_2 are two																
gilmore-lie-groups_ F02140	211	■ 0.819	$T_1^2=I$	$T_1^2=I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2=I, T_2^2=I$																
gilmore-lie-groups_ F02141	211	■ 0.819	$T_2^2=I$	$T_2^2=I$	metric-preserving mappings of the Lie algebra onto itself that obey $T_1^2=I, T_2^2=I$																
gilmore-lie-groups_ F02142	211	■ 1.000	$T_1 \neq T_2$	$T_1 \neq T_2$	(cf. Section 11.3) and $T_1 \neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm, \pm}$:																
gilmore-lie-groups_ F02143	211	■ 1.000	$\mathfrak{g}_{\pm, \pm}$	$\mathfrak{g}_{\pm, \pm}$	(cf. Section 11.3) and $T_1 \neq T_2$. Define the eigenspaces of $\mathfrak{g}$ under T_1, T_2 as $\mathfrak{g}_{\pm, \pm}$:																
gilmore-lie-groups_ F02144	212	■ 1.000	$\mathfrak{h}^{\{\prime \prime\}}, \mathfrak{p}^{\{\prime \prime\}}$	$\mathfrak{h}'', \mathfrak{p}''$	The subspaces $\mathfrak{h}'', \mathfrak{p}''$ obey commutation relations (12.20). The Cartan-Killing inner																
gilmore-lie-groups_ F02145	212	■ 0.996	$\mathfrak{p}^{\{\prime \prime\}}$	$\mathfrak{p}''$	product is indefinite on both $\mathfrak{h}''$ and $\mathfrak{p}''$ as long as $T_1 \neq T_2$.																
gilmore-lie-groups_ F02146	212	■ 0.999	$T_1=$	$T_1=$	For $\mathfrak{su}(2)$ the only two mappings are $T_1=$ block diagonal decomposition and																
gilmore-lie-groups_ F02147	212	■ 1.000	$T_2=$	$T_2=$	$T_2=$ complex conjugation. The eigenspace decomposition is																
gilmore-lie-groups_ F02148	212	■ 1.000	$i \sigma_1$	$i \sigma_1$	<table><tr><th>Operation</th><th>$i \sigma_1$</th><th>$i \sigma_2$</th><th>$i \sigma_3$</th></tr><tr><td>$T_1=$ block matrix decomposition</td><td>−1</td><td>−1</td><td>+1</td></tr><tr><td>$T_2=$ complex conjugation</td><td>−1</td><td>+1</td><td>−1</td></tr><tr><td>$T_3=T_1 T_2$</td><td>+1</td><td>−1</td><td>−1</td></tr></table>	Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$	$T_1=$ block matrix decomposition	−1	−1	+1	$T_2=$ complex conjugation	−1	+1	−1	$T_3=T_1 T_2$	+1	−1	−1
Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$																		
$T_1=$ block matrix decomposition	−1	−1	+1																		
$T_2=$ complex conjugation	−1	+1	−1																		
$T_3=T_1 T_2$	+1	−1	−1																		
gilmore-lie-groups_ F02149	212	■ 1.000	$i \sigma_2$	$i \sigma_2$	<table><tr><th>Operation</th><th>$i \sigma_1$</th><th>$i \sigma_2$</th><th>$i \sigma_3$</th></tr><tr><td>$T_1=$ block matrix decomposition</td><td>−1</td><td>−1</td><td>+1</td></tr><tr><td>$T_2=$ complex conjugation</td><td>−1</td><td>+1</td><td>−1</td></tr><tr><td>$T_3=T_1 T_2$</td><td>+1</td><td>−1</td><td>−1</td></tr></table>	Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$	$T_1=$ block matrix decomposition	−1	−1	+1	$T_2=$ complex conjugation	−1	+1	−1	$T_3=T_1 T_2$	+1	−1	−1
Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$																		
$T_1=$ block matrix decomposition	−1	−1	+1																		
$T_2=$ complex conjugation	−1	+1	−1																		
$T_3=T_1 T_2$	+1	−1	−1																		
gilmore-lie-groups_ F02150	212	■ 1.000	$i \sigma_3$	$i \sigma_3$	<table><tr><th>Operation</th><th>$i \sigma_1$</th><th>$i \sigma_2$</th><th>$i \sigma_3$</th></tr><tr><td>$T_1=$ block matrix decomposition</td><td>−1</td><td>−1</td><td>+1</td></tr><tr><td>$T_2=$ complex conjugation</td><td>−1</td><td>+1</td><td>−1</td></tr><tr><td>$T_3=T_1 T_2$</td><td>+1</td><td>−1</td><td>−1</td></tr></table>	Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$	$T_1=$ block matrix decomposition	−1	−1	+1	$T_2=$ complex conjugation	−1	+1	−1	$T_3=T_1 T_2$	+1	−1	−1
Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$																		
$T_1=$ block matrix decomposition	−1	−1	+1																		
$T_2=$ complex conjugation	−1	+1	−1																		
$T_3=T_1 T_2$	+1	−1	−1																		
gilmore-lie-groups_ F02151	212	■ 1.000	$T_3=T_1 T_2$	$T_3=T_1 T_2$	<table><tr><th>Operation</th><th>$i \sigma_1$</th><th>$i \sigma_2$</th><th>$i \sigma_3$</th></tr><tr><td>$T_1=$ block matrix decomposition</td><td>−1</td><td>−1</td><td>+1</td></tr><tr><td>$T_2=$ complex conjugation</td><td>−1</td><td>+1</td><td>−1</td></tr><tr><td>$T_3=T_1 T_2$</td><td>+1</td><td>−1</td><td>−1</td></tr></table>	Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$	$T_1=$ block matrix decomposition	−1	−1	+1	$T_2=$ complex conjugation	−1	+1	−1	$T_3=T_1 T_2$	+1	−1	−1
Operation	$i \sigma_1$	$i \sigma_2$	$i \sigma_3$																		
$T_1=$ block matrix decomposition	−1	−1	+1																		
$T_2=$ complex conjugation	−1	+1	−1																		
$T_3=T_1 T_2$	+1	−1	−1																		

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gilmore-lie-groups_ F02152	212	■ 1.000	$\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i \sigma_3, \mathfrak{g}_{-,-}=i \sigma_1, \mathfrak{g}_{-,+}=i \sigma_2, \mathfrak{g}_{+,-}=i \sigma_3, \mathfrak{g}_{-,-}=i \sigma_2, \mathfrak{g}_{-,+}=i \sigma_1$	$\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i \sigma_3, \mathfrak{g}_{-,-}=i \sigma_2, \mathfrak{g}_{-,+}=i \sigma_1$	This gives $\mathfrak{g}_{+,+}=0, \mathfrak{g}_{+,-}=i \sigma_3, \mathfrak{g}_{-,-}=i \sigma_2, \mathfrak{g}_{-,+}=i \sigma_1$. Note that each mapping
gilmore-lie-groups_ F02153	212	■ 1.000	$T_{\{i\}}$	T_i	T_i has one positive and two negative eigenvalues, and chooses a different generator
gilmore-lie-groups_ F02154	212	■ 0.999	$\mathfrak{h}^{\{\prime\}}$	$\mathfrak{h}'$	for the maximal compact subalgebra $\mathfrak{h}'$ of the noncompact real form $\mathfrak{g}'$.
gilmore-lie-groups_ F02155	212	■ 0.984	$P=G / K$	$P=G / K$	Globally symmetric spaces have the form $P=G / K$, where $\mathfrak{g}$ is a real form of
gilmore-lie-groups_ F02156	212	■ 1.000	$\phi_{\{j\}}\left(B, B^{\dagger}\right)$	$\phi_j\left(B, B^{\dagger}\right)$	1. Show that the invariant polynomials $\phi_j(B, B^{\dagger})$ in (12.4) actually depend on the invari-
gilmore-lie-groups_ F02157	212	■ 0.585	$(p-q)$	$(p-q)$	The remaining $(p-q)$ or $(q-p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02158	212	■ 0.585	$(q-p)$	$(q-p)$	The remaining $(p-q)$ or $(q-p)$ (whichever is positive) eigenvalues of the larger
gilmore-lie-groups_ F02159	213	■ 1.000	Δ^2	Δ^2	2. The second order Laplace–Beltrami operator Δ^2 is constructed from the second order
gilmore-lie-groups_ F02160	213	■ 1.000	$G / H=P$	$G / H=P$	$G / H=P$.
gilmore-lie-groups_ F02161	213	■ 1.000	$\Delta^2=\mathfrak{g}^{\{i j\}}\left(\partial_i \partial_j-\Gamma_{i j}^k \partial_k\right)$	$\Delta^2=g^{ij}\left(\partial_i \partial_j-\Gamma_{ij}^k \partial_k\right)$	tensor on P as $\Delta^2=g^{ij}\left(\partial_i \partial_j-\Gamma_{ij}^k \partial_k\right)$.
gilmore-lie-groups_ F02162	213	■ 1.000	$x^2+y^2 \leq 1$	$x^2+y^2 \leq 1$	unit disk $x^2+y^2 \leq 1$.
gilmore-lie-groups_ F02163	213	■ 1.000	$S^{\{n\}}$	S^n	e. Extend these results to the sphere S^n and its dual, $H^n, n>2$.
gilmore-lie-groups_ F02164	213	■ 1.000	$H^{\{n\}}, n>2$	$H^n, n>2$	e. Extend these results to the sphere S^n and its dual, $H^n, n>2$.
gilmore-lie-groups_ F02165	213	■ 1.000	$T_{\{1\}}^2=T_{\{2\}}^2=I$	$T_1^2=T_2^2=I$	3. Show that the two metric-preserving mappings T_1 and T_2 that satisfy $T_1^2=T_2^2=I$
gilmore-lie-groups_ F02166	213	■ 1.000	$T_{\{1\}} T_{\{2\}}=T_{\{2\}} T_{\{1\}}$	$T_1 T_2=T_2 T_1$	generate a third, $T_3=T_1 T_2$ and that $T_1 T_2=T_2 T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups_ F02167	213	■ 1.000	$T_{\{3\}} \neq I$	$T_3 \neq I$	generate a third, $T_3=T_1 T_2$ and that $T_1 T_2=T_2 T_1$. Show that $T_3 \neq I$ if $T_1 \neq T_2$. Show
gilmore-lie-groups_ F02168	213	■ 1.000	$\mathfrak{g}=\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{+,+}$	$\mathfrak{g}=\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{-,-}$	compact Lie algebra $\mathfrak{g}=\mathfrak{g}_{+,+}+\mathfrak{g}_{+,-}+\mathfrak{g}_{-,+}+\mathfrak{g}_{-,-}$ that is available by choosing
gilmore-lie-groups_ F02169	213	■ 1.000	$3 ! / 1 !=6$	$3 ! / 1 !=6$	first, one of these three involutions, and then a second (there are $3 ! / 1 !=6$ choices).
gilmore-lie-groups_ F02170	213	■ 1.000	$b_{\{3\}}=0$	$b_3=0$	from (11.2) by setting $b_3=0$:
gilmore-lie-groups_ F02171	213	■ 1.000	$\phi_{\{j\}}\left(\mathfrak{p}\right)$	$\phi_j(\mathfrak{p})$	5. Show that the coefficients $\phi_j(\mathfrak{p})$ in the secular equation for a symmetric space are
gilmore-lie-groups_ F02172	213	■ 0.992	$\phi_{\{j\}}\left(\mathfrak{h}, \mathfrak{p}\right)$	$\phi_j(\mathfrak{h}, \mathfrak{p})$	obtained from the coefficients $\phi_j(\mathfrak{h}, \mathfrak{p})$ in the secular equation for the parent Lie
gilmore-lie-groups_ F02173	213	■ 0.998	$\mathfrak{h}=0$	$\mathfrak{h}=0$	algebra (Eq. (12.3)) by setting $\mathfrak{h}=0$.
gilmore-lie-groups_ F02174	213	■ 0.891	$x_0^2-x_1^2-x_2^2=1$	$x_0^2-x_1^2-x_2^2=1$	by the condition $x_0^2-x_1^2-x_2^2=1$ in the Lorentz 3-space with signature (1, 2).
gilmore-lie-groups_ F02175	213	■ 1.000	$-d s^2=d x_0^2-d x_1^2-$	$-d s^2=d x_0^2-d x_1^2-$	b. Show that the invariant metric is induced from the metric $-d s^2=d x_0^2-d x_1^2-$
gilmore-lie-groups_ F02176	213	■ 0.973	$d x_2^2$	$d x_2^2$	$d x_2^2$ in this Lorentz 3-space.
gilmore-lie-groups_ F02177	214	■ 1.000	$x_{\{1\}}, x_{\{2\}}$	x_1, x_2	c. Use coordinates x_1, x_2 to parameterize the points in H_2^2 , and show
gilmore-lie-groups_ F02178	214	■ 1.000	$(r, \theta), x_{\{1\}}=r \cos (\theta), x_{\{2\}}=r \sin (\theta)$	$(r, \theta), x_1=r \cos (\theta), x_2=r \sin (\theta)$	e. Introduce polar coordinates $(r, \theta), x_1=r \cos (\theta), x_2=r \sin (\theta)$. Show that

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gilmore-lie-groups_ F02179	214	■ 1.000	$S O(1,2)$	$SO(1, 2)$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.
gilmore-lie-groups_ F02180	214	■ 1.000	$\left(x_{\{1\}}, x_{\{2\}}\right) \in H_{\{2\}}^{\{2\}}$	$(x_1, x_2) \in H_2^2$	f. Determine the action of a group operation in $SO(1, 2)$ on the point $(x_1, x_2) \in H_2^2$.
gilmore-lie-groups_ F02181	214	■ 1.000	$g_{\{i\} \{j\}}(x)$	$g_{ij}(x)$	7. The metric on a pseudo-Riemannian symmetric space is $g_{ij}(x)$.
gilmore-lie-groups_ F02182	214	■ 1.000	$X_{\{r\} \{s\}}=g_{\{r\} \{t\}} x^{\{t\}} \partial_{\{s\}}-$	$X_{rs} = g_{rt} x^t \partial_s -$	a. Show that the generators of infinitesimal rotations at a point are $X_{rs} = g_{rt} x^t \partial_s -$
gilmore-lie-groups_ F02183	214	■ 1.000	$g_{\{s\} \{t\}} x^{\{t\}} \partial_{\{r\}}$	$g_{st} x^t \partial_r$	$g_{st} x^t \partial_r$.
gilmore-lie-groups_ F02184	214	■ 1.000	$\left[X_{\{a\} \{b\}}, \Delta\right]=0$	$[X_{ab}, \Delta] = 0$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs} X_{ab} X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups_ F02185	214	■ 1.000	$\Delta=G^{\{a\} \{b\} ; \{r\} \{s\}} X_{\{a\} \{b\}} X_{\{r\} \{s\}}$	$\Delta = G^{ab;rs} X_{ab} X_{rs}$	b. Show $[X_{ab}, \Delta] = 0$, where $\Delta = G^{ab;rs} X_{ab} X_{rs}$ is the Laplace–Beltrami operator
gilmore-lie-groups_ F02186	214	■ 0.546	$G_{\{a\} \{b\} ; \{r\} \{s\}}=\operatorname{tr}\left\{\left(\frac{d e f}{\left(X_{\{a\} \{b\}}\right) \frac{d e f}{\left(X_{\{r\} \{s\}}\right)}}\right)\right\}$	$G_{ab;rs} = \operatorname{tr} \{ \operatorname{def}(X_{ab}) \operatorname{def}(X_{rs}) \}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \operatorname{def}(X_{ab}) \operatorname{def}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_ F02187	214	■ 0.546	$G^{\{a\} \{b\} ; \{r\} \{s\}}$	$G^{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \operatorname{def}(X_{ab}) \operatorname{def}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_ F02188	214	■ 0.546	$G_{\{a\} \{b\} ; \{r\} \{s\}}$	$G_{ab;rs}$	on this space, $G_{ab;rs} = \operatorname{tr} \{ \operatorname{def}(X_{ab}) \operatorname{def}(X_{rs}) \}$, and $G^{ab;rs}$ is the inverse of $G_{ab;rs}$.
gilmore-lie-groups_ F02189	214	■ 0.989	Δ	Δ	c. Show that Δ consists of terms that are both quadratic and linear in the operators
gilmore-lie-groups_ F02190	214	■ 1.000	$\partial_{\{r\}}$	∂_r	∂_r , and that
gilmore-lie-groups_ F02191	214	■ 0.765	$\Gamma_{\{r\} \{s\}}^{\{t\}}$	Γ_{rs}^t	The function Γ_{rs}^t is <i>not</i> a tensor. The components of the Christoffel symbol are
gilmore-lie-groups_ F02192	214	■ 1.000	$\left(r,\phi_{\{2\}}, \phi_{\{3\}}, \ldots, \phi_{\{n\}}\right)$	$(r, \phi_2, \phi_3, \dots, \phi_n)$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.
gilmore-lie-groups_ F02193	214	■ 1.000	$S^{\{n\}} \subset R^{\{n+1\}}$	$S^n \subset R^{n+1}$	8. Use radial coordinates $(r, \phi_2, \phi_3, \dots, \phi_n)$ on the sphere $S^n \subset R^{n+1}$.
gilmore-lie-groups_ F02194	215	■ 0.995	$S O(n+1)$	$SO(n+1)$	c. Compare this with the second order Casimir operator for $SO(n+1)$:
gilmore-lie-groups_ F02195	215	■ 1.000	$f_{\{1\}}(\phi)$	$f_1(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_ F02196	215	■ 1.000	$f_{\{2\}}(\phi)$	$f_2(\phi)$	Compute $f_1(\phi)$ and $f_2(\phi)$.
gilmore-lie-groups_ F02197	215	■ 0.994	$a_{\{i\}}^{\dagger} a_{\{j\}} \left(\mathcal{H} = h_{ij}(t) a_i^{\dagger} a_j, 1 \leq i, j \leq \right.$	$a_i^{\dagger} a_j \left(\mathcal{H} = h_{ij}(t) a_i^{\dagger} a_j, 1 \leq i, j \leq \right.$	is a linear superposition of the bilinear products $a_i^{\dagger} a_j$ ($\mathcal{H} = h_{ij}(t) a_i^{\dagger} a_j$, $1 \leq i, j \leq$
gilmore-lie-groups_ F02198	215	■ 0.985	$i \mathcal{H}$	$i\mathcal{H}$	n), so that $i\mathcal{H}$ is a time-dependent element in the Lie algebra $\mathfrak{u}(n)$. Assume the
gilmore-lie-groups_ F02199	215	■ 1.000	$S U(n) / U(n-1)$	$SU(n)/U(n-1)$	trajectory exists in the rank one symmetric space $SU(n)/U(n-1)$. Write down the
gilmore-lie-groups_ F02200	215	■ 1.000	$V^{\{n\}}$	$V^{(n)}$	10. Conformal group The inner product on an n -dimensional linear vector space $V^{(n)}$
gilmore-lie-groups_ F02201	215	■ 0.999	$(x, x)_{\{m\}}=m_{\{i\} \{j\}} x^{\{i\}} x^{\{j\}}$	$(x, x)_m = m_{ij} x^i x^j$	is defined by $(x, x)_m = m_{ij} x^i x^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_ F02202	215	■ 0.999	$n+2$	$n+2$	is defined by $(x, x)_m = m_{ij} x^i x^j$. Define coordinates y in an $n+2$ dimensional linear
gilmore-lie-groups_ F02203	215	■ 1.000	$W^{\{n+2\}}$	$W^{(n+2)}$	vector space $W^{(n+2)}$ as follows
gilmore-lie-groups_ F02204	215	■ 1.000	$(y, y)_{\{M\}}=M_{\{\mu\nu\}} y^{\{\mu\}} y^{\{\nu\}}=(s x, s x)_{\{m\}}-\frac{2}{s} s\left[s(x, x)_{\{m\}}\right]=0$	$(y, y)_M = M_{\mu\nu} y^{\mu} y^{\nu} = (sx, sx)_m - \frac{2}{s} s[s(x, x)_m] = 0$	a. Show $(y, y)_M = M_{\mu\nu} y^{\mu} y^{\nu} = (sx, sx)_m - \frac{2}{s} s[s(x, x)_m] = 0$.

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gilmore-lie-groups_F02205	215	■ 1.000	$G=O(n)$	$G = O(n)$	$G = O(n)$.
gilmore-lie-groups_F02206	215	■ 1.000	$O(n+1,1)$	$O(n+1,1)$	c. Show that the Lie group H that preserves inner products in $W^{(n+2)}$ is $O(n+1,1)$.
gilmore-lie-groups_F02207	215	■ 1.000	$n_{\{1\}}, n_{\{2\}} \backslash \text{left}(n_{\{1\}}+n_{\{2\}}=n \backslash \text{right})$	$n_1, n_2 (n_1 + n_2 = n)$	d. If the metric m has signature $n_1, n_2 (n_1 + n_2 = n)$, show that $G = O(n_1, n_2)$ and
gilmore-lie-groups_F02208	215	■ 1.000	$G=O \backslash \text{left}(n_{\{1\}}, n_{\{2\}} \backslash \text{right})$	$G = O(n_1, n_2)$	d. If the metric m has signature $n_1, n_2 (n_1 + n_2 = n)$, show that $G = O(n_1, n_2)$ and
gilmore-lie-groups_F02209	215	■ 1.000	$H=O \backslash \text{left}(n_{\{1\}}+1, n_{\{2\}}+1 \backslash \text{right})$	$H = O(n_1 + 1, n_2 + 1)$	$H = O(n_1 + 1, n_2 + 1)$.
gilmore-lie-groups_F02210	215	■ 1.000	$SO \backslash \text{left}(n_{\{1\}}+1, n_{\{2\}}+1 \backslash \text{right}) / SO \backslash \text{left}(n_{\{1\}}, n_{\{2\}} \backslash \text{right})$	$SO(n_1 + 1, n_2 + 1) / SO(n_1, n_2)$	f. Construct the quotient space $SO(n_1 + 1, n_2 + 1) / SO(n_1, n_2)$.
gilmore-lie-groups_F02211	215	■ 0.996	$y \rightarrow y^{\prime}$	$y \rightarrow y'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{m+1}$.
gilmore-lie-groups_F02212	215	■ 0.996	$x \rightarrow x^{\prime}$	$x \rightarrow x'$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{m+1}$.
gilmore-lie-groups_F02213	215	■ 0.996	$x^{\prime i}=y^{\prime i} / y^{\prime m+1}$	$x'^i = y'^i / y'^{m+1}$	g. Under a conformal transformation $y \rightarrow y'$ and $x \rightarrow x'$. Show $x'^i = y'^i / y'^{m+1}$.
gilmore-lie-groups_F02214	215	■ 0.997	$(+1,-1,-1,-1)$	$(+1, -1, -1, -1)$	h. The Lorentz metric $(+1, -1, -1, -1)$ leaves the four-momentum invariant:
gilmore-lie-groups_F02215	216	■ 0.988	$L_{\mu \nu}$	$L_{\mu \nu}$	The operators $L_{\mu \nu}$ are the infinitesimal generators of the Lorentz group $SO(3, 1)$
gilmore-lie-groups_F02216	216	■ 1.000	P_{μ}	P_{μ}	and P_{μ} generate translations. Taken together $L_{\mu \nu}$ and P_{μ} generate the Poincaré
gilmore-lie-groups_F02217	216	■ 1.000	K_{μ}	K_{μ}	group. The operator S generates dilations and the four operators K_{μ} generate
gilmore-lie-groups_F02218	216	■ 1.000	$x_{\mu}=g_{\mu \nu} x^{\nu}$	$x_{\mu} = g_{\mu \nu} x^{\nu}$	conformal transformations. Above $x_{\mu} = g_{\mu \nu} x^{\nu}$.
gilmore-lie-groups_F02219	216	■ 0.989	$e^{c^{\mu} K_{\mu}} \backslash \text{left}(x^{\nu} \backslash \text{right})=x^{\prime \nu}=\frac{x^{\nu}+c^{\nu}(x, x)}{1+2(c, x)+(c, c)(x, x)}$	$e^{c^{\mu} K_{\mu}}(x^{\nu}) = x'^{\nu} = \frac{x^{\nu} + c^{\nu}(x, x)}{1 + 2(c, x) + (c, c)(x, x)}$	k. Show that $e^{c^{\mu} K_{\mu}}(x^{\nu}) = x'^{\nu} = \frac{x^{\nu} + c^{\nu}(x, x)}{1 + 2(c, x) + (c, c)(x, x)}$.
gilmore-lie-groups_F02220	216	■ 0.999	$P_{\mu} \rightarrow P_{\mu}^{\prime}=x_{\mu}$	$P_{\mu} \rightarrow P'_{\mu} = x_{\mu}$	m. Discuss the duality created by $P_{\mu} \rightarrow P'_{\mu} = x_{\mu}$ and $K_{\mu} \rightarrow K'_{\mu} = 2(x, \partial) \partial_{\mu} -$
gilmore-lie-groups_F02221	216	■ 0.999	$K_{\mu} \rightarrow K_{\mu}^{\prime}=2(x, \partial) \partial_{\mu}-\partial_{\mu}$	$K_{\mu} \rightarrow K'_{\mu} = 2(x, \partial) \partial_{\mu} -$	m. Discuss the duality created by $P_{\mu} \rightarrow P'_{\mu} = x_{\mu}$ and $K_{\mu} \rightarrow K'_{\mu} = 2(x, \partial) \partial_{\mu} -$
gilmore-lie-groups_F02222	216	■ 0.719	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$	$(\partial, \partial) x_{\mu}$.
gilmore-lie-groups_F02223	216	■ 1.000	$z=x+i y$	$z = x + i y$	11. The upper half of the complex plane has coordinates $z = x + i y$. This upper half-
gilmore-lie-groups_F02224	216	■ 1.000	$P S L(2, \mathbb{R})$	$PSL(2, \mathbb{R})$	and denoted $PSL(2, \mathbb{R})$.
gilmore-lie-groups_F02225	216	■ 1.000	$M,-M \in S L(2 ; \mathbb{R})$	$M, -M \in SL(2; \mathbb{R})$	a. Show that $M, -M \in SL(2; \mathbb{R})$ generate identical transformations. The group
gilmore-lie-groups_F02226	216	■ 1.000	$y^{\prime}>0$	$y' > 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02227	216	■ 1.000	$y>0$	$y > 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02228	216	■ 1.000	$y^{\prime}=0$	$y' = 0$	In particular, show that $y' > 0$ if $y > 0$ and $y' = 0$ if $y = 0$. The transformation
gilmore-lie-groups_F02229	216	■ 0.997	$(y=0)$	$(y = 0)$	$(y = 0)$, onto itself.
gilmore-lie-groups_F02230	217	■ 1.000	$d z^{\prime}=d z / c z+d ^2$	$d z' = d z / c z + d ^2$	d. Show $d z' = d z / c z + d ^2$
gilmore-lie-groups_F02231	217	■ 1.000	$d \mu=d x d y / y^2$	$d \mu = d x d y / y^2$	e. Show that the invariant measure is $d \mu = d x d y / y^2$

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gilmore-lie-groups_F02258	221	■ 1.000	$L_{\{1\}}$	L_1	finitesimal generator L_3) the operators L_1 and L_2 go to
gilmore-lie-groups_F02259	221	■ 1.000	$ISO(2)=E(2)$	$ISO(2) = E(2)$	The commutation relations of the contracted algebra, $ISO(2) = E(2)$, are
gilmore-lie-groups_F02260	221	■ 1.000	$L_{\{3\}}, P_{\{1\}}, P_{\{2\}}$	L_3, P_1, P_2	The three operators L_3, P_1, P_2 generate the group of Euclidean motions of the plane,
gilmore-lie-groups_F02261	221	■ 0.983	$R^{\{2\}}, ISO(2)$	$R^2, ISO(2)$	$E(2)$, or inhomogeneous orthogonal transformations in the plane $R^2, ISO(2)$. This
gilmore-lie-groups_F02262	221	■ 1.000	$P_{\{1\}}=\partial_1$	$P_1 = \partial_1$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_F02263	221	■ 1.000	$P_{\{2\}}=\partial_2$	$P_2 = \partial_2$	the origin in the x - and y -directions, generated by $P_1 = \partial_1$ and $P_2 = \partial_2$.
gilmore-lie-groups_F02264	221	■ 0.698	$x^{\{2\}}+y^{\{2\}}+z^{\{2\}}=R^{\{2\}}$	$x^2 + y^2 + z^2 = R^2$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_F02265	221	■ 0.698	$0, 0, R$	$0, 0, R$	$x^2 + y^2 + z^2 = R^2$ in the neighborhood of the north pole $(0, 0, R)$, as shown in
gilmore-lie-groups_F02266	221	■ 1.000	$R \rightarrow \infty$	$R \rightarrow \infty$	In the limit $R \rightarrow \infty$ we find
gilmore-lie-groups_F02267	221	■ 0.992	$-P_{\{2\}}, +P_{\{1\}}$	$-P_2, +P_1$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_F02268	221	■ 0.992	$-y$	$-y$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_F02269	221	■ 0.992	$+x$	$+x$	large radius are operators $-P_2, +P_1$ describing displacements in the $-y$ and $+x$
gilmore-lie-groups_F02270	221	■ 1.000	$\theta_{\{1\}}, \theta_{\{2\}}$	θ_1, θ_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_F02271	221	■ 1.000	$d_{\{1\}}, d_{\{2\}}$	d_1, d_2	directions. In addition, the parameters θ_1, θ_2 and d_1, d_2 are related by
gilmore-lie-groups_F02272	221	■ 1.000	$R \theta_{\{i\}}(i=1,2)$	$R\theta_i(i = 1, 2)$	with the product $R\theta_i(i = 1, 2)$ approaching a well-defined limit. This corresponds
gilmore-lie-groups_F02273	221	■ 1.000	$\theta_{\{2\}}=d_{\{1\}} / R$	$\theta_2 = d_1/R$	to a rotation through an angle $\theta_2 = d_1/R$ about the y -axis producing a displacement
gilmore-lie-groups_F02274	221	■ 1.000	$d_{\{1\}}$	d_1	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_F02275	221	■ 1.000	$\theta_{\{1\}}=d_{\{2\}} / R$	$\theta_1 = d_2/R$	of d_1 in the x -direction, and a rotation through an angle $\theta_1 = d_2/R$ about the x -axis
gilmore-lie-groups_F02276	221	■ 1.000	$-d_{\{2\}}$	$-d_2$	producing a displacement of $-d_2$ in the y -direction.
gilmore-lie-groups_F02277	222	■ 0.737	$\mathrm{SO}(4) \rightarrow \operatorname{ISO}(3)$	$SO(4) \rightarrow ISO(3)$	13.3.2 The contraction $SO(4) \rightarrow ISO(3)$
gilmore-lie-groups_F02278	223	■ 0.418	$P_{\{i\}}$	P_i	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_F02279	223	■ 0.418	$i=1,2,3$	$i = 1, 2, 3$	The operators P_i describe displacements in the x -, y -, and z -directions ($i = 1, 2, 3$).
gilmore-lie-groups_F02280	223	■ 1.000	$ISO(3)$	$ISO(3)$	The contracted group is $ISO(3)$, the Euclidean, or inhomogeneous orthogonal
gilmore-lie-groups_F02281	223	■ 1.000	$SO(3) \rightarrow ISO(2)$	$SO(3) \rightarrow ISO(2)$	As in the case $SO(3) \rightarrow ISO(2)$, we can contract the second order Casimir
gilmore-lie-groups_F02282	223	■ 1.000	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	$\mathbf{P} \cdot \mathbf{P} = \nabla^2$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_F02283	223	■ 1.000	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	$\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$	The two invariant operators $\mathbf{P} \cdot \mathbf{P} = \nabla^2$ and $\mathbf{L} \cdot \mathbf{P} = -\mathbf{L} \cdot \nabla$ form a complete set of
gilmore-lie-groups_F02284	224	■ 0.994	$\mathrm{SO}(4,1) \rightarrow \operatorname{ISO}(3,1)$	$SO(4, 1) \rightarrow ISO(3, 1)$	13.3.3 The contraction $SO(4, 1) \rightarrow ISO(3, 1)$

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gilmore-lie-groups_F02285	224	0.592	$[S_0(3,1)]$	$[SO(3,1)]$	The group $ISO(3, 1)$ consists of proper Lorentz transformations $[SO(3, 1)]$ that
gilmore-lie-groups_F02286	224	0.997	$S_0(3,2)$	$SO(3, 2)$	This group can be contracted from either $SO(4, 1)$ or $SO(3, 2)$.
gilmore-lie-groups_F02287	224	0.998	$\epsilon^{\alpha\beta\gamma\mu\nu}$	$\epsilon^{\alpha\beta\gamma\mu\nu}$	where $\epsilon^{\alpha\beta\gamma\mu\nu}$ is the Levi–Civita symbol (antisymmetric tensor) on five symbols,
gilmore-lie-groups_F02288	224	1.000	W_{α}	W_{α}	and W_{α} is similarly defined. The contracted limit of W^{α} is nonzero only if one of
gilmore-lie-groups_F02289	224	1.000	W^{α}	W^{α}	and W_{α} is similarly defined. The contracted limit of W^{α} is nonzero only if one of
gilmore-lie-groups_F02290	224	0.615	v	v	the four remaining symbols (e.g., v) is 5:
gilmore-lie-groups_F02291	225	1.000	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} \left(\partial / \partial x^{\mu} \right)$	$\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial / \partial x^{\mu})$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial / \partial x^{\mu})$ is fairly complicated. Since $W^{\alpha} W_{\alpha}$ is invariant,
gilmore-lie-groups_F02292	225	1.000	$W^{\alpha} W_{\alpha}$	$W^{\alpha} W_{\alpha}$	The four vector $\epsilon^{\alpha\beta\gamma\mu 5} X_{\beta\gamma} (\partial / \partial x^{\mu})$ is fairly complicated. Since $W^{\alpha} W_{\alpha}$ is invariant,
gilmore-lie-groups_F02293	225	1.000	$\mathbf{P} \cdot \mathbf{P} = \sum P_{\mu} P^{\mu} = -(mc)^2$	$\mathbf{P} \cdot \mathbf{P} = \sum P_{\mu} P^{\mu} = -(mc)^2$	with $\mathbf{P} \cdot \mathbf{P} = \sum P_{\mu} P^{\mu} = -(mc)^2$.
gilmore-lie-groups_F02294	225	0.862	$\mathfrak{u}(2)$	$\mathfrak{u}(2)$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_{\pm}, J_0$ with com-
gilmore-lie-groups_F02295	225	0.862	$J_3, J_{\pm}, J_0$	$J_3, J_{\pm}, J_0$	The Lie algebra $\mathfrak{u}(2)$ is spanned by infinitesimal generators $J_3, J_{\pm}, J_0$ with com-
gilmore-lie-groups_F02296	225	0.996	$h_3, h_{\pm}, h_0$	$h_3, h_{\pm}, h_0$	The operators $h_3, h_{\pm}, h_0$ are related to $J_3, J_{\pm}, J_0$ by the following change of basis
gilmore-lie-groups_F02297	226	1.000	$c \rightarrow 0$	$c \rightarrow 0$	In the limit $c \rightarrow 0$ the change of basis transformation becomes singular but the
gilmore-lie-groups_F02298	226	1.000	$J_0 \rightarrow h_0$	$J_0 \rightarrow h_0$	Under contraction $J_0 \rightarrow h_0$ but the second Casimir operator has a more interesting
gilmore-lie-groups_F02299	226	1.000	$(h_0/2c)^2$	$(h_0/2c)^2$	The operator $(h_0/2c)^2$ is proportional to the square of the first Casimir operator. It
gilmore-lie-groups_F02300	226	1.000	$c \rightarrow 0, (ch_3)^2 \rightarrow 0$	$c \rightarrow 0, (ch_3)^2 \rightarrow 0$	in the Lie algebra. In the limit $c \rightarrow 0$, $(ch_3)^2 \rightarrow 0$ and the remaining operators go
gilmore-lie-groups_F02301	227	1.000	$\hat{n} + \frac{1}{2}I, a^{\dagger}, a$	$\hat{n} + \frac{1}{2}I, a^{\dagger}, a$	This is a quadratic operator in the generators $\hat{n} + \frac{1}{2}I, a^{\dagger}, a$, and I of H_4 . The value
gilmore-lie-groups_F02302	227	1.000	$\mathfrak{h}_4$	$\mathfrak{h}_4$	An arbitrary element in the Lie algebra $\mathfrak{u}(2)$ and its counterpart in the algebra $\mathfrak{h}_4$
gilmore-lie-groups_F02303	227	1.000	θ_0	θ_0	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_F02304	227	1.000	$\theta_0 - \theta_3/2c^2$	$\theta_0 - \theta_3/2c^2$	exist. In addition, θ_0 should diverge so that $\theta_0 - \theta_3/2c^2$ remains well defined.
gilmore-lie-groups_F02305	227	0.999	h_3	h_3	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_F02306	227	0.999	$ J, M\rangle$	$ J, M\rangle$	The action of the operators h_3 on the angular momentum state $ J, M\rangle$ is
gilmore-lie-groups_F02307	227	1.000	$ J, -J\rangle$	$ J, -J\rangle$	It is useful to measure states from the “lowest” state $ J, -J\rangle$ in the angular mo-
gilmore-lie-groups_F02308	227	1.000	$M = -J$	$M = -J$	$M = -J$, and the n th state when
gilmore-lie-groups_F02309	227	0.985	J	J	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-
gilmore-lie-groups_F02310	227	0.985	$(2J + 1)$	$(2J + 1)$	representations J of dimension $(2J + 1)$ as c becomes smaller and smaller. Specif-

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gilmore-lie-groups_F02311	228	■ 1.000	n=J+M	$n = J + M$	applying the angular momentum shift up operator $n = J + M$ times to the ground
gilmore-lie-groups_F02312	229	■ 1.000	$\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$	$\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$	contracts under $\lim_{c \rightarrow 0} \zeta / c \rightarrow \alpha$ to the BCH formula for H_4
gilmore-lie-groups_F02313	229	■ 1.000	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$	$P_m^l(\cos \theta)$ are related by (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b)
gilmore-lie-groups_F02314	229	■ 1.000	$u \rightarrow$	$u \rightarrow$	These polynomials are contracted to harmonic oscillator wavefunctions under $u \rightarrow$
gilmore-lie-groups_F02315	229	■ 1.000	$x / \sqrt{l}$	$x / \sqrt{l}$	$x / \sqrt{l}$ and $l + m = n$:
gilmore-lie-groups_F02316	229	■ 1.000	$l+m=n$	$l + m = n$	$x / \sqrt{l}$ and $l + m = n$:
gilmore-lie-groups_F02317	230	■ 1.000	$c \rightarrow 0, l \rightarrow \infty, l+m=n, 2lc^2=1$	$c \rightarrow 0, l \rightarrow \infty, l+m=n, 2lc^2=1$	The limit is taken as $c \rightarrow 0, l \rightarrow \infty, l+m=n, 2lc^2=1$. The limit inside the
gilmore-lie-groups_F02318	230	■ 1.000	$1 / \sqrt{\pi}$	$1 / \sqrt{\pi}$	first square root is $1 / \sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_F02319	230	■ 1.000	$\left(2^n n!\right)^{-1}$	$(2^n n!)^{-1}$	first square root is $1 / \sqrt{\pi}$, that within the second is $(2^n n!)^{-1}$. The result of this
gilmore-lie-groups_F02320	230	■ 1.000	$H_n(x)$	$H_n(x)$	and $H_n(x)$ is the n th Hermite polynomial.
gilmore-lie-groups_F02321	231	■ 1.000	$ISO(2)$	$ISO(2)$	representations of $ISO(2)$. Since $ISO(2)$ is a noncompact group it has no faith-
gilmore-lie-groups_F02322	231	■ 1.000	$P_{\pm}$	$P_{\pm}$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_F02323	231	■ 1.000	$\mathfrak{iso}(2)$	$\mathfrak{iso}(2)$	a. Compute the matrix elements of the operators $P_{\pm}$ in the algebra $\mathfrak{iso}(2)$ and show
gilmore-lie-groups_F02324	231	■ 1.000	$J_k(x)$	$J_k(x)$	where $J_k(x)$ is the k th Bessel function (Arecchi <i>et al.</i> , 1972; Gilmore, 1974b).
gilmore-lie-groups_F02325	231	■ 0.992	A_k^l	A_k^l	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_F02326	231	■ 0.992	$\theta^{\prime}, \phi^{\prime}$	θ', ϕ'	compute the coefficients A_k^l and the arguments θ', ϕ' explicitly. Contract these
gilmore-lie-groups_F02327	231	■ 0.998	$\mathbf{L} \cdot \mathbf{L}$	$\mathbf{L} \cdot \mathbf{L}$	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.
gilmore-lie-groups_F02328	231	■ 0.998	∇^2	∇^2	f. Show that the operator $\mathbf{L} \cdot \mathbf{L}$ contracts to ∇^2 in the plane.
gilmore-lie-groups_F02329	231	■ 1.000	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	$\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$	2. Under the contraction $\mathfrak{u}(2) \rightarrow \mathfrak{h}_4$ the representations of the unitary group $U(2)$ con-
gilmore-lie-groups_F02330	232	■ 1.000	N_n	N_n	$H_n(x)$ is the n th Hermite polynomial, and N_n is the usual normalization coefficient,
gilmore-lie-groups_F02331	232	■ 0.993	$N_n=1 / \sqrt{2^n n! \sqrt{\pi}}$	$N_n = 1 / \sqrt{2^n n! \sqrt{\pi}}$	$N_n = 1 / \sqrt{2^n n! \sqrt{\pi}}$.
gilmore-lie-groups_F02332	232	■ 0.984	$J_3, J_{\pm} \left(\left[J_3, J_{\pm} \right] = \pm J_{\pm}, \left[J_+, J_- \right] = 2 J_3 \right)$	$J_3, J_{\pm} ([J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3)$	3. Contract the Lie algebra $\mathfrak{su}(2)$ spanned by $J_3, J_{\pm}$ ($[J_3, J_{\pm}] = \pm J_{\pm}, [J_+, J_-] = 2J_3$)
gilmore-lie-groups_F02333	232	■ 0.993	$\left(P^{\prime}, T^{\prime}, V^{\prime} \right)$	(P', T', V')	operators on the right (P', T', V') satisfy an isomorphic set of commutation relations.
gilmore-lie-groups_F02334	232	■ 0.991	$\partial_x, \partial_t, t \partial_x$	$\partial_x, \partial_t, t \partial_x$	The operators $\partial_x, \partial_t, t \partial_x$ generate the Galilean group in one dimension. Conclude
gilmore-lie-groups_F02335	232	■ 1.000	$\mathfrak{a}_1$	$\mathfrak{a}_1$	that if the Lie algebra $\mathfrak{a}_1$ is contracted with respect to one of its shift operators the
gilmore-lie-groups_F02336	232	■ 0.985	$\mathfrak{gal}(1)$	$\mathfrak{gal}(1)$	Galilean algebra $\mathfrak{gal}(1)$ results.
gilmore-lie-groups_F02337	232	■ 1.000	$S^n = SO(n+1) / SO(n)$	$S^n = SO(n+1) / SO(n)$	metric and measure on the sphere $S^n = SO(n+1) / SO(n)$ reduce to the familiar

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gilmore-lie-groups_ F02338	232	■ 1.000	$R^n = \text{ISO}(n) / \text{SO}(n)$	$R^n = \text{ISO}(n)/\text{SO}(n)$	metric and measure on $R^n = \text{ISO}(n)/\text{SO}(n)$.
gilmore-lie-groups_ F02339	233	■ 1.000	$\tau = (\zeta/ \zeta) \tan(\zeta)$	$\tau = (\zeta/ \zeta) \tan(\zeta)$	and show $\tau = (\zeta/ \zeta) \tan(\zeta)$.
gilmore-lie-groups_ F02340	233	■ 1.000	$\alpha = \lim_{c \rightarrow 0} \zeta / c$	$\alpha = \lim_{c \rightarrow 0} \zeta / c$	$\alpha = \lim_{c \rightarrow 0} \zeta / c$.
gilmore-lie-groups_ F02341	233	■ 1.000	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$
gilmore-lie-groups_ F02342	233	■ 1.000	$\langle e^{\alpha X} \rangle =$	$\langle e^{\alpha X} \rangle =$	$\langle X \rangle = \text{tr} X e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$, and a generating function for expectation values is $\langle e^{\alpha X} \rangle =$
gilmore-lie-groups_ F02343	233	■ 1.000	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$	$\text{tr} e^{\alpha X} e^{-\beta \mathcal{H}} / \text{tr} e^{-\beta \mathcal{H}}$. When the operators X and $\mathcal{H}$ are elements in a finite dimensional
gilmore-lie-groups_ F02344	233	■ 0.998	$\mathcal{H} = \epsilon J_3$	$\mathcal{H} = \epsilon J_3$	a. Assume $\mathcal{H} = \epsilon J_3$ and X is in the Lie algebra $\mathfrak{su}(2)$. Show that in the 2×2 defining
gilmore-lie-groups_ F02345	233	■ 1.000	$2j + 1 = 2$	$2j + 1 = 2$	c. Show that in the 2×2 matrix representation with $j = \frac{1}{2}$ and $2j + 1 = 2$,
gilmore-lie-groups_ F02346	234	■ 0.974	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	$\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_ F02347	234	■ 0.974	$\mathcal{C}^4 =$	$\mathcal{C}^4 =$	are analytic continuations of $\mathcal{C}^2 = \sum_{ij} X_{ij}^2, \mathcal{C}^3 = \epsilon^{abcdef} X_{ab} X_{cd} X_{ef}$, and $\mathcal{C}^4 =$
gilmore-lie-groups_ F02348	234	■ 0.812	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $\text{SO}(6)$.
gilmore-lie-groups_ F02349	234	■ 0.812	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $\text{SO}(6)$.
gilmore-lie-groups_ F02350	234	■ 0.812	$\text{SO}(6)$	$\text{SO}(6)$	$\sum_{ij} Y_{ij}^2$, where $Y_{ij} = \epsilon^{ijcdef} X_{cd} X_{ef}$ of the group $\text{SO}(6)$.
gilmore-lie-groups_ F02351	234	■ 0.995	$\text{SO}(4) \otimes \text{SO}(2)$	$\text{SO}(4) \otimes \text{SO}(2)$	b. Contract $\text{SO}(4, 2)$ with respect to the subgroup $\text{SO}(4) \otimes \text{SO}(2)$.
gilmore-lie-groups_ F02352	234	■ 0.933	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	$A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$	of part a . If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_ F02353	234	■ 0.933	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	$B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$	of part a . If we define $A_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i5}$ and $B_i = \lim_{\epsilon \rightarrow 0} \epsilon X_{i6}$, then show that
gilmore-lie-groups_ F02354	234	■ 1.000	$\text{SO}(4, 2) / [\text{SO}(4) \otimes \text{SO}(2)]$	$\text{SO}(4, 2) / [\text{SO}(4) \otimes \text{SO}(2)]$	$\text{SO}(4, 2) / [\text{SO}(4) \otimes \text{SO}(2)]$ and $I[\text{SO}(4) \otimes \text{SO}(2)] / [\text{SO}(4) \otimes \text{SO}(2)]$.
gilmore-lie-groups_ F02355	234	■ 1.000	$I[\text{SO}(4) \otimes \text{SO}(2)] / [\text{SO}(4) \otimes \text{SO}(2)]$	$I[\text{SO}(4) \otimes \text{SO}(2)] / [\text{SO}(4) \otimes \text{SO}(2)]$	$\text{SO}(4, 2) / [\text{SO}(4) \otimes \text{SO}(2)]$ and $I[\text{SO}(4) \otimes \text{SO}(2)] / [\text{SO}(4) \otimes \text{SO}(2)]$.
gilmore-lie-groups_ F02356	234	■ 1.000	X_α	X_α	Operators X_i span $\mathfrak{h}$ and X_α span $\mathfrak{p}$.
gilmore-lie-groups_ F02357	234	■ 0.987	$Y_\alpha = \lim_{\epsilon \rightarrow 0} \epsilon X_\alpha$	$Y_\alpha = \lim_{\epsilon \rightarrow 0} \epsilon X_\alpha$	b. Show that in the contracted limit $Y_\alpha = \lim_{\epsilon \rightarrow 0} \epsilon X_\alpha$ a metric tensor on $\mathfrak{p}$ is well
gilmore-lie-groups_ F02358	236	■ 1.000	$ \psi\rangle$	$ \psi\rangle$	state $ \psi\rangle$ in their respective coordinate systems. They describe the state by mathe-
gilmore-lie-groups_ F02359	236	■ 1.000	$\langle S \psi \rangle$	$\langle S \psi \rangle$	matical functions $\langle S \psi \rangle$ and $\langle S' \psi \rangle$. The two observers know the relation between
gilmore-lie-groups_ F02360	236	■ 1.000	$\langle S' \psi \rangle$	$\langle S' \psi \rangle$	matical functions $\langle S \psi \rangle$ and $\langle S' \psi \rangle$. The two observers know the relation between
gilmore-lie-groups_ F02361	236	■ 0.998	$\langle S' S \rangle$	$\langle S' S \rangle$	tions from one coordinate system to the other is $\langle S' S \rangle$. The set of transformations
gilmore-lie-groups_ F02362	236	■ 1.000	$\langle S S' \rangle$	$\langle S S' \rangle$	S has seen, he applies the appropriate transformation, $\langle S S' \rangle$, to his mathematical
gilmore-lie-groups_ F02363	237	■ 1.000	$ \psi'\rangle$	$ \psi'\rangle$	by S' to describe a <i>new</i> physical state $ \psi'\rangle$
gilmore-lie-groups_ F02364	237	■ 0.997	$\text{SO}(3)$	$\text{SO}(3)$	example, a rotation in $\text{SO}(3)$ is expressed by $\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ F02365	237	■ 0.997	$\left\langle S^{\prime} \mid H \mid S^{\prime}\right\rangle=\left\langle S^{\prime} \mid S\right\rangle\left\langle S \mid H \mid S\right\rangle\left\langle S \mid S^{\prime}\right\rangle$	$\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$	example, a rotation in $SO(3)$) is expressed by $\langle S' H S' \rangle = \langle S' S \rangle \langle S H S \rangle \langle S S' \rangle$,
gilmore-lie-groups_ F02366	237	■ 1.000	$\left\langle S^{\prime} \mid H \mid S^{\prime}\right\rangle=\left\langle S^{\prime} \mid S\right\rangle\left\langle S \mid H \mid S\right\rangle\left\langle S \mid S^{\prime}\right\rangle$	$\langle S' H S' \rangle =$	the invariance under the transformation $\langle S' S \rangle$ is represented by $\langle S' H S' \rangle =$
gilmore-lie-groups_ F02367	237	■ 1.000	$\left\langle S^{\prime} \mid H \mid S^{\prime}\right\rangle=\left\langle S^{\prime} \mid S\right\rangle\left\langle S \mid H \mid S\right\rangle\left\langle S \mid S^{\prime}\right\rangle$	$\langle S H S \rangle$	$\langle S H S \rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups_ F02368	237	■ 1.000	$2 p_z$	$2p_z$	$\langle S H S \rangle$, and the existence of a $2p_z$ state in a system with spherical symmetry
gilmore-lie-groups_ F02369	237	■ 1.000	$2 p_x$	$2p_x$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_ F02370	237	■ 1.000	$2 p_y$	$2p_y$	implies the existence (by the Principle of Equivalence) of $2p_x$ and $2p_y$ states, as
gilmore-lie-groups_ F02371	237	■ 0.996	$p^{\mu} p_{\mu}=g_{\mu \nu} p^{\mu} p^{\nu}=(m c)^2$	$p^{\mu} p_{\mu}=g_{\mu \nu} p^{\mu} p^{\nu}=(m c)^2$	the relativistic dispersion relation for the free particle: $p^{\mu} p_{\mu}=g_{\mu \nu} p^{\mu} p^{\nu}=(m c)^2$.
gilmore-lie-groups_ F02372	237	■ 1.000	$\mathbf{p}$	$\mathbf{p}$	In terms of the energy E and the three-momentum $\mathbf{p}$ this is
gilmore-lie-groups_ F02373	237	■ 1.000	$p_{\mu} \rightarrow \pi_{\mu}=p_{\mu}-(q / c) A_{\mu}$	$p_{\mu} \rightarrow \pi_{\mu}=p_{\mu}-(q / c) A_{\mu}$	principle of minimal electromagnetic coupling: $p_{\mu} \rightarrow \pi_{\mu}=p_{\mu}-(q / c) A_{\mu}$, where
gilmore-lie-groups_ F02374	237	■ 0.974	Φ	Φ	the four-vector potential A consists of the scalar potential Φ and the vector potential
gilmore-lie-groups_ F02375	237	■ 0.997	$\mathbf{B}=\nabla \times \mathbf{A}$	$\mathbf{B}=\nabla \times \mathbf{A}$	$\mathbf{A}$. These obey $\mathbf{B}=\nabla \times \mathbf{A}$ and $\mathbf{E}=-\nabla \Phi-(1 / c)(\partial \mathbf{A} / \partial t)$. For an electron $q =$
gilmore-lie-groups_ F02376	237	■ 0.997	$\mathbf{E}=-\nabla \Phi-(1 / c)(\partial \mathbf{A} / \partial t)$	$\mathbf{E}=-\nabla \Phi-(1 / c)(\partial \mathbf{A} / \partial t)$	$\mathbf{A}$. These obey $\mathbf{B}=\nabla \times \mathbf{A}$ and $\mathbf{E}=-\nabla \Phi-(1 / c)(\partial \mathbf{A} / \partial t)$ For an electron $q =$
gilmore-lie-groups_ F02377	237	■ 0.997	$q =$	$q =$	$\mathbf{A}$. These obey $\mathbf{B}=\nabla \times \mathbf{A}$ and $\mathbf{E}=-\nabla \Phi-(1 / c)(\partial \mathbf{A} / \partial t)$. For an electron $q =$
gilmore-lie-groups_ F02378	237	■ 0.928	$-e$	$-e$	$-e$, where e is the charge on the proton, positive by convention. In the Coulomb
gilmore-lie-groups_ F02379	237	■ 1.000	$\Phi=e / r$	$\Phi=e / r$	field established by a proton, $\Phi=e / r$ and $\mathbf{A}=\mathbf{0}$, so that $E \rightarrow E+e^2 / r$. Here
gilmore-lie-groups_ F02380	237	■ 1.000	$\mathbf{A}=\mathbf{0}$	$\mathbf{A}=\mathbf{0}$	field established by a proton, $\Phi=e / r$ and $\mathbf{A}=\mathbf{0}$, so that $E \rightarrow E+e^2 / r$. Here
gilmore-lie-groups_ F02381	237	■ 1.000	$E \rightarrow E+e^2 / r$	$E \rightarrow E+e^2 / r$	field established by a proton, $\Phi=e / r$ and $\mathbf{A}=\mathbf{0}$, so that $E \rightarrow E+e^2 / r$. Here
gilmore-lie-groups_ F02382	237	■ 1.000	$\mathbf{p} \rightarrow(\hbar / i) \nabla$	$\mathbf{p} \rightarrow(\hbar / i) \nabla$	dispersion relation to a wave equation is to replace $\mathbf{p} \rightarrow(\hbar / i) \nabla$ and allow the
gilmore-lie-groups_ F02383	237	■ 1.000	$\psi(\mathbf{x})$	$\psi(\mathbf{x})$	resulting equation to act on a spacial function $\psi(\mathbf{x})$. This prescription results in the
gilmore-lie-groups_ F02384	238	■ 0.999	$\left\langle S^{\prime} \mid H \mid S^{\prime}\right\rangle=\left\langle S \mid H \mid S\right\rangle$	$\langle S' H S' \rangle = \langle S H S \rangle$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_ F02385	238	■ 0.999	$\left\langle S^{\prime} \mid S\right\rangle \in S O(3)$	$\langle S' S \rangle \in SO(3)$	variant) in form under rotations: $\langle S' H S' \rangle = \langle S H S \rangle$, where $\langle S' S \rangle \in SO(3)$.
gilmore-lie-groups_ F02386	238	■ 0.623	$m c^2$	$m c^2$	Since the binding energy is about 13.6 eV and the electron rest energy $m c^2$ is about
gilmore-lie-groups_ F02387	238	■ 0.906	$E=m c^2+W$	$E=m c^2+W$	510 000 eV, it makes sense to write $E=m c^2+W$, where the principal part of the
gilmore-lie-groups_ F02388	238	■ 1.000	W	W	relativistic energy E is the electron rest energy and the nonrelativistic energy W is
gilmore-lie-groups_ F02389	238	■ 0.884	$(\simeq 0.0025 \%)$	$(\simeq 0.0025 \%)$	a small perturbation of either $\simeq 0.0025 \%$. Under this substitution, and neglecting
gilmore-lie-groups_ F02390	238	■ 1.000	$\left(W+e^2 / r\right)^2 / m c^2$	$\left(W+e^2 / r\right)^2 / m c^2$	terms of order $\left(W+e^2 / r\right)^2 / m c^2$, we obtain the nonrelativistic form of Eq. (14.4),

Identifier	Page	Conf.	LaTeX source	Rendered	Image												
gilmore-lie-groups_F02391	238	0.727	$[A, B] / i \hbar=\{A, B\}$	$[A, B]/i\hbar = \{A, B\}$	tor brackets ($[A, B]/i\hbar = \{A, B\}$) is an attempt to fit quantum mechanics into the												
gilmore-lie-groups_F02392	238	0.860	(r, θ, ϕ)	(r, θ, ϕ)	useful to introduce spherical coordinates (r, θ, ϕ) . In this coordinate system the												
gilmore-lie-groups_F02393	239	1.000	$\mathcal{L}^2\left(S^2\right)$	$\mathcal{L}^2\left(S^2\right)$	The second order differential operator $\mathcal{L}^2(S^2)$ is the Laplacian on the sphere S^2 . Its												
gilmore-lie-groups_F02394	239	0.980	$\mathcal{L}^2\left(S^2\right) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)$	$\mathcal{L}^2\left(S^2\right) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)$	$s \mathcal{L}^2(S^2) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)$ The integers (l, m) satisfy $l=0, 1, 2, \dots$												
gilmore-lie-groups_F02395	239	0.980	(l, m)	(l, m)	is $\mathcal{L}^2(S^2) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $l=0, 1, 2, \dots$												
gilmore-lie-groups_F02396	239	0.980	$l=0, 1, 2, \ldots$	$l=0, 1, 2, \dots$	is $\mathcal{L}^2(S^2) Y_m^l(\theta, \phi)=-l(l+1) Y_m^l(\theta, \phi)$. The integers (l, m) satisfy $=0, 1, 2, \dots$												
gilmore-lie-groups_F02397	239	0.999	$-l \leq m \leq+l$	$-l \leq m \leq+l$	and $-l \leq m \leq+l$. The negative sign and discrete spectrum characteristically indi-												
gilmore-lie-groups_F02398	239	1.000	A, B, C	A, B, C	The values of the coefficients A, B, C that are obtained for the Klein–Gordon												
gilmore-lie-groups_F02399	239	0.424	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+\left(e^2 / \hbar c\right)^2$</td><td>$2 E e^2 /(\hbar c)^2$</td><td>$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2 m e^2 / \hbar^2$</td><td>$2 m W / \hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$
Equation	A	B	C														
Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$														
Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$														
gilmore-lie-groups_F02400	239	0.424	$2 E e^2 /(\hbar c)^2$	$2 E e^2 /(\hbar c)^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+\left(e^2 / \hbar c\right)^2$</td><td>$2 E e^2 /(\hbar c)^2$</td><td>$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2 m e^2 / \hbar^2$</td><td>$2 m W / \hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$
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Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$														
Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$														
gilmore-lie-groups_F02401	239	0.424	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+\left(e^2 / \hbar c\right)^2$</td><td>$2 E e^2 /(\hbar c)^2$</td><td>$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2 m e^2 / \hbar^2$</td><td>$2 m W / \hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$
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Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$														
Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$														
gilmore-lie-groups_F02402	239	0.424	$2 m e^2 / \hbar^2$	$2 m e^2 / \hbar^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+\left(e^2 / \hbar c\right)^2$</td><td>$2 E e^2 /(\hbar c)^2$</td><td>$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2 m e^2 / \hbar^2$</td><td>$2 m W / \hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$
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gilmore-lie-groups_F02403	239	0.424	$2 \mathrm{~m} W / \hbar^2$	$2 \mathrm{~m} W / \hbar^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+\left(e^2 / \hbar c\right)^2$</td><td>$2 E e^2 /(\hbar c)^2$</td><td>$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2 m e^2 / \hbar^2$</td><td>$2 m W / \hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+\left(e^2 / \hbar c\right)^2$	$2 E e^2 /(\hbar c)^2$	$\left[E^2-\left(m c^2\right)^2\right] /(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$
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Schrödinger	$-l(l+1)$	$2 m e^2 / \hbar^2$	$2 m W / \hbar^2$														
gilmore-lie-groups_F02404	239	1.000	$n=$	$n=$	relativistic problems are expressed in terms of the radial quantum number $n=$												
gilmore-lie-groups_F02405	239	1.000	$0, 1, 2, \ldots$	$0, 1, 2, \dots$	$0, 1, 2, \dots$ and the angular momentum quantum number $l=0, 1, 2, \dots$, mass m												
gilmore-lie-groups_F02406	239	1.000	$m_{\text {red }}^{-1}=m_e^{-1}+M_p^{-1}$	$m_{\text {red }}^{-1}=m_e^{-1}+M_p^{-1}$	$m_{red}^{-1}=m_e^{-1}+M_p^{-1}$, and the fine structure constant (Gabrielse <i>et al.</i> , 2006)												

Identifier	Page	Conf.	LaTeX source	Rendered	Image																				
gilmore-lie-groups_ F02497	240	■ 0.655	$0, \infty$	$0, \infty$	<table><thead><tr><th>Procedure</th><th>Result</th></tr></thead><tbody><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$\left[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$</td></tr></tbody></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$\left[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)\right] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$
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gilmore-lie-groups_ F02414	240	■ 0.655	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{\gamma(j+1)} f_j$	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$	<table><thead><tr><th>Procedure</th><th>Result</th></tr></thead><tbody><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2+C=0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma=\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}, \lambda=-\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R=r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$\left[\left(r D^2+2 \gamma D\right)+(2 \lambda \gamma+B+2 \lambda r D)\right] f(r)=0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E=\frac{m c^2}{\sqrt{1+(\alpha / N')^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N'=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1$</td></tr></tbody></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0: R \simeq r^\gamma, \gamma(\gamma-1)+A=0$ $r \rightarrow \infty: R \simeq e^{\lambda r}, \lambda^2+C=0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma=\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}, \lambda=-\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R=r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$\left[\left(r D^2+2 \gamma D\right)+(2 \lambda \gamma+B+2 \lambda r D)\right] f(r)=0$	6 Construct recursion relation	$f_{j+1}=-\frac{2 \lambda(j+\gamma)+B}{j(j+1)+2 \gamma(j+1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)	8 Construct quantization condition	$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$	9 Construct explicit solutions	$E=\frac{m c^2}{\sqrt{1+(\alpha / N')^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N'=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1$
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gilmore-lie-groups_ F02415	240	—	$\begin{aligned} & \& f \simeq e^{-2 \lambda r} \text { if series does not terminate } \\ & \& \simeq e^{+1 \lambda r} \text { if series does terminate } (\lambda<0) \end{aligned}$	$f \simeq e^{-2 \lambda r}$ if series does not terminate $\simeq e^{+1 \lambda r}$ if series does terminate ($\lambda<0$)	—																				
gilmore-lie-groups_ F02416	240	—	$\begin{aligned} & \& 2 \lambda(n+\gamma)+B=0 \text { or } \\ & \& n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}} \end{aligned}$	$2 \lambda(n+\gamma)+B=0$ or $n+\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}=\frac{B}{2 \sqrt{-C}}$	—																				
gilmore-lie-groups_ F02417	240	—	$\begin{aligned} & \& E=\frac{m c^2}{\sqrt{1+\left(\alpha / N^{\prime}\right)^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2} \\ & \& N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1 \end{aligned}$	$E=\frac{m c^2}{\sqrt{1+(\alpha / N')^2}}, W=-\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$ $N'=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}, N=n+l+1$	—																				
gilmore-lie-groups_ F02418	240	■ 1.000	$\hbar$	$\hbar$	three “different” areas of physics: e (electromagnetism), $\hbar$ (quantum mechanics),																				
gilmore-lie-groups_ F02419	241	■ 1.000	$Z=50$	$Z=50$	tic spectrum has been computed for $Z=50$. These energies are computed by																				
gilmore-lie-groups_ F02420	241	■ 1.000	$\alpha \rightarrow Z \alpha$	$\alpha \rightarrow Z \alpha$	replacing $\alpha \Rightarrow Z \alpha$ everywhere.																				
gilmore-lie-groups_ F02421	241	■ 1.000	$\left W_1\right =\frac{1}{2} m c^2 \alpha^2$	$\left W_1\right =\frac{1}{2} m c^2 \alpha^2$	ground state energy $\left W_1\right =\frac{1}{2} m c^2 \alpha^2$. These normalized energy levels decrease to																				
gilmore-lie-groups_ F02422	241	■ 0.999	$1 / N^2$	$1 / N^2$	zero like $1 / N^2$, where $N=n+l+1$ is the principal quantum number. The ener-																				
gilmore-lie-groups_ F02423	241	■ 0.999	$N=n+l+1$	$N=n+l+1$	zero like $1 / N^2$, where $N=n+l+1$ is the principal quantum number. The ener-																				
gilmore-lie-groups_ F02424	241	■ 1.000	$N^{\prime}$	$N^{\prime}$	ergy $m c^2$ as $N^{\prime}$ increases. When this limit is removed these energies (also rescaled																				
gilmore-lie-groups_ F02425	241	■ 1.000	$\frac{1}{2} m c^2 \alpha^2$	$\frac{1}{2} m c^2 \alpha^2$	by dividing by $\frac{1}{2} m c^2 \alpha^2$) can be plotted on the same graph. At the resolution shown,																				

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gilmore-lie-groups_F02426	241	■ 1.000	$\frac{1}{2} m c^2 (Z \alpha)^2$	$\frac{1}{2} m c^2 (Z \alpha)^2$	energy for the same ion: $\frac{1}{2} m c^2 (Z \alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_F02427	241	■ 1.000	$1 s$	$1s$	energy for the same ion: $\frac{1}{2} m c^2 (Z \alpha)^2$. The energy difference between the $1s$ ground
gilmore-lie-groups_F02428	242	■ 1.000	$g_i \psi\rangle$	$g_i \psi\rangle$	eigenstate of H with eigenvalue E , then $g_i \psi\rangle$ is also an eigenstate of H with the
gilmore-lie-groups_F02429	242	■ 1.000	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	$ \psi\rangle = \psi_{2p_z}(\mathbf{x})$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_F02430	242	■ 1.000	$\pi / 2$	$\pi/2$	To illustrate this idea, assume that $ \psi\rangle = \psi_{2p_z}(\mathbf{x})$. A rotation by $\pi/2$ radians about
gilmore-lie-groups_F02431	242	■ 1.000	$\psi_{2p_x}(\mathbf{x})$	$\psi_{2p_x}(\mathbf{x})$	the y -axis maps this state to $\psi_{2p_x}(\mathbf{x})$ and a rotation by $\pi/2$ radians about the x -axis
gilmore-lie-groups_F02432	242	■ 1.000	$-\psi_{2p_y}(\mathbf{x})$	$-\psi_{2p_y}(\mathbf{x})$	maps this state to $-\psi_{2p_y}(\mathbf{x})$. By invariance (of the hamiltonian) under the rotation
gilmore-lie-groups_F02433	242	■ 1.000	$H = \mathbf{p} \cdot \mathbf{p} / 2m - e^2 / r$	$H = \mathbf{p} \cdot \mathbf{p} / 2m - e^2 / r$	$H = \mathbf{p} \cdot \mathbf{p} / 2m - e^2 / r$. The scalar $\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$ is invariant under rotations, as
gilmore-lie-groups_F02434	242	■ 1.000	$\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$	$\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$	$H = \mathbf{p} \cdot \mathbf{p} / 2m - e^2 / r$. The scalar $\mathbf{p} \cdot \mathbf{p} = -\hbar^2 \nabla^2$ is invariant under rotations, as
gilmore-lie-groups_F02435	242	■ 1.000	$-e^2 / r$	$-e^2 / r$	is also the potential energy term $-e^2 / r$. Rotation operators can be expressed in
gilmore-lie-groups_F02436	242	■ 1.000	$i: \epsilon_{ijk} x_j \partial_k$	$i: \epsilon_{ijk} x_j \partial_k$	terms of the infinitesimal generators of rotations about axis $i: \epsilon_{ijk} x_j \partial_k$. These ge-
gilmore-lie-groups_F02437	242	■ 1.000	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$	$\mathbf{L}_i = (\mathbf{r} \times \mathbf{p})_i = (\hbar/i) \epsilon_{ijk} x_j \partial_k$. Finite rotations can be expressed as exponentials
gilmore-lie-groups_F02438	242	■ 0.998	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	The angular momentum operators $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ share the same commutation relations
gilmore-lie-groups_F02439	242	■ 0.992	$\mathbf{r} \times \nabla$	$\mathbf{r} \times \nabla$	as the infinitesimal generators of rotations $\mathbf{r} \times \nabla$, up to the proportionality factor
gilmore-lie-groups_F02440	242	■ 1.000	$\hbar / i$	$\hbar / i$	$\hbar / i$. The commutation relations are
gilmore-lie-groups_F02441	242	■ 1.000	(L_+)	(L_+)	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm i L_y$. The commutation
gilmore-lie-groups_F02442	242	■ 1.000	(L_-)	(L_-)	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm i L_y$. The commutation
gilmore-lie-groups_F02443	242	■ 1.000	$L_{\pm} = L_x \pm i L_y$	$L_{\pm} = L_x \pm i L_y$	raising (L_+) and lowering (L_-) operators by $L_{\pm} = L_x \pm i L_y$. The commutation
gilmore-lie-groups_F02444	242	■ 1.000	$L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$	$L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$	follows: $L_z = \hbar \frac{1}{2} (a_1^\dagger a_1 - a_2^\dagger a_2)$, $L_+ = \hbar a_1^\dagger a_2$, $L_- = \hbar a_2^\dagger a_1$. As a result, the angu-
gilmore-lie-groups_F02445	242	■ 1.000	$ n_1 n_2\rangle$	$ n_1 n_2\rangle$	lar momentum operators have matrix representations with basis vectors $ n_1 n_2\rangle$
gilmore-lie-groups_F02446	242	■ 0.838	$ m^j\rangle$	$ m^j\rangle$	$= m^j\rangle$, with $n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$,
gilmore-lie-groups_F02447	242	■ 0.838	$n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$	$n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$	$= m^j\rangle$, with $n_1 = 0, 1, 2, \dots$, $n_2 = 0, 1, 2, \dots$, $n_1 + n_2 = 2j$, $n_1 - n_2 = 2m$.
gilmore-lie-groups_F02448	243	■ 1.000	$-j \leq m \leq +j$	$-j \leq m \leq +j$	$-j \leq m \leq +j$. These basis vectors describe the finite-dimensional irreducible
gilmore-lie-groups_F02449	243	■ 0.994	$((x, y, z) \rightarrow (r, \theta, \phi))$	$((x, y, z) \rightarrow (r, \theta, \phi))$	operators. In spherical coordinates $((x, y, z) \rightarrow (r, \theta, \phi))$ with $x = r \sin \theta \cos \phi$
gilmore-lie-groups_F02450	243	■ 0.994	$x = r \sin \theta \cos \phi$	$x = r \sin \theta \cos \phi$	operators. In spherical coordinates $((x, y, z) \rightarrow (r, \theta, \phi))$ with $x = r \sin \theta \cos \phi$
gilmore-lie-groups_F02451	243	■ 1.000	$Y_{-l}^l(\theta, \phi)$	$Y_{-l}^l(\theta, \phi)$	construct the extremal function $Y_{-l}^l(\theta, \phi)$ by solving the equation $L_- Y_{-l}^l(\theta, \phi) = 0$

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gilmore-lie-groups_F02452	243	█ 1.000	$L_{-}\,Y_{-l}^{\,l}(\theta,\phi)=0$	$L_{-}Y_{-l}^l(\theta,\phi)=0$	construct the extremal function $Y_{-l}^l(\theta,\phi)$ by solving the equation $L_{-}Y_{-l}^l(\theta,\phi)=0$																									
gilmore-lie-groups_F02453	244	█ 0.998	$l=0$	$l=0$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$
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gilmore-lie-groups_F02454	244	█ 0.998	$l=2$	$l=2$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$
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gilmore-lie-groups_F02455	244	█ 0.998	$l=3$	$l=3$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$
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± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$																										
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± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$																										
gilmore-lie-groups_F02456	244	█ 0.998	$\sqrt[4]{\frac{1}{4\pi}}$	$\sqrt{\frac{1}{4\pi}}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$
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gilmore-lie-groups_F02457	244	█ 0.998	$\sqrt[4]{\frac{3}{4\pi}}\,\cos\theta$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\,e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)\,e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta\,e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta\,e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta\,e^{\pm 3i\phi}$
m	$l=0$	$l=1$	$l=2$	$l=3$																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																									
gilmore-lie-groups_F02458	244	■0.998	$\sqrt{\frac{5}{16\pi}}\left(3\cos^2\theta-1\right)$	$\sqrt{\frac{5}{16\pi}}\left(3\cos^2\theta-1\right)$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$																										
gilmore-lie-groups_F02459	244	■0.998	$\sqrt{\frac{7}{16\pi}}\left(5\cos^3\theta-3\cos\theta\right)$	$\sqrt{\frac{7}{16\pi}}\left(5\cos^3\theta-3\cos\theta\right)$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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gilmore-lie-groups_F02460	244	■0.998	$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$																										
gilmore-lie-groups_F02461	244	■0.998	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
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gilmore-lie-groups_F02462	244	■0.998	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta\left(5\cos^2\theta-1\right)e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta\left(5\cos^2\theta-1\right)e^{\pm i\phi}$	<table><tr><th>m</th><th>$l=0$</th><th>$l=1$</th><th>$l=2$</th><th>$l=3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}}\cos\theta$</td><td>$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$</td><td>$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$</td><td>$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$</td><td>$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$</td></tr></table>	m	$l=0$	$l=1$	$l=2$	$l=3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}}\cos\theta$	$\sqrt{\frac{5}{16\pi}}(3\cos^2\theta-1)$	$\sqrt{\frac{7}{16\pi}}(5\cos^3\theta-3\cos\theta)$	± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$	± 2			$\sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{\pm 2i\phi}$	± 3				$\mp\sqrt{\frac{35}{64\pi}}\sin^3\theta e^{\pm 3i\phi}$
m	$l=0$	$l=1$	$l=2$	$l=3$																										
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± 1		$\mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta e^{\pm i\phi}$	$\mp\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta-1)e^{\pm i\phi}$																										
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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gilmore-lie-groups_ F02464	244	■ 0.998	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i\phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i\phi}$	
gilmore-lie-groups_ F02465	244	■ 0.998	$\sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i\phi}$	$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i\phi}$	
gilmore-lie-groups_ F02466	244	■ 1.000	L_+	L_+	of the shift up operator L_+
gilmore-lie-groups_ F02467	244	■ 0.815	$(l=0, 1, 2, 3)$	$(l = 0, 1, 2, 3)$	The lowest spherical harmonics $l = 0, 1, 2, 3$ are collected in Table 14.2.
gilmore-lie-groups_ F02468	244	■ 0.662	θ, ϕ	θ, ϕ	generators acting on the sphere parameterized by coordinates (θ, ϕ) , is $-l(l+1)$,
gilmore-lie-groups_ F02469	244	■ 1.000	$\hbar^2 l(l+1)$	$\hbar^2 l(l+1)$	of the angular momentum operator, $\mathbf{L} \cdot \mathbf{L}$, is $\hbar^2 l(l+1)$. This means physically that
gilmore-lie-groups_ F02470	244	■ 1.000	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	$E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$	number $N = n + l + 1$ are degenerate, with energy $E_N = -\frac{1}{2} m c^2 \alpha^2 \frac{1}{N^2}$. There are
gilmore-lie-groups_ F02471	244	■ 1.000	$\sum_{l=0}^{N-1} (2l+1) = N^2$	$\sum_{l=0}^{N-1} (2l+1) = N^2$	$\sum_{l=0}^{N-1} (2l+1) = N^2$ states with this energy. This N^2 -fold degeneracy is larger
gilmore-lie-groups_ F02472	244	■ 1.000	$1/r^2$	$1/r^2$	related to a constant of motion that is peculiar to $1/r^2$ force laws. This constant of
gilmore-lie-groups_ F02473	244	■ 1.000	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	$d\mathbf{p}/dt = -K\mathbf{r}/r^3$	planetary motion, for which the force law has the form $d\mathbf{p}/dt = -K\mathbf{r}/r^3$, where
gilmore-lie-groups_ F02474	245	■ 1.000	$K = G M m, G$	$K = G M m, G$	$K = G M m, G$ is the universal gravitational constant, M and m are the two attracting
gilmore-lie-groups_ F02475	245	■ 1.000	$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$	$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$	masses, and $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ is the vector from one mass to the other. The time
gilmore-lie-groups_ F02476	245	■ 1.000	$\mathbf{p} \times \mathbf{L}$	$\mathbf{p} \times \mathbf{L}$	derivative of the vector $\mathbf{p} \times \mathbf{L}$ is
gilmore-lie-groups_ F02477	245	■ 0.999	$\mathbf{L}$	$\mathbf{L}$	In going from the first line in Eq. (14.22) to the second, we use the fact that $\mathbf{L}$ is

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gilmore-lie-groups_F02478	245	■ 1.000	$1 / r$	$1/r$	law for a $1/r$ potential. In going from the second line to the third, we express the
gilmore-lie-groups_F02479	245	■ 0.967	$\mathbf{r} \times \mathbf{L}$	$\mathbf{r} \times \mathbf{L}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_F02480	245	■ 0.967	$\dot{\mathbf{r}}$	$\dot{\mathbf{r}}$	cross product $\mathbf{r} \times \mathbf{L}$ in terms of (generally) nonparallel vectors $\mathbf{r}$ and $\dot{\mathbf{r}}$. We also
gilmore-lie-groups_F02481	245	■ 1.000	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$	$(d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3$	use the identity $\boxed{d/dt)(\mathbf{r}/r) = \dot{\mathbf{r}}/r - (\dot{\mathbf{r}} \cdot \mathbf{r})\mathbf{r}/r^3}$. The result is that the Laplace–
gilmore-lie-groups_F02482	245	■ 0.985	$d\mathbf{M}/dt = 0$	$d\mathbf{M}/dt = 0$	Runge–Lenz vector $\mathbf{M}$ is a constant of motion: $d\mathbf{M}/dt = 0$, where
gilmore-lie-groups_F02483	245	■ 1.000	$[H, \mathbf{M}] = 0$	$[H, \mathbf{M}] = 0$	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_F02484	245	■ 1.000	L_i, M_j	L_i, M_j	with the nonrelativistic hamiltonian: $[H, \mathbf{M}] = 0$. The six operators L_i, M_j obey
gilmore-lie-groups_F02485	245	■ 0.993	$(E < 0)$	$(E < 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_F02486	245	■ 0.993	$(E > 0)$	$(E > 0)$	bound states ($E < 0$) or $SO(3, 1)$ for excited states ($E > 0$). The operators $\mathbf{L}$ and
gilmore-lie-groups_F02487	246	■ 1.000	$\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$	$\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$	Laplace–Runge–Lenz vector by a scale factor as follows: $\mathbf{M}' = (-m/2H)^{1/2}\mathbf{M}$.
gilmore-lie-groups_F02488	246	■ 1.000	$E > 0$	$E > 0$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_F02489	246	■ 1.000	$- \rightarrow +$	$- \rightarrow +$	(For $E > 0$ change $- \rightarrow +$ and $SO(4) \rightarrow SO(3, 1)$.) The commutation relations
gilmore-lie-groups_F02490	246	■ 1.000	$SO(4) \rightarrow SO(3, 1)$	$SO(4) \rightarrow SO(3, 1)$	(For $E > 0$ change $- \rightarrow +$ and $\boxed{SO(4) \rightarrow SO(3, 1)}$.) The commutation relations
gilmore-lie-groups_F02491	246	■ 0.834	$\mathbf{A}$	$\mathbf{A}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_F02492	246	■ 0.834	$\mathbf{B}$	$\mathbf{B}$	Figs. 10.3, 10.8(b)). It is useful to introduce two vector operators $\mathbf{A}$ and $\mathbf{B}$ as
gilmore-lie-groups_F02493	246	■ 0.996	$A_3 = \frac{1}{2}(a_1^\dagger a_1 -$	$A_3 = \frac{1}{2}(a_1^\dagger a_1 -$	momentum operators $\mathbf{A}$ in terms of two independent boson modes: $A_3 = \frac{1}{2}(a_1^\dagger a_1 -$
gilmore-lie-groups_F02494	246	■ 0.910	$a_2^\dagger a_2)$	$a_2^\dagger a_2)$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2$, $A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_F02495	246	■ 0.910	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1)$	$A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1)$	$a_2^\dagger a_2)$, $\boxed{A_+ = a_1^\dagger a_2, A_- = a_2^\dagger a_1}$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_F02496	246	■ 0.910	$\hbar \rightarrow 1$	$\hbar \rightarrow 1$	$a_2^\dagger a_2)$, $A_+ = a_1^\dagger a_2$, $A_- = a_2^\dagger a_1$ (for simplicity, set $\hbar \rightarrow 1$). A similar representation
gilmore-lie-groups_F02497	246	■ 1.000	b_1, b_2	b_1, b_2	b_1, b_2 and their creation operators is also introduced.
gilmore-lie-groups_F02498	246	■ 1.000	$ p_1, p_2\rangle$	$ p_1, p_2\rangle$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02499	246	■ 1.000	$p_1 + p_2 = 2j_a$	$p_1 + p_2 = 2j_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02500	246	■ 1.000	$p_1 - p_2 = m_a$	$p_1 - p_2 = m_a$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02501	246	■ 1.000	$2j_a + 1$	$2j_a + 1$	form $ p_1, p_2\rangle$, with $p_1 + p_2 = 2j_a$ constant and $p_1 - p_2 = m_a$. The $2j_a + 1$ basis
gilmore-lie-groups_F02502	246	■ 0.968	$p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$	$p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1$	states correspond to $\boxed{p_1 = 2j_a, p_2 = 0; p_1 = 2j_a - 1, p_2 = 1}$; etc. For $\mathbf{B}$ the basis
gilmore-lie-groups_F02503	246	■ 1.000	$ q_1, q_2\rangle$	$ q_1, q_2\rangle$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02504	246	■ 1.000	$q_1 + q_2 = 2j_b$	$q_1 + q_2 = 2j_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant

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gilmore-lie-groups_F02505	246	■ 1.000	$q_{\{1\}}-q_{\{2\}}=m_{\{b\}}$	$q_1 - q_2 = m_b$	states are $ q_1, q_2\rangle$, with $q_1 + q_2 = 2j_b$ constant and $q_1 - q_2 = m_b$. The invariant
gilmore-lie-groups_F02506	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	$\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02507	246	■ 1.000	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	$\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02508	246	■ 1.000	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	$\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$	operators are $\mathbf{A} \cdot \mathbf{A} = j_a(j_a + 1)$ and $\mathbf{B} \cdot \mathbf{B} = j_b(j_b + 1)$. Since $\mathbf{A} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{B}$ (cf.
gilmore-lie-groups_F02509	246	■ 1.000	$j_{\{a\}}=j_{\{b\}}$	$j_a = j_b$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02510	246	■ 1.000	$(2j +$	$(2j +$	Problem 14.15), $j_a = j_b$ and the set of states related by the shift operators is $(2j +$
gilmore-lie-groups_F02511	246	■ 1.000	$2j + 1 = N = n + l + 1$	$2j + 1 = N = n + l + 1$	$1)^2$ fold degenerate, where $2l + 1 = N = n + l + 1$.
gilmore-lie-groups_F02512	246	■ 0.996	$\operatorname{good} l$	good l	States with good l and m quantum numbers can be constructed from these states
gilmore-lie-groups_F02513	247	■ 1.000	$M_{+}^{\prime} =$	$M_{+}^{\prime} =$	spherical harmonics, is determined in a straightforward way. For example, $M_{+}^{\prime} =$
gilmore-lie-groups_F02514	247	■ 1.000	$A_{+} - B_{+} = a_1^{\dagger} a_2 - b_1^{\dagger} b_2$	$A_{+} - B_{+} = a_1^{\dagger} a_2 - b_1^{\dagger} b_2$	$A_{+} - B_{+} = a_1^{\dagger} a_2 - b_1^{\dagger} b_2$, so that
gilmore-lie-groups_F02515	247	■ 0.998	$\mathbf{M}^{\prime}$	$\mathbf{M}^{\prime}$	The operators $\mathbf{L}$ and $\mathbf{M}^{\prime}$ are infinitesimal generators for the orthogonal group $SO(4)$.
gilmore-lie-groups_F02516	247	■ 0.608	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	$\mathbf{W} = \mathbf{L} \times \mathbf{M}$	the vectors $\mathbf{L}$ and $\mathbf{M}$. Their cross product $\mathbf{W} = \mathbf{L} \times \mathbf{M}$ is orthogonal to both and
gilmore-lie-groups_F02517	247	■ 1.000	$\mathbf{r} \cdot \mathbf{L} = 0$	$\mathbf{r} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02518	247	■ 1.000	$\mathbf{p} \cdot \mathbf{L} = 0$	$\mathbf{p} \cdot \mathbf{L} = 0$	since $\mathbf{r} \cdot \mathbf{L} = 0$. The momentum vector moves in the same plane, since $\mathbf{p} \cdot \mathbf{L} = 0$.
gilmore-lie-groups_F02519	248	■ 0.986	$\mathbf{W}$	$\mathbf{W}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02520	248	■ 0.986	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	$p_z = 0, p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02521	248	■ 0.986	$p_y =$	$p_y =$	of $\mathbf{M}$ and $\mathbf{W}$. In this coordinate system $p_z = 0$, $p_x = \mathbf{p} \cdot \mathbf{M} / \sqrt{\mathbf{M} \cdot \mathbf{M}}$ and $p_y =$
gilmore-lie-groups_F02522	248	■ 1.000	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$	$\mathbf{p} \cdot \mathbf{W} / \sqrt{\mathbf{W} \cdot \mathbf{W}}$. The two nonzero components of the momentum vector are not
gilmore-lie-groups_F02523	248	■ 1.000	mK / L	mK / L	momentum moves in the same plane on a circular orbit (radius mK / L) with the
gilmore-lie-groups_F02524	248	■ 1.000	mM / L	mM / L	center displaced from the origin by mM / L .
gilmore-lie-groups_F02525	248	■ 1.000	$p_0 = \sqrt{-2E / m}$	$p_0 = \sqrt{-2E / m}$	With $p_0 = \sqrt{-2E / m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-
gilmore-lie-groups_F02526	248	■ 1.000	$\hat{\mathbf{u}} \in S^3 \subset R^4$	$\hat{\mathbf{u}} \in S^3 \subset R^4$	With $p_0 = \sqrt{-2E / m}$, define the unit vector $\hat{\mathbf{u}} \in S^3 \subset R^4$ by the projective trans-
gilmore-lie-groups_F02527	248	■ 1.000	$\hat{\mathbf{w}}$	$\hat{\mathbf{w}}$	Here $\hat{\mathbf{w}}$ is a unit vector in R^4 that is orthogonal to all vectors in the physical space R^3 .
gilmore-lie-groups_F02528	248	■ 0.995	$\hat{\mathbf{u}}$	$\hat{\mathbf{u}}$	preserves angles (conformal). It is a simple matter to check that $\hat{\mathbf{u}}$ is a unit vector.
gilmore-lie-groups_F02529	248	■ 1.000	S^3	S^3	The circular trajectory in R^3 (Eq. (14.33)) lifts to a circle in S^3 . Reversibly, circles in
gilmore-lie-groups_F02530	248	■ 0.803	$SO(4) / SO(3)$	$SO(4) / SO(3)$	$SO(4) / SO(3)$ act to change the lengths of both $\mathbf{L}$ and $\mathbf{M}$. All group operations in

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gilmore-lie-groups_F02531	248	■ 1.000	$p_{\{0\}}$	p_0	$SO(4)$ keep p_0 fixed. In this way the group $SO(4)$ maps states with principal quan-
gilmore-lie-groups_F02532	249	■ 1.000	$N^2=(n+l+1)^2$	$N^2 = (n + l + 1)^2$	irreducible representations of dimension $N^2 = (n + l + 1)^2$.
gilmore-lie-groups_F02533	249	■ 1.000	$-N^3$	$-N^3$	plied each energy eigenvalue by $-N^3$, where N is the principal quantum number.
gilmore-lie-groups_F02534	249	■ 1.000	$\mathbf{x}^{\prime} = M \mathbf{x}$	$\mathbf{x}' = M\mathbf{x}$	linear vector space: $\mathbf{x}' = M\mathbf{x}$, with M a transformation in the group and the inner
gilmore-lie-groups_F02535	249	■ 1.000	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	$(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$	product defined by $(\mathbf{x}, \mathbf{x})_N = \mathbf{x}^t \mathbf{g} \mathbf{x} = x_i g_{ij} x_j$. As always, the metric-preserving
gilmore-lie-groups_F02536	249	■ 0.749	$\mathbf{y}$	$\mathbf{y}$	It is useful to define a new N -vector $\mathbf{y}$ as a scaled version of the original vector:
gilmore-lie-groups_F02537	249	■ 1.000	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02538	249	■ 1.000	$z_1 = \lambda$	$z_1 = \lambda$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02539	249	■ 1.000	$z_2 =$	$z_2 =$	$\mathbf{y} = \lambda \mathbf{x}$. We introduce two additional coordinates by defining $z_1 = \lambda$ and $z_2 =$
gilmore-lie-groups_F02540	249	■ 1.000	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$	$\lambda(\mathbf{x}, \mathbf{x})_N$. With these definitions we find the conformal condition
gilmore-lie-groups_F02541	249	■ 1.000	$N+2$	$N + 2$	The conformal condition defines an inner product in the $N + 2$ dimensional linear
gilmore-lie-groups_F02542	249	■ 0.998	$\mathbf{y}, z_1, z_2$	$\mathbf{y}, z_1, z_2$	vector space that is nondiagonal in the coordinates $\mathbf{y}, z_1, z_2$ but diagonal in the
gilmore-lie-groups_F02543	249	■ 0.986	$\mathbf{y}, y_{N+1}, y_{N+2}$	$\mathbf{y}, y_{N+1}, y_{N+2}$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02544	249	■ 0.986	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	$y_{N+1} = \frac{1}{2}(z_1 + z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02545	249	■ 0.986	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	$y_{N+2} = \frac{1}{2}(z_1 - z_2)$	coordinates $\mathbf{y}, y_{N+1}, y_{N+2}$, with $y_{N+1} = \frac{1}{2}(z_1 + z_2)$ and $y_{N+2} = \frac{1}{2}(z_1 - z_2)$:
gilmore-lie-groups_F02546	249	■ 1.000	R^N	R^N	sional space. If the group that preserves the metric G in R^N is $SO(p, q)$, the group
gilmore-lie-groups_F02547	250	■ 0.999	R^{N+2}	R^{N+2}	that preserves the metric in R^{N+2} is $SO(p + 1, q + 1)$. We seek to construct a
gilmore-lie-groups_F02548	250	■ 0.999	$SO(p+1, q+1)$	$SO(p + 1, q + 1)$	that preserves the metric in R^{N+2} is $SO(p + 1, q + 1)$. We seek to construct a
gilmore-lie-groups_F02549	250	■ 1.000	$S^3 \subset R^4 (E < 0)$	$S^3 \subset R^4 (E < 0)$	lifted to the coordinates on a sphere $S^3 \subset R^4 (E < 0)$ or a two-sheeted hyperboloid
gilmore-lie-groups_F02550	250	■ 1.000	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$	$H^3 \subset R^4 (E > 0)$ by the following projective transformations:
gilmore-lie-groups_F02551	250	■ 1.000	$\mathbf{u}^t G \mathbf{u} = 1$	$\mathbf{u}^t G \mathbf{u} = 1$	condition $\mathbf{u}^t G \mathbf{u} = 1$.
gilmore-lie-groups_F02552	251	■ 0.999	$\text{diag}(1, \pm I_3, -1, +1)$	$\text{diag}(1, \pm I_3, -1, +1)$	lows. The group that leaves invariant the conformal metric $\text{diag}(1, \pm I_3, -1, +1)$ is
gilmore-lie-groups_F02553	251	■ 0.984	$SO(5, 1)$	$SO(5, 1)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02554	251	■ 0.984	$E < 0$	$E < 0$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02555	251	■ 0.984	$SO(2, 4)$	$SO(2, 4)$	$SO(5, 1)$ for $E < 0$ and $SO(2, 4)$ for $E > 0$. On the surfaces (sphere, hyperboloid)
gilmore-lie-groups_F02556	251	■ 1.000	$z_1 = z_2, y_4 = \lambda$	$z_1 = z_2, y_4 = \lambda$	the condition $\mathbf{u}^t G \mathbf{u} = 1$ is satisfied, so that $z_1 = z_2, y_4 = \lambda$ and $y_5 = 0$ (the six

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gilmore-lie-groups_F02557	251	■ 1.000	$y_5=0$	$y_5 = 0$	the condition $\mathbf{u}'G\mathbf{u} = 1$ is satisfied, so that $z_1 = z_2$, $y_4 = \lambda$ and $y_5 = 0$ (the six
gilmore-lie-groups_F02558	251	■ 0.934	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2)\right)$	$\left(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2)\right)$	coordinates are labeled $(y_0, \mathbf{y} = \lambda \mathbf{u}, y_4 = \frac{1}{2}(z_1 + z_2), y_5 = \frac{1}{2}(z_1 - z_2))$. Transform-
gilmore-lie-groups_F02559	251	■ 0.908	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array}\right]$	$\left[\begin{array}{cc} M & 0 \\ 0 & 1 \end{array}\right]$	$SO(5, 1)$ or $SO(2, 4)$ of the form $\left[\begin{smallmatrix} M & 0 \\ 0 & 1 \end{smallmatrix}\right]$, with M a 5×5 matrix that preserves the
gilmore-lie-groups_F02560	251	■ 0.997	$\operatorname{diag}(1, \pm I_3, -1)$	$\operatorname{diag}(1, \pm I_3, -1)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02561	251	■ 0.997	R^5	R^5	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02562	251	■ 0.997	$S O(1,4)$	$SO(1, 4)$	metric $\operatorname{diag}(1, \pm I_3, -1)$ in R^5 . This is $SO(4, 1)$ for $E < 0$ and $SO(1, 4)$ for $E > 0$.
gilmore-lie-groups_F02563	251	■ 0.761	$\mathbf{u}'$	$\mathbf{u}'$	The inner product of $\mathbf{u}'$ with itself satisfies
gilmore-lie-groups_F02564	251	■ 1.000	$A^tGA - C^tC = G$	$A^tGA - C^tC = G$	condition (e.g., $A^tGA - C^tC = G$, etc.) it is a simple matter to show that this
gilmore-lie-groups_F02565	251	■ 1.000	u	u	follows (Bander and Itzykson, 1966a, 1966b). For $E < 0$ introduce a four-vector u
gilmore-lie-groups_F02566	251	■ 1.000	$(u_0 \rightarrow u_4)$	$(u_0 \rightarrow u_4)$	as usual $(u_0 \rightarrow u_4)$
gilmore-lie-groups_F02567	252	■ 0.967	$1/\sqrt{2m E }$	$1/\sqrt{2m E }$	vector $\mathbf{L}$ and the scaled (by $1/\sqrt{2m E }$) Runge–Lenz vector $\mathbf{M}'$ as follows:
gilmore-lie-groups_F02568	252	■ 1.000	L_i, M_i'	L_i, M_i'	The operators L_i, M_i' , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_F02569	252	■ 1.000	B_μ	B_μ	The operators L_i, M_i' , and B_μ are the infinitesimal generators of $SO(4, 1)$ as follows,
gilmore-lie-groups_F02570	252	■ 0.993	$a_i, a_j^\dagger$	$a_i, a_j^\dagger$	That is, for the operators $\mathbf{A}$ we introduce annihilation and creation operators $a_i, a_j^\dagger$
gilmore-lie-groups_F02571	252	■ 1.000	$b_i, b_j^\dagger$	$b_i, b_j^\dagger$	for two independent modes, and similarly we introduce operators $b_i, b_j^\dagger$ to describe
gilmore-lie-groups_F02572	252	■ 1.000	$ m_1, m_2; n_1, n_2\rangle$	$ m_1, m_2; n_1, n_2\rangle$	B. Basis states on which these operators act have the form $ m_1, m_2; n_1, n_2\rangle$ where,
gilmore-lie-groups_F02573	252	■ 1.000	$j_a = \frac{1}{2}(m_1 + m_2)$	$j_a = \frac{1}{2}(m_1 + m_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02574	252	■ 1.000	$j_b = \frac{1}{2}(n_1 + n_2)$	$j_b = \frac{1}{2}(n_1 + n_2)$	directly to the condition $j_a = j_b$, where $j_a = \frac{1}{2}(m_1 + m_2)$ and $j_b = \frac{1}{2}(n_1 + n_2)$.
gilmore-lie-groups_F02575	253	■ 1.000	$N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$	$N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$	principal quantum number $N = 2j_a + 1 = 2j_b + 1 = (j_a + j_b) + 1$ must change
gilmore-lie-groups_F02576	253	■ 1.000	j_a	j_a	$j_a = j_b$. Operators that change j_a have the form $a_i^\dagger$ or $a_i^\dagger a_j^\dagger$, but they do not simul-
gilmore-lie-groups_F02577	253	■ 1.000	j_b	j_b	taneously change j_b . Only operators that simultaneously add or subtract one exci-
gilmore-lie-groups_F02578	253	■ 0.991	$(H_1, H_2, H_3, H_4) = (a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1, b_2^\dagger b_2)$	$(H_1, H_2, H_3, H_4) = (a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1, b_2^\dagger b_2)$	number operators for the four boson modes: $(H_1, H_2, H_3, H_4) = (a_1^\dagger a_1, a_2^\dagger a_2, b_1^\dagger b_1, b_2^\dagger b_2)$.
gilmore-lie-groups_F02579	253	■ 0.829	$b_2^\dagger b_2$	$b_2^\dagger b_2$	$b_2^\dagger b_2$). The remaining twelve operators have eigenoperator commutation relations
gilmore-lie-groups_F02580	253	■ 1.000	$\pm \frac{1}{2}$	$\pm \frac{1}{2}$	All these roots have equal length, and inner products among these roots are all $\pm \frac{1}{2}$
gilmore-lie-groups_F02581	253	■ 1.000	$\pi/4$	$\pi/4$	or make angles of $\pi/4$ or $3\pi/4$ radians with each other. The algebra is uniquely a

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gilmore-lie-groups_F02582	253	■ 1.000	$3 \pi / 4$	$3\pi/4$	or make angles of $\pi/4$ or $3\pi/4$ radians with each other. The algebra is uniquely a
gilmore-lie-groups_F02583	253	■ 1.000	$A_3=D_3$	$A_3 = D_3$	real form of $A_3 \equiv D_3$.
gilmore-lie-groups_F02584	254	■ 1.000	$a_{1}^{\dagger} a_{1}+a_{2}^{\dagger} a_{2}+b_{1}^{\dagger} b_{1}+b_{2}^{\dagger} b_{2}$	$a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02585	254	■ 1.000	$\mathfrak{so}(4)+\mathfrak{so}(2)$	$\mathfrak{so}(4) + \mathfrak{so}(2)$	onal operator $a_1^\dagger a_1 + a_2^\dagger a_2 + b_1^\dagger b_1 + b_2^\dagger b_2$. This is the algebra $\mathfrak{so}(4) + \mathfrak{so}(2)$. The
gilmore-lie-groups_F02586	254	■ 0.890	$\mathfrak{so}(4,2)=\mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) = \mathfrak{su}(2,2)$	$\mathfrak{so}(4,2) \equiv \mathfrak{su}(2,2)$. This is the conformal algebra.
gilmore-lie-groups_F02587	254	■ 1.000	(q_1, q_2, q_3, q_4)	(q_1, q_2, q_3, q_4)	Coordinates (q_1, q_2, q_3, q_4) in the fictitious space R^4 are related to coordinates
gilmore-lie-groups_F02588	254	■ 0.559	(Q_1, Q_2, Q_3)	(Q_1, Q_2, Q_3)	(Q_1, Q_2, Q_3) in the real space by the 4×4 transformation
gilmore-lie-groups_F02589	254	■ 1.000	Q_4	Q_4	The transformation is constructed so that the “fourth” real coordinate Q_4 is identi-
gilmore-lie-groups_F02590	254	■ 1.000	$q_1^2+q_2^2+q_3^2+q_4^2 \neq 0$	$q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$	cally zero. This transformation is invertible provided $q_1^2 + q_2^2 + q_3^2 + q_4^2 \neq 0$. The
gilmore-lie-groups_F02591	254	■ 1.000	$R=\sqrt{Q_1^2+Q_2^2+Q_3^2}$	$R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02592	254	■ 1.000	$q=\sqrt{q_1^2+q_2^2+q_3^2+q_4^2}$	$q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$	distance $R = \sqrt{Q_1^2 + Q_2^2 + Q_3^2}$ in R^3 and the distance $q = \sqrt{q_1^2 + q_2^2 + q_3^2 + q_4^2}$
gilmore-lie-groups_F02593	254	■ 1.000	$R=q^2$	$R = q^2$	in R^4 are related by $R = q^2$.
gilmore-lie-groups_F02594	254	■ 1.000	$P_4=0$	$P_4 = 0$	A constraint condition must be applied to force $P_4 = 0$. This condition is
gilmore-lie-groups_F02595	254	■ 1.000	$P^2=P_1^2+P_2^2+P_3^2=\left(1 / 4 R p^2\right)-\left(\zeta^2 / 4 R^2\right) \rightarrow$	$P^2 = P_1^2 + P_2^2 + P_3^2 = (1/4Rp^2) - (\zeta^2/4R^2) \rightarrow$	With this constraint we find $P^2 = P_1^2 + P_2^2 + P_3^2 = (1/4Rp^2) - (\zeta^2/4R^2) \rightarrow$
gilmore-lie-groups_F02596	254	■ 0.980	$(1 / 4 R)\left(p_1^2+p_2^2+p_3^2+p_4^2\right)$	$(1/4R) (p_1^2 + p_2^2 + p_3^2 + p_4^2)$	$(1/4R)(p_1^2 + p_2^2 + p_3^2 + p_4^2)$. With these transformations the hamiltonian in the
gilmore-lie-groups_F02597	255	■ 1.000	$(q_1, q_2, q_3, q_4 ; p_1, p_2, p_3, p_4)$	$(q_1, q_2, q_3, q_4; p_1, p_2, p_3, p_4)$	$(q_1, q_2, q_3, q_4; p_1, p_2, p_3, p_4)$ on the left and its transpose on the right (Sadovskii
gilmore-lie-groups_F02598	255	■ 1.000	$\zeta=0$	$\zeta = 0$	(iii) preserves $\zeta = 0$.
gilmore-lie-groups_F02599	255	■ 1.000	$M^t G_1 M=G_1$	$M^t G_1 M = G_1$	symplectic metric-preserving condition: $M^t G_1 M = G_1$. Preserving the condition
gilmore-lie-groups_F02600	255	■ 1.000	$M^t G_2 M=G_2$	$M^t G_2 M = G_2$	$M^t G_2 M = G_2$.
gilmore-lie-groups_F02601	256	■ 0.988	$Sp(8 ; \mathbb{R})$	$Sp(8; \mathbb{R})$	$(+4, -4)$. The group that preserves the antisymmetric metric is $Sp(8; \mathbb{R})$ and the
gilmore-lie-groups_F02602	256	■ 1.000	$SO(4,4)$	$SO(4,4)$	group that preserves the symmetric metric is $SO(4,4)$. The group that satisfies both
gilmore-lie-groups_F02603	256	■ 0.998	(q, p)	(q, p)	coordinates (q, p) to coordinates (s, r) :
gilmore-lie-groups_F02604	256	■ 0.998	(s, r)	(s, r)	coordinates (q, p) to coordinates (s, r) :
gilmore-lie-groups_F02605	256	■ 1.000	$U(2,2)$	$U(2,2)$	The noncompact group $U(2,2)$ preserves the constraint Eq. (14.49).
gilmore-lie-groups_F02606	256	■ 0.825	$V(r)=-e^2 / r$	$(V(r) = -e^2/r)$	charged particle interacting with an external Coulomb field $V(r) = -e^2/r$ can be
gilmore-lie-groups_F02607	257	■ 1.000	$\mathfrak{so}(4,1)$	$\mathfrak{so}(4,1)$	close under commutation to span a Lie algebra that is isomorphic with $\mathfrak{so}(4,1)$.

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gilmore-lie-groups_F02608	257	■ 1.000	$\mathfrak{so}(4,2)$	$\mathfrak{so}(4,2)$	Five additional operators can be introduced that extend the algebra to $\mathfrak{so}(4,2)$.
gilmore-lie-groups_F02609	257	■ 1.000	$J_i, M_i, A_i, \Gamma_i (i=1,2,3)$	$J_i, M_i, A_i, \Gamma_i (i=1,2,3)$	The four triplets $J_i, M_i, A_i, \Gamma_i (i=1,2,3)$ have transformation properties of three-
gilmore-lie-groups_F02610	257	■ 0.996	A_4, Γ_4, Γ_5	A_4, Γ_4, Γ_5	vectors under rotations. The three additional operators A_4, Γ_4, Γ_5 close under com-
gilmore-lie-groups_F02611	257	■ 1.000	$\Phi(r)=e/r$	$\Phi(r)=e/r$	in the Coulomb field $\Phi(r)=e/r$ of a proton can be expressed in terms of operators
gilmore-lie-groups_F02612	257	■ 1.000	A_4, Γ_4	A_4, Γ_4	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups_F02613	257	■ 1.000	Γ_5	Γ_5	of type A_4, Γ_4 , and Γ_5 . These operators are displayed in Table 14.3, along with the
gilmore-lie-groups_F02614	257	■ 0.860	$(\mathbf{E}, \mathbf{B})$	$(\mathbf{E}, \mathbf{B})$	scribed by the Dirac equation. The electromagnetic field $(\mathbf{E}, \mathbf{B})$ is described by the
gilmore-lie-groups_F02615	257	■ 1.000	$A_\mu=(\phi, \mathbf{A})$	$A_\mu=(\phi, \mathbf{A})$	four-vector potential $A_\mu=(\phi, \mathbf{A})$. The electron has charge $q=-e$ (where e is
gilmore-lie-groups_F02616	257	■ 1.000	$q=-e$	$q=-e$	four-vector potential $A_\mu=(\phi, \mathbf{A})$. The electron has charge $q=-e$ (where e is
gilmore-lie-groups_F02617	257	■ 1.000	$H_D\psi=E\psi$	$H_D\psi=E\psi$	the charge on the proton) and spin $\frac{1}{2}$. The Dirac equation $H_D\psi=E\psi$ is a matrix
gilmore-lie-groups_F02618	258	■ 0.721	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\frac{p^2}{2m}-\frac{\alpha}{r}$	H $\frac{p^2}{2m}-\frac{\alpha}{r}$ A_4 $\mathbf{r}\cdot\mathbf{p}-i$ Γ_4 $\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$ Γ_5 $\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$ Θ $r(H_S-W)$ A $A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$ B $1/2m$ C $-W$ $-\alpha$ $\sqrt{p^2+m^2}-\frac{\alpha}{r}$ $\mathbf{r}\cdot\mathbf{p}-i$ $\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$ $\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$ $r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$ $A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$ 1 m^2-E^2 $-2\alpha E$

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gilmore-lie-groups_ F02620	258	■ 0.721	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	
gilmore-lie-groups_ F02621	258	■ 0.721	Γ_4	Γ_4	
gilmore-lie-groups_ F02622	258	■ 0.721	$\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}(r \mathbf{p} \cdot \mathbf{p} - r)$	

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gilmore-lie-groups_ F02623	258	■ 0.721	$\frac{1}{2}\left(r\,\mathbf{p}\cdot\mathbf{p}\cdot\mathbf{p}-r\cdot\frac{\alpha^2}{r}\right)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$	
gilmore-lie-groups_ F02624	258	■ 0.721	$\frac{1}{2}(r\,\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	
gilmore-lie-groups_ F02625	258	■ 0.721	$\frac{1}{2}\left(r\,\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	

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gilmore-lie-groups_ F02626	258	■ 0.721	\Theta	Θ	
gilmore-lie-groups_ F02627	258	■ 0.721	$r\left(H_S - W\right)$	$r\left(H_S - W\right)$	
gilmore-lie-groups_ F02628	258	■ 0.721	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$	

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gilmore-lie-groups_ F02629	258	■ 0.721	$A\left(\Gamma_5+\Gamma_4\right)+B\left(\Gamma_5-\Gamma_4\right)+C$	$A\left(\Gamma_5+\Gamma_4\right)+B\left(\Gamma_5-\Gamma_4\right)+C$	
gilmore-lie-groups_ F02630	258	■ 0.721	$1 / 2 m$	$1/2m$	
gilmore-lie-groups_ F02631	258	■ 0.721	$-W$	$-W$	

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gilmore-lie-groups_ F02632	258	0.721	$m^{\wedge\{2\}}-E^{\wedge\{2\}}$	m^2-E^2	<table><tr><td>H</td><td>$\frac{p^2}{2m}-\frac{\alpha}{r}$</td><td>$\sqrt{p^2+m^2}-\frac{\alpha}{r}$</td></tr><tr><td>$A_4$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S-W)$</td><td>$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td></tr><tr><td>A</td><td>$1/2m$</td><td>1</td></tr><tr><td>B</td><td>$-W$</td><td>m^2-E^2</td></tr><tr><td>C</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$	A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$	Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	A	$1/2m$	1	B	$-W$	m^2-E^2	C	$-\alpha$	$-2\alpha E$
H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$																														
A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$																														
Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$																														
Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$																														
Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$																														
	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$																														
A	$1/2m$	1																														
B	$-W$	m^2-E^2																														
C	$-\alpha$	$-2\alpha E$																														
gilmore-lie-groups_ F02633	258	0.721	$-2\ \alpha\ E$	$-2\alpha E$	<table><tr><td>H</td><td>$\frac{p^2}{2m}-\frac{\alpha}{r}$</td><td>$\sqrt{p^2+m^2}-\frac{\alpha}{r}$</td></tr><tr><td>$A_4$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td><td>$\mathbf{r}\cdot\mathbf{p}-i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$</td><td>$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S-W)$</td><td>$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td><td>$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$</td></tr><tr><td>A</td><td>$1/2m$</td><td>1</td></tr><tr><td>B</td><td>$-W$</td><td>m^2-E^2</td></tr><tr><td>C</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$	A_4	$\mathbf{r}\cdot\mathbf{p}-i$	$\mathbf{r}\cdot\mathbf{p}-i$	Γ_4	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}-r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}-r-\frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$	Θ	$r(H_S-W)$	$r\left\{\left(H_{KG}+\frac{\alpha}{r}\right)^2-\left(E+\frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	A	$1/2m$	1	B	$-W$	m^2-E^2	C	$-\alpha$	$-2\alpha E$
H	$\frac{p^2}{2m}-\frac{\alpha}{r}$	$\sqrt{p^2+m^2}-\frac{\alpha}{r}$																														
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Γ_5	$\frac{1}{2}(r\mathbf{p}\cdot\mathbf{p}+r)$	$\frac{1}{2}\left(r\mathbf{p}\cdot\mathbf{p}+r-\frac{\alpha^2}{r}\right)$																														
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	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$	$A(\Gamma_5+\Gamma_4)+B(\Gamma_5-\Gamma_4)+C$																														
A	$1/2m$	1																														
B	$-W$	m^2-E^2																														
C	$-\alpha$	$-2\alpha E$																														
gilmore-lie-groups_ F02634	258	1.000	$\pi=\mathbf{p}-\frac{q}{c}\ \mathbf{A}$	$\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	replaced by $\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$. Under this condition the operators still close under																											
gilmore-lie-groups_ F02635	258	0.971	γ_i	γ_i	The 4×4 matrices β and γ_i can be chosen as																											
gilmore-lie-groups_ F02636	258	0.982	σ_i	σ_i	Here σ_i are the standard Pauli 2×2 spin matrices (cf., Eq. (3.39), Problem 3.1).																											
gilmore-lie-groups_ F02637	259	1.000	$\mathbf{p}\rightarrow\pi=\mathbf{p}-\frac{q}{c}\ \mathbf{A}$	$\mathbf{p}\rightarrow\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$	As before, the substitution $\mathbf{p}\rightarrow\pi=\mathbf{p}-\frac{q}{c}\mathbf{A}$ is in order in the event there is a																											
gilmore-lie-groups_ F02638	259	1.000	$J_i,\ M_i,\ A_i,\ \Gamma_i$	J_i,M_i,A_i,Γ_i	three-vectors $J_i,\ M_i,\ A_i,\ \Gamma_i$ defined in Table 14.3. The Dirac hamiltonian is ex-																											
gilmore-lie-groups_ F02639	259	0.990	$\Gamma_4,\ \Gamma_5$	Γ_4,Γ_5	Gordon operator (see Table 14.3). In short, the operators $\Gamma_4,\ \Gamma_5$ are modified but																											
gilmore-lie-groups_ F02640	259	1.000	$D=d\ / \ d\ r$	$D=d/dr$	with $D=d/dr$. The operators r and D behave under commutation like the boson																											
gilmore-lie-groups_ F02641	259	1.000	$r\ D^{\wedge\{2\}}+r$	rD^2+r	The linear combinations rD^2+r and rD^2-r are compact and noncompact, re-																											
gilmore-lie-groups_ F02642	259	1.000	$r\ D^{\wedge\{2\}}-r$	rD^2-r	The linear combinations rD^2+r and rD^2-r are compact and noncompact, re-																											
gilmore-lie-groups_ F02643	260	1.000	$C^{\wedge\{2\}}=\Gamma_5^{\wedge\{2\}}-\Gamma_4^{\wedge\{2\}}-M_4^{\wedge\{2\}}=-a$	$C^2=\Gamma_5^2-\Gamma_4^2-M_4^2=-a$	The Casimir operator for this algebra is $C^2=\Gamma_5^2-\Gamma_4^2-M_4^2=-a$. The represen-																											
gilmore-lie-groups_ F02644	260	0.978	$a\rightarrow A$	$a\rightarrow A$	follows ($a\rightarrow A$)																											
gilmore-lie-groups_ F02645	260	1.000	$C<0$	$C<0$	If $C<0$ we can choose $e^{-\theta}+C\ e^{\theta}=0$, so that the resulting equation becomes																											
gilmore-lie-groups_ F02646	260	1.000	$e^{\{-\backslash\theta\}}+C\ e^{\{\backslash\theta\}}=0$	$e^{-\theta}+Ce^{\theta}=0$	If $C<0$ we can choose $e^{-\theta}+C\ e^{\theta}=0$, so that the resulting equation becomes																											

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gilmore-lie-groups_ F02647	260	■ 1.000	$u(r)=e^{\{\theta M_4\}} R(r)$	$u(r) = e^{\theta M_4} R(r)$	where $u(r) = e^{\theta M_4} R(r)$. If A is the Casimir invariant of this representation
gilmore-lie-groups_ F02648	260	■ 0.798	$N=-\frac{1}{2}+$	$N = -\frac{1}{2}+$	of $\mathfrak{su}(1,1)$, the discrete spectrum of the compact operator Γ_5 is $N = -\frac{1}{2} +$
gilmore-lie-groups_ F02649	260	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}+1+n, n=0,1,2, \ldots$	$\sqrt{\left(\frac{1}{2}\right)^2 - A + 1 + n, n = 0, 1, 2, \dots}$	$\sqrt{(\frac{1}{2})^2 - A + 1 + n, n = 0, 1, 2, \dots}$. This result leads directly to the eigenvalue
gilmore-lie-groups_ F02650	261	■ 1.000	$W=-\frac{1}{2} m c^2 \alpha^2 \left(1 / N^2\right)$	$W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$.
gilmore-lie-groups_ F02651	261	■ 1.000	$N=l+1+k, k=0,1,2, \ldots$	$N = l + 1 + k, k = 0, 1, 2, \dots$	relativistic case) $W = -\frac{1}{2} m c^2 \alpha^2 (1/N^2)$, where $N = l + 1 + k, k = 0, 1, 2, \dots$.
gilmore-lie-groups_ F02652	261	■ 1.000	$m_{\{l\}}$	m_l	tum operators, the quantum number m_l (eigenvalue of L_z) is also invariant under
gilmore-lie-groups_ F02653	261	■ 0.449	$ N, \ln \rangle \rightarrow N \pm 1, l m\rangle$	$ N, \ln\rangle \leftrightarrow N \pm 1, lm\rangle$	algebra are $ N, lm\rangle \leftrightarrow N \pm 1, lm\rangle$. The states on which these operators act are
gilmore-lie-groups_ F02654	261	■ 1.000	$L_{\{z\}}, L_{\{ \pm \}}$	$L_z, L_{\pm}$	Remark The angular momentum operators $L_z, L_{\pm}$ act on multiplets shown as
gilmore-lie-groups_ F02655	261	■ 1.000	$M_{\{z\}}, M_{\{ \pm \}}$	$M_z, M_{\pm}$	a single horizontal line in Figs. 14.1 and 14.2. The operators $M_z, M_{\pm}$ associated
gilmore-lie-groups_ F02656	261	■ 0.997	$\Gamma_{\{z\}}, \Gamma_{\{ \pm \}}=\Gamma_4 \pm i M_4$	$\Gamma_z, \Gamma_{\pm} = \Gamma_4 \pm i M_4$	two figures. The operators $\Gamma_{\pm} = \Gamma_4 \pm i M_4$ act vertically on the levels shown
gilmore-lie-groups_ F02657	261	■ 1.000	$[\mathbf{L}, \Gamma]=0$	$[\mathbf{L}, \Gamma] = 0$	in these figures. Since $[\mathbf{L}, \Gamma] = 0$, the operators Γ do not change the m values of
gilmore-lie-groups_ F02658	261	■ 1.000	$\Gamma_{\{-\}}$	Γ_-	Remark The shift down operator Γ_- annihilates the ground state in a given an-
gilmore-lie-groups_ F02659	261	■ 0.535	$\Gamma_{\{-\}}\left\langle r \mid N^{l=N-1}\right\rangle=0$	$\Gamma_- \langle r N^{l=N-1} \rangle = 0$	gular momentum tower: $\Gamma_- \langle r N^{l=N-1} \rangle = 0$. Since the differential operators are
gilmore-lie-groups_ F02660	261	■ 1.000	$L_{\{-\}} Y_{m=-l}^l(\theta, \phi)=0$	$L_- Y_{m=-l}^l(\theta, \phi) = 0$	known, this relation can be used, as was the relation $L_- Y_{m=-l}^l(\theta, \phi) = 0$, to deter-
gilmore-lie-groups_ F02661	261	■ 1.000	$\langle r \mid N, l=N-1\rangle$	$\langle r N, l = N - 1 \rangle$	mine the radial wavefunction $\langle r N, l = N - 1 \rangle$.
gilmore-lie-groups_ F02662	261	■ 1.000	$C>0$	$C > 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_ F02663	261	■ 1.000	$e^{\{-\theta\}}-C e^{\{\theta\}}=0$	$e^{-\theta} - C e^{\theta} = 0$	If $C > 0$ we can choose θ so that $e^{-\theta} - C e^{\theta} = 0$. Equation (14.66) reduces to
gilmore-lie-groups_ F02664	261	■ 1.000	$k \simeq e^{-\theta}$	$k \simeq e^{-\theta}$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2 m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_ F02665	261	■ 1.000	$E=\hbar^2 k^2 / 2 m$	$E = \hbar^2 k^2 / 2 m$	$k \simeq e^{-\theta}$ with $E = \hbar^2 k^2 / 2 m$. The asymptotic form of the wave function is (Gilmore
gilmore-lie-groups_ F02666	262	■ 1.000	$\delta(j)=\arg [\Gamma(j+1-i(\alpha / k))]$	$\delta(j) = \arg [\Gamma(j + 1 - i(\alpha/k))]$	where $\delta(j) = \arg [\Gamma(j + 1 - i(\alpha/k))]$ is part of the scattering phase shift, and the
gilmore-lie-groups_ F02667	262	■ 1.000	$j=-\frac{1}{2}+\sqrt{\left(\frac{1}{2}\right)^2-A}$	$j = -\frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 - A}$	expression for j is given by $j = -\frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}$.
gilmore-lie-groups_ F02668	262	■ 1.000	$-1 / r$	$-1/r$	attracted than a simple $-1/r$ potential suggests. It is possible to represent this
gilmore-lie-groups_ F02669	262	■ 1.000	$-1 / r^2$	$-1/r^2$	extra attraction by adding a term of the form $-1/r^2$ to the potential to represent
gilmore-lie-groups_ F02670	262	■ 1.000	$V(r)=-e^2 / r \rightarrow-e^2 / r-\mu_l\left(\hbar^2 / 2 m\right) / r^2$	$V(r) = -e^2 / r \rightarrow -e^2 / r - \mu_l (\hbar^2 / 2 m) / r^2$	and Klein–Gordon equations is $V(r) = -e^2 / r \rightarrow -e^2 / r - \mu_l (\hbar^2 / 2 m) / r^2$. This
gilmore-lie-groups_ F02671	262	■ 1.000	$j \rightarrow j^{\prime}=j+\Delta j$	$j \rightarrow j' = j + \Delta j$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$
gilmore-lie-groups_ F02672	262	■ 1.000	$\Delta j=$	$\Delta j =$	This change produces a change in the value of $j \rightarrow j' = j + \Delta j$, where $\Delta j =$

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gilmore-lie-groups_ F02673	262	■ 1.000	$-\mu_l/(2l+1)$	$-\mu_l/(2l+1)$	$-\mu_l/(2l+1)$ in the nonrelativistic case. This change produces a change in the
gilmore-lie-groups_ F02674	262	■ 1.000	Δj	Δj	The quantum defect Δj causes the Rydberg states to be bound more strongly than in
gilmore-lie-groups_ F02675	263	■ 1.000	$G \supset S O(3)$	$G \supset SO(3)$	group, we are prodded to search for a larger group $G \supset SO(3)$ that explains the
gilmore-lie-groups_ F02676	264	■ 0.987	$3^{\text{\text{deg}}}$	3^{deg}	of relativity. Show that the existence of the 3^{deg} microwave background radiation is
gilmore-lie-groups_ F02677	264	■ 1.000	$\mathbf{F}=m \mathbf{g}$	$\mathbf{F} = m\mathbf{g}$	tional force $\mathbf{F} = m\mathbf{g}$ in the “down” direction. Assume that the other elevator is in
gilmore-lie-groups_ F02678	264	■ 1.000	$\mathbf{A}=\frac{1}{2} \mathbf{B} \times \mathbf{r}$	$\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$	taken as $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$, so that $\mathbf{B} = \nabla \times \mathbf{A}$. Derive the Klein–Gordon equation for an
gilmore-lie-groups_ F02679	264	■ 0.996	$(r, \theta, \phi)=\left(\theta_3, \theta_2, \theta_1\right)$	$(r, \theta, \phi) = (\theta_3, \theta_2, \theta_1)$	4. Introduce spherical coordinates as follows: $(r, \theta, \phi) = (\theta_3, \theta_2, \theta_1)$ and
gilmore-lie-groups_ F02680	264	■ 1.000	$\mathcal{L}^2\left(S^1\right)=\partial^2 / \partial \theta_1^2$	$\mathcal{L}^2\left(S^1\right)=\partial^2 / \partial \theta_1^2$	Show that $\mathcal{L}^2(S^1) = \partial^2 / \partial \theta_1^2$. Show that
gilmore-lie-groups_ F02681	264	■ 1.000	$\mathcal{L}^2\left(S^3\right)$	$\mathcal{L}^2\left(S^3\right)$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial / \partial \cos \theta_3)^2$. Do this
gilmore-lie-groups_ F02682	264	■ 1.000	$\left(\partial / \partial \cos \theta_3\right)^2$	$(\partial / \partial \cos \theta_3)^2$	Generalize this result to $\mathcal{L}^2(S^3)$ recursively using $\mathcal{L}^2(S^2)$ and $(\partial / \partial \cos \theta_3)^2$. Do this
gilmore-lie-groups_ F02683	264	■ 1.000	$\mathcal{L}^2\left(S^n\right)$	$\mathcal{L}^2\left(S^n\right)$	more generally for $\mathcal{L}^2(S^n)$.
gilmore-lie-groups_ F02684	265	■ 1.000	$r \rightarrow \infty$	$r \rightarrow \infty$	a. Show that the singular points of Eq. (14.9) occur at $r = 0$ and $r \rightarrow \infty$.
gilmore-lie-groups_ F02685	265	■ 1.000	$\gamma=\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2-A}$	$\gamma = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 - A}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm \sqrt{-C}$.
gilmore-lie-groups_ F02686	265	■ 1.000	$\lambda=\pm \sqrt{-C}$	$\lambda = \pm \sqrt{-C}$	Show that $\gamma = \frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - A}$ and $\lambda = \pm \sqrt{-C}$.
gilmore-lie-groups_ F02687	265	■ 1.000	$\sqrt{\left(\frac{1}{2}\right)^2-A}$	$\sqrt{\left(\frac{1}{2}\right)^2 - A}$	c. Show that if $\sqrt{(\frac{1}{2})^2 - A}$ is real, the solution with the positive sign is always square
gilmore-lie-groups_ F02688	265	■ 1.000	$\pm \sqrt{-C}$	$\pm \sqrt{-C}$	with the negative sign square integrable? Show that if $C < 0$ the solution $\pm \sqrt{-C}$
gilmore-lie-groups_ F02689	265	■ 1.000	$R(r)=r^{\gamma} e^{\lambda r} f(r)$	$R(r) = r^\gamma e^{\lambda r} f(r)$	d. Show that a solution of the form $R(r) = r^\gamma e^{\lambda r} f(r)$ can be found where the function
gilmore-lie-groups_ F02690	265	■ 1.000	$f(r)=\sum_{j=0}^{\infty} f_j r^j$	$f(r) = \sum_{j=0}^{\infty} f_j r^j$	f. Represent the function $f(r)$ as an ascending power series: $f(r) = \sum_{j=0}^{\infty} f_j r^j$.
gilmore-lie-groups_ F02691	265	■ 1.000	$j \rightarrow \infty$	$j \rightarrow \infty$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_ F02692	265	■ 1.000	$f(r) \rightarrow e^{-2 \lambda r}$	$f(r) \rightarrow e^{-2\lambda r}$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_ F02693	265	■ 1.000	$\lambda<0$	$\lambda < 0$	determined from the behavior of f_j as $j \rightarrow \infty$, is $f(r) \rightarrow e^{-2\lambda r}$. Since $\lambda < 0$ this
gilmore-lie-groups_ F02694	265	■ 0.963	r^n	r^n	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_ F02695	265	■ 0.963	$f_n \neq 0$	$f_n \neq 0$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} = 0 (\Rightarrow f_{n+2} =$
gilmore-lie-groups_ F02696	265	■ 0.963	$f_{n+1}=0 \Rightarrow f_{n+2}=\dots$	$f_{n+1} = 0 (\Rightarrow f_{n+2} =$	highest nonzero degree term present is r^n , so that $f_n \neq 0$ but $f_{n+1} \equiv 0 (\Rightarrow f_{n+2} \equiv$
gilmore-lie-groups_ F02697	265	■ 1.000	$f_{n+3}=\dots=0$	$f_{n+3} = \dots = 0$	$f_{n+3} = \dots = 0$), show that the quantization conditon $2\lambda(n+\gamma)+B=0$ must be
gilmore-lie-groups_ F02698	265	■ 1.000	$2 \lambda(n+\gamma)+B=0$	$2\lambda(n+\gamma)+B=0$	$f_{n+3} = \dots = 0$), show that the quantization conditon $2\lambda(n+\gamma)+B \equiv 0$ must be

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gilmore-lie-groups_F02699	266	1.000	$\frac{1}{r} r^{\gamma} f(r) e^{\lambda r}$	$\frac{1}{r} r^{\gamma} f(r) e^{\lambda r}$	The radial part of the wavefunction $\frac{1}{r} r^{\gamma} f(r) e^{\lambda r}$ has exactly n nodes in the open																						
gilmore-lie-groups_F02700	266	0.907	$(0, \infty)$	$(0, \infty)$	interval $[0, \infty]$.																						
gilmore-lie-groups_F02701	266	1.000	$Z e / r$	$Z e / r$	show that the potential is $Z e / r$ and the solutions of the relativistic and nonrelativistic																						
gilmore-lie-groups_F02702	266	1.000	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$	$N^{\prime}=n+\frac{1}{2}+\sqrt{\left(l+\frac{1}{2}\right)^2-\alpha^2}$ and $N=n+l+1$																						
gilmore-lie-groups_F02703	266	1.000	$e^{\lambda r}$	$e^{\lambda r}$	8. The radial part of the wavefunction dies off like $e^{\lambda r}$ for large r , where $\lambda < 0$ for bound																						
gilmore-lie-groups_F02704	266	1.000	λ^{-1}	λ^{-1}	states. The parameter λ^{-1} has the dimensions of length, and $a \simeq 1/ \lambda $ characterizes																						
gilmore-lie-groups_F02705	266	1.000	$a \simeq 1 / \lambda $	$a \simeq 1/ \lambda $	states. The parameter λ^{-1} has the dimensions of length, and $a \simeq 1/ \lambda $ characterizes																						
gilmore-lie-groups_F02706	266	0.910	(n, l)	(n, l)	the size of a bound state orbit. Show that bound states with quantum numbers (n, l)																						
gilmore-lie-groups_F02707	267	0.953	$e^{\pm} \text{ \texttt{\textbackslash pm}}$	$e^{\pm}$	<table><thead><tr><th>Particle</th><th>Rest energy (MeV)</th></tr></thead><tbody><tr><td>electron $e^{\pm}$</td><td>0.511</td></tr><tr><td>mu meson $\mu^{\pm}$</td><td>105.7</td></tr><tr><td>tau meson $\tau^{\pm}$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^{\pm}$</td><td>938.26</td></tr><tr><td>deuteron d^{+}</td><td>1875.6</td></tr><tr><td>tritium t^{+}</td><td>2809.4</td></tr><tr><td>pi meson $\pi^{\pm}$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^{\pm}$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^{-}</td><td>1533.0</td></tr><tr><td>omega Ω^{-}</td><td>1672.0</td></tr></tbody></table>	Particle	Rest energy (MeV)	electron $e^{\pm}$	0.511	mu meson $\mu^{\pm}$	105.7	tau meson $\tau^{\pm}$	1784.0	proton, antiproton $p^{\pm}$	938.26	deuteron d^{+}	1875.6	tritium t^{+}	2809.4	pi meson $\pi^{\pm}$	139.6	sigma meson $\Sigma^{\pm}$	1385.0	cascade meson Ξ^{-}	1533.0	omega Ω^{-}	1672.0
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gilmore-lie-groups_ F02710	267	■ 0.953	$p^{\pm}$	$p^{\pm}$	<table><thead><tr><th>Particle</th><th>Rest energy (MeV)</th></tr></thead><tbody><tr><td>electron $e^{\pm}$</td><td>0.511</td></tr><tr><td>mu meson $\mu^{\pm}$</td><td>105.7</td></tr><tr><td>tau meson $\tau^{\pm}$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^{\pm}$</td><td>938.26</td></tr><tr><td>deuteron d^+</td><td>1875.6</td></tr><tr><td>tritium t^+</td><td>2809.4</td></tr><tr><td>pi meson $\pi^{\pm}$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^{\pm}$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^-</td><td>1533.0</td></tr><tr><td>omega Ω^-</td><td>1672.0</td></tr></tbody></table>	Particle	Rest energy (MeV)	electron $e^{\pm}$	0.511	mu meson $\mu^{\pm}$	105.7	tau meson $\tau^{\pm}$	1784.0	proton, antiproton $p^{\pm}$	938.26	deuteron d^+	1875.6	tritium t^+	2809.4	pi meson $\pi^{\pm}$	139.6	sigma meson $\Sigma^{\pm}$	1385.0	cascade meson Ξ^-	1533.0	omega Ω^-	1672.0
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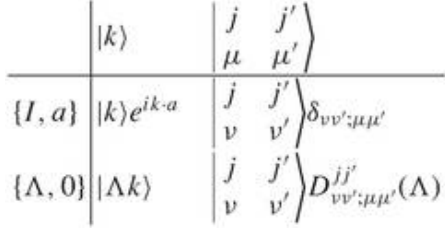
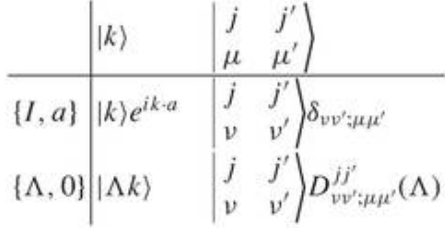
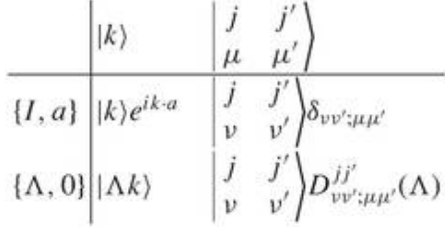
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gilmore-lie-groups_ F02717	267	■ 1.000	$a_{\rm B}=\hbar^2 / m \, e^2=0.529 \times 10^{-8} \, \mathrm{cm}$	$a_B = \hbar^2 / m e^2 = 0.529 \times 10^{-8} \, \mathrm{cm}$	In these expressions $a_B = \hbar^2 / m e^2 = 0.529 \times 10^{-8} \, \mathrm{cm}$ is the Bohr radius: the char-																						
gilmore-lie-groups_ F02718	267	■ 0.906	$n, \, l$	n, l	sizes of the atoms in the (n, l) states differ between the relativistic and nonrelativistic																						
gilmore-lie-groups_ F02719	267	■ 1.000	$W=-\frac{1}{2} \, m \, c^2 \, \alpha^2 / N^2$	$W = -\frac{1}{2} m c^2 \alpha^2 / N^2$	binding energy $W = -\frac{1}{2} m c^2 \alpha^2 / N^2$ is the reduced mass: $1/m = 1/m_1 + 1/m_2$ of																						
gilmore-lie-groups_ F02720	267	■ 1.000	$1 \, / \, m=1 \, / \, m_{\rm 1}+1 \, / \, m_{\rm 2}$	$1/m = 1/m_1 + 1/m_2$	binding energy $W = -\frac{1}{2} m c^2 \alpha^2 / N^2$ is the reduced mass: $1/m = 1/m_1 + 1/m_2$ of																						
gilmore-lie-groups_ F02721	267	■ 1.000	$(E=0)$	$(E = 0)$	$(E > 0)$; and parabolic at the separatrix $(E = 0)$. If the radial force field includes a																						
gilmore-lie-groups_ F02722	267	■ 1.000	$1 \, / \, r^3$	$1/r^3$	radial $1/r^3$ perturbation																						
gilmore-lie-groups_ F02723	267	■ 1.000	$\alpha=\sqrt{1-\eta}$, $\eta=C \, / \, K \, a$	$\alpha = \sqrt{1-\eta}, \eta = C / K a$	where $\alpha = \sqrt{1-\eta}, \eta = C / K a$. This can be treated as an ellipse that is slowly																						
gilmore-lie-groups_ F02724	267	■ 1.000	$\alpha \, \simeq \, 1$	$\alpha \simeq 1$	rotating, $\alpha \simeq 1$. In this case the parameters a and ϵ have their usual meanings																						
gilmore-lie-groups_ F02725	267	■ 1.000	η	η	η is a measure of the strength of the perturbation to the strength of the Coulomb																						

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gilmore-lie-groups_ F02726	268	■ 0.651	$E=\sqrt{\left(m\ c^2\right)^2+(\mathbf{p}\ c)^2}-K / r$	$E=\sqrt{(mc^2)^2+(\mathbf{p}c)^2}-K/r$	a. Expand the relativistic energy $E=\sqrt{(mc^2)^2+(\mathbf{p}c)^2}-K/r$ to fourth order in $\mathbf{p}$
gilmore-lie-groups_ F02727	268	■ 1.000	$E=\left(m\ c^2\right)+\left(p^2 / 2\ m\right)-\left(p^2 / 2\ m\right)^2 /\left(2\ m\ c^2\right)-K / r=m\ c^2+W$	$E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$	and show $E=\left(m c^2\right)+\left(p^2 / 2 m\right)-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)-K / r=m c^2+W$.
gilmore-lie-groups_ F02728	268	■ 1.000	$-\left(p^2 / 2\ m\right)^2 /\left(2\ m\ c^2\right)$	$-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-
gilmore-lie-groups_ F02729	268	■ 1.000	$-\left(W+K / r\right)^2 /\left(2\ m\ c^2\right)$	$-\left(W+K / r\right)^2 /\left(2 m c^2\right)$	b. Replace the quartic term $-\left(p^2 / 2 m\right)^2 /\left(2 m c^2\right)$ by $-\left(W+K / r\right)^2 /\left(2 m c^2\right)$ and ex-
gilmore-lie-groups_ F02730	268	■ 1.000	$K^{\prime}$	K'	Evaluate K' and C' and show $K'=K\left(1+W / m c^2\right)$ and $C'=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_ F02731	268	■ 1.000	$C^{\prime}$	C'	Evaluate K' and C' and show $K'=K\left(1+W / m c^2\right)$ and $C'=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_ F02732	268	■ 1.000	$K^{\prime}=K\left(1+W / m\ c^2\right)$	$K'=K\left(1+W / m c^2\right)$	Evaluate K' and C' and show $K'=K\left(1+W / m c^2\right)$ and $C'=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_ F02733	268	■ 1.000	$C^{\prime}=-K^2 /\left(2\ m\ c^2\right)$	$C'=-K^2 /\left(2 m c^2\right)$	Evaluate K' and C' and show $K'=K\left(1+W / m c^2\right)$ and $C'=-K^2 /\left(2 m c^2\right)$.
gilmore-lie-groups_ F02734	268	■ 1.000	$K \rightarrow K^{\prime}$	$K \rightarrow K'$	c. Argue that the classical motion involves a renormalized coupling $K \rightarrow K'$ as well
gilmore-lie-groups_ F02735	268	■ 1.000	$C=2\ C^{\prime}$	$C=2 C'$	as a $1 / r^3$ component to the force, with $C=2 C'$.
gilmore-lie-groups_ F02736	268	■ 1.000	$\delta \theta \simeq \eta / 2$	$\delta \theta \simeq \eta / 2$	d. Show that the advance in the perihelion of the orbit is $\delta \theta \simeq \eta / 2$ per period.
gilmore-lie-groups_ F02737	268	■ 1.000	$\epsilon=0.206$	$\epsilon=0.206$	e. Evaluate η for the planet Mercury, for which $\epsilon=0.206$ and the period is $T=0.24$
gilmore-lie-groups_ F02738	268	■ 1.000	$T=0.24$	$T=0.24$	e. Evaluate η for the planet Mercury, for which $\epsilon=0.206$ and the period is $T=0.24$
gilmore-lie-groups_ F02739	268	■ 1.000	$m=$	$m=$	correct and the mass of the particle varies with its state of motion according to $m=$
gilmore-lie-groups_ F02740	268	■ 1.000	$m_0 /\sqrt{1-(v / c)^2}$	$m_0 /\sqrt{1-(v / c)^2}$	$m_0 /\sqrt{1-(v / c)^2}$. (2) The mass of a particle is a constant of nature and Newton's
gilmore-lie-groups_ F02741	268	■ 1.000	$\mathbf{F}(r)=\left(-K / r^2+p(r)\right) \hat{\mathbf{r}}$	$\mathbf{F}(r)=\left(-K / r^2+p(r)\right) \hat{\mathbf{r}}$	force has the form $\mathbf{F}(r)=\left(-K / r^2+p(r)\right) \hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_ F02742	268	■ 1.000	$p(r)$	$p(r)$	force has the form $\mathbf{F}(r)=\left(-K / r^2+p(r)\right) \hat{\mathbf{r}}$, where $p(r)$ is a small perturbation. The
gilmore-lie-groups_ F02743	268	■ 1.000	$1 / r=\left(m K / L^2\right)\left(1+(M / m K) \cos \theta\right)$	$1 / r=\left(m K / L^2\right)\left(1+(M / m K) \cos \theta\right)$	with $1 / r=\left(m K / L^2\right)\left(1+(M / m K) \cos \theta\right)$. Here L is the particle's orbital angular
gilmore-lie-groups_ F02744	268	■ 1.000	L	L	with $1 / r=\left(m K / L^2\right)\left(1+(M / m K) \cos \theta\right)$. Here L is the particle's orbital angular
gilmore-lie-groups_ F02745	268	■ 1.000	C / r^3	C / r^3	momentum and T is its period. If the perturbing term is of the form C / r^3 the integral
gilmore-lie-groups_ F02746	268	■ 1.000	$C \times 2 \pi \frac{m K}{L^2}$	$C \times 2 \pi \frac{m K}{L^2}$	is $C \times 2 \pi \frac{m K}{L^2}$. The perturbations due to special and General Relativity are
gilmore-lie-groups_ F02747	268	■ 1.000	$K=G M m$	$K=G M m$	For planetary motion $K=G M m$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_ F02748	268	■ 1.000	$M \gg m, \omega$	$M \gg m, \omega$	For planetary motion $K=G M m$. When $M \gg m$, ω is (almost) independent of m .
gilmore-lie-groups_ F02749	269	■ 1.000	$\omega=42^{\prime \prime}$	$\omega=42''$	planetary distance from the Sun. What is the precession for the Earth? Use $\omega=42''$
gilmore-lie-groups_ F02750	269	■ 1.000	L_{-}	L_{-}	12. The action of the angular momentum shift down operator L_{-} on the lowest m -value
gilmore-lie-groups_ F02751	269	■ 1.000	$Y_m^l(\theta, \phi)=P_{-l}^l(\theta) e^{-i l \phi}$	$Y_m^l(\theta, \phi)=P_{-l}^l(\theta) e^{-i l \phi}$	a. Write $Y_m^l(\theta, \phi)=P_{-l}^l(\theta) e^{-i l \phi}$ and show
gilmore-lie-groups_ F02752	269	■ 1.000	$P_{-l}^l=(\sin \theta)^l$	$P_{-l}^l=(\sin \theta)^l$	b. Show $P_{-l}^l=(\sin \theta)^l$ satisfies this equation.

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gilmore-lie-groups_ F02779	270	■ 0.997	$x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{N\}}$	$x_1, x_2, \dots, x_N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_ F02780	270	■ 0.997	$3 \rightarrow N$	$3 \rightarrow N$	l in $x_1, x_2, \dots, x_N$ is obtained by replacing $3 \rightarrow N$ in this expression. This is the
gilmore-lie-groups_ F02781	270	■ 1.000	$r^l Y_m^l(\theta, \phi)$	$r^l Y_m^l(\theta, \phi)$	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_ F02782	270	■ 1.000	x, y, z	x, y, z	a. Show that the functions $r^l Y_m^l(\theta, \phi)$ are homogeneous polynomials in x, y, z of
gilmore-lie-groups_ F02783	270	■ 1.000	$l-2$	$l-2$	ference between the number of homogeneous polynomials of degree l and $l-2$
gilmore-lie-groups_ F02784	270	■ 1.000	$\operatorname{dim}\left\{Y_m^l\right\}=N(l, 3)-N(l-2, 3)=2 l+1$	$\dim \left\{Y_m^l\right\}=N(l, 3)-N(l-2, 3)=2 l+1$	on three variables $\dim \left\{Y_m^l\right\}=N(l, 3)-N(l-2, 3)=2 l+1$.
gilmore-lie-groups_ F02785	270	■ 0.999	$\operatorname{dim}\left\{\mathcal{Y}_{l m}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$	$\dim \left\{\mathcal{Y}_{l m}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$	degree n is $\dim \left\{\mathcal{Y}_{l m}^n\right\}=N(n, 4)-N(n-2, 4)=(n+1)^2$.
gilmore-lie-groups_ F02786	270	■ 0.995	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$	$\psi(\mathbf{x})_{nlm}$ for $n=0, 1, 2; l=0, \dots, n-1$; and $-l \leq m \leq +l$.
gilmore-lie-groups_ F02787	270	■ 0.995	$n=0, 1, 2 ; l=0, \ldots, n-1$	$n=0, 1, 2; l=0, \dots, n-1$	$\psi(\mathbf{x})_{nlm}$ for $n=0, 1, 2; l=0, \dots, n-1$; and $-l \leq m \leq +l$.
gilmore-lie-groups_ F02788	271	■ 1.000	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$	$\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$	where a and b are half odd integers: $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$.
gilmore-lie-groups_ F02789	271	■ 1.000	$N(l, d)=$	$N(l, d)=$	f. Show homogeneous polynomials satisfy the recursion relation: $N(l, d)=$
gilmore-lie-groups_ F02790	271	■ 1.000	$N(l, d-1)+N(l-1, d)$	$N(l, d-1)+N(l-1, d)$	$N(l, d-1)+N(l-1, d)$.
gilmore-lie-groups_ F02791	271	■ 1.000	S^{n-1}	S^{n-1}	spaces of spherical harmonics on spheres S^n and S^{n-1} :
gilmore-lie-groups_ F02792	271	■ 1.000	$(l+1)^2=l^2+(2 l+1)$	$(l+1)^2=l^2+(2 l+1)$	For the case $n=3$ this gives $(l+1)^2=l^2+(2 l+1)$. The initialization for all n
gilmore-lie-groups_ F02793	271	■ 1.000	$\mathcal{Y}^0\left(S^n\right)=1=\operatorname{dim} \mathcal{Y}^0\left(S^n\right)$	$\mathcal{Y}^0\left(S^n\right)=1=\dim \mathcal{Y}^0\left(S^n\right)$	is $\mathcal{Y}^0\left(S^n\right)=1=\dim \mathcal{Y}^0\left(S^n\right)$
gilmore-lie-groups_ F02794	271	■ 1.000	$\operatorname{dim} \mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2)!}{l!(n-1)!}(2 l+n-1)$	$\dim \mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2)!}{l!(n-1)!}(2 l+n-1)$	f. $\dim \mathcal{Y}^l\left(S^n\right)=\frac{(l+n-2)!}{l!(n-1)!}(2 l+n-1)$.
gilmore-lie-groups_ F02795	271	■ 1.000	S^{D-1}	S^{D-1}	The angular part of the Laplacian operator, $\mathcal{L}^2$, acts on spherical harmonics on S^{D-1} ,
gilmore-lie-groups_ F02796	271	■ 1.000	$\mathcal{Y}^l\left(S^{D-1}\right)$	$\mathcal{Y}^l\left(S^{D-1}\right)$	$\mathcal{Y}^l\left(S^{D-1}\right)$. These spherical harmonics are eigenfunctions of this (Laplace–Beltrami)
gilmore-lie-groups_ F02797	271	■ 1.000	$-\left[(l+\alpha)^2-\alpha^2\right]$	$-[(l+\alpha)^2-\alpha^2]$	operator with eigenvalue $-[(l+\alpha)^2-\alpha^2]$, and α is a quantity that depends on the
gilmore-lie-groups_ F02798	271	■ 0.709	$SO(D)$	$SO(D)$	Lie algebra of $SO(D)$: it is half the sum over all positive roots of the algebra. For
gilmore-lie-groups_ F02799	271	■ 1.000	$\alpha=D-2$	$\alpha=D-2$	$\alpha=D-2$.
gilmore-lie-groups_ F02800	271	■ 1.000	$\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l$	$\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l$	a. Show that $\psi(\mathbf{x})=\left(1 / r^{(D-1) / 2}\right) \mathcal{Y}^l(\text { angles })$ is a clever ansatz that reduces the
gilmore-lie-groups_ F02801	272	■ 1.000	$\mathrm{a}-1 / r^2$	$a-1 / r^2$	$a-1 / r^2$ perturbation. Show that the bound state energy and scattering phase shifts
gilmore-lie-groups_ F02802	272	■ 0.999	$l(l+1) \rightarrow l(l+1)-\mu_l$	$l(l+1) \rightarrow l(l+1)-\mu_l$	are given by the substitution $l(l+1) \rightarrow l(l+1)-\mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02803	272	■ 0.999	s	s	are given by the substitution $l(l+1) \rightarrow l(l+1)-\mu_l$. Argue that electrons in the s
gilmore-lie-groups_ F02804	272	■ 0.953	$\mu_0 \gg \mu_1 > \cdots$	$\mu_0 \gg \mu_1 > \cdots$	d -state electrons, $\dots$) so that $\mu_0 \gg \mu_1 > \cdots$.
gilmore-lie-groups_ F02805	272	■ 1.000	$L_{ij}=a_i^{\dagger} a_j-a_j^{\dagger} a_i=-L_{ji}$	$L_{ij}=a_i^{\dagger} a_j-a_j^{\dagger} a_i=-L_{ji}$	angular momentum operators $L_{ij}=a_i^{\dagger} a_j-a_j^{\dagger} a_i=-L_{ji}$.

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gilmore-lie-groups_F02886	272	■ 1.000	$Q_{ij} =$	$Q_{ij} =$	angular momentum operators together with the quadrupole tensor operators $Q_{ij} =$
gilmore-lie-groups_F02887	272	■ 1.000	$a_{i}^{\dagger} a_{j} + a_{j}^{\dagger} a_{i} = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$	$a_i^\dagger a_j + a_j^\dagger a_i = +Q_{ji}$.
gilmore-lie-groups_F02888	272	■ 1.000	$\left[a_{i}, a_{j}^{\dagger}\right]=1$	$\left[a_i, a_j^\dagger\right] = 1$	as the single boson operators $a_i^\dagger$ and a_j , as well as their commutator $[a_i, a_j^\dagger] = 1$.
gilmore-lie-groups_F02889	272	■ 1.000	$a_{i} a_{j}$	$a_i a_j$	$a_i a_j$. Show that this algebra is simple and describe its structure. Show that this
gilmore-lie-groups_F02810	273	■ 0.999	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t), \mathbf{B}(\mathbf{x}, t)$	The electromagnetic field $\mathbf{E}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$ is determined by Maxwell's
gilmore-lie-groups_F02811	273	■ 0.935	$\mathbf{E}(\mathbf{k}), \mathbf{B}(\mathbf{k})$	$\mathbf{E}(k), \mathbf{B}(k)$	four linear constraints on the six amplitudes $\mathbf{E}(k)$, $\mathbf{B}(k)$. Why? At a more
gilmore-lie-groups_F02812	274	■ 1.000	$\mathbf{E}(\mathbf{x}, t)$	$\mathbf{E}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-
gilmore-lie-groups_F02813	274	■ 1.000	$\mathbf{B}(\mathbf{x}, t)$	$\mathbf{B}(\mathbf{x}, t)$	and magnetic fields, $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, or their components after Fourier transfor-
gilmore-lie-groups_F02814	274	■ 1.000	$\mathbf{E}(\mathbf{k})$	$\mathbf{E}(k)$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_F02815	274	■ 1.000	$\mathbf{B}(\mathbf{k})$	$\mathbf{B}(k)$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_F02816	274	■ 1.000	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	$k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$	mation, $\mathbf{E}(k)$ and $\mathbf{B}(k)$, where k is a four-vector that obeys $k \cdot k = \mathbf{k} \cdot \mathbf{k} - k_4 k_4 = 0$.
gilmore-lie-groups_F02817	274	■ 0.798	k_4	k_4	Here $\mathbf{k}$ is essentially a three-momentum vector and k_4 is essentially an energy.
gilmore-lie-groups_F02818	274	■ 1.000	$j=1$	$j = 1$	straints so derived reduce, for $j = 1$, to Maxwell's equations. This derivation is
gilmore-lie-groups_F02819	274	■ 1.000	$k \cdot k = 0$	$k \cdot k = 0$	ton four-vector k no longer obeys $k \cdot k = 0$. In this case the manifestly covariant
gilmore-lie-groups_F02820	275	■ 0.999	$(x, y, z, i c t)$	$(x, y, z, i c t)$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_F02821	275	■ 0.999	$(x, y, z, i c t)^{\prime}$	$(x, y, z, i c t)'$	This requires that the coordinates $(x, y, z, i c t)$ and $(x, y, z, i c t)'$ for observers S
gilmore-lie-groups_F02822	275	■ 0.865	Λ	Λ	The 4×4 matrix transformations Λ belong to the Lie group $O(3, 1)$. The infinites-
gilmore-lie-groups_F02823	275	■ 1.000	$k \cdot a = \Lambda k \cdot a$	$k \cdot a = \Lambda k \cdot a$	Homogeneous Lorentz transformations leave invariant inner products: $k \cdot a = \Lambda k \cdot a$.
gilmore-lie-groups_F02824	275	■ 1.000	Λa	Λa	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_F02825	275	■ 1.000	$\Lambda \in O(3, 1)$	$\Lambda \in O(3, 1)$	Λa , where k and a are four vectors and $\Lambda \in O(3, 1)$. The infinitesimal generators
gilmore-lie-groups_F02826	275	■ 0.936	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$	$\mathbf{J}, \mathbf{K}$ satisfy the following commutation relations:
gilmore-lie-groups_F02827	276	■ 0.988	$a=(x, y, z, c t)$	$a = (x, y, z, c t)$	tion can be written as a 5×5 matrix, in terms of the 4-vector $a = (x, y, z, c t)$:
gilmore-lie-groups_F02828	276	■ 0.995	$(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$	$(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$	subgroup are $(\partial / \partial x, \partial / \partial y, \partial / \partial z, i \partial / \partial (c t))$.
gilmore-lie-groups_F02829	276	■ 1.000	$\{\Lambda, 0\}$	$\{\Lambda, 0\}$	groups. These are the subgroup of homogeneous Lorentz transformations $\{\Lambda, 0\}$
gilmore-lie-groups_F02830	276	■ 1.000	$\{I, a\}$	$\{I, a\}$	and the invariant subgroup of translations $\{I, a\}$. Both their representations play a
gilmore-lie-groups_F02831	276	■ 1.000	Γ^k	Γ^k	We may define a basis state for the one-dimensional representation Γ^k of $\{I, a\}$ as
gilmore-lie-groups_F02832	276	■ 1.000	$ k\rangle$	$ k\rangle$	$ k\rangle$:

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gilmore-lie-groups_ F02833	277	■ 1.000	$D_{\{2\}}=A_{\{1\}}+A_{\{1\}}$	$D_2 = A_1 + A_1$	The Lie algebra $\mathfrak{D}_2 \equiv \mathfrak{A}_1 + \mathfrak{A}_1$ is semisimple: it is the direct sum of two simple									
gilmore-lie-groups_ F02834	277	■ 1.000	$\mathbf{J}^{\{1\}}$	$\mathbf{J}^{(1)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$									
gilmore-lie-groups_ F02835	277	■ 1.000	$D^{\{j\}}$	D^j	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$									
gilmore-lie-groups_ F02836	277	■ 1.000	$\mathbf{J}^{\{2\}}$	$\mathbf{J}^{(2)}$	The algebra $\mathbf{J}^{(1)}$ has $2j + 1$ dimensional irreducible representations D^j while $\mathbf{J}^{(2)}$									
gilmore-lie-groups_ F02837	277	■ 0.970	$2\ j^{\backslash\prime}+1$	$2j' + 1$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$									
gilmore-lie-groups_ F02838	277	■ 0.970	$D^{\{j^{\backslash\prime}\}}$	$D^{j'}$	has $2j' + 1$ dimensional irreducible representations $D^{j'}$. Any element in $SO(3, 1)$									
gilmore-lie-groups_ F02839	277	■ 1.000	$(2\ j+1)\backslash\left(2\ j^{\backslash\prime}+1\right)$	$(2j + 1) (2j' + 1)$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows									
gilmore-lie-groups_ F02840	277	■ 1.000	$D^{\{j\ j^{\backslash\prime}\}}$	$D^{jj'}$	can be expressed in a $(2j + 1)(2j' + 1)$ dimensional representation $D^{jj'}$ as follows									
gilmore-lie-groups_ F02841	278	■ 1.000	$\mathbf{J}^{\{(j)\}}$	$\mathbf{J}^{(j)}$	where $\mathbf{J}^{(j)}$ are the three $(2j + 1) \times (2j + 1)$ angular momentum matrices. The									
gilmore-lie-groups_ F02842	278	■ 0.990	$D^{\{j\ j^{\backslash\prime}\}}(\Lambda)$	$D^{jj'}(\Lambda)$	Basis states for the action of Λ through the representation $D^{jj'}(\Lambda)$ can be com-									
gilmore-lie-groups_ F02843	278	■ 1.000	$S\ 0(3)\ \backslash\subset S\ 0(3,1)$	$SO(3) \subset SO(3, 1)$	Under restriction to the subgroup $SO(3) \subset SO(3, 1)$ this representation is reducible									
gilmore-lie-groups_ F02844	278	■ 1.000	$j^{\backslash\prime}=0$	$j' = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.									
gilmore-lie-groups_ F02845	278	■ 1.000	$j=0$	$j = 0$	This representation remains irreducible only if $j' = 0$ or $j = 0$.									
gilmore-lie-groups_ F02846	278	■ 1.000	$T_{\backslash\mu\ \backslash\nu}(x)$	$T_{\mu\nu}(x)$	A field $T_{\mu\nu}(x)$ is said to be manifestly covariant (obviously covariant) under									
gilmore-lie-groups_ F02847	278	■ 1.000	$\Lambda\ \backslash\in S\ 0(3,1)$	$\Lambda \in SO(3, 1)$	transformations of the homogeneous Lorentz group $\Lambda \in SO(3, 1)$ if									
gilmore-lie-groups_ F02848	278	■ 1.000	$x^{\backslash\prime}=x\ \backslash\Lambda^{\backslash-1}$	$x' = x\Lambda^{-1}$	nate system changes, the coordinates of the fixed point are changed by $x' = x\Lambda^{-1}$.									
gilmore-lie-groups_ F02849	279	■ 0.774	$\backslash\left \backslash\begin{array}{cc} j\ \& j^{\backslash\prime}\ \backslash\backslash\ \backslash\mu\ \& \backslash\mu^{\backslash\prime}\ \backslash\prime\end{array}\right\rangle\backslash\right\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$ separately.									
gilmore-lie-groups_ F02850	280	■ 0.815	$\backslash\left.\backslash\begin{array}{cc} j\ \& j^{\backslash\prime}\ \backslash\backslash\ \backslash\mu\ \& \backslash\mu^{\backslash\prime}\ \backslash\prime\end{array}\right\rangle\backslash\right\}$	$\left.\begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\}$	<table><tr><td></td><td>$k\rangle$</td><td>$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$</td></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$
	$ k\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$												
$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$												
$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$												
gilmore-lie-groups_ F02851	280	■ 0.815	$\backslash k\rangle\backslash\right\rangle\backslash\backslash e^{\backslash\mathrm{i}\ k\ \backslash\cdot\mathrm{a}}$	$ k\rangle e^{ik\cdot a}$	<table><tr><td></td><td>$k\rangle$</td><td>$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$</td></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$
	$ k\rangle$	$\left \begin{array}{cc} j & j' \\ \mu & \mu' \end{array}\right\rangle$												
$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle \delta_{\nu\nu';\mu\mu'}$												
$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left \begin{array}{cc} j & j' \\ \nu & \nu' \end{array}\right\rangle D^{jj'}_{\nu\nu';\mu\mu'}(\Lambda)$												

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gilmore-lie-groups_ F02852	280	■ 0.815	$\left.\begin{array}{cc} j & j^{\prime} \\ v & v^{\prime} \end{array}\right\rangle \right\rangle_{vv^{\prime}; \mu \mu^{\prime}} \backslash \operatorname{end}{array} \backslash \operatorname{right} \backslash \operatorname{rangle}_{v \ v^{\prime}} ; \ \mu \ \mu^{\prime} \backslash \operatorname{prime} \}$	$\left.\begin{array}{cc} j & j^{\prime} \\ v & v^{\prime} \end{array}\right\rangle \right\rangle_{vv^{\prime}; \mu \mu^{\prime}}$	
gilmore-lie-groups_ F02853	280	■ 0.815	$ \Lambda k\rangle$	$ \Lambda k\rangle$	
gilmore-lie-groups_ F02854	280	■ 0.815	$\left.\begin{array}{cc} j & j^{\prime} \\ v & v^{\prime} \end{array}\right\rangle \right\rangle_{vv^{\prime}; \mu \mu^{\prime}} \backslash \operatorname{end}{array} \backslash \operatorname{right} \backslash \operatorname{rangle}_{D_{\{v \ v^{\prime}\}} ; \ \mu \ \mu^{\prime} \backslash \operatorname{prime} \}}^{\{j \ j^{\prime}\}}(\Lambda)$	$\left.\begin{array}{cc} j & j^{\prime} \\ v & v^{\prime} \end{array}\right\rangle \right\rangle_{vv^{\prime}; \mu \mu^{\prime}}^{\{j \ j^{\prime}\}} D_{vv^{\prime}; \mu \mu^{\prime}}(\Lambda)$	
gilmore-lie-groups_ F02855	280	■ 1.000	$\{\Lambda, a\}$	$\{\Lambda, a\}$	Suppose we have a representation of $\{\Lambda, a\}$ that is unitary and irreducible. Un-
gilmore-lie-groups_ F02856	280	■ 1.000	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02857	280	■ 1.000	$ k ; \xi \rangle$	$ k; \xi\rangle$	$\Gamma^k(\{I, a\})$ of $\{I, a\}$. The basis states are $ k; \xi\rangle$, where k is defined by the action of
gilmore-lie-groups_ F02858	280	■ 1.000	ξ	ξ	and ξ is a helicity index that distinguishes different states with the same four-
gilmore-lie-groups_ F02859	280	■ 0.999	$k^{\prime}=\Lambda k$	$k' = \Lambda k$	subspace of states parameterized by $k' = \Lambda k$
gilmore-lie-groups_ F02860	280	■ 1.000	$M_{\xi^{\prime} \xi}(\Lambda)$	$M_{\xi' \xi}(\Lambda)$	where $M_{\xi' \xi}(\Lambda)$ is a matrix that remains to be determined.
gilmore-lie-groups_ F02861	280	■ 1.000	$k^{\prime}$	k'	k' with
gilmore-lie-groups_ F02862	281	■ 1.000	$M(\Lambda)$	$M(\Lambda)$	are present also. To construct the matrix $M(\Lambda)$, we choose one particular four-vector
gilmore-lie-groups_ F02863	281	■ 1.000	$k^{\{0\}}$	k^0	k^0 for each of the possible cases
gilmore-lie-groups_ F02864	281	■ 1.000	$\left k^{\{0\}} ; \xi\right\rangle$	$ k^0; \xi\rangle$	The effect of a homogeneous Lorentz transformation on the state $ k^0; \xi\rangle$ is de-
gilmore-lie-groups_ F02865	281	■ 0.822	$H_{\{k^{\{0\}}\}}$	H_{k^0}	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02866	281	■ 0.822	$C_{\{k\}}$	C_k	That is, H_{k^0} is the stability subgroup of the little vector k^0 and C_k is a coset repre-
gilmore-lie-groups_ F02867	282	■ 1.000	$\theta_1=\theta_2=b_3=0, \theta_3, b_1, b_2$	$\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$	The subalgebra leaving k^0 fixed is defined by $\theta_1 = \theta_2 = b_3 = 0, \theta_3, b_1, b_2$ arbi-
gilmore-lie-groups_ F02868	282	■ 1.000	$(0,0,1,-i)=T(0,0,1,+i)$	$(0,0,1,-i) = T(0,0,1,+i)$	(0,0,1,-i)=T(0,0,1,+i) the infinitesimal generators are $Y_1 = J_1 - K_2, Y_2 =$

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gilmore-lie-groups_F02869	282	■ 1.000	$Y_{\{1\}}=J_{\{1\}}-K_{\{2\}}, \ Y_{\{2\}}=$	$Y_1 = J_1 - K_2, Y_2 =$	$(0, 0, 1, -i) = T(0, 0, 1, +i)$ the infinitesimal generators are $Y_1 = J_1 - K_2, \ Y_2 =$
gilmore-lie-groups_F02870	282	■ 1.000	$J_{\{2\}}+K_{\{1\}}, \ Y_{\{3\}}=J_{\{3\}}$	$J_2 + K_1, Y_3 = J_3$	$J_2 + K_1, \ Y_3 = J_3.$
gilmore-lie-groups_F02871	282	■ 1.000	$\mathbf{b}=0$	$\mathbf{b} = 0$	The subalgebra defined by $\mathbf{b} = 0$ is spanned by the angular momentum operators
gilmore-lie-groups_F02872	283	■ 1.000	$D_{\xi'\xi}(\xi)\left(H_{k^0}\right)$	$D_{\xi'\xi}(H_{k^0})$	irreducible if and only if the representation $D_{\xi'\xi}(H_{k^0})$ of the little group is unitary
gilmore-lie-groups_F02873	283	■ 0.999	$k \cdot k > 0$	$k \cdot k > 0$	were described in Problem 5 of Chapter 11. Since $k \cdot k > 0$ describes negative mass
gilmore-lie-groups_F02874	283	■ 1.000	$j=0, \ \frac{1}{2}, 1, \ \frac{3}{2}, \ \ldots$	$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$	They are described by an integer or half-integer: $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. The angular
gilmore-lie-groups_F02875	283	■ 1.000	$\kappa=\left(\kappa_{\{1\}}, \ \kappa_{\{2\}}\right)$	$\kappa = (\kappa_1, \kappa_2)$	can be labeled by a vector $\kappa = (\kappa_1, \kappa_2)$ in a two-dimensional Euclidean space,
gilmore-lie-groups_F02876	283	■ 1.000	$\kappa \in R^2, \ \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_F02877	283	■ 1.000	$ \kappa\rangle$	$ \kappa\rangle$	$\kappa \in R^2, \kappa \cdot \kappa \geq 0$. If a state $ \kappa\rangle$ is in one such representation, so are all states
gilmore-lie-groups_F02878	283	■ 0.999	$\left \kappa^{\prime}\right\rangle$	$ \kappa'\rangle$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_F02879	283	■ 0.999	$\kappa^{\prime} \cdot \kappa^{\prime}=\kappa \cdot \kappa$	$\kappa' \cdot \kappa' = \kappa \cdot \kappa$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_F02880	283	■ 0.999	$\kappa^{\prime}=\left(\kappa_{\{1\}}^{\prime}, \ \kappa_{\{2\}}^{\prime}\right)$	$\kappa' = (\kappa'_1, \kappa'_2)$	$ \kappa'\rangle$ for which $\kappa' \cdot \kappa' = \kappa \cdot \kappa$. That is, $\kappa' = (\kappa'_1, \kappa'_2)$ is related to $\kappa = (\kappa_1, \kappa_2)$ by a
gilmore-lie-groups_F02881	283	■ 1.000	$\kappa^{\prime}=R(\theta) \kappa$	$\kappa' = R(\theta)\kappa$	rotation: $\kappa' = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_F02882	283	■ 1.000	$\kappa \cdot \kappa$	$\kappa \cdot \kappa$	rotation: $\kappa' = R(\theta)\kappa$. The invariant length $\kappa \cdot \kappa$ parameterizes the representation.
gilmore-lie-groups_F02883	283	■ 0.999	$\kappa \cdot \kappa>0 \quad$	$\kappa \cdot \kappa > 0$	(i) $\kappa \cdot \kappa > 0$ little group = Identity
gilmore-lie-groups_F02884	283	■ 0.996	$\kappa \cdot \kappa=0 \quad$	$\kappa \cdot \kappa = 0$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_F02885	283	■ 0.996	$=I S O(2)$	$= ISO(2)$	(ii) $\kappa \cdot \kappa = 0$ little group = $ISO(2)$
gilmore-lie-groups_F02886	284	■ 1.000	κ^2	κ^2	The first case presents us with two problems. First, κ^2 is a continuous quantum
gilmore-lie-groups_F02887	284	■ 1.000	$\kappa^2>0$	$\kappa^2 > 0$	$\kappa^2 > 0$ there must be an infinite number of states with this same continuous index,
gilmore-lie-groups_F02888	284	■ 1.000	$\kappa=0$	$\kappa = 0$	for each four-momentum value. Therefore we require $\kappa = 0$. This leaves us with the
gilmore-lie-groups_F02889	284	■ 0.961	$\left(Y_{\{1\}} \rightarrow 0, \ Y_{\{2\}} \rightarrow 0\right)$	$(Y_1 \rightarrow 0, Y_2 \rightarrow 0)$	following physically allowable representations of the little group $Y_1 \rightarrow 0, Y_2 \rightarrow 0$
gilmore-lie-groups_F02890	284	■ 1.000	$\left(j, \ j^{\prime}\right)$	(j, j')	particle with transformation indices (j, j') contains all states of the form
gilmore-lie-groups_F02891	285	■ 1.000	$\left\{H_{k^0}, \ 0\right\}$	$\{H_{k^0}, 0\}$	A. $\{H_{k^0}, 0\}$ on $ k^0; \xi\rangle$
gilmore-lie-groups_F02892	285	■ 1.000	$H_{k^0}=\operatorname{EXP}\left(\Theta J_3+\theta_1 Y_1+\theta_2 Y_2\right)$	$H_{k^0} = \text{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2)$	where $H_{k^0} = \text{EXP}(\Theta J_3 + \theta_1 Y_1 + \theta_2 Y_2).$
gilmore-lie-groups_F02893	285	■ 0.494	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j \ j^{\prime}}\right)_{\mu^{\prime}}^j\right\rangle$	$ k^0\rangle _{\mu}^{j \ j'}\rangle_{\mu'}^j\rangle$	B. $\{H_{k^0}, 0\}$ on $ k^0\rangle _{\mu}^{j \ j'}\rangle$ The little group maps k^0 to k^0 but acts in a nontrivial way
gilmore-lie-groups_F02894	285	■ 0.119	$\left.\left.\left.\left k^0\right\rangle\right _{\mu}^{j \ j^{\prime}}\right)_{\mu^{\prime}}^j\right\rangle$	$ k^0\rangle _{\mu}^{j \ j'}\rangle_{\mu'}^j\rangle$	The state $ k^0\rangle _{\mu}^{j \ j'}\rangle$ transforms identically to $ k^0; \xi\rangle$ if $\xi > 0$ and $j = +\xi$.

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gilmore-lie-groups_F02895	285	■ 0.119	$\xi > 0$	$\xi > 0$	The state $ k^0\rangle _{j\ 0}^j\rangle$ transforms identically to $ k^0;\xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_F02896	285	■ 0.119	$j = +\xi$	$j = +\xi$	The state $ k^0\rangle _{j\ 0}^j\rangle$ transforms identically to $ k^0;\xi\rangle$ if $\xi > 0$ and $j = +\xi$.
gilmore-lie-groups_F02897	285	■ 0.983	$\left k^0\right\rangle\left \begin{array}{cc} 0 & j^{\prime} \\ 0 & -j^{\prime} \end{array}\right\rangle$	$ k^0\rangle\left \begin{array}{cc} 0 & j^{\prime} \\ 0 & -j^{\prime} \end{array}\right\rangle$	The state $ k^0\rangle _{j\ 0}^j\rangle$ transforms identically to $ k^0;\xi\rangle$ if $\xi < 0$ and $j' = -\xi$.
gilmore-lie-groups_F02898	285	■ 0.983	$\xi < 0$	$\xi < 0$	The state $ k^0\rangle _{j\ 0}^j\rangle$ transforms identically to $ k^0;\xi\rangle$ if $\xi < 0$ and $j' = -\xi$.
gilmore-lie-groups_F02899	285	■ 0.983	$j^{\prime}=-\xi$	$j' = -\xi$	The state $ k^0\rangle _{j\ 0}^j\rangle$ transforms identically to $ k^0;\xi\rangle$ if $\xi < 0$ and $j' = -\xi$.
gilmore-lie-groups_F02900	286	■ 0.560	$ k\rangle\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array}\right\rangle$	$ k\rangle\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array}\right\rangle$	basis states $ k;\xi\rangle$ or the direct product states $ k\rangle _{\mu\ \mu'}^{j\ j'}\rangle$:
gilmore-lie-groups_F02901	286	■ 1.000	$\langle k ; \xi \mid \psi\rangle$	$\langle k ; \xi \mid \psi\rangle$	The amplitudes of the projection of $ \psi\rangle$ onto the basis states are $\langle k ; \xi \psi \rangle$ in the first
gilmore-lie-groups_F02902	286	■ 0.741	$\langle k ; \xi \mid \psi\rangle$	$\langle k ; \xi \mid \psi\rangle$	case and $\langle k ; \xi \mid \psi\rangle$ in the second. In both cases the sum extends over all k vectors
gilmore-lie-groups_F02903	286	■ 1.000	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	$\Lambda k \cdot \Lambda k = 0, k \neq 0$	for which $\Lambda k \cdot \Lambda k = 0, k \neq 0$. In the first case the sum extends over the appropriate
gilmore-lie-groups_F02904	286	■ 1.000	$\xi = \pm 1$	$\xi = \pm 1$	helicity states ξ ($\xi = \pm 1$ for photons). In the second case the sum extends over the
gilmore-lie-groups_F02905	286	■ 0.712	$\mu, \mu': -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$	$\mu, \mu': -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$	appropriate values of $\mu, \mu': -j \leq \mu \leq +j, -j' \leq \mu' \leq +j'$.
gilmore-lie-groups_F02906	286	■ 1.000	$\xi = j > 0$	$\xi = j > 0$	We discuss the positive helicity state $\xi = j > 0$ first. The amplitude $\langle k^0 ; j \psi \rangle$
gilmore-lie-groups_F02907	286	■ 1.000	$\langle k^0 ; j \mid \psi\rangle$	$\langle k^0 ; j \mid \psi\rangle$	We discuss the positive helicity state $\xi = j > 0$ first. The amplitude $\langle k^0 ; j \psi \rangle$
gilmore-lie-groups_F02908	286	■ 1.000	$ k^0 ; j\rangle$	$ k^0 ; j\rangle$	of the state $ k^0 ; j\rangle$ in any physical state $ \psi\rangle$ may be arbitrary. This is simply the
gilmore-lie-groups_F02909	286	■ 0.993	$\langle k^0 ; j_0^0 \mid \psi\rangle$	$\langle k^0 ; j_0^0 \mid \psi\rangle$	$\langle k^0 ; j_0^0 \psi \rangle$ in the same physical state $ \psi\rangle$ is the same. The amplitudes of the
gilmore-lie-groups_F02910	286	■ 0.999	$\langle k^0 ; j_m^0 \mid \psi\rangle, m \neq j$	$\langle k^0 ; j_m^0 \mid \psi\rangle, m \neq j$	states $\langle k^0 ; j_m^0 \psi \rangle, m \neq j$, must all vanish. These states are all superfluous –
gilmore-lie-groups_F02911	286	■ 0.719	$\{\cdot\}$	$\{\cdot\}$	The matrix within the bracket $\{\cdot\}$ is diagonal, with the coefficient $(j-j)k_3^0 = 0$
gilmore-lie-groups_F02912	286	■ 0.719	$(j-j) k_3^0=0$	$(j-j) k_3^0 = 0$	The matrix within the bracket $\{\cdot\}$ is diagonal, with the coefficient $(j-j)k_3^0 = 0$
gilmore-lie-groups_F02913	286	■ 0.999	$(m-j) k_3^0$	$(m-j) k_3^0$	multiplying the allowed amplitude $\langle k^0 ; j_0^0 \psi \rangle$ and nonzero coefficients $(m-j)k_3^0$
gilmore-lie-groups_F02914	286	■ 1.000	$\langle k^0 ; j_m^0 \mid \psi\rangle$	$\langle k^0 ; j_m^0 \mid \psi\rangle$	multiplying the amplitudes $\langle k^0 ; j_m^0 \psi \rangle$. Since $(m-j)k_3^0 \neq 0$, the amplitudes that
gilmore-lie-groups_F02915	286	■ 1.000	$(m-j) k_3^0 \neq 0$	$(m-j) k_3^0 \neq 0$	multiplying the amplitudes $\langle k^0 ; j_m^0 \psi \rangle$. Since $m-j)k_3^0 \neq 0$, the amplitudes that
gilmore-lie-groups_F02916	286	■ 0.961	$m \neq j$	$m \neq j$	are absent in the description of a physical state ($m \neq j$) must vanish.
gilmore-lie-groups_F02917	286	■ 1.000	$\xi = -j$	$\xi = -j$	For the negative helicity states $\xi = -j$ we have by a completely similar argument
gilmore-lie-groups_F02918	286	■ 0.556	$ k^0\rangle\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array}\right\rangle$	$ k^0\rangle\left \begin{array}{cc} j & j^{\prime} \\ \mu & \mu^{\prime} \end{array}\right\rangle$	and the subspace $ k^0\rangle _{\mu\ \mu'}^{j\ j'}\rangle$ into the subspace $ k\rangle _{\nu\ \nu'}^{j\ j'}\rangle$ through the following non-

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gilmore-lie-groups_F02942	290	■1.000	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
	Electric	Magnetic															
Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$															
Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$															
gilmore-lie-groups_F02943	290	■1.000	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
	Electric	Magnetic															
Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
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Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$															
gilmore-lie-groups_F02944	290	■1.000	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
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Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$															
Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$															
gilmore-lie-groups_F02945	290	■1.000	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
	Electric	Magnetic															
Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$															
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gilmore-lie-groups_F02946	290	■1.000	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
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Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$															
Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$															
gilmore-lie-groups_F02947	290	■1.000	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$	<table><thead><tr><th></th><th>Electric</th><th>Magnetic</th></tr></thead><tbody><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$</td></tr></tbody></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$
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Charge density	$\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)$															
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Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t)+\frac{\partial \rho_e(\mathbf{x}, t)}{\partial t}=0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t)+\frac{\partial \rho_m(\mathbf{x}, t)}{\partial t}=0$															
gilmore-lie-groups_F02948	290	■0.997	T, P, T P	T, P, TP	T, P, TP. The result is unique up to scale factor:												

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gilmore-lie-groups_ F02949	290	■ 1.000	$1 / c$	$1/c$	The factor 4π is the surface area of the unit sphere in R^3 , and the factor $1/c$ on the
gilmore-lie-groups_ F02950	291	■ 0.867	$\phi=\pi / 2$	$\phi=\pi / 2$	In particular, show that for $\phi=\pi / 2$ this is the dual transformation $(\mathbf{B}, \mathbf{E}) \rightarrow$
gilmore-lie-groups_ F02951	291	■ 0.867	$(\mathbf{B}, \mathbf{E}) \rightarrow$	$(\mathbf{B}, \mathbf{E}) \rightarrow$	In particular, show that for $\phi=\pi / 2$ this is the dual transformation $(\mathbf{B}, \mathbf{E}) \rightarrow$
gilmore-lie-groups_ F02952	291	■ 1.000	$(*)=0$	$(*)=0$	$(*)=0$, for $*$ = anyvector. Show
gilmore-lie-groups_ F02953	291	■ 1.000	$*=$	$*=$	$(*)=0$, for $*$ = anyvector. Show
gilmore-lie-groups_ F02954	291	■ 0.998	$D^{\{j-j^{\prime}\}}(\Lambda)$	$D^{j j^{\prime}+j^{\prime} j}(\Lambda)$	resentation $D^{j j^{\prime}+j^{\prime} j}(\Lambda)$, with $j-j^{\prime}=\pm 2$. In the case with $(j, j^{\prime})=(2,0)$ a curl
gilmore-lie-groups_ F02955	291	■ 0.998	$j-j^{\prime}=\pm 2$	$j-j^{\prime}=\pm 2$	resentation $D^{j j^{\prime}+j^{\prime} j}(\Lambda)$, with $j-j^{\prime}=\pm 2$. In the case with $(j, j^{\prime})=(2,0)$ a curl
gilmore-lie-groups_ F02956	291	■ 0.998	$(j, j^{\prime})=(2,0)$	$(j, j^{\prime})=(2,0)$	resentation $D^{j j^{\prime}+j^{\prime} j}(\Lambda)$, with $j-j^{\prime}=\pm 2$. In the case with $(j, j^{\prime})=(2,0)$ a curl
gilmore-lie-groups_ F02957	291	■ 0.990	$\mathbf{G}_e$	$\mathbf{G}_e$	The fields $\mathbf{G}_e$ and $\mathbf{G}_m$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02958	291	■ 0.990	$\mathbf{G}_m$	$\mathbf{G}_m$	The fields $\mathbf{G}_e$ and $\mathbf{G}_m$ are called the gravitoelectric and gravitomagnetic fields. These
gilmore-lie-groups_ F02959	291	■ 0.866	$\mathbf{J} \cdot \nabla$	$\mathbf{J} \cdot \nabla$	latter case the curl operator is $\mathbf{J} \cdot \nabla$, where $\mathbf{J}$ is the 5×5 angular momentum operator:
gilmore-lie-groups_ F02960	292	■ 1.000	$F_{ij}=F_{ji}, \sum_i F_{ii}=0$	$F_{ij}=F_{ji}, \sum_i F_{ii}=0$	The matrix components obey $F_{ij}=F_{ji}, \sum_i F_{ii}=0$, and $\partial^i F_{ij}=0$. The gravitoelec-
gilmore-lie-groups_ F02961	292	■ 1.000	$\partial^i F_{ij}=0$	$\partial^i F_{ij}=0$	The matrix components obey $F_{ij}=F_{ji}, \sum_i F_{ii}=0$, and $\partial^i F_{ij}=0$. The gravitoelec-
gilmore-lie-groups_ F02962	292	■ 0.999	$U_{ij}=\sum_k m_k(\mathbf{x}_k(t) \mathbf{x}_k(t))_{ij}$	$U_{ij}=\sum_k m_k(\mathbf{x}_k(t) \mathbf{x}_k(t))_{ij}$	dinates the source term for the gravitoelectric field is $U_{ij}=\sum_k m_k(\mathbf{x}_k(t) \mathbf{x}_k(t))_{ij}$,
gilmore-lie-groups_ F02963	292	■ 0.999	$\mathbf{G}_m+i \mathbf{G}_e \rightarrow e^{i \phi}\left(\mathbf{G}_m+i \mathbf{G}_e\right)$	$\mathbf{G}_m+i \mathbf{G}_e \rightarrow e^{i \phi}\left(\mathbf{G}_m+i \mathbf{G}_e\right)$	$\mathbf{G}_m+i \mathbf{G}_e \rightarrow e^{i \phi}\left(\mathbf{G}_m+i \mathbf{G}_e\right)$ and $\mathbf{J}_m+i \mathbf{J}_e \rightarrow e^{i \phi}\left(\mathbf{J}_m+i \mathbf{J}_e\right)$. Show this.
gilmore-lie-groups_ F02964	292	■ 0.999	$\mathbf{J}_m+i \mathbf{J}_e \rightarrow e^{i \phi}\left(\mathbf{J}_m+i \mathbf{J}_e\right)$	$\mathbf{J}_m+i \mathbf{J}_e \rightarrow e^{i \phi}\left(\mathbf{J}_m+i \mathbf{J}_e\right)$	$\mathbf{G}_m+i \mathbf{G}_e \rightarrow e^{i \phi}\left(\mathbf{G}_m+i \mathbf{G}_e\right)$ and $\mathbf{J}_m+i \mathbf{J}_e \rightarrow e^{i \phi}\left(\mathbf{J}_m+i \mathbf{J}_e\right)$. Show this.
gilmore-lie-groups_ F02965	292	■ 1.000	$D^{jj}(\Lambda)$	$D^{jj}(\Lambda)$	5. Construct the source-free field equations for gravitons for the $D^{jj}(\Lambda)$ representation,
gilmore-lie-groups_ F02966	292	■ 0.871	(J, M)	(J, M)	with $j=1$. Show that there are seven constraints that correspond to (J, M) with
gilmore-lie-groups_ F02967	292	■ 1.000	$(J, M)=(0,0),(1,0),(1, \pm 1),(2,0),(2 \pm 1)$	$(J, M)=(0,0),(1,0),(1, \pm 1),(2,0),(2 \pm 1)$	$(J, M)=(0,0),(1,0),(1, \pm 1),(2,0),(2 \pm 1)$. What are these equations in the stan-
gilmore-lie-groups_ F02968	293	■ 1.000	$N \lambda$	$N \lambda$	two (distance $N \lambda$), as time goes on and the distance increases the wavelength must
gilmore-lie-groups_ F02969	293	■ 1.000	$N \lambda^{\prime}$	$N \lambda^{\prime}$	also increase to $N \lambda^{\prime}$. This redshift source is sometimes confused with the Döppler
gilmore-lie-groups_ F02970	293	■ 1.000	$\rho(\mathbf{x}, t)=\rho(t)$	$\rho(\mathbf{x}, t)=\rho(t)$	a. Assume that the energy density in the universe has the form $\rho(\mathbf{x}, t)=\rho(t)$ (time
gilmore-lie-groups_ F02971	293	■ 1.000	$m_e(t) / M_p(t)$	$m_e(t) / M_p(t)$	b. Estimate the mass dependence of the electron–proton mass ratio $m_e(t) / M_p(t)$.
gilmore-lie-groups_ F02972	293	■ 0.981	$v=\frac{1}{2}\left(m c^2 / \hbar\right) \times\left \left(1 / n_1^2-1 / n_2^2\right)\right $	$v=\frac{1}{2}\left(m c^2 / \hbar\right) \times\left \left(1 / n_1^2-1 / n_2^2\right)\right $	the hydrogen atom has frequency $v=\frac{1}{2}\left(m c^2 / \hbar\right) \times\left \left(1 / n_1^2-1 / n_2^2\right)\right $ where n_1 and
gilmore-lie-groups_ F02973	293	■ 1.000	H_{α}	H_{α}	and m is the reduced mass of the electron–proton system. Show that H_{α} photons
gilmore-lie-groups_ F02974	294	■ 1.000	$\Phi(x)$	$\Phi(x)$	Here $\Phi(x)$ is the local Newtonian gravitational field. Find a locally linear coordinate

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gilmore-lie-groups_F02975	294	■ 1.000	$S^{\text{t}} g_{\text{grav}} S = g_{\text{flat}}$	$S^t g_{\text{grav}} S = g_{\text{flat}}$	transformation S that brings the curved metric to flat form: $S^t g_{\text{grav}} S = g_{\text{flat}}$. Interpret
gilmore-lie-groups_F02976	294	■ 1.000	V	V	The integral on the left is over the surface bounding the volume V over which the
gilmore-lie-groups_F02977	294	■ 0.867	$\mathbf{E}$	$\mathbf{E}$	integral on the right extends, $\mathbf{E}$ is the electric field and ρ is the charge density. For a
gilmore-lie-groups_F02978	294	■ 0.523	$a, p(x)=q \delta(x)$	$a, p(x) = q\delta(x)$	charge q at the origin of a sphere of radius a , $\rho(x) = q\delta(x)$, The $\mathbf{E}$ field is spherically
gilmore-lie-groups_F02979	294	■ 1.000	$ \mathbf{E}(a) =q/ \mathbf{a} $	$ \mathbf{E}(a) = q/ \mathbf{a} $	By completely similar arguments Gauss' Law in the plane R^2 gives $ \mathbf{E}(a) = q/ \mathbf{a} $.
gilmore-lie-groups_F02980	294	■ 0.808	$\oint \mathbf{E} \cdot d\mathbf{S} = \int 2\pi \rho dA$	$\left(\oint \mathbf{E} \cdot d\mathbf{S} = \int 2\pi \rho dA\right)$	Assume a Gauss law ($\oint \mathbf{E} \cdot d\mathbf{S} = \int 2\pi \rho dA$) holds on the sphere S^2 . Place a charge
gilmore-lie-groups_F02981	294	■ 1.000	$a=R \sin \theta$	$a = R\theta$	sphere S^2 (c.f., Fig. 15.1). Show that its distance a from the north pole is $a = R\theta$
gilmore-lie-groups_F02982	295	■ 1.000	$2\pi R \sin \theta$	$2\pi R \sin \theta$	and the circumference of a circle of latitude through this point is $2\pi R \sin \theta$. Use
gilmore-lie-groups_F02983	295	■ 1.000	q/a	q/a	Conclude that the field is stronger than the q/a form it would have in a plane.
gilmore-lie-groups_F02984	295	■ 1.000	$a(a=R \sin \theta)$	$a(a = R\theta)$	where a ($a = R\theta$) is the distance from the charge to the observation point.
gilmore-lie-groups_F02985	295	■ 1.000	$a=ct$	$a = ct$	charge depends on the distance from the observation point. In particular, if $a = ct$,
gilmore-lie-groups_F02986	295	■ 1.000	$H^n = SO(n, 1) / SO(n), R^n =$	$H^n = SO(n,1)/SO(n), R^n =$	sure on the three Riemannian symmetric spaces $H^n \equiv SO(n, 1)/SO(n), R^n =$
gilmore-lie-groups_F02987	295	■ 0.999	$ISO(n) / SO(n)$	$ISO(n)/SO(n)$	$ISO(n)/SO(n)$, and $S^n = SO(n+1)/SO(n)$ are
gilmore-lie-groups_F02988	295	■ 1.000	$k=(-1,0,+1)$	$k = (-1, 0, +1)$	where $k \equiv (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used;
gilmore-lie-groups_F02989	295	■ 1.000	H^n, R^n, S^n	H^n, R^n, S^n	where $k = (-1, 0, +1)$ for H^n, R^n, S^n and radial coordinates are used;
gilmore-lie-groups_F02990	295	■ 1.000	H^n, S^n	H^n, S^n	a. Derive the metric for H^n, S^n from Eq. (12.9) and the coordinate transformation
gilmore-lie-groups_F02991	295	■ 1.000	Ω	Ω	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02992	295	■ 1.000	$S^{n-1} \subset H^n, R^n$	$S^{n-1} \subset H^n, R^n$	Compute Ω , the surface area of the unit sphere $S^{n-1} \subset H^n, R^n$, or S^n . (Hint: use
gilmore-lie-groups_F02993	295	■ 1.000	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$	$\int e^{-x^2} dx = \sqrt{\pi}$, carry the n -fold integral out in Cartesian and radial coordinates,
gilmore-lie-groups_F02994	295	■ 1.000	$\Omega=2\pi^{n/2}/\Gamma(n/2)$	$\Omega = 2\pi^{n/2}/\Gamma(n/2)$	and show $\Omega = 2\pi^{n/2}/\Gamma(n/2)$.)
gilmore-lie-groups_F02995	296	■ 1.000	H^n	H^n	S^n , downward in H^n , with lookback time. Give a physical interpretation involving
gilmore-lie-groups_F02996	296	■ 1.000	R, c, t	R, c, t	on $[R, c, t]$?
gilmore-lie-groups_F02997	296	■ 0.839	$(l, b) = (263^\circ, 48^\circ)$	$(l, b) = (263^\circ, 48^\circ)$	the direction with galactic coordinates $(l, b) = (263^\circ, 48^\circ)$.
gilmore-lie-groups_F02998	296	■ 1.000	$T(\theta, \phi; t) =$	$T(\theta, \phi; t) =$	b. Assume the temperature distribution of the microwave background is $T(\theta, \phi; t) =$
gilmore-lie-groups_F02999	296	■ 1.000	$\sum_{l,m} A_m^l(t) Y_m^l(\theta, \phi)$	$\sum_{l,m} A_m^l(t) Y_m^l(\theta, \phi)$	$\sum_{l,m} A_m^l(t) Y_m^l(\theta, \phi)$. How do you use this information to determine a frame that
gilmore-lie-groups_F03000	296	■ 0.998	$\Delta U \Delta(1/T) \geq k$	$\Delta U \Delta(1/T) \geq k$	the uncertainty relations of statistical mechanics: $\Delta U \Delta(1/T) \geq k$ in the entropy
gilmore-lie-groups_F03001	297	■ 1.000	$j = \frac{1}{2}, 1, 2$	$j = \frac{1}{2}, 1, 2$	why, or why not, the special frames defined by $j = \frac{1}{2}, 1, 2$ are the same. What

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gilmore-lie-groups_F03002	297	■ 1.000	$v(t)$	$v(t)$	that it moves through the microwave background radiation with a velocity $v(t)$.
gilmore-lie-groups_F03003	297	■ 1.000	$v(t) \simeq v\left(t_0\right) e^{\{-\left(t-t_0\right) / \tau\}}$	$v(t) \simeq v\left(t_0\right) e^{-\left(t-t_0\right) / \tau}$	Show that its velocity decays to zero according to $v(t) \simeq v\left(t_0\right) e^{-\left(t-t_0\right) / \tau}$ because
gilmore-lie-groups_F03004	297	■ 1.000	τ	τ	it is moving through a viscous medium. Estimate τ and present your answer in
gilmore-lie-groups_F03005	297	■ 0.990	τ / T_p	τ / T_p	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03006	297	■ 0.990	T_p	T_p	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03007	297	■ 0.990	$T_p \simeq 13.7 \mathrm{BY}$	$T_p \simeq 13.7 \mathrm{BY}$	the form τ / T_p , where T_p is the present age of the universe ($T_p \simeq 13.7$ BY). To
gilmore-lie-groups_F03008	297	■ 1.000	T_{BH}	T_{BH}	in fact, assume it is a black hole with mass M , radius R at temperature T_{BH} .
gilmore-lie-groups_F03009	297	■ 1.000	$R=2 G M / c^2$	$R=2 G M / c^2$	Use the standard relations for a neutral nonrotating black hole $R=2 G M / c^2$,
gilmore-lie-groups_F03010	297	■ 1.000	$T_{BH}=\hbar c^3 / 8 \pi k G M$	$T_{BH}=\hbar c^3 / 8 \pi k G M$	$T_{BH}=\hbar c^3 / 8 \pi k G M$. You can assume that the mass M is sufficiently large that
gilmore-lie-groups_F03011	297	■ 1.000	$\gamma \pi R^2$	$\gamma \pi R^2$	(geometric) cross section for radiation on a black hole is $\gamma \pi R^2$, where $\gamma=3^3 / 2^2$.
gilmore-lie-groups_F03012	297	■ 1.000	$\gamma=3^3 / 2^2$	$\gamma=3^3 / 2^2$	(geometric) cross section for radiation on a black hole is $\gamma \pi R^2$, where $\gamma=3^3 / 2^2$.
gilmore-lie-groups_F03013	298	■ 1.000	$d y / d x=g(x)$	$dy/dx=g(x)$	constant that can be added to any indefinite integral of $dy/dx=g(x)$ is in
gilmore-lie-groups_F03014	298	■ 1.000	$y: d y / d x=g(x)$	$y: dy/dx=g(x)$	dependent variable y : $dy/dx=g(x)$. This is treated in Section 16.1. In that section
gilmore-lie-groups_F03015	298	■ 1.000	$d y / d x=p$	$dy/dx=p$	we set up the general formulation in terms of a constraint equation $dy/dx=p$ and
gilmore-lie-groups_F03016	299	■ 1.000	$F(x, y, p)=0$	$F(x, y, p)=0$	a surface equation $F(x, y, p)=0$. The special forms of the surface and constraint
gilmore-lie-groups_F03017	299	■ 1.000	R, S, T	R, S, T	metry is. Transformation to a set of canonical variables R, S, T is algorithmic. The
gilmore-lie-groups_F03018	299	■ 1.000	$R(x, y)$	$R(x, y)$	canonical variable $R(x, y)$ is the new independent variable (like x), $S(x, y)$ is the
gilmore-lie-groups_F03019	299	■ 1.000	$S(x, y)$	$S(x, y)$	canonical variable $R(x, y)$ is the new independent variable (like x), $S(x, y)$ is the
gilmore-lie-groups_F03020	299	■ 1.000	$T(x, y, p)$	$T(x, y, p)$	new dependent variable (like y), and $T(x, y, p)$ is the new constraint between S and
gilmore-lie-groups_F03021	299	■ 1.000	$d y / d x$	dy/dx	R (like dy/dx). In this new coordinate system the surface and constraint equations
gilmore-lie-groups_F03022	299	■ 0.999	$F(R,-, T)=0$	$F(R,-, T)=0$	assume the desired forms $F(R,-, T)=0$ and $d S / d R=f(R,-, T)$. The system
gilmore-lie-groups_F03023	299	■ 0.999	$d S / d R=f(R,-, T)$	$d S / d R=f(R,-, T)$	assume the desired forms $F(R,-, T)=0$ and $d S / d R=f(R,-, T)$. The system
gilmore-lie-groups_F03024	299	■ 1.000	$y-G(x)=0$	$y-G(x)=0$	If we write the solution in the form $y-G(x)=0$, then the surface $y+c-G(x)=$
gilmore-lie-groups_F03025	299	■ 1.000	$y+c-G(x)=$	$y+c-G(x)=$	If we write the solution in the form $y-G(x)=0$, then the surface $y+c-G(x)=$
gilmore-lie-groups_F03026	299	■ 1.000	$e^{c \partial / \partial y}$	$e^{c \partial / \partial y}$	represented by the Taylor series displacement operator $e^{c \partial / \partial y}$, for
gilmore-lie-groups_F03027	300	■ 1.000	$F(x, y, p)=p-g(x)$	$F(x, y, p)=p-g(x)$	$F(x, y, p)=p-g(x)$ for the particular case at hand. There are two relations
gilmore-lie-groups_F03028	300	■ 1.000	x, y, p	x, y, p	among the three variables x, y, p . They are given by the surface equation and

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gilmore-lie-groups_ F03029	300	■ 1.000	$\partial / \partial x, \partial / \partial y, \partial / \partial p$	$\partial / \partial x, \partial / \partial y, \partial / \partial p$	It is useful to express the action of the three partial derivatives $\partial / \partial x, \partial / \partial y, \partial / \partial p$
gilmore-lie-groups_ F03030	300	■ 1.000	$F(x, y, p)$	$F(x, y, p)$	on the surface $F(x, y, p)$ defining the ordinary differential equation. It is also
gilmore-lie-groups_ F03031	300	■ 0.798	$\left(d^n y / d x^n, n=1 \right)$	$(d^n y / d x^n, n = 1)$	Estabrook, 1976). We do this for equations of first order ($d^n y / d x^n, n = 1$) and
gilmore-lie-groups_ F03032	300	■ 1.000	$p^m = (d y / d x)^m, m=1$	$p^m = (d y / d x)^m, m = 1$	first degree (depends on $p^m = (d y / d x)^m, m = 1$). The results are independent of
gilmore-lie-groups_ F03033	300	■ 1.000	$p - g(x)$	$p - g(x)$	$F(x, y, p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y} F(x, y, p) \neq 0$, then we
gilmore-lie-groups_ F03034	300	■ 1.000	$\frac{\partial}{\partial y} F(x, y, p) \neq 0$	$\frac{\partial}{\partial y} F(x, y, p) \neq 0$	$F(x, y, p) = 0$, is not of the form $p - g(x)$, so that $\frac{\partial}{\partial y} F(x, y, p) \neq 0$, then we
gilmore-lie-groups_ F03035	300	■ 0.999	(R, S, T)	(R, S, T)	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$
gilmore-lie-groups_ F03036	300	■ 0.999	$R = R(x, y)$	$R = R(x, y)$	(ii) A new “canonical” coordinate system (R, S, T) . In this coordinate system $R = R(x, y)$
gilmore-lie-groups_ F03037	300	■ 1.000	$S = S(x, y)$	$S = S(x, y)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03038	300	■ 1.000	$T = T(x, y, p)$	$T = T(x, y, p)$	is the independent variable, $S = S(x, y)$ is the dependent variable, and $T = T(x, y, p)$
gilmore-lie-groups_ F03039	301	■ 1.000	$x \rightarrow x$	$x \rightarrow x$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03040	301	■ 1.000	$y \rightarrow y + \epsilon$	$y \rightarrow y + \epsilon$	In the simplest case Eq. (16.1), this one-parameter group is $x \rightarrow x$ and $y \rightarrow y + \epsilon$,
gilmore-lie-groups_ F03041	301	■ 1.000	$\xi = 0, \eta = 1$	$\xi = 0, \eta = 1$	so that $\xi = 0, \eta = 1$, and $\zeta = 0$.
gilmore-lie-groups_ F03042	301	■ 1.000	$\zeta(x, y, p)$	$\zeta(x, y, p)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03043	301	■ 1.000	$\xi(x, y)$	$\xi(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03044	301	■ 1.000	$\eta(x, y)$	$\eta(x, y)$	The function $\zeta(x, y, p)$ is not independent of the functions $\xi(x, y)$ and $\eta(x, y)$. The
gilmore-lie-groups_ F03045	301	■ 1.000	$\eta_x = \partial \eta / \partial x$	$\eta_x = \partial \eta / \partial x$	to first order in ϵ , where $\eta_x = \partial \eta / \partial x$, etc. As a result
gilmore-lie-groups_ F03046	302	■ 1.000	$\xi(x, y), \eta(x, y)$	$\xi(x, y), \eta(x, y)$	Specifically, these equations are used to determine the functions $\xi(x, y), \eta(x, y)$,
gilmore-lie-groups_ F03047	302	■ 1.000	$y: p = p(x, y)$	$y: p = p(x, y)$	x and $y: p = p(x, y)$. This expression is substituted into the determining equation
gilmore-lie-groups_ F03048	302	■ 1.000	$X F(x, y, p(x, y)) = 0$	$X F(x, y, p(x, y)) = 0$	$X F(x, y, p(x, y)) = 0$, so that this equation depends only on two independent vari-
gilmore-lie-groups_ F03049	302	■ 1.000	d_ξ, d_η	d_ξ, d_η	and similarly for η . These representations are truncated at finite degrees d_ξ, d_η . The
gilmore-lie-groups_ F03050	302	■ 0.999	$X F = 0$	$X F = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03051	302	■ 0.999	$\sum C_{ij} x^i y^j = 0$	$\sum C_{ij} x^i y^j = 0$	determining equation $X F = 0$ is expanded into the form $\sum C_{ij} x^i y^j = 0$. Each
gilmore-lie-groups_ F03052	302	■ 1.000	C_{ij}	C_{ij}	coefficient C_{ij} must vanish separately, by standard linear independence arguments.
gilmore-lie-groups_ F03053	302	■ 0.999	ξ_{ij}, η_{ij}	ξ_{ij}, η_{ij}	ξ_{ij}, η_{ij} . In general, there are more equations than unknowns. Since the equations
gilmore-lie-groups_ F03054	303	■ 1.000	$X(x, y, p) S(x, y) = 1$	$X(x, y, p) S(x, y) = 1$	$X(x, y, p) S(x, y) = 1$. We require S to be independent of p , so the condition
gilmore-lie-groups_ F03055	303	■ 1.000	$X S = +1$	$X S = +1$	added to the solution. Further, it is not important that $X S = +1$: we could just as

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gilmore-lie-groups_F03056	303	■ 1.000	$X \ S=-1$	$XS = -1$	well choose a solution satisfying $XS = -1$ or, for that matter, $XS = k \neq 0$, where
gilmore-lie-groups_F03057	303	■ 1.000	$X \ S=k \ \neq \ 0$	$XS = k \neq 0$	well choose a solution satisfying $XS = -1$ or, for that matter, $XS = k \neq 0$, where
gilmore-lie-groups_F03058	303	■ 1.000	$X \ R=0$	$XR = 0$	transformation group. These functions obey $XR = 0$ and $XT = 0$, which are ex-
gilmore-lie-groups_F03059	303	■ 1.000	$X \ T=0$	$XT = 0$	transformation group. These functions obey $XR = 0$ and $XT = 0$, which are ex-
gilmore-lie-groups_F03060	304	■ 1.000	$R: \ T=T(R)$	$R : T = T(R)$	as a function of R : $T = T(R)$. This expression is used in the constraint equation,
gilmore-lie-groups_F03061	304	■ 1.000	$x=x(R, \ S), \ y=y(R, \ S)$	$x = x(R, S), y = y(R, S)$	The inverse relation $x = x(R, S), y = y(R, S)$ is used to express the solution
gilmore-lie-groups_F03062	305	■ 1.000	$x \ p+y-x \ y^2=0$	$xp + y - xy^2 = 0$	Figure 16.1. The first order ordinary differential equation $xp + y - xy^2 = 0$. Here
gilmore-lie-groups_F03063	305	■ 0.980	$0.1 \ \leq x \ \leq 1.1$	$0.1 \leq x \leq 1.1$	p (vertical) is plotted over the (x, y) plane for $0.1 \leq x \leq 1.1$ and $-2 \leq y \leq +2$.
gilmore-lie-groups_F03064	305	■ 0.980	$-2 \ \leq y \ \leq +2$	$-2 \leq y \leq +2$	p (vertical) is plotted over the (x, y) plane for $0.1 \leq x \leq 1.1$ and $-2 \leq y \leq +2$.
gilmore-lie-groups_F03065	305	■ 1.000	$y \ \rightarrow \alpha \ y$	$y \rightarrow \alpha y$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-
gilmore-lie-groups_F03066	305	■ 1.000	$x \ \rightarrow \beta \ x$	$x \rightarrow \beta x$	formation $y \rightarrow \alpha y$ and $x \rightarrow \beta x$. Under this transformation the equation trans-
gilmore-lie-groups_F03067	305	■ 1.000	$\alpha \left(x \ p+y-(\alpha \ \beta) \ x \ y^2 \right) = 0$	$\alpha (xp + y - (\alpha\beta)xy^2) = 0$	forms to $\alpha(xp + y - (\alpha\beta)xy^2) = 0$. The equation is invariant provided $\alpha\beta = 1$.
gilmore-lie-groups_F03068	305	■ 1.000	$\alpha \beta = 1$	$\alpha\beta = 1$	forms to $\alpha(xp + y - (\alpha\beta)xy^2) = 0$. The equation is invariant provided $\alpha\beta = 1$.
gilmore-lie-groups_F03069	305	■ 1.000	$x \ \rightarrow \lambda \ x, \ y \ \rightarrow \lambda^{-1} \ y, \ p \ \rightarrow \lambda^{-2} \ p$	$x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$	ant is $x \rightarrow \lambda x, y \rightarrow \lambda^{-1} y, p \rightarrow \lambda^{-2} p$. Since there is a one-parameter invariance
gilmore-lie-groups_F03070	305	■ 1.000	$X(x, \ y, \ p)$	$X(x, y, p)$	it is possible to construct the infinitesimal generator $X(x, y, p)$ from this group
gilmore-lie-groups_F03071	305	■ 1.000	$p=y^2-y \ / \ x$	$p = y^2 - y/x$	The surface $p = y^2 - y/x$ is shown in Fig. 16.1. The value of p clearly depends
gilmore-lie-groups_F03072	306	■ 1.000	$p: \ p(x, \ y)=$	$p : p(x, y) =$	we use the surface equation $F(x, y, p) = 0$ to find an expression for p : $p(x, y) =$
gilmore-lie-groups_F03073	306	■ 1.000	$-y \ / \ x+y^2$	$-y/x + y^2$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_F03074	306	■ 1.000	$X \ F(x, \ y, \ p)=0$	$XF(x, y, p) = 0$	$-y/x + y^2$. This is substituted into the determining equation $XF(x, y, p) = 0$ to
gilmore-lie-groups_F03075	306	■ 1.000	$\eta: \ \xi=\xi_{00}, \ \eta=\eta_{00}$	$\eta : \xi = \xi_{00}, \eta = \eta_{00}$	We first attempt zeroth degree expressions for ξ and η : $\xi = \xi_{00}, \eta = \eta_{00}$. When these
gilmore-lie-groups_F03076	306	■ 0.999	$y \ / \ x, \ 1$	$y/x, 1$	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_F03077	306	■ 0.999	$x \ y$	xy	coefficients of the monomials $y/x, 1$, and xy depend on the unknown parameters
gilmore-lie-groups_F03078	306	■ 1.000	$\xi_{00}, \ \eta_{00}$	ξ_{00}, η_{00}	ξ_{00}, η_{00} as follows:
gilmore-lie-groups_F03079	307	■ 1.000	$\xi_{01}=\xi_{00}=\eta_{00}=\eta_{10}=0$	$\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$	four equations we determine $\xi_{01} = \xi_{00} = \eta_{00} = \eta_{10} = 0$, and from the coefficient
gilmore-lie-groups_F03080	307	■ 1.000	$x \ y^2$	xy^2	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_F03081	307	■ 1.000	$-\xi_{10}-\eta_{01}=0$	$-\xi_{10} - \eta_{01} = 0$	of xy^2 we learn $-\xi_{10} - \eta_{01} = 0$ so that, up to some overall scaling factor, we can
gilmore-lie-groups_F03082	307	■ 1.000	$\xi(x, \ y)=x$	$\xi(x, y) = x$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution

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gilmore-lie-groups_ F03083	307	■ 1.000	$\eta(x, y)=-y$	$\eta(x, y) = -y$	take $\xi(x, y) = x$ and $\eta(x, y) = -y$. Since we have found one nontrivial solution
gilmore-lie-groups_ F03084	307	■ 1.000	$\zeta=-2\ p$	$\zeta = -2p$	(16.8) gives $\zeta = -2p$, so that the generator of infinitesimal displacements is
gilmore-lie-groups_ F03085	307	■ 1.000	$-y\ d\ S(y)\ /\ d\ y=1$	$-y dS(y)/dy = 1$	y. Such a solution can be found if the equation $-y dS(y)/dy = 1$ can be solved.
gilmore-lie-groups_ F03086	307	■ 1.000	$-\ln\ (y)$	$-\ln(y)$	The solution, up to an additive constant, is $-\ln(y)$. We will adopt this solution,
gilmore-lie-groups_ F03087	307	■ 1.000	$S(x, y)=\ln\ (y)$	$S(x, y) = \ln(y)$	neglecting the negative sign: $S(x, y) = \ln(y)$.
gilmore-lie-groups_ F03088	307	■ 1.000	$y\ d\ x=-x\ d\ y$	$ydx = -xdy$	The first equation for the new independent variable simplifies to $ydx = -xdy$ or
gilmore-lie-groups_ F03089	307	■ 1.000	$d(x\ y)=0$	$d(xy) = 0$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03090	307	■ 1.000	$R(x, y)=x\ y$	$R(x, y) = xy$	$d(xy) = 0$, from which we conclude that $R(x, y) = xy$ is an invariant coordinate
gilmore-lie-groups_ F03091	307	■ 1.000	$-d\ p\ /\ 2\ p$	$-dp/2p$	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to
gilmore-lie-groups_ F03092	307	■ 1.000	$d\ x\ /\ x$	dx/x	$-dp/2p$ equal to either of the other two differentials. We set it equal to dx/x to
gilmore-lie-groups_ F03093	307	■ 1.000	$d\ x\ /\ x=-d\ p\ /\ 2\ p$	$dx/x = -dp/2p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03094	307	■ 1.000	$(1\ /\ x)\ d\left(x^{\{2\}\ p\right)=0$	$(1/x)d\left(x^2 p\right) = 0$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03095	307	■ 1.000	$T(x, y, p)=x^{\{2\}\ p}$	$T(x, y, p) = x^2 p$	$dx/x = -dp/2p$ and the solution is $(1/x)d(x^2 p) = 0$, so that $T(x, y, p) = x^2 p$.
gilmore-lie-groups_ F03096	308	■ 1.000	$T\ /\ R+1-$	$T/R + 1-$	Figure 16.2. The surface $xp + y - xy^2 = 0$ transforms to the surface $T/R + 1 -$
gilmore-lie-groups_ F03097	308	■ 0.669	$R=0$	$R = 0$	$R = 0$ in canonical coordinates. Here T (vertical) is plotted over the (R, S) plane
gilmore-lie-groups_ F03098	308	■ 0.669	R, S	R, S	$R = 0$ in canonical coordinates. Here T (vertical) is plotted over the (R, S) plane
gilmore-lie-groups_ F03099	308	■ 1.000	$-3\ \leq R\ \leq +4$	$-3 \leq R \leq +4$	for $-3 \leq R \leq +4$ and $-2 \leq S \leq +2$. The function is a simple ruled surface,
gilmore-lie-groups_ F03100	308	■ 1.000	$-2\ \leq S\ \leq +2$	$-2 \leq S \leq +2$	for $-3 \leq R \leq +4$ and $-2 \leq S \leq +2$. The function is a simple ruled surface,
gilmore-lie-groups_ F03101	308	■ 1.000	$T=T(R, S)$	$T = T(R, S)$	pendent of S . This surface $T = T(R, S)$ is plotted in Fig. 16.2. It has the desired
gilmore-lie-groups_ F03102	308	■ 1.000	$R:\ T(R)=R^{\{2\}}-R$	$R : T(R) = R^2 - R$	The surface equation is solved for T as a function of R : $T(R) = R^2 - R$. This
gilmore-lie-groups_ F03103	309	■ 0.999	(R, S)	(R, S)	The inverse transformation, Eq. (16.30), from (R, S) to (x, y) is finally used to
gilmore-lie-groups_ F03104	309	■ 1.000	$x\ d\ /\ d\ x$	xd/dx	Remarks The operator $x\ d/dx$ is the infinitesimal generator for scaling transfor-
gilmore-lie-groups_ F03105	309	■ 1.000	$e^{\{\lambda x\ d\ /\ d\ x\}}\ x=e^{\{\lambda\}}\ x$	$e^{\lambda xd/dx}x = e^{\lambda}x$	mations, since $e^{\lambda xd/dx}x = e^{\lambda}x$. As a result, the infinitesimal generator X has the
gilmore-lie-groups_ F03106	309	■ 0.766	(x, y, p)	(x, y, p)	following effect on the coordinates (x, y, p) :
gilmore-lie-groups_ F03107	309	■ 1.000	$\ln\ (y)$	$\ln(y)$	From this scaling behavior, it is easy to see that $\ln(y)$ is linear in the Lie translation
gilmore-lie-groups_ F03108	309	■ 1.000	$\ln\ \left(e^{-\{\lambda\}}\ y\right)=\ln\ (y)-\{\lambda\}$	$\ln\left(e^{-\lambda}y\right) = \ln(y) - \lambda$	group parameter: $\ln(e^{-\lambda}y) = \ln(y) - \lambda$. The invariant operators come right out of
gilmore-lie-groups_ F03109	309	■ 1.000	$x^{\{2\}\ p}$	$x^2 p$	the scaling transformations: xy and $x^2 p$ are unchanged by the scaling transforma-

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gilmore-lie-groups_ F03110	309	■ 1.000	$\ln \left(x y^2 \right)$	$\ln \left(x y^2 \right)$	tion. None of these operators is unique. The operator $\ln(xy^2)$ is linear and x^3yp
gilmore-lie-groups_ F03111	309	■ 1.000	$x^3 y p$	x^3yp	tion. None of these operators is unique. The operator $\ln(xy^2)$ is linear and x^3yp
gilmore-lie-groups_ F03112	309	■ 1.000	$F \left(x y, x^2 p \right) = 0$	$F \left(xy, x^2p \right) = 0$	Eq. (16.36) has the form $F(R, -, T) = 0$ or more simply $F(xy, x^2p) = 0$. The
gilmore-lie-groups_ F03113	309	■ 1.000	$x^2 p = h(x y)$	$x^2p = h(xy)$	$x^2p = h(xy)$ or $dy/dx = x^{-2}h(xy)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03114	309	■ 1.000	$d y / d x = x^{-2} h(x y)$	$dy/dx = x^{-2}h(xy)$	$x^2p = h(xy)$ or $dy/dx = x^{-2}h(xy)$. For the equation studied in Section 16.3,
gilmore-lie-groups_ F03115	309	■ 1.000	$h(z) = -z + z^2$	$h(z) = -z + z^2$	$h(z) = -z + z^2$. For the Riccati equation $dy/dx + y^2 - 2/x^2 = 0$, $h(z) = z^2 - 2$.
gilmore-lie-groups_ F03116	309	■ 1.000	$d y / d x + y^2 - 2 / x^2 = 0, h(z) = z^2 - 2$	$dy/dx + y^2 - 2/x^2 = 0, h(z) = z^2 - 2$	$h(z) = -z + z^2$. For the Riccati equation $dy/dx + y^2 - 2/x^2 = 0$, $h(z) = z^2 - 2$.
gilmore-lie-groups_ F03117	309	■ 1.000	$y'^2 + y^4 - x^{-4} = 0$	$y'^2 + y^4 - x^{-4} = 0$	For example, the first order, second degree equation $y'^2 + y^4 - x^{-4} = 0$ has
gilmore-lie-groups_ F03118	309	■ 1.000	$R^4 + T^2 = 1$	$R^4 + T^2 = 1$	canonical form $R^4 + T^2 = 1$. The original equation has two solution branches
gilmore-lie-groups_ F03119	310	■ 1.000	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$	$p = \pm \sqrt{x^{-4} - y^4}$, corresponding to the two solution branches in the canonical
gilmore-lie-groups_ F03120	310	■ 1.000	$T = \pm \sqrt{1 - R^4}$	$T = \pm \sqrt{1 - R^4}$	coordinate system $T = \pm \sqrt{1 - R^4}$
gilmore-lie-groups_ F03121	310	■ 1.000	$\zeta(x, y, p) = \eta^{(1)}(x, y, p)$	$\zeta(x, y, p) = \eta^{(1)}(x, y, p)$	ulated, as well as the first prolongation $\zeta(x, y, p) = \eta^{(1)}(x, y, p)$. We also present
gilmore-lie-groups_ F03122	310	■ 1.000	$d S / d R$	dS/dR	the canonical coordinates (R, S, T) . Since the constraint equation dS/dR depends
gilmore-lie-groups_ F03123	310	■ 1.000	$S \rightarrow S' = S + c, R' = R$	$S \rightarrow S' = S + c, R' = R$	$S \rightarrow S' = S + c, R' = R$. In the latter case we find $\ln(y) \rightarrow \ln(y) + c = \ln(e^c y) =$
gilmore-lie-groups_ F03124	310	■ 1.000	$\ln (y) \rightarrow \ln (y) + c = \ln \left(e^c y \right) =$	$\ln(y) \rightarrow \ln(y) + c = \ln(e^c y) =$	$S \rightarrow S' = S + c, R' = R$. In the latter case we find $\ln(y) \rightarrow \ln(y) + c = \ln(e^c y) =$
gilmore-lie-groups_ F03125	310	■ 0.995	$\ln (\bar{y}(c))$	$\ln(\bar{y}(c))$	$\ln(\bar{y}(c))$ and since $xy = \bar{x}(c)\bar{y}(c)$, the transformation is $\bar{x}(c) = e^{-c}x$ and $\bar{y}(c) =$
gilmore-lie-groups_ F03126	310	■ 0.995	$x y = \bar{x}(c) \bar{y}(c)$	$xy = \bar{x}(c)\bar{y}(c)$	$\ln(\bar{y}(c))$ and since $xy = \bar{x}(c)\bar{y}(c)$, the transformation is $\bar{x}(c) = e^{-c}x$ and $\bar{y}(c) =$
gilmore-lie-groups_ F03127	310	■ 0.995	$\bar{x}(c) = e^{-c} x$	$\bar{x}(c) = e^{-c}x$	$\ln(\bar{y}(c))$ and since $xy = \bar{x}(c)\bar{y}(c)$, the transformation is $\bar{x}(c) = e^{-c}x$ and $\bar{y}(c) =$
gilmore-lie-groups_ F03128	310	■ 0.995	$\bar{y}(c) =$	$\bar{y}(c) =$	$\ln(\bar{y}(c))$ and since $xy = \bar{x}(c)\bar{y}(c)$, the transformation is $\bar{x}(c) = e^{-c}x$ and $\bar{y}(c) =$
gilmore-lie-groups_ F03129	310	■ 0.999	$e^{+c} y$	$e^{+c}y$	$e^{+c}y$.
gilmore-lie-groups_ F03130	310	■ 0.998	$y^{(2)}$	$y^{(2)}$	involves derivatives with respect to $y^{(2)}$ and is given by
gilmore-lie-groups_ F03131	311	■ 1.000	ξ, η, ζ	ξ, η, ζ	Table 16.1. Infinitesimal generators ξ, η, ζ , canonical coordinates R, S, T , and

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gilmore-lie-groups_ F03132	311	■ 0.852	1 / T	1/T	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03133	311	■ 0.852	1 / a	1/a	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03136	311	■ 0.852	b x-a y	bx − ay	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>−1/b</td><td>0</td><td>ax + by</td><td>bx − ay</td><td>p</td><td>(b − aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>−p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b − 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b−1)</td><td>(bT/R)/(bT − aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T − R)</td></tr><tr><td>x</td><td>−y</td><td>−2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>−p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T − R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T − 2R)</td></tr><tr><td>y</td><td>0</td><td>−p²</td><td>y</td><td>x/y</td><td>x − y/p</td><td>−T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp − y</td><td>T/R²</td></tr><tr><td>−y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan^{−1}(y/x)</td><td>(y − xp)/(x + yp)</td><td>−T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px − y)/x²</td><td>y/x</td><td>x</td><td>(xp − y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² − 2ay</td><td>x/a</td><td>x − ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x − a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 − aT)</td></tr><tr><td>x</td><td>b</td><td>−p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{−1} / [1 − b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>−p²</td><td>y² − 2bx</td><td>y/b</td><td>y − b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p − yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y − xp</td><td>y/x</td><td>1/x</td><td>xp − y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp − xp²</td><td>y/x</td><td>1/y</td><td>y/p − x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>−yp − xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) − ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y − ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>−g'p²</td><td>y</td><td>x/g</td><td>1/p − xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp − f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>−f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x − p)</td><td>y/x^k</td><td>1/x^k</td><td>xp − ky</td><td>−k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p − k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p − kx</td><td>−k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	−1/b	0	ax + by	bx − ay	p	(b − aT)/(a + bT)	x	0	−p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b − 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b−1)	(bT/R)/(bT − aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T − R)	x	−y	−2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	−p	y ² /x	ln y	yp	(T/R)/(2T − R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T − 2R)	y	0	−p ²	y	x/y	x − y/p	−T/R ²	0	x	1	x	y/x	xp − y	T/R ²	−y	x	1 + p ²	√x ² + y ²	tan ^{−1} (y/x)	(y − xp)/(x + yp)	−T/R	1	y/x	(px − y)/x ²	y/x	x	(xp − y)/x ²	1/T	a	x	1	x ² − 2ay	x/a	x − ap	1/(2aT)	a	y	p	x − a ln y	x/a	p/y	(1/a)/(1 − aT)	x	b	−p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ^{−1} / [1 − b(R/T) ^(1/b)]	y	b	−p ²	y ² − 2bx	y/b	y − b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p − yf'	T/e ^{f(R)}	x ²	xy	y − xp	y/x	1/x	xp − y	1/T	xy	y ²	yp − xp ²	y/x	1/y	y/p − x	1/(TR ²)	xy	0	−yp − xp ²	y	(ln x)/y	y/(xp) − ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y − ln y	T/R ²	g(y)	0	−g'p ²	y	x/g	1/p − xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp − f'y	T/f ² (R)	f(x)	0	−f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x − p)	y/x ^k	1/x ^k	xp − ky	−k/T	kxy ^k	y ^{k+1}	y ^k (p − k ² xp ² /y)	x/y ^k	1/y ^k	y/p − kx	−k/T
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gilmore-lie-groups_ F03137	311	■ 0.852	(b-a T)/(a+b T)	(b − aT)/(a + bT)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>−1/b</td><td>0</td><td>ax + by</td><td>bx − ay</td><td>p</td><td>(b − aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>−p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b − 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b−1)</td><td>(bT/R)/(bT − aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T − R)</td></tr><tr><td>x</td><td>−y</td><td>−2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>−p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T − R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T − 2R)</td></tr><tr><td>y</td><td>0</td><td>−p²</td><td>y</td><td>x/y</td><td>x − y/p</td><td>−T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp − y</td><td>T/R²</td></tr><tr><td>−y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan^{−1}(y/x)</td><td>(y − xp)/(x + yp)</td><td>−T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px − y)/x²</td><td>y/x</td><td>x</td><td>(xp − y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² − 2ay</td><td>x/a</td><td>x − ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x − a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 − aT)</td></tr><tr><td>x</td><td>b</td><td>−p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{−1} / [1 − b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>−p²</td><td>y² − 2bx</td><td>y/b</td><td>y − b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p − yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y − xp</td><td>y/x</td><td>1/x</td><td>xp − y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp − xp²</td><td>y/x</td><td>1/y</td><td>y/p − x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>−yp − xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) − ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y − ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>−g'p²</td><td>y</td><td>x/g</td><td>1/p − xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp − f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>−f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x − p)</td><td>y/x^k</td><td>1/x^k</td><td>xp − ky</td><td>−k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p − k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p − kx</td><td>−k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	−1/b	0	ax + by	bx − ay	p	(b − aT)/(a + bT)	x	0	−p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b − 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b−1)	(bT/R)/(bT − aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T − R)	x	−y	−2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	−p	y ² /x	ln y	yp	(T/R)/(2T − R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T − 2R)	y	0	−p ²	y	x/y	x − y/p	−T/R ²	0	x	1	x	y/x	xp − y	T/R ²	−y	x	1 + p ²	√x ² + y ²	tan ^{−1} (y/x)	(y − xp)/(x + yp)	−T/R	1	y/x	(px − y)/x ²	y/x	x	(xp − y)/x ²	1/T	a	x	1	x ² − 2ay	x/a	x − ap	1/(2aT)	a	y	p	x − a ln y	x/a	p/y	(1/a)/(1 − aT)	x	b	−p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ^{−1} / [1 − b(R/T) ^(1/b)]	y	b	−p ²	y ² − 2bx	y/b	y − b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p − yf'	T/e ^{f(R)}	x ²	xy	y − xp	y/x	1/x	xp − y	1/T	xy	y ²	yp − xp ²	y/x	1/y	y/p − x	1/(TR ²)	xy	0	−yp − xp ²	y	(ln x)/y	y/(xp) − ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y − ln y	T/R ²	g(y)	0	−g'p ²	y	x/g	1/p − xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp − f'y	T/f ² (R)	f(x)	0	−f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x − p)	y/x ^k	1/x ^k	xp − ky	−k/T	kxy ^k	y ^{k+1}	y ^k (p − k ² xp ² /y)	x/y ^k	1/y ^k	y/p − kx	−k/T
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gilmore-lie-groups_ F03139	311	0.852	\ln x	\ln x	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03140	311	0.852	$x \ p$	xp	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
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0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03143	311	■ 0.852	x / a	x/a	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03144	311	■ 0.852	y / b	y/b	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03145	311	■ 0.852	(1 / b-1 / a) p	(1/b - 1/a)p	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R																																																																																																																																																																																																																								
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xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)																																																																																																																																																																																																																								
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03147	311	0.852	$b \ln y$	$b \ln y$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - 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R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
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a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
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$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f' e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F \ (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G \ (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03151	311	0.852	$(T / R) / (T-R)$	$(T/R)/(T - R)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f' e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F \ (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G \ (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03154	311	0.852	2 x	2x	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√(x² + y²)</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√(x ² + y ²)	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]																																																																																																																																																																																																																								
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0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03155	311	0.852	y^{2} / x	y ² /x	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√(x² + y²)</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√(x ² + y ²)	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03157	311	0.852	(T / R) / (2 T-R)	(T/R)/(2T - R)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√(x² + y²)</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√(x ² + y ²)	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_F03159	311	0.852	y / x^{2}	y/x^2	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax+by</td><td>bx-ay</td><td>p</td><td>(b-aT)/(a+bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b-1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT-aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T-R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T+R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T-R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T-2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x-y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp-y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1+p^2</td><td>sqrt(x^2+y^2)</td><td>tan^-1(y/x)</td><td>(y-xp)/(x+yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px-y)/x^2</td><td>y/x</td><td>x</td><td>(xp-y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>x-ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x-a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1-aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^-1/[1-b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2-2bx</td><td>y/b</td><td>y-b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^f(x)</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p-yf'</td><td>T/e^f(R)</td></tr><tr><td>x^2</td><td>xy</td><td>y-xp</td><td>y/x</td><td>1/x</td><td>xp-y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp-xp^2</td><td>y/x</td><td>1/y</td><td>y/p-x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp-xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp)-ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y+xp</td><td>x</td><td>(ln y)/x</td><td>xp/y-ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p-xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp-f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f=1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g=1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x-p)</td><td>y/x^k</td><td>1/x^k</td><td>xp-ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p-k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p-kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax+by	bx-ay	p	(b-aT)/(a+bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b-1/a)p	y^b/x^a	b ln y	p/x^(a/b-1)	(bT/R)/(bT-aR^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T-R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T+R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T-R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T-2R)	y	0	-p^2	y	x/y	x-y/p	-T/R^2	0	x	1	x	y/x	xp-y	T/R^2	-y	x	1+p^2	sqrt(x^2+y^2)	tan^-1(y/x)	(y-xp)/(x+yp)	-T/R	1	y/x	(px-y)/x^2	y/x	x	(xp-y)/x^2	1/T	a	x	1	x^2-2ay	x/a	x-ap	1/(2aT)	a	y	p	x-a ln y	x/a	p/y	(1/a)/(1-aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^-1/[1-b(R/T)^(1/b)]	y	b	-p^2	y^2-2bx	y/b	y-b/p	1/(2bT)	0	e^f(x)	f'e^f	x	y/e^f	p-yf'	T/e^f(R)	x^2	xy	y-xp	y/x	1/x	xp-y	1/T	xy	y^2	yp-xp^2	y/x	1/y	y/p-x	1/(TR^2)	xy	0	-yp-xp^2	y	(ln x)/y	y/(xp)-ln x	T/R^2	0	xy	y+xp	x	(ln y)/x	xp/y-ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p-xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp-f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f=1)	pf	1/T	0	g(y)	g'p	x	G (G'g=1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x-p)	y/x^k	1/x^k	xp-ky	-k/T	kxy^k	y^{k+1}	y^k(p-k^2 xp^2/y)	x/y^k	1/y^k	y/p-kx	-k/T
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x/a	y/b	(1/b-1/a)p	y^b/x^a	b ln y	p/x^(a/b-1)	(bT/R)/(bT-aR^(1/b))																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03160	311	0.852	$p \ / \ x$	p/x	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03161	311	0.852	$(T \ / \ R) / (T-2 \ R)$	$(T/R)/(T - 2R)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T/R)/(T + R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1 + p^2$</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1} / [1 - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03164	311	0.852	-T / R^{2}	-T/R^2	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>√x^2 + y^2</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	√x^2 + y^2	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03165	311	0.852	x p-y	xp - y	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>√x^2 + y^2</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	√x^2 + y^2	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_F03168	311	0.852	\sqrt{x^2+y^2}	$\sqrt{x^2+y^2}$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_F03169	311	0.852	\tan ^{-1}(y / x)	$\tan^{-1}(y/x)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03170	311	■0.852	(y-x p) /(x+y p)	(y - xp)/(x + yp)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03171	311	■0.852	-T / R	-T/R	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))																																																																																																																																																																																																																								
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-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03172	311	■ 0.852	(p x-y) / x^{2}	(px - y)/x^2	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1}/[1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1}/[1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(T R^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03173	311	■ 0.852	(x p-y) / x^{2}	(xp - y)/x^2	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1}/[1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1}/[1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(T R^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03175	311	■ 0.852	$x - a p$	$x - ap$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
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a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03176	311	0.852	$1/(2aT)$	$1/(2aT)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03177	311	0.852	$x-a \ln y$	$x-a \ln y$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03178	311	■ 0.852	(1 / a) /(1-a T)	(1/a)/(1 - aT)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>$y^2 - 2bx$</td><td>y/b</td><td>$y - b/p$</td><td>1/(2bT)</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x²</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>1/x</td><td>$xp - y$</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>$yp - xp^2$</td><td>y/x</td><td>1/y</td><td>$y/p - x$</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>$y/(xp) - \ln x$</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>(ln y)/x</td><td>$xp/y - \ln y$</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>1/x^k</td><td>$xp - ky$</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>1/y^k</td><td>$y/p - kx$</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	1/a	-1/b	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	-p	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	-y	-2p	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	2x	y	-p	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	2y	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	-p ²	y	x/y	$x - y/p$	-T/R ²	0	x	1	x	y/x	$xp - y$	T/R ²	-y	x	1 + p ²	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	-T/R	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	1/T	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	1/(2aT)	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p ²	$y^2 - 2bx$	y/b	$y - b/p$	1/(2bT)	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x ²	xy	$y - xp$	y/x	1/x	$xp - y$	1/T	xy	y ²	$yp - xp^2$	y/x	1/y	$y/p - x$	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	$y/(xp) - \ln x$	T/R ²	0	xy	$y + xp$	x	(ln y)/x	$xp/y - \ln y$	T/R ²	g(y)	0	-g'p ²	y	x/g	$1/p - xg'/g$	T/g(R)	0	f(x)	f'	x	y/f	$fp - f'y$	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	$x^k(k^2y/x - p)$	y/x^k	1/x ^k	$xp - ky$	-k/T	kxy ^k	y ^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	1/y ^k	$y/p - kx$	-k/T
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gilmore-lie-groups_ F03179	311	■ 0.852	e^{y} / x^{b}	e ^y /x ^b	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - R)$</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T + R)$</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T - R)$</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T - 2R)$</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>$x - y/p$</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp - y$</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y - xp)/(x + yp)$</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>$(px - y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp - y)/x^2$</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2 - 2ay$</td><td>x/a</td><td>$x - ap$</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x - a \ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1 - aT)$</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>$y^2 - 2bx$</td><td>y/b</td><td>$y - b/p$</td><td>1/(2bT)</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x²</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>1/x</td><td>$xp - y$</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>$yp - xp^2$</td><td>y/x</td><td>1/y</td><td>$y/p - x$</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>$y/(xp) - \ln x$</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>(ln y)/x</td><td>$xp/y - \ln y$</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>1/x^k</td><td>$xp - ky$</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>1/y^k</td><td>$y/p - kx$</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	1/a	-1/b	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	-p	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	-y	-2p	xy	$\ln y$	x^2p	$(T/R)/(T + R)$	2x	y	-p	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	2y	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	-p ²	y	x/y	$x - y/p$	-T/R ²	0	x	1	x	y/x	$xp - y$	T/R ²	-y	x	1 + p ²	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	-T/R	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	1/T	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	1/(2aT)	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	-p	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	-p ²	$y^2 - 2bx$	y/b	$y - b/p$	1/(2bT)	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x ²	xy	$y - xp$	y/x	1/x	$xp - y$	1/T	xy	y ²	$yp - xp^2$	y/x	1/y	$y/p - x$	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	$y/(xp) - \ln x$	T/R ²	0	xy	$y + xp$	x	(ln y)/x	$xp/y - \ln y$	T/R ²	g(y)	0	-g'p ²	y	x/g	$1/p - xg'/g$	T/g(R)	0	f(x)	f'	x	y/f	$fp - f'y$	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	$x^k(k^2y/x - p)$	y/x^k	1/x ^k	$xp - ky$	-k/T	kxy ^k	y ^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	1/y ^k	$y/p - kx$	-k/T
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03181	311	0.852	$(bR)^{-1}/$	$(bR)^{-1}/$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03182	311	■0.852	\left[1-b(R / T)^{\{(1 / b)\}}\right]	$\left[1-b(R / T)^{(1 / b)}\right]$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>$d S / d R$</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1 / T$</td></tr><tr><td>$1 / a$</td><td>$-1 / b$</td><td>0</td><td>$a x+b y$</td><td>$b x-a y$</td><td>p</td><td>$(b-a T) /(a+b T)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>$x p$</td><td>$1 / T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p / y</td><td>T</td></tr><tr><td>x / a</td><td>y / b</td><td>$(1 / b-1 / a) p$</td><td>y^b / x^a</td><td>$b \ln y$</td><td>$p / x^{(a / b-1)}$</td><td>$(b T / R) /(b T-a R^{(1 / b)})$</td></tr><tr><td>$x$</td><td>$y$</td><td>0</td><td>$y / x$</td><td>$\ln y$</td><td>$p$</td><td>$(T / R) /(T-R)$</td></tr><tr><td>$x$</td><td>$-y$</td><td>$-2 p$</td><td>$x y$</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T / R) /(T+R)$</td></tr><tr><td>$2 x$</td><td>$y$</td><td>$-p$</td><td>$y^2 / x$</td><td>$\ln y$</td><td>$y p$</td><td>$(T / R) /(2 T-R)$</td></tr><tr><td>$x$</td><td>$2 y$</td><td>$p$</td><td>$y / x^2$</td><td>$\ln y$</td><td>$p / x$</td><td>$(T / R) /(T-2 R)$</td></tr><tr><td>$y$</td><td>0</td><td>$-p^2$</td><td>$y$</td><td>$x / y$</td><td>$x-y / p$</td><td>$-T / R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y / x</td><td>$x p-y$</td><td>T / R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan ^{-1}(y / x)$</td><td>$(y-x p) /(x+y p)$</td><td>$-T / R$</td></tr><tr><td>1</td><td>y / x</td><td>$(p x-y) / x^2$</td><td>y / x</td><td>x</td><td>$(x p-y) / x^2$</td><td>$1 / T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>$x^2-2 a y$</td><td>x / a</td><td>$x-a p$</td><td>$1 /(2 a T)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a \ln y$</td><td>x / a</td><td>p / y</td><td>$(1 / a) /(1-a T)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y / x^b</td><td>y / b</td><td>$e^y * p^b$</td><td>$(b R)^{-1} /$ $[1-b(R / T)^{(1 / b)}]$</td></tr><tr><td>y</td><td>b</td><td>$-p^2$</td><td>$y^2-2 b x$</td><td>y / b</td><td>$y-b / p$</td><td>$1 /(2 b T)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f' e^f$</td><td>x</td><td>y / e^f</td><td>$p-y f'$</td><td>$T / e^{f(R)}$</td></tr><tr><td>x^2</td><td>$x y$</td><td>$y-x p$</td><td>y / x</td><td>$1 / x$</td><td>$x p-y$</td><td>$1 / T$</td></tr><tr><td>$x y$</td><td>y^2</td><td>$y p-x p^2$</td><td>y / x</td><td>$1 / y$</td><td>$y / p-x$</td><td>$1 /(T R^2)$</td></tr><tr><td>$x y$</td><td>0</td><td>$-y p-x p^2$</td><td>y</td><td>$(\ln x) / y$</td><td>$y /(x p)-\ln x$</td><td>T / R^2</td></tr><tr><td>0</td><td>$x y$</td><td>$y+x p$</td><td>x</td><td>$(\ln y) / x$</td><td>$x p / y-\ln y$</td><td>T / R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x / g</td><td>$1 / p-x g' / g$</td><td>$T / g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y / f</td><td>$f p-f'y$</td><td>$T / f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F\left(F' f=1\right)$</td><td>$p f$</td><td>$1 / T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G\left(G' g=1\right)$</td><td>p / g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$k x^k y$</td><td>$x^k\left(k^2 y / x-p\right)$</td><td>$y / x^k$</td><td>$1 / x^k$</td><td>$x p-k y$</td><td>$-k / T$</td></tr><tr><td>$k x y^k$</td><td>$y^{k+1}$</td><td>$y^k\left(p-k^2 x p^2 / y\right)$</td><td>$x / y^k$</td><td>$1 / y^k$</td><td>$y / p-k x$</td><td>$-k / T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	$d S / d R$	0	1	0	x	y	p	T	1	0	0	y	x	p	$1 / T$	$1 / a$	$-1 / b$	0	$a x+b y$	$b x-a y$	p	$(b-a T) /(a+b T)$	x	0	$-p$	y	$\ln x$	$x p$	$1 / T$	0	y	p	x	$\ln y$	p / y	T	x / a	y / b	$(1 / b-1 / a) p$	y^b / x^a	$b \ln y$	$p / x^{(a / b-1)}$	$(b T / R) /(b T-a R^{(1 / b)})$	x	y	0	y / x	$\ln y$	p	$(T / R) /(T-R)$	x	$-y$	$-2 p$	$x y$	$\ln y$	$x^2 p$	$(T / R) /(T+R)$	$2 x$	y	$-p$	y^2 / x	$\ln y$	$y p$	$(T / R) /(2 T-R)$	x	$2 y$	p	y / x^2	$\ln y$	p / x	$(T / R) /(T-2 R)$	y	0	$-p^2$	y	x / y	$x-y / p$	$-T / R^2$	0	x	1	x	y / x	$x p-y$	T / R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan ^{-1}(y / x)$	$(y-x p) /(x+y p)$	$-T / R$	1	y / x	$(p x-y) / x^2$	y / x	x	$(x p-y) / x^2$	$1 / T$	a	x	1	$x^2-2 a y$	x / a	$x-a p$	$1 /(2 a T)$	a	y	p	$x-a \ln y$	x / a	p / y	$(1 / a) /(1-a T)$	x	b	$-p$	e^y / x^b	y / b	$e^y * p^b$	$(b R)^{-1} /$ $[1-b(R / T)^{(1 / b)}]$	y	b	$-p^2$	$y^2-2 b x$	y / b	$y-b / p$	$1 /(2 b T)$	0	$e^{f(x)}$	$f' e^f$	x	y / e^f	$p-y f'$	$T / e^{f(R)}$	x^2	$x y$	$y-x p$	y / x	$1 / x$	$x p-y$	$1 / T$	$x y$	y^2	$y p-x p^2$	y / x	$1 / y$	$y / p-x$	$1 /(T R^2)$	$x y$	0	$-y p-x p^2$	y	$(\ln x) / y$	$y /(x p)-\ln x$	T / R^2	0	$x y$	$y+x p$	x	$(\ln y) / x$	$x p / y-\ln y$	T / R^2	$g(y)$	0	$-g' p^2$	y	x / g	$1 / p-x g' / g$	$T / g(R)$	0	$f(x)$	f'	x	y / f	$f p-f'y$	$T / f^2(R)$	$f(x)$	0	$-f' p$	y	$F\left(F' f=1\right)$	$p f$	$1 / T$	0	$g(y)$	$g' p$	x	$G\left(G' g=1\right)$	p / g	T	x^{k+1}	$k x^k y$	$x^k\left(k^2 y / x-p\right)$	y / x^k	$1 / x^k$	$x p-k y$	$-k / T$	$k x y^k$	y^{k+1}	$y^k\left(p-k^2 x p^2 / y\right)$	x / y^k	$1 / y^k$	$y / p-k x$	$-k / T$
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a	y	p	$x-a \ln y$	x / a	p / y	$(1 / a) /(1-a T)$																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03183	311	■0.852	$y^{\{2\}}-2 \text { b } x$	$y^2-2 b x$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>$d S / d R$</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1 / T$</td></tr><tr><td>$1 / a$</td><td>$-1 / b$</td><td>0</td><td>$a x+b y$</td><td>$b x-a y$</td><td>p</td><td>$(b-a T) /(a+b T)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>$x p$</td><td>$1 / T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p / y</td><td>T</td></tr><tr><td>x / a</td><td>y / b</td><td>$(1 / b-1 / a) p$</td><td>y^b / x^a</td><td>$b \ln y$</td><td>$p / x^{(a / b-1)}$</td><td>$(b T / R) /(b T-a R^{(1 / b)})$</td></tr><tr><td>$x$</td><td>$y$</td><td>0</td><td>$y / x$</td><td>$\ln y$</td><td>$p$</td><td>$(T / R) /(T-R)$</td></tr><tr><td>$x$</td><td>$-y$</td><td>$-2 p$</td><td>$x y$</td><td>$\ln y$</td><td>$x^2 p$</td><td>$(T / R) /(T+R)$</td></tr><tr><td>$2 x$</td><td>$y$</td><td>$-p$</td><td>$y^2 / x$</td><td>$\ln y$</td><td>$y p$</td><td>$(T / R) /(2 T-R)$</td></tr><tr><td>$x$</td><td>$2 y$</td><td>$p$</td><td>$y / x^2$</td><td>$\ln y$</td><td>$p / x$</td><td>$(T / R) /(T-2 R)$</td></tr><tr><td>$y$</td><td>0</td><td>$-p^2$</td><td>$y$</td><td>$x / y$</td><td>$x-y / p$</td><td>$-T / R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y / x</td><td>$x p-y$</td><td>T / R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan ^{-1}(y / x)$</td><td>$(y-x p) /(x+y p)$</td><td>$-T / R$</td></tr><tr><td>1</td><td>y / x</td><td>$(p x-y) / x^2$</td><td>y / x</td><td>x</td><td>$(x p-y) / x^2$</td><td>$1 / 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y)$	$S(x, y)$	$T(x, y, p)$	$d S / d R$	0	1	0	x	y	p	T	1	0	0	y	x	p	$1 / T$	$1 / a$	$-1 / b$	0	$a x+b y$	$b x-a y$	p	$(b-a T) /(a+b T)$	x	0	$-p$	y	$\ln x$	$x p$	$1 / T$	0	y	p	x	$\ln y$	p / y	T	x / a	y / b	$(1 / b-1 / a) p$	y^b / x^a	$b \ln y$	$p / x^{(a / b-1)}$	$(b T / R) /(b T-a R^{(1 / b)})$	x	y	0	y / x	$\ln y$	p	$(T / R) /(T-R)$	x	$-y$	$-2 p$	$x y$	$\ln y$	$x^2 p$	$(T / R) /(T+R)$	$2 x$	y	$-p$	y^2 / x	$\ln y$	$y p$	$(T / R) /(2 T-R)$	x	$2 y$	p	y / x^2	$\ln y$	p / x	$(T / R) /(T-2 R)$	y	0	$-p^2$	y	x / y	$x-y / p$	$-T / R^2$	0	x	1	x	y / x	$x p-y$	T / R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan ^{-1}(y / x)$	$(y-x p) /(x+y p)$	$-T / R$	1	y / x	$(p x-y) / x^2$	y / x	x	$(x p-y) / x^2$	$1 / T$	a	x	1	$x^2-2 a y$	x / a	$x-a p$	$1 /(2 a T)$	a	y	p	$x-a \ln y$	x / a	p / y	$(1 / a) /(1-a T)$	x	b	$-p$	e^y / x^b	y / b	$e^y * p^b$	$(b R)^{-1} /$ $[1-b(R / T)^{(1 / b)}]$	y	b	$-p^2$	$y^2-2 b x$	y / b	$y-b / p$	$1 /(2 b T)$	0	$e^{f(x)}$	$f' e^f$	x	y / e^f	$p-y f'$	$T / e^{f(R)}$	x^2	$x y$	$y-x p$	y / x	$1 / x$	$x p-y$	$1 / T$	$x y$	y^2	$y p-x p^2$	y / x	$1 / y$	$y / p-x$	$1 /(T R^2)$	$x y$	0	$-y p-x p^2$	y	$(\ln x) / y$	$y /(x p)-\ln x$	T / R^2	0	$x y$	$y+x p$	x	$(\ln y) / x$	$x p / y-\ln y$	T / R^2	$g(y)$	0	$-g' p^2$	y	x / g	$1 / p-x g' / g$	$T / g(R)$	0	$f(x)$	f'	x	y / f	$f p-f'y$	$T / f^2(R)$	$f(x)$	0	$-f' p$	y	$F\left(F' f=1\right)$	$p f$	$1 / T$	0	$g(y)$	$g' p$	x	$G\left(G' g=1\right)$	p / g	T	x^{k+1}	$k x^k y$	$x^k\left(k^2 y / x-p\right)$	y / x^k	$1 / x^k$	$x p-k y$	$-k / T$	$k x y^k$	y^{k+1}	$y^k\left(p-k^2 x p^2 / y\right)$	x / y^k	$1 / y^k$	$y / p-k x$	$-k / T$
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$k x y^k$	y^{k+1}	$y^k\left(p-k^2 x p^2 / y\right)$	x / y^k	$1 / y^k$	$y / p-k x$	$-k / T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03184	311	0.852	$y-b / p$	$y-b/p$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03185	311	0.852	$1/(2bT)$	$1/(2bT)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\ (F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\ (G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\ (F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\ (G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03186	311	0.852	e^{\{f(x)\}}	e^{f(x)}	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03187	311	0.852	f^{\{\prime\}} e^{\{f\}}	f'e^f	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03189	311	■0.852	p-y f^{\prime}	p - yf'	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>√x^2 + y^2</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	√x^2 + y^2	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f' p$	y	$F \ (F' f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g' p$	x	$G \ (G' g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03192	311	■ 0.852	y-x p	$y - xp$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]																																																																																																																																																																																																																								
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x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03193	311	■ 0.852	1 / x	$1/x$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03196	311	■0.852	1 / y	1/y	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]																																																																																																																																																																																																																								
y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)																																																																																																																																																																																																																								
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x ²	xy	y - xp	y/x	1/x	xp - y	1/T																																																																																																																																																																																																																								
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x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03197	311	■0.852	y / p-x	y/p - x	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03198	311	0.852	$1/\left(T R^2\right)$	$1/\left(T R^2\right)$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03199	311	0.852	$-y p-x p^2$	$-yp-xp^2$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F(F'f=1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G(G'g=1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F(F'f=1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G(G'g=1)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f' e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(T R^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g' p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f' p$</td><td>y</td><td>$F (F' f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g' p$</td><td>x</td><td>$G (G' g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$	0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$																																																																																																																																																																																																																								
1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$																																																																																																																																																																																																																								
a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$																																																																																																																																																																																																																								
a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$																																																																																																																																																																																																																								
x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1} / [1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$																																																																																																																																																																																																																								
0	$e^{f(x)}$	$f' e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$																																																																																																																																																																																																																								
x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$																																																																																																																																																																																																																								
xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(T R^2)$																																																																																																																																																																																																																								
xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2																																																																																																																																																																																																																								
0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2																																																																																																																																																																																																																								
$g(y)$	0	$-g' p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$																																																																																																																																																																																																																								
0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$																																																																																																																																																																																																																								
$f(x)$	0	$-f' p$	y	$F (F' f = 1)$	pf	$1/T$																																																																																																																																																																																																																								
0	$g(y)$	$g' p$	x	$G (G' g = 1)$	p/g	T																																																																																																																																																																																																																								
x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$																																																																																																																																																																																																																								
kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03282	311	0.852	y+x p	$y + xp$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03283	311	0.852	(\ln y) / x	$(\ln y)/x$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03285	311	■0.852	g(y)	g(y)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03209	311	■0.852	T / g(R)	T/g(R)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1}/[1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f' e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g' p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f' p</td><td>y</td><td>F (F' f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g' p</td><td>x</td><td>G (G' g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1}/[1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f' e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g' p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f' p	y	F (F' f = 1)	pf	1/T	0	g(y)	g' p	x	G (G' g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03211	311	0.852	f^{\prime}	f'	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(TR ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03212	311	0.852	y / f	y/f	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03213	311	0.852	$f \, p - f^{\prime} y$	$fp - f'y$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03215	311	■ 0.852	$-f^{\{\prime\}} p$	$-f'p$	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2 - 2bx$</td><td>$y/b$</td><td>$y - b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p - yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y - xp$</td><td>y/x</td><td>$1/x$</td><td>$xp - y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp - xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p - x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp - xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp) - \ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y + xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y - \ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p - xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp - f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F \ (F'f = 1)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G \ (G'g = 1)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>$kx^k y$</td><td>$x^k(k^2 y/x - p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$																																																																																																																																																																																																																								

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gilmore-lie-groups_03217	311	0.852	$p\ f$	pf	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>e^y*p^b</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\left(F'f=1\right)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\left(G'g=1\right)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	e^y*p^b	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\left(F'f=1\right)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\left(G'g=1\right)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03218	311	■ 0.852	g^{\prime} p	g'p	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>sqrt(x^2 + y^2)</td><td>tan^-1(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^-1/[1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^f(x)</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^f(R)</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^(k+1)</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^(k+1)</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^(a/b-1)	(bT/R)/(bT - aR^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	sqrt(x^2 + y^2)	tan^-1(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^-1/[1 - b(R/T)^(1/b)]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^f(x)	f'e^f	x	y/e^f	p - yf'	T/e^f(R)	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^(k+1)	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^(k+1)	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03219	311	■ 0.852	G\left(G^{\prime} g=1\right)	G (G'g = 1)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>sqrt(x^2 + y^2)</td><td>tan^-1(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^-1/[1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^f(x)</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^f(R)</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^(k+1)</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^(k+1)</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^(a/b-1)	(bT/R)/(bT - aR^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	sqrt(x^2 + y^2)	tan^-1(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^-1/[1 - b(R/T)^(1/b)]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^f(x)	f'e^f	x	y/e^f	p - yf'	T/e^f(R)	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^(k+1)	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^(k+1)	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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-y	x	1 + p^2	sqrt(x^2 + y^2)	tan^-1(y/x)	(y - xp)/(x + yp)	-T/R																																																																																																																																																																																																																								
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y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03221	311	0.852	x^{k+1}	x ^{k+1}	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03223	311	■ 0.852	x^{\wedge\{k\}}\backslash\mathrm{left}(k^{\wedge\{2\}} y / x\cdot p\backslash\mathrm{right})	x^k (k^2 y/x - p)	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>√x^2 + y^2</td><td>tan^{-1}(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1}/[1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	√x^2 + y^2	tan^{-1}(y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1}/[1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(T R^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03224	311	■ 0.852	$y / x^{\{k\}}$	y/x^k	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)																																																																																																																																																																																																																								
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gilmore-lie-groups_ F03227	311	■ 0.852	-k / T	-k/T	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>ξ(x, y)</th><th>η(x, y)</th><th>ζ(x, y, p)</th><th>R(x, y)</th><th>S(x, y)</th><th>T(x, y, p)</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^(a/b-1)</td><td>(bT/R)/(bT - aR^(1/b))</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x²p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y²/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x²</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p²</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R²</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R²</td></tr><tr><td>-y</td><td>x</td><td>1 + p²</td><td>√x² + y²</td><td>tan⁻¹(y/x)</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x²</td><td>y/x</td><td>x</td><td>(xp - y)/x²</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x² - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)⁻¹ / [1 - b(R/T)^(1/b)]</td></tr><tr><td>y</td><td>b</td><td>-p²</td><td>y² - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x²</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y²</td><td>yp - xp²</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(T R²)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp²</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R²</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R²</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p²</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f²(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>x^k(k²y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k²xp²/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	ξ(x, y)	η(x, y)	ζ(x, y, p)	R(x, y)	S(x, y)	T(x, y, p)	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y ^b /x ^a	b ln y	p/x ^(a/b-1)	(bT/R)/(bT - aR ^(1/b))	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x ² p	(T/R)/(T + R)	2x	y	-p	y ² /x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x ²	ln y	p/x	(T/R)/(T - 2R)	y	0	-p ²	y	x/y	x - y/p	-T/R ²	0	x	1	x	y/x	xp - y	T/R ²	-y	x	1 + p ²	√x ² + y ²	tan ⁻¹ (y/x)	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x ²	y/x	x	(xp - y)/x ²	1/T	a	x	1	x ² - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e ^y /x ^b	y/b	e ^y * p ^b	(bR) ⁻¹ / [1 - b(R/T) ^(1/b)]	y	b	-p ²	y ² - 2bx	y/b	y - b/p	1/(2bT)	0	e ^{f(x)}	f'e ^f	x	y/e ^f	p - yf'	T/e ^{f(R)}	x ²	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y ²	yp - xp ²	y/x	1/y	y/p - x	1/(T R ²)	xy	0	-yp - xp ²	y	(ln x)/y	y/(xp) - ln x	T/R ²	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R ²	g(y)	0	-g'p ²	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f ² (R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x ^{k+1}	kx ^k y	x ^k (k ² y/x - p)	y/x ^k	1/x ^k	xp - ky	-k/T	kxy ^k	y ^{k+1}	y ^k (p - k ² xp ² /y)	x/y ^k	1/y ^k	y/p - kx	-k/T
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p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp - ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p - k^2 xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p - kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax + by$	$bx - ay$	p	$(b - aT)/(a + bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b - 1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT - aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T - R)$	x	$-y$	$-2p$	xy	$\ln y$	$x^2 p$	$(T/R)/(T + R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T - R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T - 2R)$	y	0	$-p^2$	y	x/y	$x - y/p$	$-T/R^2$	0	x	1	x	y/x	$xp - y$	T/R^2	$-y$	x	$1 + p^2$	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	$(y - xp)/(x + yp)$	$-T/R$	1	y/x	$(px - y)/x^2$	y/x	x	$(xp - y)/x^2$	$1/T$	a	x	1	$x^2 - 2ay$	x/a	$x - ap$	$1/(2aT)$	a	y	p	$x - a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03229	311	0.852	$y^{\{k+1\}}$	y^{k+1}	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax + by$</td><td>$bx - ay$</td><td>p</td><td>$(b - aT)/(a + bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b - 1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT - aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T - 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a \ln y$	x/a	p/y	$(1/a)/(1 - aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1 - b(R/T)^{(1/b)}]$	y	b	$-p^2$	$y^2 - 2bx$	y/b	$y - b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p - yf'$	$T/e^{f(R)}$	x^2	xy	$y - xp$	y/x	$1/x$	$xp - y$	$1/T$	xy	y^2	$yp - xp^2$	y/x	$1/y$	$y/p - x$	$1/(TR^2)$	xy	0	$-yp - xp^2$	y	$(\ln x)/y$	$y/(xp) - \ln x$	T/R^2	0	xy	$y + xp$	x	$(\ln y)/x$	$xp/y - \ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p - xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp - f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \ (F'f = 1)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \ (G'g = 1)$	p/g	T	x^{k+1}	$kx^k y$	$x^k(k^2 y/x - p)$	y/x^k	$1/x^k$	$xp - ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p - k^2 xp^2/y)$	x/y^k	$1/y^k$	$y/p - kx$	$-k/T$
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gilmore-lie-groups_ F03231	311	■ 0.852	$x / y^{\{k\}}$	x/y^k	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x,y)$</th><th>$\eta(x,y)$</th><th>$\zeta(x,y,p)$</th><th>$R(x,y)$</th><th>$S(x,y)$</th><th>$T(x,y,p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b\ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln y$</td><td>p</td><td>$(T/R)/(T-R)$</td></tr><tr><td>x</td><td>$-y$</td><td>$-2p$</td><td>xy</td><td>$\ln y$</td><td>x^2p</td><td>$(T/R)/(T+R)$</td></tr><tr><td>$2x$</td><td>y</td><td>$-p$</td><td>y^2/x</td><td>$\ln y$</td><td>yp</td><td>$(T/R)/(2T-R)$</td></tr><tr><td>x</td><td>$2y$</td><td>p</td><td>y/x^2</td><td>$\ln y$</td><td>p/x</td><td>$(T/R)/(T-2R)$</td></tr><tr><td>y</td><td>0</td><td>$-p^2$</td><td>y</td><td>x/y</td><td>$x-y/p$</td><td>$-T/R^2$</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>$xp-y$</td><td>T/R^2</td></tr><tr><td>$-y$</td><td>x</td><td>$1+p^2$</td><td>$\sqrt{x^2+y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>$(y-xp)/(x+yp)$</td><td>$-T/R$</td></tr><tr><td>1</td><td>y/x</td><td>$(px-y)/x^2$</td><td>y/x</td><td>x</td><td>$(xp-y)/x^2$</td><td>$1/T$</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2-2ay</td><td>x/a</td><td>$x-ap$</td><td>$1/(2aT)$</td></tr><tr><td>a</td><td>y</td><td>p</td><td>$x-a\ln y$</td><td>x/a</td><td>p/y</td><td>$(1/a)/(1-aT)$</td></tr><tr><td>x</td><td>b</td><td>$-p$</td><td>e^y/x^b</td><td>y/b</td><td>$e^y * p^b$</td><td>$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$</td></tr><tr><td>$y$</td><td>$b$</td><td>$-p^2$</td><td>$y^2-2bx$</td><td>$y/b$</td><td>$y-b/p$</td><td>$1/(2bT)$</td></tr><tr><td>0</td><td>$e^{f(x)}$</td><td>$f'e^f$</td><td>x</td><td>y/e^f</td><td>$p-yf'$</td><td>$T/e^{f(R)}$</td></tr><tr><td>x^2</td><td>xy</td><td>$y-xp$</td><td>y/x</td><td>$1/x$</td><td>$xp-y$</td><td>$1/T$</td></tr><tr><td>xy</td><td>y^2</td><td>$yp-xp^2$</td><td>y/x</td><td>$1/y$</td><td>$y/p-x$</td><td>$1/(TR^2)$</td></tr><tr><td>xy</td><td>0</td><td>$-yp-xp^2$</td><td>y</td><td>$(\ln x)/y$</td><td>$y/(xp)-\ln x$</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>$y+xp$</td><td>x</td><td>$(\ln y)/x$</td><td>$xp/y-\ln y$</td><td>T/R^2</td></tr><tr><td>$g(y)$</td><td>0</td><td>$-g'p^2$</td><td>y</td><td>x/g</td><td>$1/p-xg'/g$</td><td>$T/g(R)$</td></tr><tr><td>0</td><td>$f(x)$</td><td>f'</td><td>x</td><td>y/f</td><td>$fp-f'y$</td><td>$T/f^2(R)$</td></tr><tr><td>$f(x)$</td><td>0</td><td>$-f'p$</td><td>y</td><td>$F\left(F'f=1\right)$</td><td>pf</td><td>$1/T$</td></tr><tr><td>0</td><td>$g(y)$</td><td>$g'p$</td><td>x</td><td>$G\left(G'g=1\right)$</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^ky</td><td>$x^k(k^2y/x-p)$</td><td>y/x^k</td><td>$1/x^k$</td><td>$xp-ky$</td><td>$-k/T$</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>$y^k(p-k^2xp^2/y)$</td><td>x/y^k</td><td>$1/y^k$</td><td>$y/p-kx$</td><td>$-k/T$</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x,y)$	$\eta(x,y)$	$\zeta(x,y,p)$	$R(x,y)$	$S(x,y)$	$T(x,y,p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	$1/T$	$1/a$	$-1/b$	0	$ax+by$	$bx-ay$	p	$(b-aT)/(a+bT)$	x	0	$-p$	y	$\ln x$	xp	$1/T$	0	y	p	x	$\ln y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b\ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a\ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$	y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F\left(F'f=1\right)$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G\left(G'g=1\right)$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ F03232	311	■ 0.852	1 / y^{k}	1/y^k	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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gilmore-lie-groups_ F03233	311	■ 0.852	y / p-k x	y/p - kx	<table><tr><th colspan="3">Infinitesimal generators</th><th colspan="3">Canonical coordinates</th><th>Constraint</th></tr><tr><th>$\xi(x, y)$</th><th>$\eta(x, y)$</th><th>$\zeta(x, y, p)$</th><th>$R(x, y)$</th><th>$S(x, y)$</th><th>$T(x, y, p)$</th><th>dS/dR</th></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>1/T</td></tr><tr><td>1/a</td><td>-1/b</td><td>0</td><td>ax + by</td><td>bx - ay</td><td>p</td><td>(b - aT)/(a + bT)</td></tr><tr><td>x</td><td>0</td><td>-p</td><td>y</td><td>ln x</td><td>xp</td><td>1/T</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>ln y</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>(1/b - 1/a)p</td><td>y^b/x^a</td><td>b ln y</td><td>p/x^{(a/b-1)}</td><td>(bT/R)/(bT - aR^{(1/b)})</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>ln y</td><td>p</td><td>(T/R)/(T - R)</td></tr><tr><td>x</td><td>-y</td><td>-2p</td><td>xy</td><td>ln y</td><td>x^2 p</td><td>(T/R)/(T + R)</td></tr><tr><td>2x</td><td>y</td><td>-p</td><td>y^2/x</td><td>ln y</td><td>yp</td><td>(T/R)/(2T - R)</td></tr><tr><td>x</td><td>2y</td><td>p</td><td>y/x^2</td><td>ln y</td><td>p/x</td><td>(T/R)/(T - 2R)</td></tr><tr><td>y</td><td>0</td><td>-p^2</td><td>y</td><td>x/y</td><td>x - y/p</td><td>-T/R^2</td></tr><tr><td>0</td><td>x</td><td>1</td><td>x</td><td>y/x</td><td>xp - y</td><td>T/R^2</td></tr><tr><td>-y</td><td>x</td><td>1 + p^2</td><td>$\sqrt{x^2 + y^2}$</td><td>$\tan^{-1}(y/x)$</td><td>(y - xp)/(x + yp)</td><td>-T/R</td></tr><tr><td>1</td><td>y/x</td><td>(px - y)/x^2</td><td>y/x</td><td>x</td><td>(xp - y)/x^2</td><td>1/T</td></tr><tr><td>a</td><td>x</td><td>1</td><td>x^2 - 2ay</td><td>x/a</td><td>x - ap</td><td>1/(2aT)</td></tr><tr><td>a</td><td>y</td><td>p</td><td>x - a ln y</td><td>x/a</td><td>p/y</td><td>(1/a)/(1 - aT)</td></tr><tr><td>x</td><td>b</td><td>-p</td><td>e^y/x^b</td><td>y/b</td><td>e^y * p^b</td><td>(bR)^{-1} / [1 - b(R/T)^{(1/b)}]</td></tr><tr><td>y</td><td>b</td><td>-p^2</td><td>y^2 - 2bx</td><td>y/b</td><td>y - b/p</td><td>1/(2bT)</td></tr><tr><td>0</td><td>e^{f(x)}</td><td>f'e^f</td><td>x</td><td>y/e^f</td><td>p - yf'</td><td>T/e^{f(R)}</td></tr><tr><td>x^2</td><td>xy</td><td>y - xp</td><td>y/x</td><td>1/x</td><td>xp - y</td><td>1/T</td></tr><tr><td>xy</td><td>y^2</td><td>yp - xp^2</td><td>y/x</td><td>1/y</td><td>y/p - x</td><td>1/(TR^2)</td></tr><tr><td>xy</td><td>0</td><td>-yp - xp^2</td><td>y</td><td>(ln x)/y</td><td>y/(xp) - ln x</td><td>T/R^2</td></tr><tr><td>0</td><td>xy</td><td>y + xp</td><td>x</td><td>(ln y)/x</td><td>xp/y - ln y</td><td>T/R^2</td></tr><tr><td>g(y)</td><td>0</td><td>-g'p^2</td><td>y</td><td>x/g</td><td>1/p - xg'/g</td><td>T/g(R)</td></tr><tr><td>0</td><td>f(x)</td><td>f'</td><td>x</td><td>y/f</td><td>fp - f'y</td><td>T/f^2(R)</td></tr><tr><td>f(x)</td><td>0</td><td>-f'p</td><td>y</td><td>F (F'f = 1)</td><td>pf</td><td>1/T</td></tr><tr><td>0</td><td>g(y)</td><td>g'p</td><td>x</td><td>G (G'g = 1)</td><td>p/g</td><td>T</td></tr><tr><td>x^{k+1}</td><td>kx^k y</td><td>x^k(k^2 y/x - p)</td><td>y/x^k</td><td>1/x^k</td><td>xp - ky</td><td>-k/T</td></tr><tr><td>kxy^k</td><td>y^{k+1}</td><td>y^k(p - k^2 xp^2/y)</td><td>x/y^k</td><td>1/y^k</td><td>y/p - kx</td><td>-k/T</td></tr></table>	Infinitesimal generators			Canonical coordinates			Constraint	$\xi(x, y)$	$\eta(x, y)$	$\zeta(x, y, p)$	$R(x, y)$	$S(x, y)$	$T(x, y, p)$	dS/dR	0	1	0	x	y	p	T	1	0	0	y	x	p	1/T	1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)	x	0	-p	y	ln x	xp	1/T	0	y	p	x	ln y	p/y	T	x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})	x	y	0	y/x	ln y	p	(T/R)/(T - R)	x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)	2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)	x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)	y	0	-p^2	y	x/y	x - y/p	-T/R^2	0	x	1	x	y/x	xp - y	T/R^2	-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R	1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T	a	x	1	x^2 - 2ay	x/a	x - ap	1/(2aT)	a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)	x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]	y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)	0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}	x^2	xy	y - xp	y/x	1/x	xp - y	1/T	xy	y^2	yp - xp^2	y/x	1/y	y/p - x	1/(TR^2)	xy	0	-yp - xp^2	y	(ln x)/y	y/(xp) - ln x	T/R^2	0	xy	y + xp	x	(ln y)/x	xp/y - ln y	T/R^2	g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)	0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)	f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T	0	g(y)	g'p	x	G (G'g = 1)	p/g	T	x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T	kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T
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1/a	-1/b	0	ax + by	bx - ay	p	(b - aT)/(a + bT)																																																																																																																																																																																																																								
x	0	-p	y	ln x	xp	1/T																																																																																																																																																																																																																								
0	y	p	x	ln y	p/y	T																																																																																																																																																																																																																								
x/a	y/b	(1/b - 1/a)p	y^b/x^a	b ln y	p/x^{(a/b-1)}	(bT/R)/(bT - aR^{(1/b)})																																																																																																																																																																																																																								
x	y	0	y/x	ln y	p	(T/R)/(T - R)																																																																																																																																																																																																																								
x	-y	-2p	xy	ln y	x^2 p	(T/R)/(T + R)																																																																																																																																																																																																																								
2x	y	-p	y^2/x	ln y	yp	(T/R)/(2T - R)																																																																																																																																																																																																																								
x	2y	p	y/x^2	ln y	p/x	(T/R)/(T - 2R)																																																																																																																																																																																																																								
y	0	-p^2	y	x/y	x - y/p	-T/R^2																																																																																																																																																																																																																								
0	x	1	x	y/x	xp - y	T/R^2																																																																																																																																																																																																																								
-y	x	1 + p^2	$\sqrt{x^2 + y^2}$	$\tan^{-1}(y/x)$	(y - xp)/(x + yp)	-T/R																																																																																																																																																																																																																								
1	y/x	(px - y)/x^2	y/x	x	(xp - y)/x^2	1/T																																																																																																																																																																																																																								
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a	y	p	x - a ln y	x/a	p/y	(1/a)/(1 - aT)																																																																																																																																																																																																																								
x	b	-p	e^y/x^b	y/b	e^y * p^b	(bR)^{-1} / [1 - b(R/T)^{(1/b)}]																																																																																																																																																																																																																								
y	b	-p^2	y^2 - 2bx	y/b	y - b/p	1/(2bT)																																																																																																																																																																																																																								
0	e^{f(x)}	f'e^f	x	y/e^f	p - yf'	T/e^{f(R)}																																																																																																																																																																																																																								
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g(y)	0	-g'p^2	y	x/g	1/p - xg'/g	T/g(R)																																																																																																																																																																																																																								
0	f(x)	f'	x	y/f	fp - f'y	T/f^2(R)																																																																																																																																																																																																																								
f(x)	0	-f'p	y	F (F'f = 1)	pf	1/T																																																																																																																																																																																																																								
0	g(y)	g'p	x	G (G'g = 1)	p/g	T																																																																																																																																																																																																																								
x^{k+1}	kx^k y	x^k(k^2 y/x - p)	y/x^k	1/x^k	xp - ky	-k/T																																																																																																																																																																																																																								
kxy^k	y^{k+1}	y^k(p - k^2 xp^2/y)	x/y^k	1/y^k	y/p - kx	-k/T																																																																																																																																																																																																																								

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gilmore-lie-groups_ F03260	315	■ 1.000	x^i, t	x^i, t	An invariant coordinate depending on the x^i, t and u is $R = ut^{n/2}e^{ x ^2/4t}$, from
gilmore-lie-groups_ F03261	315	■ 1.000	$R=ut^{n/2}e^{ x ^2/4t}$	$R = ut^{n/2}e^{ x ^2/4t}$	An invariant coordinate depending on the x^i, t and u is $R = ut^{n/2}e^{ x ^2/4t}$, from
gilmore-lie-groups_ F03262	315	■ 1.000	$F(R, -, T)=$	$F(R, -, T) =$	3. Under what conditions is it possible to solve the canonical surface equation $F(R, -, T) =$
gilmore-lie-groups_ F03263	315	■ 0.980	$\int f(R, -, T(R)) dR$	$\int f(R, -, T(R))dR$	$\int f(R, -, T(R))dR$
gilmore-lie-groups_ F03264	316	■ 1.000	ξ, η	ξ, η	1830 Galois solve algebraic equations 1874 Lie solve differential equations 1969 Risch integrate in closed form ? – solve determining equations for ξ, η ? – solve determining equations for R, S, T ? – solve $F(R, -, T) = 0$ for R .
gilmore-lie-groups_ F03265	316	■ 1.000	$\zeta(x, y, p)=\eta^{(1)}\left(x, y, y^{(1)}\right)$	$\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)})$	(ii) Construct the first prolongation $\zeta(x, y, p) = \eta^{(1)}(x, y, y^{(1)})$ from the functions defin-
gilmore-lie-groups_ F03266	316	■ 1.000	$X=\xi \partial / \partial x+\eta \partial / \partial y+\zeta \partial / \partial p$	$X = \xi \partial / \partial x + \eta \partial / \partial y + \zeta \partial / \partial p$	(iii) Introduce the operator $X = \xi \partial / \partial x + \eta \partial / \partial y + \zeta \partial / \partial p$. This describes a Taylor series
gilmore-lie-groups_ F03267	316	■ 1.000	$F=0$	$F = 0$	(iv) Solve the determining equation $XF = 0$ when $F = 0$ for the functions $\xi(x, y)$ and
gilmore-lie-groups_ F03268	316	■ 1.000	$X R=0, X S=1, X T=0$	$XR = 0, XS = 1, XT = 0$	(v) Solve the determining equations $XR = 0, XS = 1, XT = 0$ for the canonical coor-
gilmore-lie-groups_ F03269	316	■ 1.000	$F \rightarrow F(R, -, T)=0$	$F \rightarrow F(R, -, T) = 0$	equation is independent of the new dependent variable: $F \rightarrow F(R, -, T) = 0$.
gilmore-lie-groups_ F03270	317	■ 1.000	$S: S=\int f(R, -, T(R))+c$	$S: S = \int f(R, -, T(R)) + c$	(viii) Solve the constraint equation for $S: S = \int f(R, -, T(R)) + c$.
gilmore-lie-groups_ F03271	317	■ 1.000	$x=x(R, S)$	$x = x(R, S)$	(ix) Backsubstitute the original coordinates for the new coordinates, $x = x(R, S)$ and
gilmore-lie-groups_ F03272	317	■ 1.000	$y=y(R, S)$	$y = y(R, S)$	$y = y(R, S)$, to obtain the solution of the original equation.
gilmore-lie-groups_ F03273	317	■ 1.000	$n=0,1$	$n = 0, 1$	Show that the first two terms $n = 0, 1$ are exactly the determining equations
gilmore-lie-groups_ F03274	317	■ 1.000	$V(\mathbf{x})$	$V(\mathbf{x})$	mass m in the presence of a potential $V(\mathbf{x})$ is
gilmore-lie-groups_ F03275	317	■ 1.000	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$	$m \rightarrow \alpha m), \mathbf{x} \rightarrow \beta \mathbf{x}, t \rightarrow \gamma t$. Assume also that the potential is homogeneous of degree
gilmore-lie-groups_ F03276	317	■ 1.000	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$	$k: V(\beta \mathbf{x}) \rightarrow \beta^k V(\mathbf{x})$ (Landau and Lifshitz, 1960). Under this scaling transformation
gilmore-lie-groups_ F03277	317	■ 1.000	$\alpha^1 \beta^{2-k}$	$\alpha^1 \beta^{2-k}$	a. Show that the scaled equation is identical to the original provided $\alpha^1 \beta^{2-k}$
gilmore-lie-groups_ F03278	317	■ 1.000	$\gamma^{-2}=1$	$\gamma^{-2} = 1$	$\gamma^{-2} = 1$.
gilmore-lie-groups_ F03279	317	■ 1.000	$\alpha=1$	$\alpha = 1$	b. Set $\alpha = 1$. Show that trajectories are invariant under the scaling transformation with
gilmore-lie-groups_ F03280	317	■ 1.000	$\gamma^2=\beta^{2-k}$	$\gamma^2 = \beta^{2-k}$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k, = +1, k = +2$ the following
gilmore-lie-groups_ F03281	317	■ 1.000	$k=-1, k=0, k, =+1, k=+2$	$k = -1, k = 0, k, = +1, k = +2$	$\gamma^2 = \beta^{2-k}$. Show that in the cases $k = -1, k = 0, k, = +1, k = +2$ the following

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gilmore-lie-groups_ F03282	318	1.000	$\gamma^2=\beta^3$	$\gamma^2 = \beta^3$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03283	318	1.000	$\gamma^2=\beta^2$	$\gamma^2 = \beta^2$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03284	318	1.000	$\gamma^2=\beta^1$	$\gamma^2 = \beta^1$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03285	318	1.000	$\gamma^2=\beta^0$	$\gamma^2 = \beta^0$	<table><tr><th>k</th><th>Potential type</th><th>Transformation</th></tr><tr><td>-1</td><td>Coulomb</td><td>$\gamma^2 = \beta^3$</td></tr><tr><td>0</td><td>no force</td><td>$\gamma^2 = \beta^2$</td></tr><tr><td>+1</td><td>local gravitational potential</td><td>$\gamma^2 = \beta^1$</td></tr><tr><td>+2</td><td>harmonic oscillator</td><td>$\gamma^2 = \beta^0$</td></tr></table>	k	Potential type	Transformation	-1	Coulomb	$\gamma^2 = \beta^3$	0	no force	$\gamma^2 = \beta^2$	+1	local gravitational potential	$\gamma^2 = \beta^1$	+2	harmonic oscillator	$\gamma^2 = \beta^0$
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gilmore-lie-groups_ F03286	318	0.785	γ^2	γ^2	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'															
gilmore-lie-groups_ F03287	318	0.785	β^3	β^3	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'															
gilmore-lie-groups_ F03288	318	0.785	$R^{\prime}$	R'	square of the period (γ^2) is proportional to the cube of the semiaxis (β^3). If R'															
gilmore-lie-groups_ F03289	318	1.000	$T^{\prime}$	T'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and															
gilmore-lie-groups_ F03290	318	1.000	$P^{\prime}$	P'	and T' are the semiaxis and period of planet P' and R and T are the semiaxis and															
gilmore-lie-groups_ F03291	318	1.000	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$	$\beta^3 \rightarrow (R'/R)^3 = (T'/T)^2 \leftarrow \gamma^2$. The second line is a statement of the integral of															
gilmore-lie-groups_ F03292	318	0.860	(β)	(β)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that															
gilmore-lie-groups_ F03293	318	0.860	(γ)	(γ)	traveled (β) is proportional to the time elapsed (γ). The third line is a statement that															
gilmore-lie-groups_ F03294	318	1.000	$V=m \text{ g } z$	$V = mgz$	in a local gravitational potential of the form $V = mgz$, the distance fallen increases															
gilmore-lie-groups_ F03295	318	1.000	$\gamma=1$	$\gamma = 1$	c. Fix $\gamma = 1$ and construct a table relating the mass and orbital scale under the four															
gilmore-lie-groups_ F03296	318	1.000	$\beta=1$	$\beta = 1$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.															
gilmore-lie-groups_ F03297	318	1.000	$\sqrt{M}$	$\sqrt{M}$	d. Fix $\beta = 1$ and show that the period scales like $\sqrt{M}$ for all homogeneous potentials.															
gilmore-lie-groups_ F03298	318	0.994	$\alpha \beta^2 \gamma^{-2}=\beta^k$	$\alpha \beta^2 \gamma^{-2} = \beta^k$	f. Show that the kinetic energy scales like $\alpha \beta^2 \gamma^{-2} = \beta^k$ (use a). Since the potential															
gilmore-lie-groups_ F03299	318	1.000	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d \mathbf{x}=0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d \mathbf{x} = 0$	$\delta \int \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}) d \mathbf{x} = 0$. Show that if a scaling transformation leaves the Lagrangian invari-															
gilmore-lie-groups_ F03300	318	1.000	v_i	v_i	Show that the following six differential operators v_i are infinitesimal generators of															

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gilmore-lie-groups_ F03301	318	1.000	$e^{\epsilon v_i} f(x, t)$	$e^{\epsilon v_i} f(x, t)$	the invariance group of this equation. Show that $e^{\epsilon v_i} f(x, t)$ has the action shown for																					
gilmore-lie-groups_ F03302	319	1.000	$e^{\epsilon v_i} f(x, t) =$	$e^{\epsilon v_i} f(x, t) =$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - xu \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - xu \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03303	319	1.000	∂_x	∂_x	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - xu \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - xu \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$																								
gilmore-lie-groups_ F03304	319	1.000	$f(x - \epsilon, t)$	$f(x - \epsilon, t)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - xu \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - xu \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03305	319	1.000	∂_t	∂_t	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - xu \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - xu \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03306	319	1.000	$f(x, t - \epsilon)$	$f(x, t - \epsilon)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - xu \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - xu \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t)u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03307	319	■1.000	$u \ \partial_u$	$u \partial_u$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03308	319	■1.000	$e^{\epsilon} f(x, t)$	$e^\epsilon f(x, t)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03309	319	■1.000	$v_{\{4\}}$	v_4	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03310	319	■1.000	$x \ \partial_x + 2t \ \partial_t$	$x \partial_x + 2t \partial_t$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03311	319	■1.000	$f\left(e^{-\epsilon} x, e^{-2 \epsilon} t\right)$	$f\left(e^{-\epsilon} x, e^{-2\epsilon} t\right)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03312	319	■1.000	$v_{\{5\}}$	v_5	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03313	319	■1.000	$2\,t\,\partial_x - x\,u\,\partial_u$	$2t\partial_x - xu\partial_u$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03314	319	■1.000	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03315	319	■1.000	v_6	v_6	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03316	319	■1.000	$4\,x\,t\,\partial_x + 4\,t^2\,\partial_t - \left(x^2 + 2\,t\,\partial_u\right) u\,\partial_u$	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03317	319	■1.000	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03318	319	■1.000	$\lambda^2 = 1/(1 + 4\epsilon t)$	$\lambda^2 = 1/(1 + 4\epsilon t)$	<table><tr><th>v_i</th><th>Infinitesimal</th><th>$e^{\epsilon v_i} f(x, t) =$</th></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u\partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x\partial_x + 2t\partial_t$</td><td>$f(e^{-\epsilon}x, e^{-2\epsilon}t)$</td></tr><tr><td>$v_5$</td><td>$2t\partial_x - xu\partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$</td><td>$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u\partial_u$	$e^\epsilon f(x, t)$	v_4	$x\partial_x + 2t\partial_t$	$f(e^{-\epsilon}x, e^{-2\epsilon}t)$	v_5	$2t\partial_x - xu\partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u$	$\lambda e^{-\epsilon\lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ F03319	319	■ 1.000	$D_{\{2\}}=u \partial_u$	$D_2 = u\partial_u$	Show that $D_2 = u\partial_u$ commutes with all remaining generators. Construct the commu-
gilmore-lie-groups_ F03320	319	■ 1.000	$S_0(2+1,1+1)=S_0(3,2)$	$SO(2+1,1+1)=SO(3,2)$	$SO(2+1,1+1)=SO(3,2)$.
gilmore-lie-groups_ F03321	319	■ 1.000	$S_0(3+1,1+1)=S_0(4,2)$	$SO(3+1,1+1)=SO(4,2)$	$SO(3+1,1+1)=SO(4,2)$ (Bateman, 1910). The extra generator commutes with
gilmore-lie-groups_ F03322	319	■ 1.000	$u_{\{x\ x\}}-u_{\{t\}}=0$	$u_{xx} - u_t = 0$	8. The heat equation in one dimension is $u_{xx} - u_t = 0$. The infinitesimal generator of
gilmore-lie-groups_ F03323	319	■ 1.000	$X=\xi^i\frac{\partial}{\partial x^i}+\eta\frac{\partial}{\partial u}+\cdots=\xi^1\frac{\partial}{\partial x}+\xi^2\frac{\partial}{\partial t}+\eta\frac{\partial}{\partial u}+\cdots$	$X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \cdots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \cdots$	symmetries for this equation is $X = \xi^i \frac{\partial}{\partial x^i} + \eta \frac{\partial}{\partial u} + \cdots = \xi^1 \frac{\partial}{\partial x} + \xi^2 \frac{\partial}{\partial t} + \eta \frac{\partial}{\partial u} + \cdots$.
gilmore-lie-groups_ F03324	320	■ 1.000	$h(x, t)$	$h(x, t)$	Here $h(x, t)$ is any function that satisfies the homogeneous heat equation. Construct
gilmore-lie-groups_ F03325	320	■ 0.986	$\left(t \rightarrow t' = T(t, x, \epsilon) = t + \right.$	$(t \rightarrow t' = T(t, x, \epsilon) = t +$	Under a one-parameter family of change of variables $(t \rightarrow t' = T(t, x, \epsilon) = t +$
gilmore-lie-groups_ F03326	320	■ 1.000	$\left.\epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x) \right)$	$\epsilon \xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x))$	$\xi(t, x), x_i \rightarrow x'_i = X_i(t, x, \epsilon) = x_i + \epsilon \eta_i(t, x)$ the action integral transforms to
gilmore-lie-groups_ F03327	320	■ 1.000	$d t' / d t = \partial T / \partial t + (\partial T / \partial x_i) d x_i / d t$	$dt'/dt = \partial T/\partial t + (\partial T/\partial x_i) dx_i/dt$	where $dt'/dt = \partial T/\partial t + (\partial T/\partial x_i) dx_i/dt$. Show that if you differentiate the action
gilmore-lie-groups_ F03328	320	■ 1.000	$\epsilon = 0$	$\epsilon = 0$	integral with respect to ϵ , then set $\epsilon = 0$ the result is
gilmore-lie-groups_ F03329	321	■ 1.000	$L[u]=$	$L[u] =$	the dynamics of a system is derivable from an action integral of the form $L[u] =$
gilmore-lie-groups_ F03330	321	■ 1.000	$\int \mathcal{L}(x, u) d x, x \in R^p, u \in R^q$	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$	$\int \mathcal{L}(x, u) dx, x \in R^p, u \in R^q$, and suppose the infinitesimal generators that leave the
gilmore-lie-groups_ F03331	321	■ 1.000	$d \rho(g)$	$d\rho(g)$	12. Representation theory G is a compact Lie group with invariant measure $d\rho(g)$
gilmore-lie-groups_ F03332	321	■ 0.995	$\operatorname{Vol}(G)=\int d \rho(g), \Gamma_{\mu \nu}^{\lambda}(g)$	$\operatorname{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$	and volume $\operatorname{Vol}(G) = \int d\rho(g), \Gamma_{\mu\nu}^\lambda(g)$ are the irreducible representations of G con-
gilmore-lie-groups_ F03333	321	■ 1.000	$\phi(g), \psi(g)$	$\phi(g), \psi(g)$	structed by reduction of tensor products (Wigner–Stone theorem), and $\phi(g), \psi(g)$
gilmore-lie-groups_ F03334	322	■ 1.000	$\psi(g)=\langle g \mid \psi\rangle$	$\psi(g) = \langle g \psi \rangle$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \psi \rangle$ and $\langle \frac{\lambda}{\mu\nu} \psi \rangle =$
gilmore-lie-groups_ F03335	322	■ 1.000	$\langle \left.\right\rangle_{\mu \nu}^{\lambda} \mid \psi\rangle=\langle g \mid \psi\rangle$	$\langle \frac{\lambda}{\mu\nu} \psi \rangle =$	c. Carry out a Fourier decomposition on the functions $\psi(g) = \langle g \psi \rangle$ and $\langle \frac{\lambda}{\mu\nu} \psi \rangle =$
gilmore-lie-groups_ F03336	322	■ 1.000	$\int d \rho(g)\langle \left.\right\rangle_{\mu \nu}^{\lambda} \mid g\rangle\langle g \mid \psi\rangle$	$\int d\rho(g) \langle \frac{\lambda}{\mu\nu} g \rangle \langle g \psi \rangle$	$\int d\rho(g) \langle \frac{\lambda}{\mu\nu} g \rangle \langle g \psi \rangle$ (and similarly for $\phi(g) = \langle g \phi \rangle$) using the Dirac represen-
gilmore-lie-groups_ F03337	322	■ 1.000	$\phi(g)=\langle g \mid \phi\rangle$	$\phi(g) = \langle g \phi \rangle$	$\int d\rho(g) \langle \frac{\lambda}{\mu\nu} g \rangle \langle g \psi \rangle$ (and similarly for $\phi(g) = \langle g \phi \rangle$) using the Dirac represen-
gilmore-lie-groups_ F03338	322	■ 1.000	$\int \phi^*(g) \psi(g) d \rho(g)$	$\int \phi^*(g) \psi(g) d\rho(g)$	tation. Write down the Parseval equality for the inner product $\int \phi^*(g) \psi(g) d\rho(g)$
gilmore-lie-groups_ F03339	323	■ 1.000	He^{+}	He^+	O. BELY (1966), Quantum defect theory III. Electron scattering by He^+ , <i>Proc. Phys. Soc.</i>
gilmore-lie-groups_ F03340	324	■ 0.868	$A_{\{n\}} ; D_{\{n\}}, B_{\{n\}}, C_{\{n\}}, J$	$A_n; D_n, B_n, C_n, J$	$A_n; D_n, B_n, C_n, J$, <i>Math. Phys.</i> 11 , 513–523.
gilmore-lie-groups_ F03341	325	■ 0.866	$\mathrm{mx} / \sim$	$\mathrm{mx}/\sim$	$\mathrm{mx}/\sim$ mcintosh/comun/symm/symm.html.
gilmore-lie-groups_ F03342	327	■ 0.995	$A(p\ q),\ 39,48$	$A(pq), 39, 48$	$A(pq), 39, 48$
gilmore-lie-groups_ F03343	327	■ 0.978	$A_{\{1\}},\ 161$	$A_1, 161$	$A_1, 161$

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gilmore-lie-groups_F03344	327	■ 0.991	$A_{\{2\}}$	A_2	A_2 , 151, 160
gilmore-lie-groups_F03345	327	■ 0.865	$A_{\{3\}}$, 46, 161, 162	A_3 , 46, 161, 162	A_3 , 46, 161, 162
gilmore-lie-groups_F03346	327	■ 0.946	$A_{\{n\}}$, 46, 49, 161, 166	A_n , 46, 49, 161, 166	A_n , 46, 49, 161, 166
gilmore-lie-groups_F03347	327	■ 0.992	$B_{\{1\}}$, 161	B_1 , 161	B_1 , 161
gilmore-lie-groups_F03348	327	■ 0.749	$B_{\{2\}}$, 151, 160, 161, 164	B_2 , 151, 160, 161, 164	B_2 , 151, 160, 161, 164
gilmore-lie-groups_F03349	327	■ 0.761	$B_{\{3\}}$, 162	B_3 , 162	B_3 , 162
gilmore-lie-groups_F03350	327	■ 0.729	$\boldsymbol{B}_{\{n\}}$, 161, 166, 168	$\boldsymbol{B}_n$, 161, 166, 168	$\boldsymbol{B}_n$, 161, 166, 168
gilmore-lie-groups_F03351	327	■ 1.000	$C_{\{1\}}$, 161	C_1 , 161	C_1 , 161
gilmore-lie-groups_F03352	327	■ 0.523	$C_{\{2\}}$, 151, 160, 161, 164	C_2 , 151, 160, 161, 164	C_2 , 151, 160, 161, 164
gilmore-lie-groups_F03353	327	■ 0.712	$C_{\{3\}}$, 162	C_3 , 162	C_3 , 162
gilmore-lie-groups_F03354	327	■ 0.997	$C_{\{n\}}$, 161, 166, 168	C_n , 161, 166, 168	C_n , 161, 166, 168
gilmore-lie-groups_F03355	327	■ 0.840	$D_{\{2\}}$, 151, 160, 162	D_2 , 151, 160, 162	D_2 , 151, 160, 162
gilmore-lie-groups_F03356	327	■ 0.435	$G\ L(1\ ;\ \mathbb{Q})$, 40, 47	$GL(1; \mathbb{Q})$, 40, 47	$GL(1; \mathbb{Q})$, 40, 47
gilmore-lie-groups_F03357	327	■ 0.979	$G\ L(2\ ;\ \mathbb{C})$, 47	$GL(2; \mathbb{C})$, 47	$GL(2; \mathbb{C})$, 47
gilmore-lie-groups_F03358	327	■ 0.788	$G\ L(2\ ;\ \mathbb{R})$	$GL(2; \mathbb{R})$	$GL(2; \mathbb{R})$, 43
gilmore-lie-groups_F03359	327	■ 0.731	$G\ L(2\ ;\ \mathbb{Z})$, 45, 49	$GL(2; \mathbb{Z})$, 45, 49	$GL(2; \mathbb{Z})$, 45, 49
gilmore-lie-groups_F03360	327	■ 0.903	$G\ L(3\ ;\ \mathbb{Z})$, 45	$GL(3; \mathbb{Z})$, 45	$GL(3; \mathbb{Z})$, 45
gilmore-lie-groups_F03361	327	■ 0.471	$G\ L(n\ ;\ \mathbb{F})$, 34, 36, 74	$GL(n; \mathbb{F})$, 34, 36, 74	$GL(n; \mathbb{F})$, 34, 36, 74
gilmore-lie-groups_F03362	327	■ 0.698	$G\ L(n\ ;\ \mathbb{Q})$	$GL(n; \mathbb{Q})$	$GL(n; \mathbb{Q})$, 47
gilmore-lie-groups_F03363	327	■ 0.973	$G_{\{2\}}$, 151, 160, 162, 165	G_2 , 151, 160, 162, 165	G_2 , 151, 160, 162, 165
gilmore-lie-groups_F03364	327	■ 0.933	$\operatorname{HT}(p,\ q)$, 37, 48	$HT(p, q)$, 37, 48	$HT(p, q)$, 37, 48
gilmore-lie-groups_F03365	327	■ 0.634	$\operatorname{Nil}(n)$, 38, 48	$\operatorname{Nil}(n)$, 38, 48	$\operatorname{Nil}(n)$, 38, 48
gilmore-lie-groups_F03366	327	■ 0.788	$O(31)$, 261	$O(31)$, 261	$O(31)$, 261
gilmore-lie-groups_F03367	327	■ 0.988	$O(3)$, 40, 78	$O(3)$, 40, 78	$O(3)$, 40, 78
gilmore-lie-groups_F03368	327	■ 1.000	$O(3\ ;\ \mathbb{Z})$, 46	$O(3; \mathbb{Z})$, 46	$O(3; \mathbb{Z})$, 46
gilmore-lie-groups_F03369	327	■ 0.991	$O(n)$, 40, 43, 145	$O(n)$, 40, 43, 145	$O(n)$, 40, 43, 145
gilmore-lie-groups_F03370	327	■ 1.000	$O(n\ ;\ \mathbb{G})$, 41	$O(n; \mathbb{G})$, 41	$O(n; \mathbb{G})$, 41
gilmore-lie-groups_F03371	327	■ 1.000	$O(n\ ;\ \mathbb{Z})$, 45, 49	$O(n; \mathbb{Z})$, 45, 49	$O(n; \mathbb{Z})$, 45, 49
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gilmore-lie-groups_F03376	327	■ 0.956	$S\,L(n\,;\,\mathbb{R})$, 30	$SL(n;\mathbb{R})$, 30	$SL(n;\mathbb{R})$, 30
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gilmore-lie-groups_F03378	327	■ 0.519	$\operatorname{SL}(n\,;\,\mathbb{Q})$	$SL(n;\mathbb{Q})$	$SL(n;\mathbb{Q})$, 43
gilmore-lie-groups_F03379	327	■ 0.747	$\operatorname{SL}(n\,;\,\mathbb{R})$	$SL(n;\mathbb{R})$	$SL(n;\mathbb{R})$, 43, 47, 164
gilmore-lie-groups_F03380	327	■ 0.880	$S\,L(n\,;\,\mathbb{Z})$, 45	$SL(n;\mathbb{Z})$, 45	$SL(n;\mathbb{Z})$, 45
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gilmore-lie-groups_F03382	327	■ 0.943	$S\,O(2,1)\,/\,S\,O(2)$, 106	$SO(2,1)/SO(2)$, 106	$SO(2,1)/SO(2)$, 106
gilmore-lie-groups_F03383	327	■ 0.820	$\operatorname{SO}(2)$, 48,164	$SO(2)$, 48, 164	$SO(2)$, 48, 164
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gilmore-lie-groups_F03386	327	■ 0.607	$S\,O(3,1)$, 263	$SO(3,1)$, 263	$SO(3,1)$, 263
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gilmore-lie-groups_F03389	327	■ 0.498	$\operatorname{SO}(4,1)$, 210	$SO(4,1)$, 210	$SO(4,1)$, 210
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gilmore-lie-groups_F03391	327	■ 0.637	$\operatorname{SU}(1\,;\,\mathbb{Q})$, 40,48	$SU(1;\mathbb{Q})$, 40, 48	$SU(1;\mathbb{Q})$, 40, 48
gilmore-lie-groups_F03392	327	■ 0.998	$\operatorname{SU}(1,1)$, 38,43,48,105	$SU(1,1)$, 38, 43, 48, 105	$SU(1,1)$, 38, 43, 48, 105
gilmore-lie-groups_F03393	327	■ 1.000	$S\,U(1,1)\,/\,U(1)$, 106	$SU(1,1)/U(1)$, 106	$SU(1,1)/U(1)$, 106
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gilmore-lie-groups_F03396	327	■ 0.983	$\operatorname{SU}(n)$, 43,90,164	$SU(n)$, 43, 90, 164	$SU(n)$, 43, 90, 164
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gilmore-lie-groups_F03399	328	■ 0.886	$S_{\{n\}}$, 45,49	S_n , 45, 49	S_n , 45, 49
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gilmore-lie-groups_F03403	328	■ 0.883	$\operatorname{Sp}(2 n ; \mathbb{R})$, 41	$\operatorname{Sp}(2n; \mathbb{R})$, 41	<i>Sp</i> (2 <i>n</i> ; $\mathbb{R}$), 41
gilmore-lie-groups_F03404	328	■ 0.739	$\operatorname{Sp}(n)$, 40, 164	$\operatorname{Sp}(n)$, 40, 164	<i>Sp</i> (<i>n</i>), 40, 164
gilmore-lie-groups_F03405	328	■ 0.999	$\operatorname{Sp}(n ; \mathbb{C})$, 41	$\operatorname{Sp}(n; \mathbb{C})$, 41	<i>Sp</i> (<i>n</i> ; $\mathbb{C}$), 41
gilmore-lie-groups_F03406	328	■ 0.944	$\operatorname{Sp}(n ; \mathbb{G})$, 41	$\operatorname{Sp}(n; \mathbb{G})$, 41	<i>Sp</i> (<i>n</i> ; $\mathbb{G}$), 41
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gilmore-lie-groups_F03408	328	■ 0.903	$\operatorname{Sp}(p, q)$, 164	$\operatorname{Sp}(p, q)$, 164	<i>Sp</i> (<i>p</i> , <i>q</i>), 164
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gilmore-lie-groups_F03410	328	■ 0.619	$U(2)$, 40, 78	$U(2)$, 40, 78	<i>U</i> (2), 40, 78
gilmore-lie-groups_F03411	328	■ 1.000	$U(2 ; \mathbb{Q})$, 164	$U(2; \mathbb{Q})$, 164	<i>U</i> (2; $\mathbb{Q}$), 164
gilmore-lie-groups_F03412	328	■ 0.804	$U(n)$, 40, 43, 90	$U(n)$, 40, 43, 90	<i>U</i> (<i>n</i>), 40, 43, 90
gilmore-lie-groups_F03413	328	■ 1.000	$U(n ; \mathbb{G})$, 41	$U(n; \mathbb{G})$, 41	<i>U</i> (<i>n</i> ; $\mathbb{G}$), 41
gilmore-lie-groups_F03414	328	■ 1.000	$U(p, q)$, 43	$U(p, q)$, 43	<i>U</i> (<i>p</i> , <i>q</i>), 43
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gilmore-lie-groups_F03416	328	■ 0.397	$\operatorname{UT}(p, q)$, 48	$\operatorname{UT}(p, q)$, 48	<i>UT</i> (<i>p</i> , <i>q</i>), 48
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gilmore-lie-groups_F03422	329	■ 0.560	$\overline{S U(1, 1) / U(1)}$, 108	$\overline{SU(1, 1) / U(1)}$, 108	<u><i>SU</i></u> (1, 1)/ <u><i>U</i></u> (1), 108
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gilmore-lie-groups_F03429	330	■ 0.997	$\mathfrak{o}(p, q)$, 78	$\mathfrak{o}(p, q)$, 78	$\mathfrak{o}(p, q)$, 78
gilmore-lie-groups_F03430	330	■ 0.630	$\mathfrak{sl}(2 ; \mathbb{R})$, 100, 102, 154, 173	$\mathfrak{sl}(2; \mathbb{R})$, 100, 102, 154, 173	$\mathfrak{sl}(2; \mathbb{R})$, 100, 102, 154, 173

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gilmore-lie-groups_F03431	330	■ 0.811	$\mathfrak{sl}(n)$, 80	$\mathfrak{sl}(n)$, 80	$\mathfrak{sl}(n)$, 80
gilmore-lie-groups_F03432	330	■ 0.637	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86	$\mathfrak{sl}(n; \mathbb{C})$, 80, 85, 86
gilmore-lie-groups_F03433	330	■ 0.940	$\mathfrak{sl}(n; \mathbb{Q})$, 80, 86	$\mathfrak{sl}(n; \mathbb{Q})$, 80, 86	$\mathfrak{sl}(n; \mathbb{Q})$, 80, 86
gilmore-lie-groups_F03434	330	■ 0.743	$\mathfrak{sl}(n; \mathbb{R})$, 80, 85, 178, 180	$\mathfrak{sl}(n; \mathbb{R})$, 80, 85, 178, 180	$\mathfrak{sl}(n; \mathbb{R})$, 80, 85, 178, 180
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gilmore-lie-groups_F03449	330	■ 0.788	$\mathfrak{sp}(G; \mathbb{R})$, 79	$\mathfrak{sp}(G; \mathbb{R})$, 79	$\mathfrak{sp}(G; \mathbb{R})$, 79
gilmore-lie-groups_F03450	330	■ 0.986	$\mathfrak{sp}(n)$, 132, 178	$\mathfrak{sp}(n)$, 132, 178	$\mathfrak{sp}(n)$, 132, 178
gilmore-lie-groups_F03451	330	■ 0.791	$\mathfrak{sp}(n; G)$, 79	$\mathfrak{sp}(n; G)$, 79	$\mathfrak{sp}(n; G)$, 79
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gilmore-lie-groups_F03455	331	■ 0.943	$\mathfrak{su}(2n)$, 180	$\mathfrak{su}(2n)$, 180	$\mathfrak{su}(2n)$, 180
gilmore-lie-groups_F03456	331	■ 0.857	$\mathfrak{su}(n)$, 80, 132, 178	$\mathfrak{su}(n)$, 80, 132, 178	$\mathfrak{su}(n)$, 80, 132, 178
gilmore-lie-groups_F03457	331	■ 1.000	$\mathfrak{su}(p, q)$, 85, 178	$\mathfrak{su}(p, q)$, 85, 178	$\mathfrak{su}(p, q)$, 85, 178
gilmore-lie-groups_F03458	331	■ 0.882	$\mathfrak{su}^*(2n)$, 180	$\mathfrak{su}^*(2n)$, 180	$\mathfrak{su}^*(2n)$, 180
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gilmore-lie-groups_ F03464	331	■ 1.000	$\mathfrak{u}(n;G)$, 79	$\mathfrak{u}(n;G)$, 79	$\mathfrak{u}(n;G)$, 79
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